

# Peer Interface Volume

For

**A-GRA**

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# 1 PEER INTERFACE

The role of the Peer interface is to provide communication between multiple MA-enabled ACPs, enabling data to be shared to facilitate collaborative perception and behaviors among one or more teams of ACPs. The Peer interface covers sharing information relating to teaming and formation, custody, contingency, and discovery via UCI messaging. ACPs send and receive off-board OTA network messages via the DMS (Decentralized Messaging Service).

## 1.1 Interface Description

The role of an MA to peer system in the operation of MA-enabled ACPs is to provide the transfer of vehicle data, facilitate direct communication among multiple ACPs, and enable collaboration between all MA platforms by simplifying the sending and receiving of messages from C2 to a team through the RBAC construct. A peer is defined as an A-GRA compliant ACP operating within the same theater as another instance of MA. This document will refer to one ACP as MA and will refer to additional MA-compliant ACPs as peers. Although most diagrams depict transfer between MA and a single peer, MA can communicate with multiple peers simultaneously.

The RBAC construct does not require peers to have custody over one another but rather relies on the team leader to authorize the message received from C2 to execute team commands. MA will verify that all messages sent by C2 are authorized via use of role and certificate information provided during mission planning. Once the team leader has verified that the message comes from an authorized role, it will send any required messages to peers in order to support the C2 command. Upon creating or joining a team, peers will automatically accept any message sent to them by the team leader. Peers should only accept messages either from the team leader or directly from C2 to be verified with an authorized role. If the message sent to the team leader is unauthorized, MA will respond to the C2 source with the according rejection response and will not send any message to peers. For more information, see the C2 RBAC section.

## 1.2 Interface Interactions

The peer interactions with MA have been scoped as shown in the below figure. Within each interaction there is a baseline set of functionalities that must be supported by MA and peer systems for a compliant system.

All off-board messaging between ACPs occurs via their respective DMS which is defined in the Mission Systems section. The core method for messages to transfer between peer MAs utilizes a UCI message wrapper (*MA\_TxDataPayloadCommandMT*) to send messages to the DMS and another UCI message wrapper (*MA\_RxDataPayloadMT*) the DMS publishes when messages are received. These wrapper messages are specified in more detail in the Mission Systems section. The "Additional Comms Capability" MUC specifies a more holistic approach to managing the OTA communications with a Data Dissemination Plan. The Data Dissemination Plan specifies how often and whether ingress/egress messages will be subscribed/published directly on the ASB or if the UCI wrappers can be used. The MAs are responsible for authentication of the received messages before processing.



Figure 1-1: Peer to Peer Interactions

1.2.1 Peer Acknowledgment Configurations Package

1.2.1.1 P2P Message Acknowledgment Configuration Status

This interaction provides the current Message Acknowledgment Configuration defining what messages one MA will respond to with an acknowledgment upon receipt. MA will report the current Message Acknowledgment Configuration on change by sending a *MA\_MessageAcknowledgmentConfigurationMT* message.

The *DefaultGroupMessageFreq* field defines the frequency at which the Grouped Acknowledgment message is set. If the field is not set, there is no grouping. The *DefaultMessagesToAck* field sets the list of messages that an Acknowledgment message will be sent for when received. If the field is not set, all messages listed in *MA\_AckMessageEnum* will receive a response. The *ReceiverSpecificSettings* field specifies how MA will respond to individual Peers if they do not fall under the default settings. If MA does not support Message Acknowledgment Configuration, the *ConfigurationSupported* field will be set to "FALSE" and it will respond with an ungrouped acknowledgment for all possible messages listed in *MA\_AckMessageEnum*.

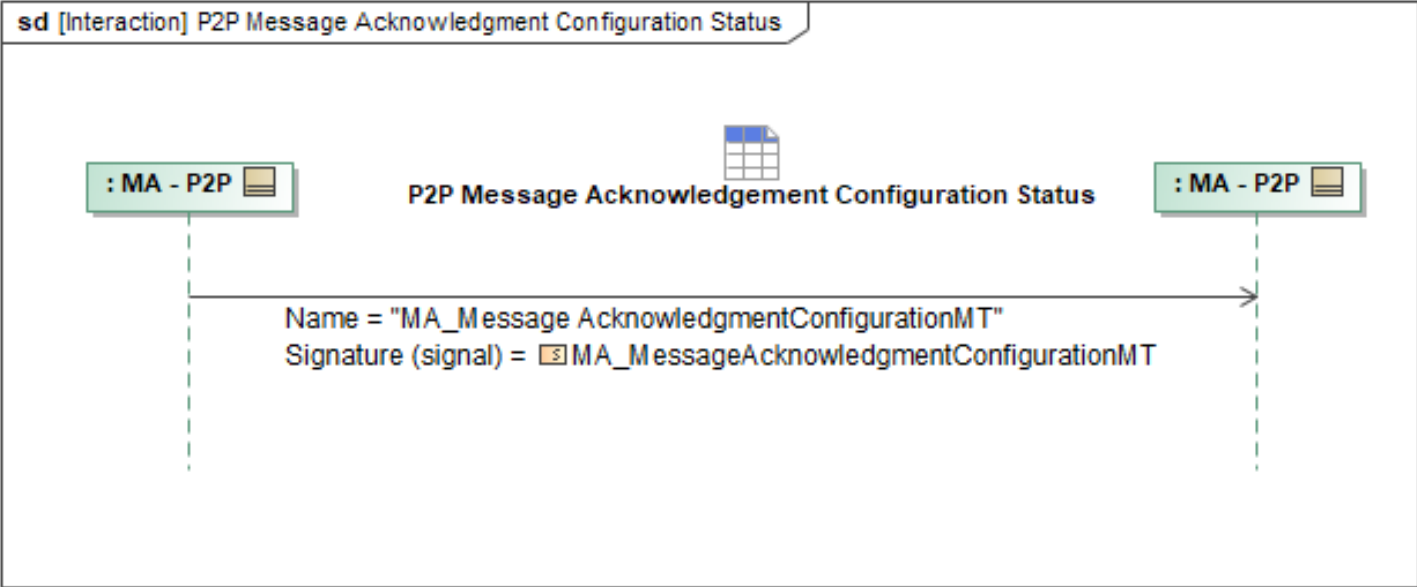


Figure 1-2: P2P Message Acknowledgment Configuration Status

Table 1-1: P2P Message Acknowledgment Configuration Status

| Message Name (Name:Type)                                                                | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------------------|-----------------------------|
| MA_Message<br>AcknowledgmentConfigurationMT:<br>MA_MessageAcknowledgmentConfigurationMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.2 P2P Update Message Acknowledgment Configurations

This interaction specifies when one MA wants to set or update the Message Acknowledgment Configuration of another MA. MA can update default responses by filling out the



*DefaultGroupMessageFreq* and *DefaultMessagesToAck*. MA can also use the *RequestedReceiverSpecificSetting* field to define the messages that MA expects a Peer MA to respond with an acknowledgment for and if the messages will be grouped. MA should only request *RequestedReceiverSpecificSetting* for itself and not other systems.

MA sends the *MA\_MessageAcknowledgmentConfigurationRequestMT* message to the Peer MA. The Peer MA may accept or reject the configuration request. The Peer MA accepts the request by sending a *MA\_MessageAcknowledgmentConfigurationRequestStatusMT\_1* with the *RequestProcessingState* set to "COMPLETED", followed by a *MA\_MessageAcknowledgmentConfigurationMT* that reflects the updated configuration.

The Peer MA may reject the configuration request for the following reasons:

- It does not support configuration. Peer MA sends a *MA\_MessageAcknowledgmentConfigurationRequestStatusMT\_2* message and sets the *RequestProcessingStateReason* to "SYSTEM\_UNAVAILABLE."
- There is limited bandwidth. Peer MA sends a *MA\_MessageAcknowledgmentConfigurationRequestStatusMT\_2* message, sets the *RequestProcessingStateReason* to "INSUFFICIENT\_RESOURCES", and fills out the description field.
- The system receives an unsupported input. Peer MA sends a *MA\_MessageAcknowledgmentConfigurationRequestStatusMT\_2* message and sets the *RequestProcessingStateReason* to "INVALID\_INPUT\_PARAMETER."
- MA requests *RequestedReceiverSpecificSetting* for another system. Peer MA sends a *MA\_MessageAcknowledgmentConfigurationRequestStatusMT\_2* message and sets the *RequestProcessingStateReason* to "INELIGIBLE\_CONTROL\_SOURCE."



Figure 1-3: P2P Update Message Acknowledgment Configurations

Table 1-2: P2P Update Message Acknowledgment Configurations

| Message Name (Name:Type)                                                                                        | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------|
| MA_MessageAcknowledgmentConfigurationMT<br>:<br>MA_MessageAcknowledgmentConfigurationMT                         | None                        |
| MA_MessageAcknowledgmentConfigurationRequestMT:<br>MA_MessageAcknowledgmentConfigurationRequestMT               | None                        |
| MA_MessageAcknowledgmentConfigurationRequestStatusMT_1:<br>MA_MessageAcknowledgmentConfigurationRequestStatusMT | None                        |
| MA_MessageAcknowledgmentConfigurationRequestStatusMT_2:<br>MA_MessageAcknowledgmentConfigurationRequestStatusMT | None                        |



Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.2 Peer CAP Package

### 1.2.2.1 Receive CAP Execution Status

This interaction specifies the CAP Execution Status message that is sent between peers. For ACPs within a Package, Followers executing a CAP will send *MA\_CAP\_ExecutionStatusReportMT* periodically to other ACPs, including the Package Leader, or when their *CAP\_States* have changed. Defined in the *MA\_CAP\_ExecutionStatusReportMT* messages is a set of package-level fields as well as a set of vehicle-level fields. At the package-level, the *PackageID* may be specified, and the *PackageRequirementID* may be defined to include the IDs the package-level CAP Task and/or action. The *AOI\_Coverage* and *PlanMetrics* fields are not intended to be specified for P2P exchanges as these fields are intended to be defined when reporting status to C2 (See "Share CAP Execution Status" for more details). At the vehicle-level, *SystemID* shall be set to the ID of the ACP, *RequirementID* shall be set to the ID of the vehicle-level CAP Task, and the *CAP\_Status* object shall be defined. Included in the *CAP\_Status* object is the required *CAP\_State* field and optional status fields. Defined in the *MA\_TaskStatusMT* message are status fields of the vehicle-level CAP Tasks. The *TaskID*, *ExecutingSystemID*, and *ExecutionState* fields shall be defined. Optionally, the *TimeWindow* field may be defined to specify the remaining time left performing the CAP Task. Within the *MA\_TaskStatusMT* message, the *TimeWindow* field may be defined to specify the remaining time-on-station for performing the CAP Task.

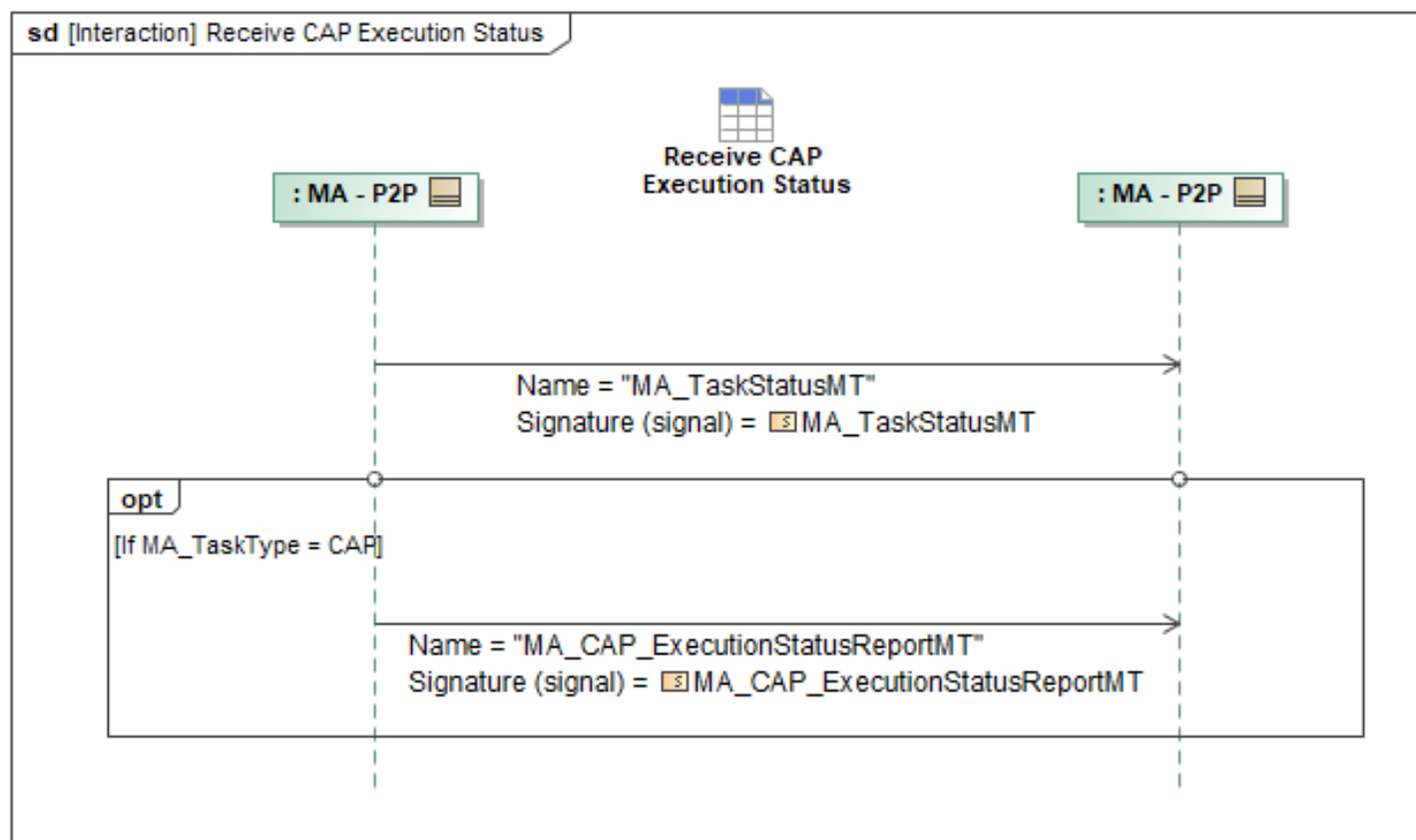


Figure 1-4: Receive CAP Execution Status

Table 1-3: Receive CAP Execution Status

| Message Name (Name:Type)                                          | Required Fields (Name:Type) |
|-------------------------------------------------------------------|-----------------------------|
| MA_CAP_ExecutionStatusReportMT:<br>MA_CAP_ExecutionStatusReportMT | None                        |
| MA_TaskStatusMT: MA_TaskStatusMT                                  | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.3 Peer Communications Package

### 1.2.3.1 Peer Overrides Data Dissemination Plan

**Mission Use Case:** Additional Comms Capabilities

In some situations a modification to the Data Dissemination Plan is required that are short term in nature and a complete update to the *MA\_DataDisseminationPlanMT* message is not required. The *MA\_DataDisseminationPlanOverrideRequestMT* can be used for these sorts of modifications. Peer-1 MA can send the *MA\_DataDisseminationPlanOverrideRequestMT* message to Peer-2 MA which forwards it to Peer-2 MS. Peer-2 MS can then accept or reject the request(s). If the request(s) are accepted, Peer-2 MS sends the *MA\_DataDisseminationPlanOverrideRequestStatusMT\_1* message to Peer-2 MA who forwards it to Peer-1 MA with the Request(s) having the Request.RequestProcessingState set to "QUEUED", "PROCESSING", "CANCELED", or

"COMPLETED" as needed. Depending on the original request, the *EffectivityActivationStatus* or *OperatorModificationStatus* fields will be filled out as needed. The interaction "Peer Updates Data Dissemination Plan Activation Status" is initiated which will provide an update to the active status and include the *OverrideModificationState* associated with the override request status. For each applicable Gateway Activity, Peer-2 MS sends the *MA\_GatewayActivityMT* message(s) to Peer-2 MA who forwards it to Peer-1 MA which will include any modifications that were made. If the request is NOT accepted, Peer-2 MS sends the *MA\_DataDisseminationPlanOverrideRequestStatusMT\_2* message to Peer-2 MA who forwards it to Peer-1 MA with the *RequestProcessingState* set to "REJECTED" or "FAILED", and the *RequestProcessingStateReason* filled out as needed. The *EffectivityActivationStatus* or *OperatorModificationStatus* fields will be filled out outlining the reason for the rejections/failures.

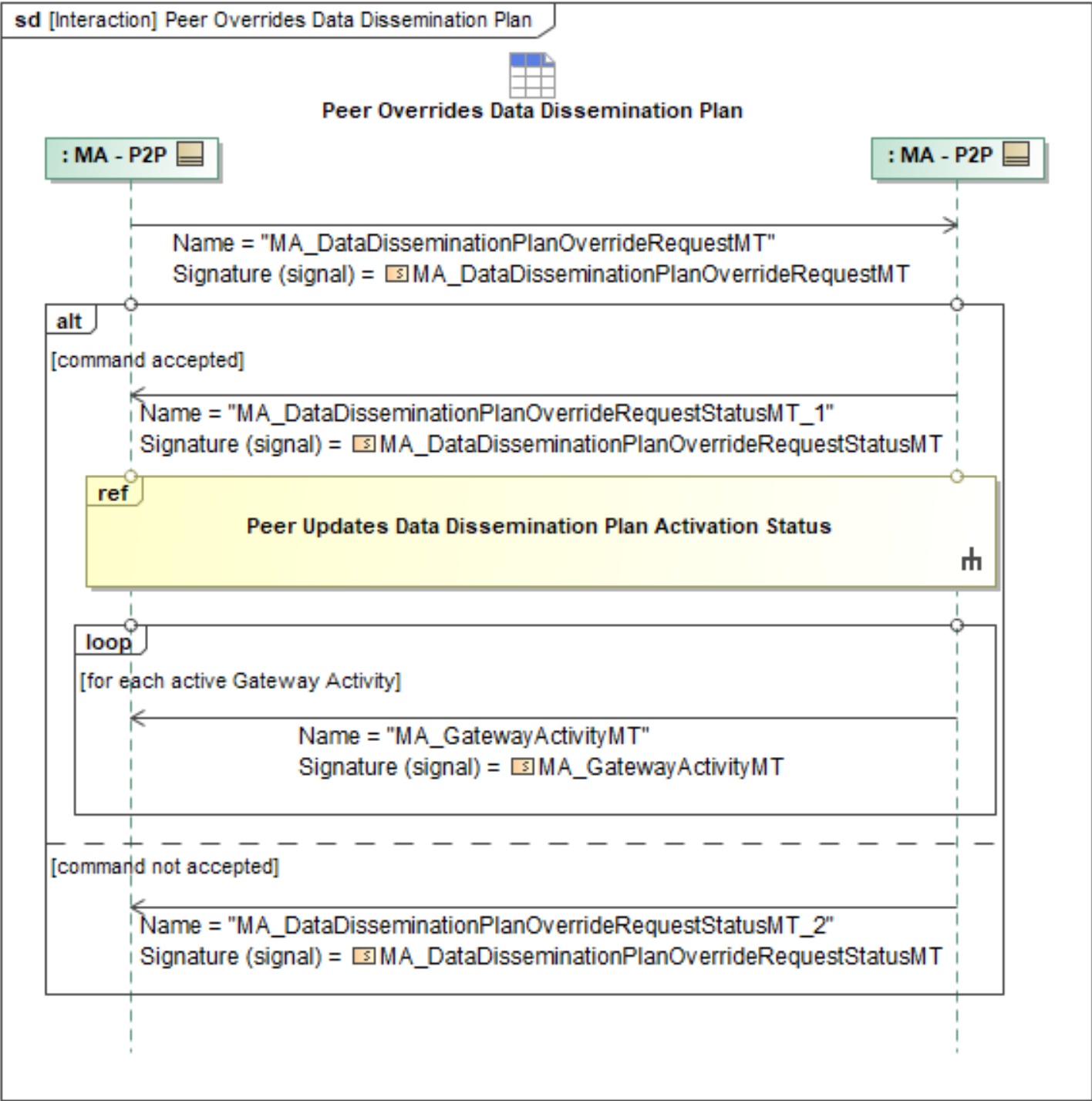


Figure 1-5: Peer Overrides Data Dissemination Plan

Table 1-4: Peer Overrides Data Dissemination Plan

| Message Name (Name:Type)                                                                    | Required Fields (Name:Type)                |
|---------------------------------------------------------------------------------------------|--------------------------------------------|
| MA_DataDisseminationPlanOverrideRequestMT<br>:<br>MA_DataDisseminationPlanOverrideRequestMT | AllocatedToSystemIDs: MA_PackageSystemType |

| Message Name (Name:Type)                                                                              | Required Fields (Name:Type)                    |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------|
| MA_DataDisseminationPlanOverrideRequestStatusMT_1:<br>MA_DataDisseminationPlanOverrideRequestStatusMT | None                                           |
| MA_DataDisseminationPlanOverrideRequestStatusMT_2:<br>MA_DataDisseminationPlanOverrideRequestStatusMT | RequestProcessingStateReason: CannotComplyType |
| MA_GatewayActivityMT:<br>MA_GatewayActivityMT                                                         | None                                           |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.3.2 Peer Updates Comms Gateway Status

**Mission Use Case:** Additional Comms Capabilities

The Gateway Capability in the local MS is how messages egress or ingress over the air. To provide information about each of the active gateways available for use, the local MA receives and forwards the *GatewayCapabilityMT* and *GatewayCapabilityStatusMT* messages from the local MS to a peer MA. Then, for each active Gateway Activity, the local MA also forwards the *MA\_GatewayActivityMT* message(s) to a peer MA. These messages provide the information needed to understand the state of the over the air messaging, they will provide context for how things are currently configured, and provide information for how to configure over the air messaging. MS typically configures the gateways, but the information can be useful when creating the plans that a MS uses.

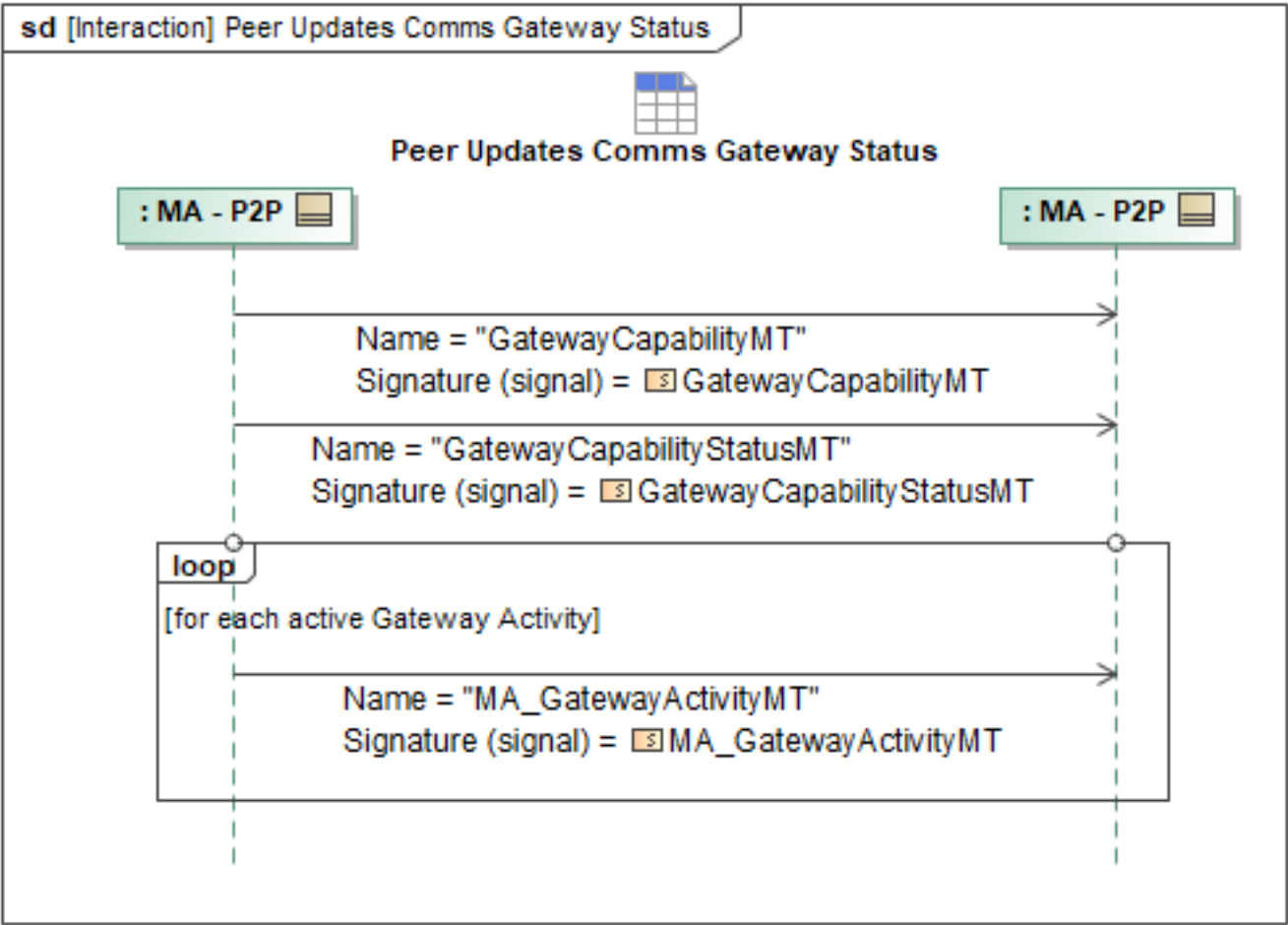


Figure 1-6: Peer Updates Comms Gateway Status

Table 1-5: Peer Updates Comms Gateway Status

| Message Name (Name:Type)                             | Required Fields (Name:Type) |
|------------------------------------------------------|-----------------------------|
| GatewayCapabilityMT: GatewayCapabilityMT             | None                        |
| GatewayCapabilityStatusMT: GatewayCapabilityStatusMT | None                        |
| MA_GatewayActivityMT: MA_GatewayActivityMT           | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3.3 Peer Updates Data Dissemination Plan

**Mission Use Case:** Additional Comms Capabilities

The *MA\_DataDisseminationPlanMT* is used to configure how messages ingress and egress over the air. MS manages/implements the Data Dissemination Plan and Mission Autonomy acts as a pass through if a Peer MA needs to update and/or activate the plan. The MS section "Update Data Dissemination Plan" details the contents of the *MA\_DataDisseminationPlanMT* message and how it can be configured.

If the Data Dissemination Plan needs update or is a new plan, Peer-1 MA sends the *MA\_DataDisseminationPlanMT* message and the *MA\_DataDisseminationPlanActivationCommandMT* message to Peer-2 MA who forwards them to Peer-2 MS. If the command is accepted by Peer-2 MS, Peer-2 MS sends the *MA\_DataDisseminationPlanActivationCommandStatusMT\_1* message to Peer-2 MA who forwards it to Peer-1 MA with the *CommandProcessingStateEnum* set to "ACCEPTED", the interaction "Peer Updates Data Dissemination Plan Activation Status" is initiated, and for each applicable Gateway Activity Peer-2 MS sends the *MA\_GatewayActivityMT* message(s) to Peer-2 MA who forwards them to Peer-1 MA. If the command is NOT accepted, Peer-2 MS send the *MA\_DataDisseminationPlanActivationCommandStatusMT\_2* message to Peer-2 MA who forwards it to Peer-1 MA with the *CommandProcessingStateEnum* set to "REJECTED" and the *PlanActivationCommandStatus* is set according to the reason it was rejected.



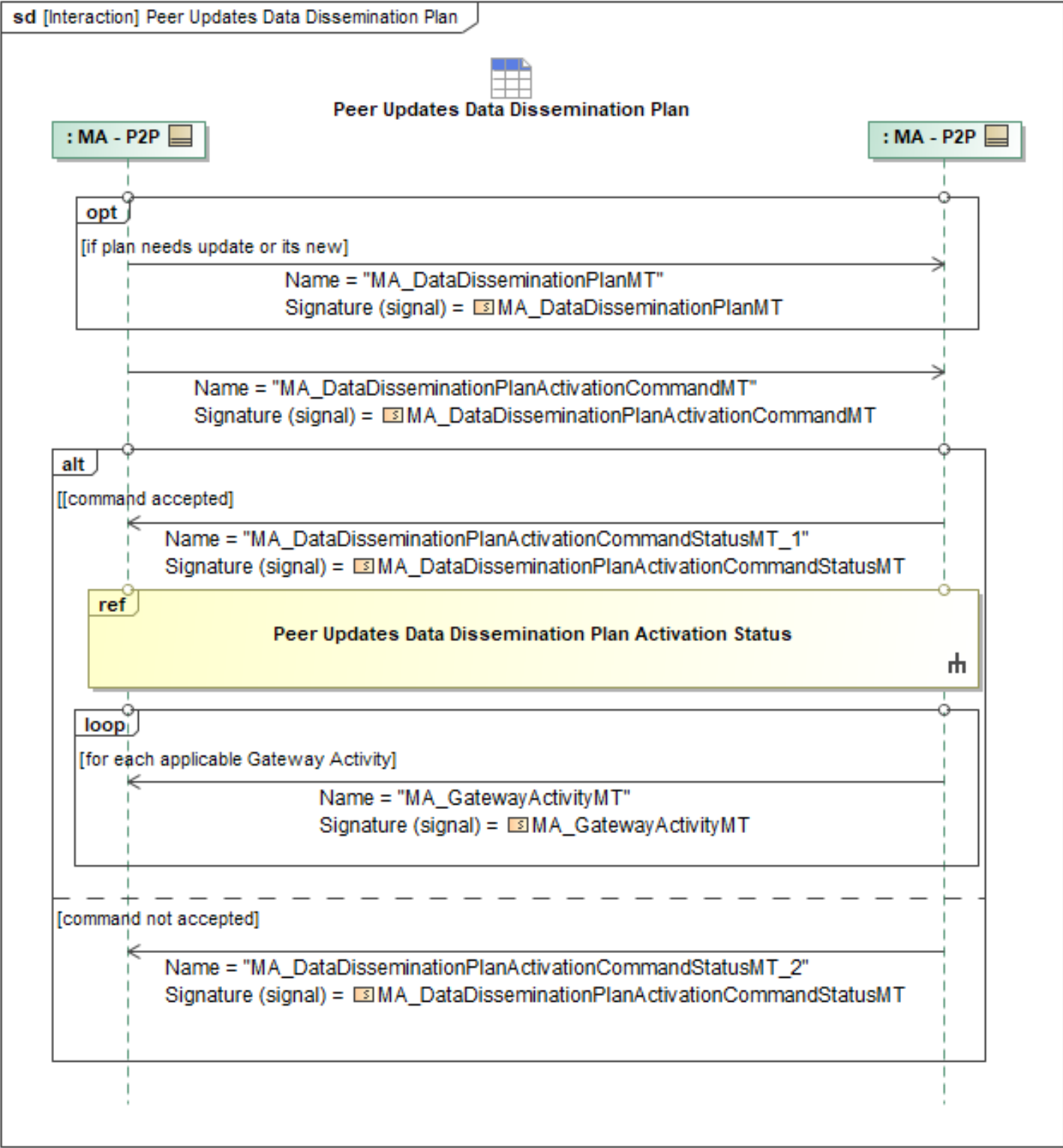


Figure 1-7: Peer Updates Data Dissemination Plan

Table 1-6: Peer Updates Data Dissemination Plan

| Message Name (Name:Type)                                                                                  | Required Fields (Name:Type)                    |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------|
| MA_DataDisseminationPlanActivationCommandMT:<br>MA_DataDisseminationPlanActivationCommandMT               | Applicability: MA_PackageSystemType            |
| MA_DataDisseminationPlanActivationCommandStatusMT_1:<br>MA_DataDisseminationPlanActivationCommandStatusMT | None                                           |
| MA_DataDisseminationPlanActivationCommandStatusMT_2:<br>MA_DataDisseminationPlanActivationCommandStatusMT | RequestProcessingStateReason: CannotComplyType |
| MA_DataDisseminationPlanMT:<br>MA_DataDisseminationPlanMT                                                 | None                                           |
| MA_GatewayActivityMT:<br>MA_GatewayActivityMT                                                             | None                                           |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.3.4 Peer Updates Data Dissemination Plan Activation Status

**Mission Use Case:** Additional Comms Capabilities

The local MS will send the *MA\_DataDisseminationPlanActivationStatusMT* message to the local MA who forwards it to a peer MA which reflect the active state of local Data Dissemination Plan. This can be sent periodically, but will always be sent upon change to the active state. The message will reflect the active Effectivities and Override Modifications of the local Data Dissemination Plan.

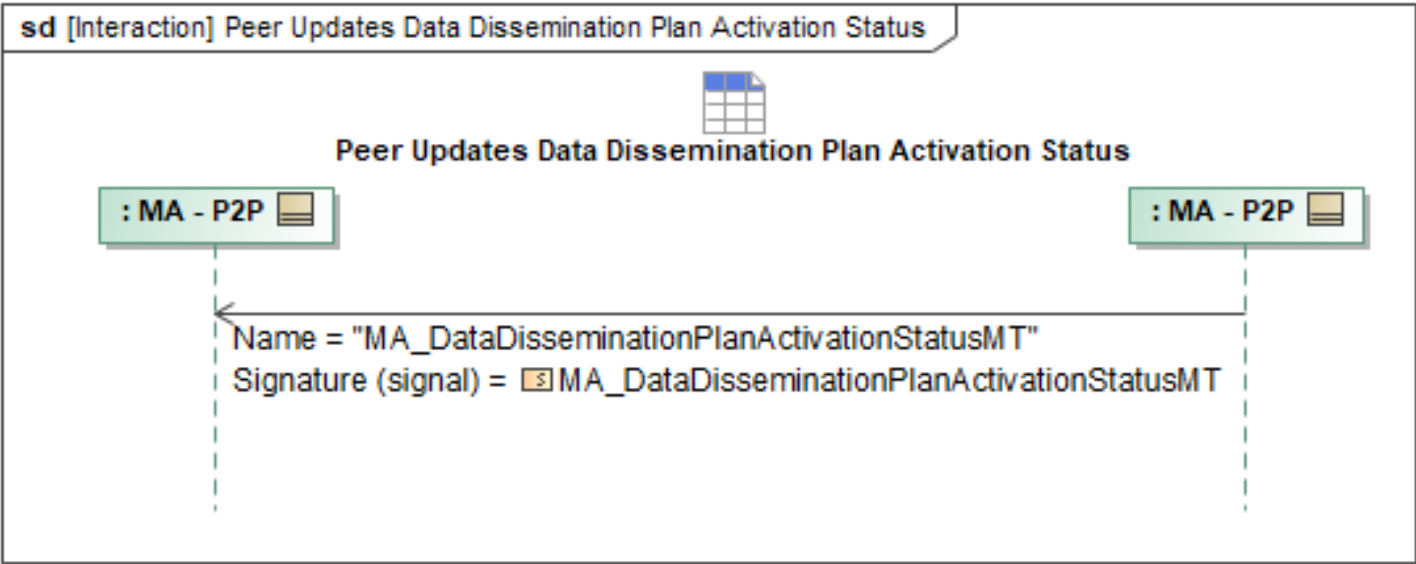


Figure 1-8: Peer Updates Data Dissemination Plan Activation Status

Table 1-7: Peer Updates Data Dissemination Plan Activation Status

| Message Name (Name:Type)                                                                      | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------------------------|-----------------------------|
| MA_DataDisseminationPlanActivationStatusMT<br>:<br>MA_DataDisseminationPlanActivationStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4 Peer Contingencies Package

In the event of a lost communications between members or a team, or a Team Leader and C2 the team will behave in the ways described below. It is important to note that MA assumes that loss of communications to a node, whether it be C2 or a peer, is reported by the Mission Systems Communications Manager or is determined by the absence of a heartbeat message past a configurable timeout interval. During planned communication outages, MA depends on MS for lost comms notification. Further material on these requirements can be found in section 4.6 Mission Systems Interface.

1.2.4.1 Define Autonomous Contingency Re-Plan

MA Team Leads can configure Peer MA to initiate autonomous replanning in the event of a contingency. Team Lead MA will send a *MA\_PlanningFunctionSettingsCommandMT* to set the contingency trigger and indicate the desired replanning that should occur if the contingency condition is met. *MissionPlanningAutonomy.AutonomousPlanningResponseChoice* field should use *AutonomousPlanningAction* or *AlertOnly* field should be "FALSE". *MissionPlanningAutonomy.AutonomousPlanningResponseChoice.AutonomousPlanningAction* is used to configure what type of replanning Peer MA should initiate. *MissionPlanningAutonomy.AutonomousPlanningResponseChoice.AutonomousPlanningAction* should include an *AutonomousPlanningActionID* to be used to reference this planning action, as well as *AllowedType* field which will configure the specifics of the planning process.

*AllowedType.PlanningProcessID* should be set to the desired ID for this planning process and will be used to reference this process in future messages. *AllowedType.ForPlanningUseOnly* should be set to "FALSE" as this plan must be executed to resolve the contingency.

When Peer MA receives *MA\_PlanningFunctionSettingsCommandMT*, it replies with *MA\_PlanningFunctionSettingsCommandStatusMT*, with *CommandedProcessingState* set to "ACCEPTED" if the system is able to apply the autonomy settings or "REJECTED" if the system is not able to apply these settings. Should it be rejected, the *CommandProcessingStateReason* field will be populated with the reason for the rejection of the *PlanningFunctionSettingsCommand*.

Peer MA will determine whether Team Lead MA is authorized to command the initial *MA\_PlanningFunctionSettingsCommandMT*. If Team Lead MA is unauthorized, Peer MA will ignore the *MA\_PlanningFunctionSettingsCommandMT* command and reply with *MA\_PlanningFunctionSettingsCommandStatusMT* where *CommandProcessingStateEnum* is set to "REJECTED" and *CommandProcessingStateReason* shall define the *CannotComplyEnum* as "INELIGIBLE\_CONTROL\_SOURCE".

In order for Peer MA to execute a generated plan, the plan must first be approved per the *ApprovalPolicy*. Mission Planning will set the criteria for what plans can be approved with the use of *MA\_ApprovalPolicyMT* by an operator vs. which plans can be automatically activated by MA.

In the event Team Lead MA wishes to modify an existing approval policy, Team Lead MA sends an *MA\_ApprovalPolicyMT* message, which includes a *DataRecordInstanceID*. For contingencies, the approval policy will have an *ApprovalPolicyID* set to the desired ID of the *ApprovalPolicy*. The *Plans* field should have *Plans.ApplicablePlanningActivity* set to \*\_PLAN, where the asterisk indicates the appropriate requirement for this planning service. Additionally, the approval policy should have *ByTriggerPolicy.TriggerSource* set to "AUTONOMOUS\_TRIGGER" to indicate this policy applies to replanning from autonomous triggers and not operator-commanded replanning.

*ByTriggerPolicy.Trigger* can be used to specify that this policy is only for a specific type of trigger. The *MA\_ApprovalPolicyMT* will also have *PlanActivation.Policy.ApprovalRequirement* set to either "AUTO\_APPROVE" if Team Lead MA wishes the contingency replan to be executed immediately, or "APPROVAL\_REQUIRED" which will ask Team Lead MA for approval of the proposed plan immediately before execution. *MA\_ApprovalPolicyMT* with the desired settings, specifically *Plans.ByTriggerPolicy.Policy* can be set to "APPROVAL\_REQUIRED" to require MA to wait for operator approval before activating an autonomous replan in response to the defined contingency trigger.

For more information on execution sequences with respect to autonomous replanning, see section Triggered Autonomous Contingency Replan.

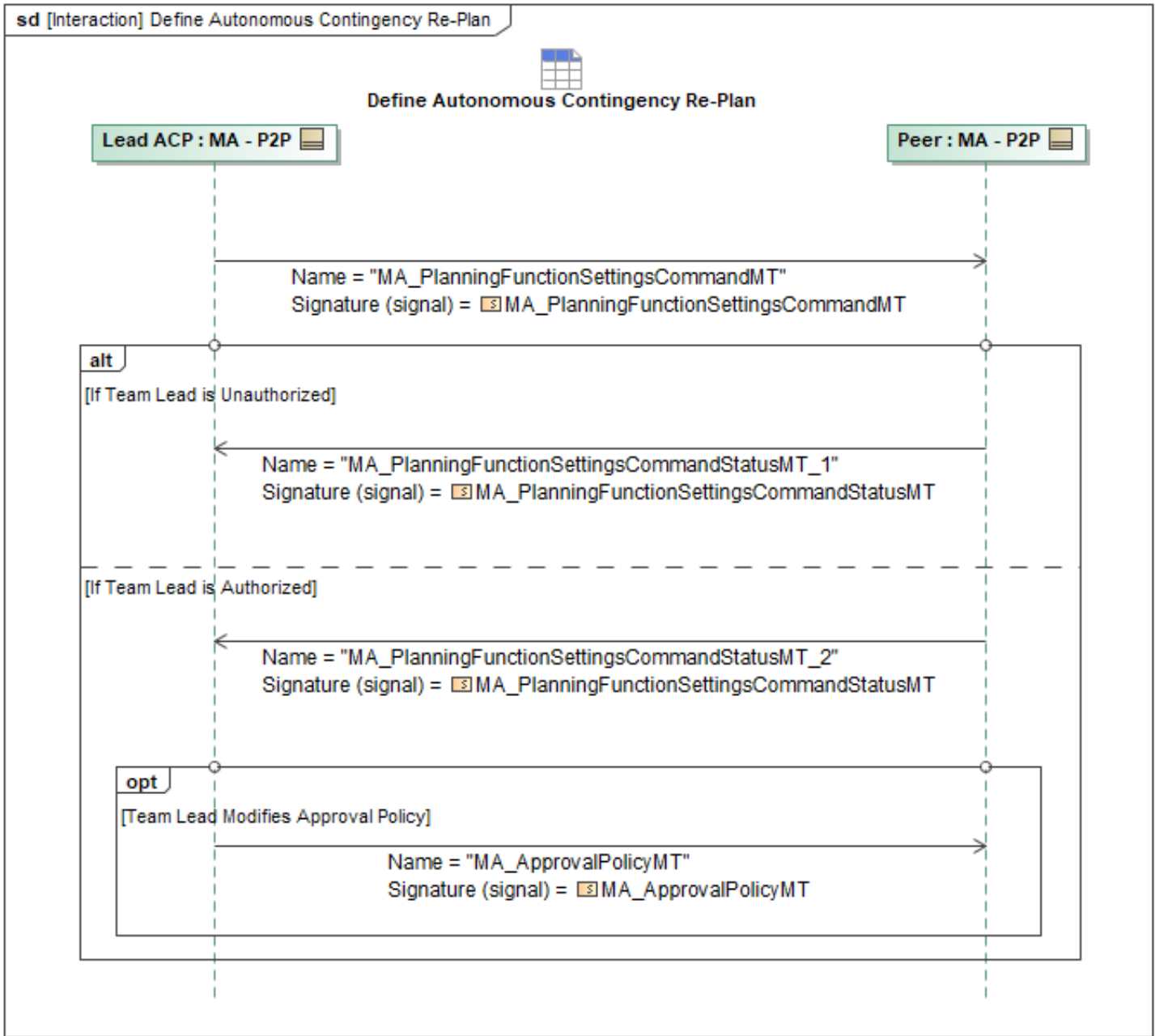


Figure 1-9: Define Autonomous Contingency Re-Plan

Table 1-8: Define Autonomous Contingency Re-Plan

| Message Name (Name:Type)                        | Required Fields (Name:Type)                                                                                                                                                                          |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_ApprovalPolicyMT: MA_ApprovalPolicyMT</b> | DataRecordInstanceID: DataRecordInstanceID_Type<br>PlanActivation: MA_PlanActivationPolicyType<br>Plans: MA_PlanPolicyType<br>ByTriggerPolicy: ByTriggerPolicyType<br>Policy: ApprovalPolicyBaseType |

| Message Name (Name:Type)                                                                                                        | Required Fields (Name:Type)                                       |
|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>MA_PlanningFunctionSettingsCommandMT:</b><br><b>MA_PlanningFunctionSettingsCommandMT</b>                                     | MissionPlanningAutonomy:<br>MA_MissionPlanningAutonomySettingType |
| <b>MA_PlanningFunctionSettingsCommandStatus</b><br><b>MT_1:</b><br><b>MA_PlanningFunctionSettingsCommandStatus</b><br><b>MT</b> | CommandProcessingStateReason: CannotComplyType                    |
| <b>MA_PlanningFunctionSettingsCommandStatus</b><br><b>MT_2:</b><br><b>MA_PlanningFunctionSettingsCommandStatus</b><br><b>MT</b> | None                                                              |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.4.2 Define Team Lost Comms Contingency

In order to set up a lost communication contingency between a peer and its team, C2 systems send a *MA\_SystemManagementRequestMT* to the Team Leader of a team. (See section *Define Team Contingency* for more details on the Team Leader's communication with C2). When the Team Leader Receives a *MA\_SystemManagementRequestMT*, it will configure itself in accordance with the message, and then send each of its team members a *MA\_SystemManagementRequestMT* instructing them to configure themselves in the same way. Once a team member successfully configures itself, it will send back a *MA\_SystemManagementRequestStatusMT* to the Team Leader with the *RequestProcessingState* set to "COMPLETED". If a team member is not able to configure themselves, it will instead set the *RequestProcessingState* to "FAILED" and send at least one *RequestProcessingStateReason* for why it cannot apply the changes.

Once the team's contingency trigger has been configured, the Team Leader will send a *ResponsePlanMT* to each of its team members. To begin execution of the new *ResponsePlanMT*, the Team Leader will send a *MissionPlanActivationCommandMT* with *Command.MissionPlanID* set to the ID of the *ResponsePlanMT*. Once activated, response monitoring will begin and each team member will start sending periodic *MA\_ResponseActivityMTs* to their Team Leader. An *ActivityStatus* of "ENABLED" means the trigger is currently being monitored. An *ActivityStatus* of "ACTIVE\_\*" indicates the response trigger was tripped and the response is currently executing the desired \*Plan. An *ActivityStatus* of "COMPLETED" means that the desired \*Plan has finished execution. (See C2 *Contingency Management* section for more details on Responses).

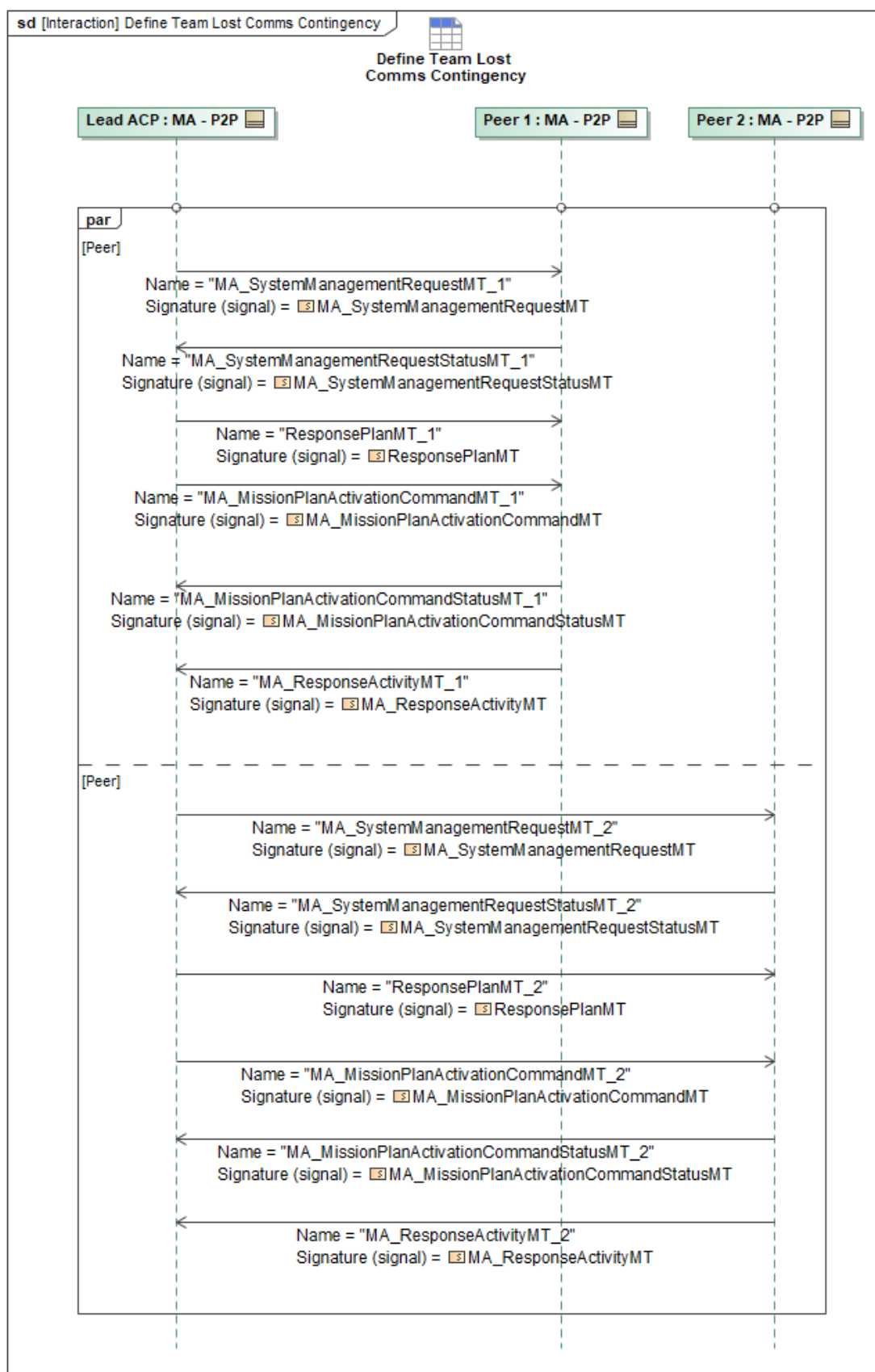


Figure 1-10: Define Team Lost Comms Contingency



Table 1-9: Define Team Lost Comms Contingency

| Message Name (Name:Type)                                                              | Required Fields (Name:Type)                                                                                                                                          |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MA_MissionPlanActivationCommandMT_1:<br>MA_MissionPlanActivationCommandMT             | ResponsePlan: ResponsePlanActivationType                                                                                                                             |
| MA_MissionPlanActivationCommandMT_2:<br>MA_MissionPlanActivationCommandMT             | ResponsePlan: ResponsePlanActivationType                                                                                                                             |
| MA_MissionPlanActivationCommandStatusMT_1:<br>MA_MissionPlanActivationCommandStatusMT | CommandStatus: ProcessingStatusEnum                                                                                                                                  |
| MA_MissionPlanActivationCommandStatusMT_2:<br>MA_MissionPlanActivationCommandStatusMT | CommandStatus: ProcessingStatusEnum                                                                                                                                  |
| MA_ResponseActivityMT_1:<br>MA_ResponseActivityMT                                     | AlertID: MissionContingencyAlertID_Type<br>ActivatedPlans: MA_PlansReferenceType<br>ActivatedPlans: MA_PlansReferenceType<br>AlertID: MissionContingencyAlertID_Type |
| MA_ResponseActivityMT_2:<br>MA_ResponseActivityMT                                     | AlertID: MissionContingencyAlertID_Type<br>ActivatedPlans: MA_PlansReferenceType<br>ActivatedPlans: MA_PlansReferenceType<br>AlertID: MissionContingencyAlertID_Type |
| MA_SystemManagementRequestMT_1:<br>MA_SystemManagementRequestMT                       | LostCommTimeout: DurationType                                                                                                                                        |
| MA_SystemManagementRequestMT_2:<br>MA_SystemManagementRequestMT                       | LostCommTimeout: DurationType                                                                                                                                        |
| MA_SystemManagementRequestStatusMT_1:<br>MA_SystemManagementRequestStatusMT           | None                                                                                                                                                                 |
| MA_SystemManagementRequestStatusMT_2:<br>MA_SystemManagementRequestStatusMT           | None                                                                                                                                                                 |
| ResponsePlanMT_1: ResponsePlanMT                                                      | AllocatedResponse: ResponseAllocationType<br>PlanCommandID: ResponsePlanCommandID_ChoiceType                                                                         |
| ResponsePlanMT_2: ResponsePlanMT                                                      | AllocatedResponse: ResponseAllocationType<br>PlanCommandID: ResponsePlanCommandID_ChoiceType                                                                         |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.4.3 Multiple Peers Lost Comms with Team Leader

In the event that multiple peers are disconnected from the Team Leader, the team will split into two separate teams. The side of the split with the original Team Leader will retain its original *PackageID*, but update C2 on who the remaining members are. See section "*Team Leader Lost Comms with Multiple Team Members*" in the C2 section for more information on Team Leaders communication with C2. If it the team leader cannot connect to a C2 system, refer to section "*C2 Lost Comms with*

*Team Leader."*

The side of the split that doesn't have a leader will immediately begin the process of forming a new team among the peers they still have communications with (see section "Create a Full Package Amongst Peers.") This includes initialization of the new package and election of a new Team Leader. Once elected, the new Team Leader will try to connect to a C2 system and update C2 on the status of the second half of the split team. See section "*Team Leader Lost Comms with Multiple Team Members*" in the C2 section for more information. In the event that the new Team Leader can't communicate with C2, see section "*Team Leader Lost Comms with C2.*"

In the event that a team was separated into two groups, and they are able to regain communication, then the two teams will stay as two separate teams until commanded to merge. See section "*Team Leader Lost Comms with Multiple Team Members*" in the C2 section for more information on notifying C2.

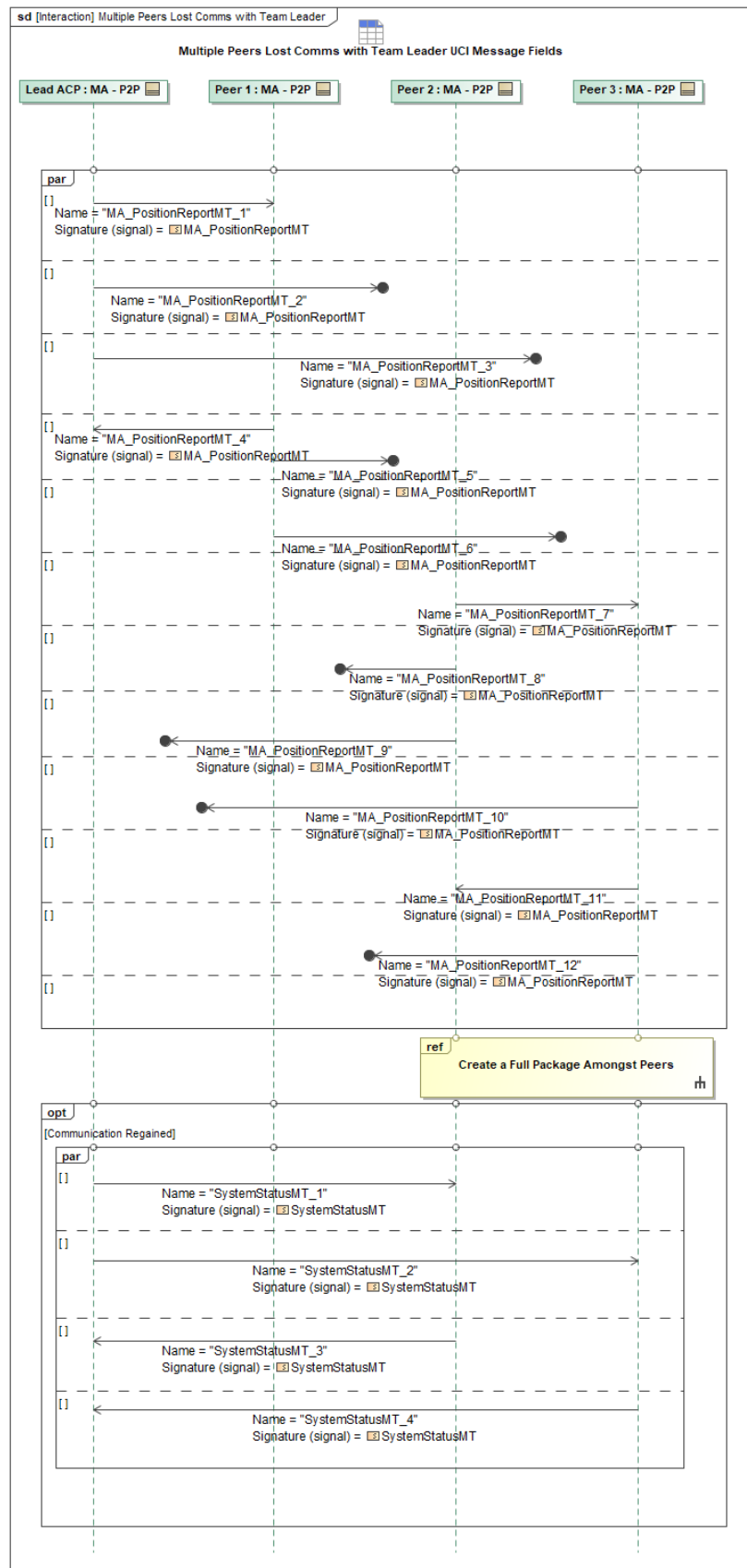


Figure 1-11: Multiple Peers Lost Comms with Team Leader

Table 1-10: Multiple Peers Lost Comms with Team Leader

| Message Name (Name:Type)                      | Required Fields (Name:Type)                                                                                           |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| MA_PositionReportMT_1:<br>MA_PositionReportMT | CalibratedAirspeed: SpeedType<br>DomainVelocity: Velocity3D_Type<br>Mach: MachType<br>GroundVelocity: Velocity2D_Type |
| MA_PositionReportMT_2:<br>MA_PositionReportMT | GroundVelocity: Velocity2D_Type<br>Mach: MachType<br>CalibratedAirspeed: SpeedType<br>DomainVelocity: Velocity3D_Type |
| MA_PositionReportMT_3:<br>MA_PositionReportMT | Mach: MachType<br>DomainVelocity: Velocity3D_Type<br>CalibratedAirspeed: SpeedType<br>GroundVelocity: Velocity2D_Type |
| MA_PositionReportMT_4:<br>MA_PositionReportMT | Mach: MachType<br>CalibratedAirspeed: SpeedType<br>GroundVelocity: Velocity2D_Type<br>DomainVelocity: Velocity3D_Type |
| MA_PositionReportMT_5:<br>MA_PositionReportMT | GroundVelocity: Velocity2D_Type<br>DomainVelocity: Velocity3D_Type<br>CalibratedAirspeed: SpeedType<br>Mach: MachType |
| MA_PositionReportMT_6:<br>MA_PositionReportMT | DomainVelocity: Velocity3D_Type<br>CalibratedAirspeed: SpeedType<br>Mach: MachType<br>GroundVelocity: Velocity2D_Type |
| MA_PositionReportMT_7:<br>MA_PositionReportMT | CalibratedAirspeed: SpeedType<br>Mach: MachType<br>GroundVelocity: Velocity2D_Type<br>DomainVelocity: Velocity3D_Type |
| MA_PositionReportMT_8:<br>MA_PositionReportMT | Mach: MachType<br>CalibratedAirspeed: SpeedType<br>DomainVelocity: Velocity3D_Type<br>GroundVelocity: Velocity2D_Type |
| MA_PositionReportMT_9:<br>MA_PositionReportMT | CalibratedAirspeed: SpeedType<br>GroundVelocity: Velocity2D_Type<br>Mach: MachType                                    |

| Message Name (Name:Type)                                     | Required Fields (Name:Type)                                                                                           |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|                                                              | DomainVelocity: Velocity3D_Type                                                                                       |
| <b>MA_PositionReportMT_10:</b><br><b>MA_PositionReportMT</b> | CalibratedAirspeed: SpeedType<br>DomainVelocity: Velocity3D_Type<br>GroundVelocity: Velocity2D_Type<br>Mach: MachType |
| <b>MA_PositionReportMT_11:</b><br><b>MA_PositionReportMT</b> | DomainVelocity: Velocity3D_Type<br>Mach: MachType<br>GroundVelocity: Velocity2D_Type<br>CalibratedAirspeed: SpeedType |
| <b>MA_PositionReportMT_12:</b><br><b>MA_PositionReportMT</b> | Mach: MachType<br>CalibratedAirspeed: SpeedType<br>GroundVelocity: Velocity2D_Type<br>DomainVelocity: Velocity3D_Type |
| <b>SystemStatusMT_1: SystemStatusMT</b>                      | None                                                                                                                  |
| <b>SystemStatusMT_2: SystemStatusMT</b>                      | None                                                                                                                  |
| <b>SystemStatusMT_3: SystemStatusMT</b>                      | None                                                                                                                  |
| <b>SystemStatusMT_4: SystemStatusMT</b>                      | None                                                                                                                  |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.4.4 P2P Incorrect Messaging Sequence Error Handling

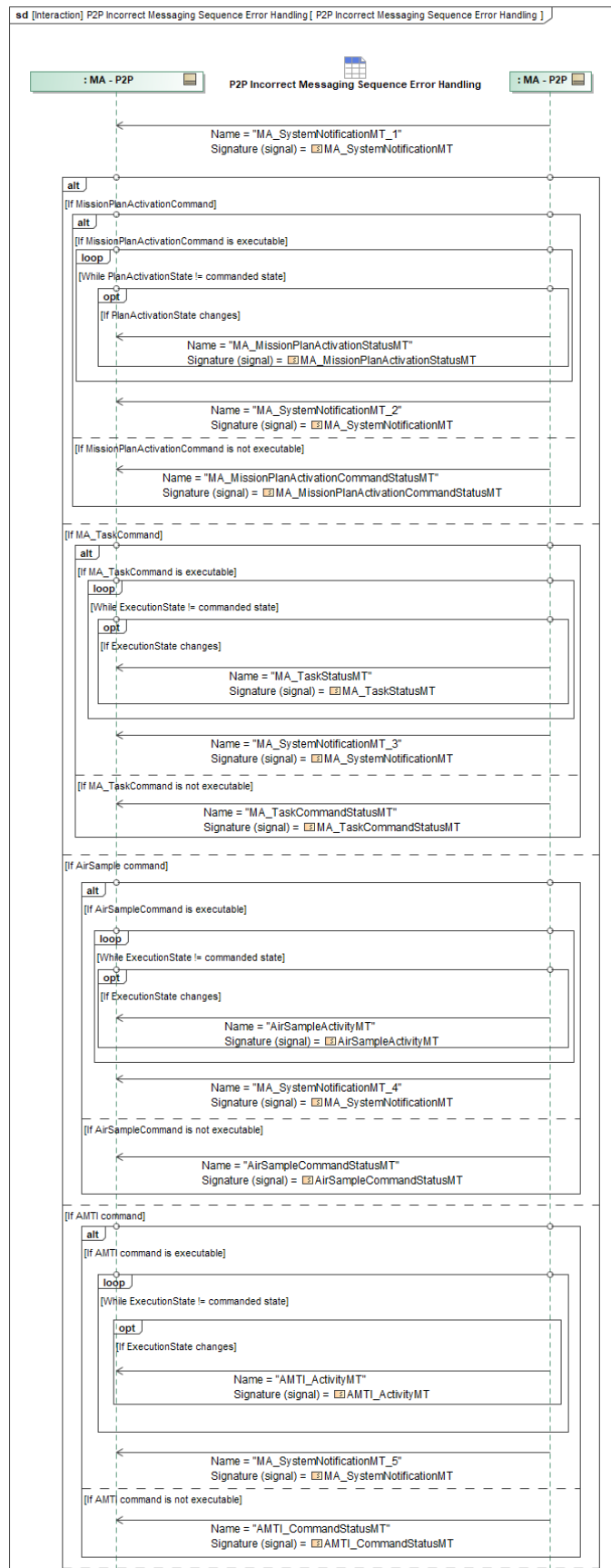
When MA receives a command, it is possible that the commanded state of an object (e.g., *MissionPlan*, *Task*, *EA\_Capability*) is more than one state away from the object's current state. If this is the case, an Incorrect Messaging Sequence error is triggered. Upon identification of an Incorrect Messaging Sequence error, MA will send *MA\_SystemNotificationMT* to Peer indicate that the commanded state is out of sequence and that corrective state transitions will be performed. Within the message, MA sets *NotificationState* to *ACTIVE* and sets *NotificationDetail.Reason* to *STATE\_OR\_SETTINGS*. *NotificationState* communicates that the error exists and *NotificationDetail.Reason* indicates that the error exists due to the object's current state. *AssociatedMessage.MessageType* identifies the command associated with the error, and *AssociatedMessage.AssociatedID* will reference the ID of the associated command.

If MA determines that the command cannot be carried out for any reason, MA will publish the corresponding *\*CommandStatus* message with *CommandProcessingState* set to *REJECTED*. If MA determines that the command can be carried out, MA will publish the corresponding *\*CommandStatus* message with *CommandProcessingState* set to *ACCEPTED*, and will progress the object's status towards the commanded state.

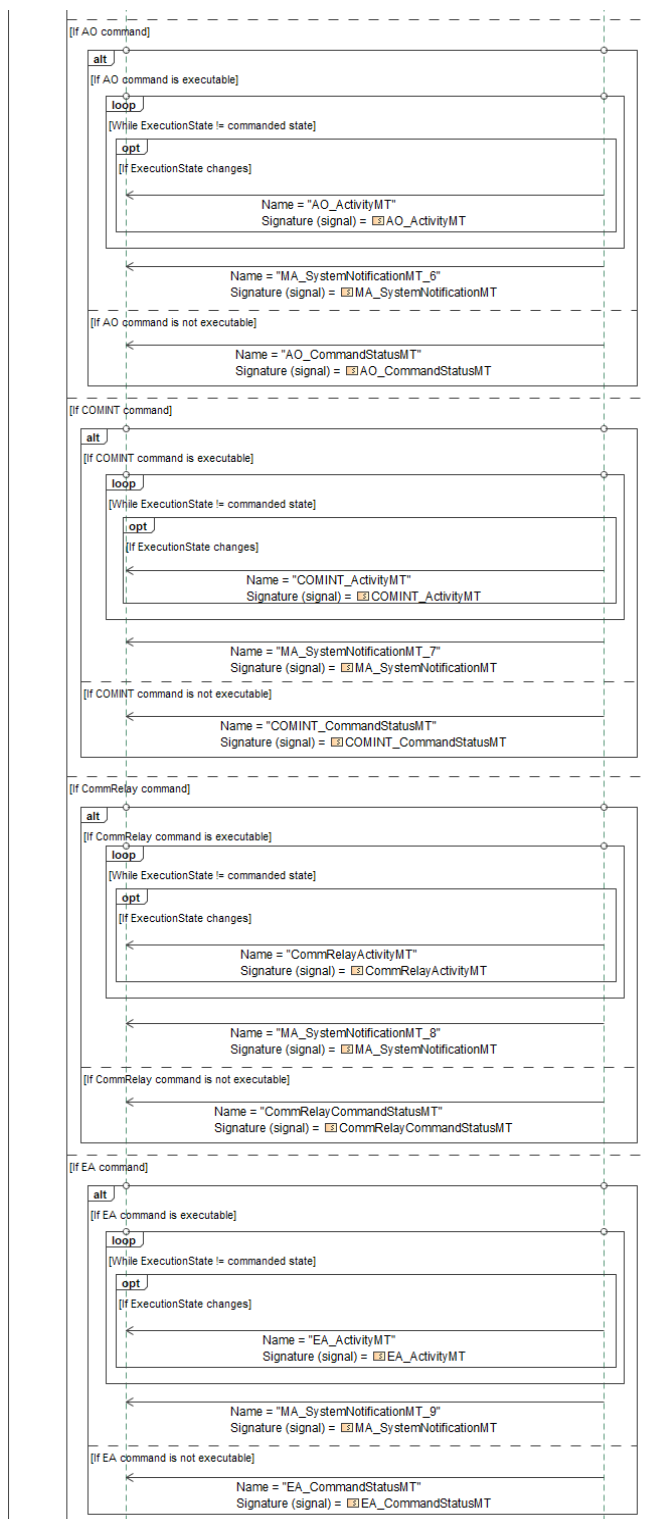
MA communicates the object's status when the object's status changes during corrective actions. Status updates are not intended to be periodic; they are sent when MA transitions the object to a different state while progressing toward the commanded state. Object status reflects the object's

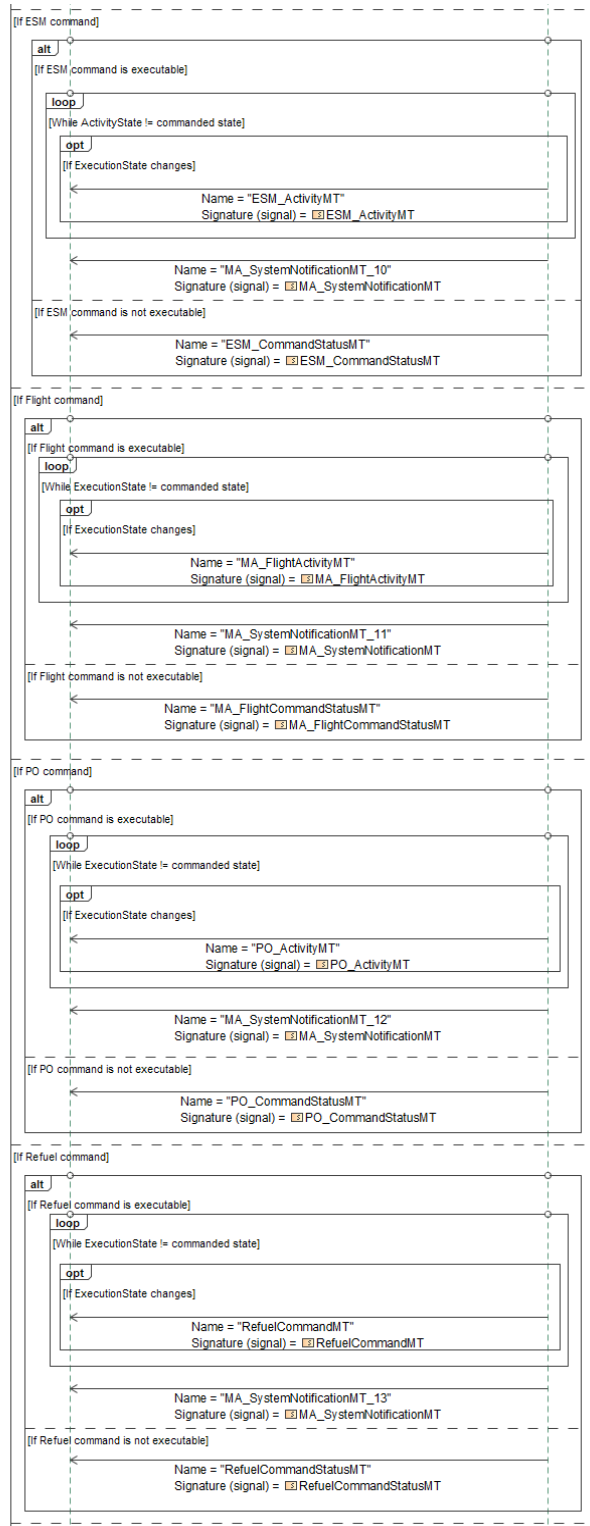
actual state at the time the status is reported and provides visibility into the intermediate transitions performed by MA during resolution of the Incorrect Messaging Sequence error. The applicable status message depends on the affected object. For mission plan activation commands, MA uses *MA\_MissionPlanActivationStatus* to communicate the current *PlanActivationState* (e.g., *READY\_FOR\_ACTIVATION*, *ACTIVATING*, *ACTIVATED*, *ACTIVE\_DEGRADED*, *DEACTIVATING*, and *DEACTIVATED*). For a TaskCommand, MA uses *MA\_TaskStatusMT* to communicate the current *ExecutionState* (e.g. *AWAITING\_PLAN\_ACTIVATION*, *EXECUTING*, *COMPLETED*, *DROPPED*, *FAILED*, *CANCELED*). For UCI capability commands, MA uses the capability's corresponding *\*Activity message* (e.g., *EA\_Activity*, *AMTI\_Activity*, *AO\_Activity*) to communicate the current *ActivityState* (e.g., *ENABLED*, *ACTIVE\_PARTIALLY\_CONSTRAINED*, *ACTIVE\_UNCONSTRAINED*, *COMPLETED*, *FAILED*).

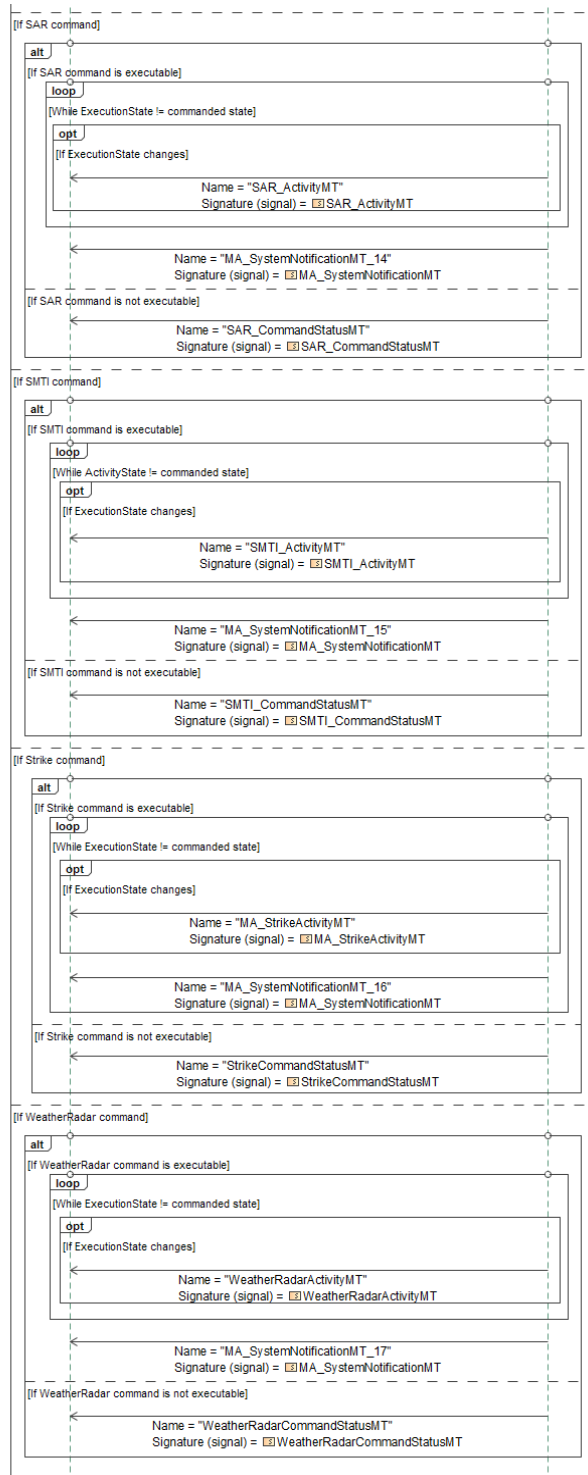
The commanded state represents the state requested by the originating command and serves as the target state against which MA evaluates progress during executing corrective actions. When the current object state matches the commanded state, MA considers the Incorrect Messaging Sequence Error to be resolved and will send updated *MA\_SystemNotificationMT* with *NotificationState* set to *CLOSED*.











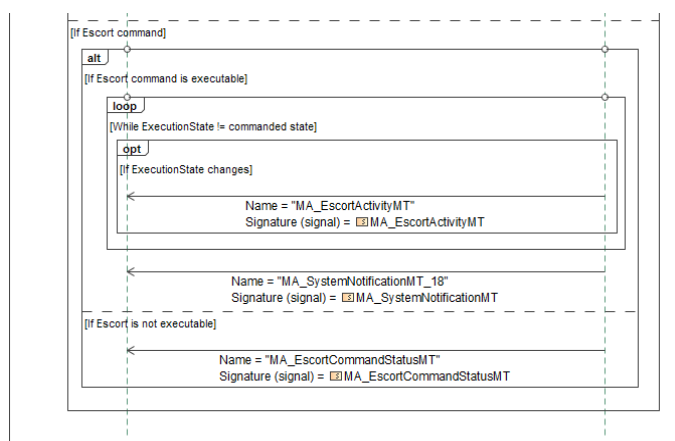


Figure 1-12: P2P Incorrect Messaging Sequence Error Handling

Table 1-11: P2P Incorrect Messaging Sequence Error Handling

| Message Name (Name:Type)                              | Required Fields (Name:Type) |
|-------------------------------------------------------|-----------------------------|
| AirSampleActivityMT: AirSampleActivityMT              | None                        |
| AirSampleCommandStatusMT:<br>AirSampleCommandStatusMT | None                        |
| AMTI_ActivityMT: AMTI_ActivityMT                      | None                        |
| AMTI_CommandStatusMT:<br>AMTI_CommandStatusMT         | None                        |
| AO_ActivityMT: AO_ActivityMT                          | None                        |
| AO_CommandStatusMT:<br>AO_CommandStatusMT             | None                        |
| COMINT_ActivityMT: COMINT_ActivityMT                  | None                        |
| COMINT_CommandStatusMT:<br>COMINT_CommandStatusMT     | None                        |
| CommRelayActivityMT: CommRelayActivityMT              | None                        |
| CommRelayCommandStatusMT:<br>CommRelayCommandStatusMT | None                        |
| EA_ActivityMT: EA_ActivityMT                          | None                        |
| EA_CommandStatusMT:<br>EA_CommandStatusMT             | None                        |
| ESM_ActivityMT: ESM_ActivityMT                        | None                        |
| ESM_CommandStatusMT:<br>ESM_CommandStatusMT           | None                        |
| MA_EscortActivityMT: MA_EscortActivityMT              | None                        |
| MA_EscortCommandStatusMT:<br>MA_EscortCommandStatusMT | None                        |
| MA_FlightActivityMT: MA_FlightActivityMT              | None                        |
| MA_FlightCommandStatusMT:<br>MA_FlightCommandStatusMT | None                        |

| Message Name (Name:Type)                                                            | Required Fields (Name:Type) |
|-------------------------------------------------------------------------------------|-----------------------------|
| MA_MissionPlanActivationCommandStatusMT:<br>MA_MissionPlanActivationCommandStatusMT | None                        |
| MA_MissionPlanActivationStatusMT:<br>MA_MissionPlanActivationStatusMT               | None                        |
| MA_StrikeActivityMT: MA_StrikeActivityMT                                            | None                        |
| MA_SystemNotificationMT_1:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_2:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_3:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_4:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_5:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_6:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_7:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_8:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_9:<br>MA_SystemNotificationMT                               | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_10:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_11:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_12:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_13:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_14:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_15:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_16:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_17:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_SystemNotificationMT_18:<br>MA_SystemNotificationMT                              | AssociatedID: ID_Type       |
| MA_TaskCommandStatusMT:<br>MA_TaskCommandStatusMT                                   | None                        |

| Message Name (Name:Type)                                    | Required Fields (Name:Type) |
|-------------------------------------------------------------|-----------------------------|
| MA_TaskStatusMT: MA_TaskStatusMT                            | None                        |
| PO_ActivityMT: PO_ActivityMT                                | None                        |
| PO_CommandStatusMT:<br>PO_CommandStatusMT                   | None                        |
| RefuelCommandMT: RefuelCommandMT                            | None                        |
| RefuelCommandStatusMT:<br>RefuelCommandStatusMT             | None                        |
| SAR_ActivityMT: SAR_ActivityMT                              | None                        |
| SAR_CommandStatusMT:<br>SAR_CommandStatusMT                 | None                        |
| SMTI_ActivityMT: SMTI_ActivityMT                            | None                        |
| SMTI_CommandStatusMT:<br>SMTI_CommandStatusMT               | None                        |
| StrikeCommandStatusMT:<br>StrikeCommandStatusMT             | None                        |
| WeatherRadarActivityMT:<br>WeatherRadarActivityMT           | None                        |
| WeatherRadarCommandStatusMT:<br>WeatherRadarCommandStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.4.5 P2P Malformed Message/Precondition Not Met Handling

Upon receipt of a malformed message, MA will send *MA\_SystemNotificationMT* to P2P, indicating that the message could not be successfully processed. Within the message, MA sets *NotificationState* to ACTIVE and populates *NotificationDetail.Reason* with *INVALID\_INPUT\_PARAMETER*. *NotificationDetail.Description* will contain a human-readable string stating which fields are malformed. *AssociatedMessage* will reference the malformed message. The *MessageType* will be set to the same type as the malformed message. *AssociatedID* will be populated with a reference ID from the malformed message if applicable, the *UUID* of that message. If additional data is required to resolve the condition and the fault remains active, MA can issue *QueryDataRequestMT* with the Query field containing the ID of the specific message. P2P would then respond with *QueryDataRequestStatusMT*, specifically the Message field within the Result field will contain the corrected message containing the originally missing data. If the fault has been addressed, then MA will only send a new *MA\_SystemNotificationMT* with the same fields as above and *NotificationState* set to CLOSED.

Upon receipt of a command from P2P for which the required inputs are not present on the platform/data store, MA will send *MA\_SystemNotificationMT* to P2P, indicating that the command could not be successfully executed. Within the message, MA sets *NotificationState* to ACTIVE and populates *NotificationDetail.Reason* with *UNKNOWN\_ID*. *NotificationDetail.Description* will contain a human-readable string stating which data is not present. *AssociatedMessage* will reference the attempted command. The *MessageType* will be set to the same type as the command. *AssociatedID*

will be populated with a reference ID from the command if applicable, the *UUID* of that command. If additional data is required to resolve the condition and the fault remains active, MA can issue *QueryDataRequestMT* with the Query field containing the ID of the specific command. P2P would then respond with *QueryDataRequestStatusMT*, specifically the Message field within the Result field will contain the corrected command containing the originally missing data. If the fault has been addressed, then MA will only send a new *MA\_SystemNotificationMT* with the same fields as above and *NotificationState* set to CLOSED.

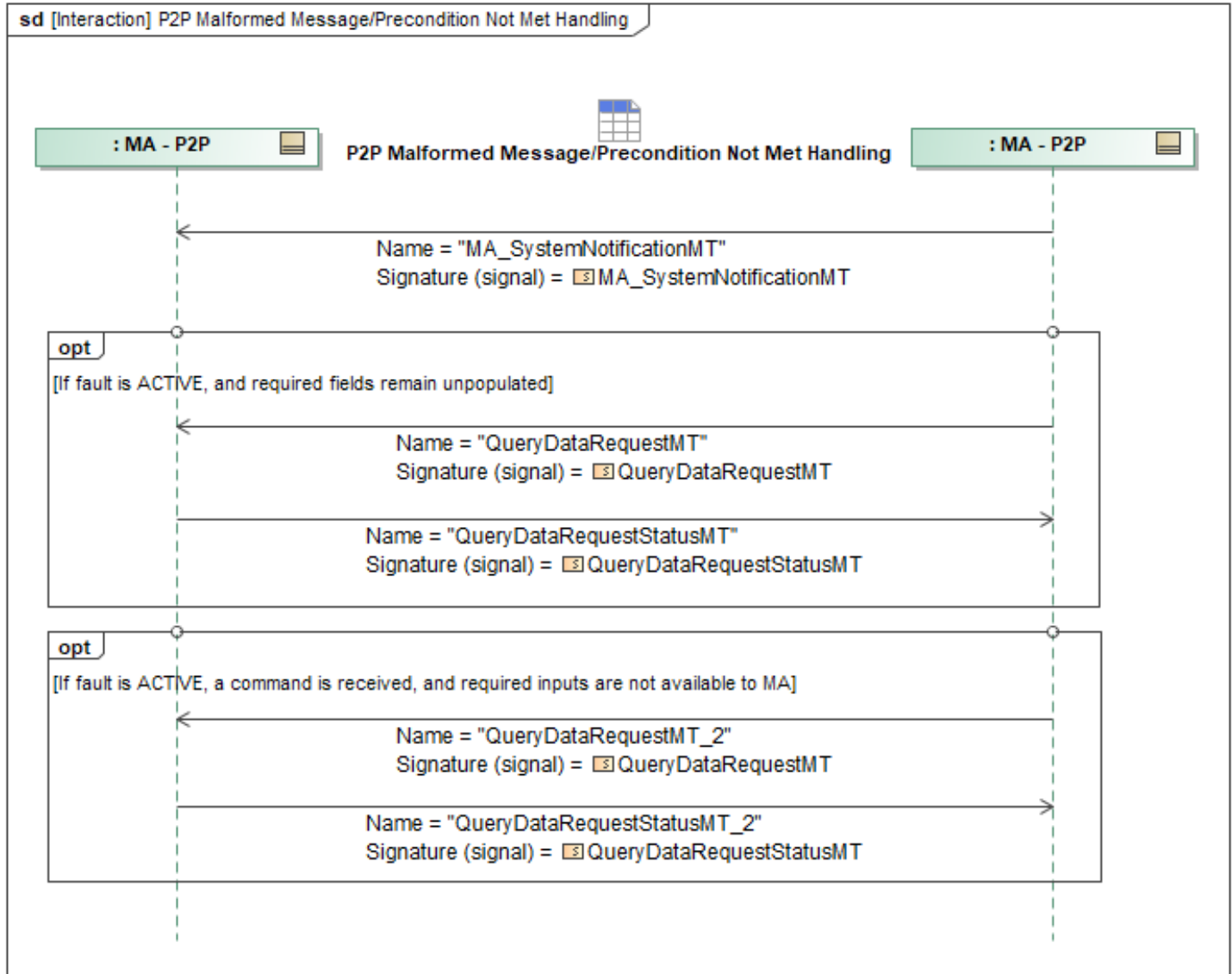


Figure 1-13: P2P Malformed Message/Precondition Not Met Handling

Table 1-12: P2P Malformed Message/Precondition Not Met Handling

| Message Name (Name:Type)                            | Required Fields (Name:Type)                                 |
|-----------------------------------------------------|-------------------------------------------------------------|
| MA_SystemNotificationMT:<br>MA_SystemNotificationMT | Description: VisibleString1024Type<br>AssociatedID: ID_Type |
| QueryDataRequestMT: QueryDataRequestMT              | Query: QueryPET                                             |

| Message Name (Name:Type)                                | Required Fields (Name:Type) |
|---------------------------------------------------------|-----------------------------|
| QueryDataRequestMT_2: QueryDataRequestMT                | None                        |
| QueryDataRequestStatusMT:<br>QueryDataRequestStatusMT   | Message: MessageType        |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.4.6 P2P Path Deconfliction Proposal

During the collaborative path deconfliction process, MA sends the current proposed deconfliction solution via an *MA\_DeconflictionSolutionRequestMT* to connected peers it has a path conflict with. This message includes the projected trajectory of the sending MA in the *RoutePlan* field, a unique *ConflictID*, and a *ConflictStatusEnum* indicating whether the conflict has been resolved.

After receipt of the *MA\_DeconflictionSolutionRequestMT*, the peer in conflict responds with an *MA\_DeconflictionSolutionRequestStatusMT*, setting the *RequestProcessingState* to one of *QUEUED*/*PROCESSING*/*COMPLETED*/*FAILED*/*CANCELED*/*REJECTED*.

The connected peers in conflict responds using *MA\_DeconflictionSolutionRequestMT\_1* with the *RequestProcessingState* field is set to *QUEUED* upon receiving a request and is updated once the request can be processed. If *RequestProcessingState* is set to *FAILED*, *CANCELED*, or *REJECTED*, the *MA\_DeconflictionSolutionRequestStatusMT\_3* message will include an accompanying *RequestProcessingStateReason*. Once the *RequestProcessingState* is set to *COMPLETED*, the *MA\_DeconflictionSolutionRequestStatusMT\_2* will include a *DeconflictionProposalStatus* field will be set to either *ACCEPTED* or *REJECTED*. If *DeconflictionProposalStatus* field is set to *ACCEPTED*, the message will include the *ConflictID* and *RoutePlanID* to indicate the specific conflict and route plan of the sequence originating ACP that are being responded to. If *DeconflictionProposalStatus* field is set to *REJECTED*, this will indicate to the sequence originating ACP that the conflict remains unresolved and that it may produce and send a new solution for peer's approval.



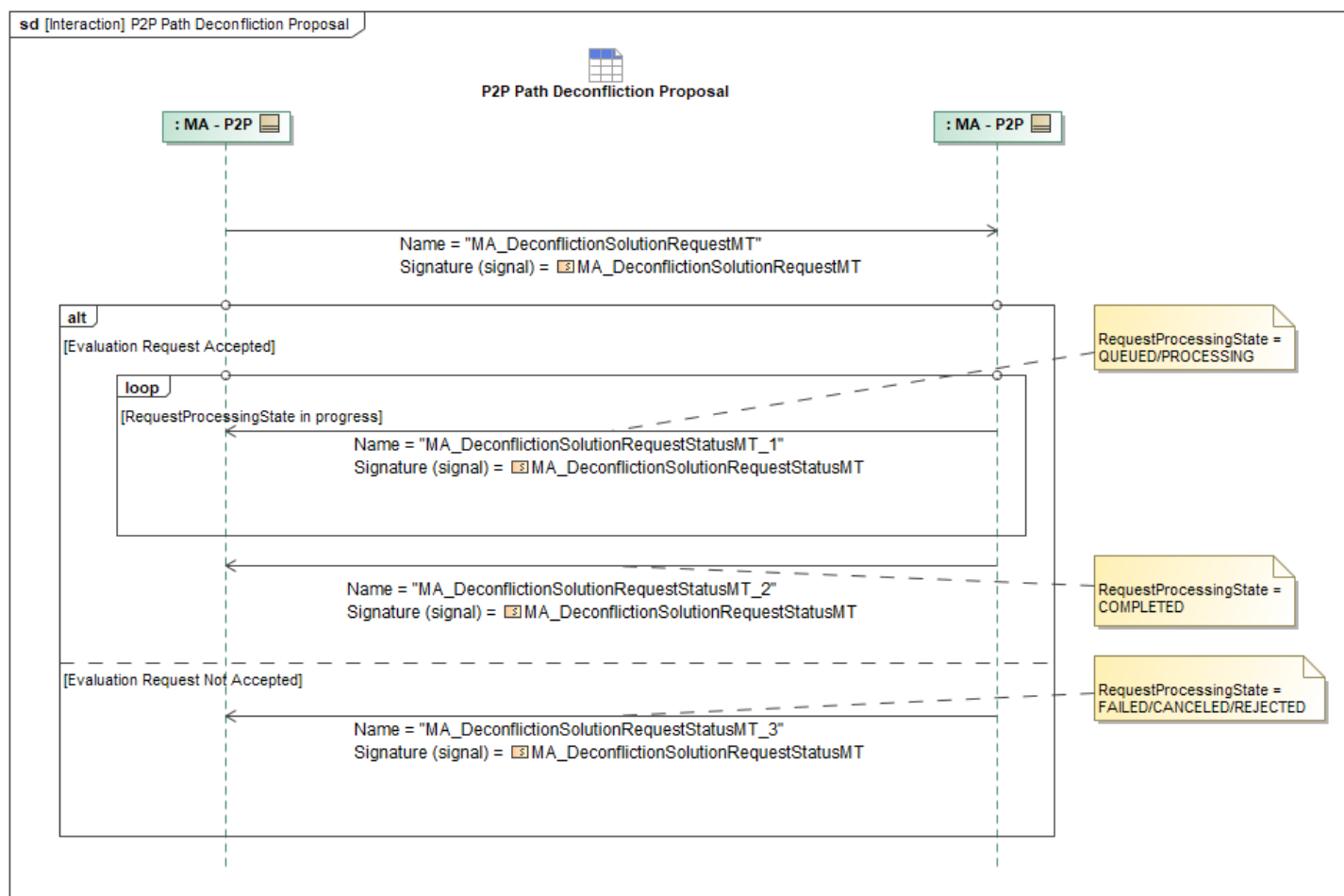


Figure 1-14: P2P Path Deconfliction Proposal

Table 1-13: P2P Path Deconfliction Proposal

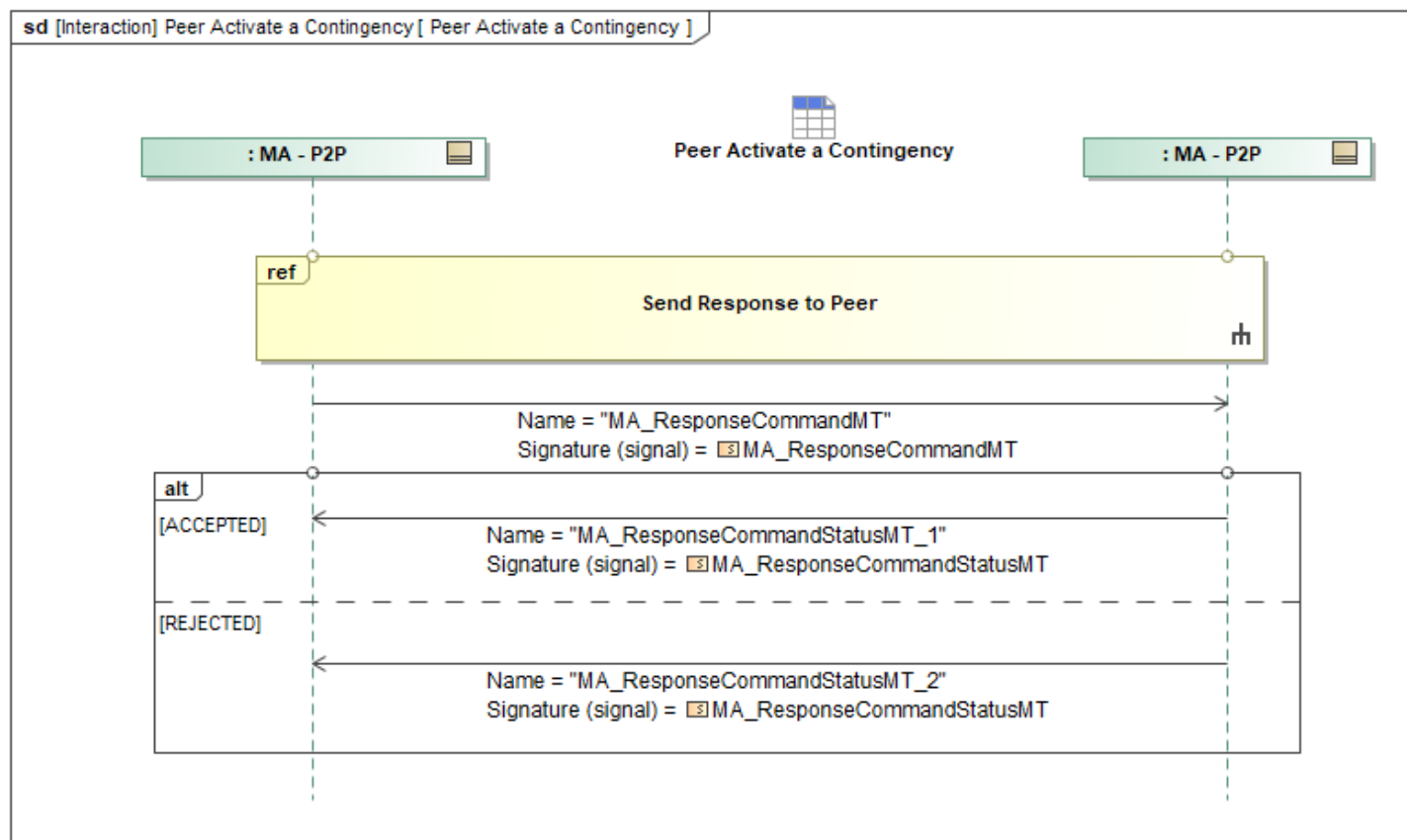
| Message Name (Name:Type)                                                              | Required Fields (Name:Type)                    |
|---------------------------------------------------------------------------------------|------------------------------------------------|
| MA_DeconflictionSolutionRequestMT:<br>MA_DeconflictionSolutionRequestMT               | None                                           |
| MA_DeconflictionSolutionRequestStatusMT_1:<br>MA_DeconflictionSolutionRequestStatusMT | None                                           |
| MA_DeconflictionSolutionRequestStatusMT_2:<br>MA_DeconflictionSolutionRequestStatusMT | None                                           |
| MA_DeconflictionSolutionRequestStatusMT_3:<br>MA_DeconflictionSolutionRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.4.7 Peer Activate a Contingency

Peer sends a *MA\_ResponseCommandMT* to activate or deactivate a previously uploaded *MA\_ResponseMT*. The *MA\_ResponseCommandMT* will set the *ConstraintOverride.Allocation.RequiredAllocation.Allocation* field to indicate the target *SystemIDs* or

*PackageID* to activate the *MA\_ResponseMT*. Optionally, the *AssociatedPlan* field can be used to link back to a plan which contains the *MA\_ResponseMT*. MA responds with a *MA\_ResponseCommandStatusMT* to indicate if the command was successful or not.



### Figure 1-15: Peer Activate a Contingency

### Table 1-14: Peer Activate a Contingency

| Message Name (Name:Type)                                    | Required Fields (Name:Type) |
|-------------------------------------------------------------|-----------------------------|
| MA_ResponseCommandMT:<br>MA_ResponseCommandMT               | None                        |
| MA_ResponseCommandStatusMT_1:<br>MA_ResponseCommandStatusMT | None                        |
| MA_ResponseCommandStatusMT_2:<br>MA_ResponseCommandStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.4.8 Peer Receive Contingency Status

MA will periodically publish *MA\_ResponseActivityMT* to convey monitoring and triggered status for all active contingencies. An *ActivityState* of *ENABLED* indicates MA is actively monitoring the trigger condition for the *Response* while *DISABLED* means monitoring has stopped. When the trigger condition has been satisfied, the *ActivityState* will change to *COMPLETED* with the

*RequirementInstantiation* or *ActivatedPlans* fields filled out if the *Response* was configured to perform a response action, otherwise only *AlertID* will be set.

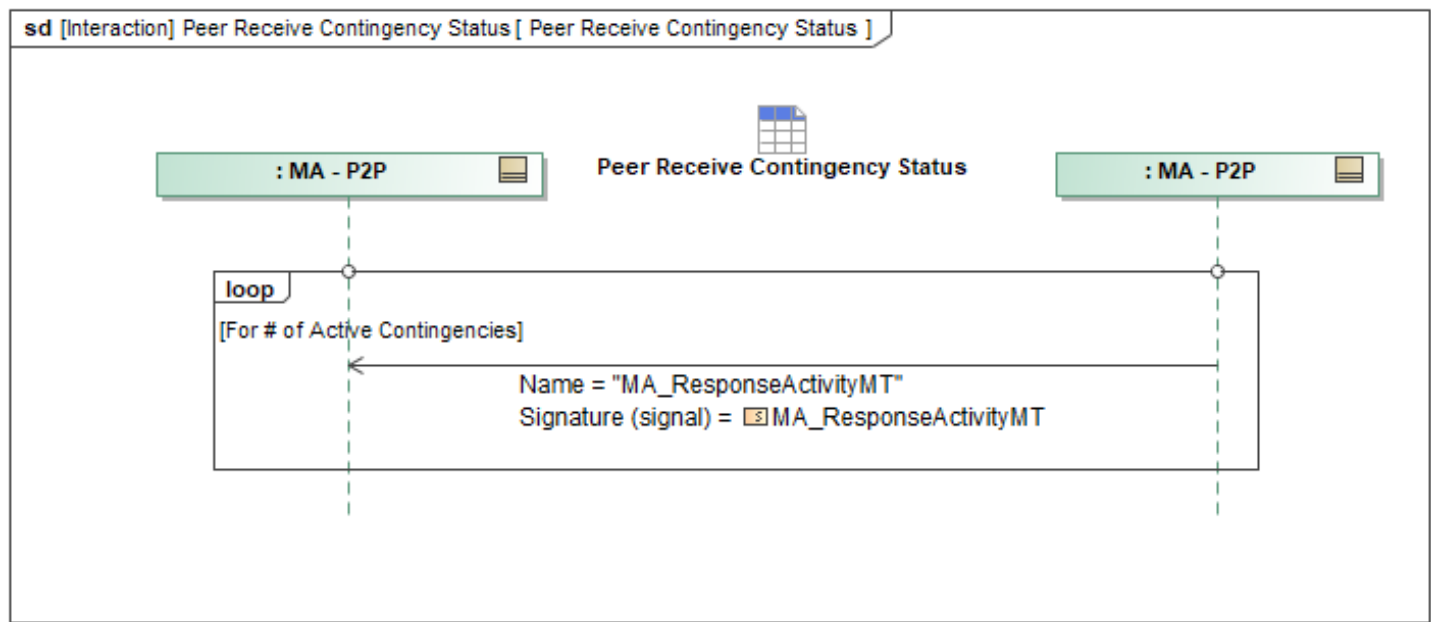


Figure 1-16: Peer Receive Contingency Status

Table 1-15: Peer Receive Contingency Status

| Message Name (Name:Type)                        | Required Fields (Name:Type) |
|-------------------------------------------------|-----------------------------|
| MA_ResponseActivityMT:<br>MA_ResponseActivityMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.9 Peer Total Comms Lost with Team

In the event that a peer stops receiving a *MA\_PositionReportMT* from its team, it will activate its previously defined lost communications contingency response plan until it has regained communications with the team (see section *C2 Contingency Management* for more information). If the peer regains communications with its team via the *MA\_PositionReportMT*, it will join the team again and continue executing the team's mission.

*MA\_PositionReportMT* 1-4 denote unique message instances between each peer attempting to report their individual positions. *MA\_PositionReportMT* 5-8 are unique message instances that show the message flows between each peer with comms established.

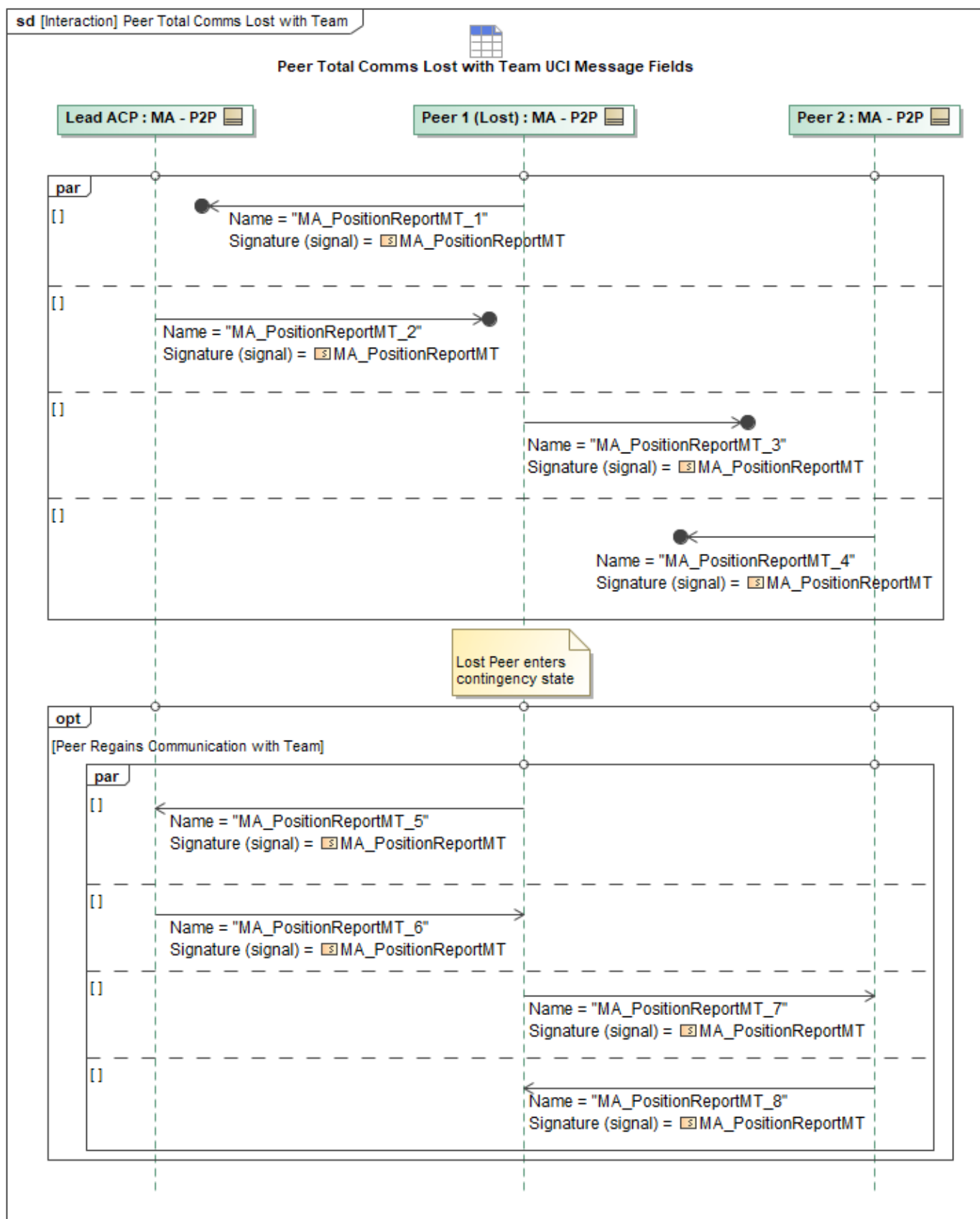


Figure 1-17: Peer Total Comms Lost with Team

Table 1-16: Peer Total Comms Lost with Team

| Message Name (Name:Type)                      | Required Fields (Name:Type)                                                                                           |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| MA_PositionReportMT_1:<br>MA_PositionReportMT | DomainVelocity: Velocity3D_Type<br>CalibratedAirspeed: SpeedType<br>GroundVelocity: Velocity2D_Type<br>Mach: MachType |
| MA_PositionReportMT_2:<br>MA_PositionReportMT | DomainVelocity: Velocity3D_Type<br>Mach: MachType<br>CalibratedAirspeed: SpeedType<br>GroundVelocity: Velocity2D_Type |
| MA_PositionReportMT_3:<br>MA_PositionReportMT | GroundVelocity: Velocity2D_Type<br>CalibratedAirspeed: SpeedType<br>Mach: MachType<br>DomainVelocity: Velocity3D_Type |
| MA_PositionReportMT_4:<br>MA_PositionReportMT | Mach: MachType<br>GroundVelocity: Velocity2D_Type<br>DomainVelocity: Velocity3D_Type<br>CalibratedAirspeed: SpeedType |
| MA_PositionReportMT_5:<br>MA_PositionReportMT | Mach: MachType<br>GroundVelocity: Velocity2D_Type<br>DomainVelocity: Velocity3D_Type<br>CalibratedAirspeed: SpeedType |
| MA_PositionReportMT_6:<br>MA_PositionReportMT | Mach: MachType<br>DomainVelocity: Velocity3D_Type<br>GroundVelocity: Velocity2D_Type<br>CalibratedAirspeed: SpeedType |
| MA_PositionReportMT_7:<br>MA_PositionReportMT | DomainVelocity: Velocity3D_Type<br>GroundVelocity: Velocity2D_Type<br>Mach: MachType<br>CalibratedAirspeed: SpeedType |
| MA_PositionReportMT_8:<br>MA_PositionReportMT | CalibratedAirspeed: SpeedType<br>DomainVelocity: Velocity3D_Type<br>GroundVelocity: Velocity2D_Type<br>Mach: MachType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.4.10 Team Leader Lost Comms with C2

In the event that the Team Leader loses comms with C2, the leader shall alert its team members via the *MissionContingencyAlertMT*. The *MissionContingencyAlertMT* contains a unique *MissionContingencyAlertID* field set to the Alert ID generated for the triggered contingency and the *ContingencyCondition.ConflictState* will be set to "CONFLICTED". Once the contingency response behavior has been activated (see *Define Team Lost Comms Contingency*), response activity monitoring shall begin between each member of the team, including the leader, whereby each team member will send *MA\_ResponseActivityMTs* to each other with *ActivityState* set to "ACTIVE\_\*". If Communication has been regained with C2, the leader shall again alert its team members via the *MissionContingencyAlertMT*. The *MissionContingencyAlertMT* will have the same fields set as when the contingency was triggered, with the exception of the *MissionContingencyAlertID* (which will be unique to this message) and the *ContingencyCondition.ConflictState* will be updated to "CLEARED".

*MA\_ResponseActivityMT\_1-6* denote unique message instances on specific message paths between each team member. Each message instance is the originating team member reporting its own activity to one individual peer.

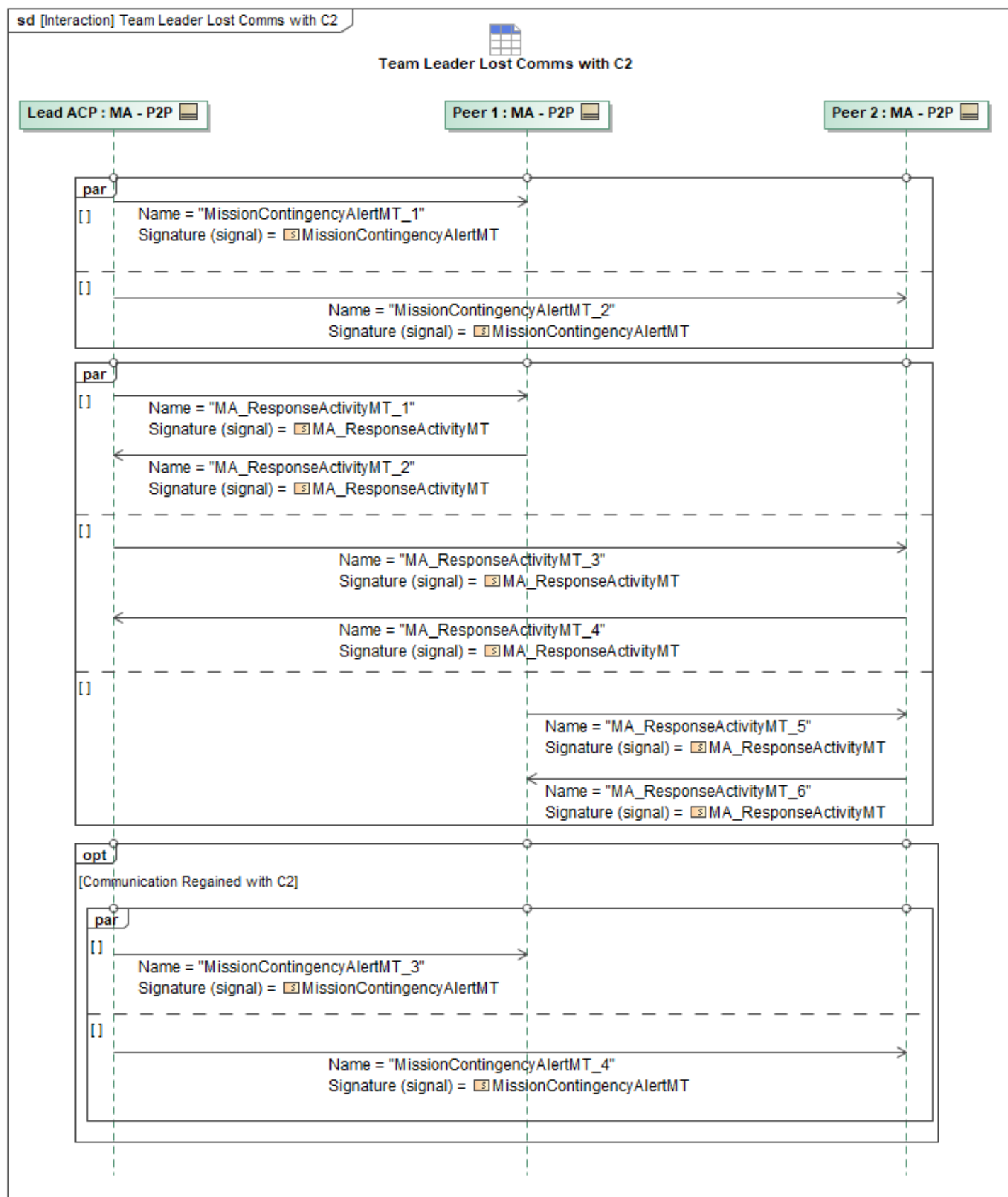


Figure 1-18: Team Leader Lost Comms with C2

Table 1-17: Team Leader Lost Comms with C2

| Message Name (Name:Type)                                                | Required Fields (Name:Type)                                                                                                                                          |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_ResponseActivityMT_1:</b><br><b>MA_ResponseActivityMT</b>         | ActivatedPlans: MA_PlansReferenceType<br>AlertID: MissionContingencyAlertID_Type<br>AlertID: MissionContingencyAlertID_Type<br>ActivatedPlans: MA_PlansReferenceType |
| <b>MA_ResponseActivityMT_2:</b><br><b>MA_ResponseActivityMT</b>         | ActivatedPlans: MA_PlansReferenceType<br>AlertID: MissionContingencyAlertID_Type<br>AlertID: MissionContingencyAlertID_Type<br>ActivatedPlans: MA_PlansReferenceType |
| <b>MA_ResponseActivityMT_3:</b><br><b>MA_ResponseActivityMT</b>         | ActivatedPlans: MA_PlansReferenceType<br>AlertID: MissionContingencyAlertID_Type<br>AlertID: MissionContingencyAlertID_Type<br>ActivatedPlans: MA_PlansReferenceType |
| <b>MA_ResponseActivityMT_4:</b><br><b>MA_ResponseActivityMT</b>         | ActivatedPlans: MA_PlansReferenceType<br>AlertID: MissionContingencyAlertID_Type<br>AlertID: MissionContingencyAlertID_Type<br>ActivatedPlans: MA_PlansReferenceType |
| <b>MA_ResponseActivityMT_5:</b><br><b>MA_ResponseActivityMT</b>         | ActivatedPlans: MA_PlansReferenceType<br>AlertID: MissionContingencyAlertID_Type<br>AlertID: MissionContingencyAlertID_Type<br>ActivatedPlans: MA_PlansReferenceType |
| <b>MA_ResponseActivityMT_6:</b><br><b>MA_ResponseActivityMT</b>         | ActivatedPlans: MA_PlansReferenceType<br>AlertID: MissionContingencyAlertID_Type<br>AlertID: MissionContingencyAlertID_Type<br>ActivatedPlans: MA_PlansReferenceType |
| <b>MissionContingencyAlertMT_1:</b><br><b>MissionContingencyAlertMT</b> | AlertOnly: EmptyType                                                                                                                                                 |
| <b>MissionContingencyAlertMT_2:</b><br><b>MissionContingencyAlertMT</b> | AlertOnly: EmptyType                                                                                                                                                 |
| <b>MissionContingencyAlertMT_3:</b><br><b>MissionContingencyAlertMT</b> | AlertOnly: EmptyType                                                                                                                                                 |
| <b>MissionContingencyAlertMT_4:</b><br><b>MissionContingencyAlertMT</b> | AlertOnly: EmptyType                                                                                                                                                 |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)



#### 1.2.4.11 Triggered Contingency Alert to Peer

When a trigger for a contingency is tripped, MA will send a *MissionContingencyAlertMT\_1* to the relevant Peer systems. The *MissionContingencyAlertMT\_1* will have the *MissionContingencyAlertID* field set to the ID of the alert, the *SourceSystemID* field set to the ID of the system that detects the contingency and is generating the alert, the *ContingencyCondition.ConflictedSystemID* field set to the ID of the system that has the contingency condition, the *ContingencyCondition.ConflictState* field set to "CONFLICTED" to indicate that the conflict that caused the trigger has not been resolved, the *ContingencyCondition.Conflict.ConflictID* set to the unique ID for this conflict, the *ContingencyCondition.Trigger.Source* field set to "AUTONOMOUS\_TRIGGER", and the *ContingencyCondition.Trigger.ByCaseTrigger* field set to the initial by case trigger for autonomous mission planning functions. If the *MissionContingencyAlertMT\_1* is not to result in any action being taken, set the *ContingencyCondition.AutonomousActionStatus.AlertOnly* field to "TRUE" to indicate that there will be no action taken.

If the contingency has been cleared, the MA will send the relevant Peer systems a *MissionContingencyAlertMT\_2* with the *MissionContingencyAlertID* field set to the ID of the alert, the *SourceSystemID* field set to the ID of the system that detected the contingency and generated the alert, the *ContingencyCondition.ConflictedSystemID* field set to the ID of the system that had the contingency condition, the *ContingencyCondition.ConflictState* field set to "CLEARED" to indicate that the conflict that caused the trigger has been resolved, the *ContingencyCondition.Conflict.ConflictID* set to the unique id for this conflict, the *ContingencyCondition.Trigger.Source* field set to "AUTONOMOUS\_TRIGGER", and the *ContingencyCondition.Trigger.ByCaseTrigger* field set to the initial by case trigger for autonomous mission planning functions. If no action was taken as a result of the initial *MissionContingencyAlertMT\_1*, set the *ContingencyCondition.AutonomousActionStatus.AlertOnly* field to "TRUE" to indicate that no was action taken.

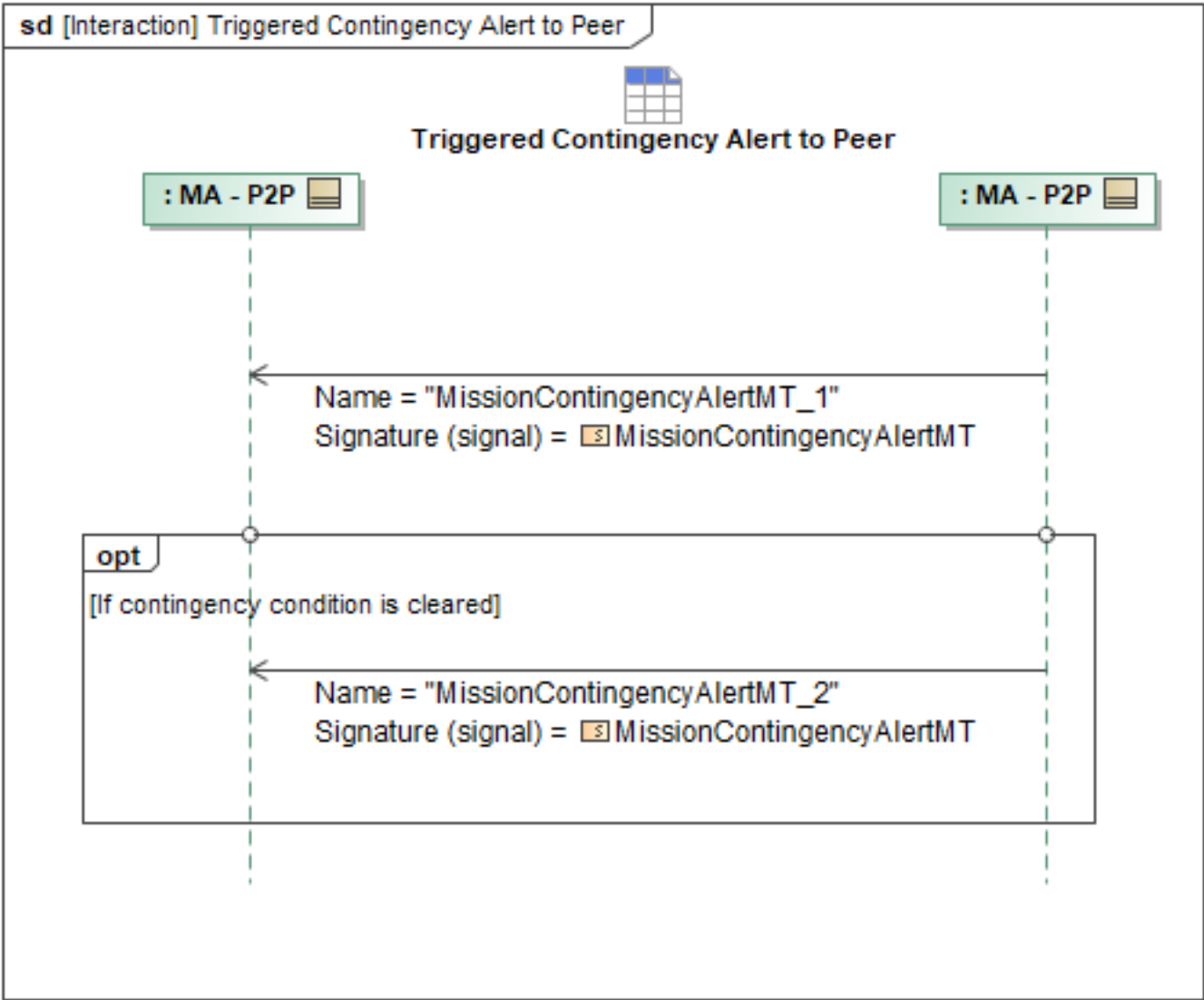


Figure 1-19: Triggered Contingency Alert to Peer

Table 1-18: Triggered Contingency Alert to Peer

| Message Name (Name:Type)                                  | Required Fields (Name:Type)              |
|-----------------------------------------------------------|------------------------------------------|
| MissionContingencyAlertMT_1:<br>MissionContingencyAlertMT | ByCaseTrigger: PlanningByCaseTriggerType |
| MissionContingencyAlertMT_2:<br>MissionContingencyAlertMT | ByCaseTrigger: PlanningByCaseTriggerType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.5 Peer COP and Coordination Package

1.2.5.1 Configure COP and Request COP Configuration Settings Status (P2P)

This interaction describes the process of the Lead ACP configuring the COP using the P2P interface.

If the Lead ACP configures or updates the COP configuration settings, the *MA\_COP\_ConfigurationSettingsCommandMT* will be sent to Mission Autonomy (MA) at the Recipient ACPs, with the *COP\_Participants*, *COP\_LeadPrioritized*, and *RequestAuthority* field set to *REQUEST*, to update their COP configuration settings accordingly. If both C2 and the Lead ACP send *MA\_COP\_ConfigurationSettingsCommandMT* to Recipient ACPs, the C2 command will be prioritized over the Lead ACP command. To update a Configuration item, set *ConfigurationID* to the UUID of the item to be updated. To add a Configuration item, the *ConfigurationID* is optional. To delete a Configuration item, set the *ConfigurationID* to the UUID of the item to be removed and omit the remaining fields in the Configuration. Once received, MA on the recipient ACP can optionally respond with *MA\_COP\_ConfigurationSettingsCommandStatusMT\_2* with *CommandProcessingState* set to *RECEIVED*. If the command is completed, the ACP will send *MA\_COP\_ConfigurationSettingsCommandStatusMT\_3* with *CommandProcessingState* set to *ACCEPTED*. If unable to execute the command, the ACP will send *MA\_COP\_ConfigurationSettingsCommandStatusMT\_4* with *CommandProcessingState* set to *REJECTED*. MA will determine whether the COP Leader is authorized to command the initial *MA\_COP\_ConfigurationSettingsCommandMT*. If unauthorized, MA will ignore the *MA\_COP\_ConfigurationSettingsCommandMT* command and reply with *MA\_COP\_ConfigurationSettingsCommandStatusMT\_1* where *CommandProcessingStateEnum* is set to *REJECTED* and *CommandProcessingStateReason* shall define the *CannotComplyEnum* as *INELIGIBLE\_CONTROL\_SOURCE*.

In addition to this, the Lead ACP may send *MA\_COP\_ConfigurationSettingsRequestMT*, requesting a status update on the COP Configuration Settings from all ACPs. While processing the request, the ACP may optionally send a *MA\_COP\_ConfigurationSettingsRequestStatusMT\_2* with *RequestProcessingState* set to *PROCESSING*. Once the request is completed, the ACP will send *MA\_COP\_ConfigurationSettingsRequestStatusMT\_3* with the *RequestProcessingState* set to *COMPLETED* and the current settings in the *ConfigurationSettings*. If unable to complete the command, the ACP will send *MA\_COP\_ConfigurationSettingsRequestStatusMT\_4* with the *RequestProcessingState* set to *FAILED*. MA will determine whether the COP Leader is authorized to command the initial *MA\_COP\_ConfigurationSettingsRequestMT*. If the COP leader is unauthorized, MA will ignore the *MA\_COP\_ConfigurationSettingsRequestMT* command and reply with *MA\_COP\_ConfigurationSettingsRequestStatusMT\_1* where *RequestProcessingStateEnum* is set to *REJECTED* and *RequestProcessingStateReason* shall define the *CannotComplyEnum* as *INELIGIBLE\_CONTROL\_SOURCE*.



Figure 1-20: Configure COP and Request COP Configuration Settings Status (P2P)

Table 1-19: Configure COP and Request COP Configuration Settings Status (P2P)

| Message Name (Name:Type)                                                                      | Required Fields (Name:Type)                    |
|-----------------------------------------------------------------------------------------------|------------------------------------------------|
| MA_COP_ConfigurationSettingsCommandMT:<br>MA_COP_ConfigurationSettingsCommandMT               | None                                           |
| MA_COP_ConfigurationSettingsCommandStatusMT_1:<br>MA_COP_ConfigurationSettingsCommandStatusMT | CommandProcessingStateReason: CannotComplyType |
| MA_COP_ConfigurationSettingsCommandStatusMT_2:<br>MA_COP_ConfigurationSettingsCommandStatusMT | None                                           |
| MA_COP_ConfigurationSettingsCommandStatusMT_3:<br>MA_COP_ConfigurationSettingsCommandStatusMT | None                                           |
| MA_COP_ConfigurationSettingsCommandStatusMT_4:<br>MA_COP_ConfigurationSettingsCommandStatusMT | CommandProcessingStateReason: CannotComplyType |
| MA_COP_ConfigurationSettingsRequestMT:<br>MA_COP_ConfigurationSettingsRequestMT               | None                                           |
| MA_COP_ConfigurationSettingsRequestStatusMT_1:<br>MA_COP_ConfigurationSettingsRequestStatusMT | RequestProcessingStateReason: CannotComplyType |
| MA_COP_ConfigurationSettingsRequestStatusMT_2:<br>MA_COP_ConfigurationSettingsRequestStatusMT | None                                           |
| MA_COP_ConfigurationSettingsRequestStatusMT_3:<br>MA_COP_ConfigurationSettingsRequestStatusMT | None                                           |
| MA_COP_ConfigurationSettingsRequestStatusMT_4:<br>MA_COP_ConfigurationSettingsRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.5.2 Coordinate Simultaneous Arrival Time

To start coordinating arrival times between team members, the Lead ACP sends *MA\_OpPointMT\_1* with an OpPoint of reference and the *Category* set to *RENDEZVOUS*. Upon receipt of the initial *MA\_OpPointMT\_1*, all Follower ACPs compute their earliest and latest arrival times that can be achieved to the referenced OpPoint. All Follower ACPs will then send *MA\_OpPointMT\_2* to the Lead ACP with *Category* set to *RENDEZVOUS* and the *AssociatedTime* fields for both the earliest and

latest arrival times populated. The *AssociatedTime* fields will have the *Function* set to *ARRIVAL* and *Value* field set to the earliest/latest arrival time respectively. The Lead ACP will then compute the coordinated arrival time that all ACPs can support and sends out *MA\_OpPointMT\_3* to all Follower ACPs with the final coordinated arrival time in the *AssociatedTime* field and *Category* set to *RENDEZVOUS*. All ACPs will execute simultaneous arrival based on the coordinated time of arrival that the Lead ACP has computed.

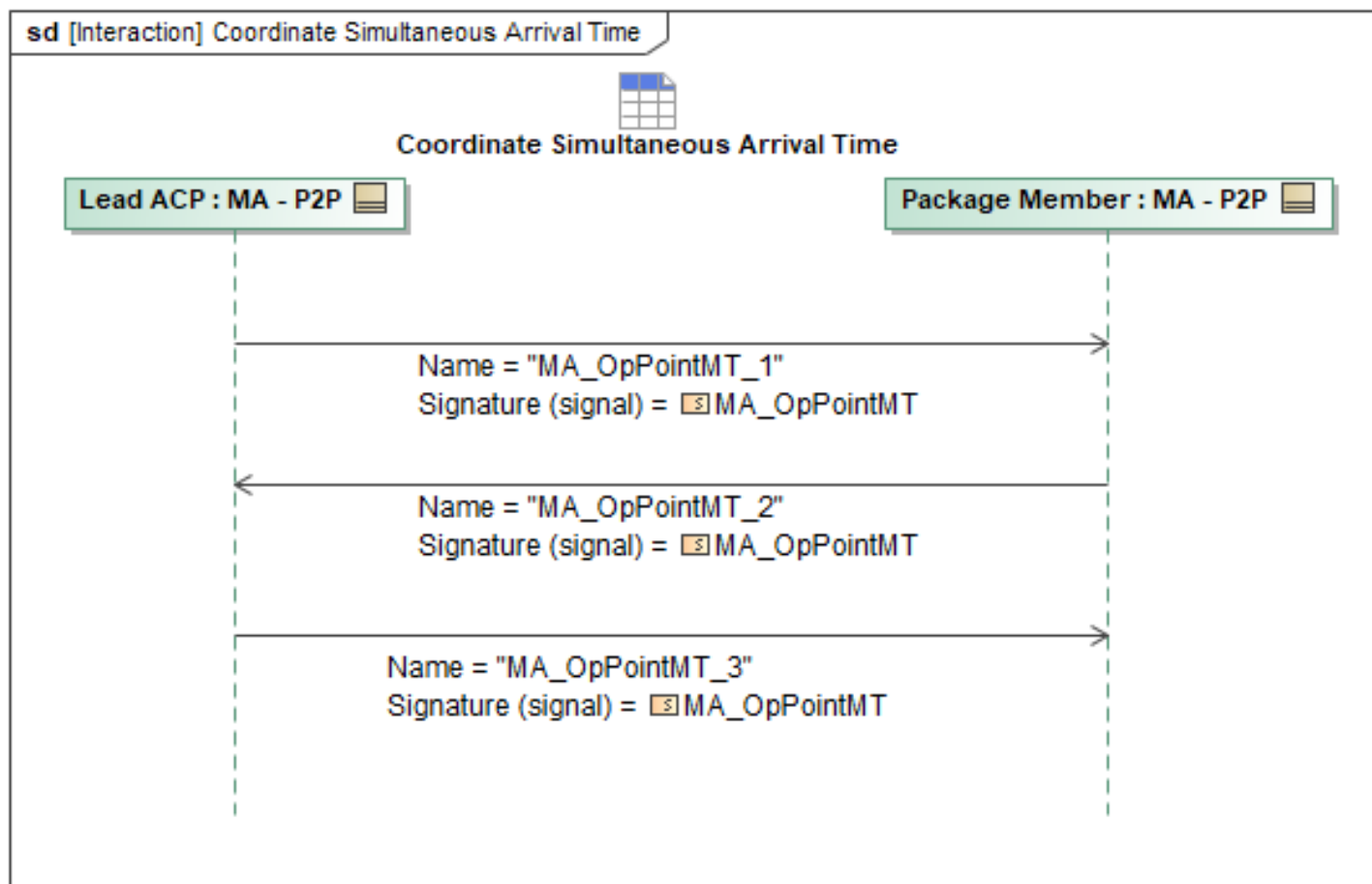


Figure 1-21: Coordinate Simultaneous Arrival Time

Table 1-20: Coordinate Simultaneous Arrival Time

| Message Name (Name:Type)     | Required Fields (Name:Type)      |
|------------------------------|----------------------------------|
| MA_OpPointMT_1: MA_OpPointMT | None                             |
| MA_OpPointMT_2: MA_OpPointMT | AssociatedTime: TimeFunctionType |
| MA_OpPointMT_3: MA_OpPointMT | AssociatedTime: TimeFunctionType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.5.3 COP Follower Configuration Update

The COP Follower Configuration Update interaction supports the use case in which an ACP determines the need to update its own COP Configuration settings. This could be triggered by a

change in capabilities or environment, but is not constrained to a specific trigger. After the trigger occurs, the COP Follower makes an update locally to the COP Configuration Settings. The COP Follower then must notify the COP Leader of their configuration settings update. The ACP sends *MA\_COP\_ConfigurationSettingsMT* to the COP Leader with the *Configuration* updates populated. Once received, the COP Leader reviews the updates and has the option to respond with additional changes or adjustments to the ACP COP Configuration Settings. If the COP Leader wants to make changes to the COP Follower Configuration Settings, they send *MA\_COP\_ConfigurationSettingsCommandMT* to the COP Follower with the new *Configuration* settings. To update a Configuration item, set *ConfigurationID* to the UUID of the item to be updated. Once received, MA on the COP Follower can optionally respond with *MA\_COP\_ConfigurationSettingsCommandStatusMT\_2* with *CommandProcessingState* set to *RECEIVED*. If the command is completed, the COP Follower will send *MA\_COP\_ConfigurationSettingsCommandStatusMT\_3* with *CommandProcessingState* set to *ACCEPTED*. If unable to execute the command, the ACP will send *MA\_COP\_ConfigurationSettingsCommandStatusMT\_4* with *CommandProcessingState* set to *REJECTED*. MA will determine whether the COP Leader is authorized to command the initial *MA\_COP\_ConfigurationSettingsCommandMT*. If unauthorized, MA will ignore the *MA\_COP\_ConfigurationSettingsCommandMT* command and reply with *MA\_COP\_ConfigurationSettingsCommandStatusMT\_1* where *CommandProcessingStateEnum* is set to *REJECTED* and *CommandProcessingStateReason* shall define the *CannotComplyEnum* as *INELIGIBLE\_CONTROL\_SOURCE*.



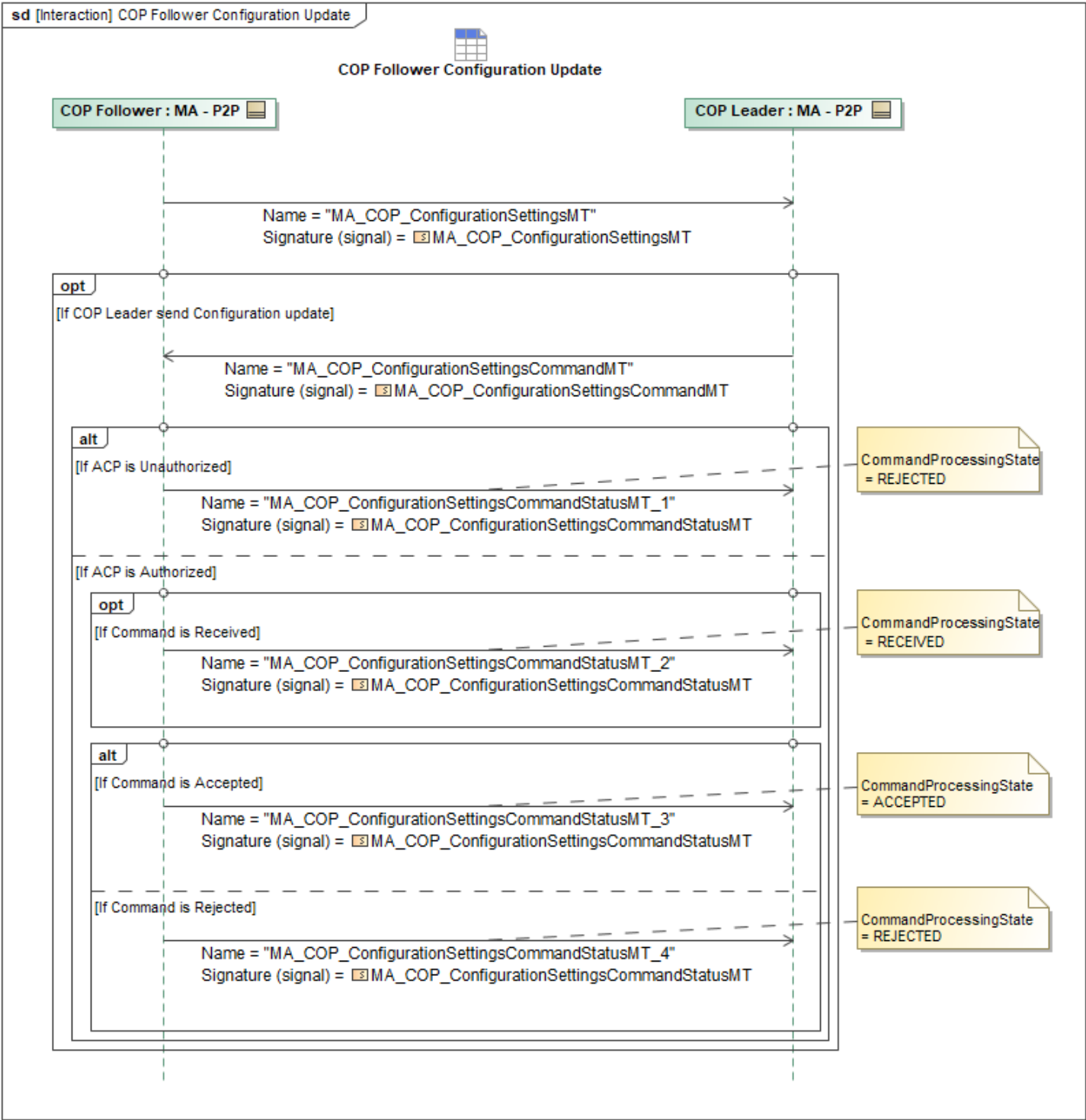


Figure 1-22: COP Follower Configuration Update

Table 1-21: COP Follower Configuration Update

| Message Name (Name:Type)                                                        | Required Fields (Name:Type)                  |
|---------------------------------------------------------------------------------|----------------------------------------------|
| MA_COP_ConfigurationSettingsCommandMT:<br>MA_COP_ConfigurationSettingsCommandMT | ConfigurationID: MA_COP_ConfigurationID_Type |



| Message Name (Name:Type)                                                                      | Required Fields (Name:Type)                    |
|-----------------------------------------------------------------------------------------------|------------------------------------------------|
| MA_COP_ConfigurationSettingsCommandStatusMT_1:<br>MA_COP_ConfigurationSettingsCommandStatusMT | CommandProcessingStateReason: CannotComplyType |
| MA_COP_ConfigurationSettingsCommandStatusMT_2:<br>MA_COP_ConfigurationSettingsCommandStatusMT | None                                           |
| MA_COP_ConfigurationSettingsCommandStatusMT_3:<br>MA_COP_ConfigurationSettingsCommandStatusMT | None                                           |
| MA_COP_ConfigurationSettingsCommandStatusMT_4:<br>MA_COP_ConfigurationSettingsCommandStatusMT | CommandProcessingStateReason: CannotComplyType |
| MA_COP_ConfigurationSettingsMT:<br>MA_COP_ConfigurationSettingsMT                             | ConfigurationID: MA_COP_ConfigurationID_Type   |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.5.4 Distribute Sensor Track Data to Facilitate Package Fusion and Update COP

For context, it is important to clarify what is meant by "un-fused" and "fused" tracks. Un-fused tracks, that is, tracks that have not yet been fused together, are the result of an onboard sensor tracker associating a series of observation measurement reports (OMRs) and linking them to an entity. This can be performed across all sensors that a given platform is equipped with. Once these sensor trackers have produced a set of entities, they may then be run through the sensor fusion engine, which will ingest all available entities and associate, fuse, and/or deconflict them. The output is what is called a fused entity (or track)

The interaction shown here describes the sharing of track data (either fused or un-fused) between ACPs in a package by sending *EntityMT* across P2P interface to facilitate package level fusion. If the network has sufficient bandwidth, *ObservationMeasurementReportMT* data may also be sent to better inform the *EntityMT* data. A kinematic and CID fusion implementation will likely want the capability that OMRs were generated from, and from which system/subsystem had the capability generating the OMR

Within the *ObservationMeasurementReportMT* message, there are two optional paths to different sets of *SubsystemIDs* and *SystemIDs*

1. *ObservationMeasurementReportMDT* → *ObservationMeasurement* → *MeasurementSource* → *ElementDetails* → *SourceElementIdentifier* → *SystemID* + *SubsystemID*

These messages provide the system/subsystem that's the source of the OMR's kinematics

2. *ObservationMeasurementReportMDT* → *ObservationMeasurement* → *MeasurementSource* → *SourceIdentity* → *CapabilityID* + *SystemID* + *SubsystemID*

These messages provide the system/subsystem and capability for the source of identity in the OMR (it's expected this will usually be the same system/subsystem as the kinematics) A client would want to know the capability type of the provided entity ID, and may have to subscribe to additional messages to understand the type (e.g. [*Capability*]*StatusMT* messages)

The *EntityMT* message contains the *MeasurementID* of all of the OMRs that have contributed to the formation of the entity. Whether the entity or track being shared between ACPs is un-fused or fused can be determined by the *SourceTypeIdentifier* field, and is represented as *EntityMT\_1* and *EntityMT\_2*, respectively. Within the *SourceTypeIdentifier* is a *Fusion* field which contains all of the fusion contributors, that is, the un-fused entities that have contributed to a fused track, meaning that an un-fused entity will not contain any fusion contributors in the *SourceTypeIdentifier* data. Once un-fused entities (and optionally OMRs) have been received, the fusion engine may produce fused entities that can then be ingested and utilized by the Common Operating Picture (COP).

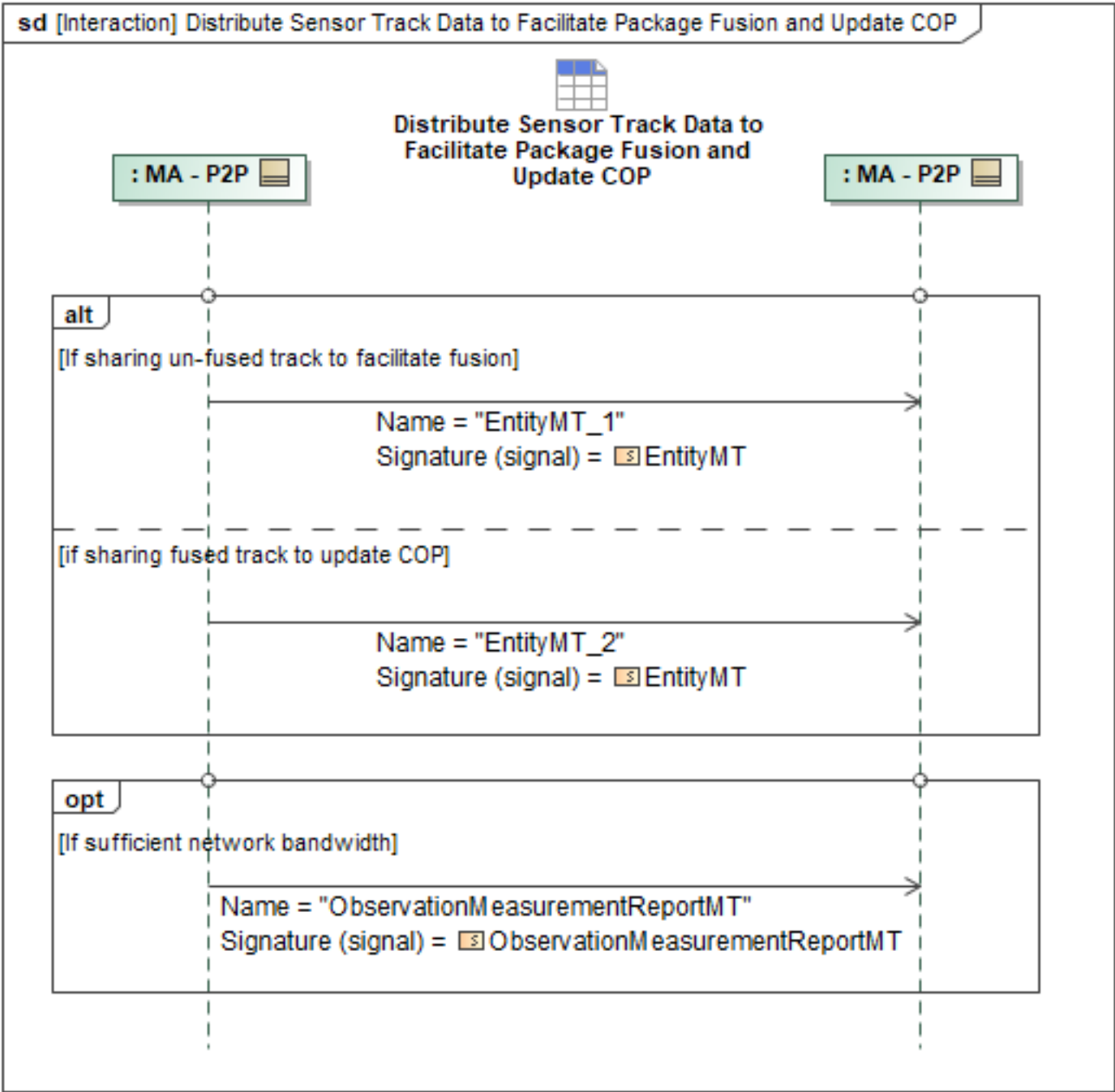


Figure 1-23: Distribute Sensor Track Data to Facilitate Package Fusion and Update COP

Table 1-22: Distribute Sensor Track Data to Facilitate Package Fusion and Update COP

| Message Name (Name:Type)                                       | Required Fields (Name:Type)                                                           |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| EntityMT_1: EntityMT                                           | MeasurementID: MeasurementID_Type                                                     |
| EntityMT_2: EntityMT                                           | MeasurementID: MeasurementID_Type<br>SourceTypeIdentifier: EntitySourceIdentifierType |
| ObservationMeasurementReportMT: ObservationMeasurementReportMT | MeasurementSource: ObservationMeasurementSourceType                                   |

| Message Name (Name:Type) | Required Fields (Name:Type)                                                                                                                                                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | ElementDetails: ElementDetailsType<br>SourceElementIdentifier: SourceKinematicsElementIdentityType<br>SystemID: SystemID_Type<br>SubsystemID: SubsystemID_Type<br>SourceIdentity: MeasurementSourceIdentityType<br>CapabilityID: CapabilityID_Type<br>SystemID: SystemID_Type<br>SubsystemID: SubsystemID_Type<br>MeasurementKinematics: MeasurementKinematicsChoiceType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.5.5 Synchronize Global COP to Peer

The Synchronize Global COP to Peer interaction supports the use case in which the COP Leader has generated a new or updated Global COP and needs to distribute the data to the COP Followers. The output of the Global COP includes updated Entity data and updated Team States. This information needs to be synchronized to each peer in the executing package each time the Global COP is updated. The COP Leader sends *EntityMT* and *PackageStatusMT* to each of the COP Followers. *EntityMT* includes the *MeasurementID* and the *SourceTypeIdentifier* fields to provide context for the updated measurements and originating sources, respectively. *PackageStatusMT* communicates package status, as well as Team States of each member in the package using *PackagePartnerStatus*.

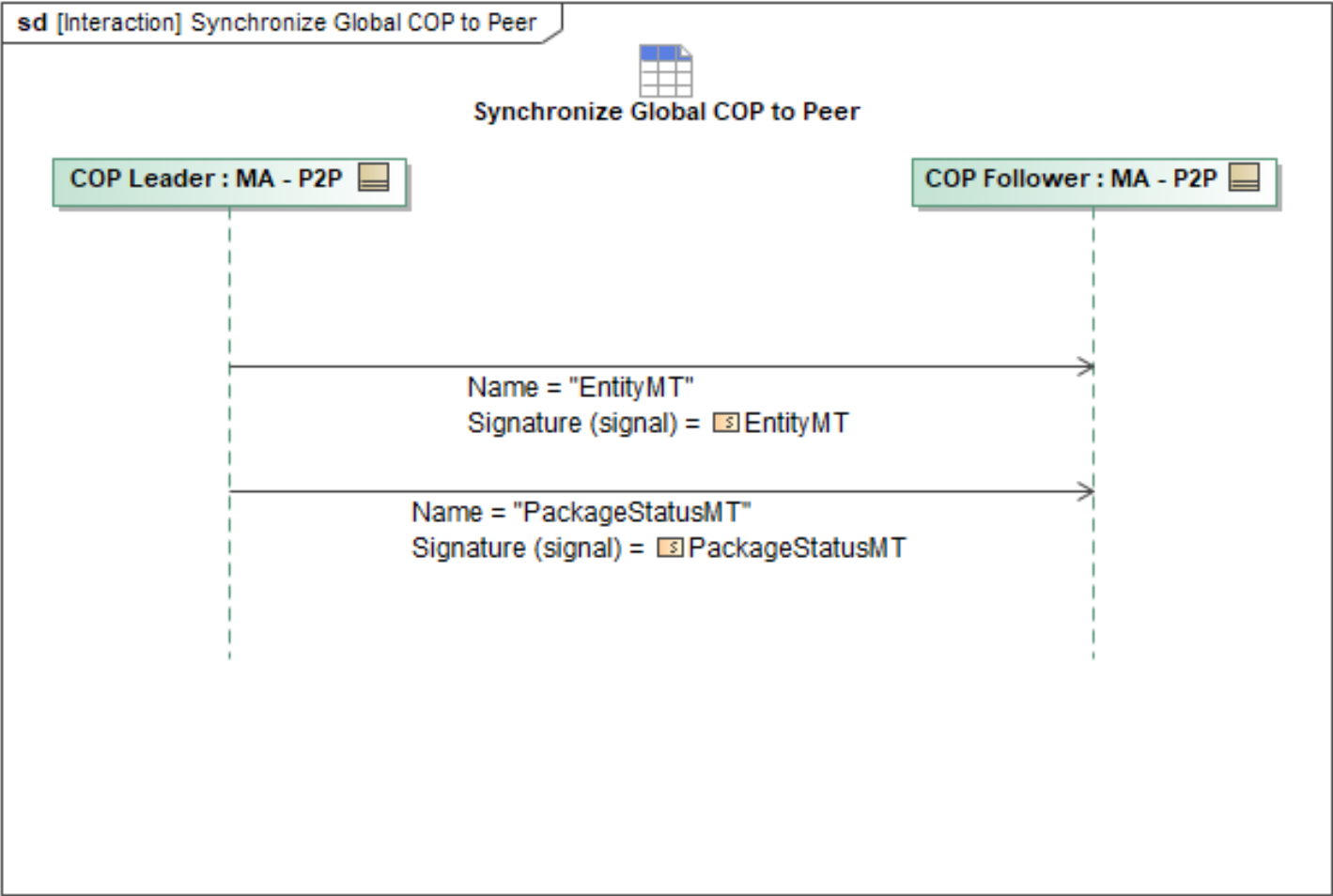


Figure 1-24: Synchronize Global COP to Peer

Table 1-23: Synchronize Global COP to Peer

| Message Name (Name:Type)         | Required Fields (Name:Type)                                                           |
|----------------------------------|---------------------------------------------------------------------------------------|
| EntityMT: EntityMT               | MeasurementID: MeasurementID_Type<br>SourceTypeIdentifier: EntitySourceIdentifierType |
| PackageStatusMT: PackageStatusMT | None                                                                                  |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.6 Peer COP Vehicle State Package

1.2.6.1 P2P Send Platform Override Status

The Peer-to-Peer (P2P) Send Platform Override Status sequence specifies the interaction of Mission Autonomy (MA) sharing the active override(s) with peers. The transmission of this message allows Autonomy Capable Platforms (ACPs) to understand the operating state of peers in the airspace. At predefined frequencies, the *MA\_PlatformOverrideStatusMT* is sent. The *MA\_PlatformOverrideStatusMT* must contain the active global override settings; the shared value will be a speed, altitude, formation, or Identification Friend or Foe (IFF) setting conveyed via the

*PlatformOverrideStatus* field with the value of each active override setting. In addition, the *ActiveOverrides* field must be set.

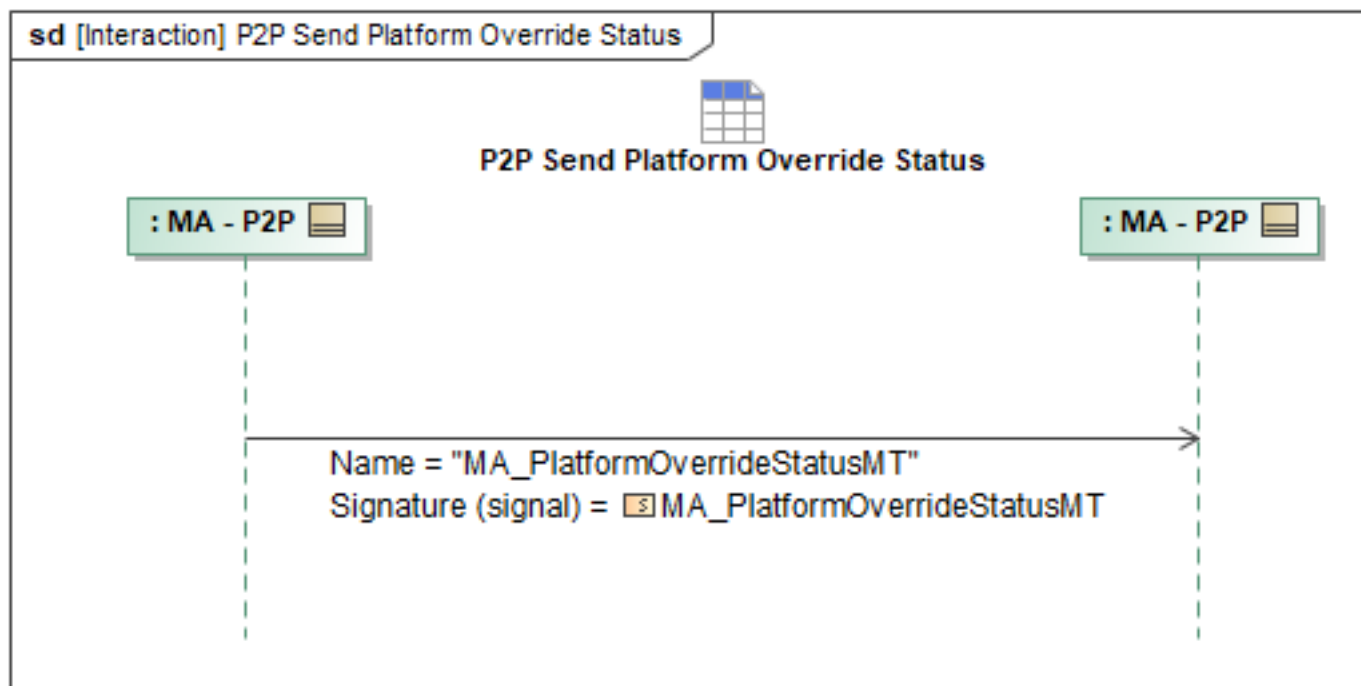


Figure 1-25: P2P Send Platform Override Status

Table 1-24: P2P Send Platform Override Status

| Message Name (Name:Type)                                    | Required Fields (Name:Type)                                      |
|-------------------------------------------------------------|------------------------------------------------------------------|
| MA_PlatformOverrideStatusMT:<br>MA_PlatformOverrideStatusMT | PlatformOverrideStatus:<br>MA_PlatformDefaultAndOverrideDataType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.6.2 P2P State Data

Vehicle State Data is reported periodically between peers. Vehicle state data is included in the messages *MA\_FlightActivityMT*, *ComponentStatusMT*, *MA\_PositionReportMT*, and *MA\_NavigationReportMT*. *ComponentStatusMT* and *MA\_TaskStatusMT* can be included to provide further tasking details when required between peers in instances such as formation tasks. *MA\_PositionReportMT* includes *InertialState*, defining *Position*, *DomainVelocity*, *GroundVelocity*, *CalibratedAirspeed*, and *Mach*. *DomainVelocity* represents the True Airspeed of the vehicle. The transmission of these messages among peers will give each peer a full picture of the package they are all part of. Vehicle State Data is reported periodically or optionally requested by MA. This may be a timed interval, like for telemetry, or during an action that requires the peers to have more information, like for waypoint following.

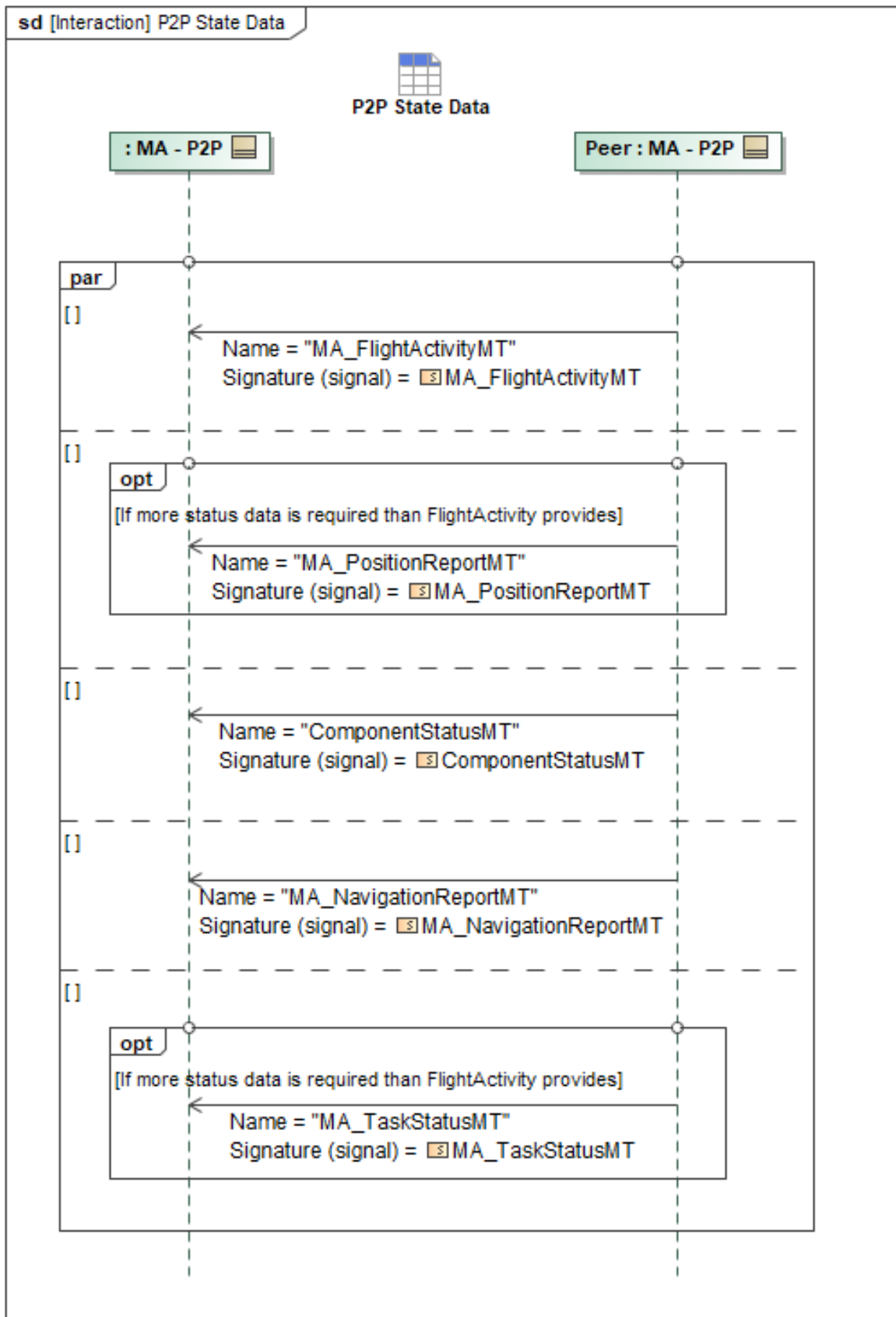


Figure 1-26: P2P State Data

Table 1-25: P2P State Data

| Message Name (Name:Type)                        | Required Fields (Name:Type)                                                                                           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| ComponentStatusMT: ComponentStatusMT            | None                                                                                                                  |
| MA_FlightActivityMT: MA_FlightActivityMT        | None                                                                                                                  |
| MA_NavigationReportMT:<br>MA_NavigationReportMT | None                                                                                                                  |
| MA_PositionReportMT: MA_PositionReportMT        | CalibratedAirspeed: SpeedType<br>Mach: MachType<br>GroundVelocity: Velocity2D_Type<br>DomainVelocity: Velocity3D_Type |
| MA_TaskStatusMT: MA_TaskStatusMT                | Metrics: MetricsType                                                                                                  |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.6.3 Peer Heartbeat

Peers are responsible for issuing periodic heartbeat messages to each other via the *SystemStatusMT* message. For the purposes of vehicle identification, the field *Identity.SpecificVehicle.VehicleUniqueIdentifier.TailNumber* shall be populated. The *TailNumber* and other system identification information is sourced from pre-mission planning data. No response is required for heartbeat messages issued.



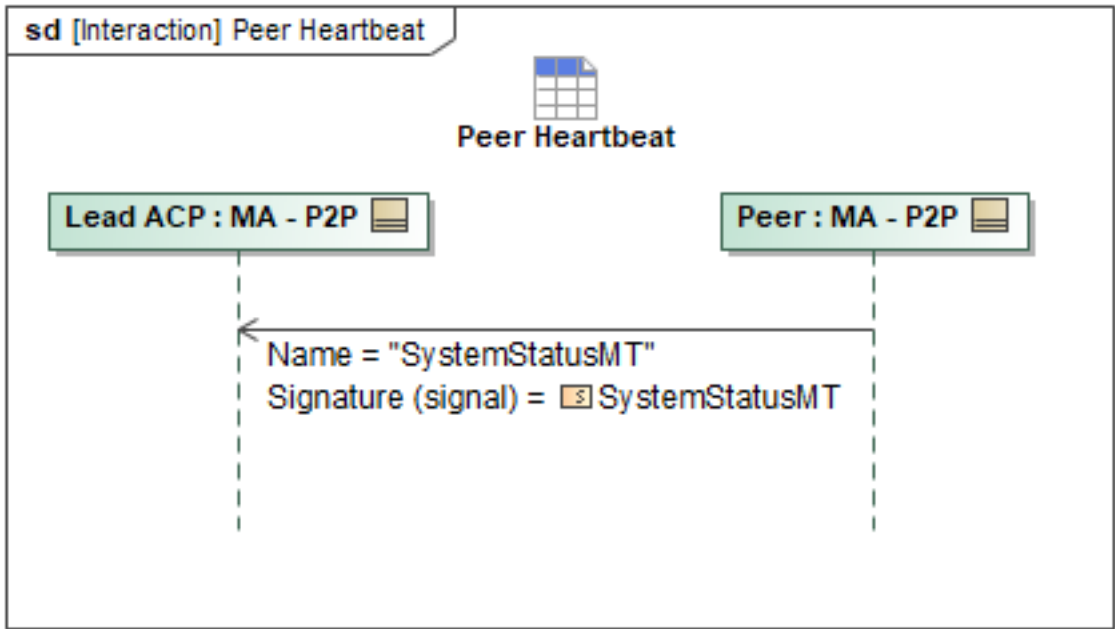


Figure 1-27: Peer Heartbeat

Table 1-26: Peer Heartbeat

| Message Name (Name:Type)       | Required Fields (Name:Type) |
|--------------------------------|-----------------------------|
| SystemStatusMT: SystemStatusMT | TailNumber: TailNumberType  |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.6.4 Peer Publish Control Status

Flight Autonomy (FA), through the Vehicle Interface (VI), will periodically publish the *ControlStatusMT* message to Mission Autonomy (MA), indicating the current controlling system and service. MA must then relay the *PRIMARY\_CONTROLLER* and *SECONDARY\_CONTROLLER* designations to Peers for monitoring. This is depicted through the Peer Publish Controls Status interaction, where the status is indicated by specifying through *ControlStatusMT/MessageData/ControlType/CapabilityControl/PrimaryController* and *ControlStatusMT/MessageData/ControlType/CapabilityControl/SecondaryController*.

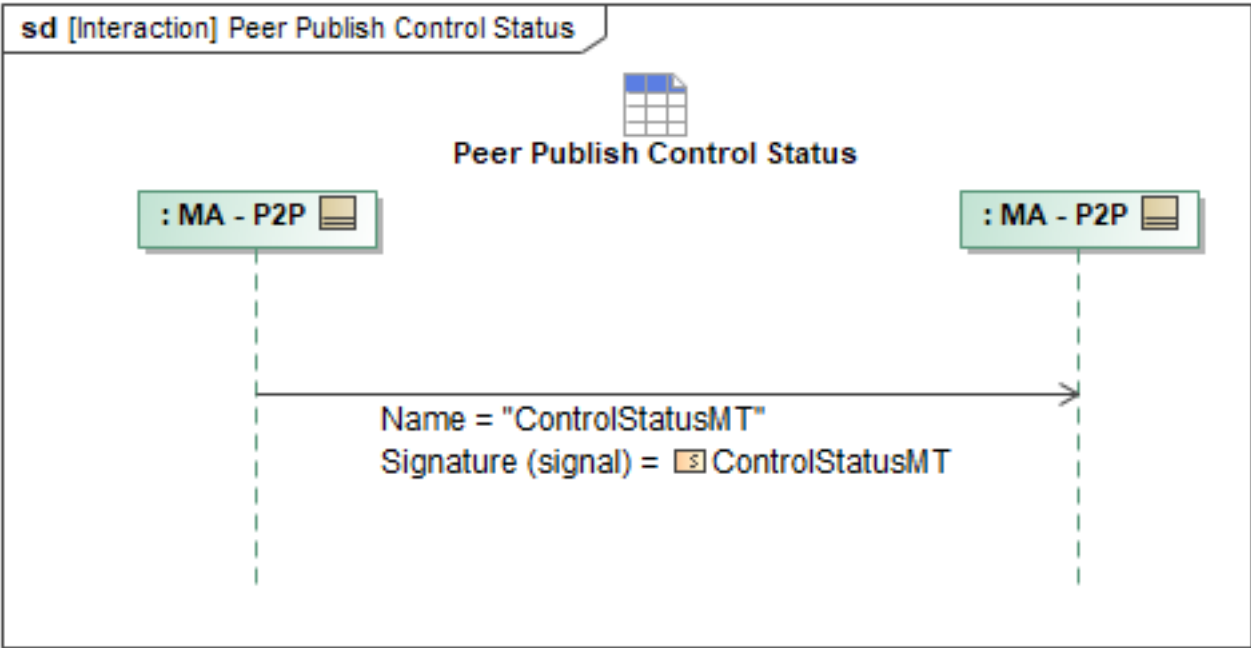


Figure 1-28: Peer Publish Control Status

Table 1-27: Peer Publish Control Status

| Message Name (Name:Type)         | Required Fields (Name:Type)                                                              |
|----------------------------------|------------------------------------------------------------------------------------------|
| ControlStatusMT: ControlStatusMT | PrimaryController: PrimaryControllerType<br>SecondaryController: SecondaryControllerType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.6.5 Send Abort to Peer

When there is an abort for an ACP, MA may forward that information to the Lead MA or to any or all it's Peers. This interaction between MA and one of it's Peer (could be the Lead MA or just a normal Peer) forwards the *MissionPlanActivationStatusMT* and *RoutePlanExecutionStatusMT* from MA to the Peer which will be used to communicate:

- RoutePlanID* of the route plan that has been aborted
- MissionPlanActivationStatus:SubPlanActivationState:ActivationState* is *DEACTIVATED*
- PlanID* of the route plan that has been aborted and a *PlanExecutionState* of *Canceled*

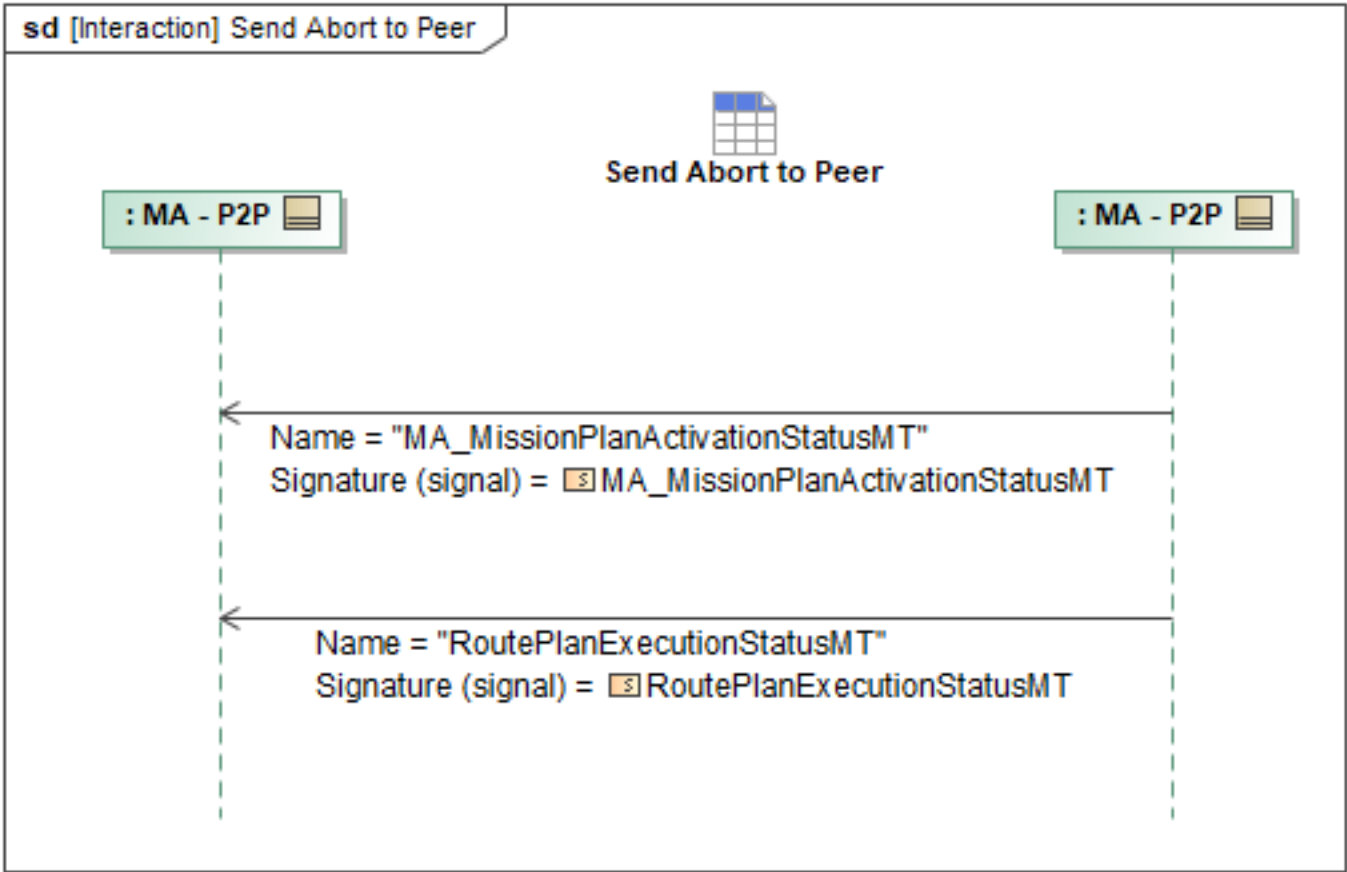


Figure 1-29: Send Abort to Peer

Table 1-28: Send Abort to Peer

| Message Name (Name:Type)                                              | Required Fields (Name:Type)                                                               |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| MA_MissionPlanActivationStatusMT:<br>MA_MissionPlanActivationStatusMT | DEACTIVATED:<br>RoutePlanID: RoutePlanID_Type<br>ActivationState: PlanActivationStateEnum |
| RoutePlanExecutionStatusMT:<br>RoutePlanExecutionStatusMT             | PlanExecutionState: PlanExecutionStateEnum<br>PlanID: RoutePlanID_Type<br>CANCELED:       |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.6.6 Send AMTI Capability State to Peer

Mission Use Case: AMTI

The “Send AMTI Capability State Data to Peer” sequence diagram specifies the interaction of the Mission Autonomy (MA) sharing Airborne Moving Target Indicator (AMTI) Capability State Data with peers. The transmission of these messages allows ACPs to understand the operational AMTI capability of peers in the airspace. In the context of a centralized Common Operating Picture (COP)

distribution setup, the AMTI capability state messages would be aggregated by the COP lead for distribution outside of the package.

If the platform has Airborne Moving Target Indicator (AMTI) sensing capabilities, the following messages are sent periodically: *MA\_AMTI\_CapabilityMT*, *AMTI\_CapabilityStatusMT*, and *AMTI\_ActivityMT*.

The *MA\_AMTI\_CapabilityMT* is published with at least one *Capability* specified.

The *AMTI\_CapabilityStatusMT* and *AMTI\_ActivityMT* report status and usage of the AMTI capability. COP uses *MA\_AMTI\_CapabilityMT* and *AMTI\_CapabilityStatusMT* to know if the AMTI sensor capability is available to command.

COP uses *AMTI\_ActivityMT* to know if the AMTI capability can effectively be used (e.g. not fully scheduled).

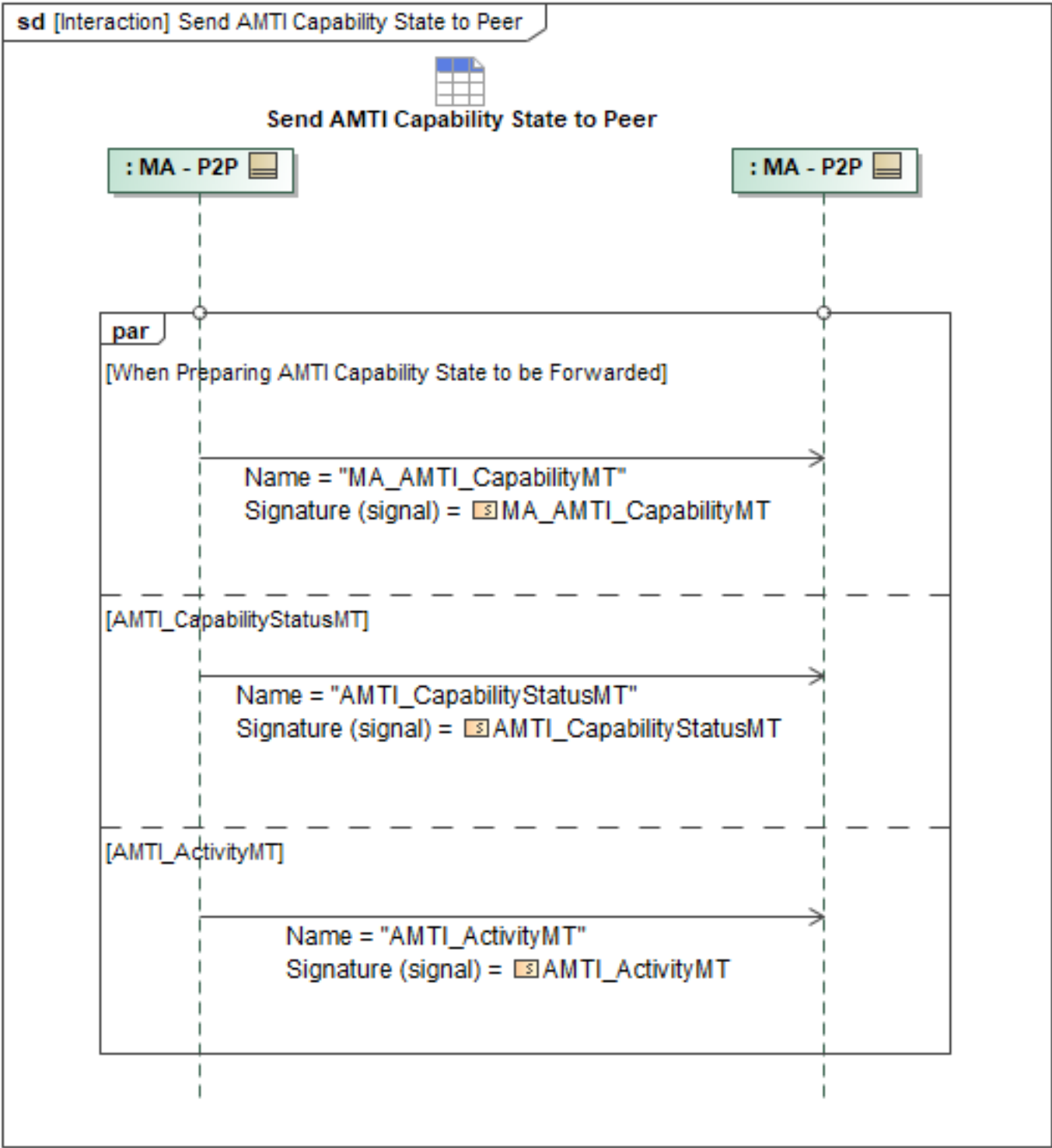


Figure 1-30: Send AMTI Capability State to Peer

Table 1-29: Send AMTI Capability State to Peer

| Message Name (Name:Type)         | Required Fields (Name:Type) |
|----------------------------------|-----------------------------|
| AMTI_ActivityMT: AMTI_ActivityMT | None                        |

| Message Name (Name:Type)                            | Required Fields (Name:Type) |
|-----------------------------------------------------|-----------------------------|
| AMTI_CapabilityStatusMT:<br>AMTI_CapabilityStatusMT | None                        |
| MA_AMTI_CapabilityMT:<br>MA_AMTI_CapabilityMT       | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.6.7 Send PO Capability State to Peer

#### Mission Use Case: PO

The “Send Passive Optical (PO) Capability State Data to Peer” sequence diagram specifies the interaction of the Mission Autonomy (MA) sharing PO Capability State Data with peers. The transmission of these messages allows ACPs to understand the operational PO capability of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the PO capability state messages would be aggregated by the COP lead for distribution outside of the package.

If the platform has Passive Optical (PO) sensor capability, the following messages are sent periodically: *PO\_CapabilityMT*, *PO\_CapabilityStatusMT*, *PO\_ActivityMT*.

The *PO\_CapabilityMT* is published with with at least one *Capability* specified.

The *PO\_CapabilityStatusMT* and *PO\_ActivityMT* report status and usage of the PO capability. COP uses *PO\_CapabilityMT* and *PO\_CapabilityStatusMT* to know if the PO sensor capability is available to command. COP uses *PO\_ActivityMT* to know if the PO capability can effectively be used (e.g. not fully scheduled).

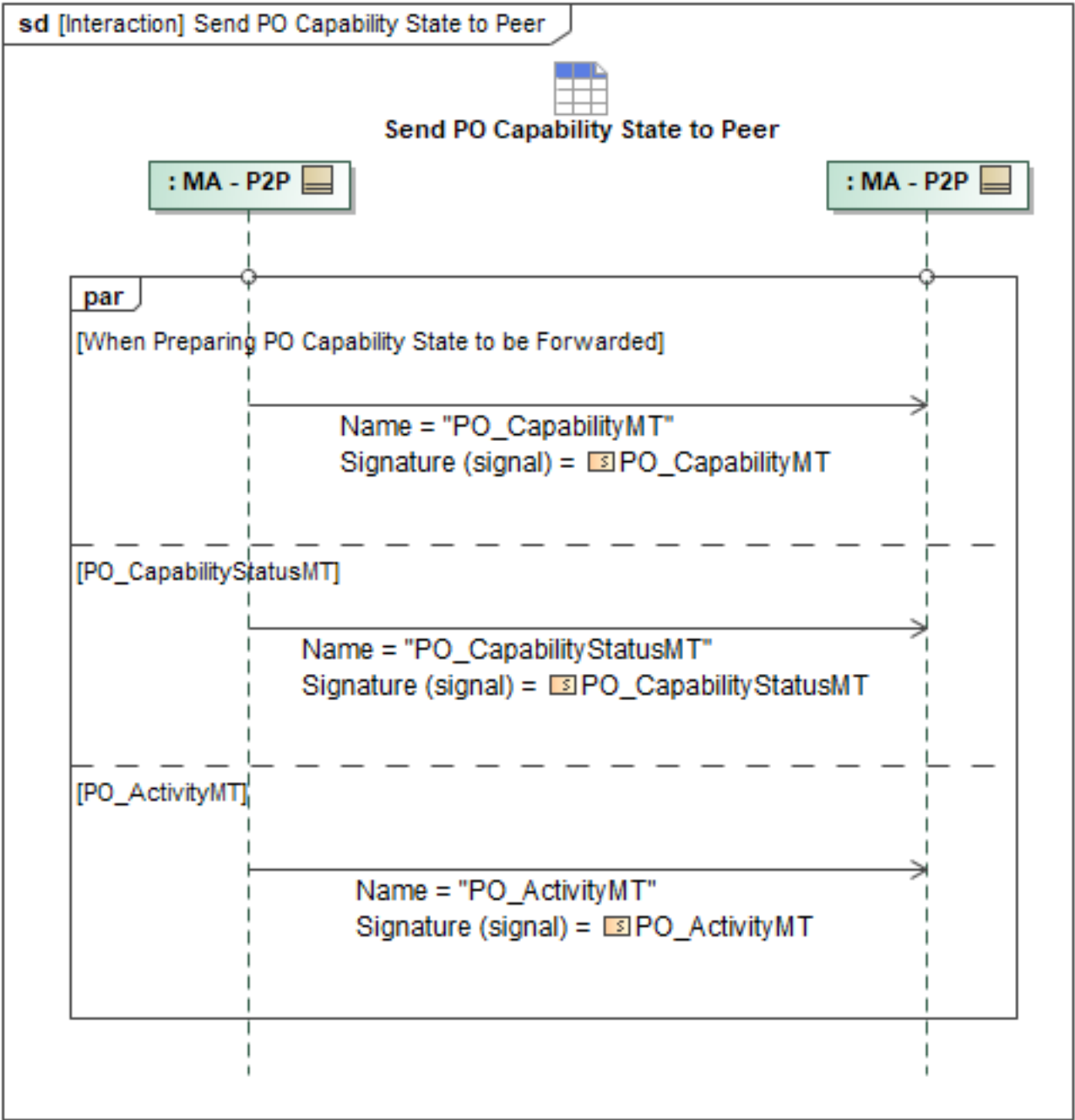


Figure 1-31: Send PO Capability State to Peer

Table 1-30: Send PO Capability State to Peer

| Message Name (Name:Type)                     | Required Fields (Name:Type) |
|----------------------------------------------|-----------------------------|
| PO_ActivityMT: PO_ActivityMT                 | None                        |
| PO_CapabilityMT: PO_CapabilityMT             | None                        |
| PO_CapabilityStatusMT: PO_CapabilityStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy

Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.6.8 Send Strike Capability State to Peer

**Mission Use Case:** Store Management

The “Send Strike Capability State Data to Peer” sequence diagram specifies the interaction of the Mission Autonomy (MA) sharing Strike Capability State Data with peers. The transmission of these messages allows ACPs to understand the operational strike capability of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the strike capability state messages would be aggregated by the COP lead for distribution outside of the package.

If the platform has a Strike Capability, the following messages are sent periodically:

*MA\_StrikeCapabilityMT*, *StrikeCapabilityStatusMT*, *StoreManagementStatusMT*, *MA\_StrikeActivityMT*.

The *MA\_StrikeCapabilityMT* message is published with at least one *Capability* required. Each *Capability* has its own unique identifier set in the *CapabilityID*. The *AcceptedInterface* field must include at least *CAPABILITY\_COMMAND*. The *StoreType* and *StoreCategory* fields are used to identify the type of weapons the *Capability* is used to status and command. MS implementation determines how the *InterruptOtherActivities* field is set. Other schema-optional fields are not required but may be specified for more information.

The *StrikeCapabilityStatusMT* messages report status of the weapon capabilities identified by the *Capability* fields in the *MA\_StrikeCapabilityMT* message. Each *CapabilityStatus* has the *CapabilityID* set to match the *CapabilityID* field of the corresponding *Capability*. The *Availability* field is used to report general usability of the weapons *Capability* by MA. The *StoreStatus* field is required with *StoreState* and *StoreStatusDetail* communicating the specific state information of the weapon(s) associated with the *Capability*. Other schema-optional fields are not required but may be specified for more information.

The *StoreManagementStatusMT* messages report the status of individual stores and allow MA to know weapon quantities and individual statuses. This is complementary to general usage status provided by *MA\_StrikeCapabilityMT* and *StrikeCapabilityStatusMT*. The *SupportCapabilityID* field specifies a unique identifier used by other messaging to manipulate store settings. Each *StoreStation* has a *StationIndex* to identify station using MS implementation specific schemes. The *CompatibleStore* field specifies the type of weapons that station can hold. The *InstalledStore* field is optional but required when the station currently holds a weapon. When specified, the *InstalledStore* field identifies the weapon type. If the *InstalledStore* field and sub-fields are omitted, it is interpreted that this station does not hold a weapon. Each station is associated with a *Capability* defined by *MA\_StrikeCapabilityMT* by specifying the *CapabilityID* field, which is required.

The *MA\_StrikeActivityMT* communicates more information about the current employment of weapons to MA. Each *Activity* specifies a unique identifier in the *ActivityID* field, corresponding to a scheduled weapon employment. The *CapabilityID* field associates the employment with a specific *Capability* defined in the *MA\_StrikeCapabilityMT* message. Interactive is set according to MS implementation. The *ActivityState* field is used to specify the state of the weapon employment activity. The *SelectedForKeyLoad*, *WeaponArmed*, *ReadyForRelease*, *SelectedForRelease*, and *SelectedForJettison* fields provide information about the current state of the weapon while it prepares for the weapon employment activity. The *AssignedTarget* field is required when the weapon is scheduled to be launched at a target and not jettisoned. The *TimeToRelease* field is required and



specifies the timing of the weapon release.

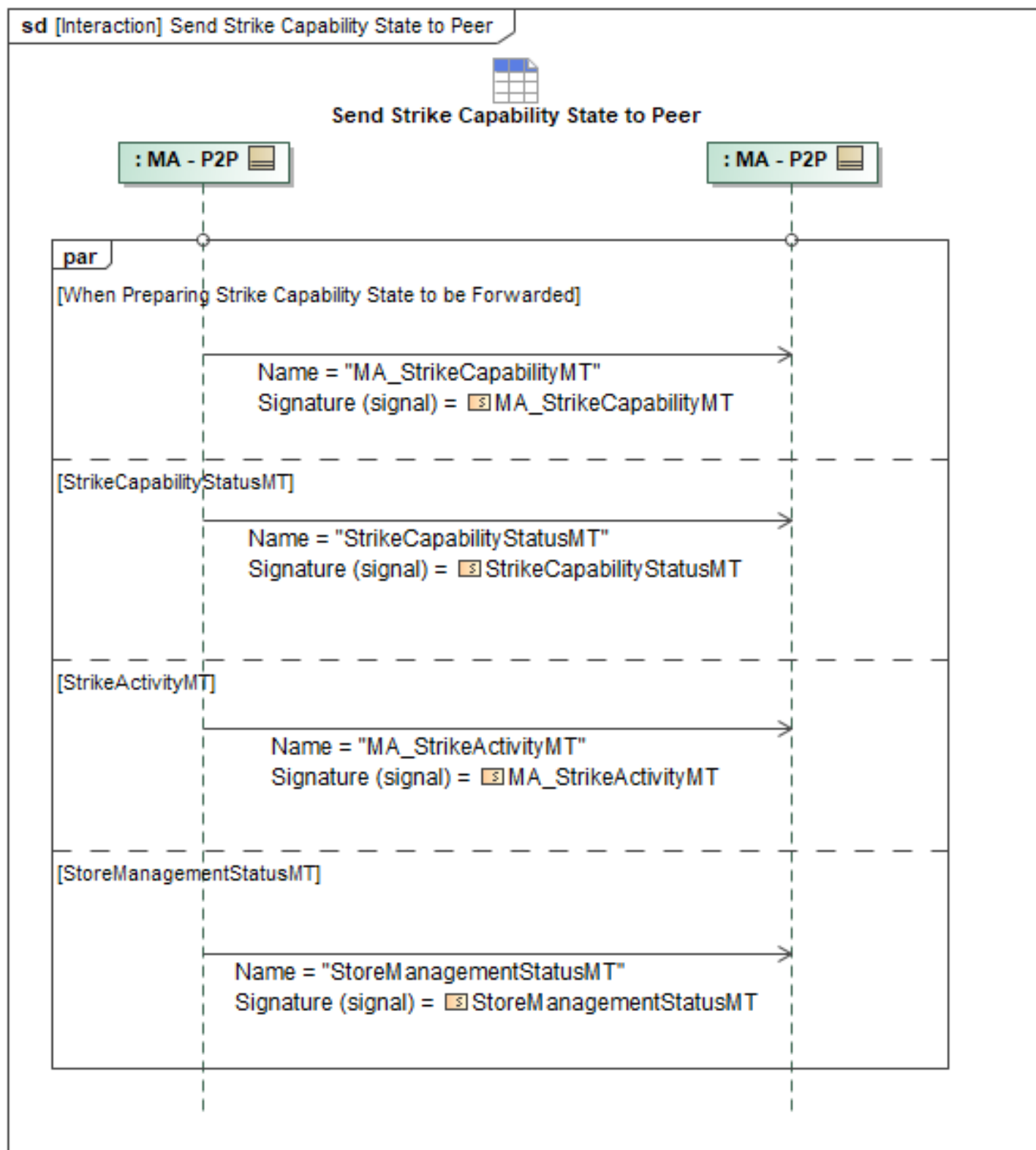


Figure 1-32: Send Strike Capability State to Peer

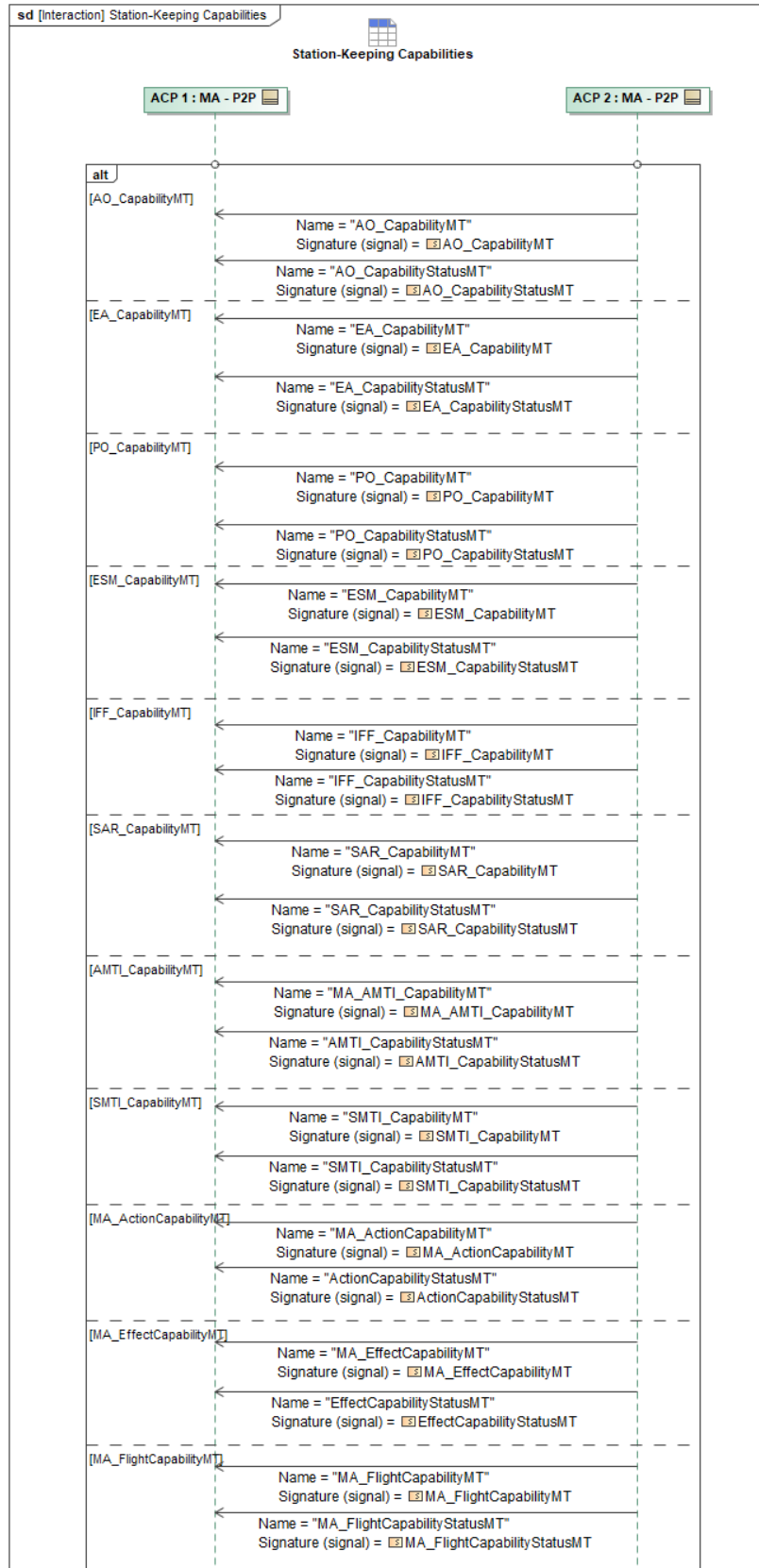
Table 1-31: Send Strike Capability State to Peer

| Message Name (Name:Type)                                  | Required Fields (Name:Type)         |
|-----------------------------------------------------------|-------------------------------------|
| <b>MA_StrikeActivityMT: MA_StrikeActivityMT</b>           | ActivityState: ActivityStateEnum    |
| <b>MA_StrikeCapabilityMT: MA_StrikeCapabilityMT</b>       | Capability: MA_StrikeCapabilityType |
| <b>StoreManagementStatusMT: StoreManagementStatusMT</b>   | CapabilityID: CapabilityID_Type     |
| <b>StrikeCapabilityStatusMT: StrikeCapabilityStatusMT</b> | StoreStatus: StoreStatusType        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.6.9 Station-Keeping Capabilities

The known set of station-keeping capabilities is presented in the following sequence diagram. A platform may advertise multiple of these capabilities (*[Capability]ActivityMT* and *[Capability]StatusMTs*) based on platform configuration, and each of which is sent periodically as a separate instance of this sequence.



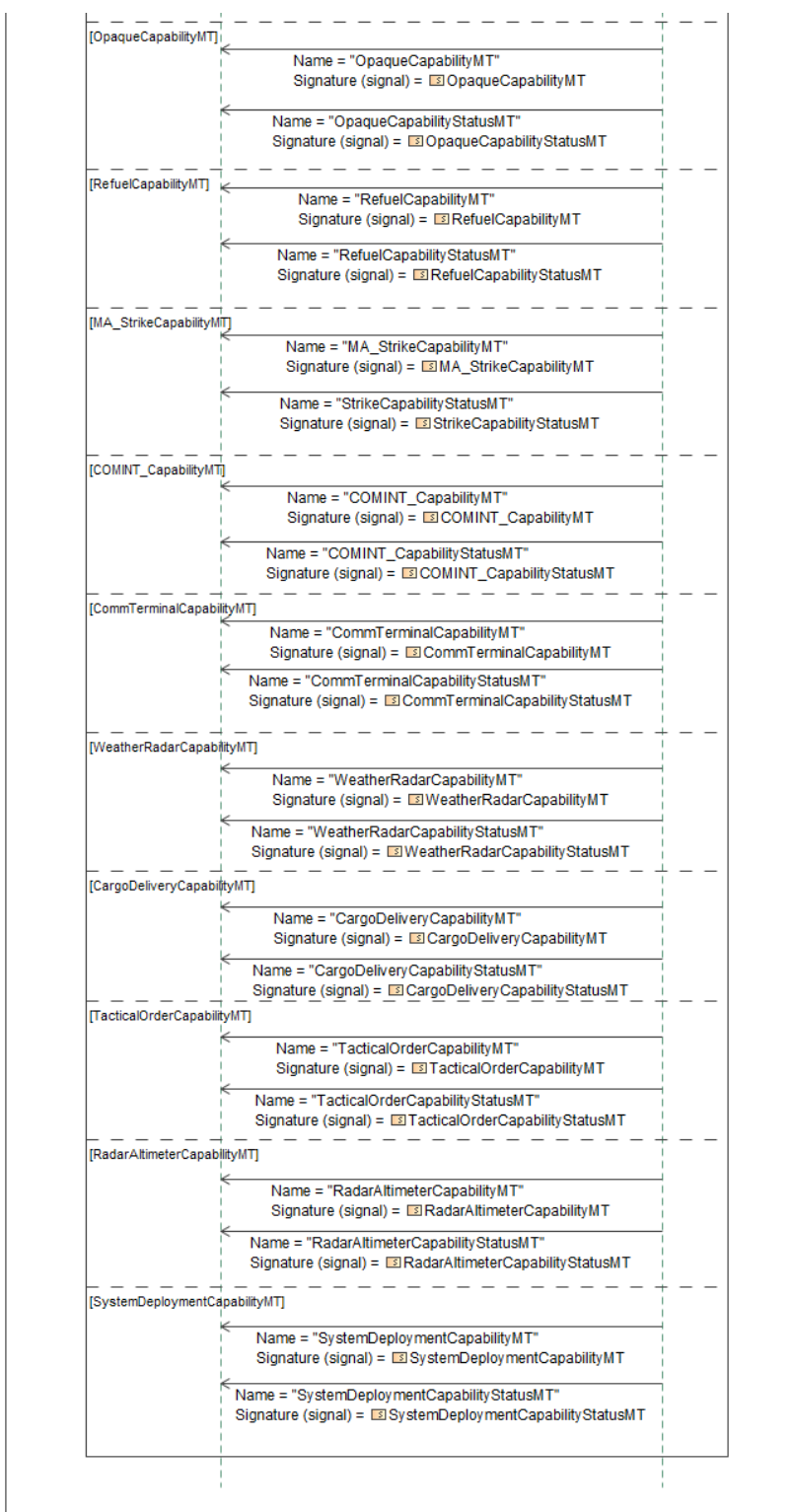


Figure 1-33: Station-Keeping Capabilities

Table 1-32: Station-Keeping Capabilities

| Message Name (Name:Type)                                            | Required Fields (Name:Type) |
|---------------------------------------------------------------------|-----------------------------|
| ActionCapabilityStatusMT:<br>ActionCapabilityStatusMT               | None                        |
| AMTI_CapabilityStatusMT:<br>AMTI_CapabilityStatusMT                 | None                        |
| AO_CapabilityMT: AO_CapabilityMT                                    | None                        |
| AO_CapabilityStatusMT:<br>AO_CapabilityStatusMT                     | None                        |
| CargoDeliveryCapabilityMT:<br>CargoDeliveryCapabilityMT             | None                        |
| CargoDeliveryCapabilityStatusMT:<br>CargoDeliveryCapabilityStatusMT | None                        |
| COMINT_CapabilityMT: COMINT_CapabilityMT                            | None                        |
| COMINT_CapabilityStatusMT:<br>COMINT_CapabilityStatusMT             | None                        |
| CommTerminalCapabilityMT:<br>CommTerminalCapabilityMT               | None                        |
| CommTerminalCapabilityStatusMT:<br>CommTerminalCapabilityStatusMT   | None                        |
| EA_CapabilityMT: EA_CapabilityMT                                    | None                        |
| EA_CapabilityStatusMT:<br>EA_CapabilityStatusMT                     | None                        |
| EffectCapabilityStatusMT:<br>EffectCapabilityStatusMT               | None                        |
| ESM_CapabilityMT: ESM_CapabilityMT                                  | None                        |
| ESM_CapabilityStatusMT:<br>ESM_CapabilityStatusMT                   | None                        |
| IFF_CapabilityMT: IFF_CapabilityMT                                  | None                        |
| IFF_CapabilityStatusMT:<br>IFF_CapabilityStatusMT                   | None                        |
| MA_ActionCapabilityMT:<br>MA_ActionCapabilityMT                     | None                        |
| MA_AMTI_CapabilityMT:<br>MA_AMTI_CapabilityMT                       | None                        |
| MA_EffectCapabilityMT:<br>MA_EffectCapabilityMT                     | None                        |
| MA_FlightCapabilityMT: MA_FlightCapabilityMT                        | None                        |
| MA_FlightCapabilityStatusMT:<br>MA_FlightCapabilityStatusMT         | None                        |
| MA_StrikeCapabilityMT:<br>MA_StrikeCapabilityMT                     | None                        |

| Message Name (Name:Type)                                                          | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------------|-----------------------------|
| <b>OpaqueCapabilityMT: OpaqueCapabilityMT</b>                                     | None                        |
| <b>OpaqueCapabilityStatusMT:<br/>OpaqueCapabilityStatusMT</b>                     | None                        |
| <b>PO_CapabilityMT: PO_CapabilityMT</b>                                           | None                        |
| <b>PO_CapabilityStatusMT:<br/>PO_CapabilityStatusMT</b>                           | None                        |
| <b>RadarAltimeterCapabilityMT:<br/>RadarAltimeterCapabilityMT</b>                 | None                        |
| <b>RadarAltimeterCapabilityStatusMT:<br/>RadarAltimeterCapabilityStatusMT</b>     | None                        |
| <b>RefuelCapabilityMT: RefuelCapabilityMT</b>                                     | None                        |
| <b>RefuelCapabilityStatusMT:<br/>RefuelCapabilityStatusMT</b>                     | None                        |
| <b>SAR_CapabilityMT: SAR_CapabilityMT</b>                                         | None                        |
| <b>SAR_CapabilityStatusMT:<br/>SAR_CapabilityStatusMT</b>                         | None                        |
| <b>SMTI_CapabilityMT: SMTI_CapabilityMT</b>                                       | None                        |
| <b>SMTI_CapabilityStatusMT:<br/>SMTI_CapabilityStatusMT</b>                       | None                        |
| <b>StrikeCapabilityStatusMT:<br/>StrikeCapabilityStatusMT</b>                     | None                        |
| <b>SystemDeploymentCapabilityMT:<br/>SystemDeploymentCapabilityMT</b>             | None                        |
| <b>SystemDeploymentCapabilityStatusMT:<br/>SystemDeploymentCapabilityStatusMT</b> | None                        |
| <b>TacticalOrderCapabilityMT:<br/>TacticalOrderCapabilityMT</b>                   | None                        |
| <b>TacticalOrderCapabilityStatusMT:<br/>TacticalOrderCapabilityStatusMT</b>       | None                        |
| <b>WeatherRadarCapabilityMT:<br/>WeatherRadarCapabilityMT</b>                     | None                        |
| <b>WeatherRadarCapabilityStatusMT:<br/>WeatherRadarCapabilityStatusMT</b>         | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.7 Peer Engagement Criteria Behaviors Package

A Package Lead manages the activation of the engagement criteria for its Followers based on the activation commanded by C2. The Package Lead will also manage delegation of Targets that were designated by a Target Authority. The Package Lead can act as a Target Authority to the package if and only if a Target Authority has already designated that Target to the package or a specific follower

in the package. Once the Target Authority has Designated a Target, the Package Lead can:

- If the Target was assigned to the Package ID:
  - The Lead can use the Centralized Computation of Target Assignments sequence to determine the allocation of Targets to ACPs and can then send a *MA\_DesignationMT* or directly command a Targeting Act to the appropriate Followers.
  - The Lead can use the Distributed Computation of Target Assignments sequence to allow the ACPs to determine which Target they should go after. The followers will send a *MA\_DesignationRequestMT* to the Leader. The leader will then send a *MA\_DesignationMT* or directly command a Targeting Act to the appropriate Follower.
- If the Target was assigned to a specific System ID that is a Follower in the Package, the Lead can send a *MA\_DesignationMT* or directly command a Targeting Act to that Follower.

Any Follower can send a *MA\_DesignationRequestMT* to the Leader, if the Target was not already approved, the leader would send a *MA\_DesignationRequestMT* to a Target Authority for approval. Each Package member can still receive Target assignments directly from a Targeting Authority.

### 1.2.7.1 Activate Peer Engagement Criteria

The Activate Peer Engagement Criteria sequence diagram specifies the interaction of the lead Autonomous Collaborative Platform (ACP) commanding its follower ACPs to activate one of its stored engagement criteria messages. First, the lead ACP sends a *MA\_EngagementCriteriaCommandMT* message where the *AppliesToID* field is defined with the System ID of the follower ACP. The *MA\_EngagementCriteriaCommandMT* message shall specify the *EngagementCriteriaID* field containing the ID of the engagement criteria message that the lead ACP wishes to be active. Any new *MA\_EngagementCriteriaCommandMT* will override any previously sent message, even if the *CommandID* has changed. When a *MA\_EngagementCriteriaCommandMT* is sent with the *CommandedState* set to *CANCEL*, this will cancel the active command and engagement criteria activation schema. In response to the command, the follower ACPs shall send *MA\_EngagementCriteriaCommandStatusMT\_1* to the lead ACP, with the *AppliesToID* field set to their System ID and the *CommandProcessingState* field set to *RECEIVED* to indicate receipt of the command. Once the follower ACPs have finished querying engagement criteria from onboard MP, the follower ACPs shall send *MA\_EngagementCriteriaCommandStatusMT\_2* to the lead ACP with the *AppliesToID* field set to their System ID and the *CommandProcessingState* set to *ACCEPTED*. The follower ACPs will send *MA\_EngagementCriteriaStatusMT* to the lead ACP containing the *EngagementCriteriaID*, the *AppliesToID* field set to their System ID, and other configured data as needed.

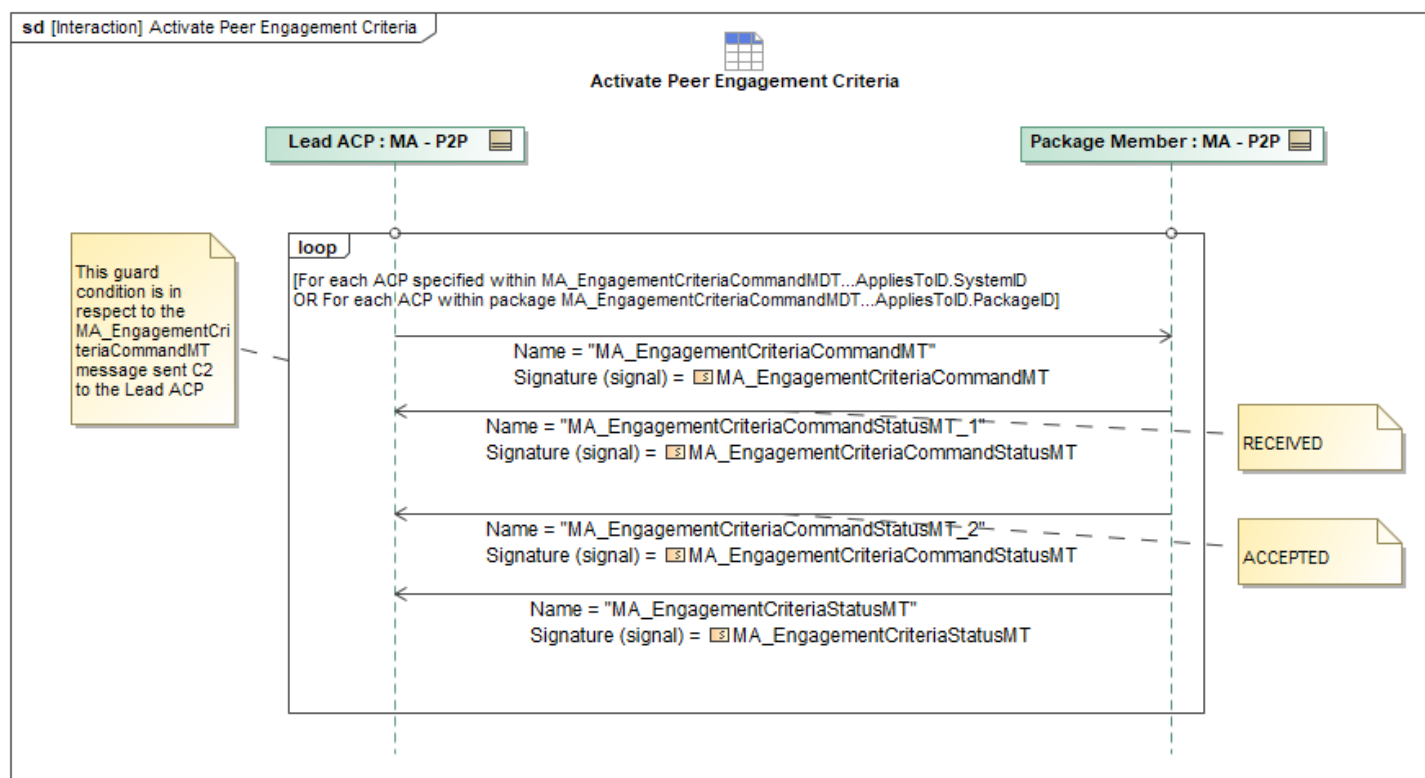


Figure 1-34: Activate Peer Engagement Criteria

Table 1-33: Activate Peer Engagement Criteria

| Message Name (Name:Type)                                                        | Required Fields (Name:Type) |
|---------------------------------------------------------------------------------|-----------------------------|
| MA_EngagementCriteriaCommandMT:<br>MA_EngagementCriteriaCommandMT               | None                        |
| MA_EngagementCriteriaCommandStatusMT_1:<br>MA_EngagementCriteriaCommandStatusMT | None                        |
| MA_EngagementCriteriaCommandStatusMT_2:<br>MA_EngagementCriteriaCommandStatusMT | None                        |
| MA_EngagementCriteriaStatusMT:<br>MA_EngagementCriteriaStatusMT                 | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.8 Peer Generic Package

### 1.2.8.1 Peer Sends Message Acknowledgment

Peer Acknowledgment Configurations sequences defines which messages MA would expect a Peer MA to respond with an acknowledgment of receipt. If a Peer MA receives a configured message, the Peer MA will send a *MA\_SystemNotificationMT* with the *NotificationState* set to *CONFIRMED*, *Source* set to its own *SystemID*, *Severity* set to *INFORMATIONAL*, *SystemSubjectIDs* set to its own *SystemID*, *SystemPerspective* set to *SOURCE*, and *AssociatedMessage* set to the *MessageType* of the received message and the *AssociatedID* of the applicable ID in the received message. The



*MA\_SystemNotificationMT* can also be sent at limited periodic rates for acknowledgments based on the Message Acknowledgment Configuration where it would group together all received *AssociatedMessages* over the periodic time that was configured. This is done by adding multiple entries to the *AssociatedMessage* field for each message. The *MA\_SystemNotificationMT* would only be sent at that periodic rate if there was a received message during that time.

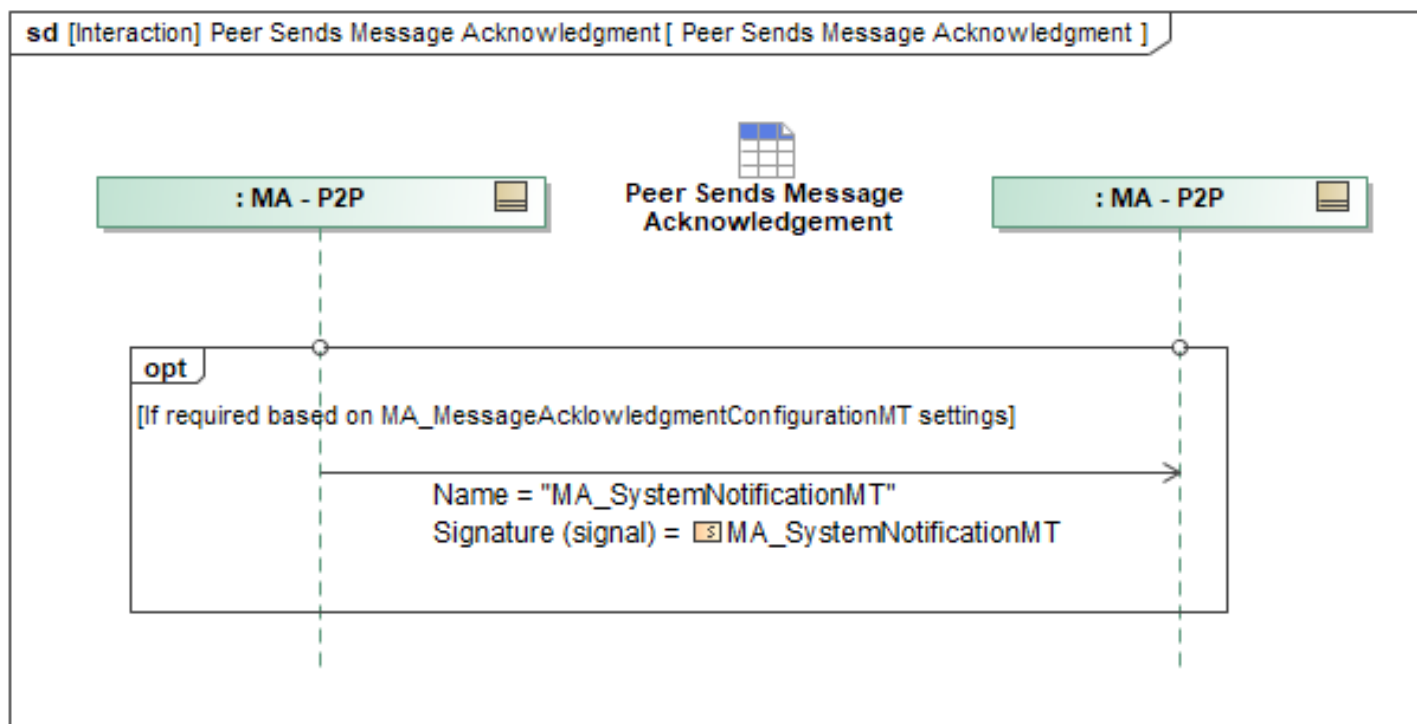


Figure 1-35: Peer Sends Message Acknowledgment

Table 1-34: Peer Sends Message Acknowledgment

| Message Name (Name:Type)                                           | Required Fields (Name:Type) |
|--------------------------------------------------------------------|-----------------------------|
| <i>MA_SystemNotificationMT</i> :<br><i>MA_SystemNotificationMT</i> | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.9 Peer Geozones Package

### 1.2.9.1 Peer Command Geozone

This diagram contains the collection of data messages used when Peer commands the state of any Op\* message (e.g., *MA\_OpZoneMT*, *MA\_OpVolumeMT*, etc.) currently stored on another Peer ACP, independent of mission plan. When the state of a given Op\* message is "ACTIVE", the constraints imposed by said message must be considered by MA.

Upon receipt of an *MA\_OpCommandMT*, the receiving Peer MA shall store the command locally. The *MA\_OpCommandMT* includes a reference to an existing Op\* message (*OpID*), an indication of whether the command is to activate or de-activate (*CommandedState*), and any relevant conditions that must be met in order for the command to execute (*ConstraintsOverride*). Additionally, the *Rank*

field allows the receiving Peer MA to resolve conflicting command messages through comparative ranking. The receiving Peer MA acknowledges a received command by sending *MA\_OpCommandStatusMT\_1* with *CommandProcessingState* of "RECEIVED". The receiving Peer MA then evaluates the conditions and ranking of the received *MA\_OpCommandMT* and sends an *MA\_OpCommandStatusMT\_2* message with *CommandProcessingState* of "ACCEPTED" if successful. Otherwise, *MA\_OpCommandStatusMT\_3* is sent with the *CommandProcessingState* of "REJECTED", and the accompanying *CommandProcessingStateReason* populated with the reason for the rejection (e.g., "RANKING").

MA continuously monitors the conditions specified in *ConstraintsOverride* for all stored *MA\_OpCommandMT* messages. If there are no conditions specified (in the *ConstraintsOverride* field), or all conditions are met, MA changes the state of the referenced Op\* message and sends an *MA\_OpStatusMT* notifying the sending Peer MA of the updated status (*CurrentState*) of the Op\* message and the command that triggered the change (*CommandID*).

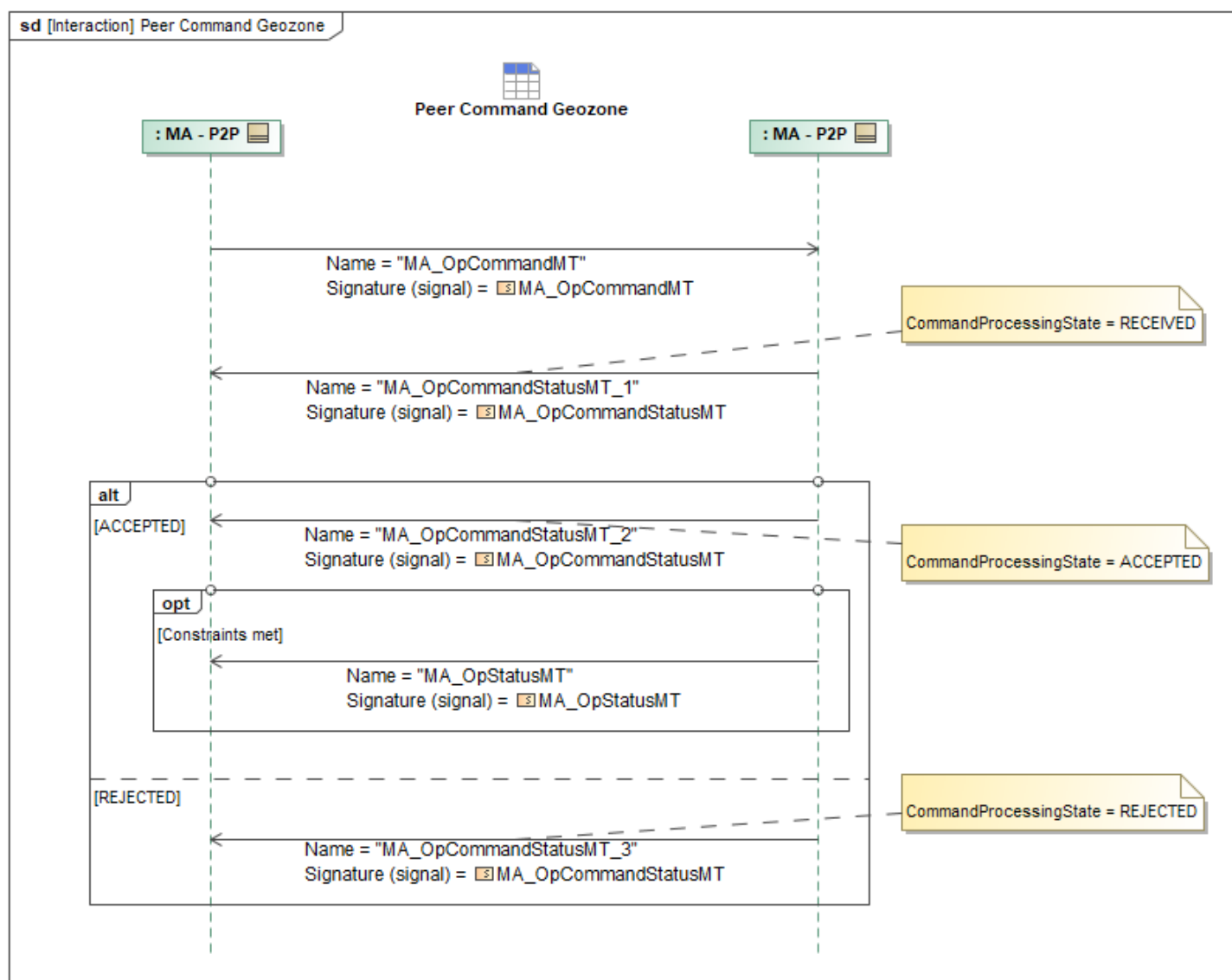


Figure 1-36: Peer Command Geozone

Table 1-35: Peer Command Geozone

| Message Name (Name:Type)                        | Required Fields (Name:Type)                    |
|-------------------------------------------------|------------------------------------------------|
| MA_OpCommandMT: MA_OpCommandMT                  | OpID: OpID_ChoiceType                          |
| MA_OpCommandStatusMT_1:<br>MA_OpCommandStatusMT | None                                           |
| MA_OpCommandStatusMT_2:<br>MA_OpCommandStatusMT | None                                           |
| MA_OpCommandStatusMT_3:<br>MA_OpCommandStatusMT | CommandProcessingStateReason: CannotComplyType |
| MA_OpStatusMT: MA_OpStatusMT                    | None                                           |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10 Peer Mission Plan Management Package

1.2.10.1 P2P Contingency Alert

This interaction describes the process of a package member alerting the package lead that a contingency has occurred. The package member sends a *MissionContingencyAlertMT* message to the lead indicating what has happened. This may trigger additional actions by the package lead which could include retasking or a replan.

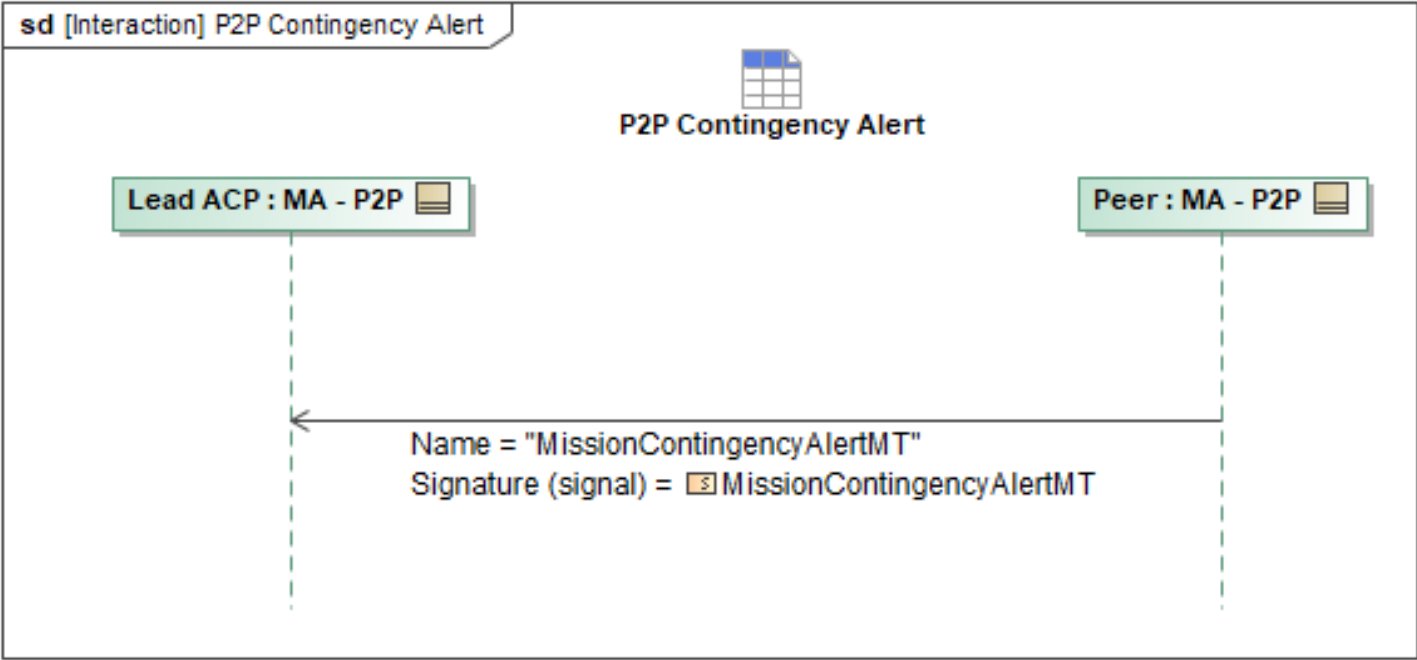


Figure 1-37: P2P Contingency Alert

Table 1-36: P2P Contingency Alert

| Message Name (Name:Type)                                | Required Fields (Name:Type) |
|---------------------------------------------------------|-----------------------------|
| MissionContingencyAlertMT:<br>MissionContingencyAlertMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.2 P2P Detect Mission Plan Route Discontinuity Response

Upon identification of a Mission Plan Route Discontinuity Error, MA will send *MA\_SystemNotificationMT* to C2 and P2P, to indicate that a Mission Plan Route Discontinuity condition exists. Within the message, MA sets *NotificationState* to ACTIVE and populates *NotificationDetail.Reason* with INFEASIBLE\_ROUTE. This indicates that the ACP can no longer continue along the nominal Mission Plan path from its current or projected position. *AssociatedMessage.MessageType* will have the ROUTE\_PLAN enumeration to indicate to P2P that the route is no longer feasible. *AssociatedMessage.AssociatedID* may reference the affected *PathSegmentID* that precedes the discontinuity, or alternatively, the *FlightCommandID* can be used if a *PathSegmentID* does not exist or cannot be referenced.

At the time of the discontinuity, MA also sends *MA\_PositionReportDetailed* so that P2P has access to the current and projected position data relevant to the error condition. The *Kinematics* field communicates the ACP's current position and velocity associated with the discontinuity. Resolution actions may include MA executing a plan modification or receiving a resolution from C2 or P2P. Upon resolution of the error, MA will send *MA\_SystemNotificationMT* to C2 and P2P with *NotificationState* set to CLOSED and the same other fields mentioned in the initial *MA\_SystemNotificationMT*, including *NotificationDetail.Reason* and AssociatedMessage details, to provide a consistent update on the resolved condition.

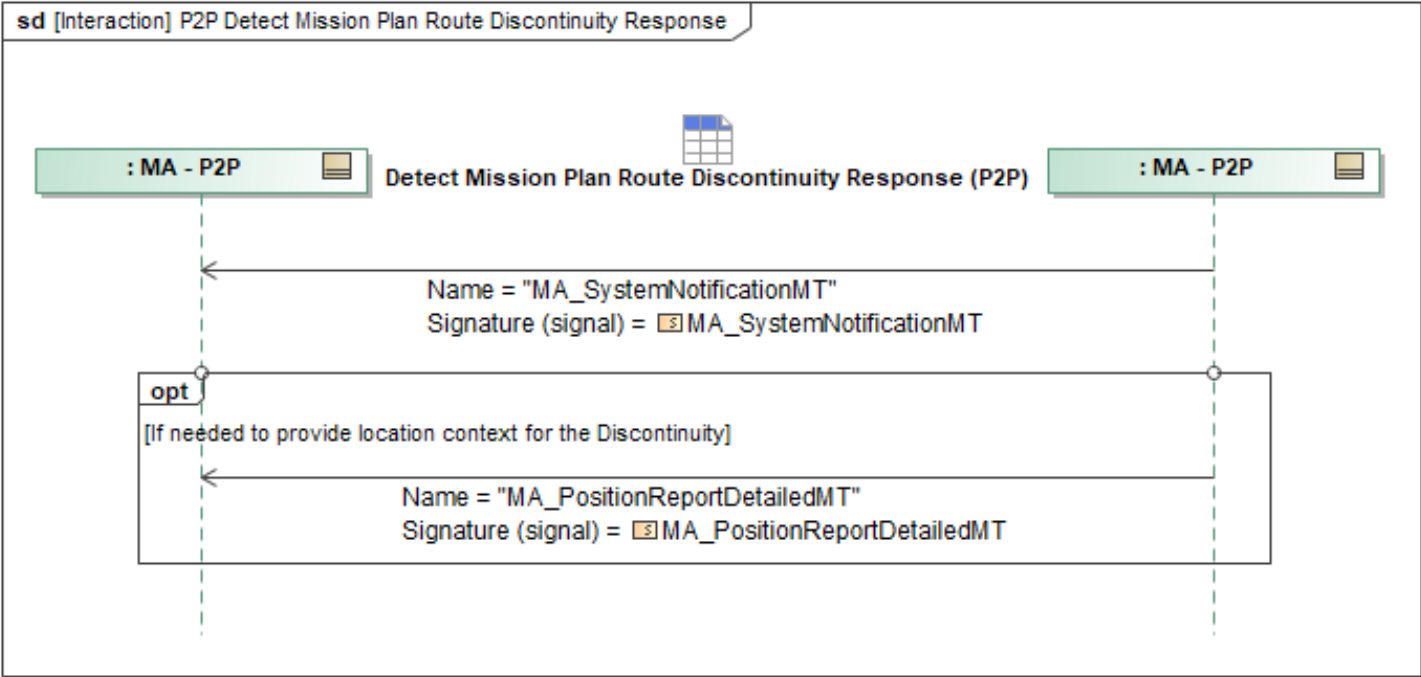


Figure 1-38: P2P Detect Mission Plan Route Discontinuity Response

Table 1-37: P2P Detect Mission Plan Route Discontinuity Response

| Message Name (Name:Type)                                    | Required Fields (Name:Type)                                 |
|-------------------------------------------------------------|-------------------------------------------------------------|
| MA_PositionReportDetailedMT:<br>MA_PositionReportDetailedMT | None                                                        |
| MA_SystemNotificationMT:<br>MA_SystemNotificationMT         | Description: VisibleString1024Type<br>AssociatedID: ID_Type |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.3 P2P Request Plan Patch

In the event that Package Lead determines that a Plan Patch is necessary, a Package Lead can request that a Follower MA perform a Plan Patch. If the desired Plan Patch is applicable to a Route Plan, Package Lead sends a Follower MA a *RouteModificationRequestMT*. The *RouteModificationRequestMT* contains a Path field and a *PathSegment* field that communicate one of three possible outcomes: the addition of a path, removal of a path, or update a path, through the *Add*, *RemoveID*, or *Update* fields, respectively. Contained in the *PathSegment* field, the *PathID*

field identifies the Path which the segment changes apply to. If the desired Plan Patch is applicable to any other type of Plan, Package Lead sends a Follower MA an *MA\_PlanModificationRequestMT*. The *MA\_PlanModificationRequest* contains a *PlanModificationDetails* field which allows for a number of potential plan modification methods. For Plan Patching, the *SubPlansModification* field will be used to either add a *SubPlan* (*SubPlanAdditions* field) or retract a *SubPlan* (*SubPlanRetractions* field). Follower MA then responds with the corresponding \**StatusMT* message (*RouteModificationRequestStatusMT* or *MA\_PlanModificationRequestStatusMT*), which contains the *RequestID* and the *RequestProcessingState* which indicates the status of the request.

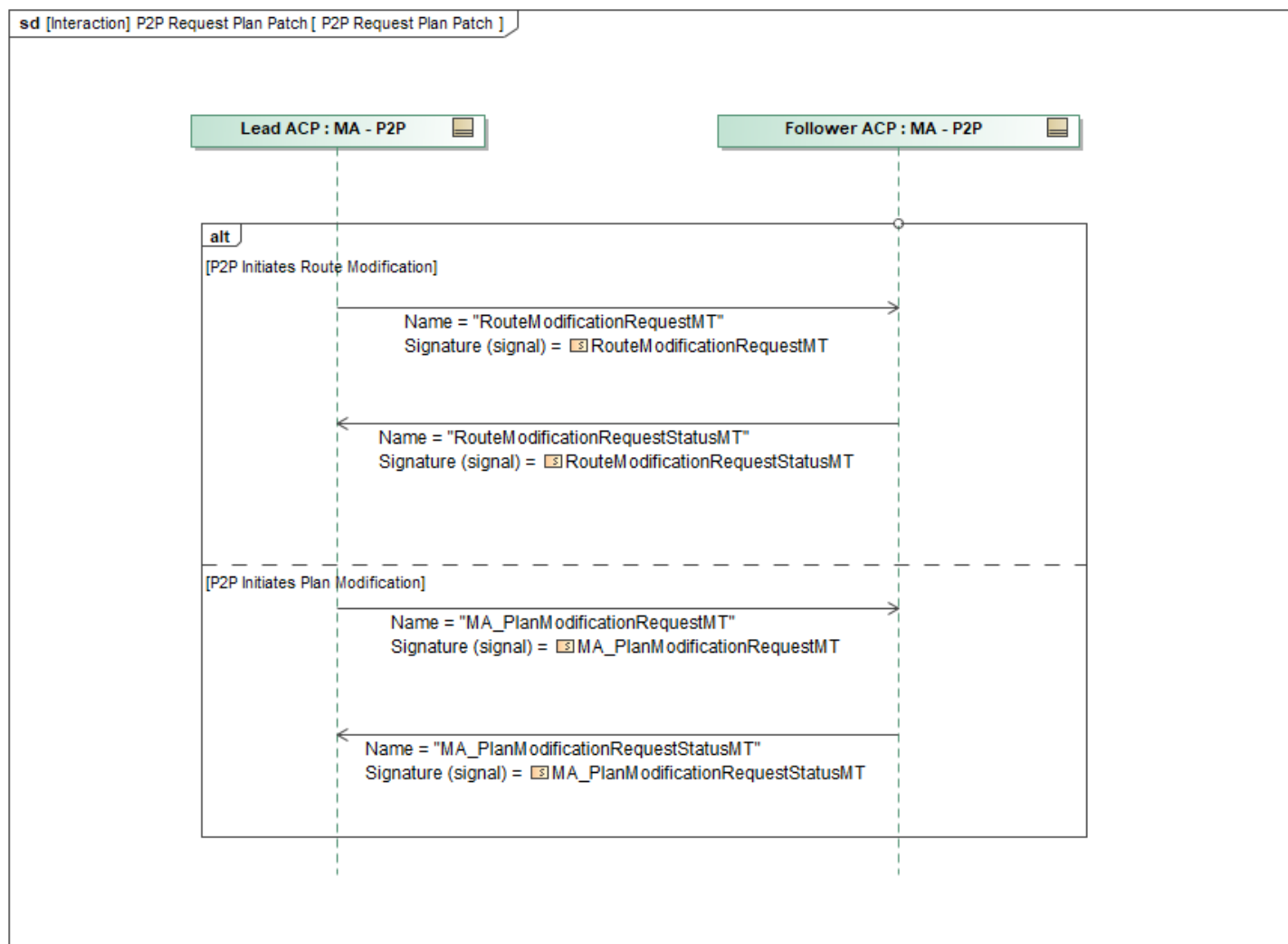


Figure 1-39: P2P Request Plan Patch

Table 1-38: P2P Request Plan Patch

| Message Name (Name:Type)                                                  | Required Fields (Name:Type) |
|---------------------------------------------------------------------------|-----------------------------|
| MA_PlanModificationRequestMT:<br>MA_PlanModificationRequestMT             | None                        |
| MA_PlanModificationRequestStatusMT:<br>MA_PlanModificationRequestStatusMT | None                        |

| Message Name (Name:Type)                                                            | Required Fields (Name:Type) |
|-------------------------------------------------------------------------------------|-----------------------------|
| <b>RouteModificationRequestMT:</b><br><b>RouteModificationRequestMT</b>             | None                        |
| <b>RouteModificationRequestStatusMT:</b><br><b>RouteModificationRequestStatusMT</b> | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.4 Peer Receive Plan Modification Notification

When MA updates a Plan, there may be Peer elements that need to be informed for situational awareness or synchronization reasons. This sequence describes the messaging that MA can use to broadcast the fact that MA has changed a Plan. If MA has modified a *MissionPlan* or a *SubPlan*, MA will send a *MA\_PlanModificationRequestMT* to Peer, requesting that Peer updates the Plan in question. Likewise, if MA has only modified a *RoutePlan*, MA will send a *RouteModificationRequestMT* to the same effect. Contained in the *MessageData* fields of these messages are fields that describe the Plan modifications which are populated as required to mirror the change that MA has already made to the Plan. Upon receipt of request, Peer will respond to MA with a *MA\_PlanModificationRequestStatusMT* or *RouteModificationStatusMT* describing the status of the Plan update.

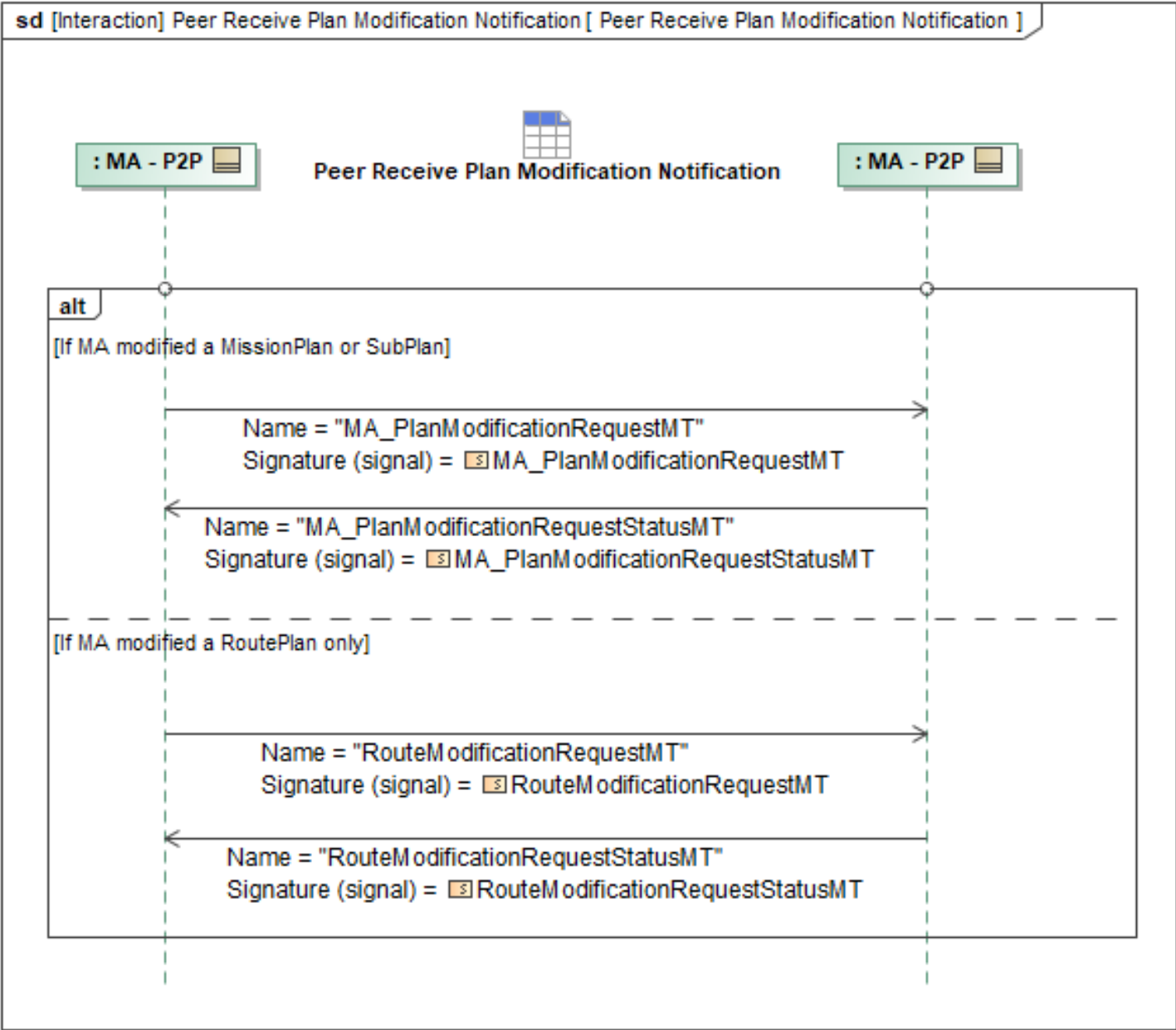


Figure 1-40: Peer Receive Plan Modification Notification

Table 1-39: Peer Receive Plan Modification Notification

| Message Name (Name:Type)                                                  | Required Fields (Name:Type) |
|---------------------------------------------------------------------------|-----------------------------|
| MA_PlanModificationRequestMT:<br>MA_PlanModificationRequestMT             | None                        |
| MA_PlanModificationRequestStatusMT:<br>MA_PlanModificationRequestStatusMT | None                        |
| RouteModificationRequestMT:<br>RouteModificationRequestMT                 | None                        |
| RouteModificationRequestStatusMT:<br>RouteModificationRequestStatusMT     | None                        |



Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.10.5 Query Deconfliction Data

This interaction describes the process of an ACP requesting deconfliction data from another ACP. Asset Deconfliction data, used to prevent the ACP from colliding with another platform, is defined by *OpLineMT*, *OpZoneMT*, and *OpRoutingMT*.

*OpLineMT* contains data used to convey a Corridor (a line with width along the designated Route) describing the passageway designated for Mission planning deconfliction for each ACP. Using the *Category* field set to *CORRIDOR*. The *LeftWidth* and *RightWidth* fields are required to describe the overall width of the corridor. The defined data is to be used when deconflicting entities routes or generating routes to ensure proper separation. These parameters are applied at the vehicle level.

*OpZoneMT*, contains data used to convey Altitude. Data in the *Zone* field captures the shape of the region, along with the Max and Min altitudes. Describing the region designated for mission planning deconfliction for each ACP. The defined data is to be used when deconflicting entities routes or generating routes to ensure proper Altitude block compliance. These parameters are applied at the vehicle level.

*OpRoutingMT* contains data used to convey a Bubble: The *SpecificBlueSeparation* and *SpecificRedSeparation* define parameters to use when deconflicting entity routes or generating routes to ensure compliance of the designated keep in and keep out zones. These parameters are applied at the vehicle type level.

Data is stored or cached in data manager services which may have a database backend. The ACP can query this data via *QueryDataRequestMT* by including three instances of *MessageType* with their enumerations set to *OP\_LINE*, *OP\_ROUTING*, and *OP\_ZONE* respectively. This will be responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may also respond with intermediate *RequestProcessingState* field such as “*QUEUED*” and “*PROCESSING*” to indicate additional in-process states. When complete, the *QueryDataRequestStatusMT* message’s *RequestProcessingState* field is set to “*COMPLETED*” and contains the requested data. When the original *QueryDataRequestMT*’s optional field *ResultsInNative* is populated, the underlying data management services publishes the requested data in its original or raw format as a separate message from the *QueryDataRequestStatusMT* message at completion.

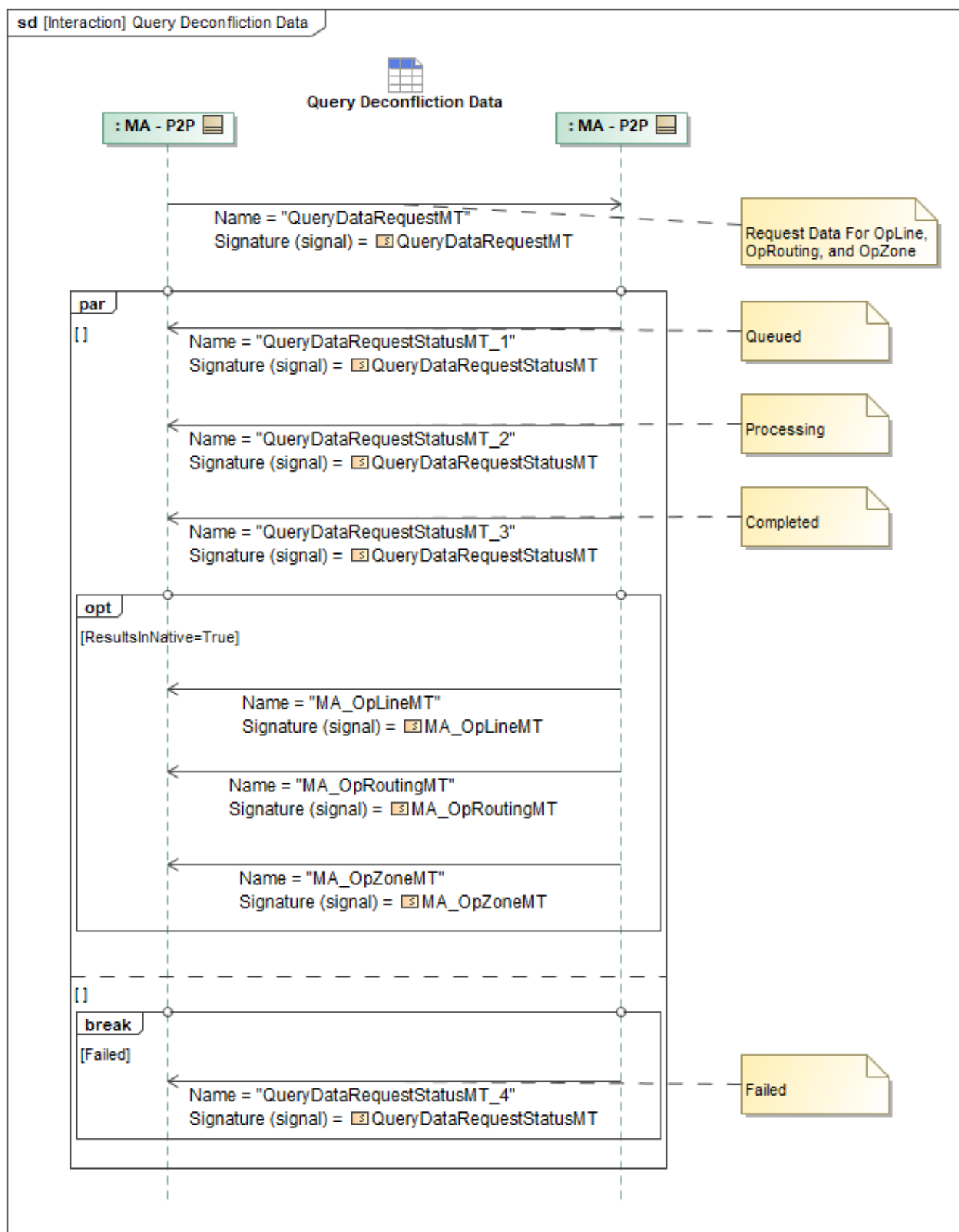


Figure 1-41: Query Deconfliction Data

Table 1-40: Query Deconfliction Data

| Message Name (Name:Type)                                | Required Fields (Name:Type)                                                                            |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| MA_OpLineMT: MA_OpLineMT                                | RightWidth: DistanceType<br>LeftWidth: DistanceType                                                    |
| MA_OpRoutingMT: MA_OpRoutingMT                          | SpecificBlueSeparation: SpecificBlueSeparationType<br>SpecificRedSeparation: SpecificRedSeparationType |
| MA_OpZoneMT: MA_OpZoneMT                                | None                                                                                                   |
| QueryDataRequestMT: QueryDataRequestMT                  | MessageType: MessageTypeEnum                                                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                                                                                   |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                                                                                   |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                                                                                |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType                                                         |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.6 Synchronize Contingency Plan

This diagram contains the messages sent between Peers to synchronize a contingency plan. The *MA\_ApprovalPolicyMT* and *ResponsePlanMT* will be sent with an *ObjectState* enumerations set to UPDATED.

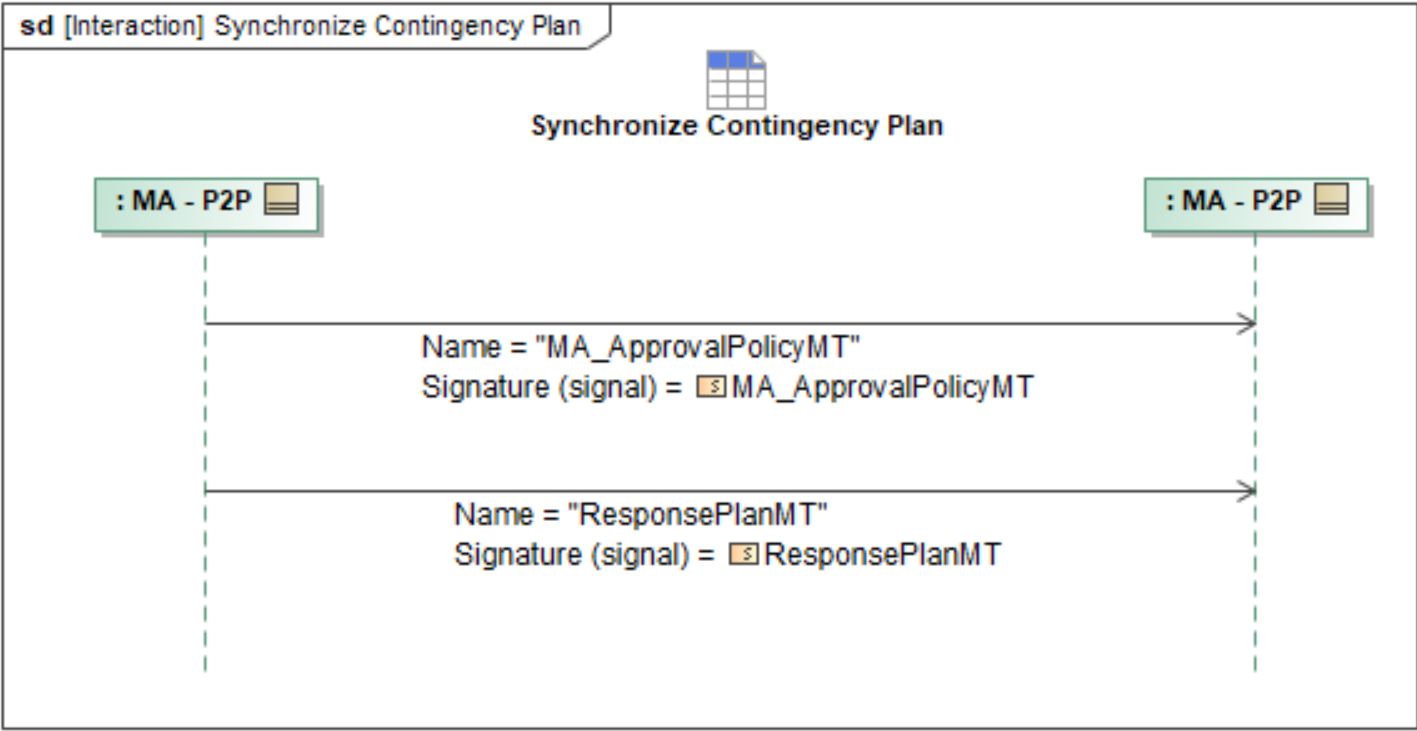


Figure 1-42: Synchronize Contingency Plan

Table 1-41: Synchronize Contingency Plan

| Message Name (Name:Type)                 | Required Fields (Name:Type)                     |
|------------------------------------------|-------------------------------------------------|
| MA_ApprovalPolicyMT: MA_ApprovalPolicyMT | DataRecordInstanceID: DataRecordInstanceID_Type |
| ResponsePlanMT: ResponsePlanMT           | ObjectState: ObjectStateEnum                    |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.7 Update Peer Airfield Data

Mission Plan Data such as Airfield Reports are updated via UCI *AirfieldReportMT* message with the Optional field *ObjectState*.

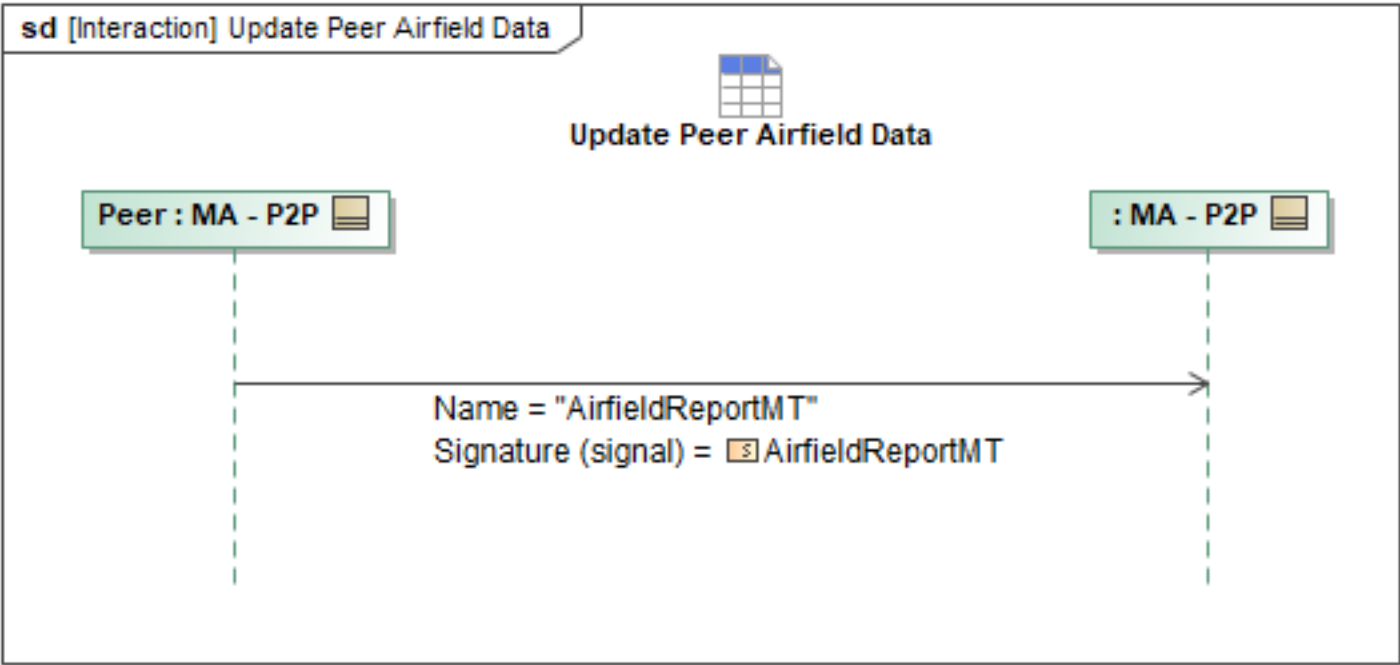


Figure 1-43: Update Peer Airfield Data

Table 1-42: Update Peer Airfield Data

| Message Name (Name:Type)           | Required Fields (Name:Type) |
|------------------------------------|-----------------------------|
| AirfieldReportMT: AirfieldReportMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.8 P2P Lead Updates Members

1.2.10.8.1 P2P Lead Updates Member with Constraint Priority List

This interaction specifies how a lead ACP can update various data stored in the follower ACPs. For each follower ACP, the lead ACP shares *MA\_ConstraintPriorityListMT* with the package member. Upon receiving this message, follower ACPs update onboard MP with new CPL.

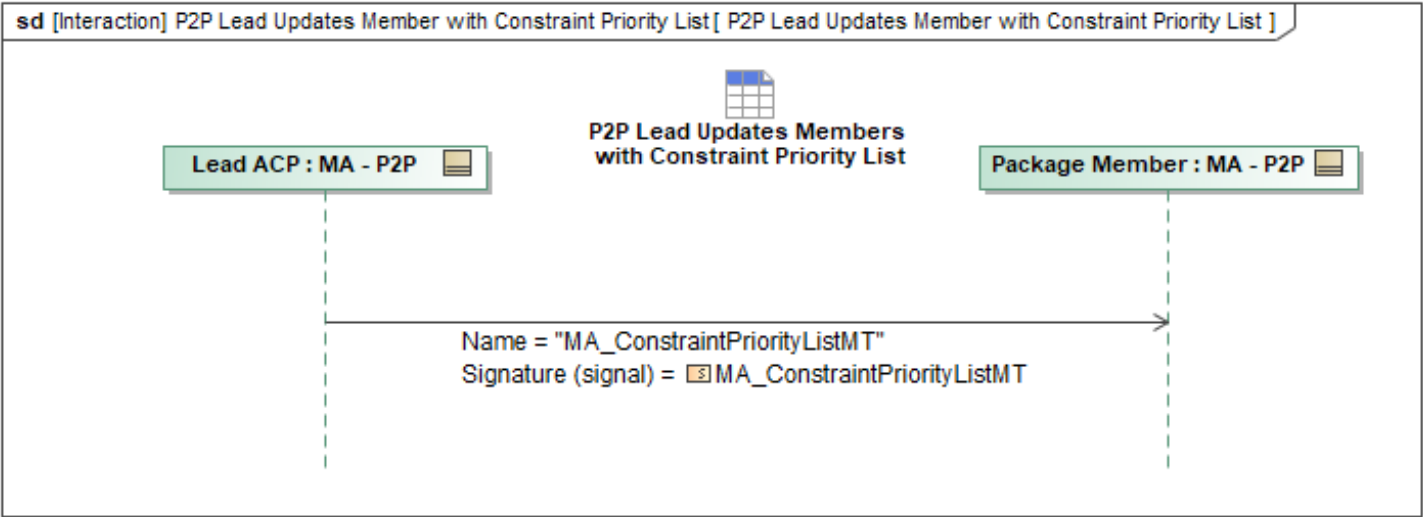


Figure 1-44: P2P Lead Updates Member with Constraint Priority List

Table 1-43: P2P Lead Updates Member with Constraint Priority List

| Message Name (Name:Type)     | Required Fields (Name:Type)                 |
|------------------------------|---------------------------------------------|
| MA_ConstraintPriorityListMT: | ObjectState: ObjectStateEnum                |
| MA_ConstraintPriorityListMT  | ListItem: MA_ConstraintPriorityListItemType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.8.2 P2P Lead Updates Member with Constraint Priority List Activation Data

This interaction specifies how a lead ACP can update various data stored in the Follower ACPs. For each follower ACP, the lead ACP shares *MA\_ConstraintPriorityListActivationDataMT* with the package member. Upon receiving this message, follower ACPs update onboard MP with new CPL activation data.

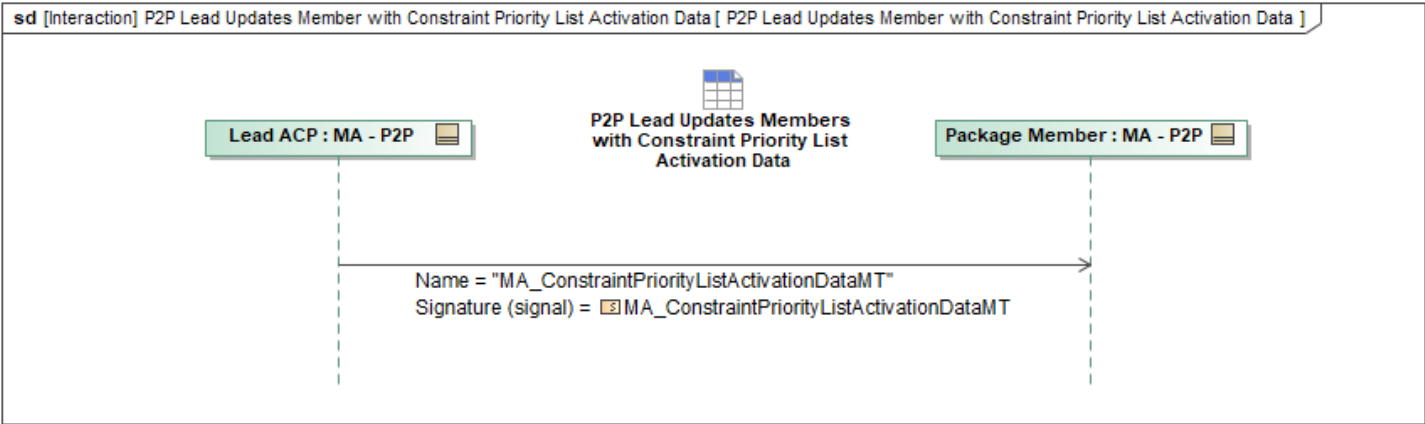


Figure 1-45: P2P Lead Updates Member with Constraint Priority List Activation Data

Table 1-44: P2P Lead Updates Member with Constraint Priority List Activation Data

| Message Name (Name:Type)                                                                | Required Fields (Name:Type)  |
|-----------------------------------------------------------------------------------------|------------------------------|
| MA_ConstraintPriorityListActivationDataMT:<br>MA_ConstraintPriorityListActivationDataMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.8.3 P2P Lead Updates Member with Engagement Criteria

This interaction specifies how a lead ACP can update various data stored in the Follower ACPs. For each follower ACP, the lead ACP shares *MA\_EngagementCriteriaMT* with the package member. Upon receiving this message, follower ACPs update onboard MP with new engagement criteria data.

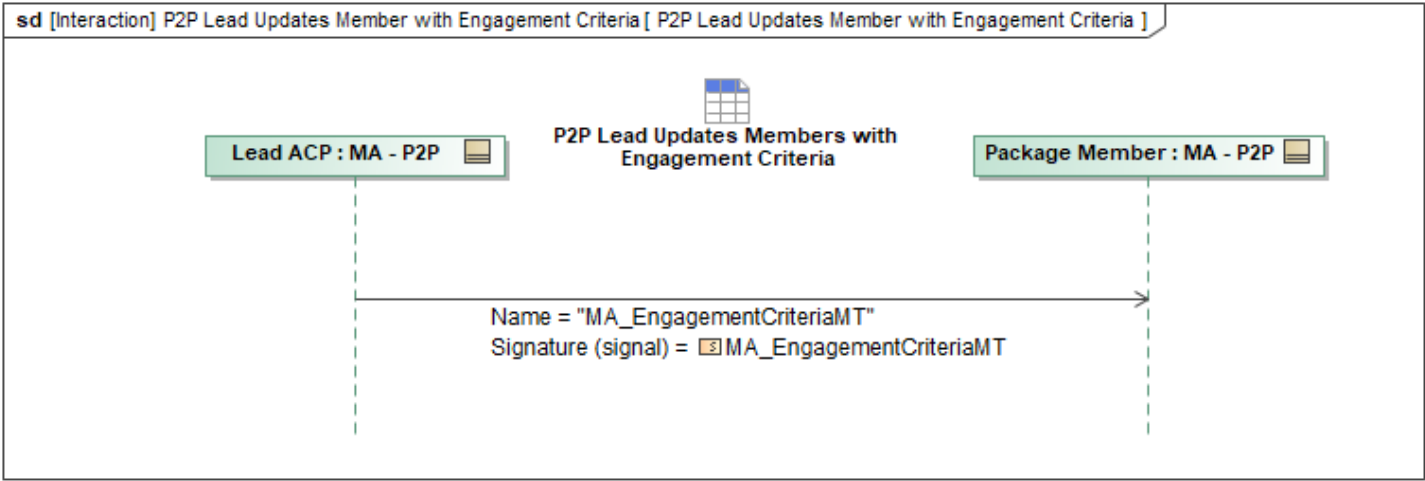


Figure 1-46: P2P Lead Updates Member with Engagement Criteria

Table 1-45: P2P Lead Updates Member with Engagement Criteria

| Message Name (Name:Type)                            | Required Fields (Name:Type)  |
|-----------------------------------------------------|------------------------------|
| MA_EngagementCriteriaMT:<br>MA_EngagementCriteriaMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.8.4 P2P Lead Updates Member with Execution Priority List

This interaction specifies how a lead ACP can update various data stored in the Follower ACPs. For each follower ACP, the lead ACP shares *MA\_ExecutionPriorityListMT* with the package member. Upon receiving this message, follower ACPs update onboard MP with new EPL.

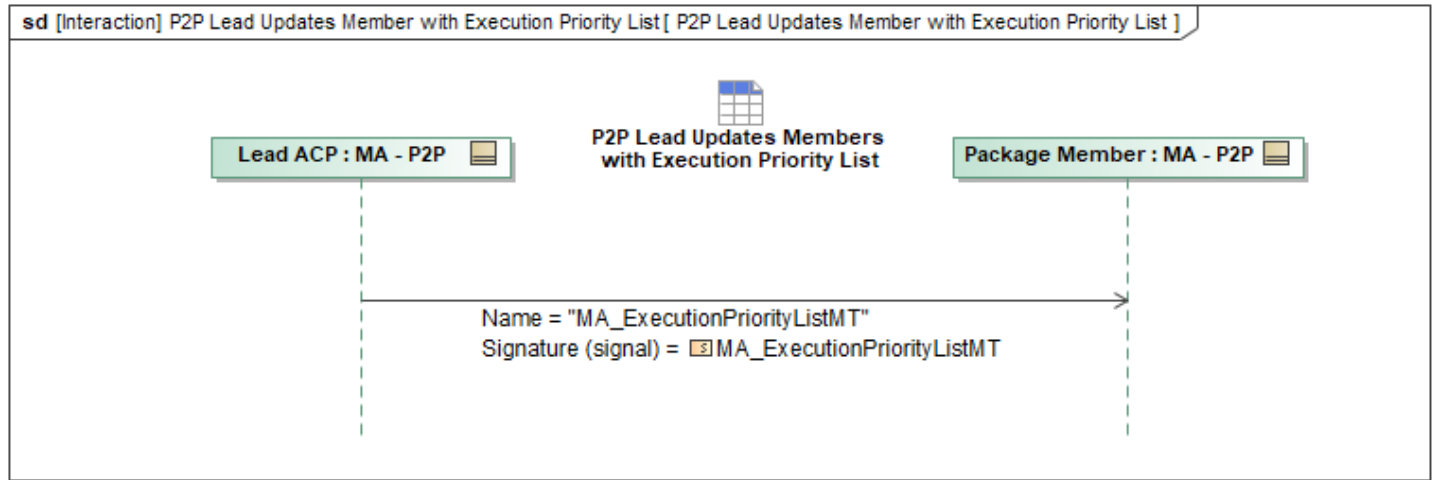


Figure 1-47: P2P Lead Updates Member with Execution Priority List

Table 1-46: P2P Lead Updates Member with Execution Priority List

| Message Name (Name:Type)    | Required Fields (Name:Type)                |
|-----------------------------|--------------------------------------------|
| MA_ExecutionPriorityListMT: | ListItem: MA_ExecutionPriorityListItemType |
| MA_ExecutionPriorityListMT  | ObjectState: ObjectStateEnum               |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.8.5 P2P Lead Updates Member with Execution Priority List Activation Data

This interaction specifies how a lead ACP can update various data stored in the Follower ACPs. For each follower ACP, the lead ACP shares *MA\_ExecutionPriorityListActivationDataMT* with the package member. Upon receiving this message, follower ACPs update onboard MP with new EPL activation data.



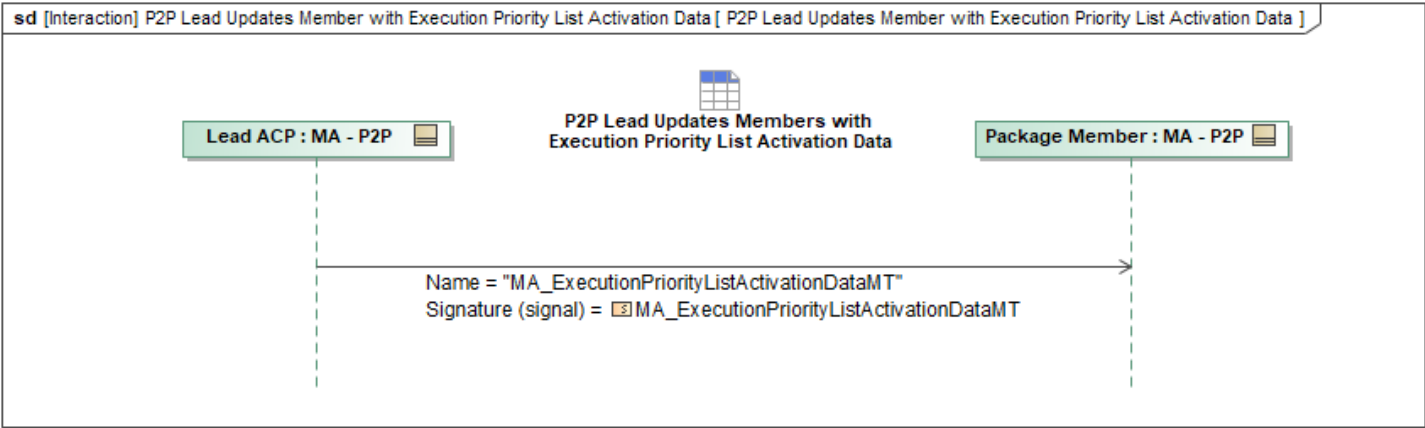


Figure 1-48: P2P Lead Updates Member with Execution Priority List Activation Data

Table 1-47: P2P Lead Updates Member with Execution Priority List Activation Data

| Message Name (Name:Type)                                                              | Required Fields (Name:Type)  |
|---------------------------------------------------------------------------------------|------------------------------|
| MA_ExecutionPriorityListActivationDataMT:<br>MA_ExecutionPriorityListActivationDataMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.9 P2P Member Updates Lead

1.2.10.9.1 P2P Member to Lead Update with Constraint Priority List Activation Status

This interaction allows the package members to update the lead regarding mission/task related data. The package member shares *MA\_ConstraintPriorityListActivationStatusMT* with the lead ACP.

Members notify lead of new active CPL after entering a new geozone by sending the lead *MA\_ConstraintPriorityListActivationStatusMT* containing the *ConstraintPriorityListActivationDataID* and *ActiveConstraintPriorityListID*. The *AppliesToID* shall be set to the *System ID* of the follower ACP.

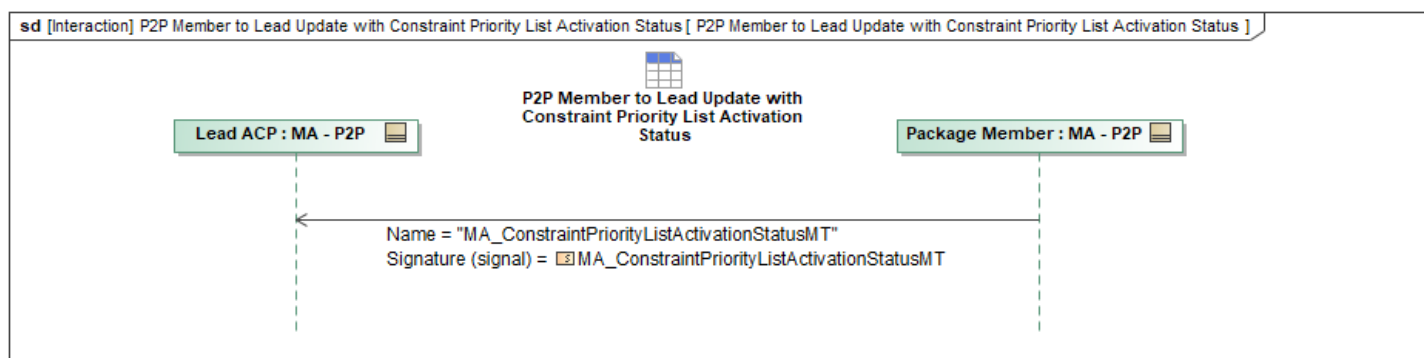


Figure 1-49: P2P Member to Lead Update with Constraint Priority List Activation Status

Table 1-48: P2P Member to Lead Update with Constraint Priority List Activation Status

| Message Name (Name:Type)                                                                    | Required Fields (Name:Type) |
|---------------------------------------------------------------------------------------------|-----------------------------|
| MA_ConstraintPriorityListActivationStatusMT:<br>MA_ConstraintPriorityListActivationStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.9.2 P2P Member to Lead Update with Execution Priority List Activation Status

This interaction allows the package members to update the lead regarding mission/task related data. The package member shares *MA\_ExecutionPriorityListActivationStatusMT* with the lead ACP.

Followers notify lead of a new active EPL after entering a geozone by sending the lead *MA\_ExecutionPriorityListActivationStatusMT* containing the *ExecutionPriorityListActivationDataID* and *ActiveExecutionPriorityListID*. The *AppliesToID* shall be set to the *System ID* of the follower ACP.

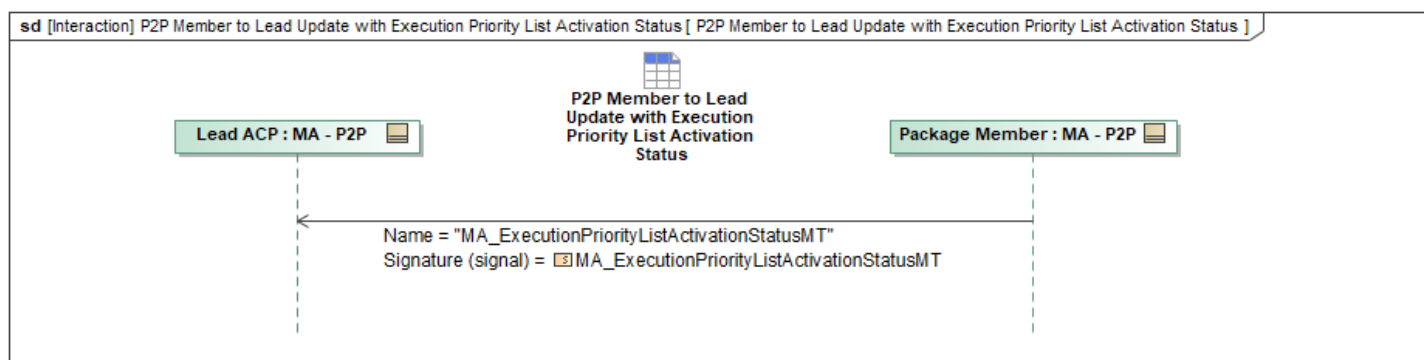


Figure 1-50: P2P Member to Lead Update with Execution Priority List Activation Status

Table 1-49: P2P Member to Lead Update with Execution Priority List Activation Status

| Message Name (Name:Type)                                                                  | Required Fields (Name:Type) |
|-------------------------------------------------------------------------------------------|-----------------------------|
| MA_ExecutionPriorityListActivationStatusMT:<br>MA_ExecutionPriorityListActivationStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.10.9.3 P2P Member to Lead Update with System Notification

This interaction allows the package members to update the lead regarding mission/task related data. The package member shares *MA\_SystemNotificationMT* with the lead ACP.

To update the lead of a member's synchronization completion status, the package members send *MA\_SystemNotificationMT* message with the *SystemSubjectID* field containing the *System ID* of the sending package member. The *AppliesTo.SystemID* should be the *SystemID* of the package leader. The *AssociatedMessage.MessageType* should be set to *MISSION\_PLAN* and the *AssociatedMessage.AssociatedID* should be set to the UUID of the Mission Plan that completed synchronization.

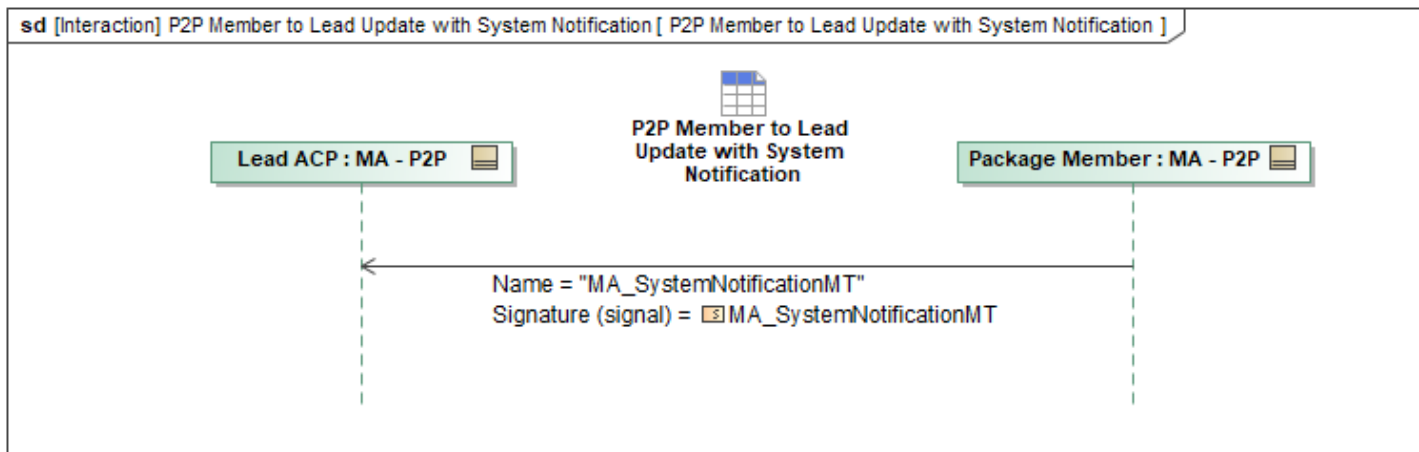


Figure 1-51: P2P Member to Lead Update with System Notification

Table 1-50: P2P Member to Lead Update with System Notification

| Message Name (Name:Type)                                   | Required Fields (Name:Type)                                                                 |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>MA_SystemNotificationMT:</b><br>MA_SystemNotificationMT | AppliesTo: SubjectType<br>AssociatedMessage: AssociatedMessageType<br>AssociatedID: ID_Type |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.10.10 P2P Query Data

#### 1.2.10.10.1 P2P Query Activity Plan

*MA\_ActivityPlanMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to

*COMPLETED*, and this message contains the requested *MA\_ActivityPlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA\_ActivityPlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA\_ActivityPlanMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA\_ActivityPlanMT*.

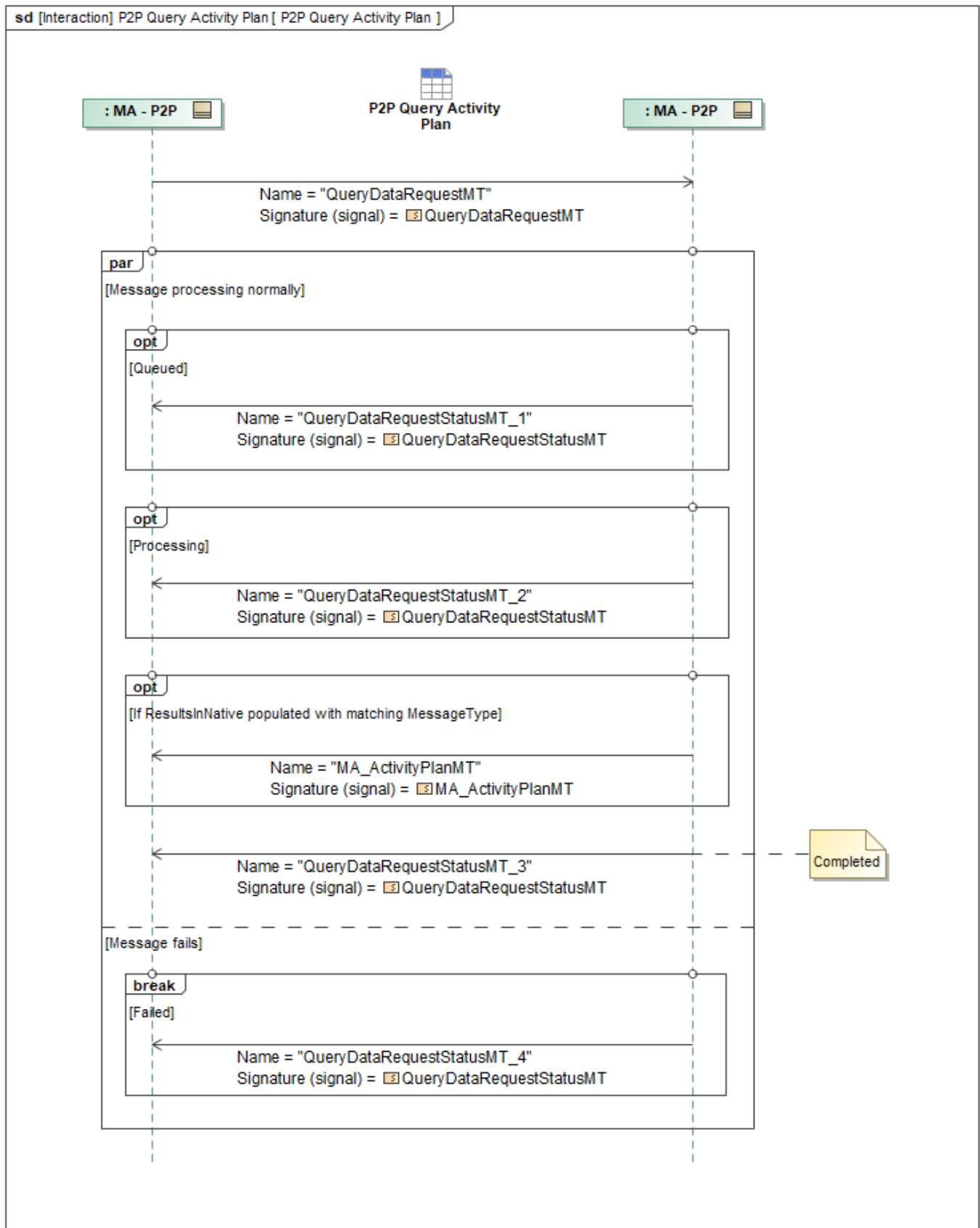


Figure 1-52: P2P Query Activity Plan

Table 1-51: P2P Query Activity Plan

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| MA_ActivityPlanMT: MA_ActivityPlanMT                    | None                                           |
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.10.2 P2P Query Airfield Report

*AirfieldReportMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *AirfieldReportMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *AirfieldReportMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *AirfieldReportMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *AirfieldReportMT*.

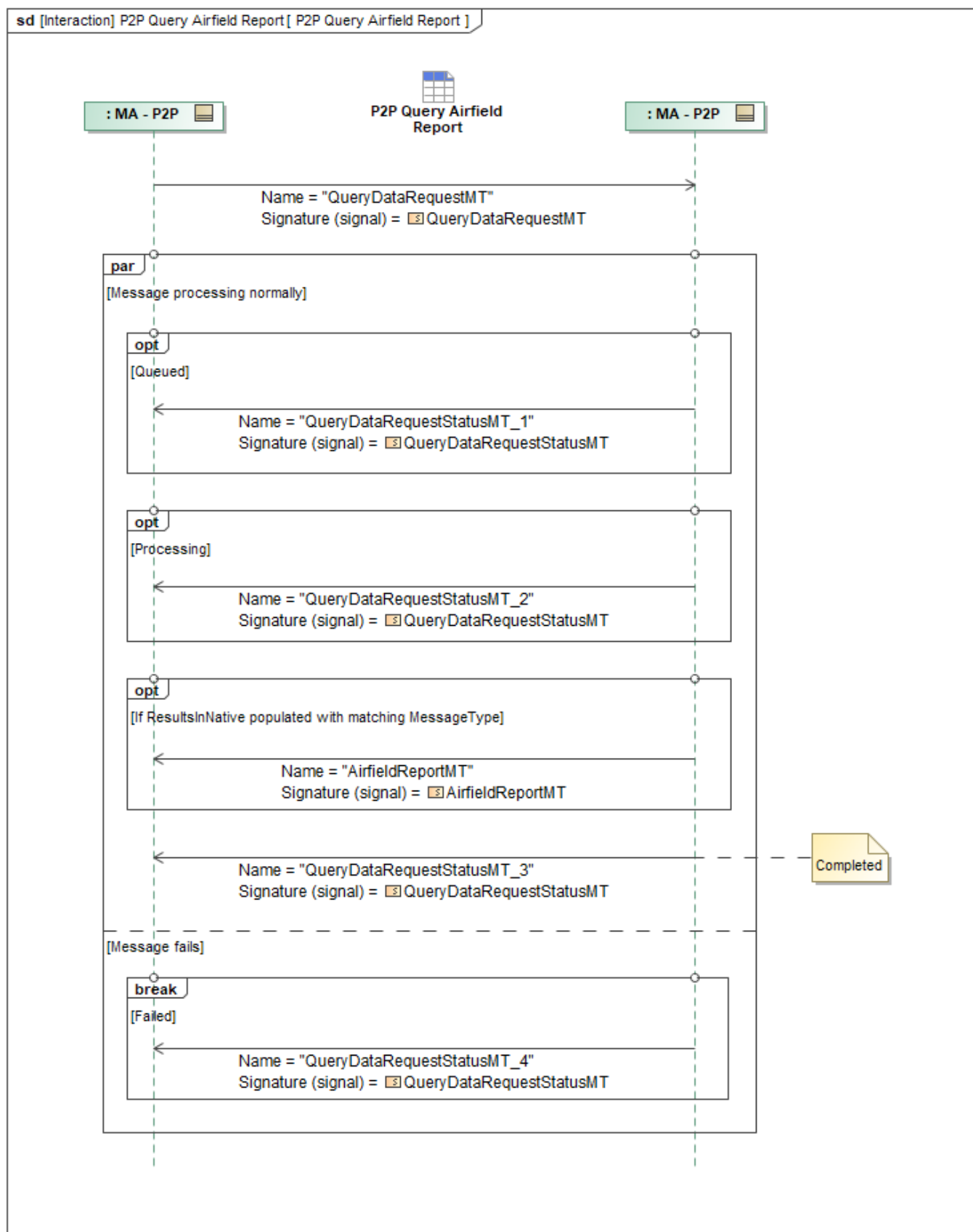


Figure 1-53: P2P Query Airfield Report

Table 1-52: P2P Query Airfield Report

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| AirfieldReportMT: AirfieldReportMT                      | None                                           |
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.10.10.3 P2P Query Mission Plan

*MA\_MissionPlanMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA\_MissionPlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA\_MissionPlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA\_MissionPlanMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA\_MissionPlanMT*.



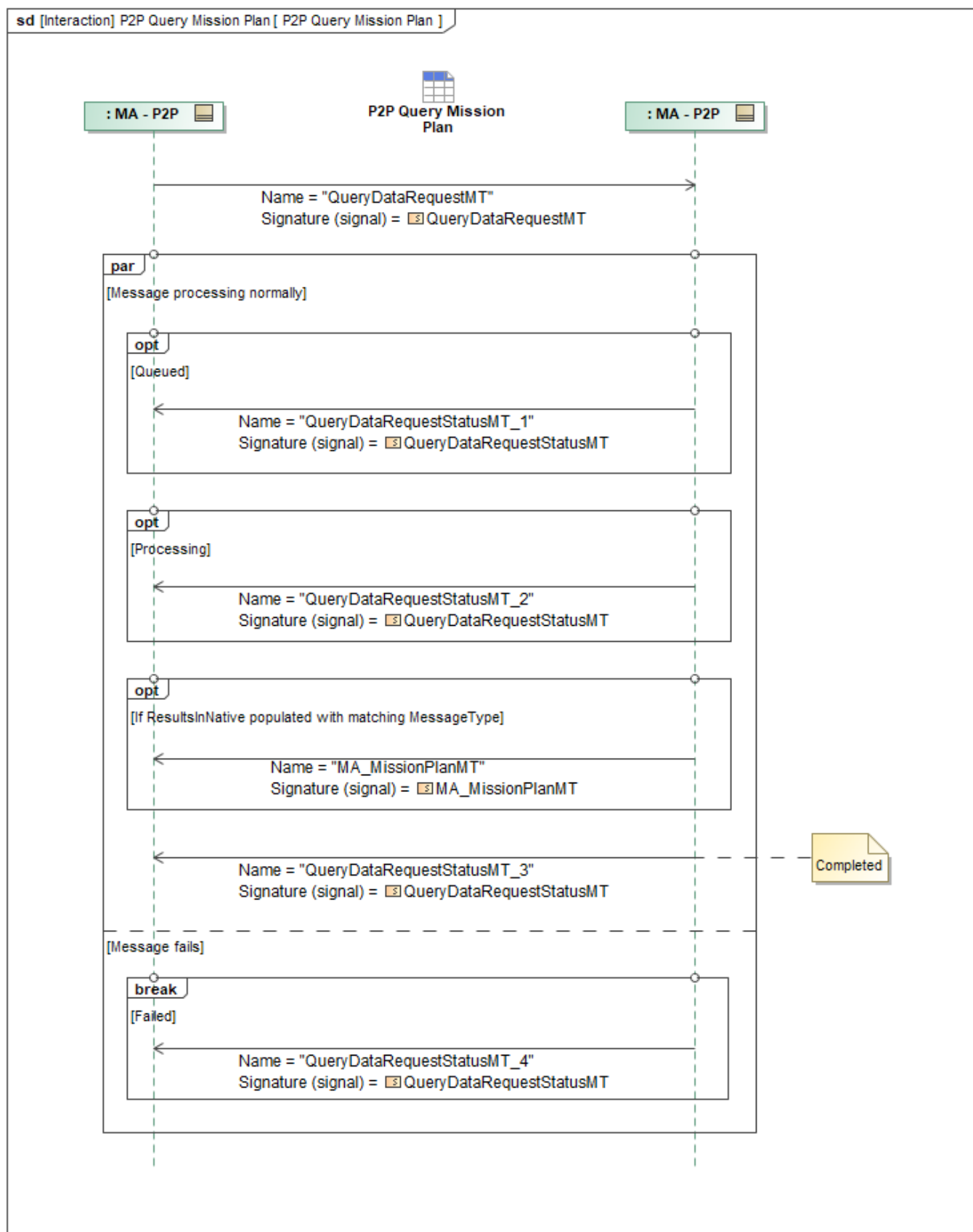


Figure 1-54: P2P Query Mission Plan

Table 1-53: P2P Query Mission Plan

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| MA_MissionPlanMT: MA_MissionPlanMT                      | None                                           |
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.10.4 P2P Query OpZone

*MA\_OpZoneMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA\_OpZoneMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA\_OpZoneMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA\_OpZoneMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA\_OpZoneMT*.

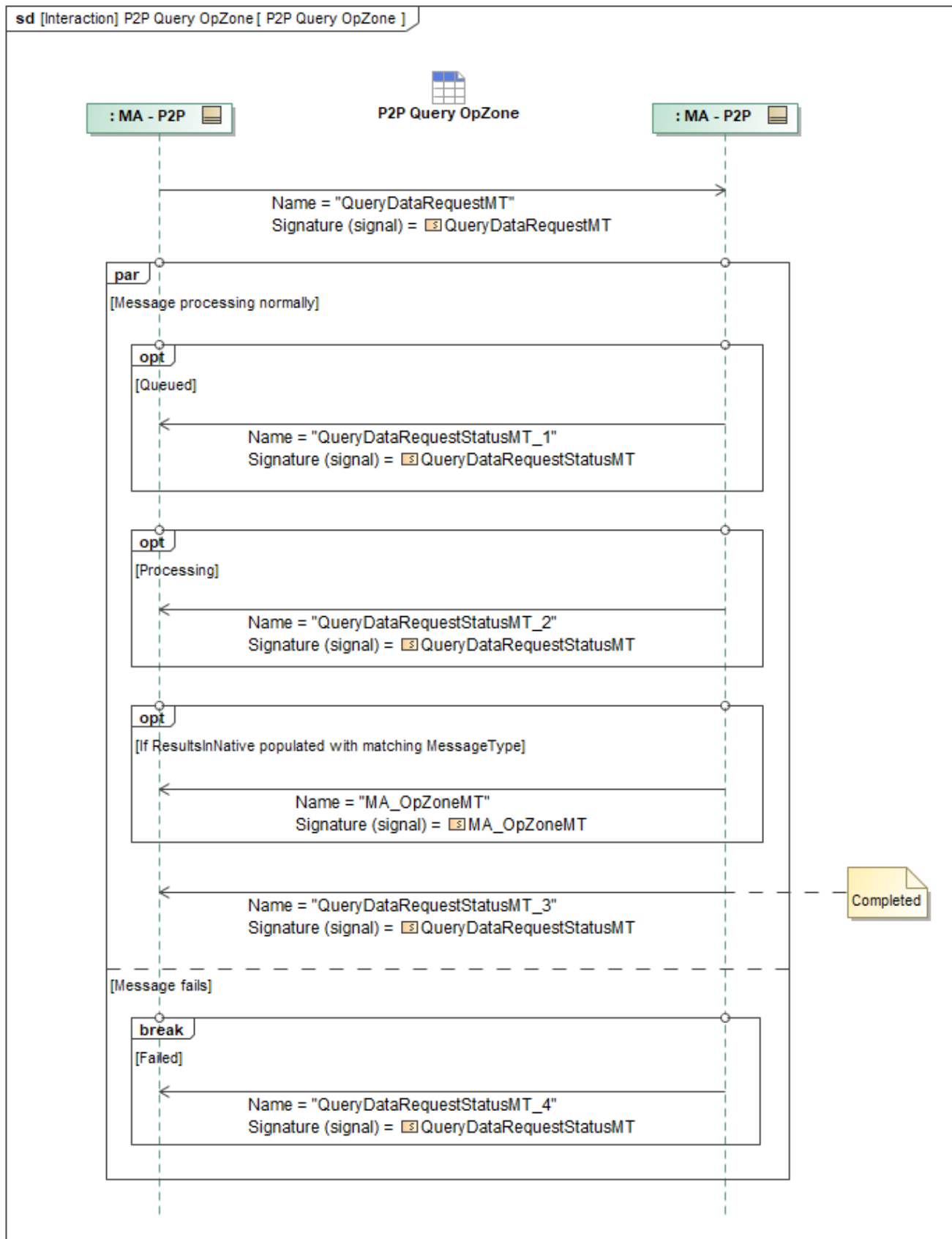


Figure 1-55: P2P Query OpZone

Table 1-54: P2P Query OpZone

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| MA_OpZoneMT: MA_OpZoneMT                                | None                                           |
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.10.5 P2P Query Response Plan

*ResponsePlanMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *ResponsePlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *ResponsePlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *ResponsePlanMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *ResponsePlanMT*.

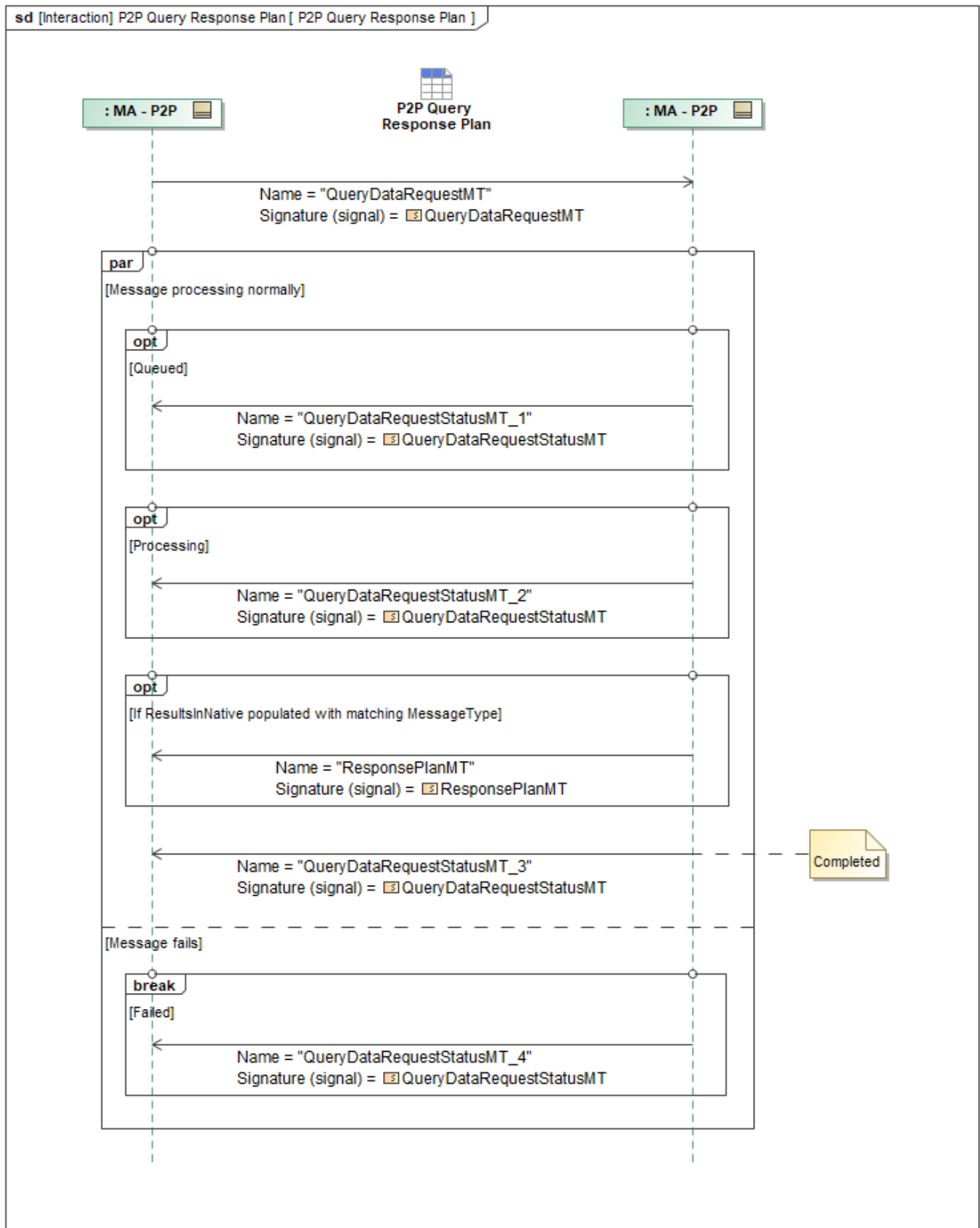


Figure 1-56: P2P Query Response Plan

Table 1-55: P2P Query Response Plan

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |
| ResponsePlanMT: ResponsePlanMT                          | None                                           |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.10.6 P2P Query Route Activity Plan

*MA\_RouteActivityPlanMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA\_RouteActivityPlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA\_RouteActivityPlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA\_RouteActivityPlanMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA\_RouteActivityPlanMT*.

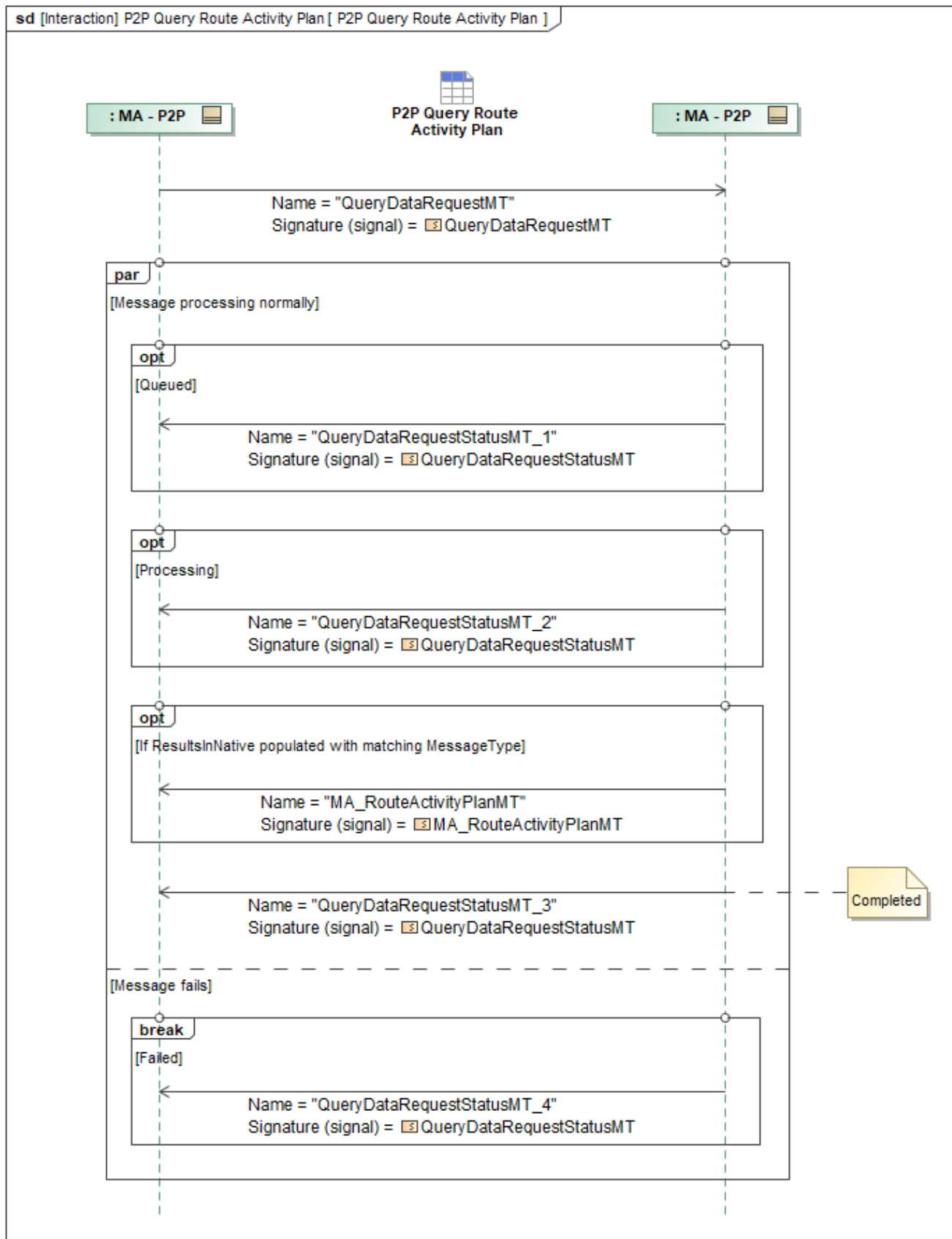


Figure 1-57: P2P Query Route Activity Plan

Table 1-56: P2P Query Route Activity Plan

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| MA_RouteActivityPlanMT:<br>MA_RouteActivityPlanMT       | None                                           |
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.10.7 P2P Query Route Plan

*MA\_RoutePlanMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA\_RoutePlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA\_RoutePlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA\_RoutePlanMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA\_RoutePlanMT*.



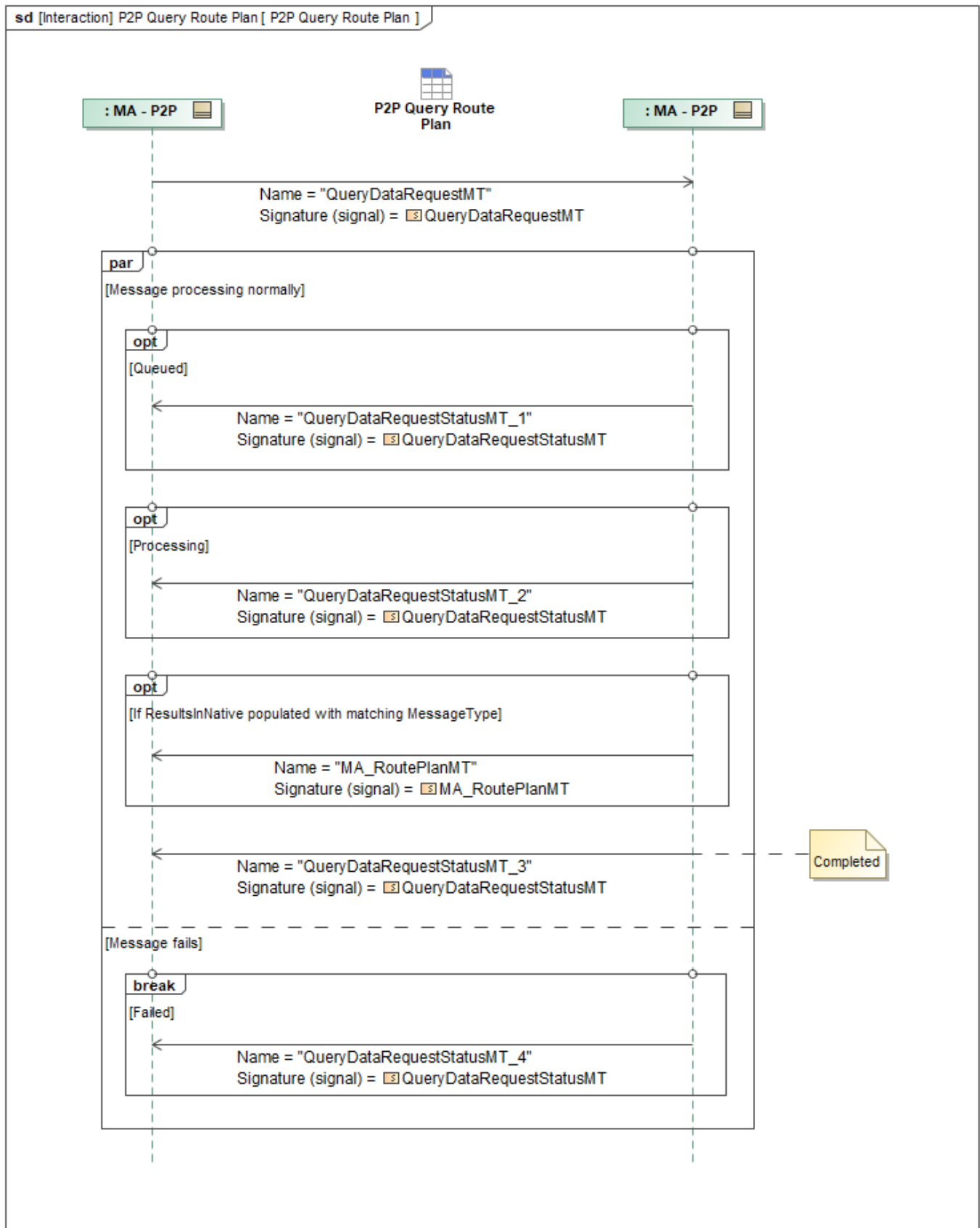


Figure 1-58: P2P Query Route Plan

Table 1-57: P2P Query Route Plan

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| MA_RoutePlanMT: MA_RoutePlanMT                          | None                                           |
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.10.8 P2P Query Task Plan

*MA\_TaskPlanMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA\_TaskPlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA\_TaskPlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA\_TaskPlanMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA\_TaskPlanMT*.

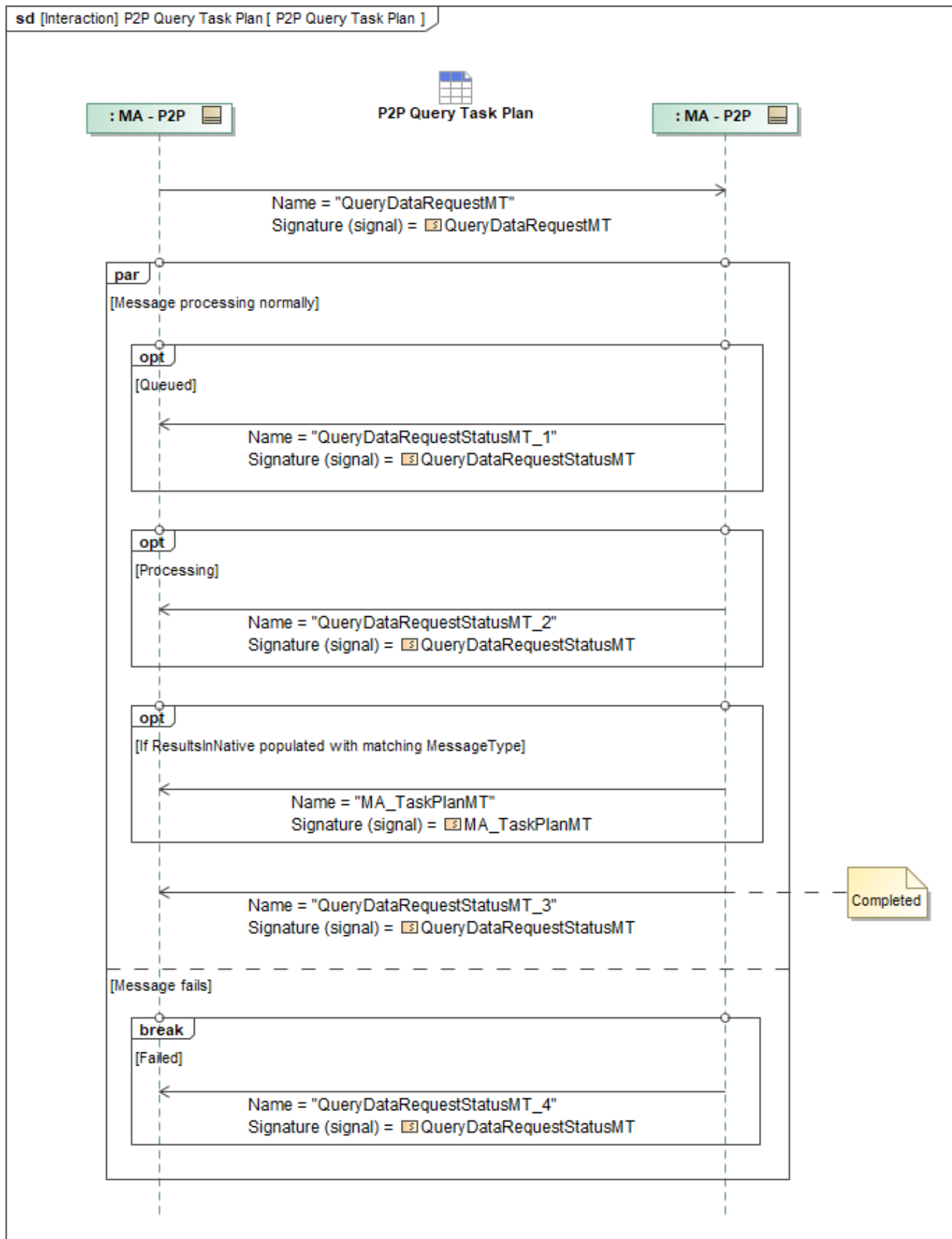


Figure 1-59: P2P Query Task Plan

Table 1-58: P2P Query Task Plan

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| MA_TaskPlanMT: MA_TaskPlanMT                            | None                                           |
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.10.10.9 P2P Query Weather

*WeatherMT* is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT\_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT\_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *WeatherMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *WeatherMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *WeatherMT* message and still send *QueryDataRequestStatusMT\_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT\_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *WeatherMT*.

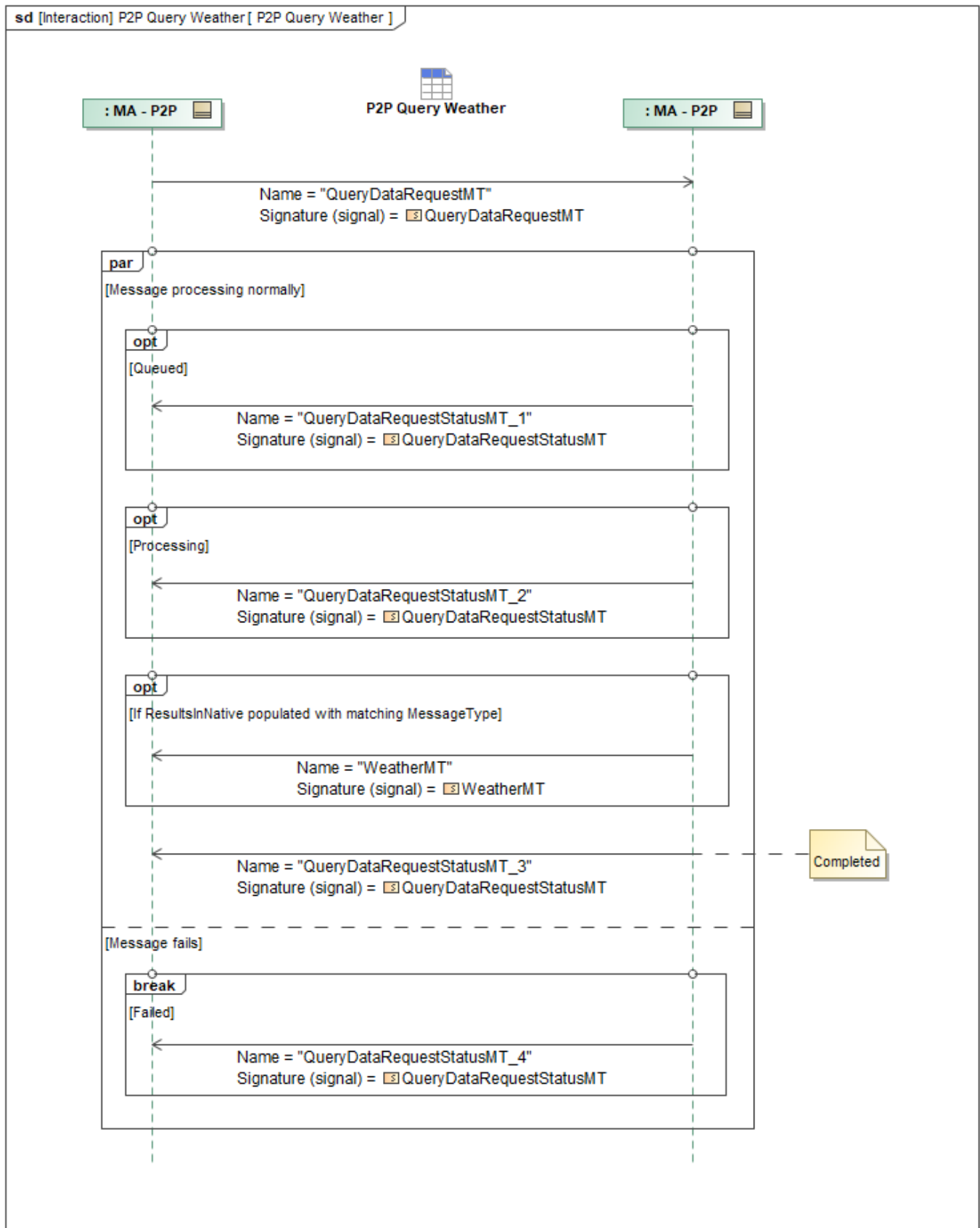


Figure 1-60: P2P Query Weather

Table 1-59: P2P Query Weather

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| QueryDataRequestMT: QueryDataRequestMT                  | None                                           |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType                        |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |
| WeatherMT: WeatherMT                                    | None                                           |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.10.11 P2P Send Data

#### 1.2.10.11.1 Send Action Plan to Peer

This diagram contains the message, *MA\_ActionPlanMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

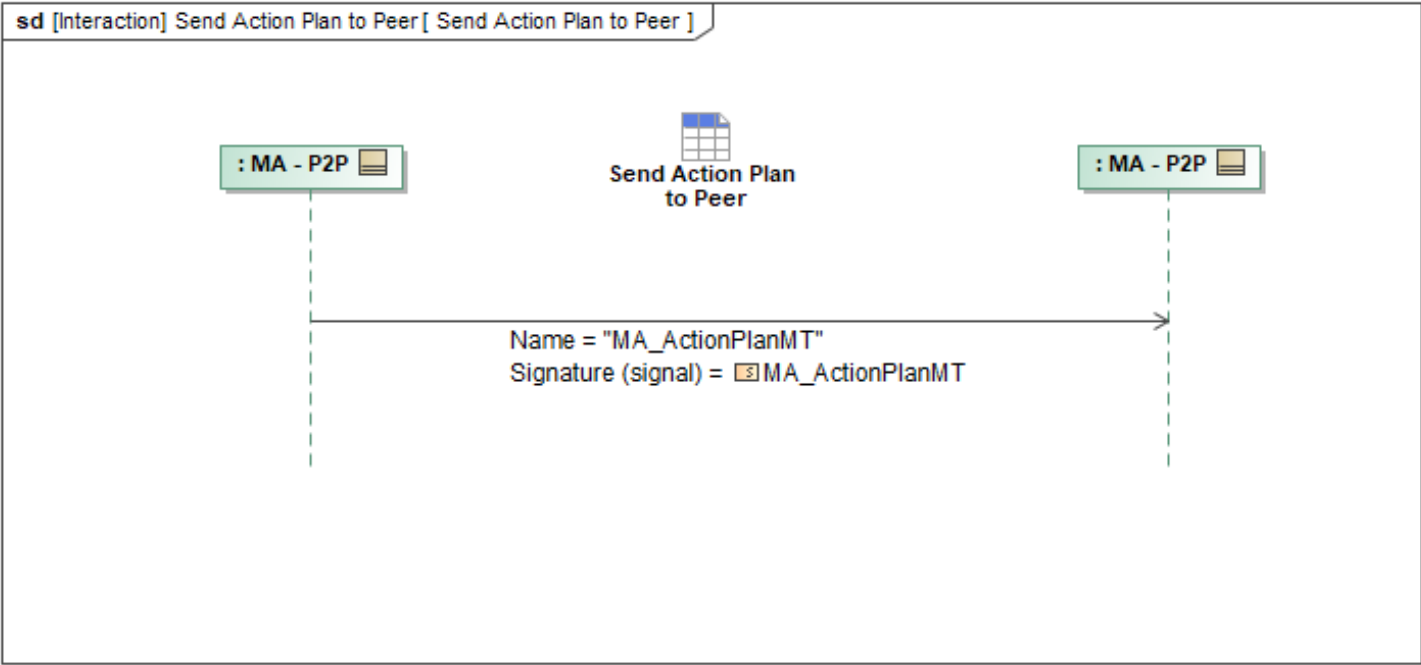


Figure 1-61: Send Action Plan to Peer

Table 1-60: Send Action Plan to Peer

| Message Name (Name:Type)         | Required Fields (Name:Type) |
|----------------------------------|-----------------------------|
| MA_ActionPlanMT: MA_ActionPlanMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.2 Send Action to Peer

This diagram contains the message, *MA\_ActionMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

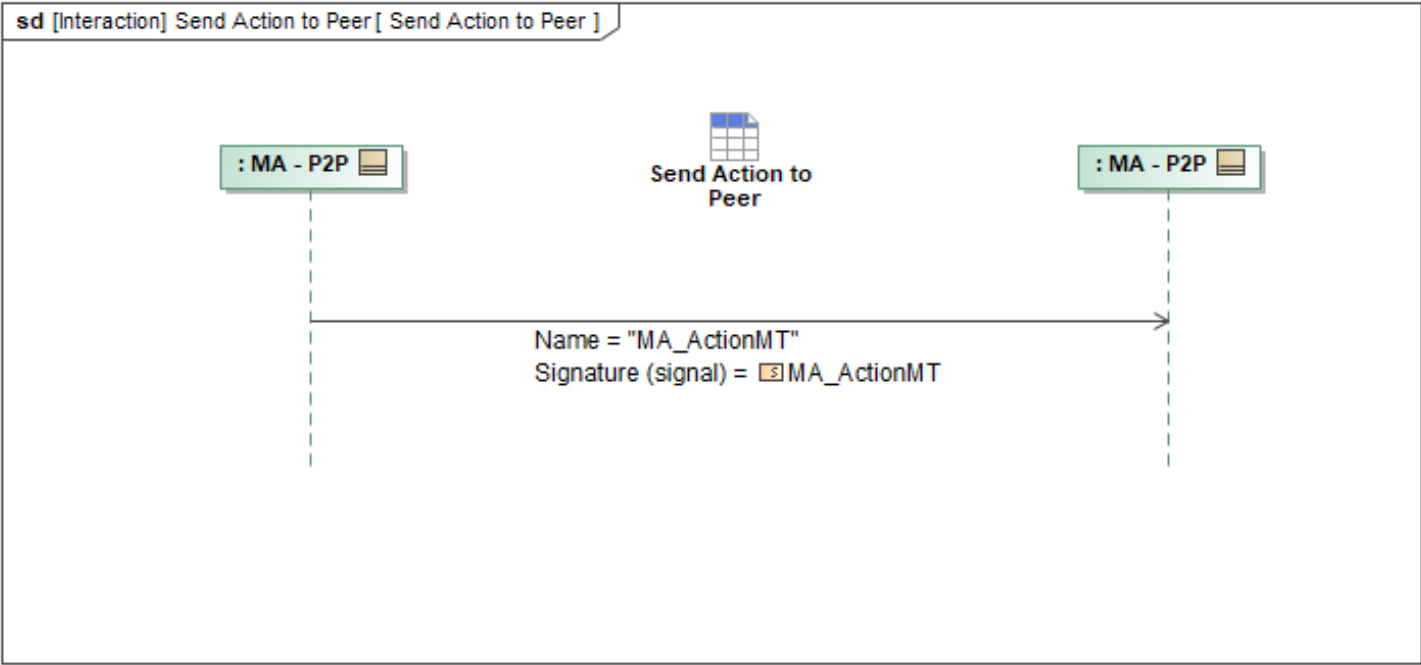


Figure 1-62: Send Action to Peer

Table 1-61: Send Action to Peer

| Message Name (Name:Type) | Required Fields (Name:Type)  |
|--------------------------|------------------------------|
| MA_ActionMT: MA_ActionMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.3 Send Activity Plan to Peer

This diagram contains the message, *MA\_ActivityPlanMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.



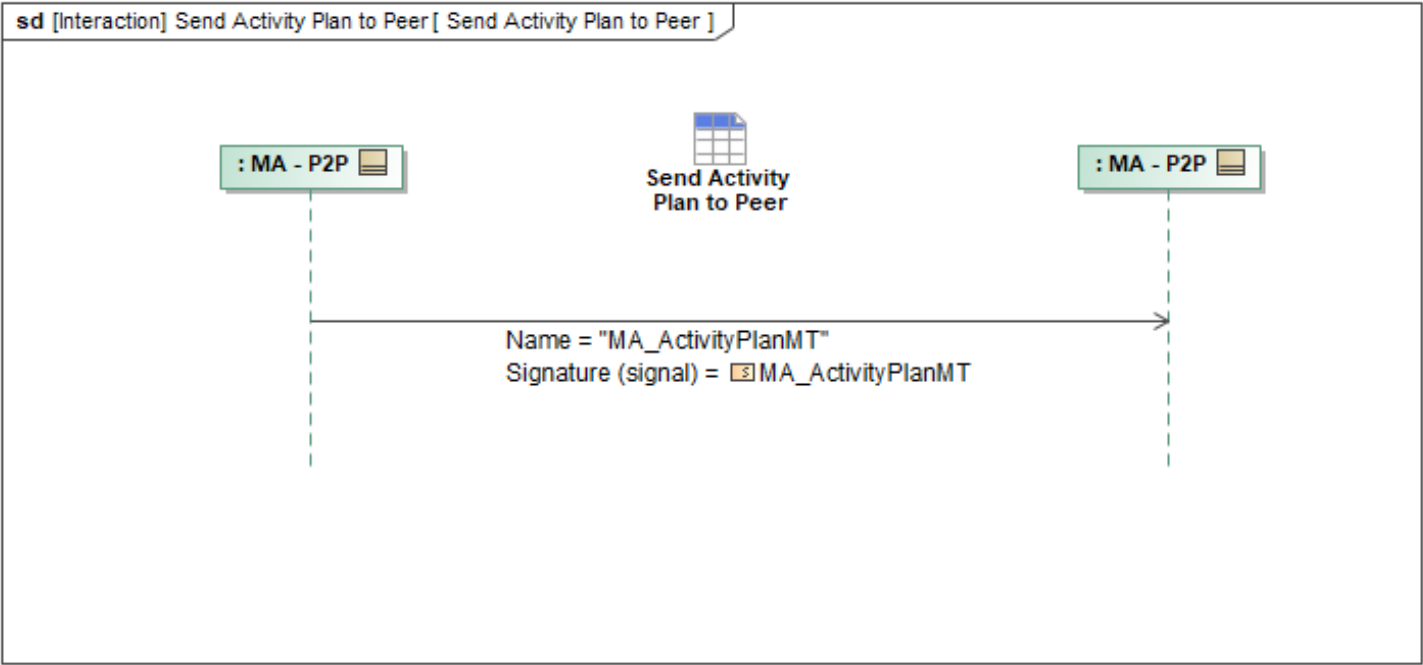


Figure 1-63: Send Activity Plan to Peer

Table 1-62: Send Activity Plan to Peer

| Message Name (Name:Type)             | Required Fields (Name:Type) |
|--------------------------------------|-----------------------------|
| MA_ActivityPlanMT: MA_ActivityPlanMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.4 Send Approval Authority to Peer

This diagram contains the message, *MA\_ApprovalAuthorityMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

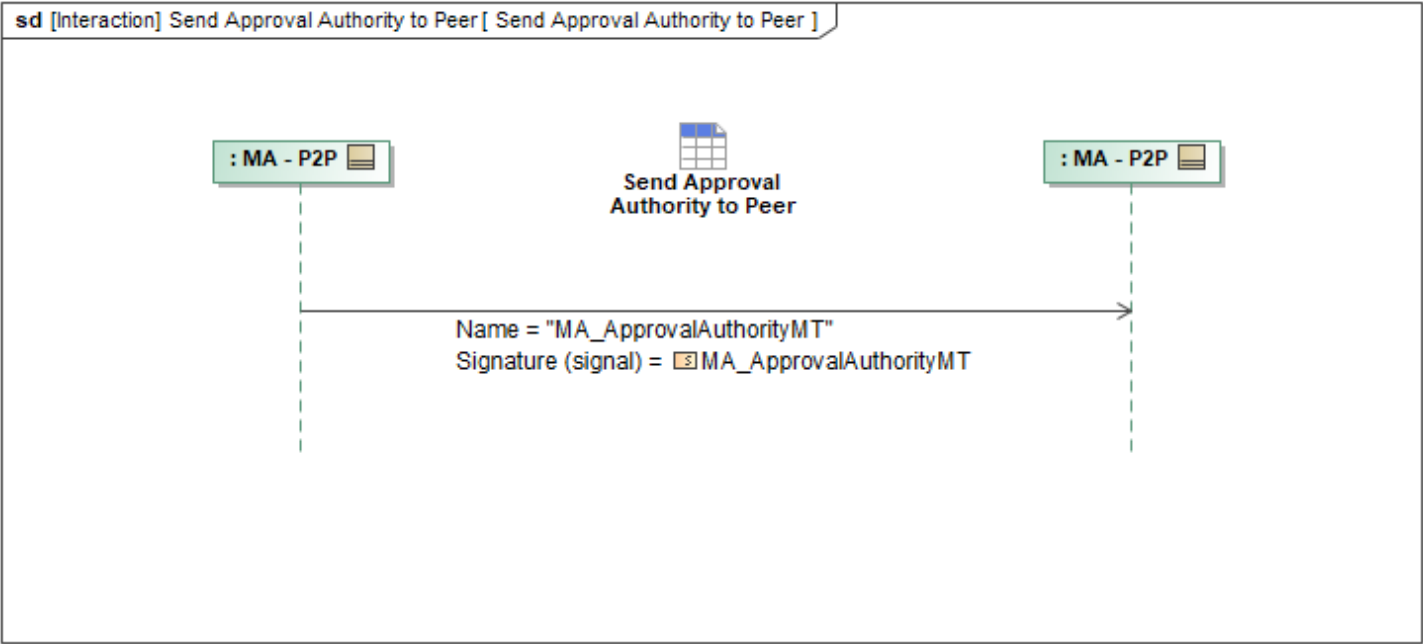


Figure 1-64: Send Approval Authority to Peer

Table 1-63: Send Approval Authority to Peer

| Message Name (Name:Type)                          | Required Fields (Name:Type)  |
|---------------------------------------------------|------------------------------|
| MA_ApprovalAuthorityMT:<br>MA_ApprovalAuthorityMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.5 Send Approval Policy to Peer

This diagram contains the message, *MA\_ApprovalPolicyMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.



Figure 1-65: Send Approval Policy to Peer

Table 1-64: Send Approval Policy to Peer

| Message Name (Name:Type)                 | Required Fields (Name:Type) |
|------------------------------------------|-----------------------------|
| MA_ApprovalPolicyMT: MA_ApprovalPolicyMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.6 Send Capability Set to Peer

This diagram contains the message, *MA\_CapabilitySetMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

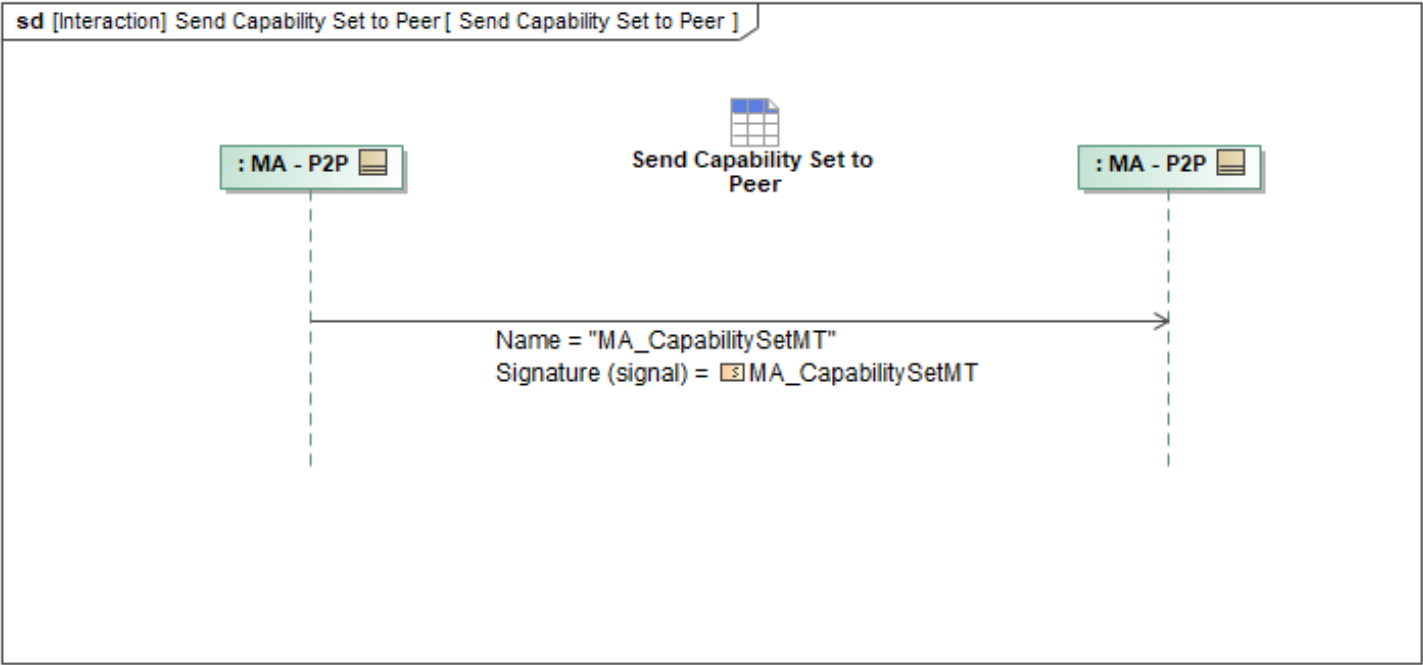


Figure 1-66: Send Capability Set to Peer

Table 1-65: Send Capability Set to Peer

| Message Name (Name:Type)               | Required Fields (Name:Type)  |
|----------------------------------------|------------------------------|
| MA_CapabilitySetMT: MA_CapabilitySetMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.7 Send Component Status to Peer

This interaction specifies how Mission Autonomy (MA) shares *ComponentStatusMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the *ComponentStatusMT* is sent. The *ComponentStatusMT* must have the *ComponentID* set to the *SystemID* and the *ComponentState* set to the Vehicle Autonomy’s state.

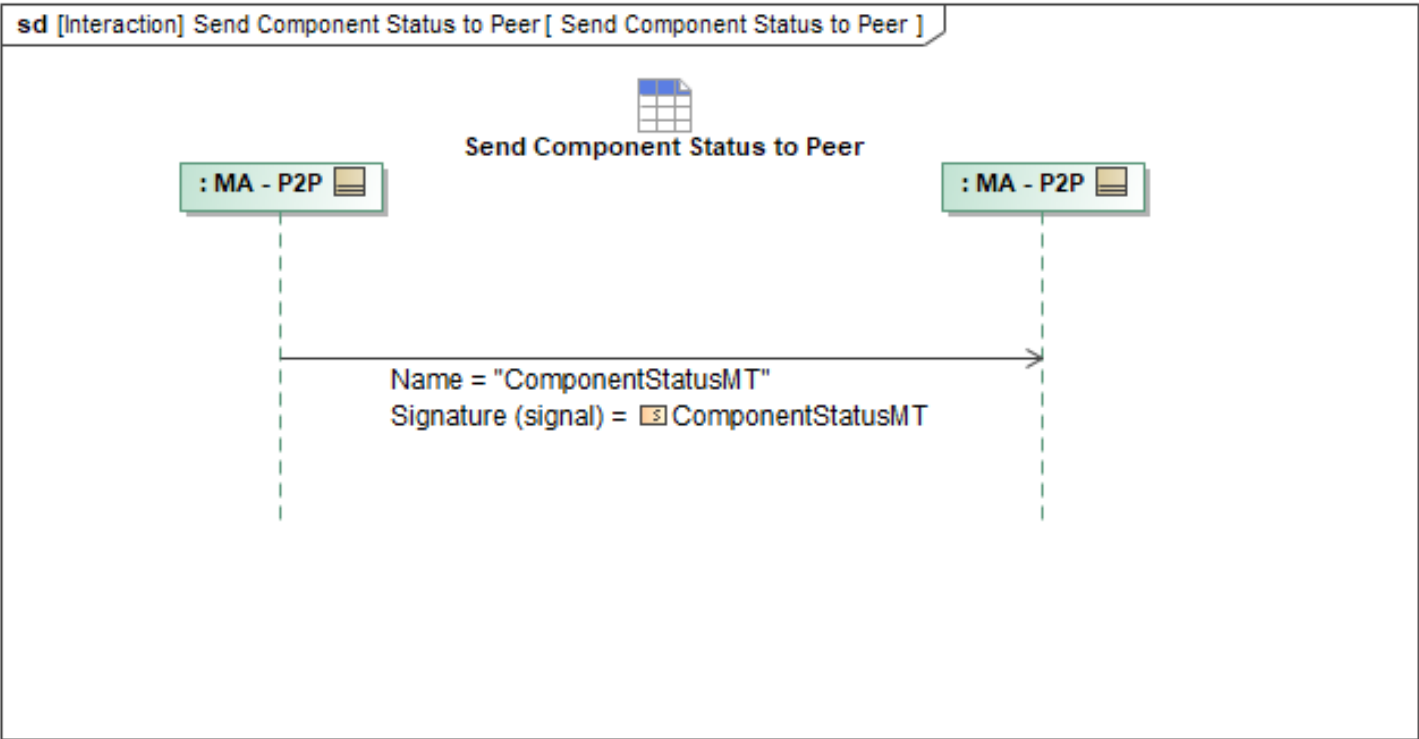


Figure 1-67: Send Component Status to Peer

Table 1-66: Send Component Status to Peer

| Message Name (Name:Type)             | Required Fields (Name:Type) |
|--------------------------------------|-----------------------------|
| ComponentStatusMT: ComponentStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.8 Send Entity to Peer

This diagram contains the message, *EntityMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

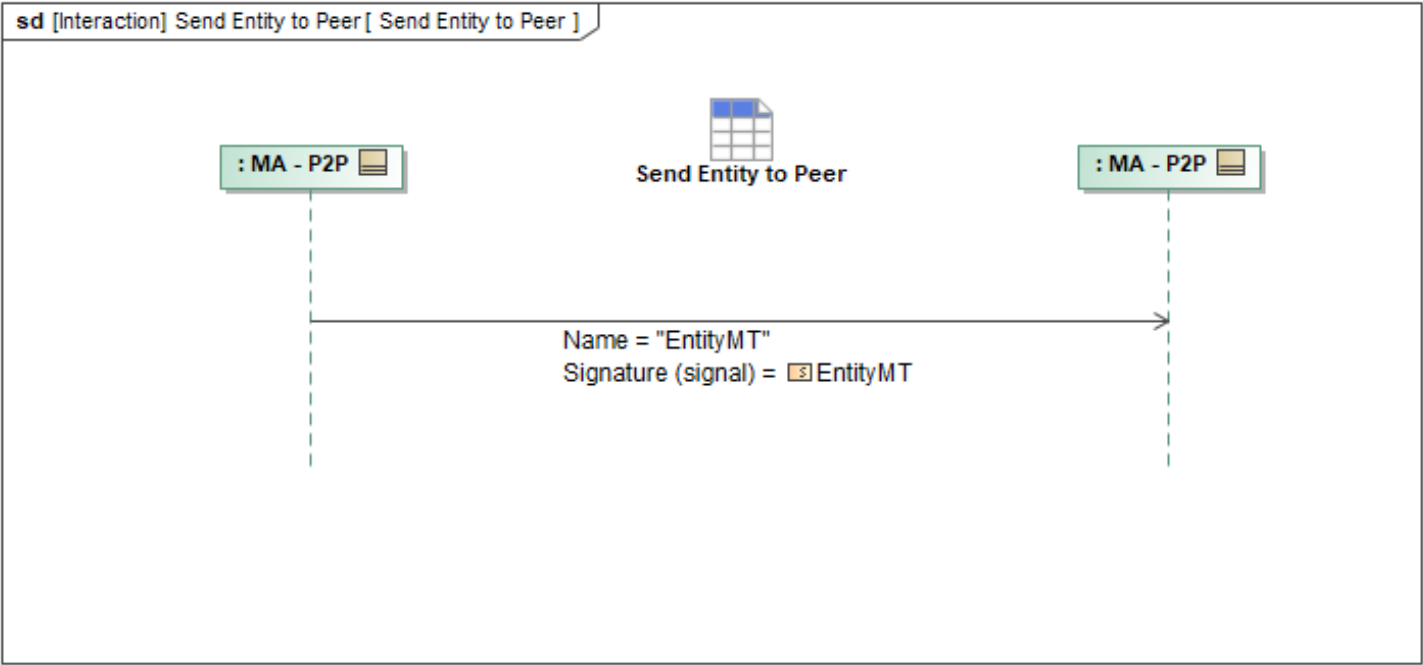


Figure 1-68: Send Entity to Peer

Table 1-67: Send Entity to Peer

| Message Name (Name:Type) | Required Fields (Name:Type)  |
|--------------------------|------------------------------|
| EntityMT: EntityMT       | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.9 Send Flight Activity to Peer

This interaction specifies how Mission Autonomy (MA) shares *MA\_FlightActivityMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the *MA\_FlightActivityMT* message is sent. The *MA\_FlightActivityMT* must contain the *CommandID* of the last flight command (if any) in the *CapabilityCommandID* and any commanded flight values in the *VehicleCommandState*. If the vehicle is flying waypoints, they must be listed in the *ActualEndPoint*.

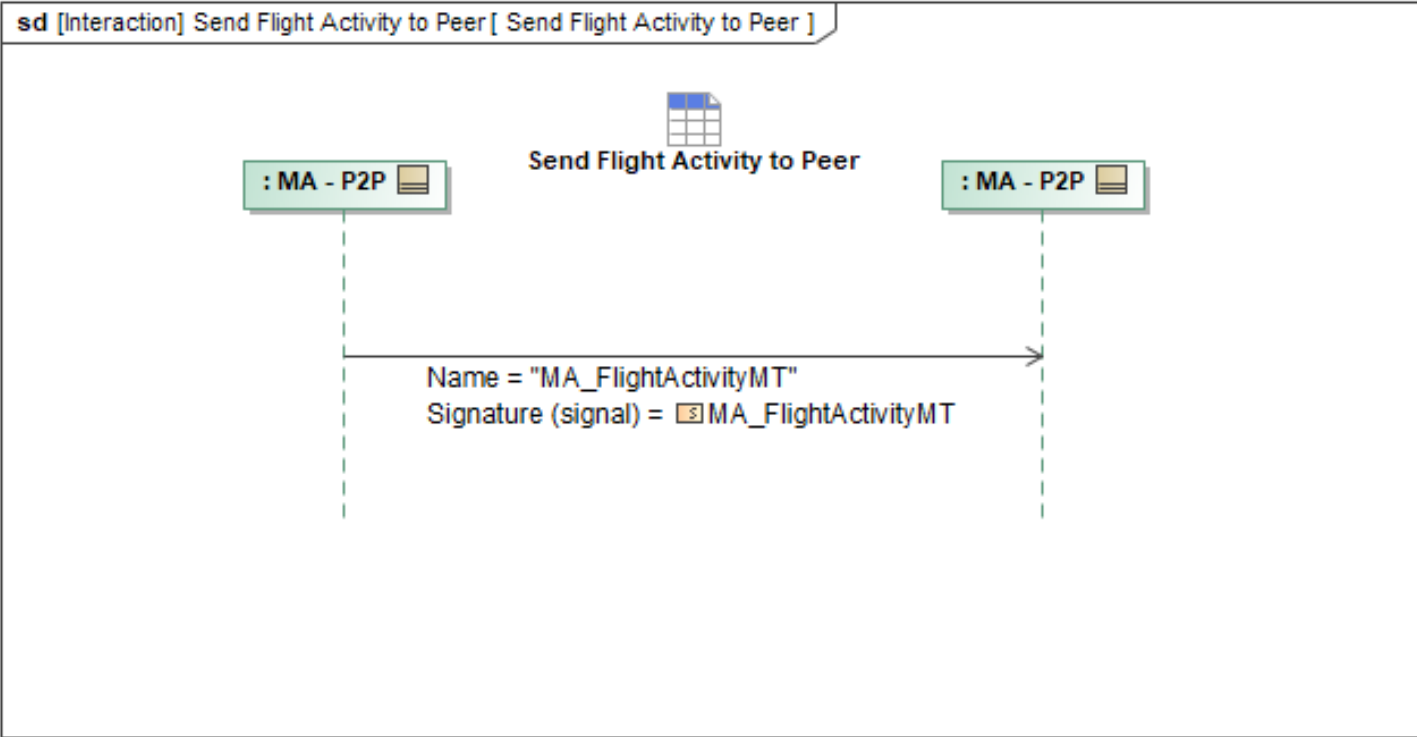


Figure 1-69: Send Flight Activity to Peer

Table 1-68: Send Flight Activity to Peer

| Message Name (Name:Type)                 | Required Fields (Name:Type) |
|------------------------------------------|-----------------------------|
| MA_FlightActivityMT: MA_FlightActivityMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.10 Send Flight Capability Status to Peer

This interaction specifies how Mission Autonomy (MA) shares *MA\_FlightCapabilityStatusMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the *MA\_FlightCapabilityStatusMT* message is sent.

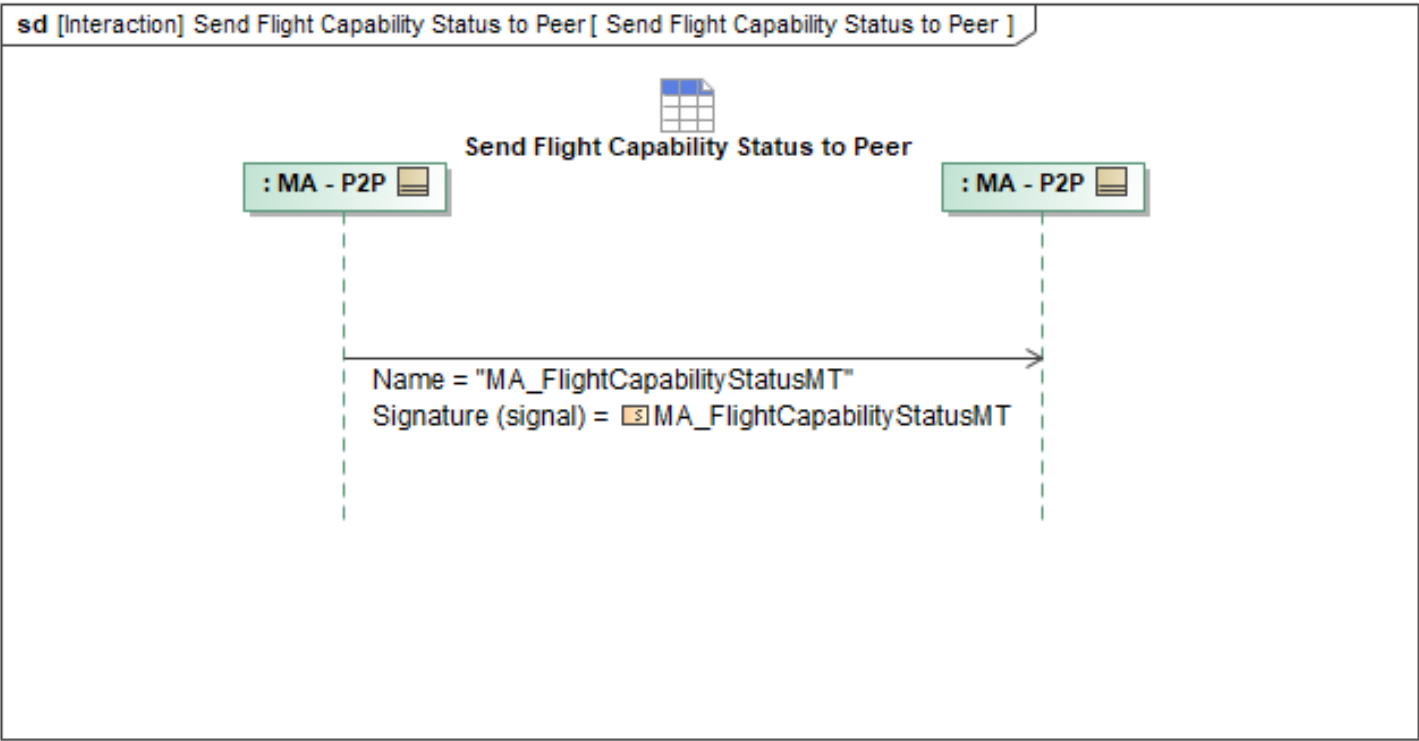


Figure 1-70: Send Flight Capability Status to Peer

Table 1-69: Send Flight Capability Status to Peer

| Message Name (Name:Type)                                    | Required Fields (Name:Type)                                                 |
|-------------------------------------------------------------|-----------------------------------------------------------------------------|
| MA_FlightCapabilityStatusMT:<br>MA_FlightCapabilityStatusMT | Availability: CapabilityAvailabilityEnum<br>CapabilityID: CapabilityID_Type |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.11 Send Flight Capability to Peer

This interaction specifies how Mission Autonomy (MA) shares *MA\_FlightCapabilityMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the *MA\_FlightCapabilityMT* message is sent. The *MA\_FlightCapabilityMT* must contain the applicable performance limits. The *MA\_FlightCapabilityStatusMT* must contain the *CapabilityID* of the flight capability and the *Availability* set the availability of Flight.



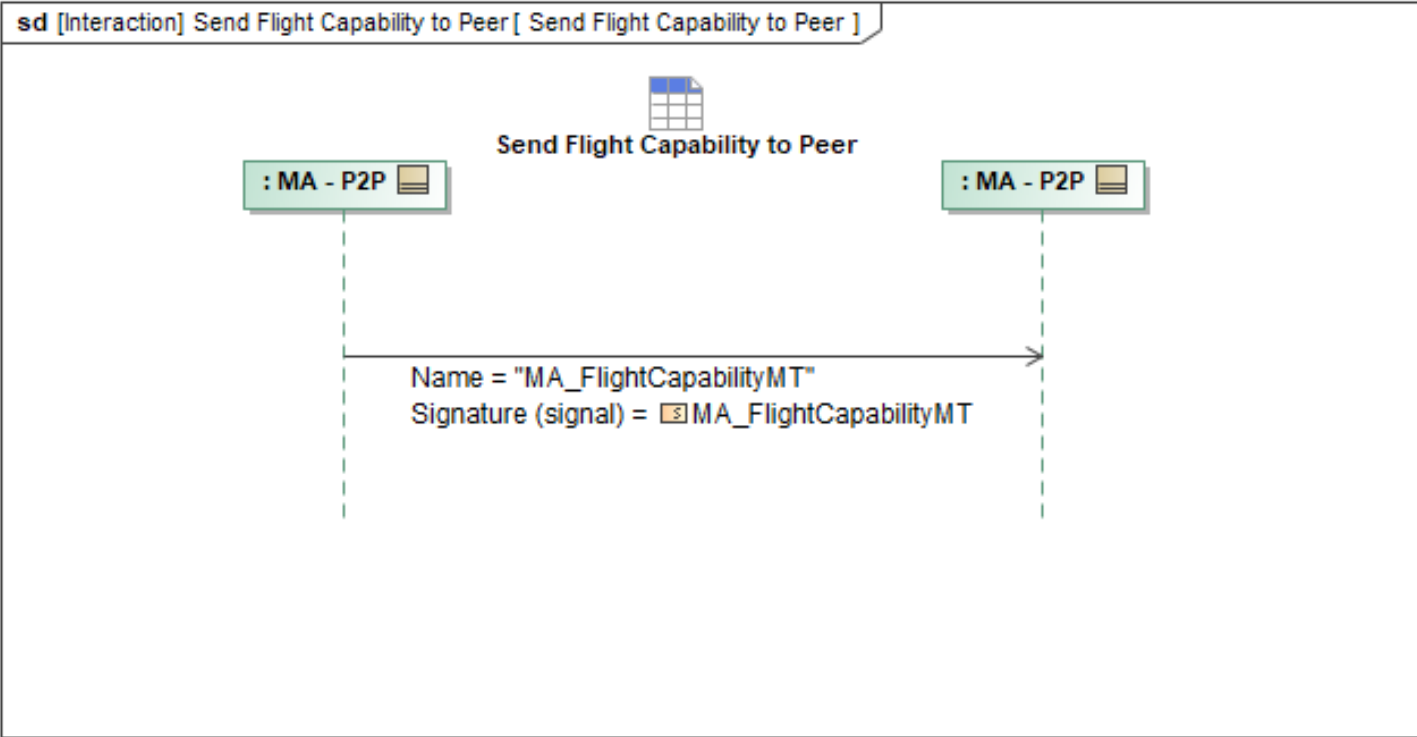


Figure 1-71: Send Flight Capability to Peer

Table 1-70: Send Flight Capability to Peer

| Message Name (Name:Type)                     | Required Fields (Name:Type) |
|----------------------------------------------|-----------------------------|
| MA_FlightCapabilityMT: MA_FlightCapabilityMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.12 Send Mission Plan Execution Status to Peer

This interaction specifies how Mission Autonomy (MA) shares *MA\_MissionPlanExecutionStatusMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the *MA\_MissionPlanExecutionStatusMT* message is sent. The *MissionPlanExecutionStatusMT* must have the *SystemID* of the vehicle and a *PlanExecutionStatus* for all active MissionPlans.

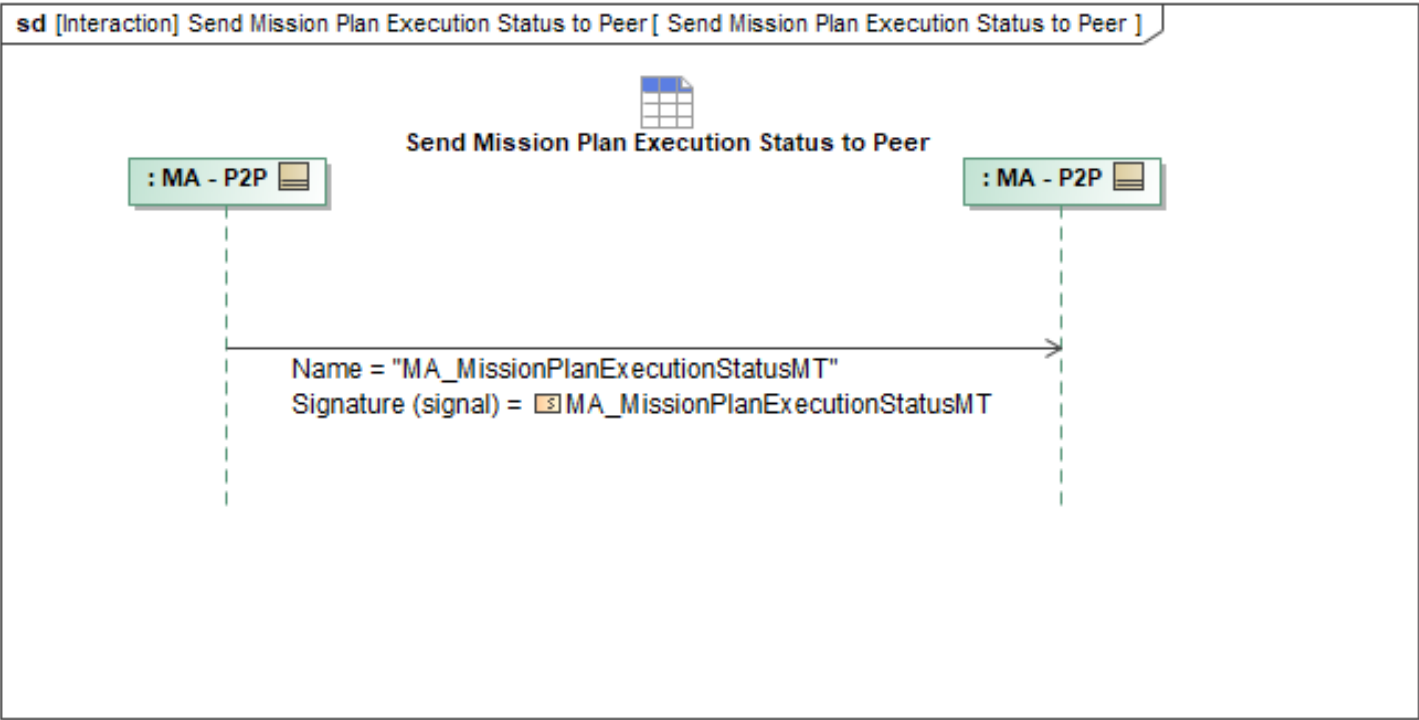


Figure 1-72: Send Mission Plan Execution Status to Peer

Table 1-71: Send Mission Plan Execution Status to Peer

| Message Name (Name:Type)                                            | Required Fields (Name:Type) |
|---------------------------------------------------------------------|-----------------------------|
| MA_MissionPlanExecutionStatusMT:<br>MA_MissionPlanExecutionStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.13 Send Mission Plan to Peer

This diagram contains the message, *MA\_MissionPlanMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

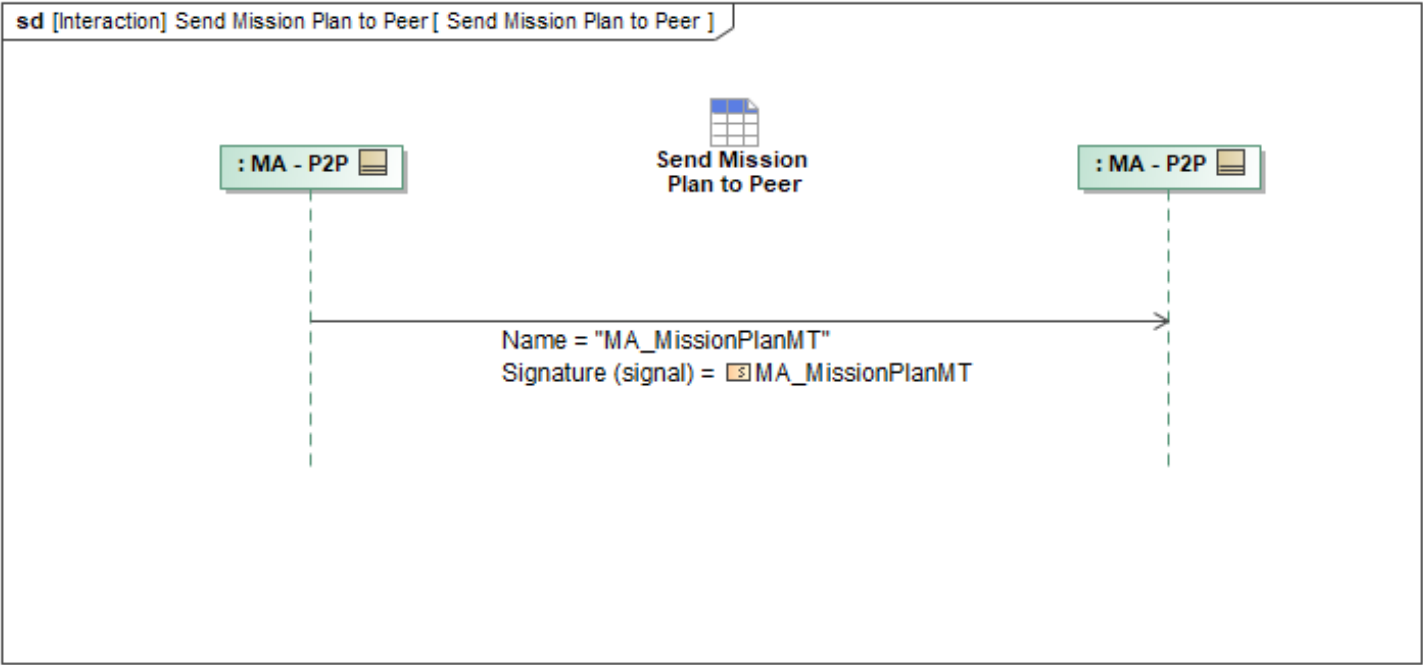


Figure 1-73: Send Mission Plan to Peer

Table 1-72: Send Mission Plan to Peer

| Message Name (Name:Type)           | Required Fields (Name:Type) |
|------------------------------------|-----------------------------|
| MA_MissionPlanMT: MA_MissionPlanMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.14 Send Navigation Report to Peer

This interaction specifies how Mission Autonomy (MA) shares *MA\_NavigationReportMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the *MA\_NavigationReportMT* message is sent.

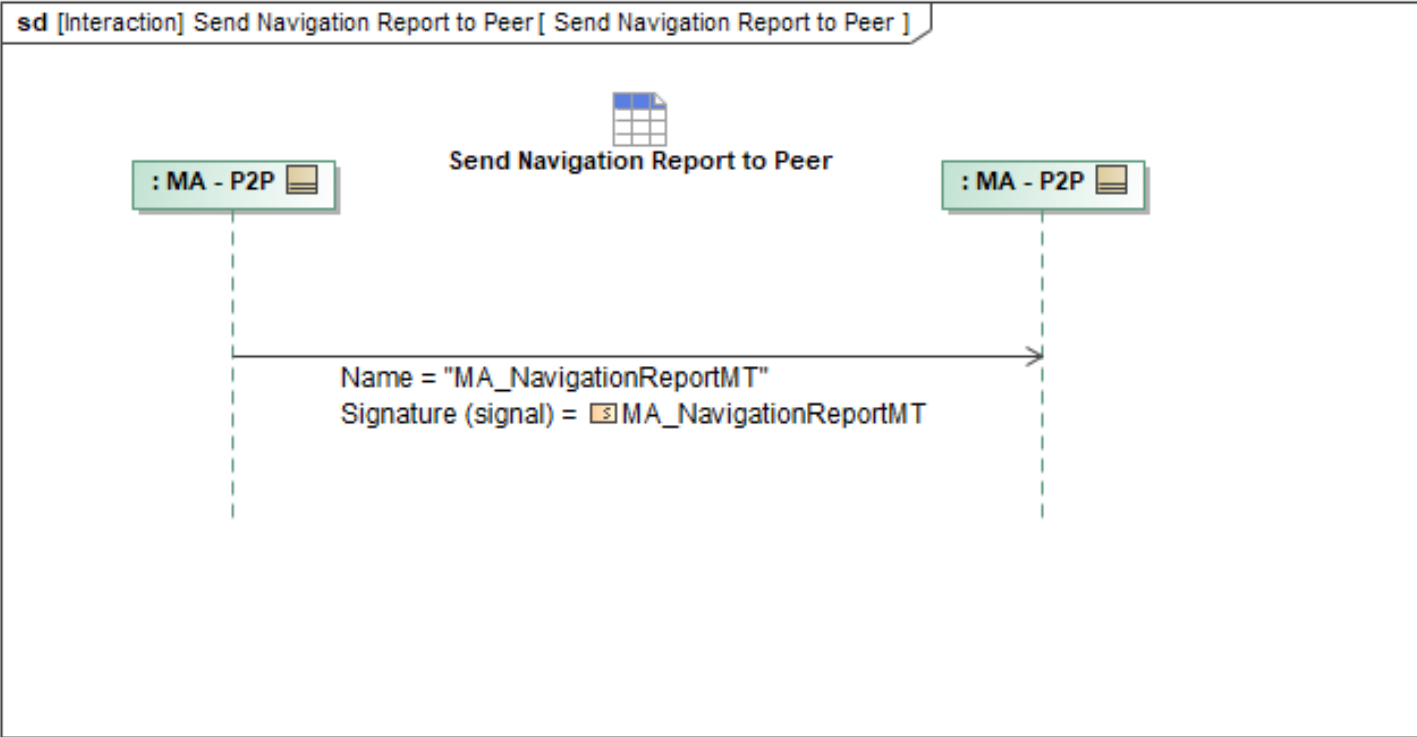


Figure 1-74: Send Navigation Report to Peer

Table 1-73: Send Navigation Report to Peer

| Message Name (Name:Type)                        | Required Fields (Name:Type)                                           |
|-------------------------------------------------|-----------------------------------------------------------------------|
| MA_NavigationReportMT:<br>MA_NavigationReportMT | Fuel: MassType<br>Duration: DurationRangeType<br>Percent: PercentType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.15 Send Operator Role to Peer

This diagram contains the message, *MA\_OperatorRoleMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

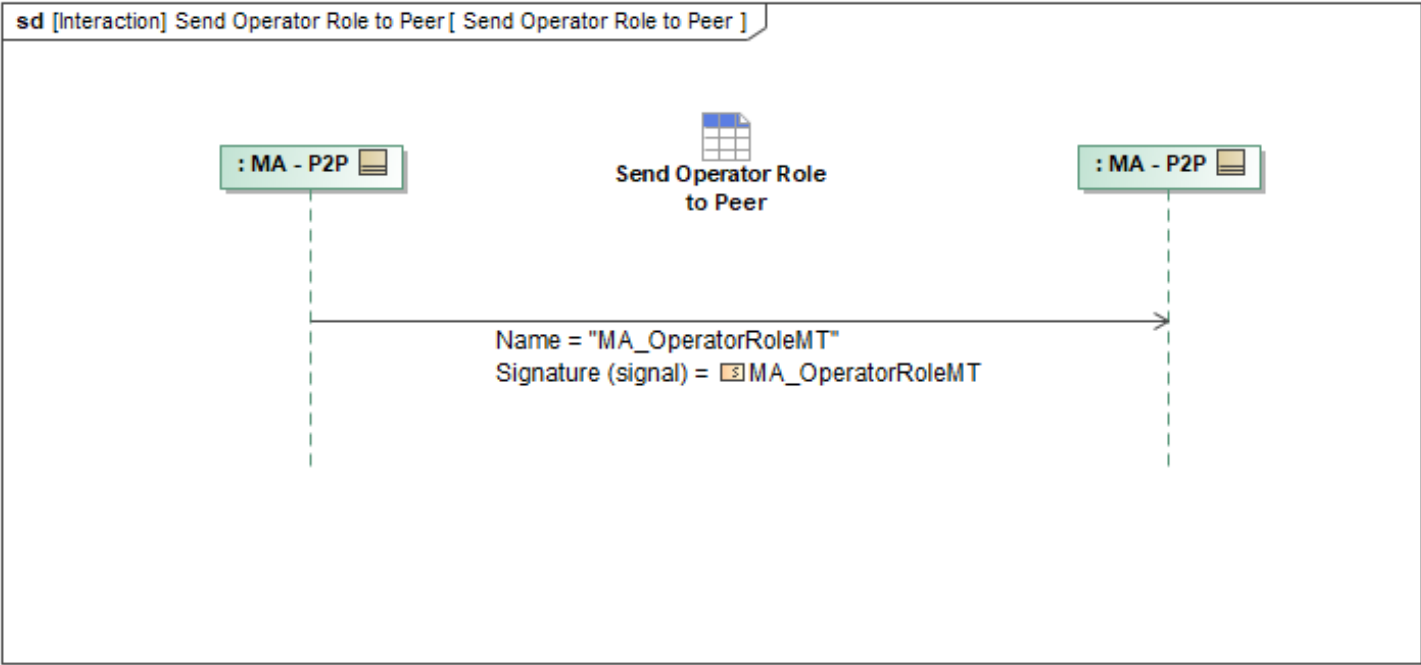


Figure 1-75: Send Operator Role to Peer

Table 1-74: Send Operator Role to Peer

| Message Name (Name:Type)             | Required Fields (Name:Type)  |
|--------------------------------------|------------------------------|
| MA_OperatorRoleMT: MA_OperatorRoleMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.16 Send Operator to Peer

This diagram contains the message, *OperatorMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

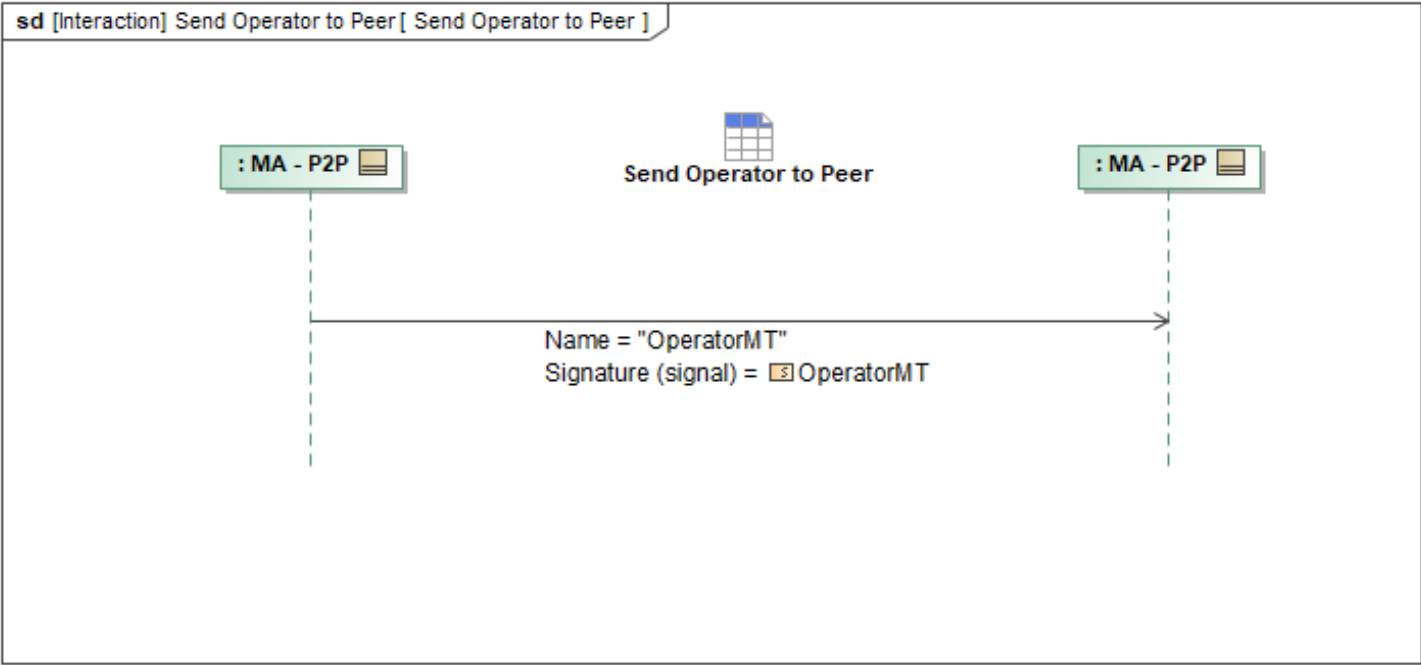


Figure 1-76: Send Operator to Peer

Table 1-75: Send Operator to Peer

| Message Name (Name:Type) | Required Fields (Name:Type)  |
|--------------------------|------------------------------|
| OperatorMT: OperatorMT   | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.17 Send OpLine to Peer

This diagram contains the message, *MA\_OpLineMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

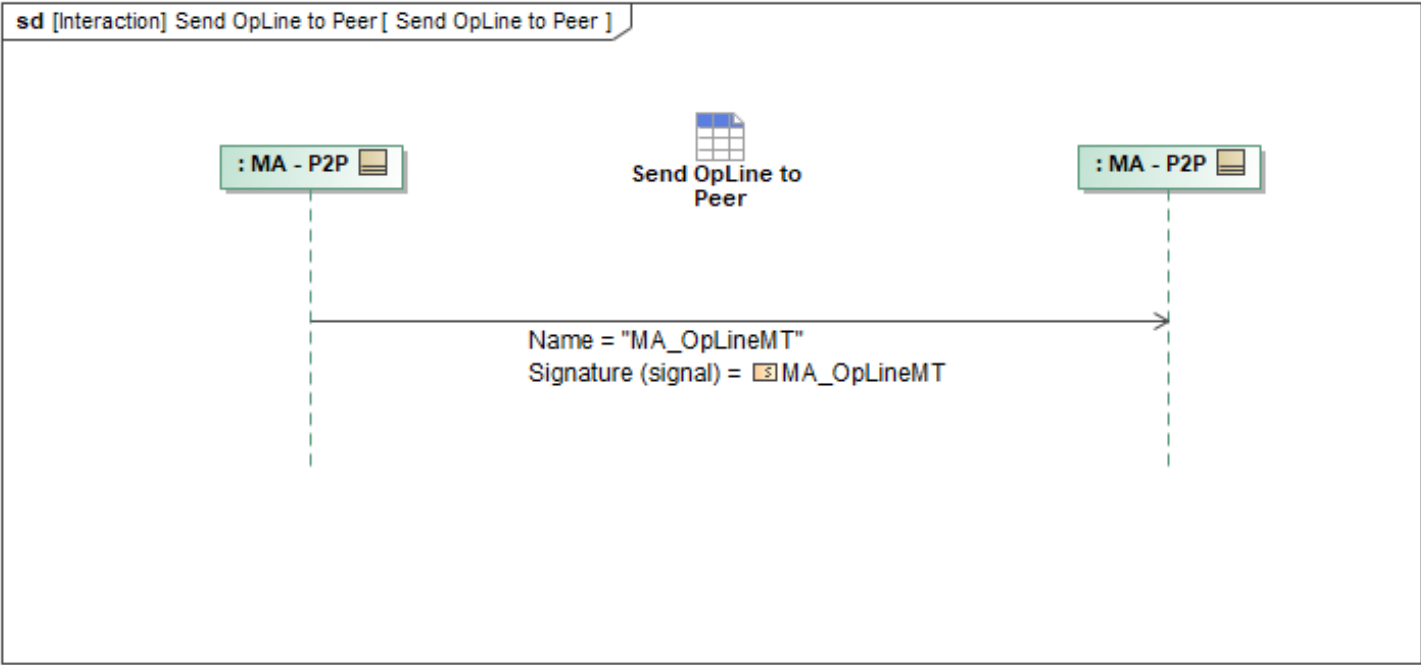


Figure 1-77: Send OpLine to Peer

Table 1-76: Send OpLine to Peer

| Message Name (Name:Type) | Required Fields (Name:Type)  |
|--------------------------|------------------------------|
| MA_OpLineMT: MA_OpLineMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.18 Send OpPlan to Peer

This diagram contains the message, *MA\_OpPlanMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

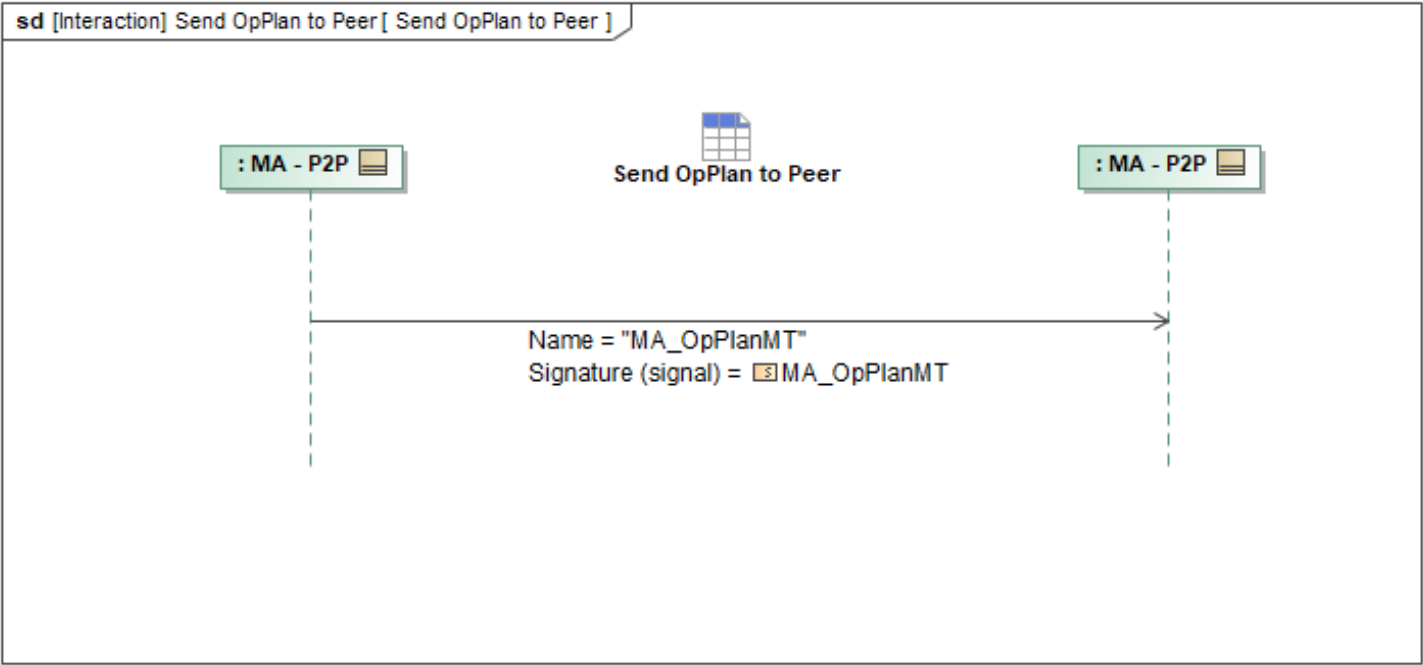


Figure 1-78: Send OpPlan to Peer

Table 1-77: Send OpPlan to Peer

| Message Name (Name:Type) | Required Fields (Name:Type)  |
|--------------------------|------------------------------|
| MA_OpPlanMT: MA_OpPlanMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.19 Send OpRouting to Peer

This diagram contains the message, *MA\_OpRoutingMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.



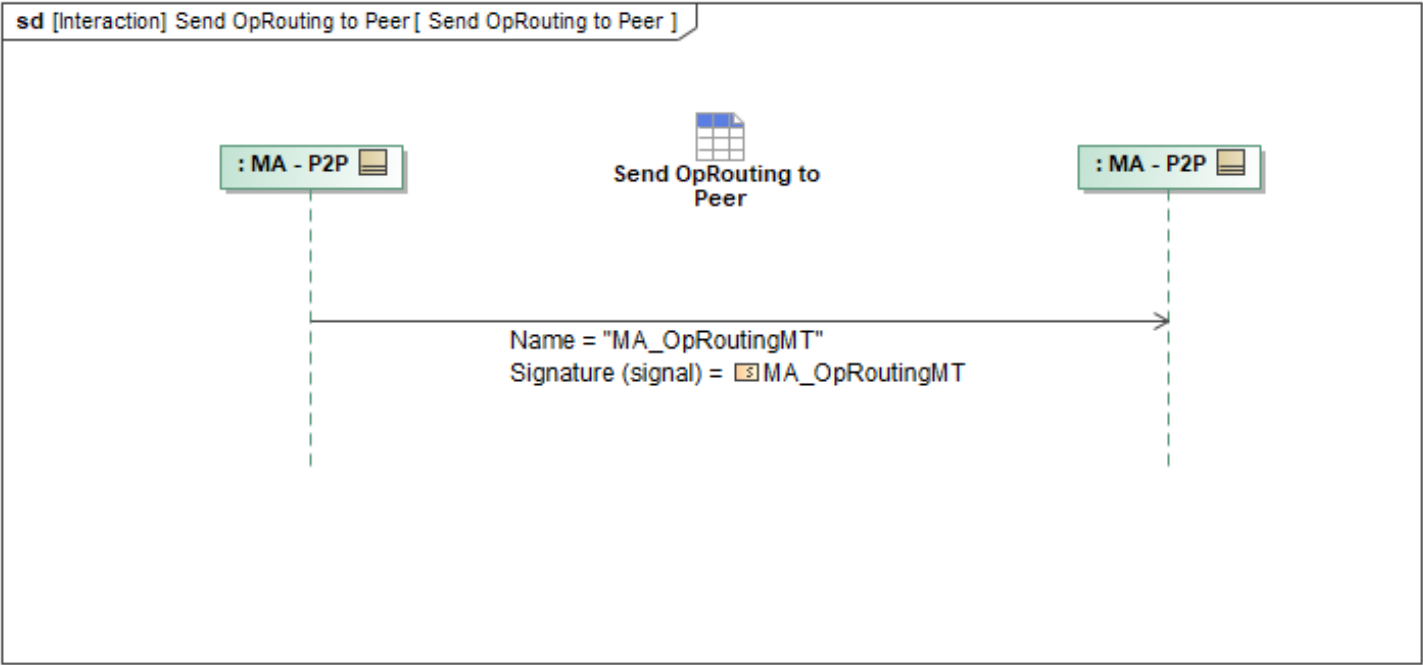


Figure 1-79: Send OpRouting to Peer

Table 1-78: Send OpRouting to Peer

| Message Name (Name:Type)       | Required Fields (Name:Type)  |
|--------------------------------|------------------------------|
| MA_OpRoutingMT: MA_OpRoutingMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.20 Send OpVolume to Peer

This diagram contains the message, *MA\_OpVolumeMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

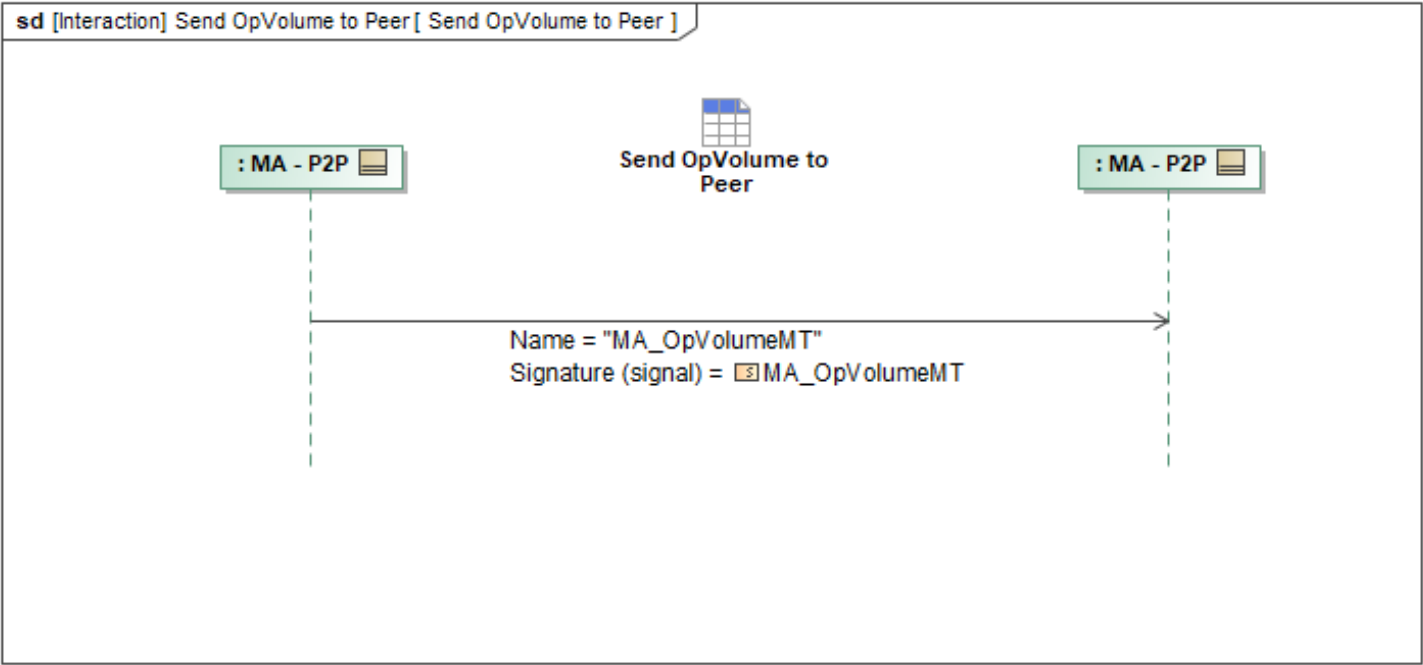


Figure 1-80: Send OpVolume to Peer

Table 1-79: Send OpVolume to Peer

| Message Name (Name:Type)     | Required Fields (Name:Type)  |
|------------------------------|------------------------------|
| MA_OpVolumeMT: MA_OpVolumeMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.21 Send OpZone to Peer

This diagram contains the message, *MA\_OpZoneMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

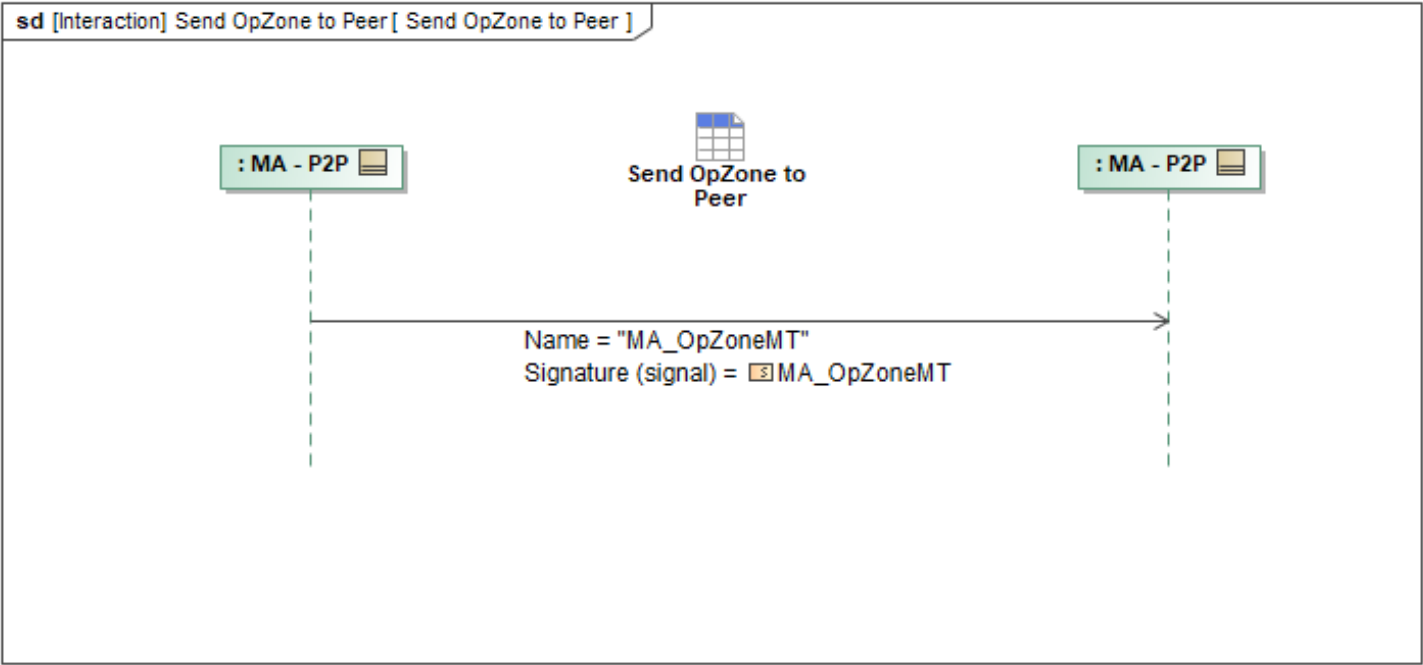


Figure 1-81: Send OpZone to Peer

Table 1-80: Send OpZone to Peer

| Message Name (Name:Type) | Required Fields (Name:Type) |
|--------------------------|-----------------------------|
| MA_OpZoneMT: MA_OpZoneMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.22 Send Position Report to Peer

This interaction specifies how Mission Autonomy (MA) shares *MA\_PositionReportMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the *MA\_PositionReportMT* message is sent. The *MA\_PositionReportMT* must have the *PositionSource* set to the *SystemID* of the vehicle. *Position* is set including the timestamp of the data. In addition, *Velocity* and *Orientation* must be set.

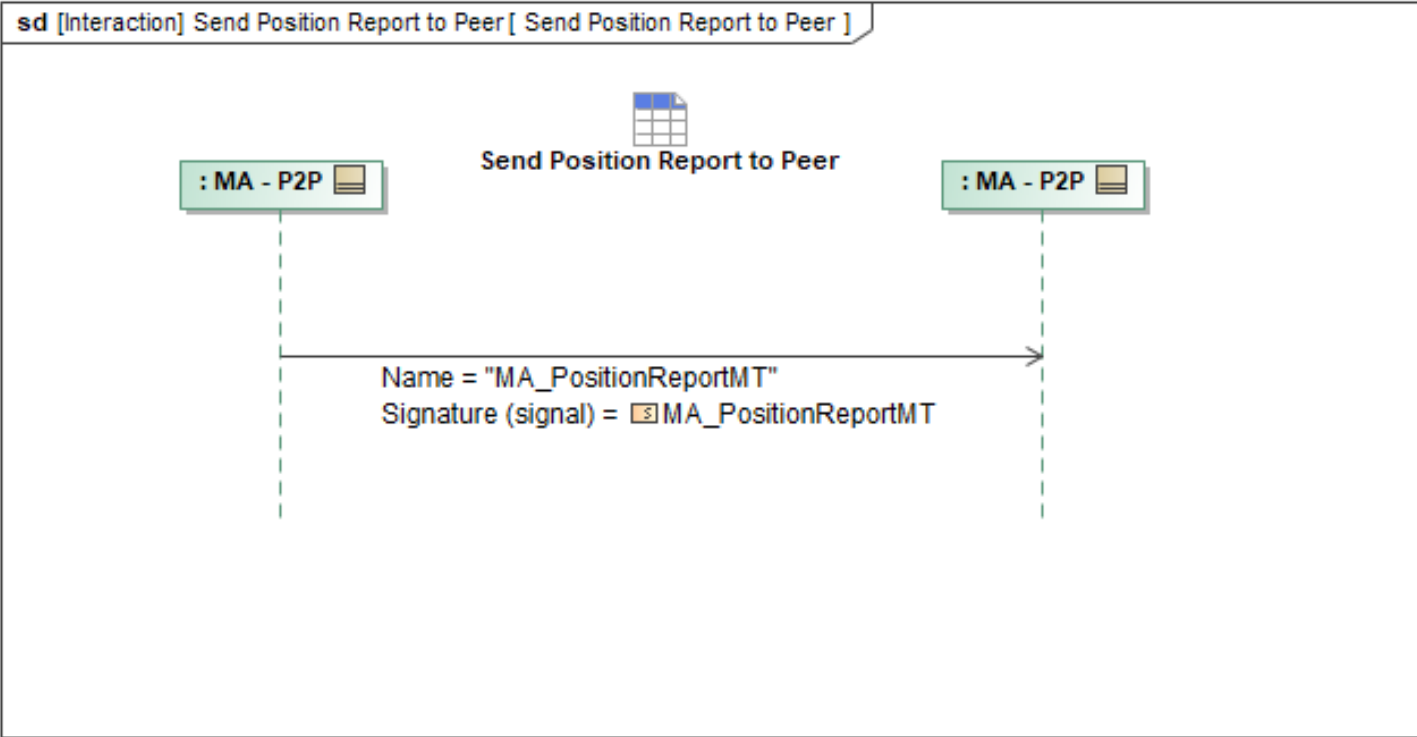


Figure 1-82: Send Position Report to Peer

Table 1-81: Send Position Report to Peer

| Message Name (Name:Type)                 | Required Fields (Name:Type)  |
|------------------------------------------|------------------------------|
| MA_PositionReportMT: MA_PositionReportMT | Orientation: OrientationType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.23 Send Response Plan to Peer

This diagram contains the message, *ResponsePlanMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

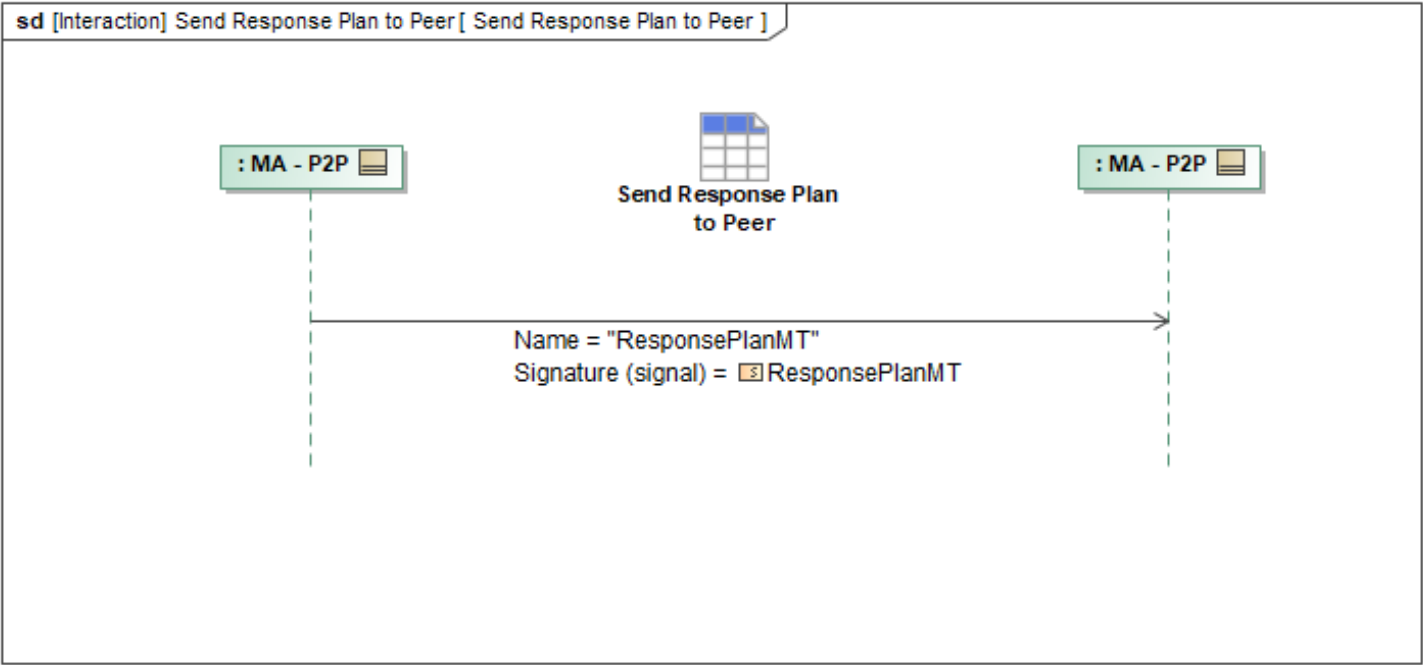


Figure 1-83: Send Response Plan to Peer

Table 1-82: Send Response Plan to Peer

| Message Name (Name:Type)       | Required Fields (Name:Type)  |
|--------------------------------|------------------------------|
| ResponsePlanMT: ResponsePlanMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.24 Send Response to Peer

This diagram contains the message, *MA\_ResponseMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

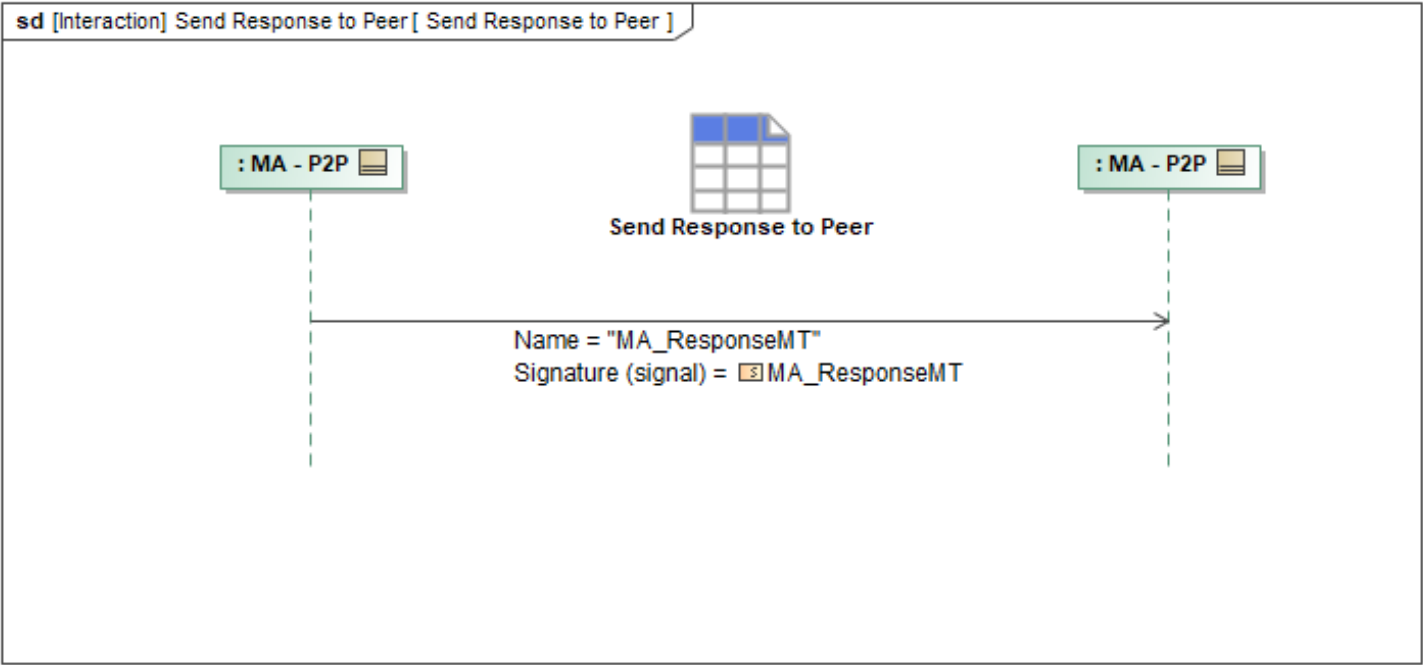


Figure 1-84: Send Response to Peer

Table 1-83: Send Response to Peer

| Message Name (Name:Type)     | Required Fields (Name:Type)  |
|------------------------------|------------------------------|
| MA_ResponseMT: MA_ResponseMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.25 Send Route Activity Plan to Peer

This diagram contains the message, *MA\_RouteActivityPlanMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

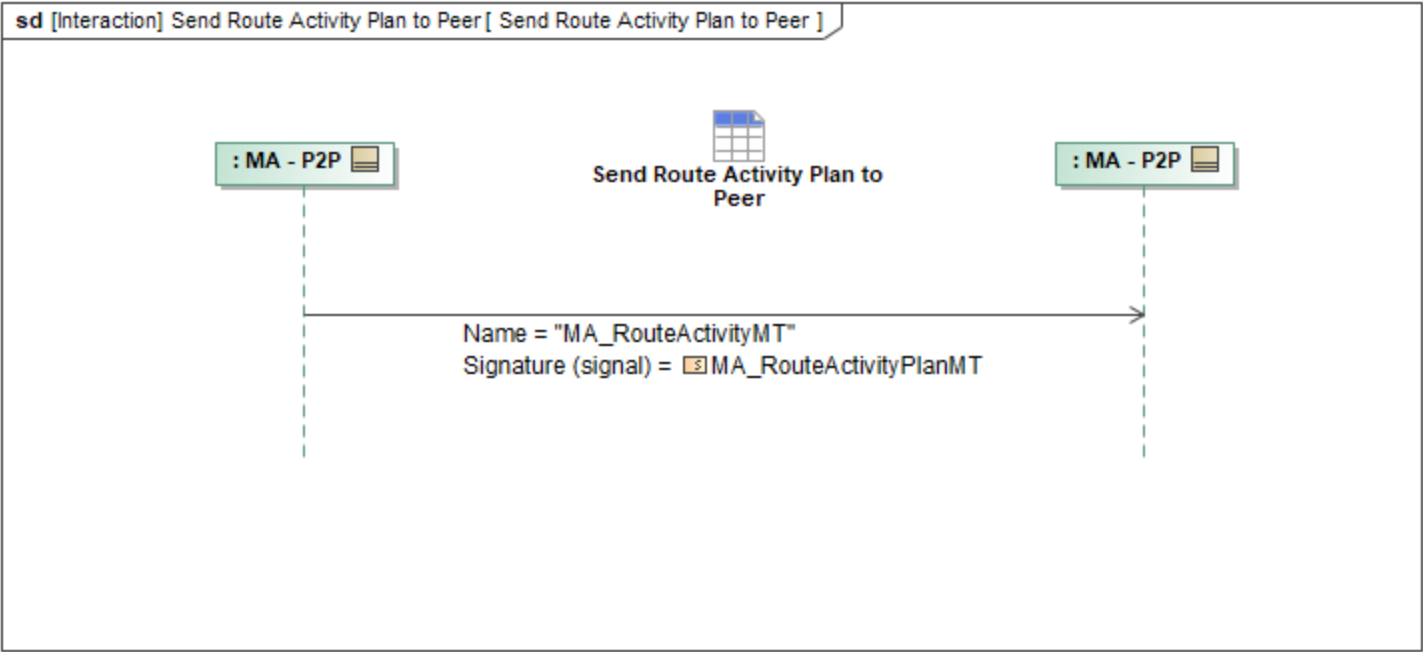


Figure 1-85: Send Route Activity Plan to Peer

Table 1-84: Send Route Activity Plan to Peer

| Message Name (Name:Type)                   | Required Fields (Name:Type) |
|--------------------------------------------|-----------------------------|
| MA_RouteActivityMT: MA_RouteActivityPlanMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.26 Send Route Deconfliction Alert to Peer

This interaction specifies how Mission Autonomy (MA) shares *MA\_RouteDeconflictionAlertMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

The *MA\_RouteDeconflictionAlertMT* is intended to alert other ACPs of a detected route conflict. This message contains a *ConflictPoint* field to represent inputs from the original request and data fields to represent specific *ConflictID* and *RoutePlanID* for potential deconfliction solutions.

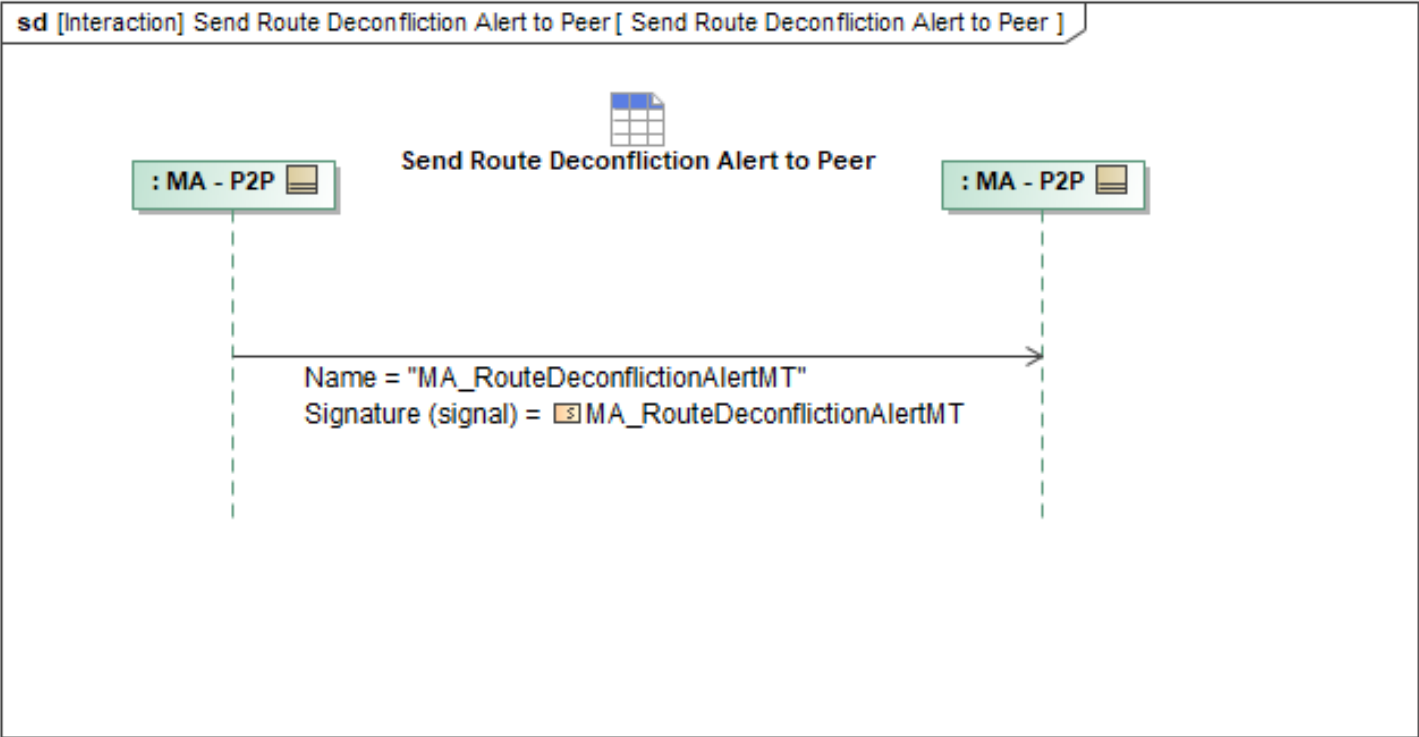


Figure 1-86: Send Route Deconfliction Alert to Peer

Table 1-85: Send Route Deconfliction Alert to Peer

| Message Name (Name:Type)                                      | Required Fields (Name:Type) |
|---------------------------------------------------------------|-----------------------------|
| MA_RouteDeconflictionAlertMT:<br>MA_RouteDeconflictionAlertMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.27 Send Route Plan Execution Status to Peer

This interaction specifies how Mission Autonomy (MA) shares *RoutePlanExecutionStatusMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the message *RoutePlanExecutionStatusMT* is sent. The *RoutePlanExecutionStatusMT* must have the *SystemID* of the vehicle and the *CurrentSegment* contain the *RoutePlanID*, *PathID*, and *SegmentID* of the current segment



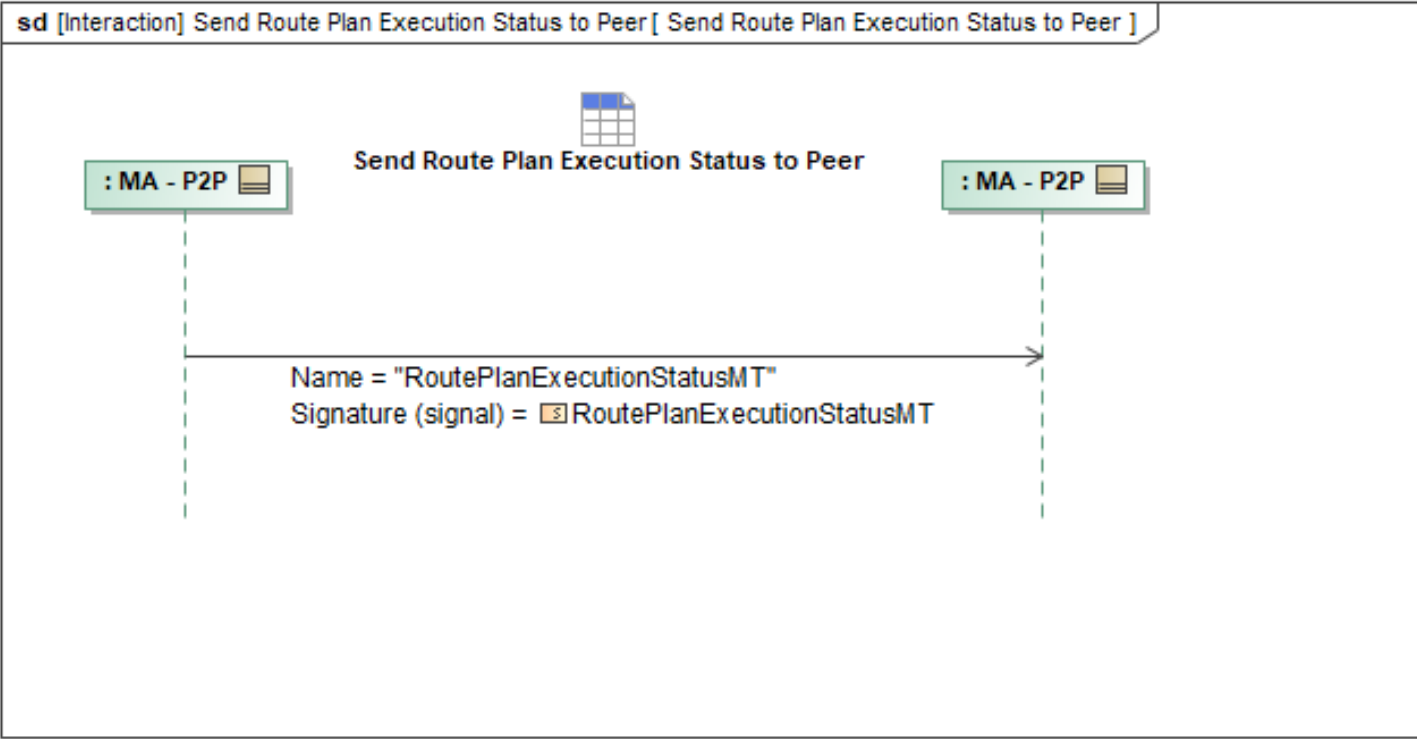


Figure 1-87: Send Route Plan Execution Status to Peer

Table 1-86: Send Route Plan Execution Status to Peer

| Message Name (Name:Type)                                  | Required Fields (Name:Type) |
|-----------------------------------------------------------|-----------------------------|
| RoutePlanExecutionStatusMT:<br>RoutePlanExecutionStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.28 Send Route Plan to Peer

This diagram contains the message, *MA\_RoutePlanMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

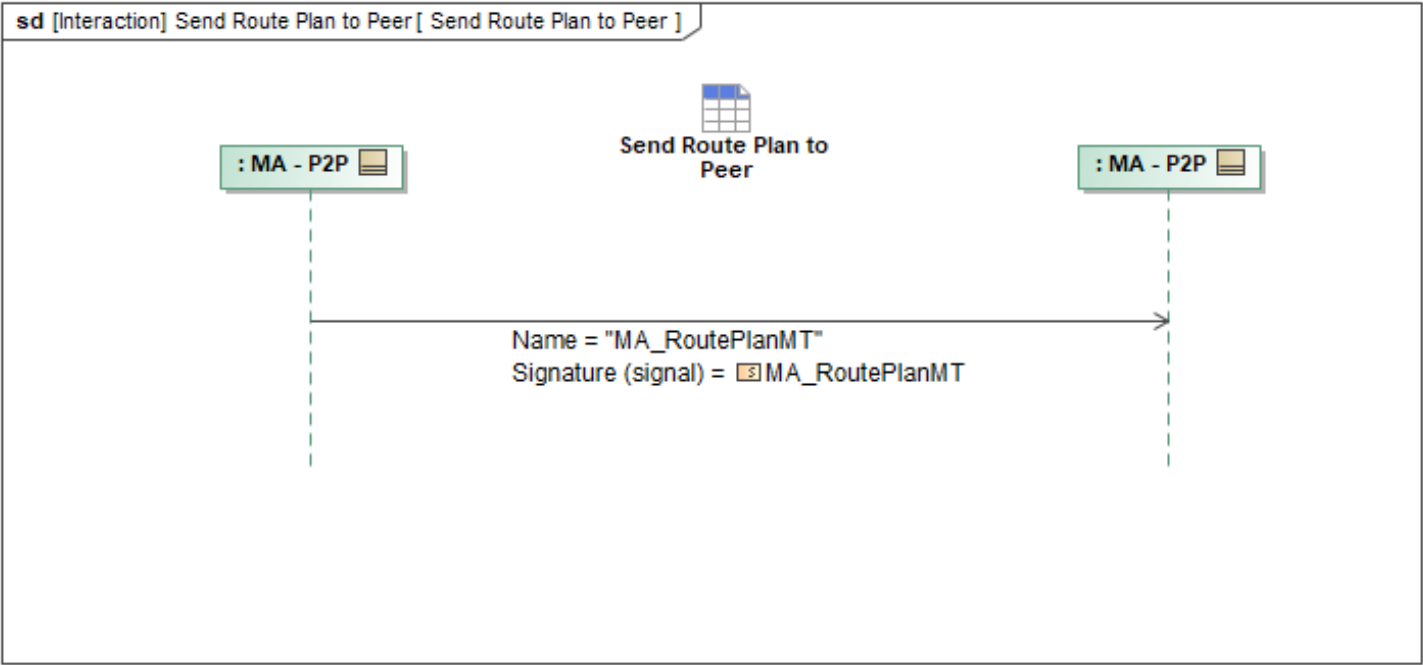


Figure 1-88: Send Route Plan to Peer

Table 1-87: Send Route Plan to Peer

| Message Name (Name:Type)       | Required Fields (Name:Type) |
|--------------------------------|-----------------------------|
| MA_RoutePlanMT: MA_RoutePlanMT | MA_RoutePlanMT:             |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.29 Send Subsystem Status to Peer

This interaction specifies how Mission Autonomy (MA) shares *SubsystemStatusMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

The *SubsystemStatusMT* must have the *SubsystemID* of the comms subsystem and *SubsystemState* set to the state of the comms subsystem and its links.

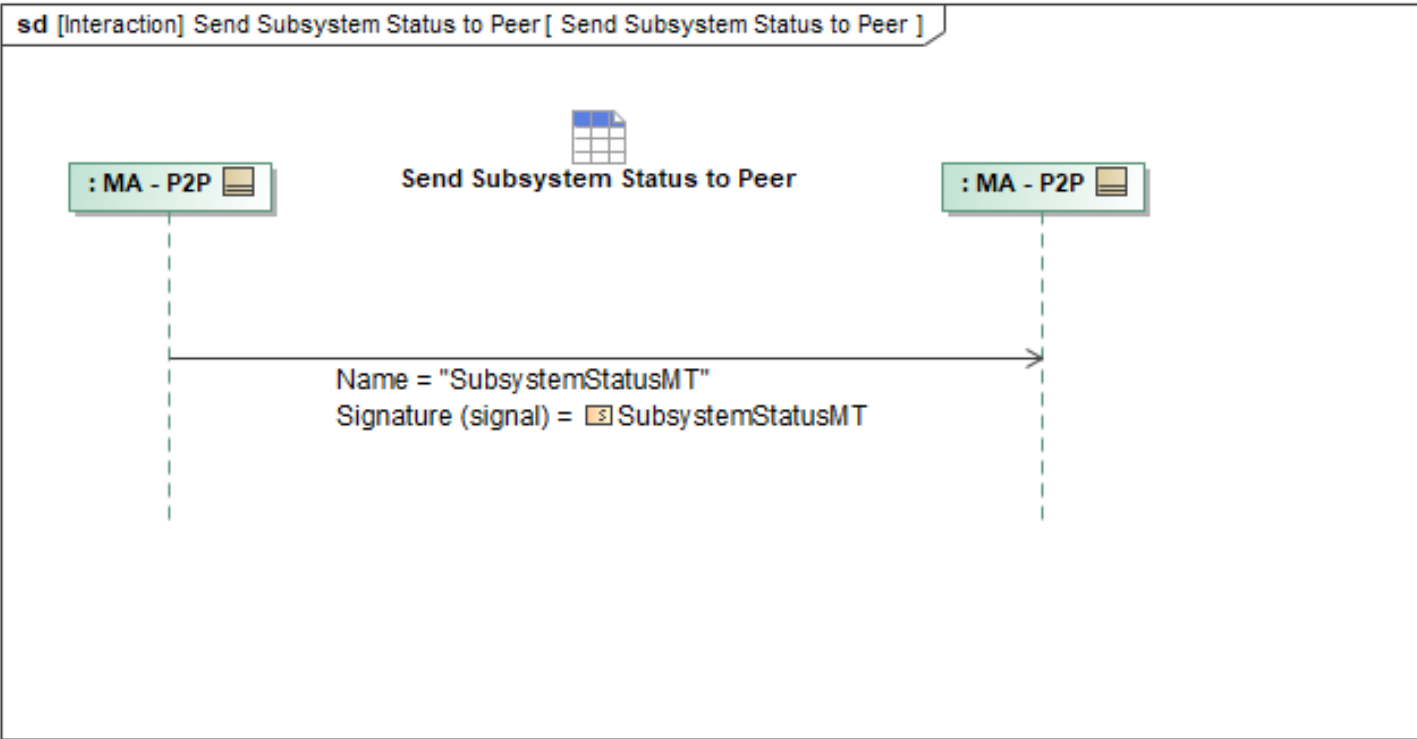


Figure 1-89: Send Subsystem Status to Peer

Table 1-88: Send Subsystem Status to Peer

| Message Name (Name:Type)             | Required Fields (Name:Type) |
|--------------------------------------|-----------------------------|
| SubsystemStatusMT: SubsystemStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.30 Send System Status to Peer

This interaction specifies how Mission Autonomy (MA) shares *SystemStatusMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the message *SystemStatusMT* is sent. The *SystemStatusMT* must have the *SystemID* of the vehicle and the *SystemState* with the overall state of the system. The *Source* must be set to the data source and the *MissionCommunicationsState* must be set to the current state of the comms. For the purposes of vehicle identification, the field *Identity.SpecificVehicle.VehicleUniqueIdentifier.TailNumber* shall be populated. The *TailNumber* and other system identification information is sourced from pre-mission planning data. In addition, the message should list the IDs of the *Subsystems*, *Capabilities*, and *Services* in the system.

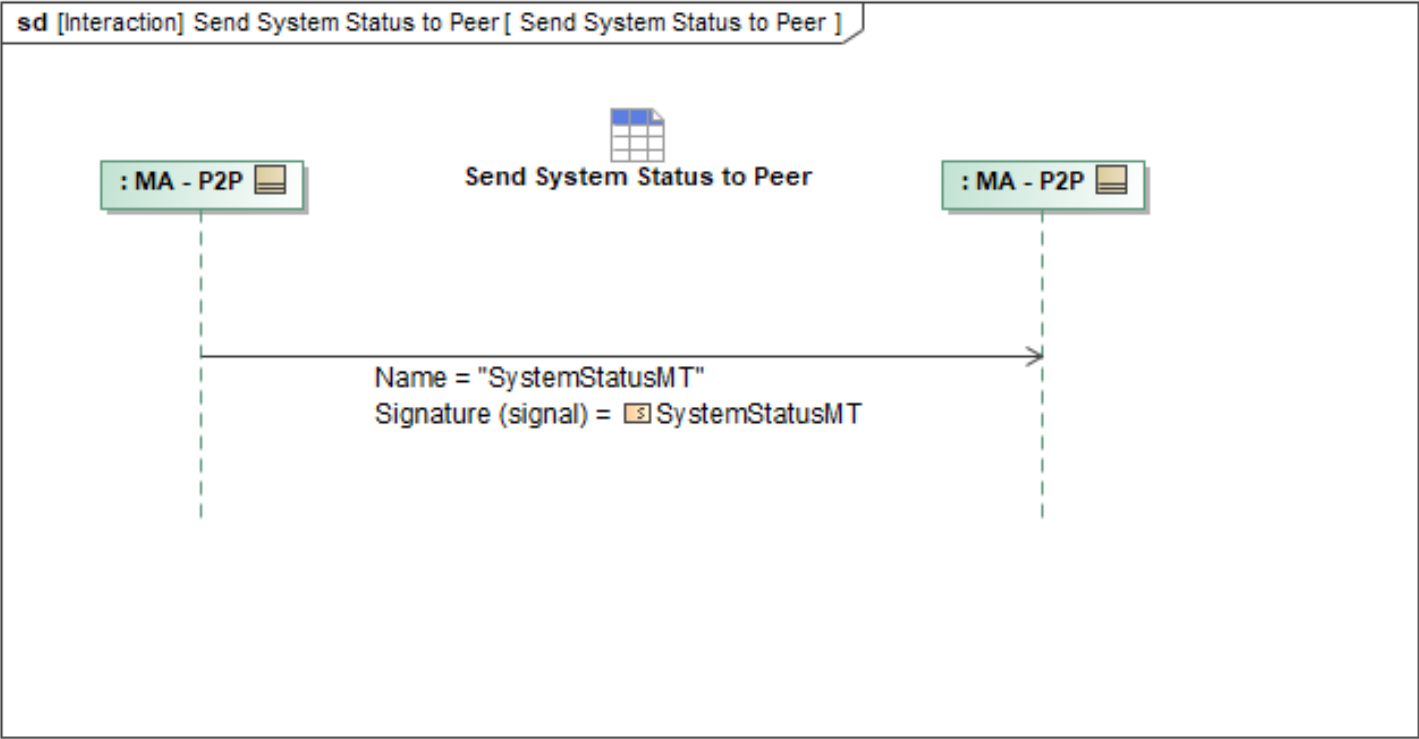


Figure 1-90: Send System Status to Peer

Table 1-89: Send System Status to Peer

| Message Name (Name:Type)       | Required Fields (Name:Type)                                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| SystemStatusMT: SystemStatusMT | TailNumber: TailNumberType<br>SubsystemID: SubsystemID_Type<br>CapabilityID: CapabilityID_Type<br>ServiceID: ServiceID_Type |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.31 Send Task Plan to Peer

This diagram contains the message, *MA\_TaskPlanMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

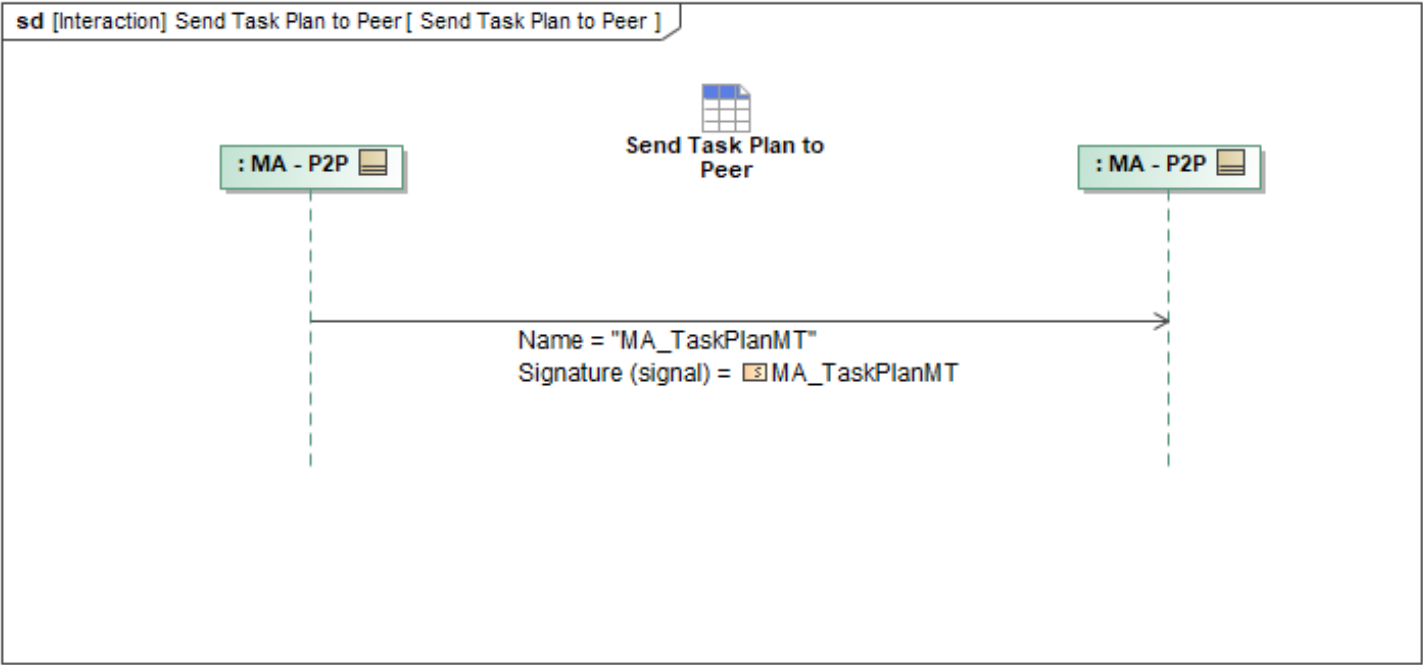


Figure 1-91: Send Task Plan to Peer

Table 1-90: Send Task Plan to Peer

| Message Name (Name:Type)     | Required Fields (Name:Type)  |
|------------------------------|------------------------------|
| MA_TaskPlanMT: MA_TaskPlanMT | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.32 Send Task Status to Peer

This interaction specifies how Mission Autonomy (MA) shares *MA\_TaskStatusMT* through the MA-P2P interface. The transmission of this message allows ACPs to understand the operating state of peers in the airspace. In the context of a centralized Common Operating Picture (COP) distribution setup, the vehicle state messages would be aggregated by the COP lead for distribution outside of the package.

At predefined frequencies, the message *TaskStatusMT* is sent. The *MA\_TaskStatusMT* must have the *ExecutingSystemID* set to the *SystemID* of the system. The *ExecutionState* will be *EXECUTING* if MA is executing the task, *COMPLETED* when the target is acquired, and *FAILED* if the execution fails. If the *ExecutionState* is *FAILED*, the *ExecutionStateReason* is required. The *TaskID* contains the ID of the task being executed.

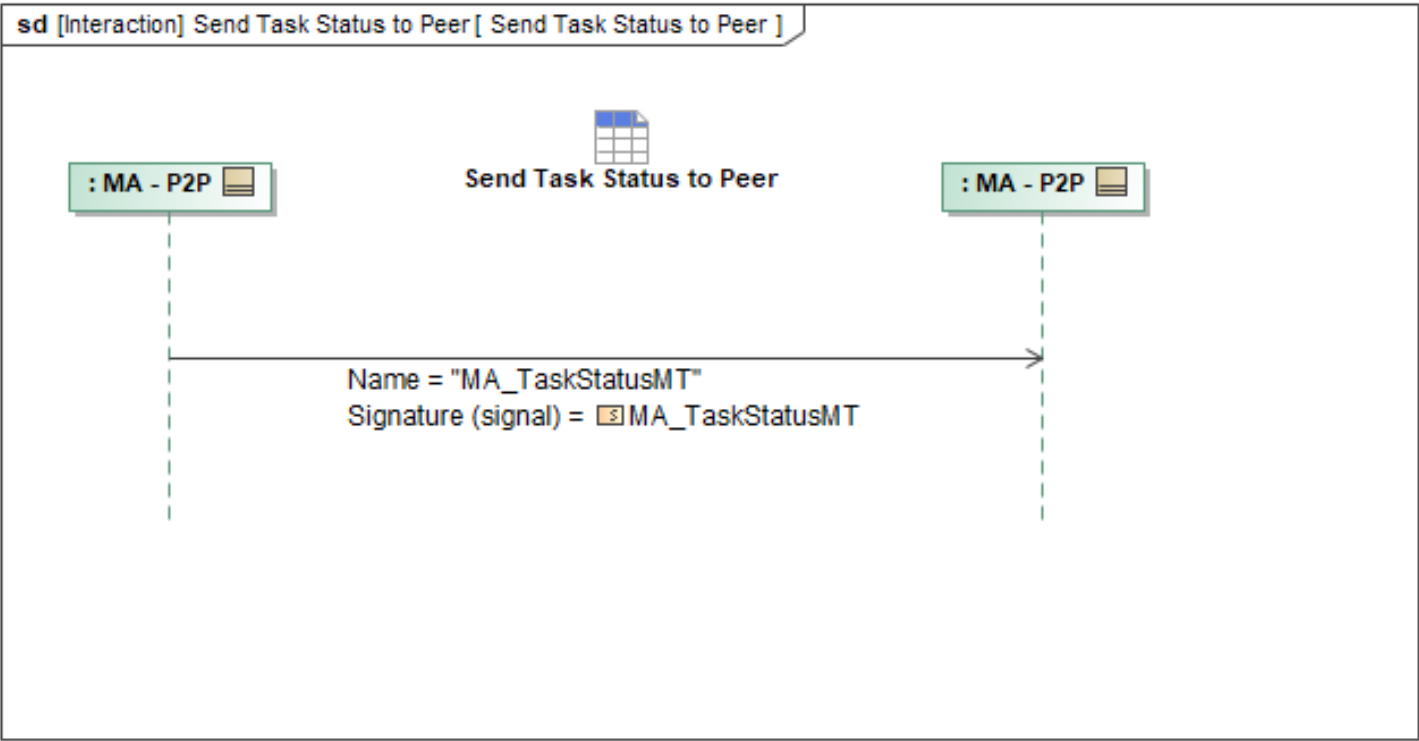


Figure 1-92: Send Task Status to Peer

Table 1-91: Send Task Status to Peer

| Message Name (Name:Type)         | Required Fields (Name:Type) |
|----------------------------------|-----------------------------|
| MA_TaskStatusMT: MA_TaskStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.10.11.33 Send Task to Peer

This diagram contains the message, *MA\_TaskMT*, exchanged between Peers. This interaction is not limited to any specific package roles. Nominally, the initiating peer may be the Leader where the leader is relaying plan data received from C2 to a Follower.

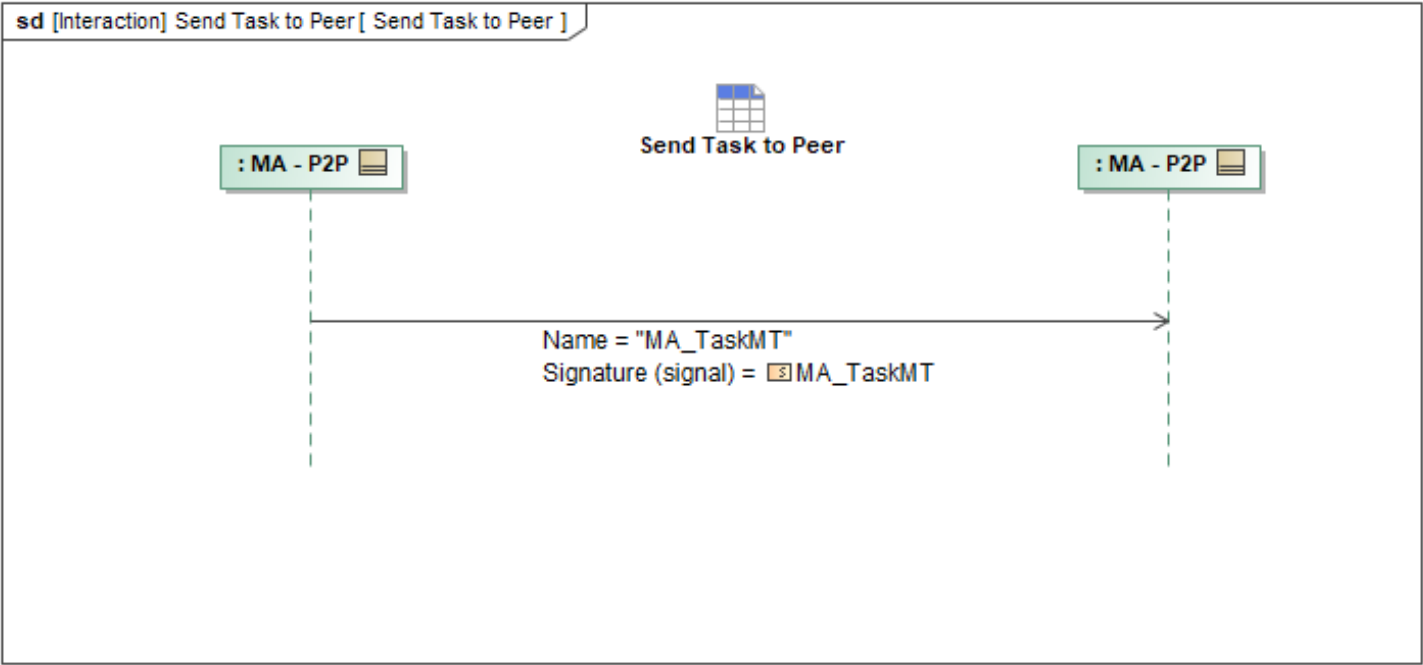


Figure 1-93: Send Task to Peer

Table 1-92: Send Task to Peer

| Message Name (Name:Type) | Required Fields (Name:Type)  |
|--------------------------|------------------------------|
| MA_TaskMT: MA_TaskMT     | ObjectState: ObjectStateEnum |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11 Peer Package Management and Leader Election Package

This section specifies the set of interactions for Peer Package Management and Leader Election. Leader Election includes multiple possible election methods (as specified by separate interactions) that can be implemented at the discretion of the implementer. All the specified election methods in this section are considered optional and which method an implementer decides to use can verify based on tradeoffs, including communication complexity, support for static vs. dynamic fitness scores and off-nominal considerations.

The ICD includes the following methods for Leader Election:

- 1. Bully Leader Election
- 2. Maximum Consensus Leader Election
- 3. Raft Leader Election
- 4. Static Fitness Score Leader Election
- 5. Off Nominal Leader Election

The implementer can decide whether to utilize a Static or Dynamic Leadership Fitness Score for electing a leader. Note: the Raft Leader Election algorithm does not require a leadership fitness score (whether static or dynamic).

Static Leadership Fitness Scores are intended to be uploaded as part of pre-Mission Planning or specified by C2. These scores are not subject to change during the Mission (unless C2 decides to upload a different Static Leadership Fitness Score for one or multiple ACPs).

Dynamic Leadership Fitness Scores are intended to change throughout the mission based on changing conditions (e.g. Comms Health). A Dynamic Leadership Fitness Score is associated with a determination method, which may be selected based on the enumerated list specified within that message field. This enumerated list also includes a "user-defined" value where an implementation may choose to determine a Dynamic Leadership Fitness Score based on a method at the discretion of the implementation. Utilizing a Dynamic Leadership Fitness is optional (for any of the election methods) for the implementation, as well as the specified method for determining how the fitness score is dynamically computed.

**Table 1-93: Leader Election Method Static/Dynamic Support**

| Method                               | Supports                        |                                  |
|--------------------------------------|---------------------------------|----------------------------------|
|                                      | Static Leadership Fitness Score | Dynamic Leadership Fitness Score |
| Bully Leader Election                | Yes                             | Yes                              |
| Maximum Consensus Leader Election    | Yes                             | Yes                              |
| Raft Leader Election                 | N/A                             | N/A                              |
| Static Fitness Score Leader Election | Yes                             | No                               |
| Off Nominal Leader Election          | Yes                             | Yes                              |

### 1.2.11.1 Bully Leader Election

#### Mission Use Case: Bully Leader Election

This leader election method is optional and implementers are expected to implement at least one of the leader election methods.

This outlines the peer interactions for the Bully Leader Election process, including the nominal case as well as some off-nominal cases. The diagram is explained, followed by a description of the general algorithm from a single ACP's perspective.

#### Diagram Description

Assumptions:

- The package has been initialized (See "Initializing a Package Amongst Peers" interaction).
- Leadership fitness scores are  $ACP\ 1 < ACP\ 2 < ACP\ 3$ . Since ACP 3 has the highest fitness score, it is the intended leader.
- ACP 1 is the initiator of the election. Note that any ACP can be the initiator, and multiple initiators would not be problematic. Once an ACP determines the package has been initialized, it will initiate the election if no other ACP has done so yet.

ACP 1 initiates the election by sending *MA\_LeaderUpdateRequestMT\_1* to ACP 2 and *MA\_LeaderUpdateRequestMT\_2* to ACP 3. The *Request* field has a blank *ProposedLeaderID* and the *Status* field is set to *UNSPECIFIED*. ACP 1 populates its own static fitness score in the *StaticLeadershipFitnessScore* field. ACP 1 may also populate its own dynamic fitness score in the *DynamicLeadershipFitnessScore* field (if applicable, based on the selected



*PackageLeadershipFitnessScoreDeterminationMethod*).

Next, a consensus block is triggered in which ACPs reach consensus on their fitness scores. ACP 3 sends *MA\_LeaderUpdateRequestMT\_3 -4* to share its fitness score with the other two ACPs by populating the *StaticLeadershipFitnessScore* field and *DynamicLeadershipFitnessScore* (if applicable). Similarly, ACP 2 sends *MA\_LeaderUpdateRequestMT\_5 -6* with the *StaticLeadershipFitnessScore* and *DynamicLeadershipFitnessScore* (if applicable) fields populated with its score(s). These messages have a blank *ProposedLeaderID* in the *Request* field, and the *Status* field set to *UNSPECIFIED*.

Once ACP 1 and ACP 2 have received all fitness scores, they each begin an election timer. In the nominal case, ACP 3 declares its leadership after receiving all fitness scores (before an election timeout is reached), and will proceed to declare leadership via "Peer Declares Package Leadership to Peers."

If ACP 3 hasn't declared leadership before one of the election timers reaches timeout, ACP 1 or 2 (whichever reaches timeout first) should send ACP 3 *MA\_LeaderUpdateRequestMT* with the *Status* field set to *ASSIGN* and ACP 3's System ID populated in *ProposedLeaderID* of the *Request* field. These messages are shown as *MA\_LeaderUpdateRequestMT\_7* and *MA\_LeaderUpdateRequestMT\_8* in the diagram. After ACP 3 receives this leader assignment message, it will declare leadership via "Peer Declares Package Leadership to Peers."

Other off-nominal cases of leader election are illustrated in the referenced sub-diagram.

### Algorithm Description (from an ACP's perspective)

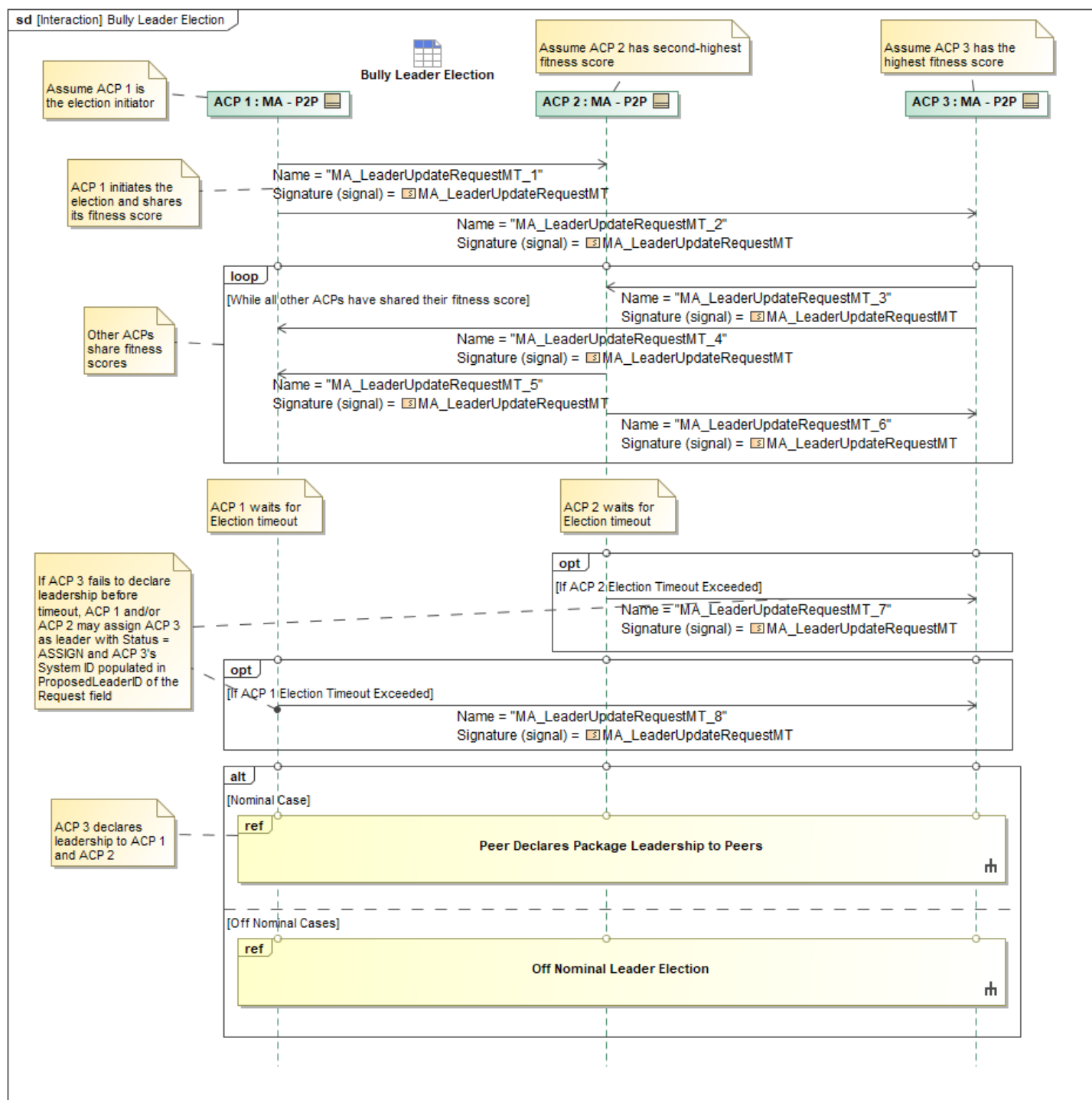
First, I undergo the process of reaching consensus on leadership fitness scores and rank ACPs from highest to lowest scores. Ties are broken by highest tail number, which is found via the heartbeat system status messages in the Peer Heartbeat and Send Vehicle State to Peer interactions. If I receive *MA\_LeaderUpdateRequestMT* assigning me as leader before I receive all fitness scores, another ACP has determined I have highest fitness score, so I will declare myself leader. If I determine that I have the highest fitness score among all peer ACPs, I will declare myself leader.

If I determine I do not have the highest fitness score, I wait until the election times out and should send *MA\_LeaderUpdateRequestMT* assigning leadership to the highest scoring ACP. I start a new election timer, and if the highest-scoring ACP does not declare itself leader before the timeout, I should send *MA\_LeaderUpdateRequestMT* to the next highest-scoring ACP with *Status* set to *NOMINATE*. I wait to receive a response of *MA\_LeaderUpdateRequestStatusMT* with *RequestProcessingState* of *PROCESSING* from the ACP, indicating the nomination was received and pending (See "Off Nominal Leader Election" sequence). If this election also times out, I should repeat the process of sending *MA\_LeaderUpdateRequestMT* to the next highest-scoring ACP in the ranked list of fitness scores until receiving *MA\_LeaderUpdateRequestStatusMT* with *RequestProcessingState* of *PROCESSING* from an ACP.

During this process:

- If a leader is declared, terminate the current election as it is no longer needed.
- If I determine myself to be the next highest-scoring ACP after an election reaches timeout, send *MA\_LeaderUpdateRequestMT* to everyone nominating myself as leader. After a majority of ACPs accept, declare my leadership and terminate the election. If rejected, wait for a leader to be declared.

- If an ACP nominates itself as leader, accept their leadership request if they have a higher fitness score than myself. Otherwise, reject the request.
- If I receive a message from another ACP nominating me for leadership, respond with *MA\_LeaderUpdateRequestStatusMT* with *RequestProcessingState* of *PROCESSING*. Continue following the election algorithm.



**Figure 1-94: Bully Leader Election**

**Table 1-94: Bully Leader Election**

| Message Name (Name:Type)                                | Required Fields (Name:Type)                                           |
|---------------------------------------------------------|-----------------------------------------------------------------------|
| MA_LeaderUpdateRequestMT_1:<br>MA_LeaderUpdateRequestMT | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |

| Message Name (Name:Type)                                              | Required Fields (Name:Type)                                           |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>MA_LeaderUpdateRequestMT_2:</b><br><b>MA_LeaderUpdateRequestMT</b> | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| <b>MA_LeaderUpdateRequestMT_3:</b><br><b>MA_LeaderUpdateRequestMT</b> | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| <b>MA_LeaderUpdateRequestMT_4:</b><br><b>MA_LeaderUpdateRequestMT</b> | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| <b>MA_LeaderUpdateRequestMT_5:</b><br><b>MA_LeaderUpdateRequestMT</b> | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| <b>MA_LeaderUpdateRequestMT_6:</b><br><b>MA_LeaderUpdateRequestMT</b> | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| <b>MA_LeaderUpdateRequestMT_7:</b><br><b>MA_LeaderUpdateRequestMT</b> | ProposedLeaderID: SystemID_Type                                       |
| <b>MA_LeaderUpdateRequestMT_8:</b><br><b>MA_LeaderUpdateRequestMT</b> | ProposedLeaderID: SystemID_Type                                       |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.2 Create a Full Package Amongst Peers

This is the peer interaction which results in a fully-formed package. First, the ACPs follow the "Initializing a Package Amongst Peers" interaction, which is triggered by an Initiator ACP sending *MA\_PackageManagementCommandMT* to all intended package member ACPs. After confirming status of all ACPs in the initialization process, the leader must be determined and declared. If the *PackageLeaderID* field of the *MA\_PackageManagementCommandMT* message was populated, the identified Lead ACP declares its leadership to all other package members via "Peer Declares Package Leadership to Peers". Otherwise, the ACPs perform leader election according to the method specified in the *PackageLeaderElectionMethod* field of *MA\_PackageManagementCommandMT*. Election methods are currently defined for "Static Fitness Score Leader Election," "Maximum Consensus Leader Election," "Raft Leader Election," and "Dynamic Fitness Score Leader Election." These leader election methods are optional, and implementors are expected to implement at least one of the leader election methods. If the election method is unspecified, the implementor will specify a default election method for leader election.

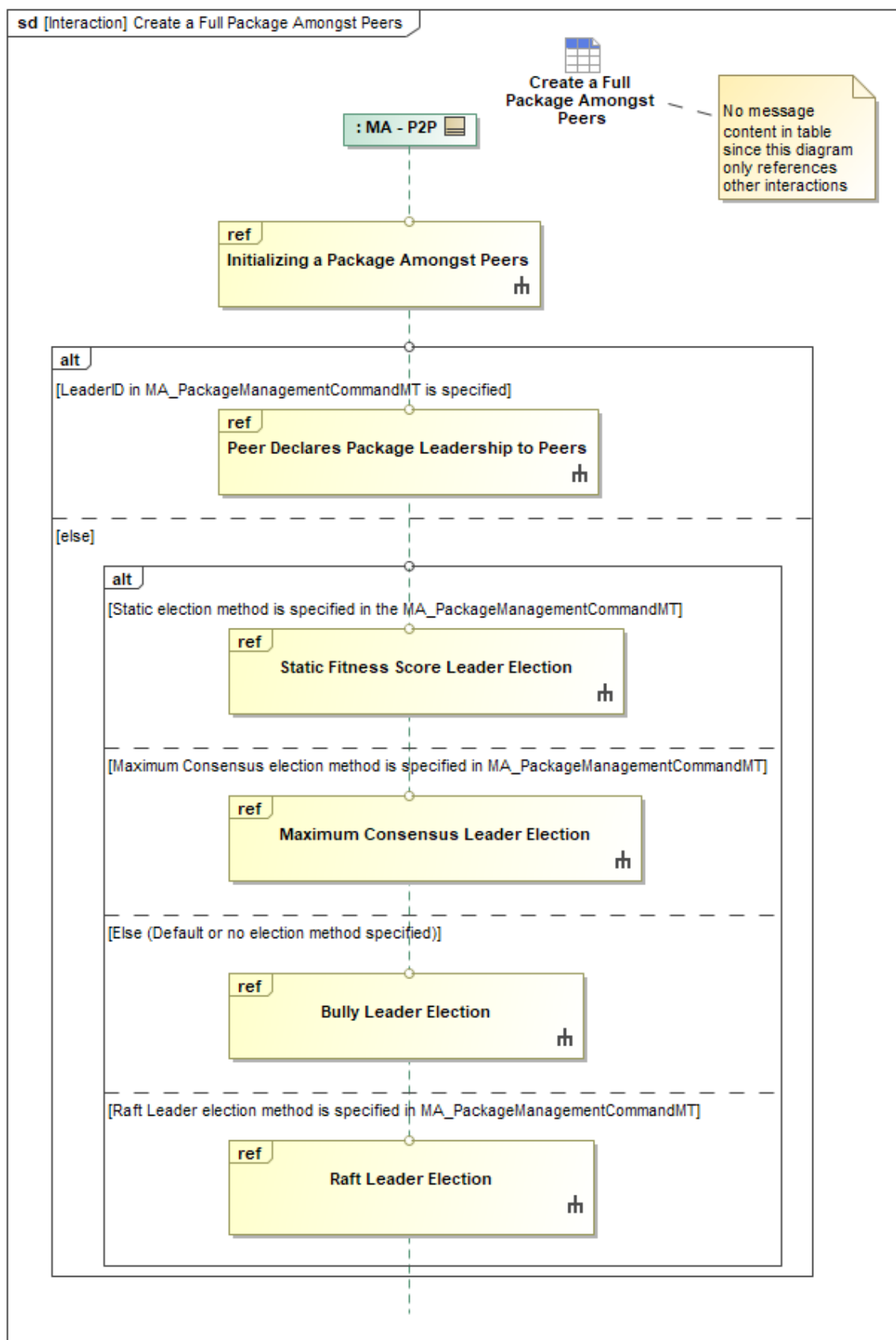


Figure 1-95: Create a Full Package Amongst Peers

### 1.2.11.3 Current Package Lead Assigns New Package Lead

This is the peer interaction where the Current Leader ACP assigns a New Leader ACP. This interaction occurs in the use case "C2 Specifies Team Lead on Existing Team" after the Current Leader receives a command from C2 to update the package leader. First, the Current Leader sends *MA\_LeaderUpdateRequestMT* to the New Leader with the *Status* field set to *ASSIGN*, and the *ProposedLeaderID* within the *Request* field populated with the New Leader's ID. In response, the New Leader sends *MA\_LeaderUpdateRequestStatusMT* with *RequestProcessingState* of *COMPLETED*. The New Leader then declares its leadership status to all followers via "Peer Declares Package Leadership to Peers."

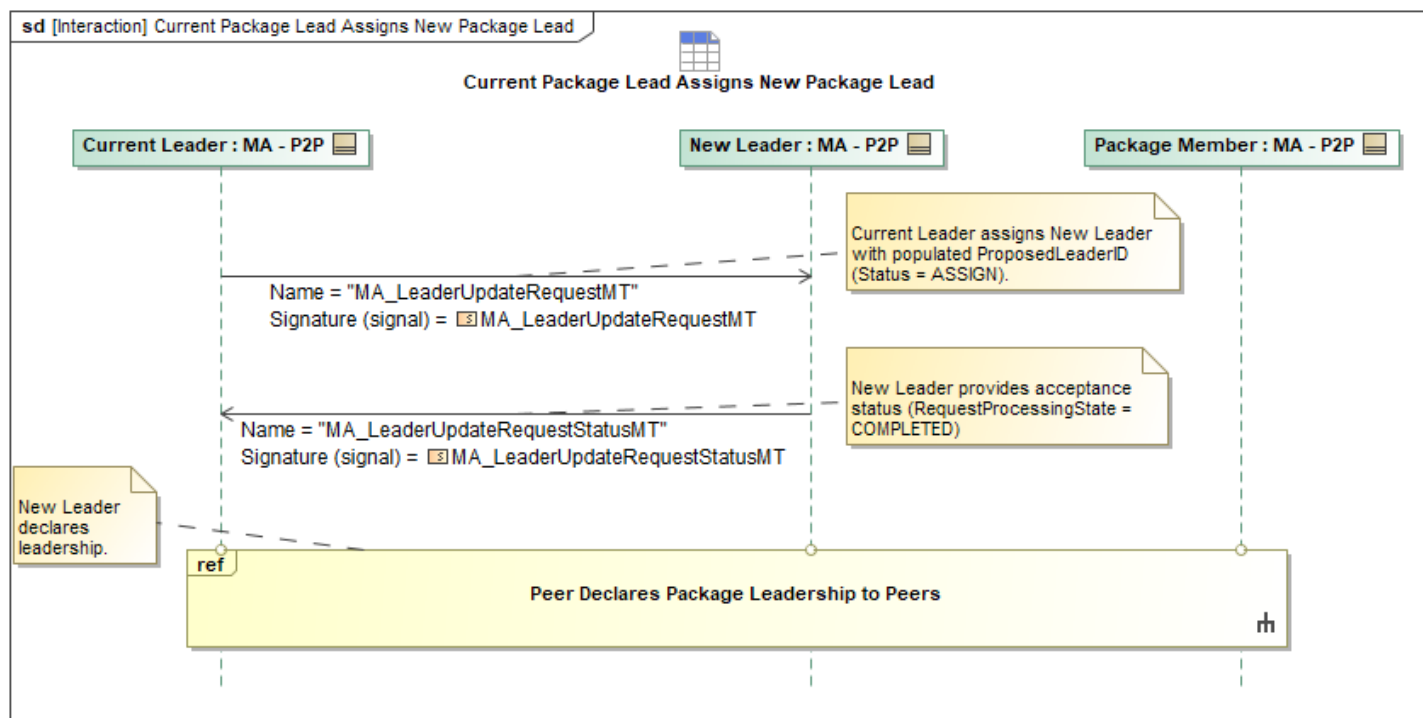


Figure 1-96: Current Package Lead Assigns New Package Lead

Table 1-95: Current Package Lead Assigns New Package Lead

| Message Name (Name:Type)                                                 | Required Fields (Name:Type)     |
|--------------------------------------------------------------------------|---------------------------------|
| <b>MA_LeaderUpdateRequestMT:</b><br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| <b>MA_LeaderUpdateRequestStatusMT:</b><br>MA_LeaderUpdateRequestStatusMT | None                            |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.4 Initializing a Package Amongst Peers

This is the peer interaction which results in an "initialized" package, where the status of all intended package members is confirmed before leader election occurs. Note that the package will not be "fully formed" yet, since a leader has not been determined. Package initialization takes place after ACPs have received a command to form a package. In this diagram, ACP 1 is the first ACP to receive the

package formation command, so it acts as the initiator. Note that if multiple ACPs attempt to initiate, it doesn't pose a problem for the interaction.

If the leader election method requires fitness scores, ACP 1 sends

*MA\_PackageManagementCommandMT\_1* to ACP 2 and *MA\_PackageManagementCommandMT\_2* to ACP 3, as the intended package members specified in the original package formation command. If the leader election method does not require fitness scores, ACP 1 sends *MA\_PackageManagementCommandMT\_3* to ACP 2 and *MA\_PackageManagementCommandMT\_4* to ACP 3, as the intended package members specified in the original package formation. This message contains all information in the original command, and the *ChangeType* field is set to *FORM\_NEW\_PACKAGE*.

ACPs 2 and 3 respond with *MA\_PackageManagementCommandStatusMT\_1* and *MA\_PackageManagementCommandStatusMT\_2*, respectively.

If the leader election method requires static fitness scores, after ACP 1 receives *MA\_PackageManagementCommandStatusMT* messages with *CommandProcessingState* of *ACCEPTED* from both ACPs, it sends *MA\_PackagePartnerStatusReportMT\_1* to ACP 2 and *MA\_PackagePartnerStatusReportMT\_2* to ACP 3, to inform them of the presence and status of all peers.

If the leader election method requires dynamic fitness scores, after ACP 1 receives *MA\_PackageManagementCommandStatusMT* messages with *CommandProcessingState* of *ACCEPTED* from both ACPs, it sends *MA\_PackagePartnerStatusReportMT\_3* to ACP 2 and *MA\_PackagePartnerStatusReportMT\_4* to ACP 3, to inform them of the presence and status of all peers.

If the leader election method does not require fitness scores, after ACP 1 receives *MA\_PackageManagementCommandStatusMT* messages with *CommandProcessingState* of *ACCEPTED* from both ACPs, it sends *MA\_PackagePartnerStatusReportMT\_5* to ACP 2 and *MA\_PackagePartnerStatusReportMT\_6* to ACP 3, to inform them of the presence and status of all peers.

Since the leader has not yet been declared, via election or otherwise, the *PackageLeaderID* field will be unpopulated.

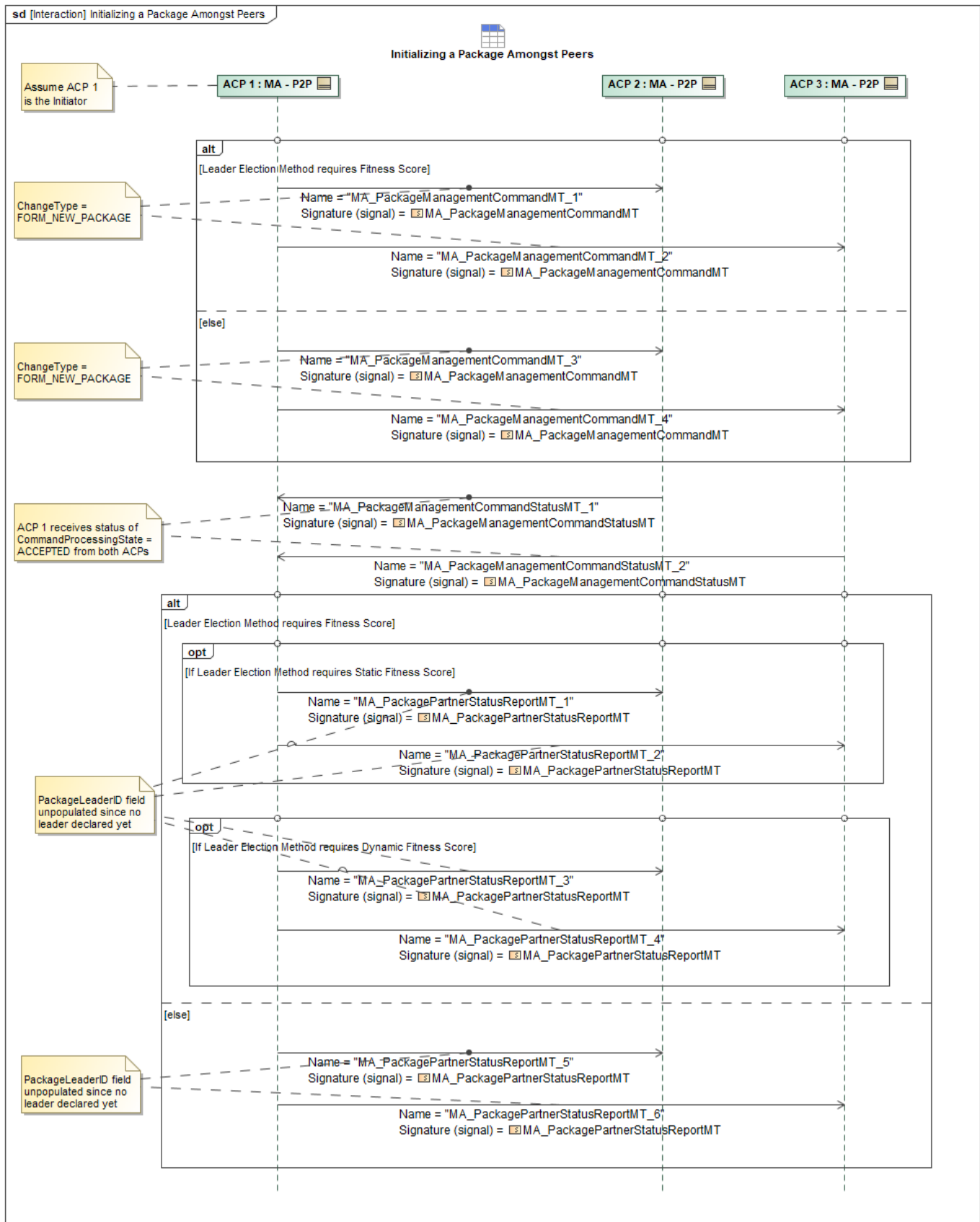


Figure 1-97: Initializing a Package Amongst Peers



Table 1-96: Initializing a Package Amongst Peers

| Message Name (Name:Type)                                                      | Required Fields (Name:Type)                                            |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------|
| MA_PackageManagementCommandMT_1:<br>MA_PackageManagementCommandMT             | PackagePartnerMetrics: MA_PartnerMetricsType                           |
| MA_PackageManagementCommandMT_2:<br>MA_PackageManagementCommandMT             | PackagePartnerMetrics: MA_PartnerMetricsType                           |
| MA_PackageManagementCommandMT_3:<br>MA_PackageManagementCommandMT             | None                                                                   |
| MA_PackageManagementCommandMT_4:<br>MA_PackageManagementCommandMT             | None                                                                   |
| MA_PackageManagementCommandStatusMT_1:<br>MA_PackageManagementCommandStatusMT | None                                                                   |
| MA_PackageManagementCommandStatusMT_2:<br>MA_PackageManagementCommandStatusMT | None                                                                   |
| MA_PackagePartnerStatusReportMT_1:<br>MA_PackagePartnerStatusReportMT         | StaticLeadershipFitnessScore: FloatMinMaxType                          |
| MA_PackagePartnerStatusReportMT_2:<br>MA_PackagePartnerStatusReportMT         | StaticLeadershipFitnessScore: FloatMinMaxType                          |
| MA_PackagePartnerStatusReportMT_3:<br>MA_PackagePartnerStatusReportMT         | DynamicLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_PackagePartnerStatusReportMT_4:<br>MA_PackagePartnerStatusReportMT         | DynamicLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_PackagePartnerStatusReportMT_5:<br>MA_PackagePartnerStatusReportMT         | None                                                                   |
| MA_PackagePartnerStatusReportMT_6:<br>MA_PackagePartnerStatusReportMT         | None                                                                   |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.5 Leader Commanded Package Addition

This peer interaction describes the process of an ACP joining an existing package via command from the Leader of the new package. In this scenario, the Lead ACP will only execute the package addition command if it was sent by an authorized role. The Leader sends

*MA\_PackageManagementCommandMT* to the new asset, with the *ChangeType* field set to *JOIN\_PACKAGE*, and the *SystemID* field populated with the new asset's ID. The new asset responds by sending the requesting Leader *MA\_PackageManagementCommandStatusMT* with *CommandProcessingState* of *ACCEPTED* to indicate the ACP has joined the package. To all followers (including the newly joined ACP), the Leader sends *PackageStatusMT*, indicating the updated membership of the package. The newly-joined ACP (and all followers) will send *MA\_PackagePartnerStatusReportMT* periodically to report status to the leader.

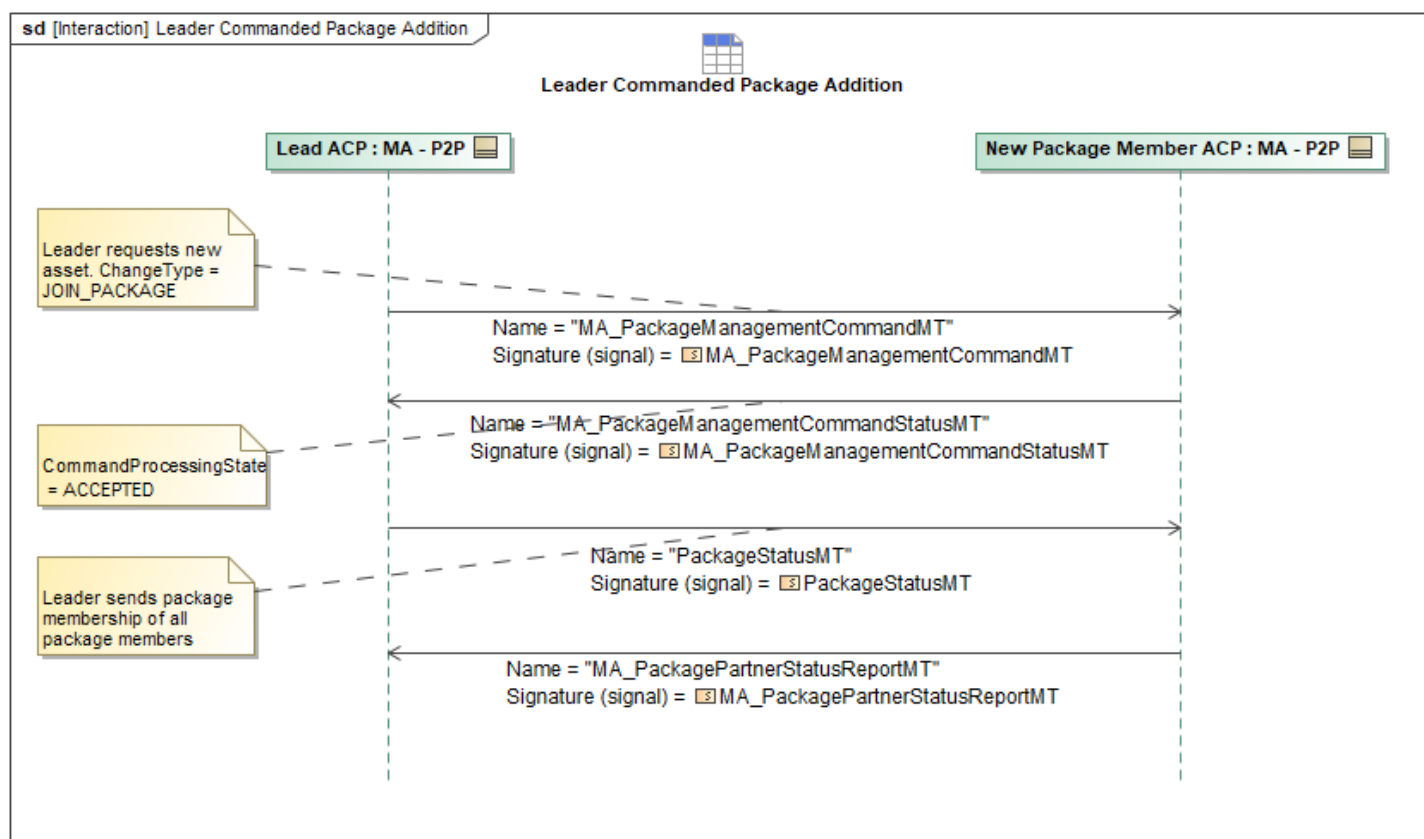


Figure 1-98: Leader Commanded Package Addition

Table 1-97: Leader Commanded Package Addition

| Message Name (Name:Type)                                                    | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------|-----------------------------|
| MA_PackageManagementCommandMT:<br>MA_PackageManagementCommandMT             | None                        |
| MA_PackageManagementCommandStatusMT:<br>MA_PackageManagementCommandStatusMT | None                        |
| MA_PackagePartnerStatusReportMT:<br>MA_PackagePartnerStatusReportMT         | None                        |
| PackageStatusMT: PackageStatusMT                                            | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.6 Maximum Consensus Leader Election

#### Mission Use Case: Max Consensus Leader Election

This leader election method is optional and implementers are expected to implement at least one of the leader election methods. This election method supports both a *StaticLeadershipFitnessScore* or a *DynamicLeadershipFitnessScore* based on the selected *PackageLeadershipFitnessScoreDeterminationMethod*. Either score can be used as the “primary fitness score” (further detailed in this section). If the *DynamicLeadershipFitnessScore* is used as the primary fitness score, the *StaticLeadershipFitnessScore* may be used as the “secondary fitness

score” rather than the *SystemID*

This interaction describes how ACPs would use the maximum consensus algorithm to reach a team consensus on which ACP has the maximum leadership fitness score, if the Maximum Consensus Leader Election method is chosen as the method to determine consensus on the ACP leader. Once the algorithm begins, each ACP's fitness score remains constant for the duration of the election. For the Maximum Consensus Method to Leader Election, each ACP initially sends out its primary leadership fitness score to its neighbors. Every ACP will do this by setting its maximum leadership fitness score equal to its primary leadership fitness score in *MA\_LeaderUpdateRequestMT* with the *RequestState* set to *NEW*, the *Status* set to *UNSPECIFIED*, the *StaticLeadershipFitnessScore* and *DynamicLeadershipFitnessScore* (if applicable). Providing its *AssociatedSystemID* as the secondary leadership fitness score in the *StaticLeadershipFitnessScore* is optional. Every ACP updates its tracked maximum leadership fitness score, if it is smaller than the largest maximum leadership fitness scores it has received from its in-neighbors. Optionally, each ACP keeps track of the *AssociatedSystemID* that corresponds to that new maximum leadership score. Any ties of primary fitness scores are broken via the *AssociatedSystemID*. Every ACP sends *only* its updated, tracked maximum leadership fitness score to its out-neighbors in *MA\_LeaderUpdateRequestMT*. Every ACP sends its updated maximum leadership fitness score *only* when it has changed. Optionally, every ACP includes the max fitness score's corresponding *AssociatedSystemID* in the *StaticLeadershipFitnessScore*. This will be repeated for a pre-defined amount of time. The ACP with the primary fitness score equal to the tracked maximum leadership fitness score (and optionally the *AssociatedSystemID* if provided) nominates itself the leader by sending a nomination *MA\_LeaderUpdateRequestMT* with the primary *LeadershipFitnessScore* and the secondary fitness score in the *AssociatedSystemID* field to all other ACPs. There may be multiple ACPs nominating themselves leaders when the algorithm terminates before it has converged, there are message dropouts, or there is a tie on the primary leadership fitness score. All other ACPs send a reply message accepting the nomination for the leader with *MA\_LeaderUpdateRequestStatusMT*. When there are multiple ACPs declaring themselves leaders, ACPs will only accept the one with the largest primary leadership fitness score and largest secondary leadership fitness score (unique *AssociatedSystemID* in *StaticLeadershipFitnessScore*) in the case there is a tie on the primary leadership fitness score. Other nomination requests will be rejected by sending *MA\_LeaderUpdateRequestStatusMT*. The ACP that received acceptance from the majority of ACPs will send a notification to all other ACPs of its leader nomination. Once the new leader ACP sends a notification of its leader nomination, any ACPs that had accepted the wrong ACP leader nomination previously will be overridden.

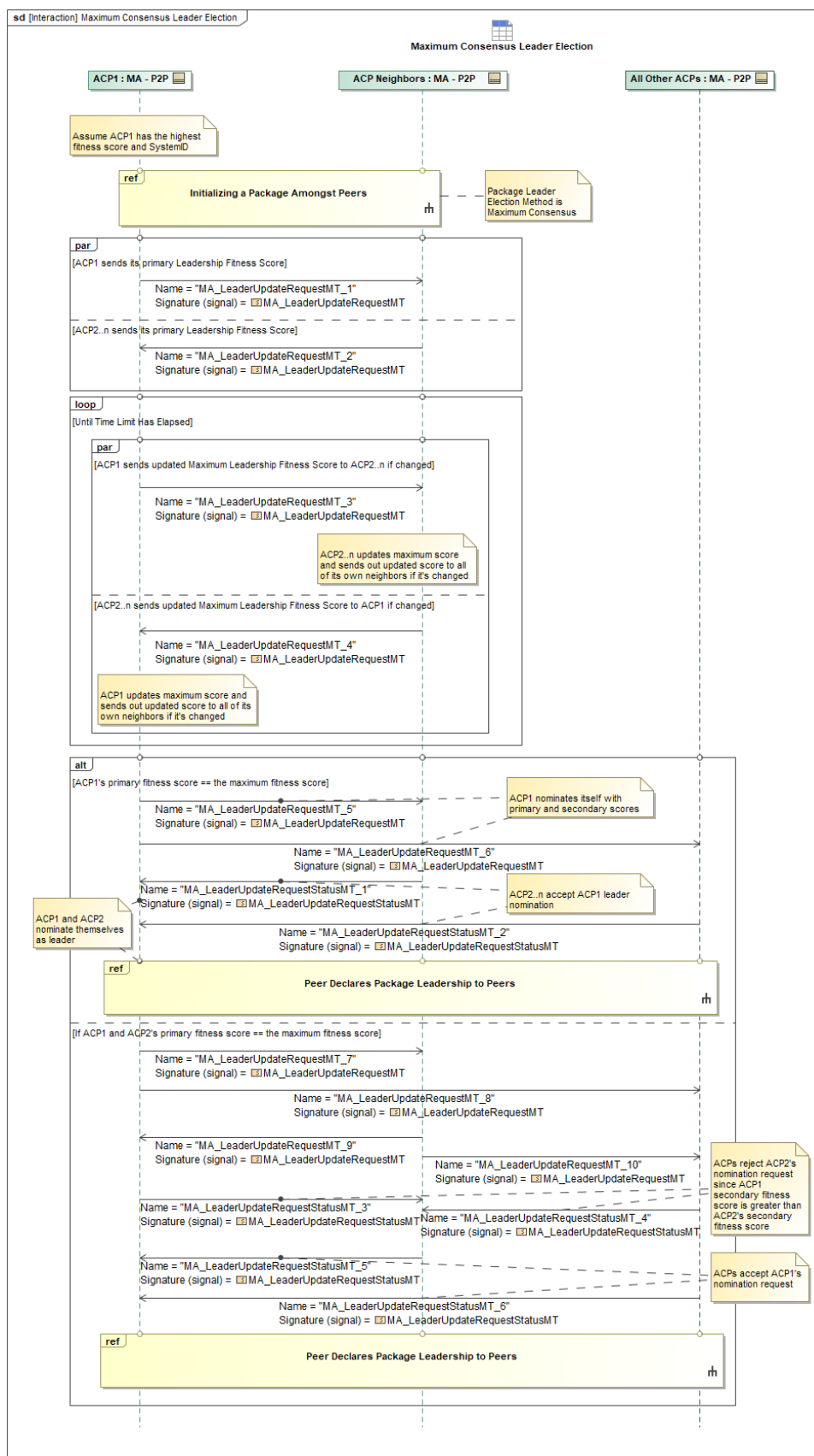


Figure 1-99: Maximum Consensus Leader Election

Table 1-98: Maximum Consensus Leader Election

| Message Name (Name:Type)                                            | Required Fields (Name:Type)                                           |
|---------------------------------------------------------------------|-----------------------------------------------------------------------|
| MA_LeaderUpdateRequestMT_1:<br>MA_LeaderUpdateRequestMT             | None                                                                  |
| MA_LeaderUpdateRequestMT_2:<br>MA_LeaderUpdateRequestMT             | None                                                                  |
| MA_LeaderUpdateRequestMT_3:<br>MA_LeaderUpdateRequestMT             | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_LeaderUpdateRequestMT_4:<br>MA_LeaderUpdateRequestMT             | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_LeaderUpdateRequestMT_5:<br>MA_LeaderUpdateRequestMT             | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_LeaderUpdateRequestMT_6:<br>MA_LeaderUpdateRequestMT             | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_LeaderUpdateRequestMT_7:<br>MA_LeaderUpdateRequestMT             | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_LeaderUpdateRequestMT_8:<br>MA_LeaderUpdateRequestMT             | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_LeaderUpdateRequestMT_9:<br>MA_LeaderUpdateRequestMT             | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_LeaderUpdateRequestMT_10:<br>MA_LeaderUpdateRequestMT            | StaticLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |
| MA_LeaderUpdateRequestStatusMT_1:<br>MA_LeaderUpdateRequestStatusMT | None                                                                  |
| MA_LeaderUpdateRequestStatusMT_2:<br>MA_LeaderUpdateRequestStatusMT | None                                                                  |
| MA_LeaderUpdateRequestStatusMT_3:<br>MA_LeaderUpdateRequestStatusMT | None                                                                  |
| MA_LeaderUpdateRequestStatusMT_4:<br>MA_LeaderUpdateRequestStatusMT | None                                                                  |
| MA_LeaderUpdateRequestStatusMT_5:<br>MA_LeaderUpdateRequestStatusMT | None                                                                  |
| MA_LeaderUpdateRequestStatusMT_6:<br>MA_LeaderUpdateRequestStatusMT | None                                                                  |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.7 Notify Peers of Package Formation Command from Unauthorized C2

This is the peer interaction component for the off-nominal use case where an unauthorized C2 operator attempts to create or initialize a package including the ACP. In response to receiving *MA\_PackageManagementCommandMT* from an unauthorized C2 operator, the Commanded Peer ACP will send *MA\_PackageManagementCommandStatusMT* to all other Peers (members of the commanded package) with *CommandProcessingState* of *REJECTED* and

*CommandProcessingStateReason* of *INELIGIBLE\_CONTROL\_SOURCE*. This provides notification to the Peers that package formation attempt has failed.

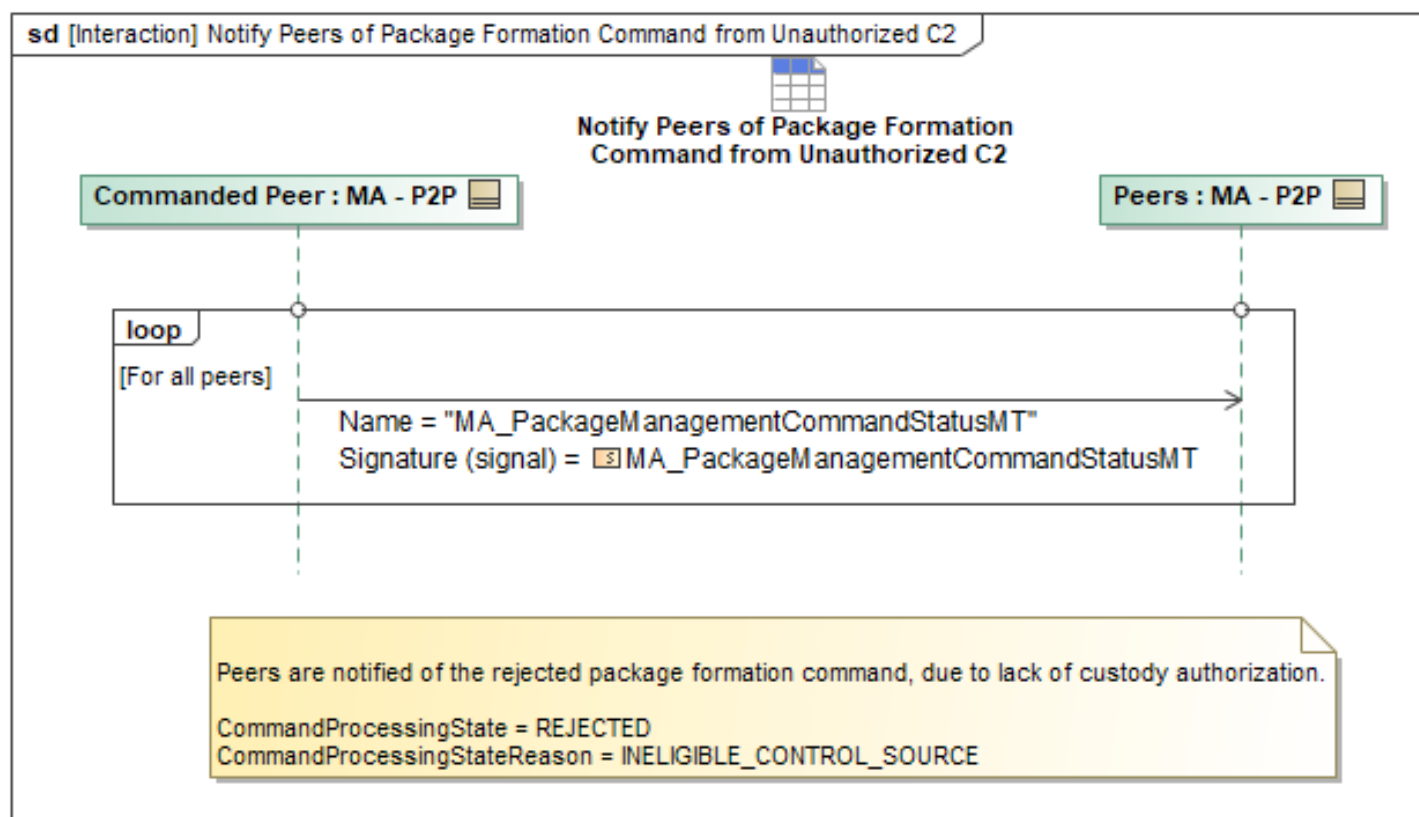


Figure 1-100: Notify Peers of Package Formation Command from Unauthorized C2

Table 1-99: Notify Peers of Package Formation Command from Unauthorized C2

| Message Name (Name:Type)                                                    | Required Fields (Name:Type)                    |
|-----------------------------------------------------------------------------|------------------------------------------------|
| MA_PackageManagementCommandStatusMT:<br>MA_PackageManagementCommandStatusMT | CommandProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.8 Off Nominal Leader Election

**Mission Use Case:** Bully Leader Election, Static Fitness Score Leader Election, Max Consensus Leader Election

The First and Second Off-Nominal Case are applicable if the Bully Leader Election Method, Maximum Consensus Leader Election, or the Static Fitness Score Leader Election Method is implemented.

This is the peer interaction describing two off-nominal cases for leader election, which may occur at the point when the ACP with highest fitness score would be declaring itself leader. Refer to the Algorithm Description section within the documentation of the "Bully Leader Election" sequence for a generalized description in the case of additional failures (e.g. if ACP 2 also fails during the first off-nominal case).



### First Off-Nominal Case: Best Leader Candidate Fails During Election

If ACP 3, which has the highest fitness score, fails before declaring its leadership, the ACPs wait for the election timeout. Assume that ACP 1 reaches its timeout first. At that point, ACP 1 should send *MA\_LeaderUpdateRequestMT\_1* with *ProposedLeaderID* within the *Request* field set to ACP 2's System ID (since ACP 2 has the second-highest fitness score) and the *Status* field set to *NOMINATE*. ACP 2 responds with *MA\_LeaderUpdateRequestStatusMT\_1* with *RequestProcessingState* of *PROCESSING*. Once ACP 2 reaches its election timeout, it sends the ACP peers *MA\_LeaderUpdateRequest\_2-3* with the *Status* field set to *NOMINATE* and *ProposedLeaderID* within the *Request* field set to its own System ID. Note that ACP 3 will not receive this message since it is undergoing a communications failure. All connected ACPs (in this case ACP 1), respond with *MA\_LeaderUpdateRequestStatusMT\_2*, with *RequestProcessingState* of *COMPLETED*. Since a majority of connected ACPs accepted the request, ACP 2 declares leadership via "Peer Declares Package Leadership to Peers."

### Second Off-Nominal Case: Peer Communications Failure

In the case where ACP 2 undergoes a communications failure and doesn't receive ACP 3's leader declaration, it may try to incorrectly assume leadership. ACP 3 performs the "Peer Declares Package Leadership to Peers" process, sending *PackageMT\_1*, and *PackageStatusMT\_1* messages which are received by ACP 1. The *PackageMT\_2*, and *PackageStatusMT\_2* messages are not received by ACP 2 due to its communication failure. ACP 2 waits for the election timeout to occur, and then nominates itself as leader by sending *MA\_LeaderUpdateRequestMT\_4-5* with the *Status* field set to *NOMINATE* and *ProposedLeaderID* within the *Request* field set to its own System ID. ACPs 1 and 3 respond by rejecting this command, sending *MA\_LeaderUpdateRequestStatusMT\_3-4* with *RequestProcessingState* of *REJECTED*. Regardless of whether ACP 2 receives these rejection messages, it will not assume leadership.

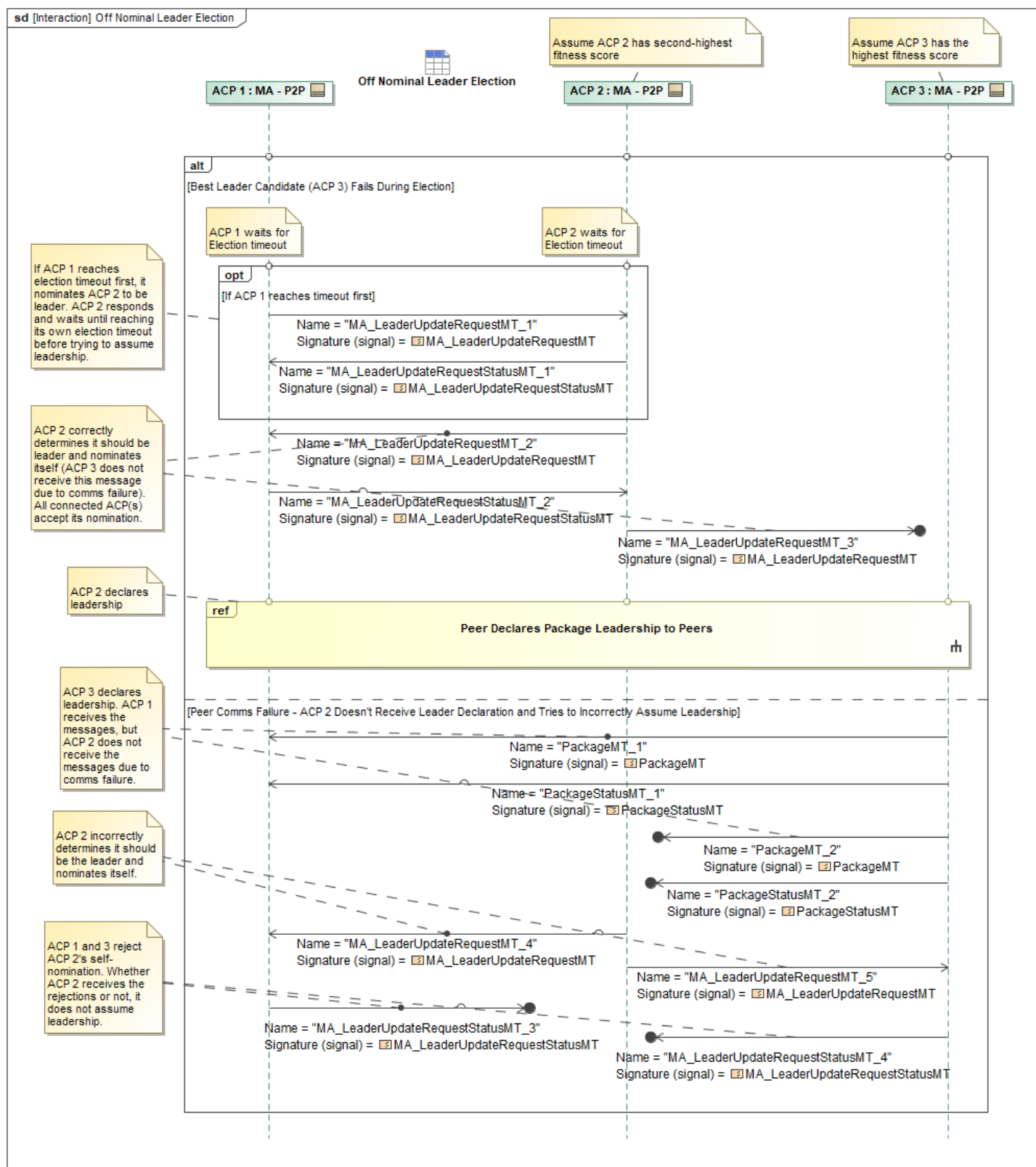


Figure 1-101: Off Nominal Leader Election

Table 1-100: Off Nominal Leader Election



| Message Name (Name:Type)                                            | Required Fields (Name:Type)     |
|---------------------------------------------------------------------|---------------------------------|
| MA_LeaderUpdateRequestMT_1:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestMT_2:<br>MA_LeaderUpdateRequestMT             | None                            |
| MA_LeaderUpdateRequestMT_3:<br>MA_LeaderUpdateRequestMT             | None                            |
| MA_LeaderUpdateRequestMT_4:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestMT_5:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestStatusMT_1:<br>MA_LeaderUpdateRequestStatusMT | None                            |
| MA_LeaderUpdateRequestStatusMT_2:<br>MA_LeaderUpdateRequestStatusMT | None                            |
| MA_LeaderUpdateRequestStatusMT_3:<br>MA_LeaderUpdateRequestStatusMT | None                            |
| MA_LeaderUpdateRequestStatusMT_4:<br>MA_LeaderUpdateRequestStatusMT | None                            |
| PackageMT_1: PackageMT                                              | LeadSystemID: SystemID_Type     |
| PackageMT_2: PackageMT                                              | LeadSystemID: SystemID_Type     |
| PackageStatusMT_1: PackageStatusMT                                  | None                            |
| PackageStatusMT_2: PackageStatusMT                                  | None                            |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.9 Peer Declares Package Leadership to Peers

This is the peer interaction where a newly elected or assigned leader declares its leadership to the followers in the package. Note that this is a sub-process within Leader Election as a whole, and this interaction is referenced by other sequences after a new leader is determined. For each follower within the package, the new leader sends *PackageMT*, with the *LeadSystemID* field containing its own System ID. The leader sends *PackageStatusMT*, containing the status of all followers. Each follower reports status to the new leader by sending *MA\_PackagePartnerStatusReportMT*.

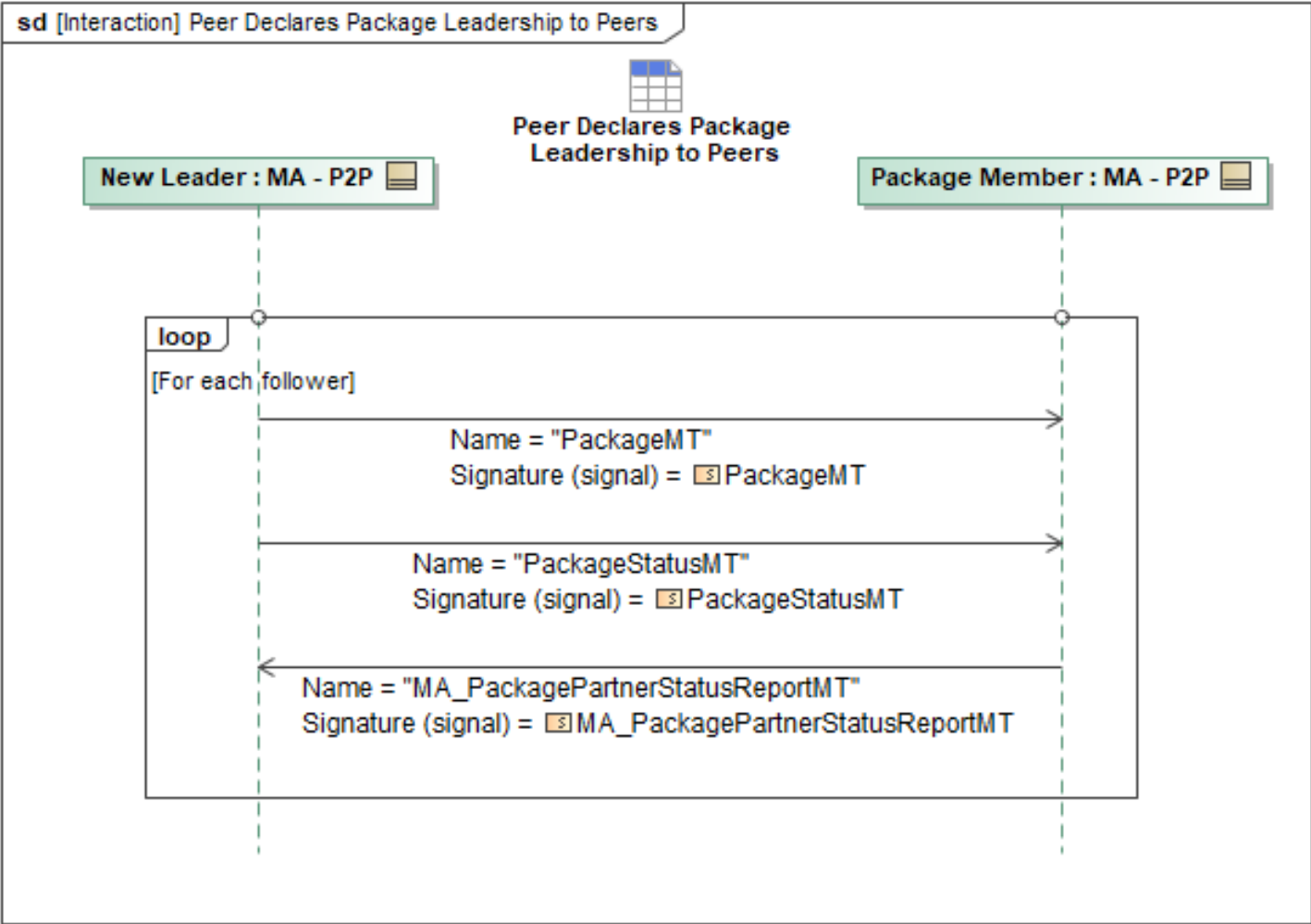


Figure 1-102: Peer Declares Package Leadership to Peers

Table 1-101: Peer Declares Package Leadership to Peers

| Message Name (Name:Type)                                         | Required Fields (Name:Type) |
|------------------------------------------------------------------|-----------------------------|
| MA_PackagePartnerStatusReportMT: MA_PackagePartnerStatusReportMT | None                        |
| PackageMT: PackageMT                                             | LeadSystemID: SystemID_Type |
| PackageStatusMT: PackageStatusMT                                 | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.10 Peer Joining a Package

This peer interaction describes the process of an ACP joining an existing package. The Leader, or an ACP that is a current member in a package, sends *MA\_PackageManagementCommandMT* to the new asset, with the *ChangeType* field set to *JOIN\_PACKAGE*, and the *SystemID* field populated with the new asset's ID. The new asset responds by sending the requesting Leader or requesting ACP *MA\_PackageManagementCommandStatusMT* with *CommandProcessingState* of *ACCEPTED* to

indicate the ACP has joined the package. To all followers (including the newly joined ACP), the Leader sends *PackageStatusMT*, indicating the updated membership of the package. The newly-joined ACP (and all followers) will send *MA\_PackagePartnerStatusReportMT* periodically to report status to the Leader.

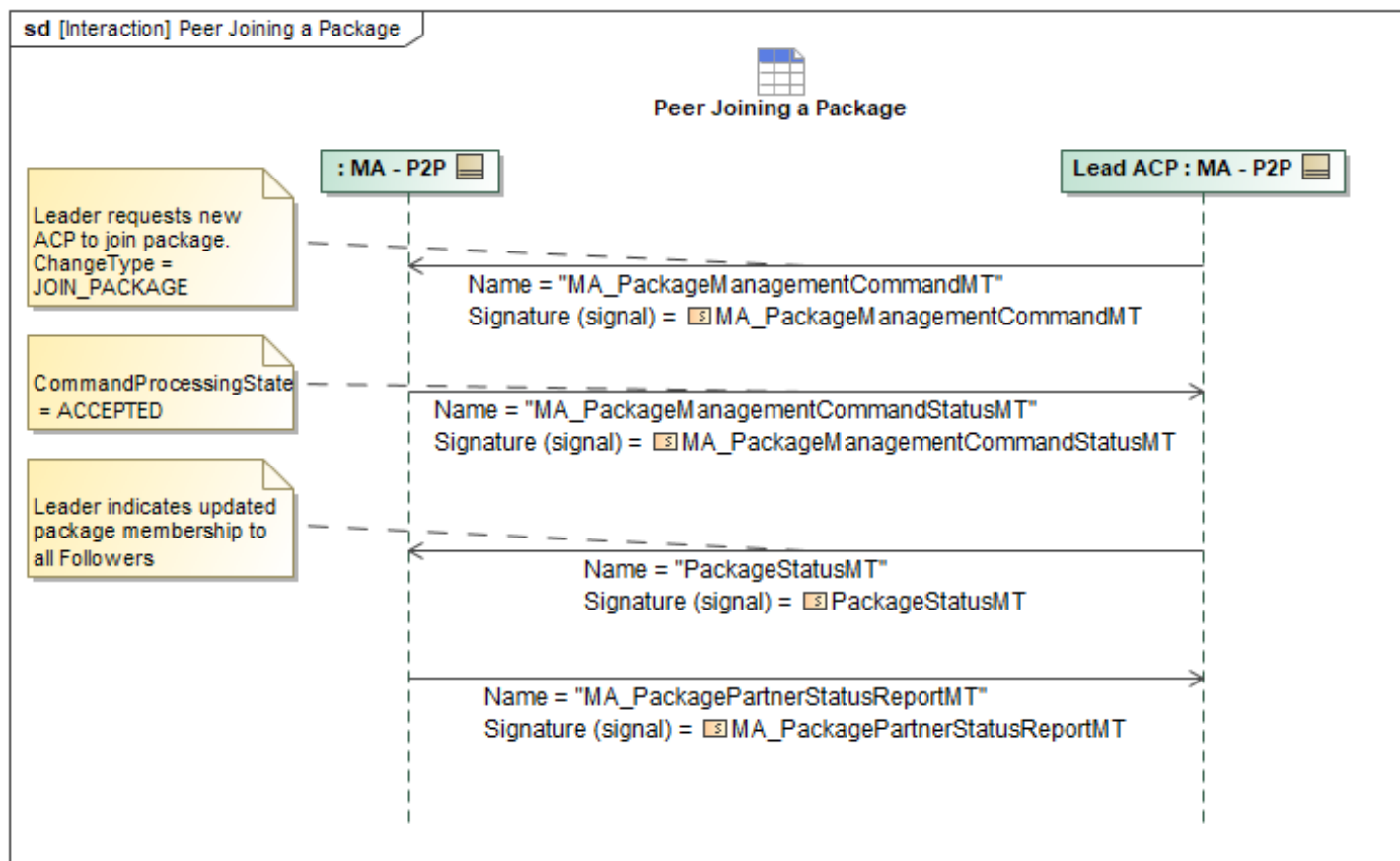


Figure 1-103: Peer Joining a Package

Table 1-102: Peer Joining a Package

| Message Name (Name:Type)                                                    | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------|-----------------------------|
| MA_PackageManagementCommandMT:<br>MA_PackageManagementCommandMT             | None                        |
| MA_PackageManagementCommandStatusMT:<br>MA_PackageManagementCommandStatusMT | None                        |
| MA_PackagePartnerStatusReportMT:<br>MA_PackagePartnerStatusReportMT         | None                        |
| PackageStatusMT: PackageStatusMT                                            | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.11 Raft Leader Election

**Mission Use Case:** Raft Leader Election

This leader election method is optional and implementors are expected to implement at least one of the leader election methods.

This describes the peer interaction for the Raft Leader Election process. If the Raft Leader Election method is chosen as the method to the package leader, this method will allow for dynamic leader election that is robust, well-tested, and easily comprehensible. As part of the Raft Consensus Algorithm, the Raft Leader Election uses a heartbeat mechanism and a randomized heartbeat timeout to maintain leaders (randomized heartbeat timeout ensures that split votes are rare and quickly resolved). There can be at most one Leader per term (Raft divides time into terms of arbitrary length, and terms are numbered with consecutive integers). Each ACP will maintain the current term value. If a peer has a later term, then the ACP will update the term and revert to the Follower state.

Each Follower expects to receive regular heartbeats from the Leader. An ACP remains in the Follower state as long as it receives a valid heartbeat (*MA\_RaftAppendEntriesRequestMT*) from a leader or *VoteRequest* (*MA\_RaftVoteRequestMT*) from a candidate. If a Follower receives no communication over a period of time called the election timeout, then it assumes there is no viable leader and begins an election to choose a new leader.

On package formation, since all ACPs are initialized as a Follower, the ACPs will timeout and enter the Candidate state (randomized heartbeat timeout). In the Candidate state, the ACP will send *MA\_RaftVoteRequestMT* to all of the other peers with an incremented term and a vote for itself as leader (CandidateID). The ACPs, upon receipt of *MA\_RaftVoteRequestMT*, will compare the term sent from the Candidate. If the term is greater than the term of the ACP, the ACP will respond with a positive vote (*MA\_RaftVoteRequestStatus*) for the Candidate to become leader (note: log replication is currently not implemented as part of the Raft Leader Election method).

While waiting for votes, a candidate may receive a *MA\_RaftAppendEntriesRequestMT* from another ACP claiming to be leader. If the proclaimed leader's term (included in its *MA\_RaftAppendEntriesRequestMT*) is at least as large as the candidate's current term, then the candidate recognizes the leader as legitimate and returns to follower state. If the term in the *MA\_RaftAppendEntriesRequestMT* is smaller than the candidate's current term, then the candidate rejects the *MA\_RaftAppendEntriesRequestMT* and continues the leader election in candidate state. When the Candidate ACP has received the majority of votes from Peers in the package, then the Candidate will enter the Leader state and begin sending periodic heartbeats to maintain package leadership (*MA\_RaftAppendEntriesRequestMT*). If the Candidate ACP did not receive the majority of votes from Peers in the package such as the case of a split vote between multiple Candidate ACPs, the Candidate ACP(s) will start a new election by sending *MA\_RaftVoteRequestMT* (the election timeout is randomized). This process repeats until a Leader has been elected.

The following section is optional and is left open for interpretation by implementers:

In instances where the Leader is synchronizing log entries with the package, the Leader will send *MA\_RaftAppendEntriesRequestMT* with *MA\_RaftLogEntry* populated and the Followers will respond with *MA\_RaftAppendEntriesRequestStatusMT* to confirm successful log commit. Within *MA\_RaftLogEntry*, the *Command* field will be populated with key/value pairs that will be open for implementers to determine.

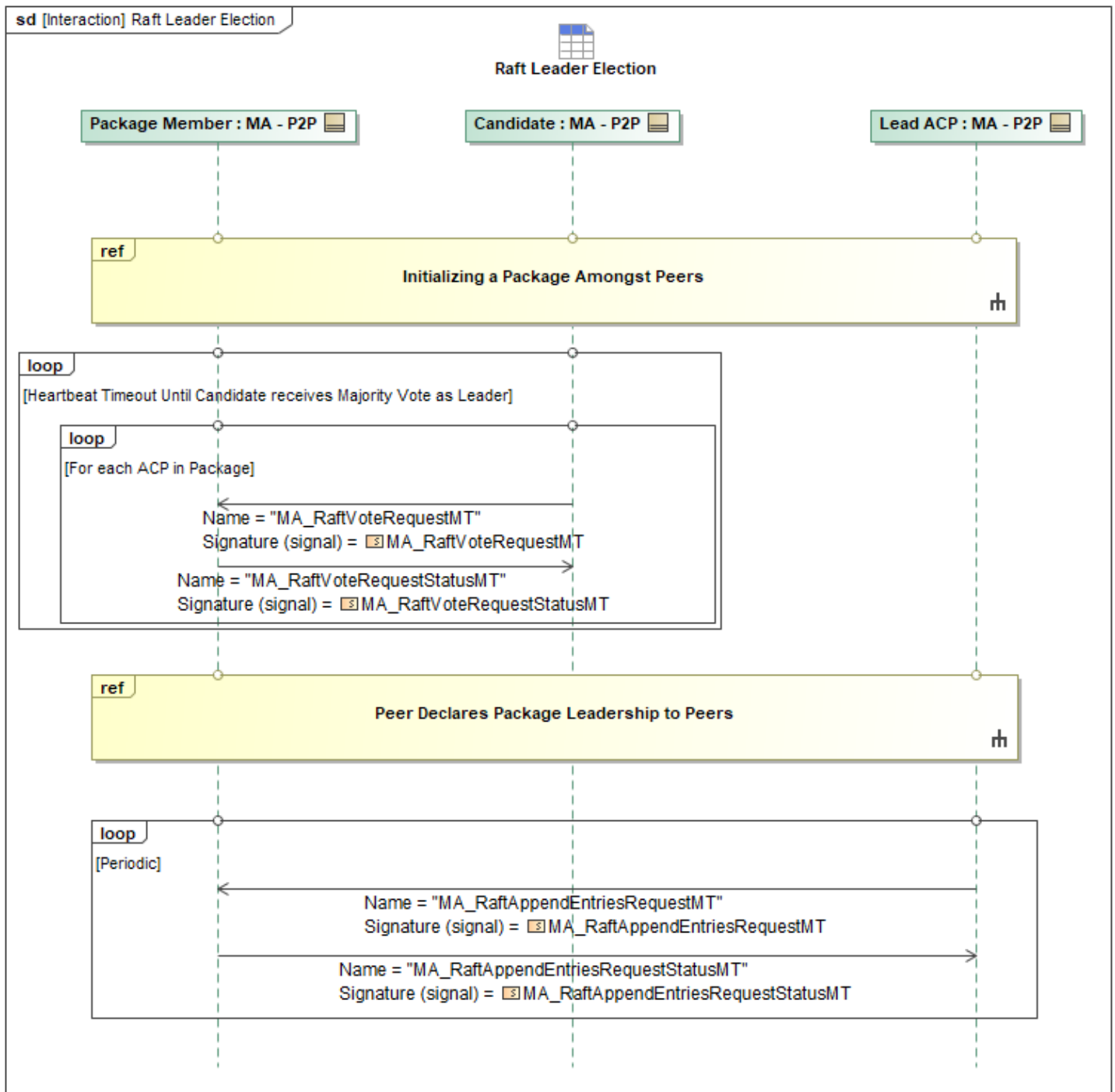


Figure 1-104: Raft Leader Election

Table 1-103: Raft Leader Election

| Message Name (Name:Type)                                                    | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------|-----------------------------|
| MA_RaftAppendEntriesRequestMT:<br>MA_RaftAppendEntriesRequestMT             | None                        |
| MA_RaftAppendEntriesRequestStatusMT:<br>MA_RaftAppendEntriesRequestStatusMT | None                        |

| Message Name (Name:Type)                                  | Required Fields (Name:Type) |
|-----------------------------------------------------------|-----------------------------|
| MA_RaftVoteRequestMT:<br>MA_RaftVoteRequestMT             | None                        |
| MA_RaftVoteRequestStatusMT:<br>MA_RaftVoteRequestStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.12 Share DLFS with Peers

The Share DLFS with Peers sequence diagram specifies the interaction between two ACP peers which is a required interaction ONLY when *MA\_LeadershipFitnessScoreDeterminationMethod* equals *DYNAMIC* or *USER\_DEFINED*, and allows an ACP to share its Dynamic Leadership Fitness Score with other ACP peers (which may trigger a new leader election depending on implementation). One ACP peer (MA) sends a *MA\_PackagePartnerStatusReportMT* to other ACP peers (MA) with the *DynamicLeadershipFitnessScore* attribute set to reflect their current (most recently calculated) dynamic leadership fitness score.

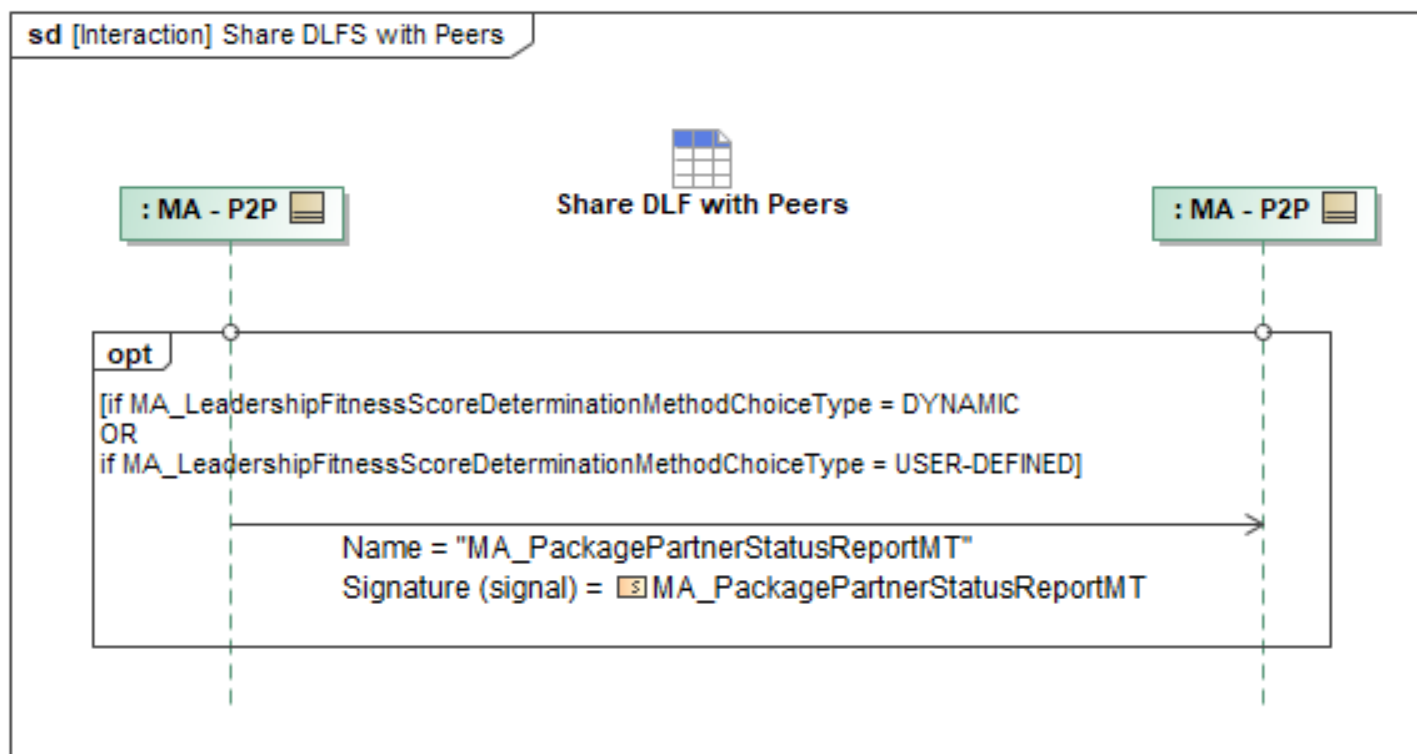


Figure 1-105: Share DLFS with Peers

Table 1-104: Share DLFS with Peers

| Message Name (Name:Type)                                            | Required Fields (Name:Type)                                            |
|---------------------------------------------------------------------|------------------------------------------------------------------------|
| MA_PackagePartnerStatusReportMT:<br>MA_PackagePartnerStatusReportMT | DynamicLeadershipFitnessScore:<br>MA_PartnerLeadershipFitnessScoreType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy

Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.11.13 Static Fitness Score Leader Election

#### **Mission Use Case:** Static Fitness Score Leader Election

This leader election method is optional and implementers are expected to implement at least one of the leader election methods.

This describes the peer interaction for the Static Fitness Score Leader Election process. This election method can be used when fitness scores are static (unchanging), based on the selected *PackageLeadershipFitnessScoreDeterminationMethod* or do not need to be shared at the time of election. This section describes using this method as the first election after package initialization, when leader communications are lost, and in other off-nominal cases.

For this interaction, assume ACP 3 has the highest fitness score. Ties are broken by highest tail number, which is found via the heartbeat system status messages in the Peer Heartbeat and Send Vehicle State to Peer interactions.

#### **First Election after Package Initialization:**

Once ACP peers determine the package has completed initialization via "Initializing a Package Amongst Peers," their election timers begin. In the nominal case, ACP 3 recognizes it has the highest *StaticLeadershipFitnessScore* and declares its leadership via "Peer Declares Package Leadership to Peers." If ACP 3 fails to declare leadership before ACP 1 or 2's election timeout is reached, ACP 1 or 2 should assign leadership to ACP 3 by sending *MA\_LeaderUpdateRequestMT\_1* or *MA\_LeaderUpdateRequestMT\_2* with the *Status* field set to *ASSIGN* and ACP 3's System ID populated in *ProposedLeaderID* of the *Request* field.

#### **Otherwise (e.g. Lost Leader Scenario):**

In the case where Package Initialization did not occur directly before the election (for instance, if the leader lost communications), the package should gain concurrence that a leader election will be held. This allows for the handling of various off-nominal communication scenarios.

Once ACP peers determine an election is underway (either by initiating the election or receiving *MA\_LeaderUpdateRequestMT* as part of the election), their election timers begin. In a typical case, ACP 3 will recognize an election is required and that it has the highest fitness score, and will nominate itself for leadership by sending *MA\_LeaderUpdateRequestMT\_5* and *MA\_LeaderUpdateRequestMT\_6* to ACP 2 and ACP 1, respectively. This message will have the *Status* field set to *NOMINATE* and ACP 3's System ID populated in *ProposedLeaderID* of the *Request* field.

If ACP 3 fails to nominate itself for leadership before ACP 1 or 2's election timeout is reached, ACP 1 or 2 should nominate ACP 3 by sending *MA\_LeaderUpdateRequestMT\_3* or *MA\_LeaderUpdateRequestMT\_4* with the *Status* field set to *NOMINATE* and ACP 3's System ID populated in *ProposedLeaderID* of the *Request* field. In recognition of the nomination, ACP 3 sends *MA\_LeaderUpdateRequestMT\_5* and *MA\_LeaderUpdateRequestMT\_6* to ACP 2 and ACP 1, respectively, so all ACPs are aware of its nomination

ACPs 2 and 1 respond with *MA\_LeaderUpdateRequestStatusMT\_1* and *MA\_LeaderUpdateRequestStatusMT\_2*, respectively, with *RequestProcessingState* of *COMPLETED* to indicate acceptance of ACP 3's leader nomination. ACP 3 then declares leadership via "Peer

Declares Package Leadership to Peers."

**Off-Nominal:**

Other off-nominal cases of leader election are illustrated in the referenced sub-diagram. These are handled in the same way as the off-nominal cases within Bully Leader Election method.



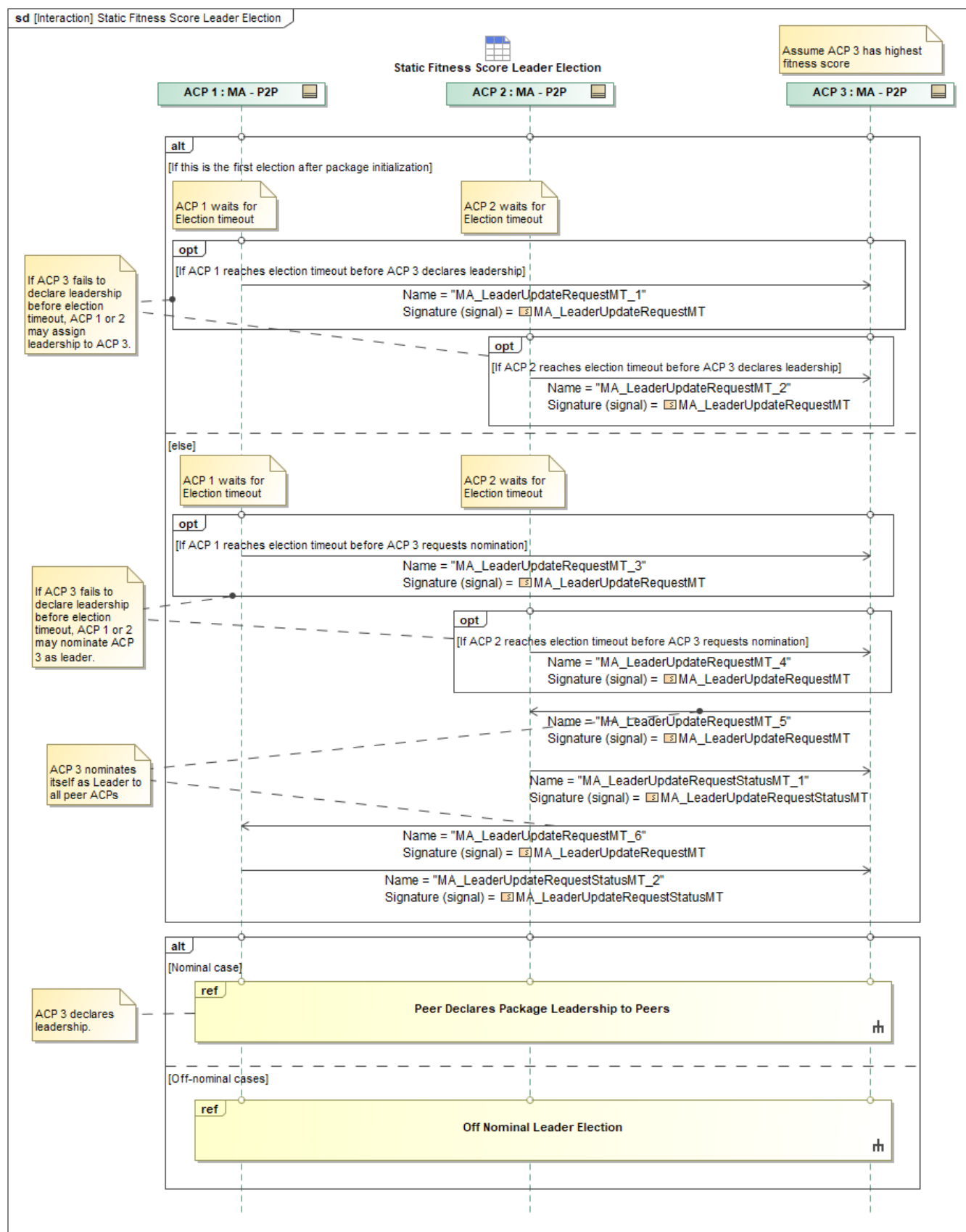


Figure 1-106: Static Fitness Score Leader Election

Table 1-105: Static Fitness Score Leader Election

| Message Name (Name:Type)                                            | Required Fields (Name:Type)     |
|---------------------------------------------------------------------|---------------------------------|
| MA_LeaderUpdateRequestMT_1:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestMT_2:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestMT_3:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestMT_4:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestMT_5:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestMT_6:<br>MA_LeaderUpdateRequestMT             | ProposedLeaderID: SystemID_Type |
| MA_LeaderUpdateRequestStatusMT_1:<br>MA_LeaderUpdateRequestStatusMT | None                            |
| MA_LeaderUpdateRequestStatusMT_2:<br>MA_LeaderUpdateRequestStatusMT | None                            |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.12 Peer Task Prioritization and Mission Context Package

### 1.2.12.1 Peer Acceptable Level of Risk Behaviors

#### 1.2.12.1.1 Peer Provides Route Risk Assessment Results

##### Mission Use Case: ALR

The optional Peer Provides Route Risk Assessment Results sequence diagram specifies the interaction where an ACP provides a Peer ACP with risk assessment results for a Route Plan. This may occur if 1) A Peer ACP requests a Route Plan Threat Assessment or 2) Updated threat(s) trigger an ACP to assess the current Route Plan and deliver the results to a Peer ACP. This interaction assumes that the Route Plan to be assessed has been previously uploaded to the ACP.

In the first case, the Requestor MA will request a risk assessment for a Route Plan, sending *MA\_AssessmentRequestMT* with a specified *AnalysisRouteID* and the *Parameters* of the *RouteThreatAssessment* field within the *Category* field populated. The Assessor MA may optionally report status on the request using *MA\_AssessmentRequestStatusMT\_1* to convey the processing state and/or estimated completion information. The Assessor MA will then deliver the risk assessment results with the *MA\_AssessmentRequestStatusMT\_2* message, populating the *Data* of the *RouteThreatAssessment* field within the *Assessment* field.

In the second case, MA internally triggers a risk assessment due to updated threats, rather than receiving a request to perform the assessment. Upon determining that the current Route Plan exceeds the specified Acceptable Level of Risk (ALR) setting, the ACP may attempt to generate a new Route Plan through the threat environment which complies with the ALR setting. (If the current Route Plan does not exceed the ALR setting, no risk assessment will be required, and this interaction

would not apply). After a new Route Plan is generated, the ACP sends *AssessmentMT* to the Peer ACP to inform of the risk assessment results, populating the *Data* of the *RouteThreatAssessment* field within the *Assessment* field of this message. The message will have a populated *AnalysisRouteID* to identify the Route Plan for which the assessment applies. If a Route Plan which complies with the ALR setting is not generated or able to be achieved, the *MA\_AssessmentMT* message will convey this by populating the *Achievability* field within *AchievabilityAssessment* as *NOT\_ACHIEVABLE*. After, if the ACP receiving the assessment is the Lead ACP, it may forward the risk assessment results to C2 as shown in the “Provide Risk Assessment Results” interaction in the C2 profile.

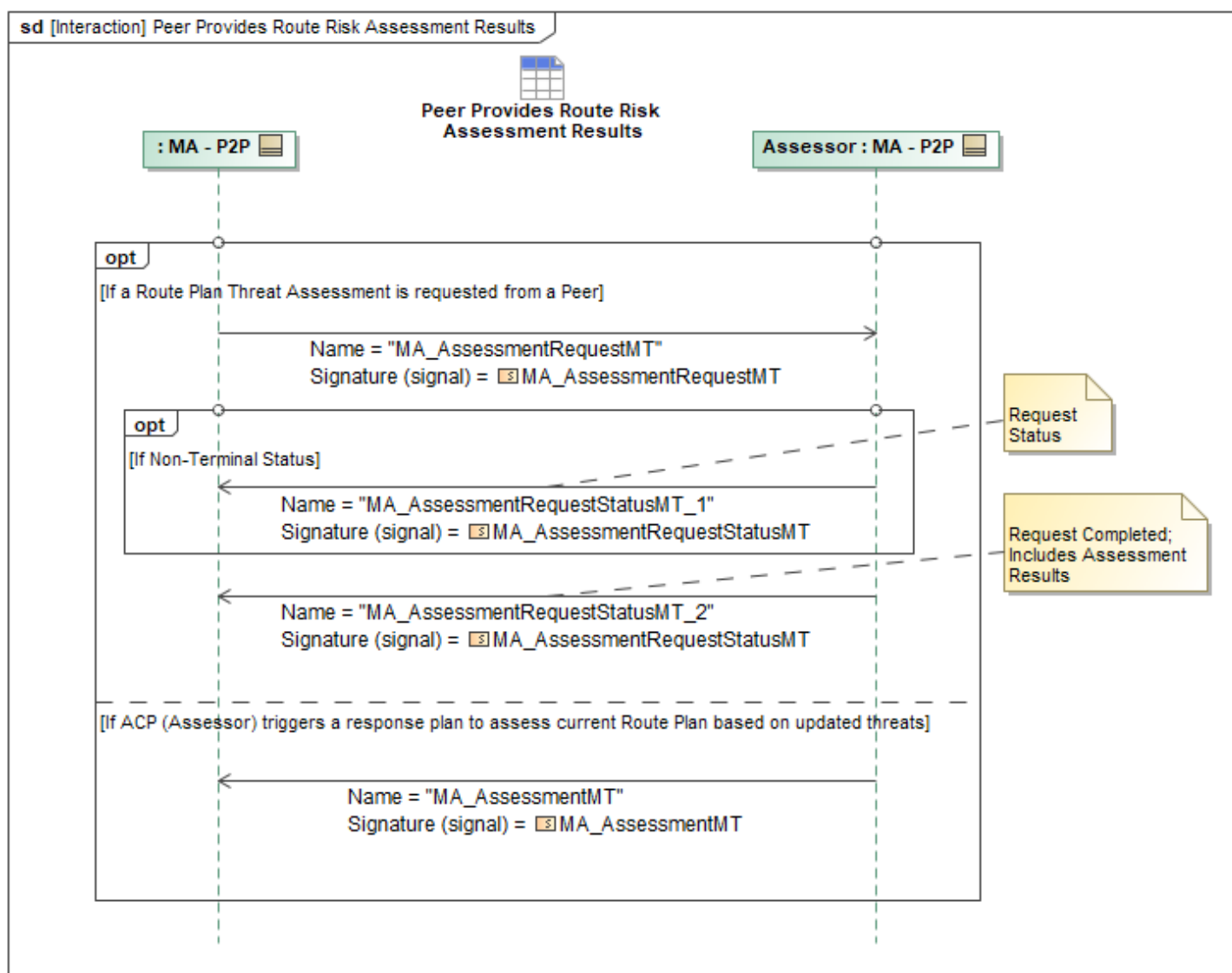


Figure 1-107: Peer Provides Route Risk Assessment Results

Table 1-106: Peer Provides Route Risk Assessment Results

| Message Name (Name:Type)         | Required Fields (Name:Type)           |
|----------------------------------|---------------------------------------|
| MA_AssessmentMT: MA_AssessmentMT | AnalysisRouteID: AnalysisRouteID_Type |

| Message Name (Name:Type)                                                      | Required Fields (Name:Type)           |
|-------------------------------------------------------------------------------|---------------------------------------|
| <b>MA_AssessmentRequestMT:</b><br><b>MA_AssessmentRequestMT</b>               | AnalysisRouteID: AnalysisRouteID_Type |
| <b>MA_AssessmentRequestStatusMT_1:</b><br><b>MA_AssessmentRequestStatusMT</b> | None                                  |
| <b>MA_AssessmentRequestStatusMT_2:</b><br><b>MA_AssessmentRequestStatusMT</b> | Assessment: AssessmentType            |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.12.1.2 Update Peer Acceptable Level of Risk

#### Mission Use Case: ALR

The Update Peer Acceptable Level of Risk (ALR) sequence diagram below specifies the interaction of the lead ACP commanding its follower ACPs to update its ALR settings. First, the lead ACP sends a *MA\_AcceptableLevelOfRiskCommandMT* message where the *AppliesToID* field is defined with the *SystemID* of the follower ACP. The *MA\_AcceptableLevelOfRiskCommandMT* message defines an *AcceptableRiskSetting*, which allows for the ALR to be set as *HIGH*, *MEDIUM*, *LOW*, or *NONE* at the package or vehicle level. This ALR setting applies to all tasks and actions for the vehicle or package. It also defines a *SurvivabilityRiskLevelID* which allows for a *RiskSetting* to be populated to describe what *HIGH*, *MEDIUM*, and *LOW* mean with respect to the *MaximumContinuousExposure* for a given Vulnerability Type. Any new *MA\_AcceptableLevelOfRiskCommandMT* will override any previously sent message, even if the *CommandID* has changed. When a *MA\_AcceptableLevelOfRiskCommandMT* is sent with the *CommandState* set to *CANCEL*, this will cancel the active command.

In response to the command, the follower ACPs shall send *MA\_AcceptableLevelOfRiskCommandStatusMT\_1* to the lead ACP, with the *AppliesToID* field set to their System ID and the *CommandProcessingState* field set to *RECEIVED* to indicate receipt of the command. The Follower ACPs proceed to update their ownship ALR setting and send *MA\_AcceptableLevelOfRiskCommandStatusMT\_2* to the lead ACP with the *AppliesToID* field set to their System ID and the *CommandProcessingState* set to *ACCEPTED*, to indicate they have completed the update to the new ALR setting. The Follower ACPs will then send *MA\_AcceptableLevelOfRiskStatusMT* to the Lead ACP to provide status on their currently activated ALR setting and Survivability Risk Level, using the *AcceptableRiskSetting* and *SurvivabilityRiskLevelID* fields, and populating their System ID in the *AppliesToID* field.

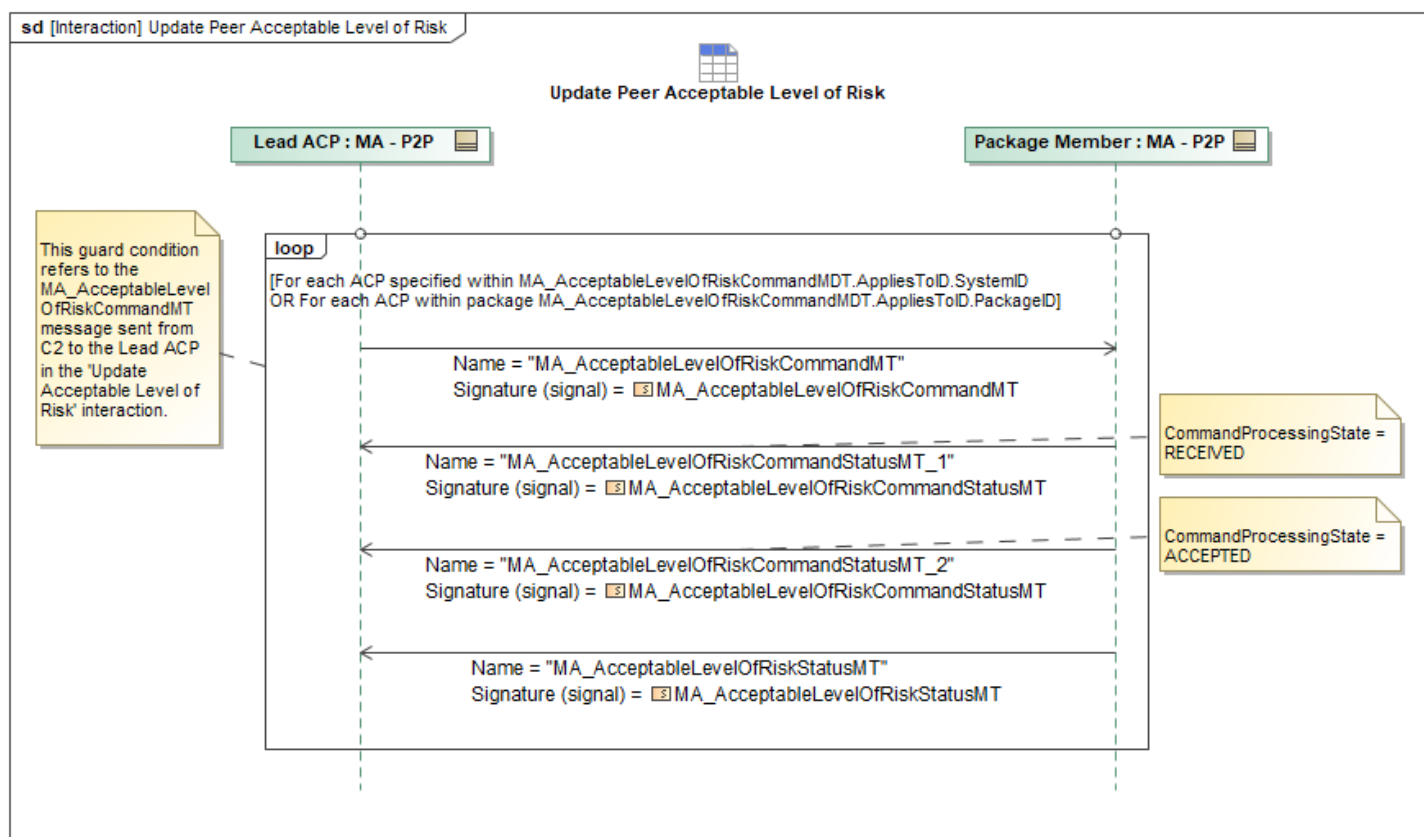


Figure 1-108: Update Peer Acceptable Level of Risk

Table 1-107: Update Peer Acceptable Level of Risk

| Message Name (Name:Type)                                                                  | Required Fields (Name:Type) |
|-------------------------------------------------------------------------------------------|-----------------------------|
| MA_AcceptableLevelOfRiskCommandMT:<br>MA_AcceptableLevelOfRiskCommandMT                   | None                        |
| MA_AcceptableLevelOfRiskCommandStatusMT<br>_1:<br>MA_AcceptableLevelOfRiskCommandStatusMT | None                        |
| MA_AcceptableLevelOfRiskCommandStatusMT<br>_2:<br>MA_AcceptableLevelOfRiskCommandStatusMT | None                        |
| MA_AcceptableLevelOfRiskStatusMT:<br>MA_AcceptableLevelOfRiskStatusMT                     | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.12.1.3 Update Peer Survivability Risk Definition

#### Mission Use Case: ALR

The optional Update Peer Survivability Risk Definition sequence diagram specifies the interaction where the Lead ACP sends Follower ACPs updated Survivability Risk Definition data. Upon receiving a new Survivability Risk Definition from C2 (shown in "Update Survivability Risk Definition" interaction

in C2 profile), the Lead ACP will send *SurvivabilityRiskLevelMT* to all applicable follower ACPs with the *ObjectState* field set to *UPDATED* and the *RiskSetting* field populated. Unless one or a subset of Follower System IDs are specified in the *AppliesToSystemID* field, the Lead ACP will send the update to all package members (i.e. if *AppliesToSystemID* is unpopulated, it is sent to entire package). Note that this does not initiate activation of the Survivability Risk Definition (See “Update Peer Acceptable Level of Risk” for interaction which commands activation).

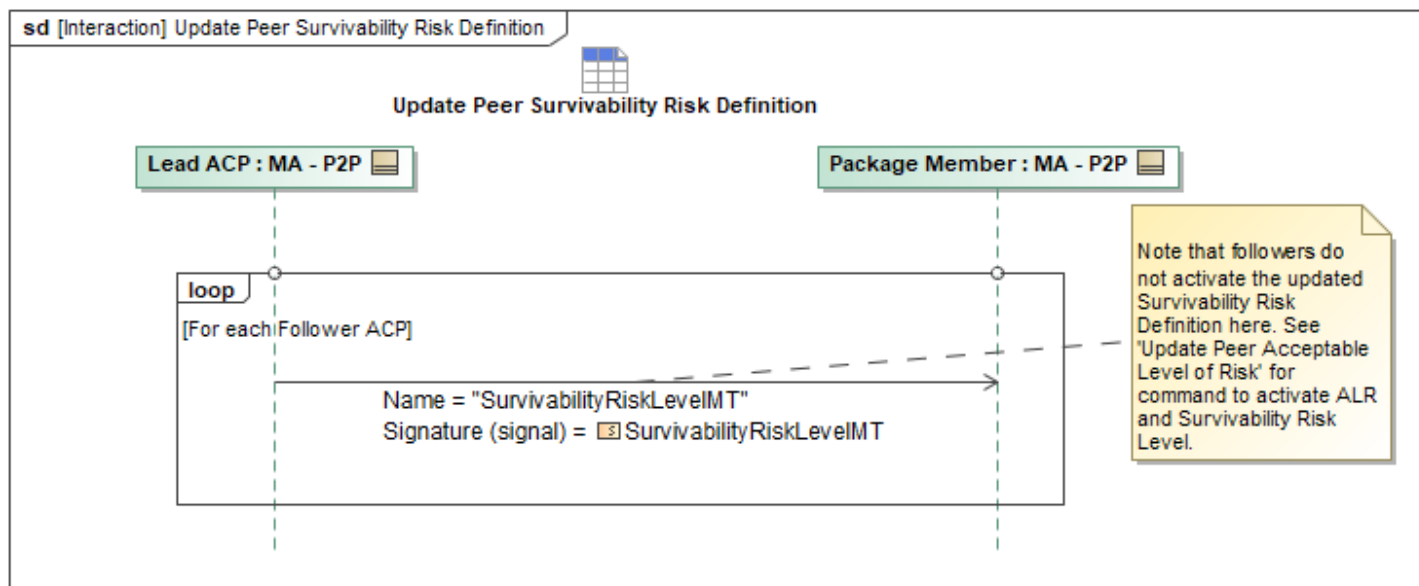


Figure 1-109: Update Peer Survivability Risk Definition

Table 1-108: Update Peer Survivability Risk Definition

| Message Name (Name:Type)                                            | Required Fields (Name:Type)                           |
|---------------------------------------------------------------------|-------------------------------------------------------|
| <b>SurvivabilityRiskLevelMT:</b><br><b>SurvivabilityRiskLevelMT</b> | ObjectState: ObjectStateEnum<br>AssociatedID: ID_Type |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.12.2 Peer Constraint Priority List Behaviors

### 1.2.12.2.1 Activate Peer Constraint Priority List

The Activate Peer Constraint Priority List (CPL) sequence diagram below specifies the interaction of the lead ACP commanding its follower ACPs to activate one of its stored CPL activation schema. First, the lead ACP sends a *MA\_ConstraintPriorityListCommandMT* message where the *AppliesToID* field is defined with the *SystemID* of the follower ACP. The *MA\_ConstraintPriorityListCommandMT* message shall specify the *ConstraintPriorityListActivationDataID* field containing the ID of the Constraint Priority List Activation Data message (or schema) that the lead ACP wishes to be active. Any new *MA\_ConstraintPriorityListCommandMT* will override any previously sent message, even if the *CommandID* has changed. When a *MA\_ConstraintPriorityListCommandMT* is sent with the *CommandedState* set to *CANCEL*, this will cancel the active command and CPL activation schema.

In response to the command, the follower ACPs shall send *MA\_ConstraintPriorityListCommandStatusMT\_1* to the lead ACP, with the *AppliesToID* field set to



their System ID and the *CommandProcessingState* field set to *RECEIVED* to indicate receipt of the command. The follower ACPs shall then use the *ConstraintPriorityListActivationDataID* to query their onboard MP to retrieve the *MA\_ConstraintPriorityListActivationDataMT* (see "Query Constraint Priority List Activation Data"). Then the follower ACPs will use the *MA\_ConstraintPriorityListActivationDataMT* to query for all the CPL messages (see "Query Constraint Priority List"). Once the follower ACPs have finished querying, the follower ACPs shall send *MA\_ConstraintPriorityListCommandStatusMT\_2* to the lead ACP with the *AppliesToID* field set to their System ID and the *CommandProcessingState* set to *ACCEPTED*.

The follower ACPs shall use their location and the *MA\_ConstraintPriorityListActivationDataMT* to determine which CPL ID should be activated (default or geozone based). Once doing so, the follower ACPs will send *MA\_ConstraintPriorityListActivationStatusMT* to the lead ACP containing the *ConstraintPriorityListActivationDataID*, *ActiveConstraintPriorityListID*, and the *AppliesToID* field set to their System ID.

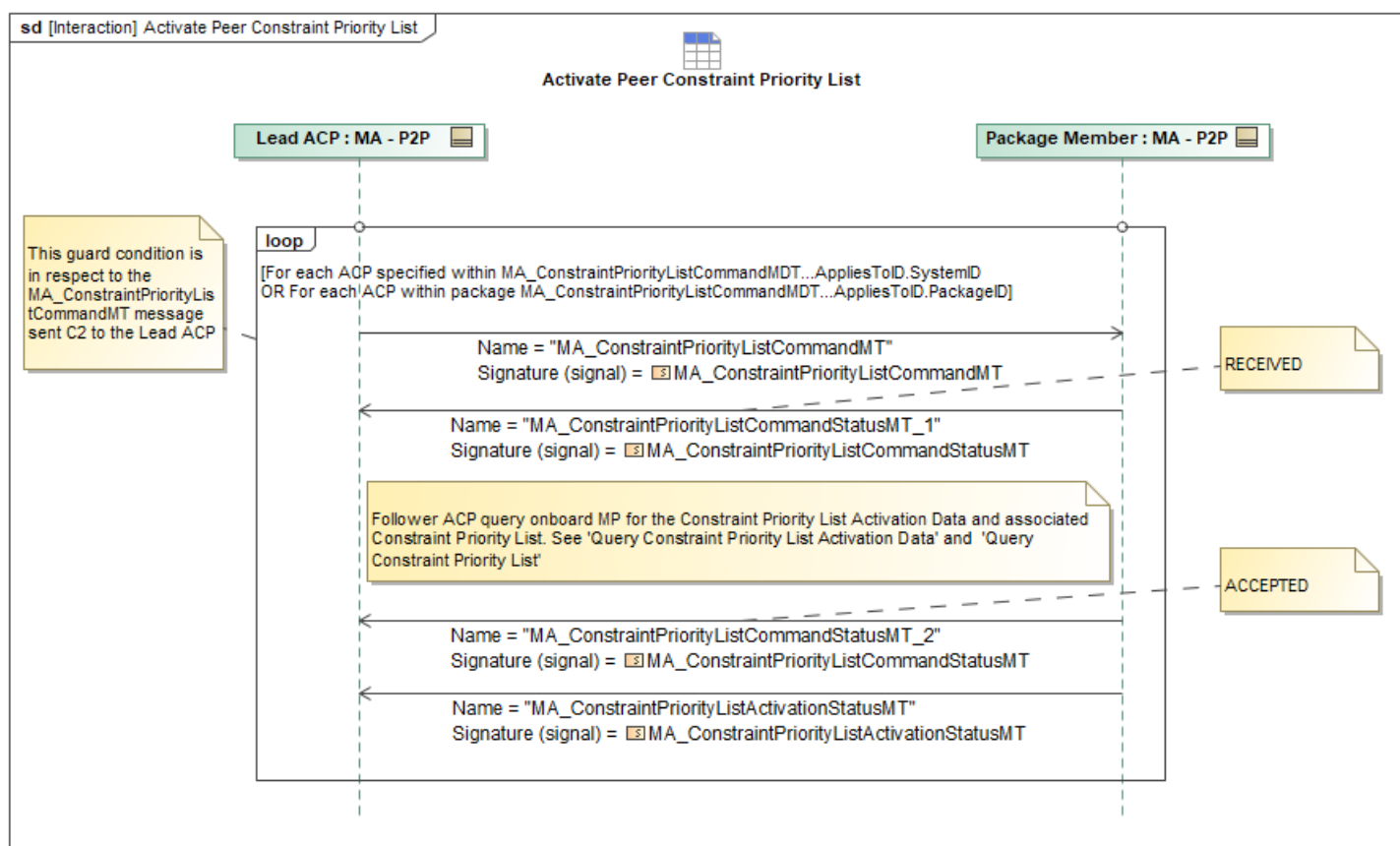


Figure 1-110: Activate Peer Constraint Priority List

Table 1-109: Activate Peer Constraint Priority List

| Message Name (Name:Type)                                                                                  | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------------------------------------|-----------------------------|
| <b>MA_ConstraintPriorityListActivationStatusMT:</b><br><b>MA_ConstraintPriorityListActivationStatusMT</b> | None                        |
| <b>MA_ConstraintPriorityListCommandMT:</b><br><b>MA_ConstraintPriorityListCommandMT</b>                   | None                        |

| Message Name (Name:Type)                                                                | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------------------|-----------------------------|
| MA_ConstraintPriorityListCommandStatusMT_1:<br>MA_ConstraintPriorityListCommandStatusMT | None                        |
| MA_ConstraintPriorityListCommandStatusMT_2:<br>MA_ConstraintPriorityListCommandStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.12.3 Peer Execution Priority List Behaviors

#### 1.2.12.3.1 Activate Peer Execution Priority List

The Activate Peer Execution Priority List sequence diagram below specifies the interaction of the lead ACP commanding its follower ACPs to activate one of its stored EPL activation schema. First, the lead ACP sends an *MA\_ExecutionPriorityListCommandMT* message where the *AppliesToID* field is defined with the *SystemID* of the follower ACP. The *MA\_ExecutionPriorityListCommandMT* message shall specify the *ExecutionPriorityListActivationDataID* field containing the ID of the Execution Priority List Activation Data message (or schema) that the lead ACP wishes to be active. Any new *MA\_ExecutionPriorityListCommandMT* will override any previously sent message, even if the *CommandID* has changed. When a *MA\_ExecutionPriorityListCommandMT* is sent with the *CommandedState* set to *CANCEL*, this will cancel the active command and EPL activation schema

In response to the command, the follower ACPs shall send

*MA\_ExecutionPriorityListCommandStatusMT\_1* to the lead ACP, with the *AppliesToID* field set to their System ID and the *CommandProcessingState* field set to *RECEIVED* to indicate receipt of the command. The follower ACPs shall then use the *ExecutionPriorityListActivationDataID* to query their onboard MP to retrieve the *MA\_ExecutionPriorityListActivationDataMT* (see "Query Execution Priority List Activation Data"). Then the follower ACPs will use the *MA\_ExecutionPriorityListActivationDataMT* to query for all the EPL messages (see "Query Execution Priority List"). Once the follower ACPs have finished querying, the follower ACPs shall send *MA\_ExecutionPriorityListCommandStatusMT\_2* to the lead ACP with the *AppliesToID* field set to their System ID and the *CommandProcessingState* set to *ACCEPTED*.

The follower ACPs shall use their location and the *MA\_ExecutionPriorityListActivationDataMT* to determine which EPL ID should be activated (default or geozone based). Once doing so, the follower ACPs will send *MA\_ExecutionPriorityListActivationStatusMT* to the lead ACP containing the *ExecutionPriorityListActivationDataID*, *ActiveExecutionPriorityListID*, and the *AppliesToID* field set to their System ID.



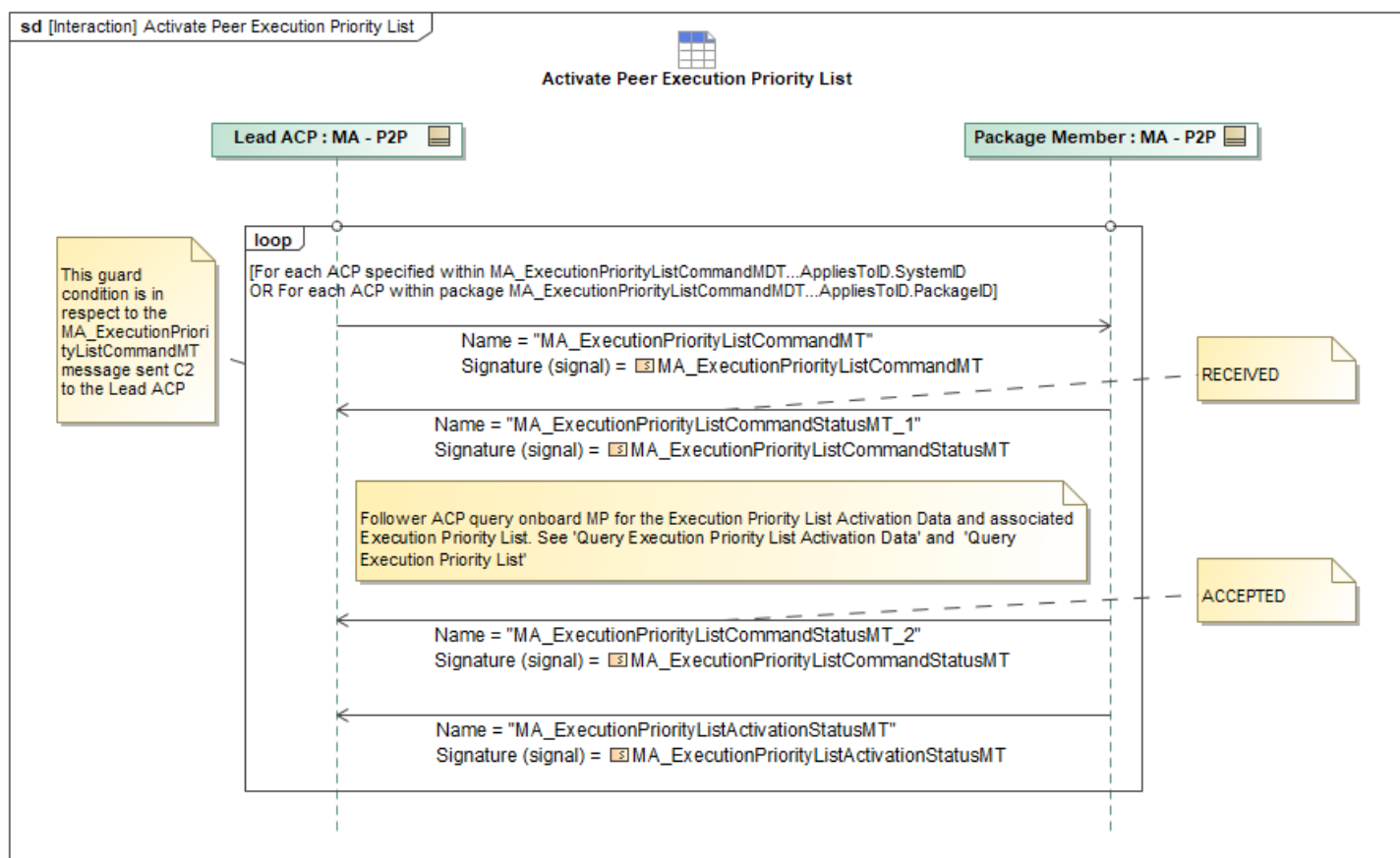


Figure 1-111: Activate Peer Execution Priority List

Table 1-110: Activate Peer Execution Priority List

| Message Name (Name:Type)                                                                  | Required Fields (Name:Type) |
|-------------------------------------------------------------------------------------------|-----------------------------|
| MA_ExecutionPriorityListActivationStatusMT:<br>MA_ExecutionPriorityListActivationStatusMT | None                        |
| MA_ExecutionPriorityListCommandMT:<br>MA_ExecutionPriorityListCommandMT                   | None                        |
| MA_ExecutionPriorityListCommandStatusMT_1<br>: MA_ExecutionPriorityListCommandStatusMT    | None                        |
| MA_ExecutionPriorityListCommandStatusMT_2<br>: MA_ExecutionPriorityListCommandStatusMT    | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.12.4 Peer Ignore Zone Constraints Behaviors

### 1.2.12.4.1 Activate Peer Ignore Zone Constraints

The Activate Peer Ignore Zone Constraints sequence diagram below specifies the interaction where the lead ACP enables a zone constraint override for one or more tasks for its followers. Note that the intent of this sequence only covers OpZone and OpVolume constraints. First, the lead ACP sends a *MA\_IgnoreConstraintCommandMT* message where the *AppliesToID* field is defined with the

*SystemID* of the follower ACP. This message contains either a list of zone constraints to ignore (*ListOfConstraintsToIgnore*), or an indicator that all constraints should be ignored (*IgnoreAllConstraints*). It also contains the ID(s) of the task(s) the command applies to (*ReferenceType*). Multiple commands can be active on the MA and each should have a unique *CommandID*. For new commands, the *CommandState* shall be set to *NEW*. For updates to active commands, the *CommandState* shall be set to *UPDATE* and the *CommandID* shall match that of the active message being updated by the leader MA. To cancel an active command, the *CommandState* shall be set to *CANCEL* with *CommandID* set to that of the message being canceled.

In response to the command, the follower ACPs shall send *MA\_IgnoreConstraintCommandStatusMT\_1* to the lead ACP, with the *AppliesToID* field set to their System ID and the *CommandProcessingState* field set to *RECEIVED* to indicate receipt of the command. The Follower ACPs proceed to execute the command by updating their ignored constraint and send *MA\_IgnoreConstraintCommandStatusMT\_2* to the lead ACP with the *AppliesToID* field set to their System ID and the *CommandProcessingState* set to *ACCEPTED*, to indicate they have completed the command.



Figure 1-112: Activate Peer Ignore Zone Constraints

Table 1-111: Activate Peer Ignore Zone Constraints

| Message Name (Name:Type)                                                           | Required Fields (Name:Type)                                                     |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>MA_IgnoreConstraintCommandMT:</b><br>MA_IgnoreConstraintCommandMT               | ListOfConstraintsToIgnore: MA_RegionChoiceType<br>IgnoreAllConstraints: boolean |
| <b>MA_IgnoreConstraintCommandStatusMT_1:</b><br>MA_IgnoreConstraintCommandStatusMT | None                                                                            |

| Message Name (Name:Type)                                                    | Required Fields (Name:Type) |
|-----------------------------------------------------------------------------|-----------------------------|
| MA_IgnoreConstraintCommandStatusMT_2:<br>MA_IgnoreConstraintCommandStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.13 Peer Tasking Package

### 1.2.13.1 Allocate an Altitude Stacked Marshall

To allocate an Altitude Stacked Marshall, the ACP designated as the Package lead sends an *MA\_FlightCommandMT\_1* to all allocated Package partners, using Capability for *MA\_FlightCommandType* with the *Capability.AltitudeStackedMarshall* field populated with the relevant kinematic parameters for a new task or with *MA\_FlightCommandMT\_2* using Activity to modify an existing flight activity. All Package members then respond with *MA\_FlightCommandStatusMT\_1* to indicate if the command was accepted or otherwise. If the command is rejected, the *MA\_FlightCommandStatusMT\_2* message will specify the *CannotComplyDetails* field to provide further details. The peer may optionally specify a suggested flight task with adjusted parameters using the *MA\_TaskMT* message. The peer ACP will forward *MA\_TaskMT* to the Package lead. The Package lead can restart the sequence with the suggestion or choose an entirely different flight command.

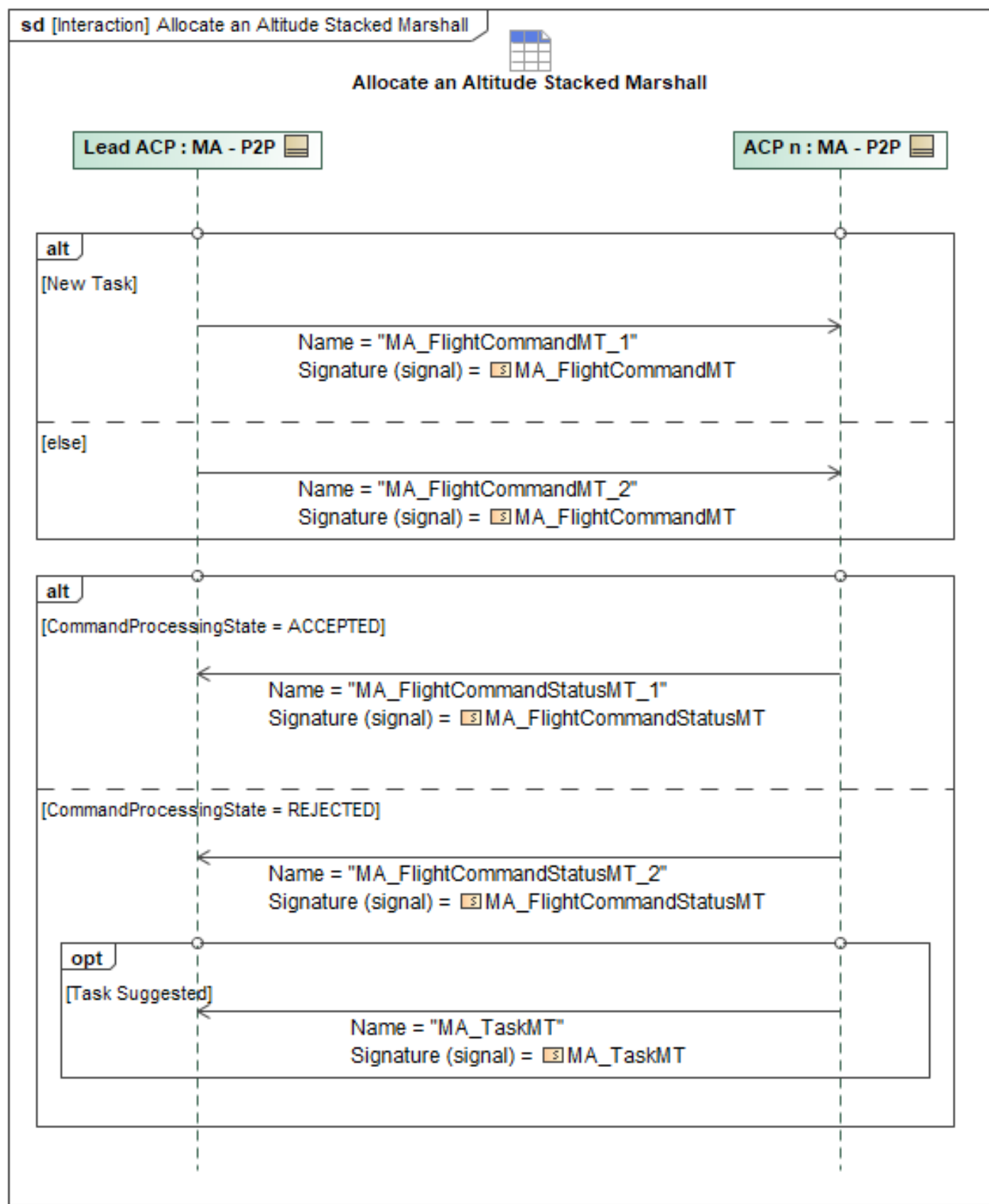


Figure 1-113: Allocate an Altitude Stacked Marshall

Table 1-112: Allocate an Altitude Stacked Marshall

| Message Name (Name:Type)                                | Required Fields (Name:Type)                                                                           |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| MA_FlightCommandMT_1:<br>MA_FlightCommandMT             | Capability: MA_FlightCapabilityCommandType<br>AltitudeStackedMarshall: MA_AltitudeStackedMarshallType |
| MA_FlightCommandMT_2:<br>MA_FlightCommandMT             | Activity: MA_FlightActivityCommandType                                                                |
| MA_FlightCommandStatusMT_1:<br>MA_FlightCommandStatusMT | None                                                                                                  |
| MA_FlightCommandStatusMT_2:<br>MA_FlightCommandStatusMT | CannotComplyDetails: MA_CannotComplyDetailsType                                                       |
| MA_TaskMT: MA_TaskMT                                    | None                                                                                                  |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.13.2 Allocating an Escort Task

To allocate an Escort Task, the ACP designated as the Package lead sends an *MA\_EscortCommandMT* to all allocated Package partners to invoke a previously advertised *MA\_EscortCapabilityMT*. This command is not necessarily sent to all members, only to those that are being engaged in the Escort command. All commanded Package members then respond with *MA\_EscortCommandStatusMT\_1* to indicate if the command was accepted or *MA\_EscortCommandStatusMT\_2* with the *CommandProcessingStateReason* field populated if rejected.

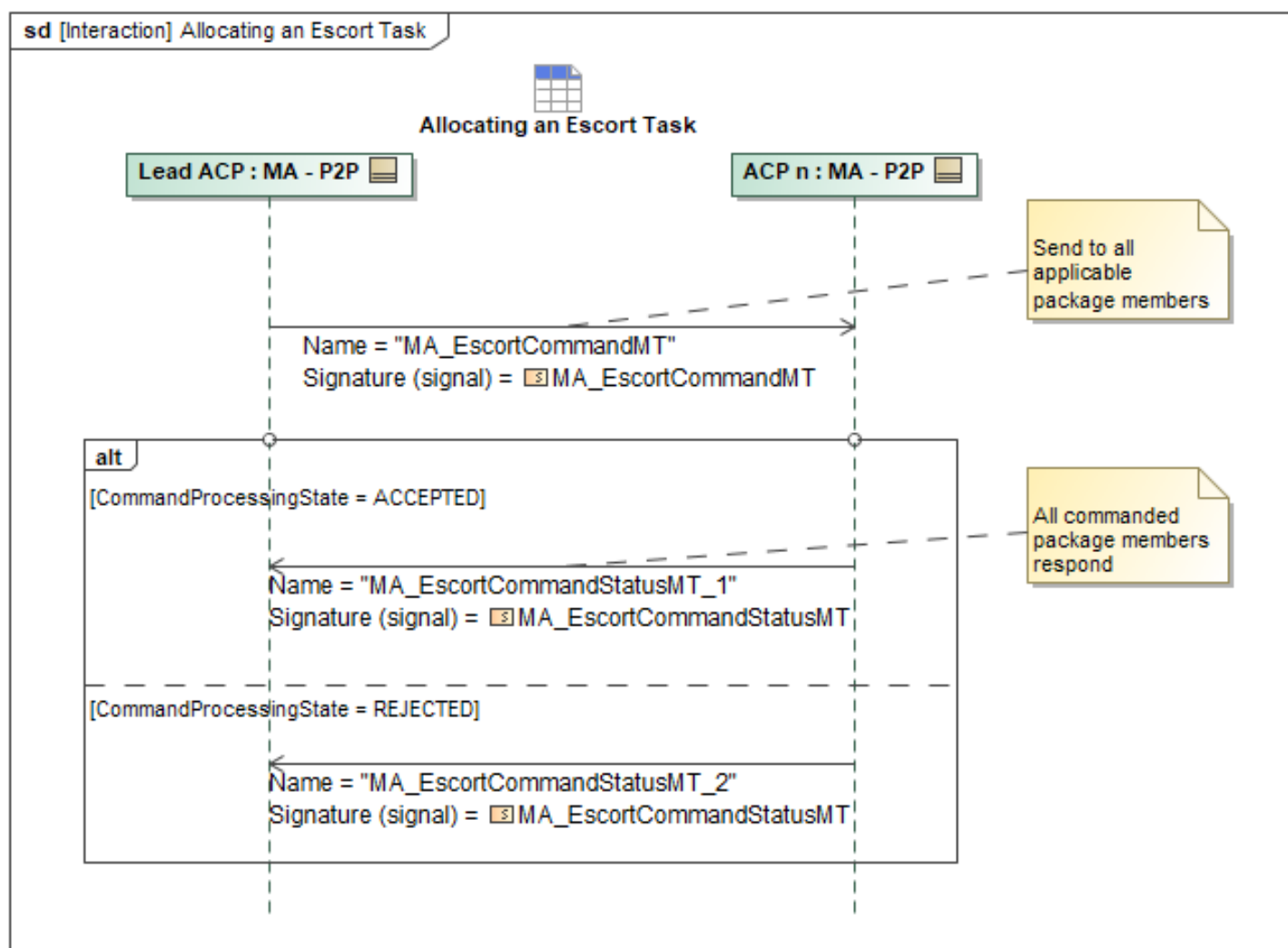


Figure 1-114: Allocating an Escort Task

Table 1-113: Allocating an Escort Task

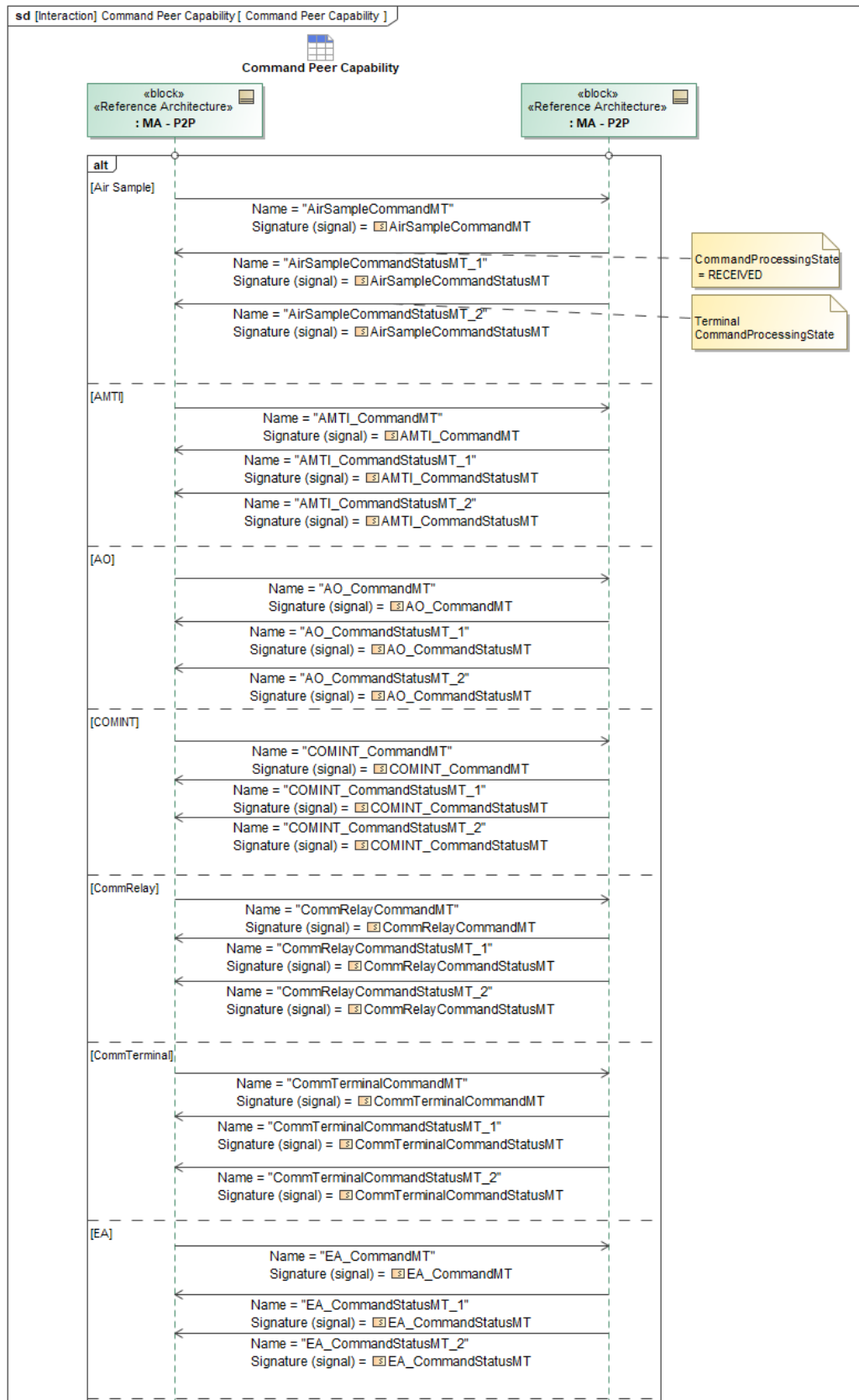
| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| MA_EscortCommandMT:<br>MA_EscortCommandMT               | None                                           |
| MA_EscortCommandStatusMT_1:<br>MA_EscortCommandStatusMT | None                                           |
| MA_EscortCommandStatusMT_2:<br>MA_EscortCommandStatusMT | CommandProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

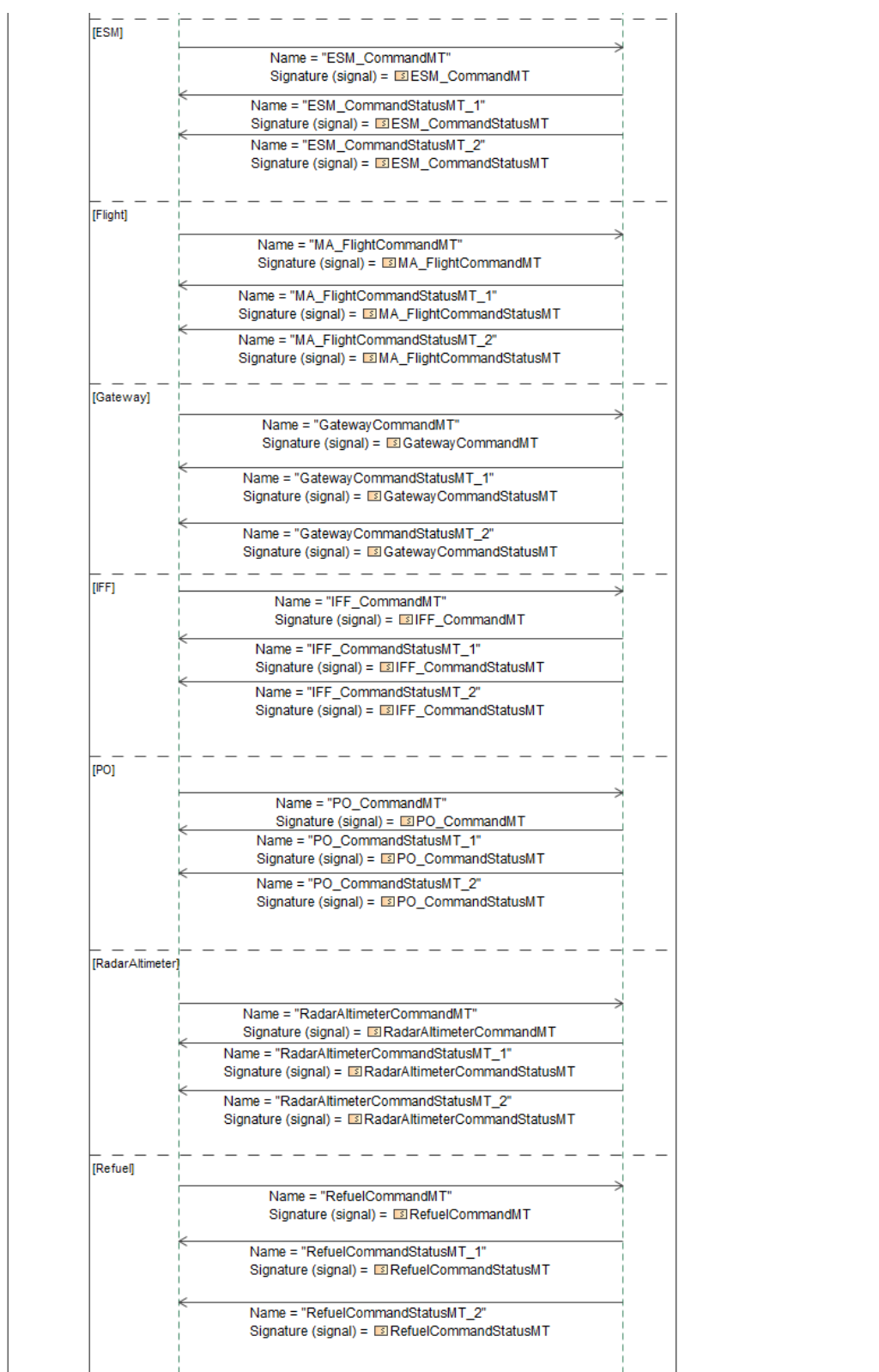
### 1.2.13.3 Command Peer Capability

An ACP can send a *[Capability]Command* to a Peer, commanding the Peer to execute the appropriate *Capability*. In response, the recipient of the command checks its *ApprovalPolicy* to verify that the commanding ACP has authority to command this specific *Capability*. The Peer will send a

*[Capability]CommandStatus ("\_CommandStatusMT\_1)* indicating that the command has been received. Once the command reaches a terminal state (*ACCEPTED*, *REJECTED*, *CANCELED*), the Peer will send a *[Capability]CommandStatus ("\_CommandStatusMT\_2)* indicating that the command has reached that terminal state. This interaction assumes that the commanding ACP is aware that the commanded ACP is able to execute the commanded Capability.







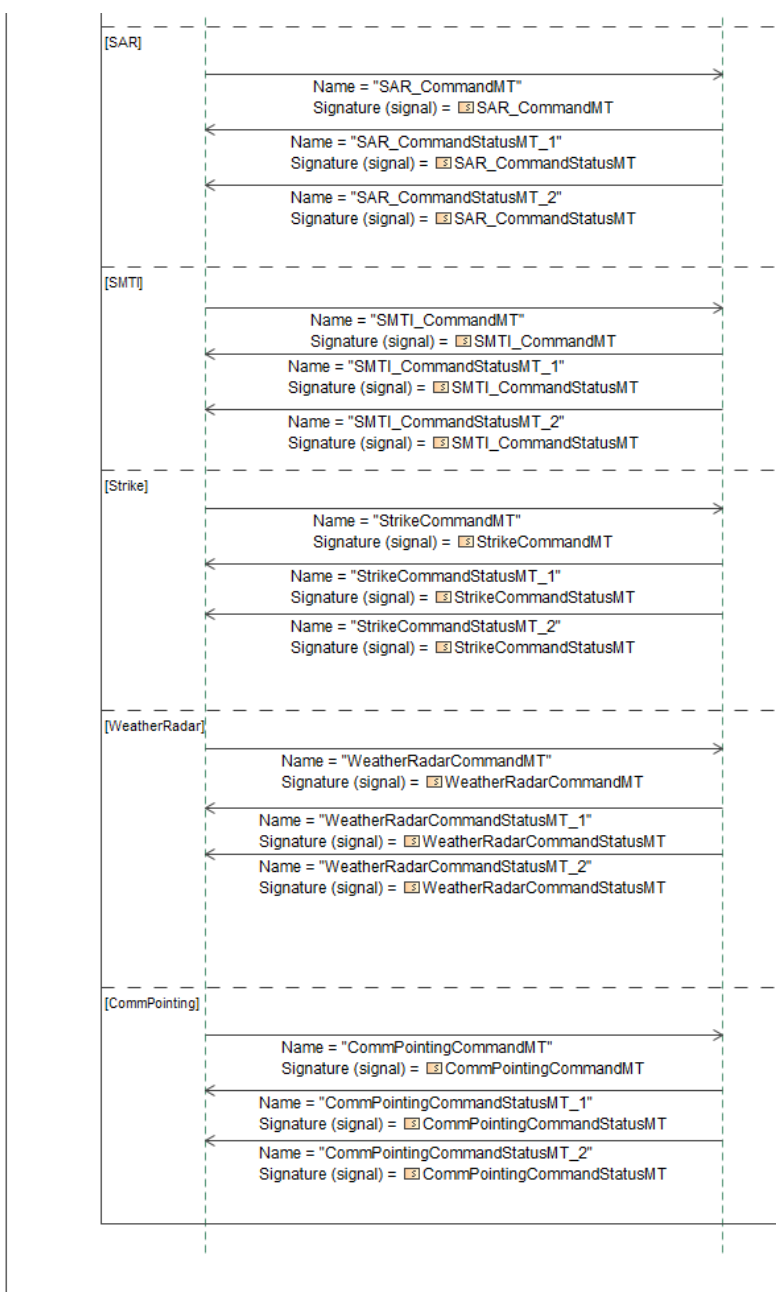


Figure 1-115: Command Peer Capability

Table 1-114: Command Peer Capability

| Message Name (Name:Type)                                       | Required Fields (Name:Type)                |
|----------------------------------------------------------------|--------------------------------------------|
| <b>AirSampleCommandMT:</b><br>AirSampleCommandMT               | Capability: AirSampleCapabilityCommandType |
| <b>AirSampleCommandStatusMT_1:</b><br>AirSampleCommandStatusMT | None                                       |
| <b>AirSampleCommandStatusMT_2:</b><br>AirSampleCommandStatusMT | None                                       |
| <b>AMTI_CommandMT:</b> AMTI_CommandMT                          | Capability: AMTI_CapabilityCommandType     |

| Message Name (Name:Type)                                      | Required Fields (Name:Type)                   |
|---------------------------------------------------------------|-----------------------------------------------|
| AMTI_CommandStatusMT_1:<br>AMTI_CommandStatusMT               | None                                          |
| AMTI_CommandStatusMT_2:<br>AMTI_CommandStatusMT               | None                                          |
| AO_CommandMT: AO_CommandMT                                    | Capability: AO_CapabilityCommandType          |
| AO_CommandStatusMT_1:<br>AO_CommandStatusMT                   | None                                          |
| AO_CommandStatusMT_2:<br>AO_CommandStatusMT                   | None                                          |
| COMINT_CommandMT: COMINT_CommandMT                            | Capability: COMINT_CapabilityCommandType      |
| COMINT_CommandStatusMT_1:<br>COMINT_CommandStatusMT           | None                                          |
| COMINT_CommandStatusMT_2:<br>COMINT_CommandStatusMT           | None                                          |
| CommPointingCommandMT:<br>CommPointingCommandMT               | None                                          |
| CommPointingCommandStatusMT_1:<br>CommPointingCommandStatusMT | None                                          |
| CommPointingCommandStatusMT_2:<br>CommPointingCommandStatusMT | None                                          |
| CommRelayCommandMT:<br>CommRelayCommandMT                     | Capability: CommRelayCapabilityCommandType    |
| CommRelayCommandStatusMT_1:<br>CommRelayCommandStatusMT       | None                                          |
| CommRelayCommandStatusMT_2:<br>CommRelayCommandStatusMT       | None                                          |
| CommTerminalCommandMT:<br>CommTerminalCommandMT               | Capability: CommTerminalCapabilityCommandType |
| CommTerminalCommandStatusMT_1:<br>CommTerminalCommandStatusMT | None                                          |
| CommTerminalCommandStatusMT_2:<br>CommTerminalCommandStatusMT | None                                          |
| EA_CommandMT: EA_CommandMT                                    | Capability: EA_CapabilityCommandType          |
| EA_CommandStatusMT_1:<br>EA_CommandStatusMT                   | None                                          |
| EA_CommandStatusMT_2:<br>EA_CommandStatusMT                   | None                                          |
| ESM_CommandMT: ESM_CommandMT                                  | Capability: ESM_CapabilityCommandType         |
| ESM_CommandStatusMT_1:<br>ESM_CommandStatusMT                 | None                                          |
| ESM_CommandStatusMT_2:<br>ESM_CommandStatusMT                 | None                                          |

| Message Name (Name:Type)                                          | Required Fields (Name:Type)                     |
|-------------------------------------------------------------------|-------------------------------------------------|
| GatewayCommandMT: GatewayCommandMT                                | Capability: GatewayCapabilityCommandType        |
| GatewayCommandStatusMT_1:<br>GatewayCommandStatusMT               | None                                            |
| GatewayCommandStatusMT_2:<br>GatewayCommandStatusMT               | None                                            |
| IFF_CommandMT: IFF_CommandMT                                      | Capability: IFF_CapabilityCommandType           |
| IFF_CommandStatusMT_1:<br>IFF_CommandStatusMT                     | None                                            |
| IFF_CommandStatusMT_2:<br>IFF_CommandStatusMT                     | None                                            |
| MA_FlightCommandMT:<br>MA_FlightCommandMT                         | Capability: MA_FlightCapabilityCommandType      |
| MA_FlightCommandStatusMT_1:<br>MA_FlightCommandStatusMT           | None                                            |
| MA_FlightCommandStatusMT_2:<br>MA_FlightCommandStatusMT           | None                                            |
| PO_CommandMT: PO_CommandMT                                        | Capability: PO_CapabilityCommandType            |
| PO_CommandStatusMT_1:<br>PO_CommandStatusMT                       | None                                            |
| PO_CommandStatusMT_2:<br>PO_CommandStatusMT                       | None                                            |
| RadarAltimeterCommandMT:<br>RadarAltimeterCommandMT               | Capability: RadarAltimeterCapabilityCommandType |
| RadarAltimeterCommandStatusMT_1:<br>RadarAltimeterCommandStatusMT | None                                            |
| RadarAltimeterCommandStatusMT_2:<br>RadarAltimeterCommandStatusMT | None                                            |
| RefuelCommandMT: RefuelCommandMT                                  | Capability: RefuelCapabilityExtendCommandType   |
| RefuelCommandStatusMT_1:<br>RefuelCommandStatusMT                 | None                                            |
| RefuelCommandStatusMT_2:<br>RefuelCommandStatusMT                 | None                                            |
| SAR_CommandMT: SAR_CommandMT                                      | Capability: SAR_CapabilityCommandType           |
| SAR_CommandStatusMT_1:<br>SAR_CommandStatusMT                     | None                                            |
| SAR_CommandStatusMT_2:<br>SAR_CommandStatusMT                     | None                                            |
| SMTI_CommandMT: SMTI_CommandMT                                    | Capability: SMTI_CapabilityCommandType          |
| SMTI_CommandStatusMT_1:<br>SMTI_CommandStatusMT                   | None                                            |
| SMTI_CommandStatusMT_2:<br>SMTI_CommandStatusMT                   | None                                            |

| Message Name (Name:Type)                                          | Required Fields (Name:Type)                   |
|-------------------------------------------------------------------|-----------------------------------------------|
| <b>StrikeCommandMT: StrikeCommandMT</b>                           | Capability: StrikeCapabilityCommandType       |
| <b>StrikeCommandStatusMT_1: StrikeCommandStatusMT</b>             | None                                          |
| <b>StrikeCommandStatusMT_2: StrikeCommandStatusMT</b>             | None                                          |
| <b>WeatherRadarCommandMT: WeatherRadarCommandMT</b>               | Capability: WeatherRadarCapabilityCommandType |
| <b>WeatherRadarCommandStatusMT_1: WeatherRadarCommandStatusMT</b> | None                                          |
| <b>WeatherRadarCommandStatusMT_2: WeatherRadarCommandStatusMT</b> | None                                          |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

#### 1.2.13.4 Command Peer via Action

This interaction allows a lead to command a peer via an action. It also allows a lead to propose an action to a peer and for the peer to accept or reject the proposed action. If the peer rejects the proposed action, it can propose an alternative.

The lead sends *MA\_ActionMT\_1* to the peer. If the lead wants the peer to evaluate but not execute the action, the *ActionGuidance.TacticalPlanningAndExecution* field is set to *BE\_PREPARED\_TO*.

The lead then sends an *MA\_ActionCommandMT* to prompt the peer to evaluate (and possibly execute) the *MA\_ActionMT\_1*.

While the command is being processed, the peer may optionally send *MA\_ActionCommandStatusMT\_1* with the *CommandProcessingState* set to *RECEIVED*.

If the *Action* can be executed, the peer will respond with an *MA\_ActionCommandStatusMT\_3* with the *CommandProcessingState* set to *ACCEPTED*. The peer then sends an *MA\_ActionStatusMT* whenever the status of the action changes.

If the peer cannot execute the *Action*, it will respond with *MA\_ActionCommandStatusMT\_2* with the *CommandProcessingState* set to *REJECTED* and set the *CommandProcessingStateReason.AssociatedID* in *MA\_ActionCommandStatusMT\_2* to the *ActionID* of *MA\_ActionMT\_1*. If the peer is able to generate an action that it can accept, it will respond with *MA\_ActionMT\_2*. The *Metadata.Traceability.ActionID* field in *MA\_ActionMT\_2* must be set to the *ActionID* of *MA\_ActionMT\_1*.

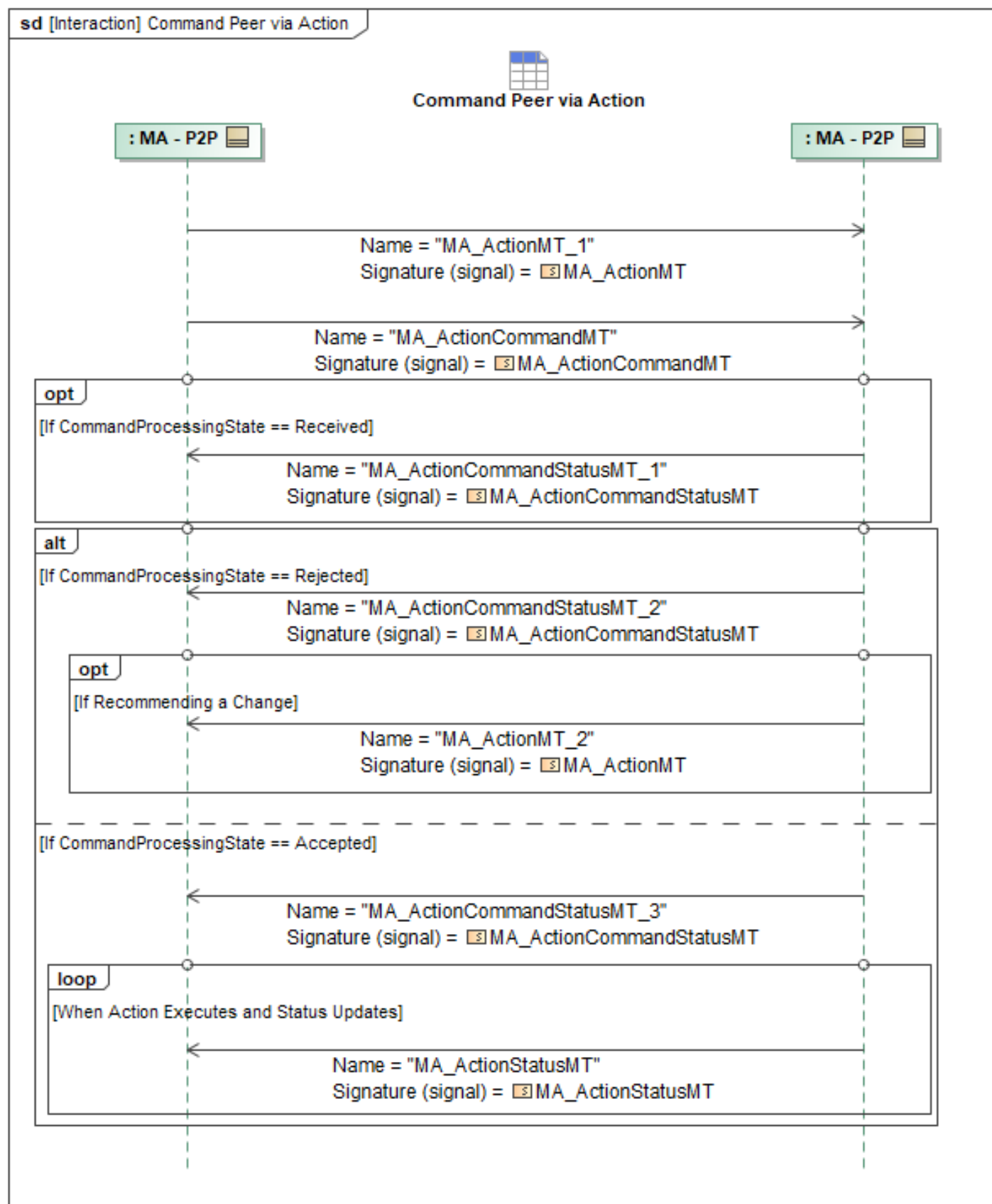


Figure 1-116: Command Peer via Action

Table 1-115: Command Peer via Action

| Message Name (Name:Type)                                | Required Fields (Name:Type)                                             |
|---------------------------------------------------------|-------------------------------------------------------------------------|
| MA_ActionCommandMT:<br>MA_ActionCommandMT               | None                                                                    |
| MA_ActionCommandStatusMT_1:<br>MA_ActionCommandStatusMT | None                                                                    |
| MA_ActionCommandStatusMT_2:<br>MA_ActionCommandStatusMT | CommandProcessingStateReason: CannotComplyType<br>AssociatedID: ID_Type |
| MA_ActionCommandStatusMT_3:<br>MA_ActionCommandStatusMT | None                                                                    |
| MA_ActionMT_1: MA_ActionMT                              | None                                                                    |
| MA_ActionMT_2: MA_ActionMT                              | Metadata: RequirementMetadataType<br>Traceability: TraceabilityType     |
| MA_ActionStatusMT: MA_ActionStatusMT                    | SystemID: SystemID_Type                                                 |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.13.5 Command Peer(s) via Task

*MA\_TaskMT* contains different tasks that a system can perform against a specific entity or geographical location. The MA Lead System sends the *MA\_TaskMT* to the peer MA immediately followed by a *MA\_TaskCommandMT* to activate it. Within the *MA\_TaskCommandMT* the *ConstraintOverride.Allocation.RequiredAllocation* is used to specify the ACP(s) by their *SystemID*. *MA\_TaskCommandStatusMT* is then sent back to indicate if the command was accepted. Each individual ACP will then report *MA\_TaskStatusMT* individually to the MA Lead system.

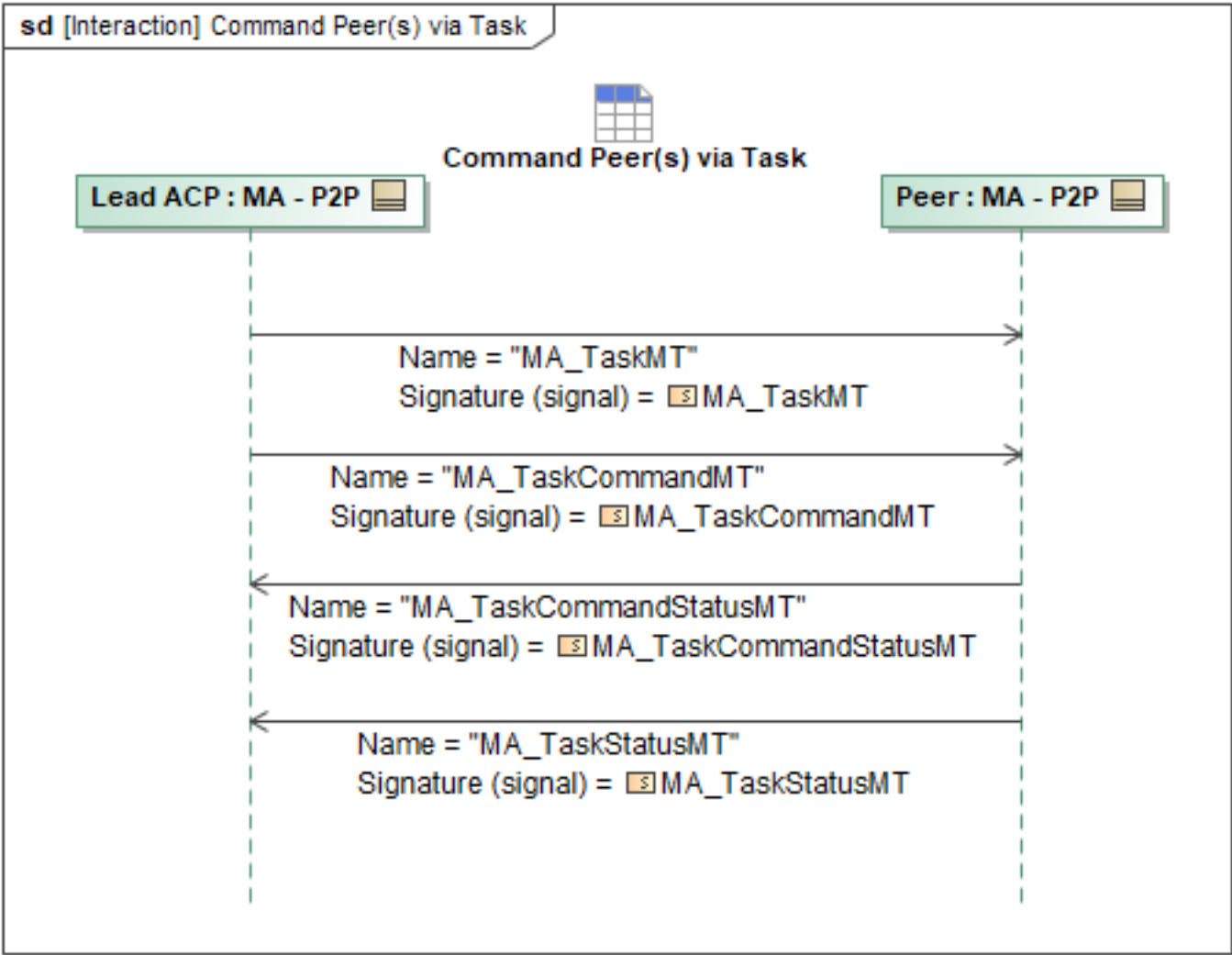


Figure 1-117: Command Peer(s) via Task

Table 1-116: Command Peer(s) via Task

| Message Name (Name:Type)                       | Required Fields (Name:Type)                                                                                                                                                       |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MA_TaskCommandMT: MA_TaskCommandMT             | RequiredAllocation: MA_RequirementAllocationConstraintType<br>SystemID: SystemID_Type<br>Allocation: MA_AllocationChoiceType<br>ConstraintOverride: MA_RequirementConstraintsType |
| MA_TaskCommandStatusMT: MA_TaskCommandStatusMT | None                                                                                                                                                                              |
| MA_TaskMT: MA_TaskMT                           | None                                                                                                                                                                              |
| MA_TaskStatusMT: MA_TaskStatusMT               | None                                                                                                                                                                              |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)



### 1.2.13.6 Control Mode Authorization

Peer systems seeking to control MA should be made aware of the capabilities that MA has been granted by Flight Autonomy. The VI communicates how MA can control the vehicle with *MA\_FlightCapabilityMT*. The *MA\_FlightCapabilityMT* message requires the *FlightCapabilityPerformanceProfile* field be populated with the appropriate profile subtypes of the capabilities advertised and any parameters needed by MA for commanding each type present. It then sends *MA\_FlightCapabilityStatusMT* message to communicate the availability of control modes to MA control (see Control Authorization in the Vehicle Interface for more details). Both of these messages should be passed to Peer nodes when connecting to MA and *MA\_FlightCapabilityStatusMT* should be forwarded along to Peer nodes whenever there is a change in flight capabilities.

Once the required capabilities are available, Peers with custody can issue commands to the MA via the Direct HSA/CSA Command use case or the Direct Waypoint Command use case. The *MA\_FlightCommandMT* message within these use cases carries control mode-specific data, as described in later sections.

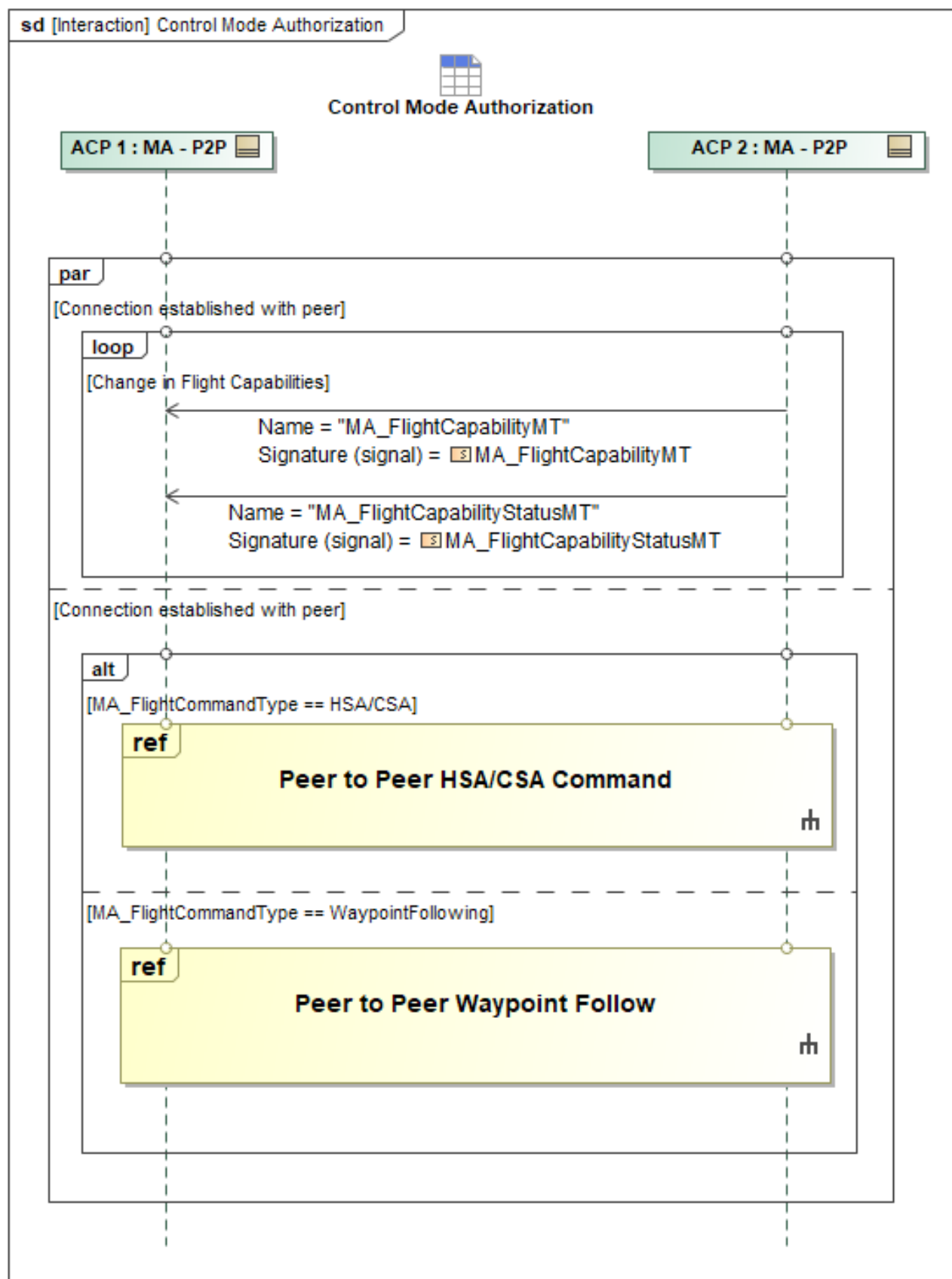


Figure 1-118: Control Mode Authorization

Table 1-117: Control Mode Authorization

| Message Name (Name:Type)                                            | Required Fields (Name:Type)                                                      |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>MA_FlightCapabilityMT: MA_FlightCapabilityMT</b>                 | FlightCapabilityPerformanceProfile:<br>MA_FlightCapabilityPerformanceProfileType |
| <b>MA_FlightCapabilityStatusMT:<br/>MA_FlightCapabilityStatusMT</b> | None                                                                             |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.13.7 Package Plan Approval Status

The Package Plan Approval Status sequence diagram specifies the interaction of two ACPs where the ACP has received approval for the plans indicated, and the ACP is synchronizing across the package that the plans have now been designated as approved. While the Approval Status specifically statuses the response of the Approval Request, the plan approval status messages are a status of the approval state of the plan itself, and they are utilized by the system to confirm individual plans approval or rejection status. As each executable plan is approved, the ACP sends the plan approval status to another ACP for awareness and tracking purposes via the *TaskPlanApprovalStatusMT*, *MissionPlanApprovalStatusMT*, *RoutePlanApprovalStatusMT*, and *ActivityPlanApprovalStatusMT* messages. For each of the messages sent, the *PlanID* and *PlanApprovalState* is populated with the applicable plan UUID and approval state, respectively.

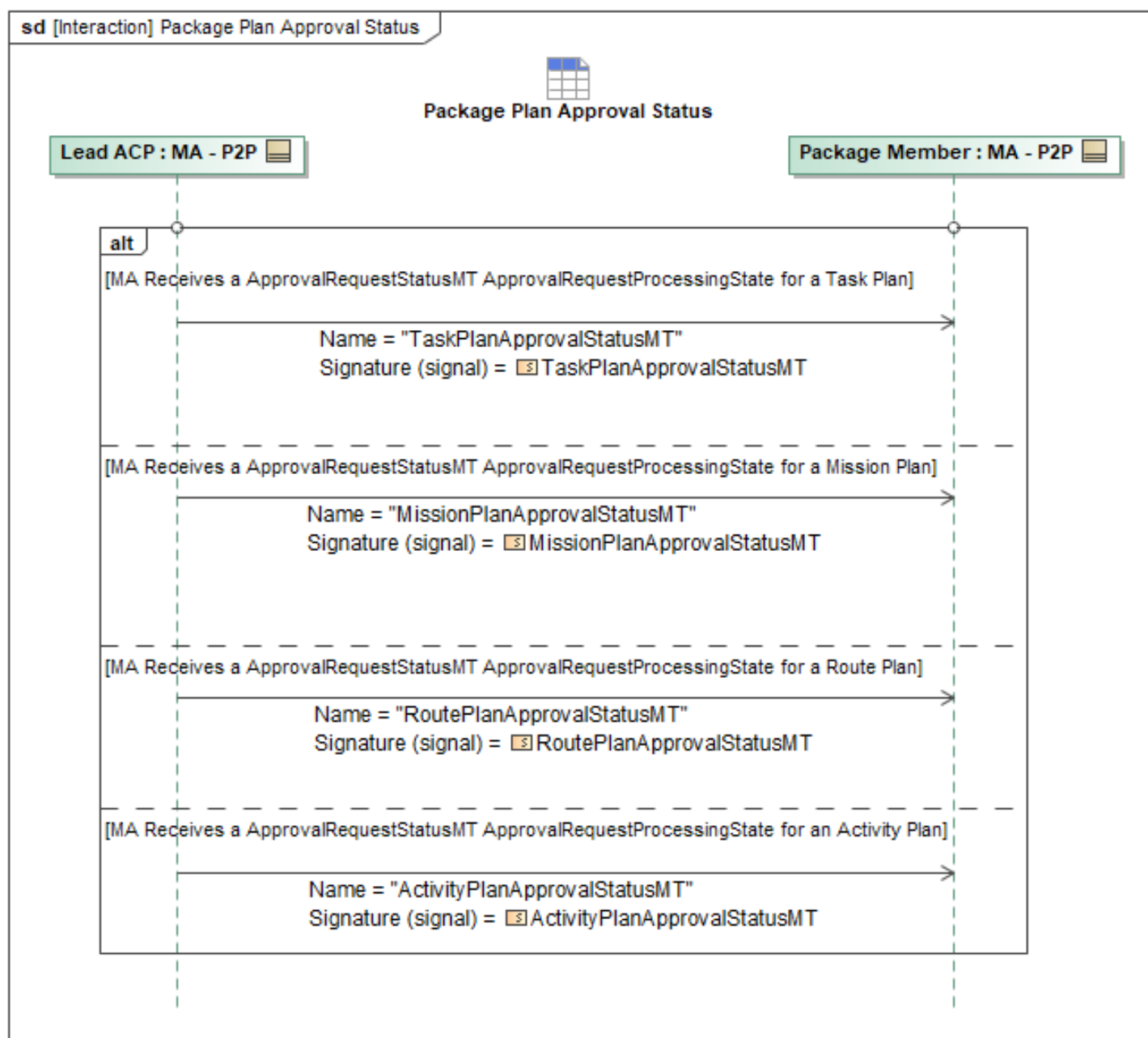


Figure 1-119: Package Plan Approval Status

Table 1-118: Package Plan Approval Status

| Message Name (Name:Type)                                      | Required Fields (Name:Type) |
|---------------------------------------------------------------|-----------------------------|
| ActivityPlanApprovalStatusMT:<br>ActivityPlanApprovalStatusMT | None                        |
| MissionPlanApprovalStatusMT:<br>MissionPlanApprovalStatusMT   | None                        |
| RoutePlanApprovalStatusMT:<br>RoutePlanApprovalStatusMT       | None                        |

| Message Name (Name:Type)                              | Required Fields (Name:Type) |
|-------------------------------------------------------|-----------------------------|
| TaskPlanApprovalStatusMT:<br>TaskPlanApprovalStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.13.8 Peer Activate or Deactivate Mission Plan

A Package Leader can activate or deactivate a mission plan by sending a *MA\_MissionPlanActivationCommandMT* message to a Follower. The *MA\_MissionPlanActivationCommandMT*'s **CommandState** field shall be *NEW*, the **Command** field shall specify which mission plan to activate or deactivate, and **CommandID** field shall be a unique identifier for the command. To activate or deactivate a mission plan and all its sub-plans, send *Command.ActivationDetails.ByMissionPlan*. **To activate an entire mission plan, set *Command.ActivationDetails.ByMissionPlan.ActivationCommand* to *ACTIVATE*. To deactivate an entire mission plan, set *Command.ActivationDetails.ByMissionPlan.ActivationCommand* to *DEACTIVATE*.** To activate or deactivate only a subset of subplans belonging to a mission plan, send *Command.ActivationDetails.BySubPlan* detailing which subplans should be activated or deactivated. **To activate a subplan, set *Command.ActivationDetails.BySubPlan.\*Plan.CommandType* to *ACTIVATE*. To deactivate a subplan, set *Command.ActivationDetails.BySubPlan.\*Plan.CommandType* to *DEACTIVATE*.** To activate or deactivate only a specific *ExecutionPlanSet* belonging to a mission plan, send *Command.ActivationDetails.ByExecutionPlanSet*. **To activate a specific *ExecutionPlanSet*, set *Command.ActivationDetails.ByExecutionPlanSet.CommandType* to *ACTIVATE*. To deactivate a specific *ExecutionPlanSet*, set *Command.ActivationDetails.ByExecutionPlanSet.CommandType* to *DEACTIVATE*.**

The Follower will send at least one *MA\_MissionPlanActivationCommandStatusMT* message, the final of which (*MissionPlanActivationCommandStatusMT\_2*) must indicate a terminal status, i.e. ***ActivationStatus.CommandProcessingState* field set to *ACCEPTED* or *REJECTED* and *ActivationStatus.CommandStatus* field set to *COMPLETED* or *FAILED*.** The Follower may optionally send *MA\_MissionPlanActivationCommandStatusMT* messages (*MA\_MissionPlanActivationCommandStatusMT\_1*) prior to the terminal status to indicate in-progress processing of the *MA\_MissionPlanActivationCommandMT*.

If a mission plan or subplan was activated or deactivated successfully, the Follower will send the Leader a *MA\_MissionPlanActivationStatusMT\_2* detailing the current state of the mission plan. If plan execution status has changed, the Follower will send the Leader a *MA\_MissionPlanExecutionStatusMT* with *Source* set to *ACTUAL* and *PlanExecutionStatus* detailing the change in execution status. For a mission plan operating with an *ExecutionSequence*, *PlanExecutionStatus.ExecutionPlanSetStatus* can be included to provide statuses on the *ExecutionPlanSets* within the mission plan.

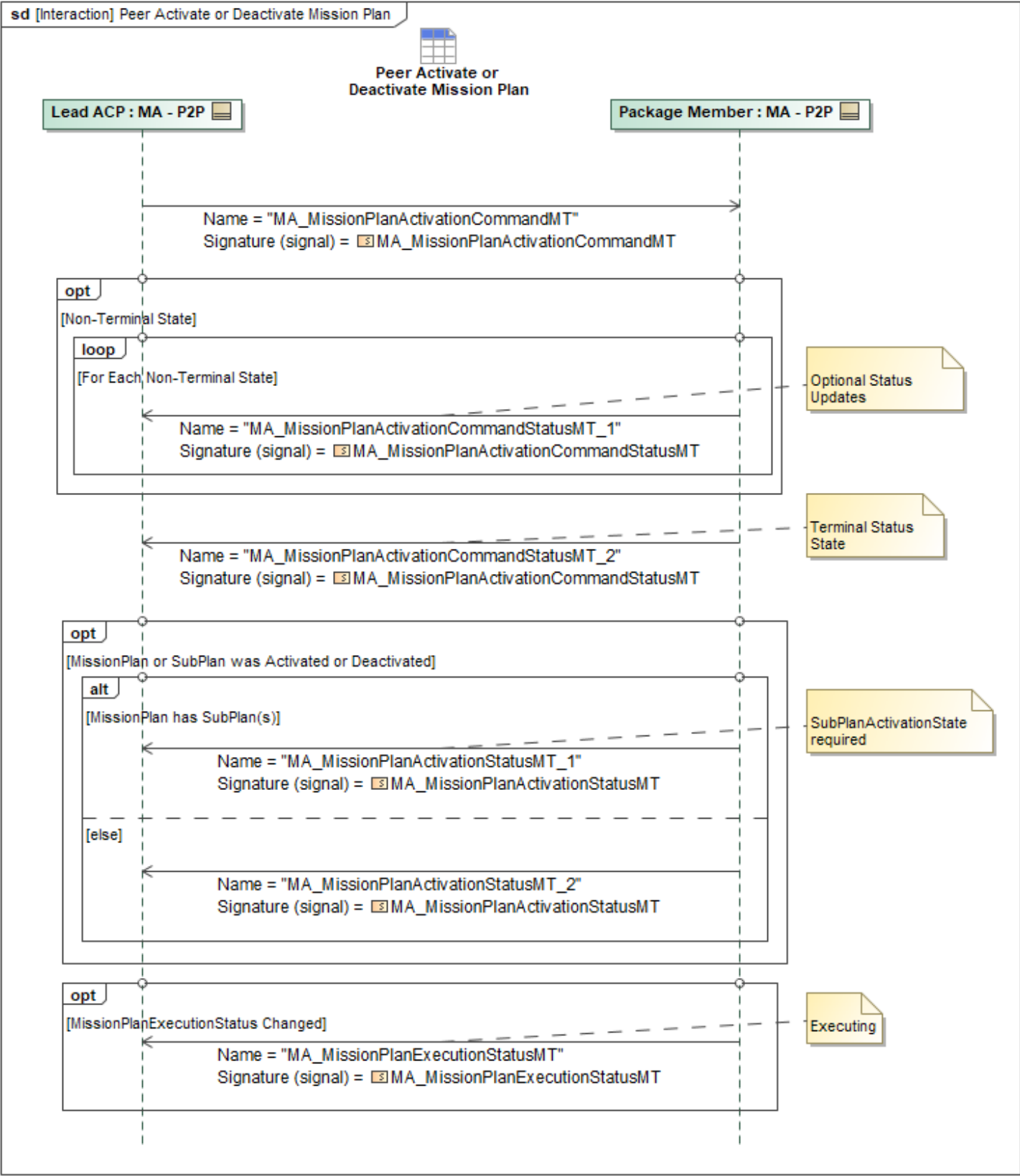


Figure 1-120: Peer Activate or Deactivate Mission Plan

Table 1-119: Peer Activate or Deactivate Mission Plan

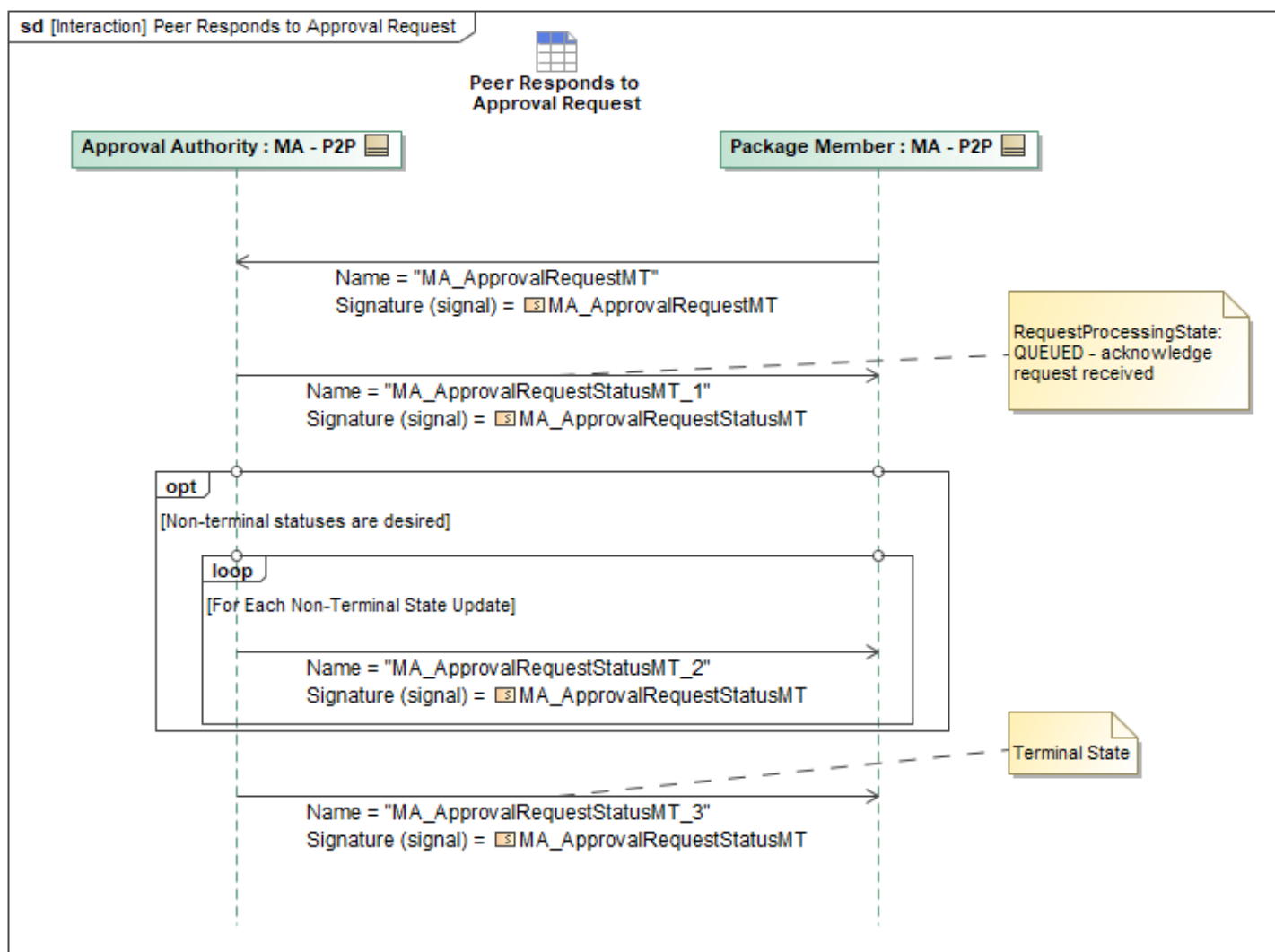
| Message Name (Name:Type)                                                              | Required Fields (Name:Type)                                                                              |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| MA_MissionPlanActivationCommandMT:<br>MA_MissionPlanActivationCommandMT               | None                                                                                                     |
| MA_MissionPlanActivationCommandStatusMT_1:<br>MA_MissionPlanActivationCommandStatusMT | None                                                                                                     |
| MA_MissionPlanActivationCommandStatusMT_2:<br>MA_MissionPlanActivationCommandStatusMT | None                                                                                                     |
| MA_MissionPlanActivationStatusMT_1:<br>MA_MissionPlanActivationStatusMT               | SubPlanActivationState: MA_PlanActivationStateType<br>SubPlanActivationState: MA_PlanActivationStateType |
| MA_MissionPlanActivationStatusMT_2:<br>MA_MissionPlanActivationStatusMT               | None                                                                                                     |
| MA_MissionPlanExecutionStatusMT:<br>MA_MissionPlanExecutionStatusMT                   | PlanExecutionStatus: MissionPlanExecutionStateType                                                       |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.13.9 Peer Responds to Approval Request

In the event that MA requires external approval from a Peer, such as when another package member is the Approval Authority for an *MA\_ApprovalPolicyMT* which contains a *Policy* field that is set to *APPROVAL\_REQUIRED*, the Approval Authority Peer can respond to the approval request sent by the Package Member. A Package Member sends *MA\_ApprovalRequestMT* to the Approval Authority ACP when it identifies, per the Approval Policy, that it must request approval for something. **The Approval Authority sends *MA\_ApprovalRequestStatusMT\_1* with *RequestProcessingState* set to *QUEUED* in order to acknowledge the request. Optionally, the Approving Authority can send non-terminal status updates using *MA\_ApprovalRequestStatusMT\_2* by setting *RequestProcessingState* as *PROCESSING*.** The Approval Authority will make an approval determination and send *MA\_ApprovalRequestStatusMT\_3* to the Package Member, populating the *ApprovalRequestProcessingState* field with the approval state.

This interaction is applicable for engagement-related approval requests when a Target Authority has already authorized a Targeting Act against a particular Target, whether or not the Leader ACP is specified as a Target Authority itself. The Leader ACP is able to approve a Follower's *MA\_ApprovalRequestMT* for performing the Targeting Act in these cases of having Target Authority pre-approval, as well as cases where the Leader ACP is listed as a Target Authority.



**Figure 1-121: Peer Responds to Approval Request**

**Table 1-120: Peer Responds to Approval Request**

| Message Name (Name:Type)                                    | Required Fields (Name:Type) |
|-------------------------------------------------------------|-----------------------------|
| MA_ApprovalRequestMT:<br>MA_ApprovalRequestMT               | None                        |
| MA_ApprovalRequestStatusMT_1:<br>MA_ApprovalRequestStatusMT | None                        |
| MA_ApprovalRequestStatusMT_2:<br>MA_ApprovalRequestStatusMT | None                        |
| MA_ApprovalRequestStatusMT_3:<br>MA_ApprovalRequestStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)



### 1.2.13.10 Peer to Peer HSA/CSA Command

Peer systems with custody can send heading, course, speed, and altitude (HSA/CSA) commands to a peer's MA via the *MA\_FlightCommandMT* UCI message. The ACP1 system sends an *MA\_FlightCommandMT* UCI message to command the system and invokes a previously advertised capability. The *MA\_FlightCommandMT* specifies the *Command.Capability.FlightControlMode.HSA\_CSA* field in order to specify the flight command mode. The peer MA responds with a *MA\_FlightCommandStatusMT* message, either approving or rejecting the command. If the command is *REJECTED*, the loop in the sequence will break and the *MA\_FlightCommandStatusMT* message will specify the *CannotComplyDetails* field to provide further details. ACP2 may optionally specify a suggested flight task with adjusted parameters that constitute a "best-effort" using the *MA\_TaskMT* message and *Flight* field. *MA\_TaskMT* is used to encapsulate a suggested flight task with the ability to define the parameters of the command that could be accepted. MA can restart the sequence with the suggestion or choose an entirely different flight command. The break may also occur if the *Activity State* is *COMPLETED* or *FAILED*.

If the command is *APPROVED*, the ACP will then begin publishing the *MA\_FlightActivityMT* report, which includes data on the commanded state of the ownship, and the execution state of the command described initially in *MA\_FlightCommandMT*. The cycle of *MA\_FlightCommandMT* > *MA\_FlightCommandStatusMT* > *MA\_FlightActivityMT* repeats until either control is taken away from MA, or MA relinquishes control.

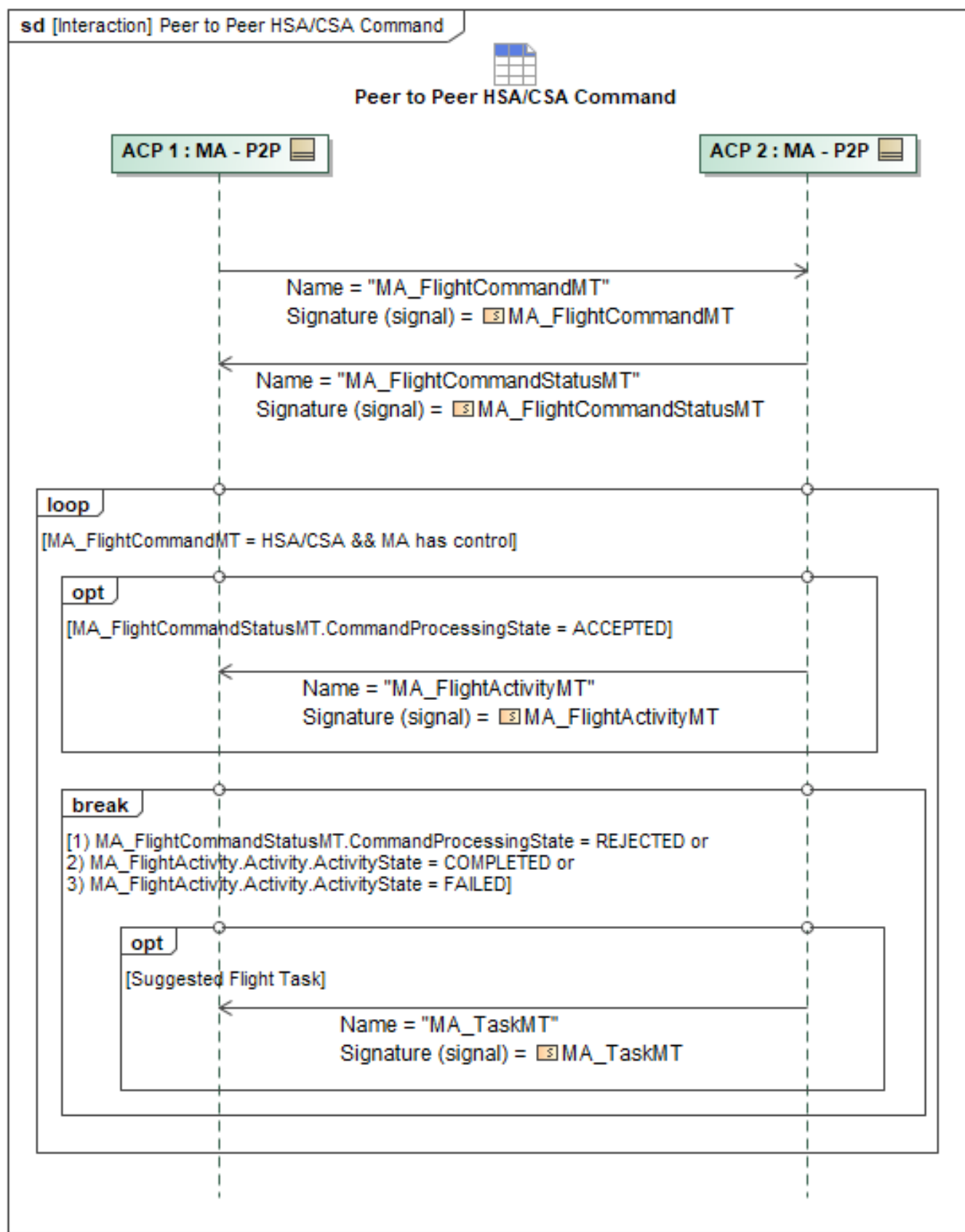


Figure 1-122: Peer to Peer HSA/CSA Command

Table 1-121: Peer to Peer HSA/CSA Command

| Message Name (Name:Type)                                      | Required Fields (Name:Type)                                                                                                  |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_FlightActivityMT: MA_FlightActivityMT</b>               | None                                                                                                                         |
| <b>MA_FlightCommandMT:<br/>MA_FlightCommandMT</b>             | HSA_CSA: MA_HSA_CSA_Type<br>Capability: MA_FlightCapabilityCommandType<br>FlightControlMode: MA_FlightControlModesChoiceType |
| <b>MA_FlightCommandStatusMT:<br/>MA_FlightCommandStatusMT</b> | None                                                                                                                         |
| <b>MA_TaskMT: MA_TaskMT</b>                                   | None                                                                                                                         |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.13.11 Peer to Peer Waypoint Follow

Waypoint following, alternatively referred to as route following, is a control mode where Mission Autonomy provides the peer a series of waypoints describing a desired flight path. Waypoint behavior may include simple commands such as fly-by or fly-through; complex commands including payload action and loiter/hold; or bounded by constraints such as vehicle attitude or timing. Route paths will be transferred between Mission Autonomy and its peers whenever the C2 system sends Mission Autonomy a new route, or an update to an existing route.

Communication of new waypoints begins with the transmission of *MA\_FlightCommandMT* from Mission Autonomy to the peer receiving waypoint following information. A well-conditioned *Command.Capability.FlightControlMode.WaypointFollowing* field will simply include a *Command.Capability.FlightControlMode.WaypointFollowing.Route* field. Detailed information on the contents of the *Route* and *Path* fields are available within the OMS UCI 2.3 schema. After this, the peer will send back *MA\_FlightCommandStatusMT* in response to the command. If the command is *APPROVED*, *MA\_FlightActivityMT* will report data on the commanded state of the ownship, and the execution state of the command described initially in *MA\_FlightCommandMT*. If the command is *REJECTED*, the loop in the sequence will break and the *MA\_FlightCommandStatusMT* message will include the *CannotComplyDetails* field to provide further details. MA can then restart the sequence or use *MA\_TaskMT* to encapsulate a suggested flight task with defined parameters. The break may also occur if the *Activity State* is *COMPLETED* or *FAILED*.

The cycle of *MA\_FlightCommandMT* > *MA\_FlightCommandStatusMT* > *MA\_FlightActivityMT* repeats until either control is taken away from MA, or MA relinquishes control.

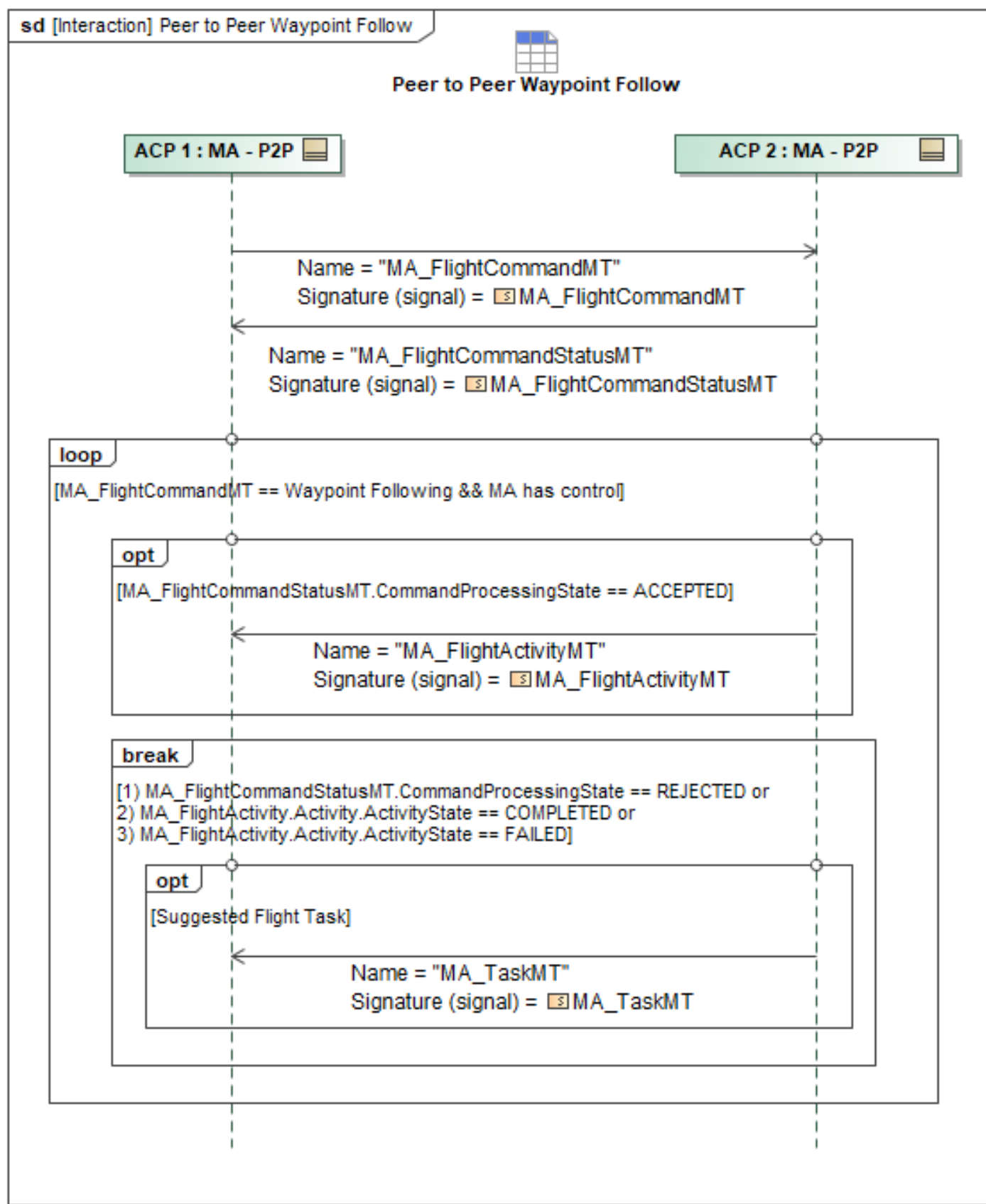


Figure 1-123: Peer to Peer Waypoint Follow

Table 1-122: Peer to Peer Waypoint Follow

| Message Name (Name:Type)                              | Required Fields (Name:Type)                                                                                                                     |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| MA_FlightActivityMT: MA_FlightActivityMT              | None                                                                                                                                            |
| MA_FlightCommandMT:<br>MA_FlightCommandMT             | WaypointFollowing: MA_WaypointFollowingType<br>Capability: MA_FlightCapabilityCommandType<br>FlightControlMode: MA_FlightControlModesChoiceType |
| MA_FlightCommandStatusMT:<br>MA_FlightCommandStatusMT | None                                                                                                                                            |
| MA_TaskMT: MA_TaskMT                                  | None                                                                                                                                            |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.13.12 Peer Validate Action Plan

This interaction allows a lead to request a peer verify an action plan. This allows a lead to propose an action plan to a peer and for the peer to accept or reject the proposed action. If the peer rejects the proposed action, it can propose an alternative.

The lead sends *ActionPlanValidationCommandMT* to the peer with the *ActionPlanID* set to the plan of the action to evaluate.

While the command is being processed, the peer may optionally send *ActionPlanValidationCommandStatusMT\_1* with the *CommandProcessingState* set to *RECEIVED*.

If the *ActionPlan* cannot execute the validation, the peer will respond with an *ActionPlanValidationCommandStatusMT\_2* with the *CommandProcessingState* set to *REJECTED*.

If the *ActionPlan* is valid, the peer will respond with an *ActionPlanValidationCommandStatusMT\_3* with the *CommandProcessingState* set to *ACCEPTED* and a *MA\_ActionPlanValidationMT\_1* with the *ValidationState* set to *VALID*.

If the *ActionPlan* is invalid, the peer will respond with *ActionPlanValidationCommandStatusMT\_4* with the *CommandProcessingState* set to *ACCEPTED* and a *MA\_ActionPlanValidationMT\_2* with the *ValidationState* set to *INVALID*. If the peer is able to generate an *ActionPlan* that it can accept, it will respond with *MA\_ActionMT* and *MA\_ActionPlanMT* and set the *ModifiedPlanID* to the *ActionPlanID* of *MA\_ActionPlanMT*.

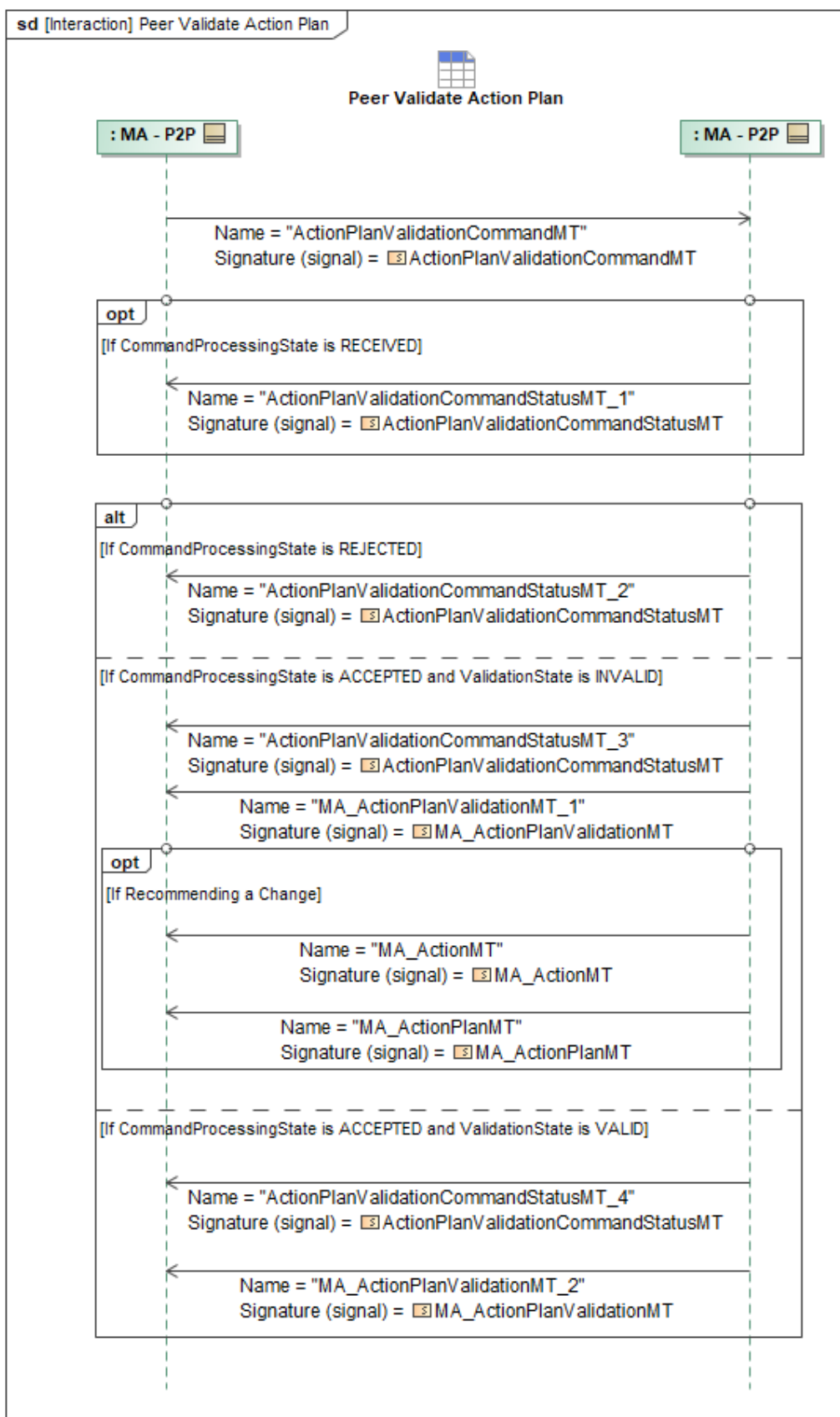


Figure 1-124: Peer Validate Action Plan

Table 1-123: Peer Validate Action Plan

| Message Name (Name:Type)                                                      | Required Fields (Name:Type) |
|-------------------------------------------------------------------------------|-----------------------------|
| ActionPlanValidationCommandMT:<br>ActionPlanValidationCommandMT               | None                        |
| ActionPlanValidationCommandStatusMT_1:<br>ActionPlanValidationCommandStatusMT | None                        |
| ActionPlanValidationCommandStatusMT_2:<br>ActionPlanValidationCommandStatusMT | None                        |
| ActionPlanValidationCommandStatusMT_3:<br>ActionPlanValidationCommandStatusMT | None                        |
| ActionPlanValidationCommandStatusMT_4:<br>ActionPlanValidationCommandStatusMT | None                        |
| MA_ActionMT: MA_ActionMT                                                      | None                        |
| MA_ActionPlanMT: MA_ActionPlanMT                                              | None                        |
| MA_ActionPlanValidationMT_1:<br>MA_ActionPlanValidationMT                     | None                        |
| MA_ActionPlanValidationMT_2:<br>MA_ActionPlanValidationMT                     | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.13.13 Provide and Receive Formation Status

Once the Team Leader and Peers are within a Formation, a transfer of *MA\_TaskStatusMT* messages between each other will take place to maintain required communication. These messages are sent periodically to each other, based on the chosen implementation.

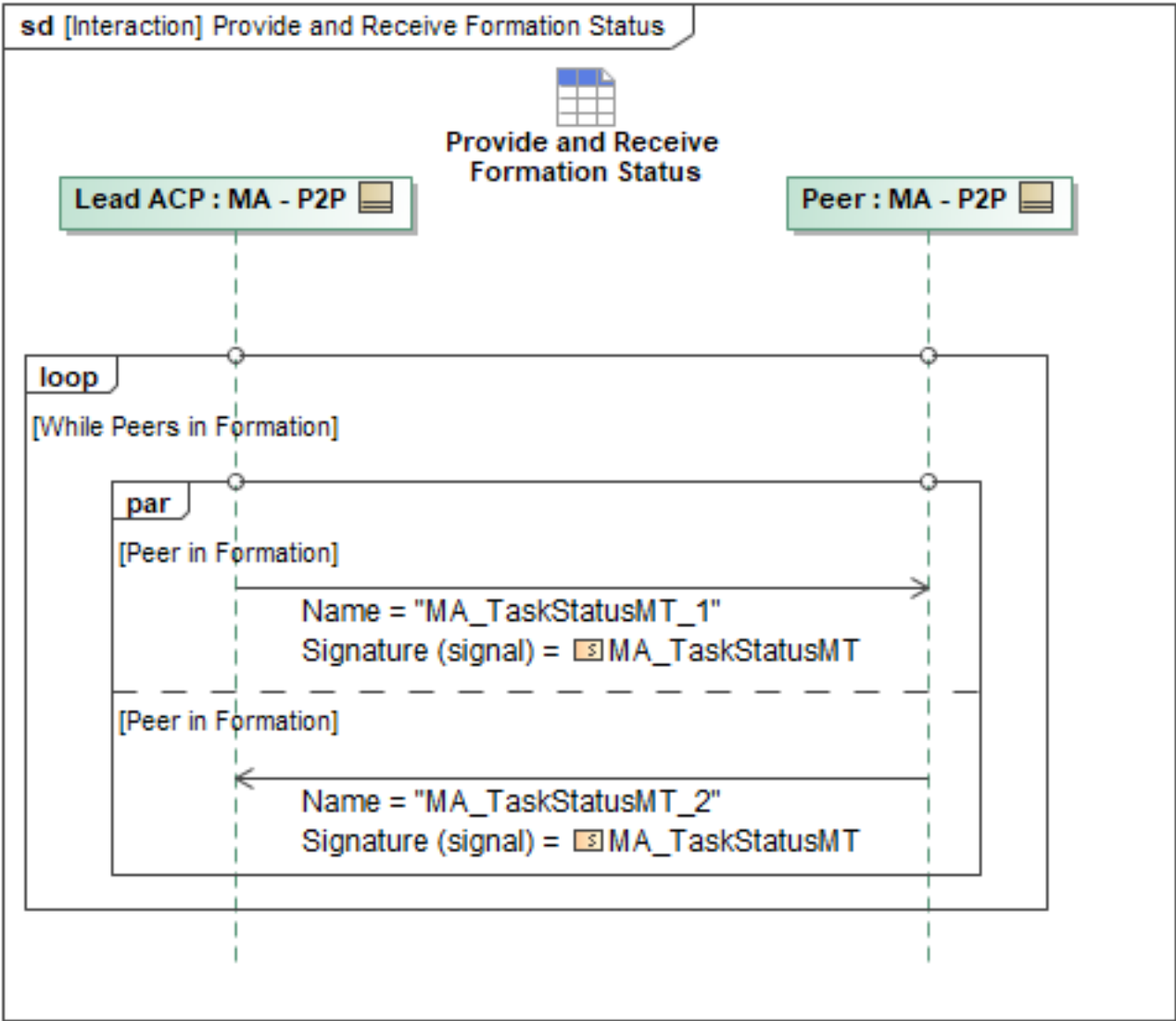


Figure 1-125: Provide and Receive Formation Status

Table 1-124: Provide and Receive Formation Status

| Message Name (Name:Type)           | Required Fields (Name:Type) |
|------------------------------------|-----------------------------|
| MA_TaskStatusMT_1: MA_TaskStatusMT | None                        |
| MA_TaskStatusMT_2: MA_TaskStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.13.14 Report Task Status to Leader

This interaction specifies the Task Status Notification between the Leader and the Follower in a Peer-to-Peer (P2P) interaction. The Follower MA periodically reports *MA\_TaskStatusMT* and *TaskPlanExecutionStatusMT* to the Leader MA. Within *MA\_TaskStatusMT* and *TaskPlanExecutionStatusMT* is message data describing the condition of the task that was carried out by the followers and the status of the task plan that is associated with the previous mentioned



task. Defined within the *MA\_TaskStatusMT* is the *TaskID* which indicates the unique ID of the task whose execution status is being reported to help with identification. Defined within the *TaskPlanExecutionStatusMT* are *SystemID*, *Source*, and *PlanExecutionStatus*. *SystemID* is the unique ID of the System whose Plan is being statused to help with plan identification. *Source* indicates the original location of the data being utilized while *PlanExecutionStatus* indicates the execution state of a *TaskPlan* of the System.

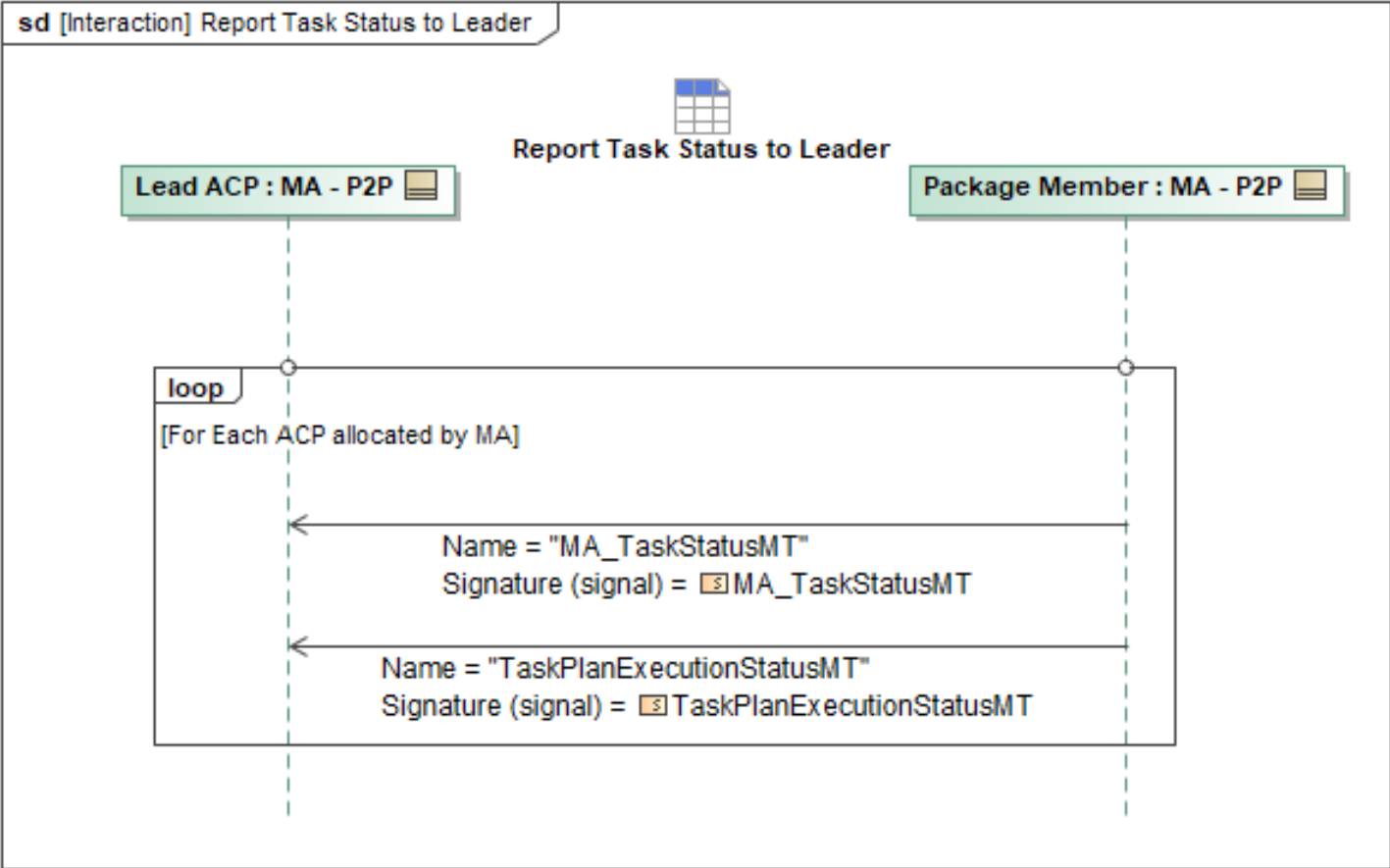


Figure 1-126: Report Task Status to Leader

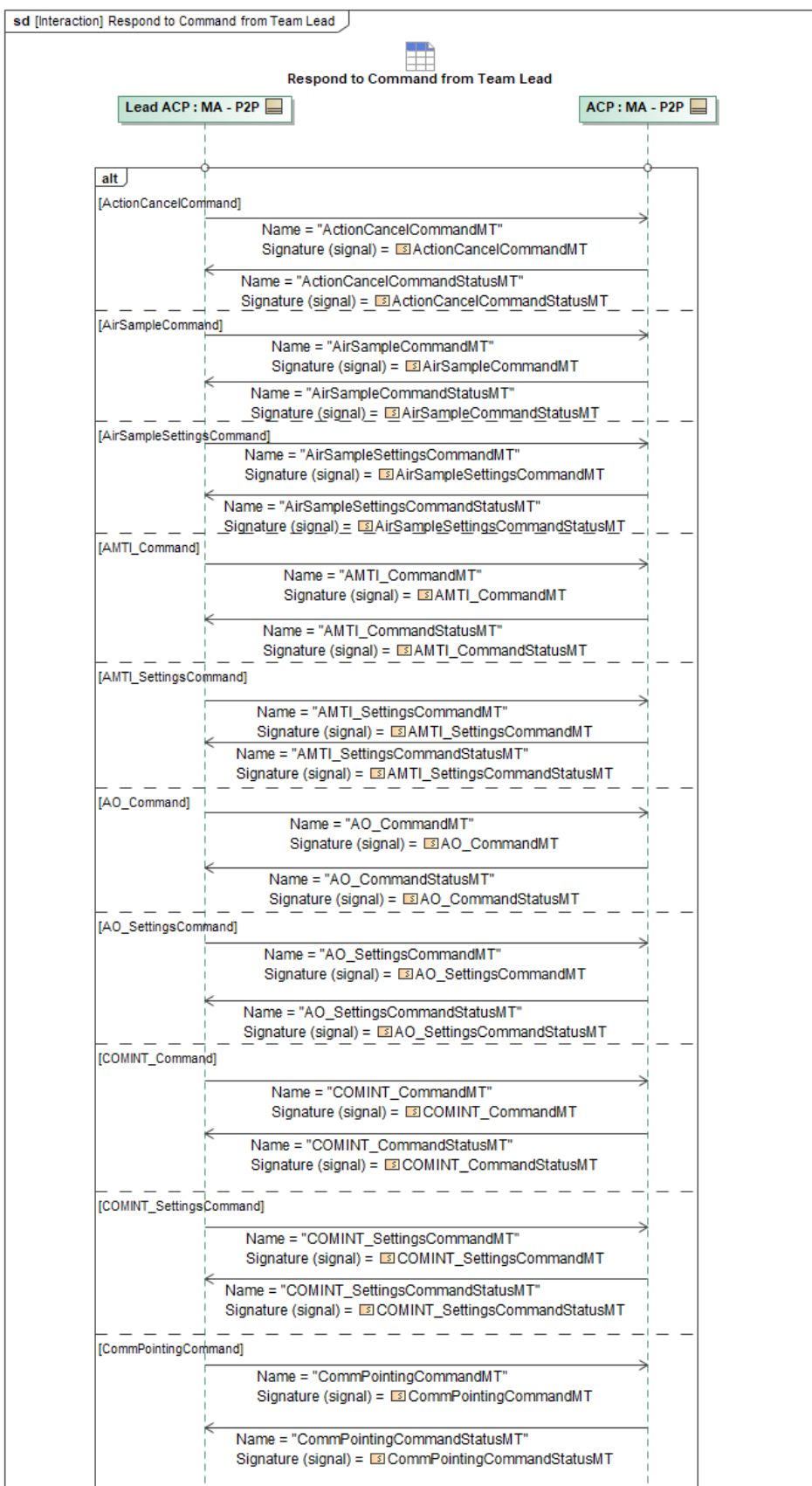
Table 1-125: Report Task Status to Leader

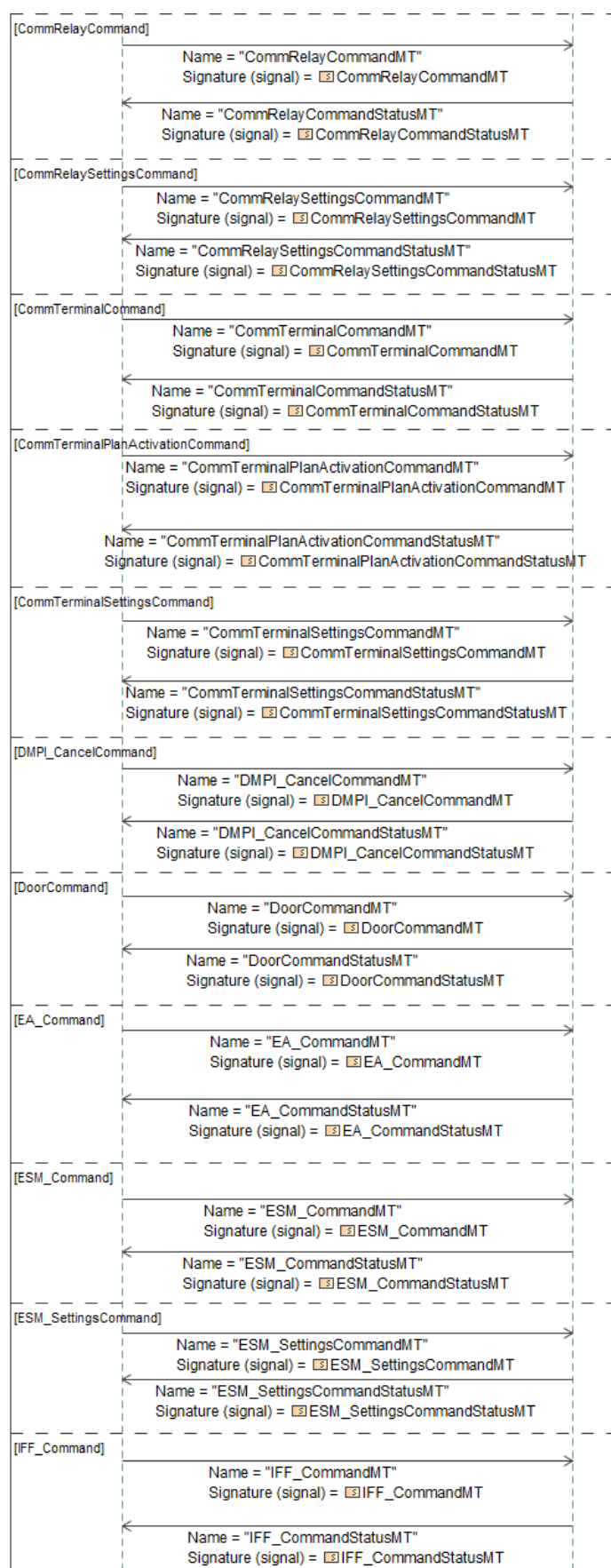
| Message Name (Name:Type)                             | Required Fields (Name:Type)                     |
|------------------------------------------------------|-------------------------------------------------|
| MA_TaskStatusMT: MA_TaskStatusMT                     | None                                            |
| TaskPlanExecutionStatusMT: TaskPlanExecutionStatusMT | PlanExecutionStatus: TaskPlanExecutionStateType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.13.15 Respond to Command from Team Lead

The lead platform that is responsible for distributing command messages to the appropriate platform(s) in the formation. The receiving platform is to then respond back with the respective status message of the original command message sent.







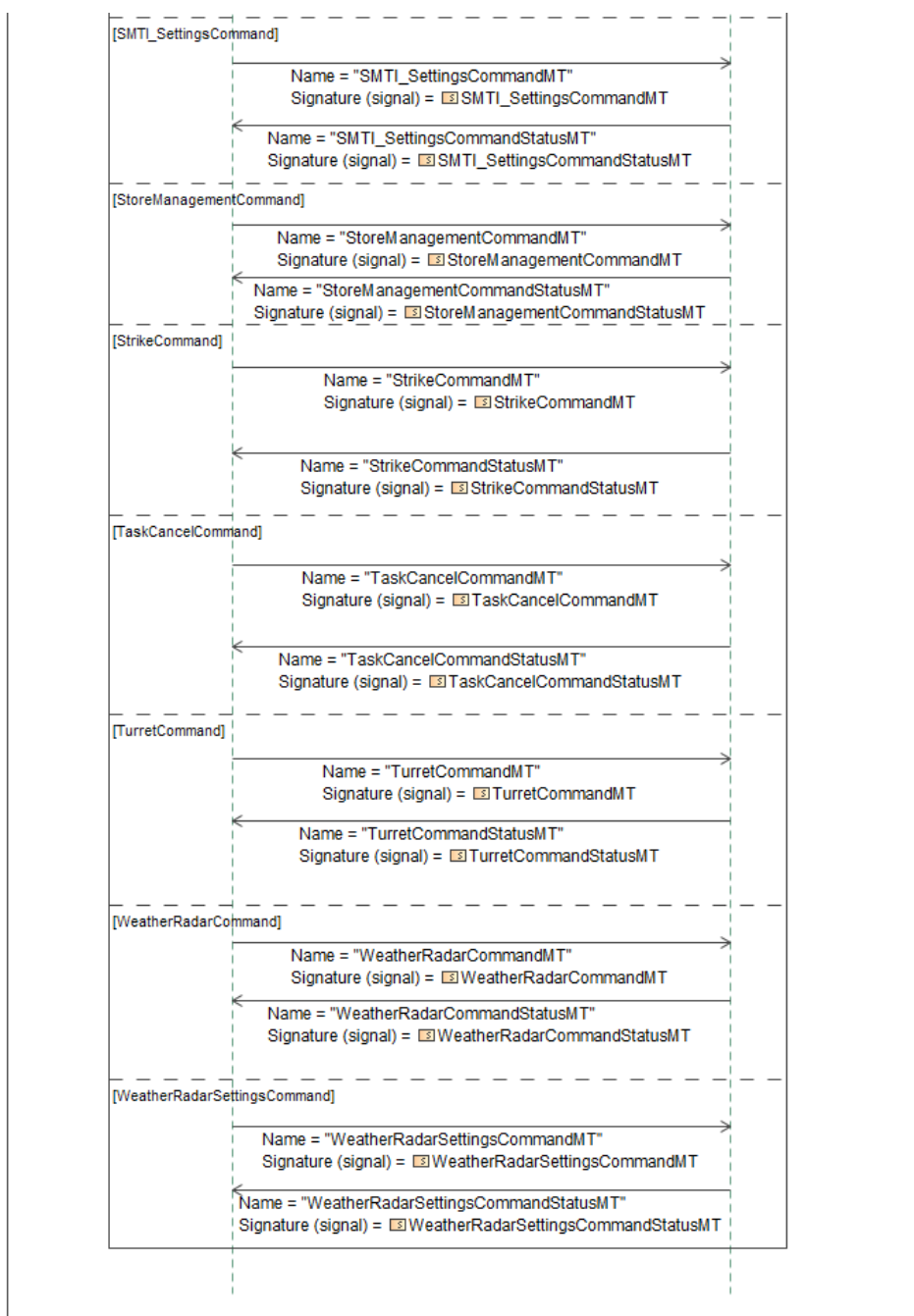


Figure 1-127: Respond to Command from Team Lead

Table 1-126: Respond to Command from Team Lead

| Message Name (Name:Type)                                    | Required Fields (Name:Type) |
|-------------------------------------------------------------|-----------------------------|
| ActionCancelCommandMT:<br>ActionCancelCommandMT             | None                        |
| ActionCancelCommandStatusMT:<br>ActionCancelCommandStatusMT | None                        |
| AirSampleCommandMT:<br>AirSampleCommandMT                   | None                        |

| Message Name (Name:Type)                                              | Required Fields (Name:Type) |
|-----------------------------------------------------------------------|-----------------------------|
| AirSampleCommandStatusMT:<br>AirSampleCommandStatusMT                 | None                        |
| AirSampleSettingsCommandMT:<br>AirSampleSettingsCommandMT             | None                        |
| AirSampleSettingsCommandStatusMT:<br>AirSampleSettingsCommandStatusMT | None                        |
| AMTI_CommandMT: AMTI_CommandMT                                        | None                        |
| AMTI_CommandStatusMT:<br>AMTI_CommandStatusMT                         | None                        |
| AMTI_SettingsCommandMT:<br>AMTI_SettingsCommandMT                     | None                        |
| AMTI_SettingsCommandStatusMT:<br>AMTI_SettingsCommandStatusMT         | None                        |
| AO_CommandMT: AO_CommandMT                                            | None                        |
| AO_CommandStatusMT:<br>AO_CommandStatusMT                             | None                        |
| AO_SettingsCommandMT:<br>AO_SettingsCommandMT                         | None                        |
| AO_SettingsCommandStatusMT:<br>AO_SettingsCommandStatusMT             | None                        |
| COMINT_CommandMT: COMINT_CommandMT                                    | None                        |
| COMINT_CommandStatusMT:<br>COMINT_CommandStatusMT                     | None                        |
| COMINT_SettingsCommandMT:<br>COMINT_SettingsCommandMT                 | None                        |
| COMINT_SettingsCommandStatusMT:<br>COMINT_SettingsCommandStatusMT     | None                        |
| CommPointingCommandMT:<br>CommPointingCommandMT                       | None                        |
| CommPointingCommandStatusMT:<br>CommPointingCommandStatusMT           | None                        |
| CommRelayCommandMT:<br>CommRelayCommandMT                             | None                        |
| CommRelayCommandStatusMT:<br>CommRelayCommandStatusMT                 | None                        |
| CommRelaySettingsCommandMT:<br>CommRelaySettingsCommandMT             | None                        |
| CommRelaySettingsCommandStatusMT:<br>CommRelaySettingsCommandStatusMT | None                        |
| CommTerminalCommandMT:<br>CommTerminalCommandMT                       | None                        |

| Message Name (Name:Type)                                                                        | Required Fields (Name:Type) |
|-------------------------------------------------------------------------------------------------|-----------------------------|
| CommTerminalCommandStatusMT:<br>CommTerminalCommandStatusMT                                     | None                        |
| CommTerminalPlanActivationCommandMT:<br>CommTerminalPlanActivationCommandMT                     | None                        |
| CommTerminalPlanActivationCommandStatus<br>MT:<br>CommTerminalPlanActivationCommandStatus<br>MT | None                        |
| CommTerminalSettingsCommandMT:<br>CommTerminalSettingsCommandMT                                 | None                        |
| CommTerminalSettingsCommandStatusMT:<br>CommTerminalSettingsCommandStatusMT                     | None                        |
| DMPI_CancelCommandMT:<br>DMPI_CancelCommandMT                                                   | None                        |
| DMPI_CancelCommandStatusMT:<br>DMPI_CancelCommandStatusMT                                       | None                        |
| DoorCommandMT: DoorCommandMT                                                                    | None                        |
| DoorCommandStatusMT:<br>DoorCommandStatusMT                                                     | None                        |
| EA_CommandMT: EA_CommandMT                                                                      | None                        |
| EA_CommandStatusMT:<br>EA_CommandStatusMT                                                       | None                        |
| ESM_CommandMT: ESM_CommandMT                                                                    | None                        |
| ESM_CommandStatusMT:<br>ESM_CommandStatusMT                                                     | None                        |
| ESM_SettingsCommandMT:<br>ESM_SettingsCommandMT                                                 | None                        |
| ESM_SettingsCommandStatusMT:<br>ESM_SettingsCommandStatusMT                                     | None                        |
| IFF_CommandMT: IFF_CommandMT                                                                    | None                        |
| IFF_CommandStatusMT:<br>IFF_CommandStatusMT                                                     | None                        |
| IFF_SettingsCommandMT:<br>IFF_SettingsCommandMT                                                 | None                        |
| IFF_SettingsCommandStatusMT:<br>IFF_SettingsCommandStatusMT                                     | None                        |
| PO_CommandMT: PO_CommandMT                                                                      | None                        |
| PO_CommandStatusMT:<br>PO_CommandStatusMT                                                       | None                        |
| PO_SettingsCommandMT:<br>PO_SettingsCommandMT                                                   | None                        |

| Message Name (Name:Type)                                                        | Required Fields (Name:Type) |
|---------------------------------------------------------------------------------|-----------------------------|
| PO_SettingsCommandStatusMT:<br>PO_SettingsCommandStatusMT                       | None                        |
| RadarAltimeterCommandMT:<br>RadarAltimeterCommandMT                             | None                        |
| RadarAltimeterCommandStatusMT:<br>RadarAltimeterCommandStatusMT                 | None                        |
| RadarAltimeterSettingsCommandMT:<br>RadarAltimeterSettingsCommandMT             | None                        |
| RadarAltimeterSettingsCommandStatusMT:<br>RadarAltimeterSettingsCommandStatusMT | None                        |
| RefuelCommandMT: RefuelCommandMT                                                | None                        |
| RefuelCommandStatusMT:<br>RefuelCommandStatusMT                                 | None                        |
| RefuelSettingsCommandMT:<br>RefuelSettingsCommandMT                             | None                        |
| RefuelSettingsCommandStatusMT:<br>RefuelSettingsCommandStatusMT                 | None                        |
| SAR_CommandMT: SAR_CommandMT                                                    | None                        |
| SAR_CommandStatusMT:<br>SAR_CommandStatusMT                                     | None                        |
| SAR_SettingsCommandMT:<br>SAR_SettingsCommandMT                                 | None                        |
| SAR_SettingsCommandStatusMT:<br>SAR_SettingsCommandStatusMT                     | None                        |
| SMTI_CommandMT: SMTI_CommandMT                                                  | None                        |
| SMTI_CommandStatusMT:<br>SMTI_CommandStatusMT                                   | None                        |
| SMTI_SettingsCommandMT:<br>SMTI_SettingsCommandMT                               | None                        |
| SMTI_SettingsCommandStatusMT:<br>SMTI_SettingsCommandStatusMT                   | None                        |
| StoreManagementCommandMT:<br>StoreManagementCommandMT                           | None                        |
| StoreManagementCommandStatusMT:<br>StoreManagementCommandStatusMT               | None                        |
| StrikeCommandMT: StrikeCommandMT                                                | None                        |
| StrikeCommandStatusMT:<br>StrikeCommandStatusMT                                 | None                        |
| TaskCancelCommandMT:<br>TaskCancelCommandMT                                     | None                        |
| TaskCancelCommandStatusMT:<br>TaskCancelCommandStatusMT                         | None                        |



| Message Name (Name:Type)                                                            | Required Fields (Name:Type) |
|-------------------------------------------------------------------------------------|-----------------------------|
| <b>TurretCommandMT: TurretCommandMT</b>                                             | None                        |
| <b>TurretCommandStatusMT:<br/>TurretCommandStatusMT</b>                             | None                        |
| <b>WeatherRadarCommandMT:<br/>WeatherRadarCommandMT</b>                             | None                        |
| <b>WeatherRadarCommandStatusMT:<br/>WeatherRadarCommandStatusMT</b>                 | None                        |
| <b>WeatherRadarSettingsCommandMT:<br/>WeatherRadarSettingsCommandMT</b>             | None                        |
| <b>WeatherRadarSettingsCommandStatusMT:<br/>WeatherRadarSettingsCommandStatusMT</b> | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.14 Peer Threat Assessment Package

### 1.2.14.1 P2P Threat Assessment

#### Mission Use Case: ALR

The optional Threat Assessment sequence diagram specifies the interactions where an ACP Peer provides ACP Lead with vehicle threat assessment results. ACP Lead may request an ACP Peer to perform vehicle threat assessment(s) for specific Entity ID's or all known red Entity ID's by sending *MA\_AssessmentRequestMT\_1* with respective *VehicleThreatAssessmentRequestType* field. ACP Lead can request the assessment to be performed once or periodically at a defined frequency by setting the *RequestFrequencyType* to PERIODIC and adding a value to *RequestFrequencyPeriod*. ACP Lead may also request what capability MA peer will use to perform the vehicle threat assessment (i.e. onboard sensors), however if ACP Lead wants MA peer to use historical tracks provided by track management function, then the *ThreatCapability* enumeration will be *OTHER\_UNSPECIFIED\_DETECTION*. The ACP will respond once or periodically with a *MA\_AssessmentRequestStatusMT\_1* to acknowledge the request and provide the assessment. If the request specified the *ResultsInAssessmentMessage* as FALSE, the assessment will be sent in a *MA\_AssessmentMT* message. If periodic assessments were requested, MA Lead will send *MA\_AssessmentRequestMT\_1* with *RequestFrequencyType* set to STOP. MA will respond with the corresponding *MA\_AssessmentRequestStatusMT\_1* to indicate request complete.

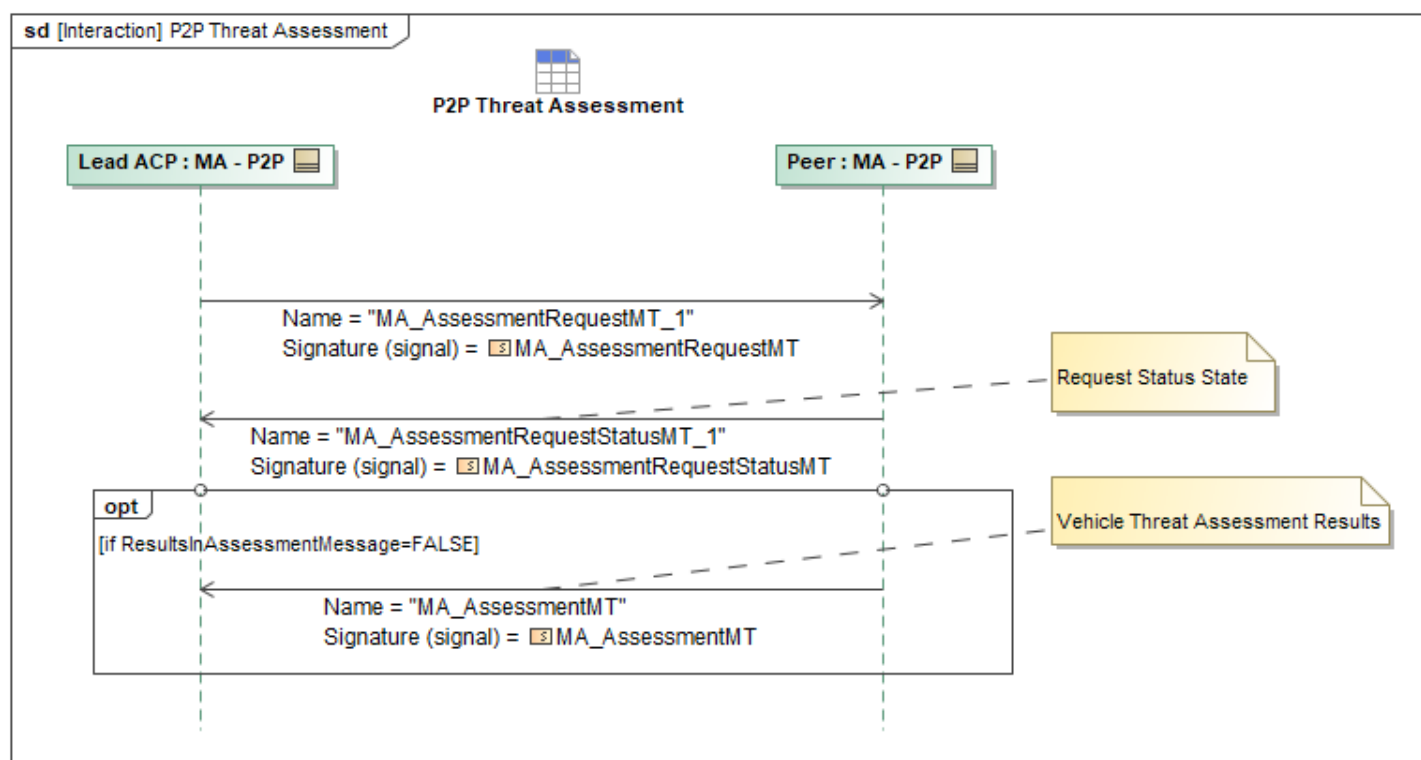


Figure 1-128: P2P Threat Assessment

Table 1-127: P2P Threat Assessment

| Message Name (Name:Type)                                               | Required Fields (Name:Type)                                                                                                 |
|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>MA_AssessmentMT:</b> MA_AssessmentMT                                | VehicleThreatAssessment: VehicleThreatAssessmentType<br>AssessmentRequestID: RequestID_Type<br>ObjectState: ObjectStateEnum |
| <b>MA_AssessmentRequestMT_1:</b><br>MA_AssessmentRequestMT             | VehicleThreatAssessment:<br>VehicleThreatAssessmentRequestType                                                              |
| <b>MA_AssessmentRequestStatusMT_1:</b><br>MA_AssessmentRequestStatusMT | None                                                                                                                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

## 1.2.15 Peer Weapon Employment Package

### 1.2.15.1 Centralized Computation of Target Assignment

This interaction describes the ACP lead computing the targeting assignments for ACP followers. First, the ACP lead sends out *AccessAssessmentRequestMT* to request all ACP followers to evaluate and score their capabilities for executing each target. Then, ACP followers send *AccessAssessmentRequestStatusMT* to the ACP lead to report their scores. Once the scores have been received by ACP Lead from all ACP followers to assist in computing targeting assignments, a Lead ACP must consider other variables that go into the assignment of Target-ACP pairings such as Shot Queue, Threat Prioritization, and targeting and intercept maneuver parameters set in a Mission

Plan or by C2.

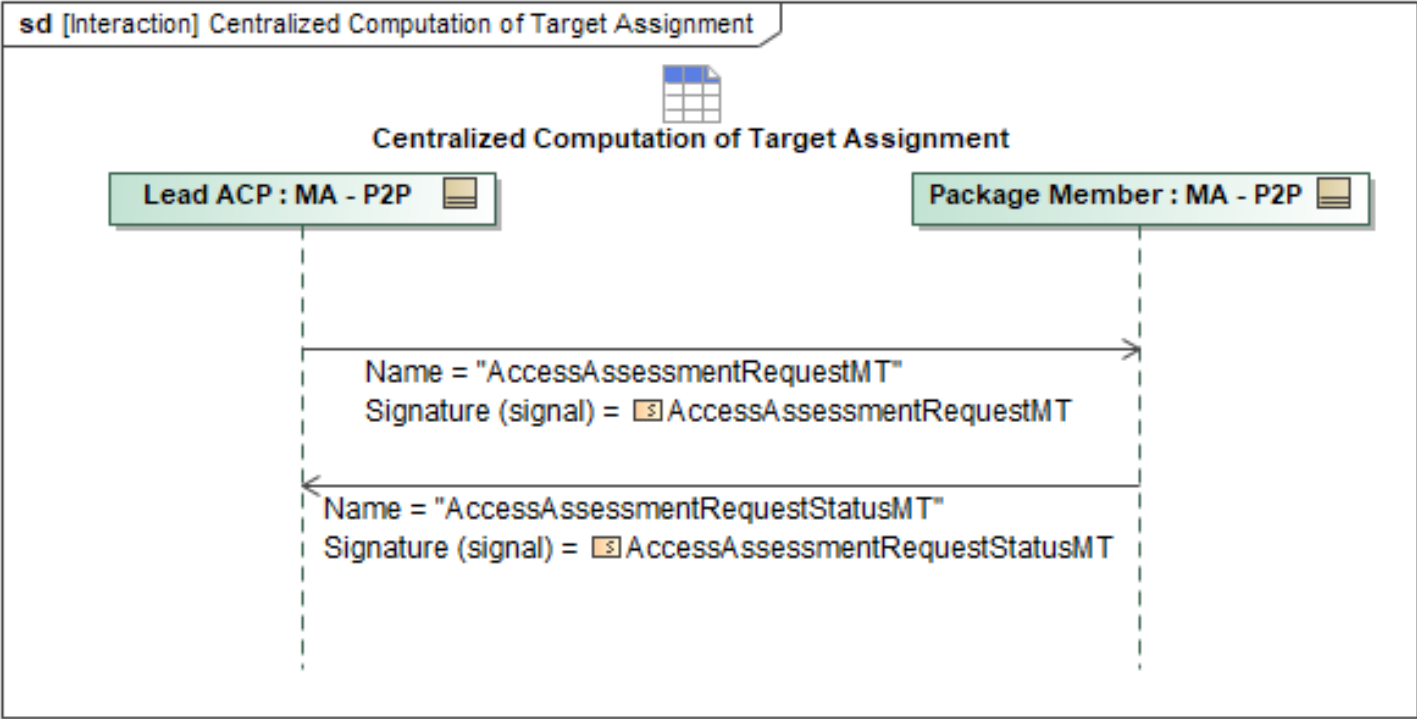


Figure 1-129: Centralized Computation of Target Assignment

Table 1-128: Centralized Computation of Target Assignment

| Message Name (Name:Type)                                            | Required Fields (Name:Type)                                                                                             |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| AccessAssessmentRequestMT:<br>AccessAssessmentRequestMT             | EffortToAssess: AccessEffortEnum<br>CapabilitiesToAssess: CapabilityTaxonomyUniversalType                               |
| AccessAssessmentRequestStatusMT:<br>AccessAssessmentRequestStatusMT | CompletionStatus: CompletionStatusType<br>ProbabilityOfAccessAttempt: PercentType<br>Result: AccessAssessmentResultType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.2 Distributed Computation of Target Assignment

This interaction describes the ACPs computing the target assignments by themselves. When all ACPs receive notice to start computing the assignments for target execution, every ACP evaluates and scores its capability for executing each target. Then, every ACP sends *AccessAssessmentRequestStatusMT* to all other ACPs to report its scores. When an ACP receives the scores from all other ACPs, it uses the scores to compute the target assignments for all other ACPs. This process leaves the ACPs with computed scores for target pairings. The assignment of targets, along with desired targeting or intercept parameters for these pairings is completed using the "Peer Assign Target Pairings" interaction.

Note:

- In the distributed case, it is critical that every ACP compute the same target assignment for all ACPs.
- This includes ensuring any arbitrary choices (e.g. choosing between targets that have the same score for an ACP) are made the same on all ACPs.

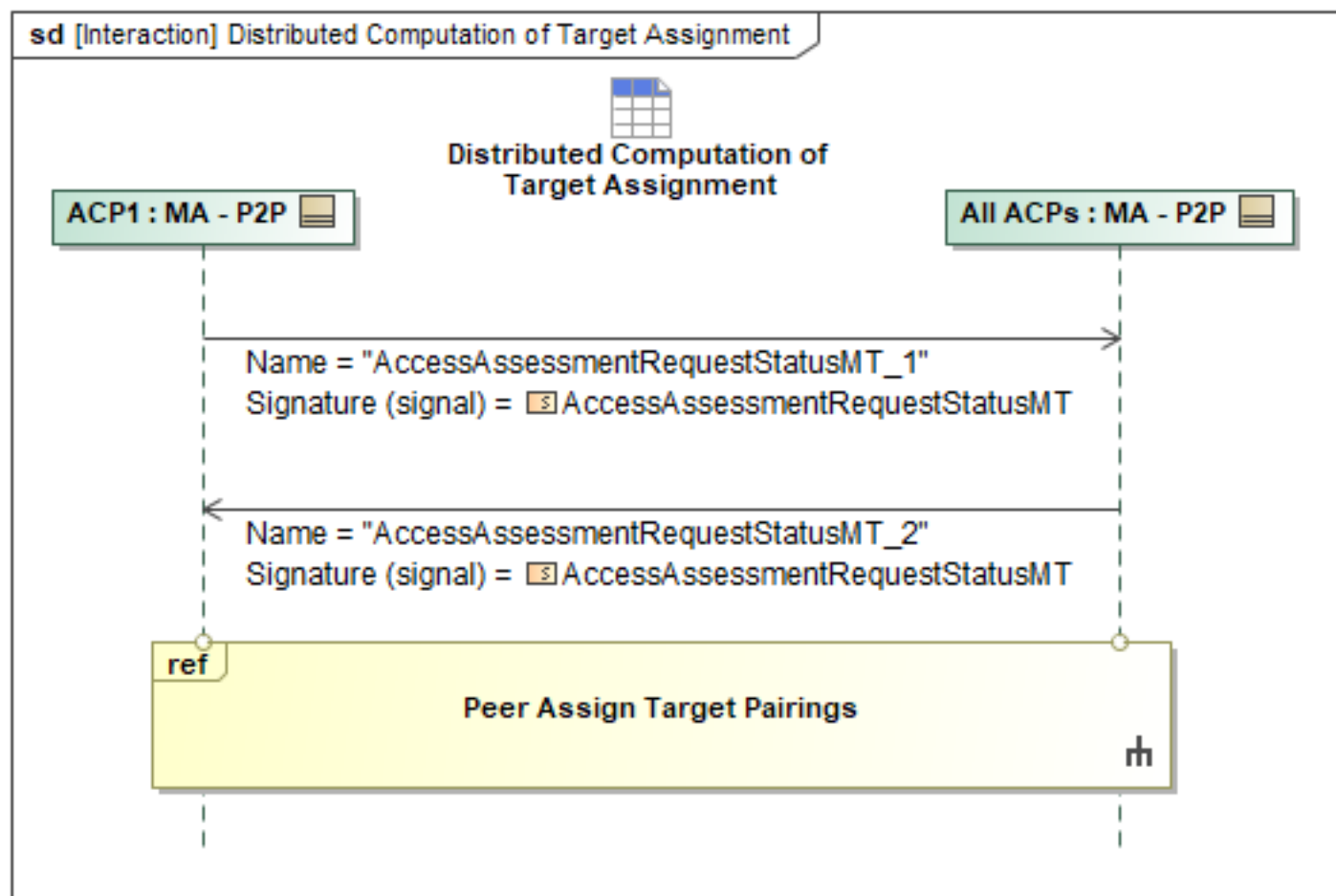


Figure 1-130: Distributed Computation of Target Assignment

Table 1-129: Distributed Computation of Target Assignment

| Message Name (Name:Type)                                                            | Required Fields (Name:Type)                                                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>AccessAssessmentRequestStatusMT_1:</b><br><b>AccessAssessmentRequestStatusMT</b> | ProbabilityOfAccessAttempt: PercentType<br>Result: AccessAssessmentResultType<br>CompletionStatus: CompletionStatusType |
| <b>AccessAssessmentRequestStatusMT_2:</b><br><b>AccessAssessmentRequestStatusMT</b> | CompletionStatus: CompletionStatusType<br>ProbabilityOfAccessAttempt: PercentType<br>Result: AccessAssessmentResultType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.15.3 DLZ Compound Convergence

DLZ Compound Convergence is a fusion of multiple DLZ from team members within a flight group. The fusion of DLZ enables P2P (Team Lead) to generate a composite view of flight group's weapon engagement capabilities to assess to allocate tasking for the best shot. Once ACPs are producing a DLZ in support of strike task, MA will send *DLZ\_MT* messages periodically to P2P (Team Lead).

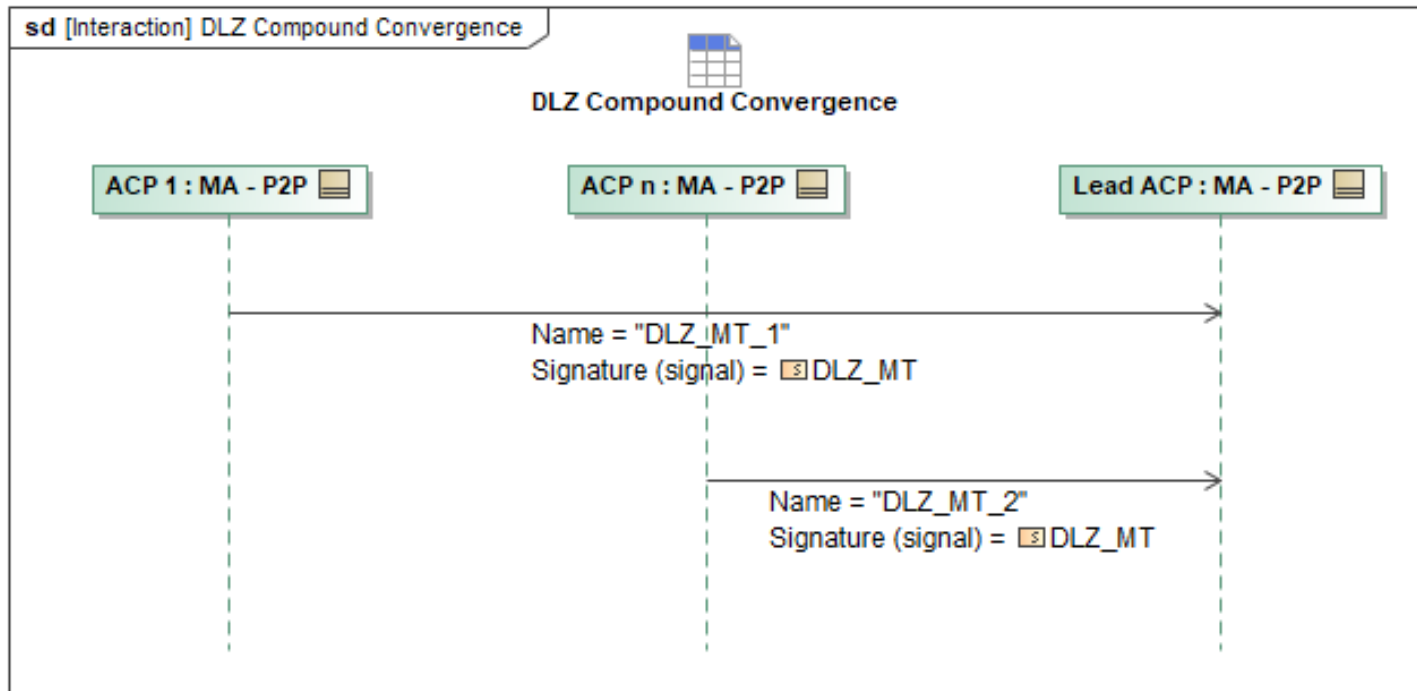


Figure 1-131: DLZ Compound Convergence

Table 1-130: DLZ Compound Convergence

| Message Name (Name:Type) | Required Fields (Name:Type) |
|--------------------------|-----------------------------|
| DLZ_MT_1: DLZ_MT         | None                        |
| DLZ_MT_2: DLZ_MT         | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.15.4 Peer Assign Target Pairings

The Assign Target Pairings sequence on the Peer Interface describes how a team leader would assign target pairings to peer ACPs, along with a specific task and accompanying constraints and parameters to engage the paired entity. C2 will likely command an ACP package, via an *ActionCommandMT* message to engage on a group of hostile entities. Lead MA is responsible for fusing various data to compute the best shot allocations and intercept pairings for all of its Peers within the package. This may include the Lead ACP utilizing either a Centralized Computation of Target Assignments, or a Distributed Computation of Target Assignments, which are Peer interactions. Along with these target assignment computation, the Lead ACP will also account for a Shot Queue and Threat Prioritization and Threat Assessment. Once this process has completed, the Lead MA would send a *RelationshipDesignationMT* message. The *RelationshipDesignationMT*

message contains information about the entity which the Peer follower is being tasked to engage. The *Destination* field should indicate the target, likely an *EntityID*, but could be any of the other types contained within the *GeoLocatedObjectType* field. The *Source* field, also a *GeoLocatedObjectType*, indicates the ACP being assigned this target pairing relationship.

The sequential *MA\_TaskMT* message is where the leader provides the type of *Task* to be carried out on the given relationship. The *TaskType* will likely be that of a *StrikeTask* for *targeting*, however there are other options given the initial higher level *MA\_ActionTypeEnum* that was received by the team leader. Intercept maneuver and shot doctrine parameters would be shared within this *MA\_TaskMT* message as applicable. The *MA\_TaskMT* message contains the same fields present within an *MA\_ActionMT* to denote parameters for *EngagementParameters* within *TaskConstraints*. This sequence does not actually command the Peer to begin executing the engagement but rather provides all of the information the Peer ACP will need to execute the engagement including the ACP-target pairing(s), intercept maneuver parameters, and shot doctrine parameters. The Peer ACP's response is a *MA\_SystemNotificationMT* message which acts as an acknowledgment that the data was received.

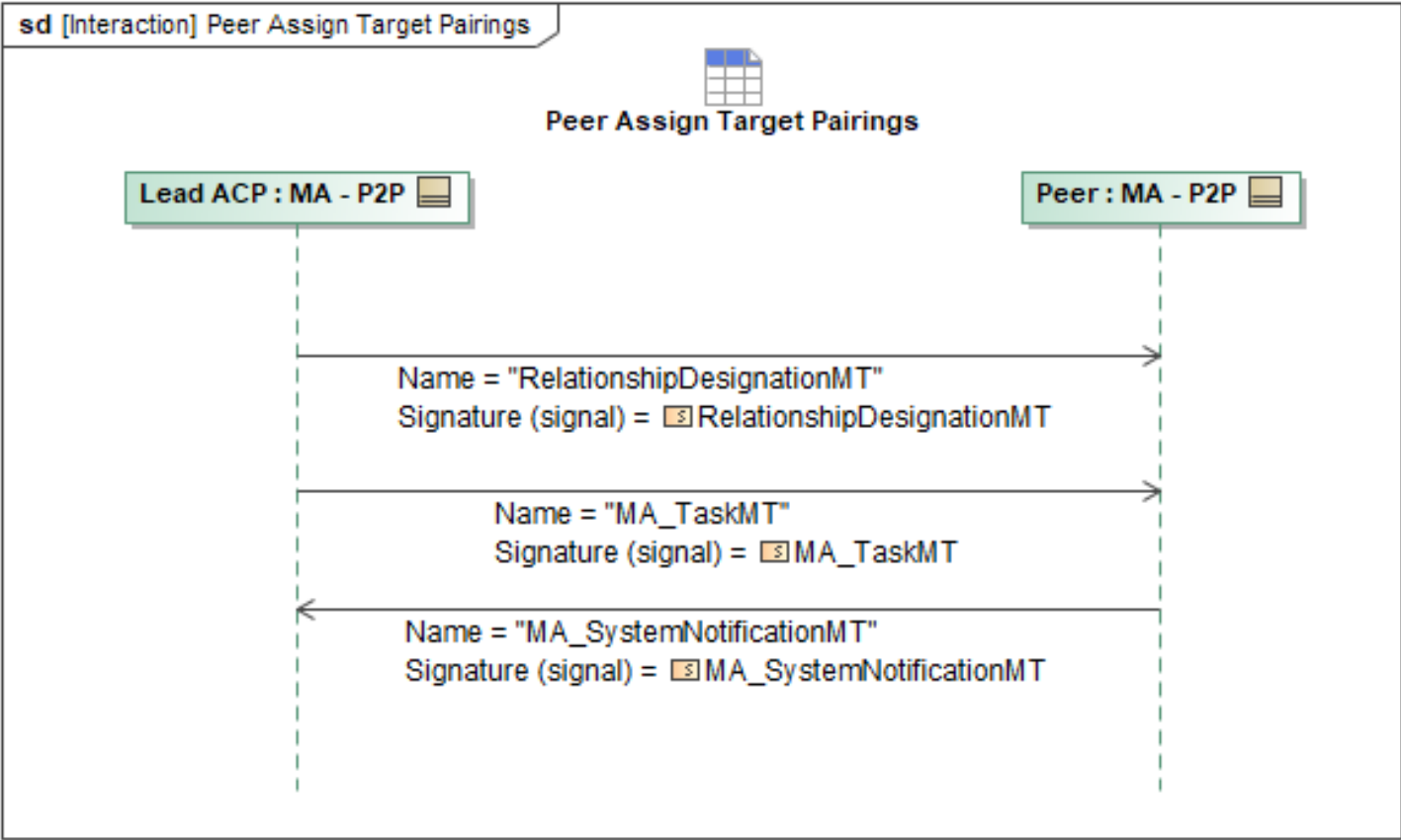


Figure 1-132: Peer Assign Target Pairings

Table 1-131: Peer Assign Target Pairings

| Message Name (Name:Type)                         | Required Fields (Name:Type)              |
|--------------------------------------------------|------------------------------------------|
| MA_SystemNotificationMT: MA_SystemNotificationMT | AssociatedMessage: AssociatedMessageType |
| MA_TaskMT: MA_TaskMT                             | TaskType: TaskType                       |

| Message Name (Name:Type)                                              | Required Fields (Name:Type)                                                                                                                                                                          |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                       | TaskID: TaskID_Type<br>TaskConstraints: RequirementConstraintsType<br>EngagementParameters: MA_EngagementParametersType                                                                              |
| <b>RelationshipDesignationMT:</b><br><b>RelationshipDesignationMT</b> | RelationshipDesignationID: RelationshipDesignationID_Type<br>Pairing: PairingRelationshipEnum<br>Source: GeoLocatedObjectType<br>Destination: GeoLocatedObjectType<br>Relationship: RelationshipType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.15.5 Peer Report Tactic Status

The Report Tactic Status sequence on the Peer Interface describes a status reporting interaction between a Lead ACP and one of it's followers. The follower will send *MA\_ActionStatusMT* periodically to report the status of its commanded tactic as a part of a package. Any time there is an update to the on-going engagement, the follower should update the Lead MA with a new *MA\_ActionStatusMT* message.

The *Metric* field within *MA\_ActionStatusMT* contains information on *EnduranceUsage*, *Expendables*, *Weapons*, and more specifically a *StoreType*, *StoreCategory*, and *StoreTypeVariant* can be specified. The *ExecutingSystemOrPackage* field will be set to a *SystemID* for peer reporting. This *MA\_ActionStatusMT* message will trace to a specific Activity using the *ActivityID* field within *Traceability*. For more information on the tactic being employed, the Peer ACP will send *MA\_StrikeActivityMT* periodically to the Lead to inform the lead about the specific Activity it is executing. Information contained in this message will inform the lead on the Peer's *ActivityState*, *EstimatedStateTime* or *EstimatedCompletionTime*, if a weapon has been *SelectedForKeyload* or not, if a weapon is armed via *WeaponArmed*, if a weapon is *SelectedForRelease* or *ReadyForRelease*, and more information on an *AssignedTarget*.



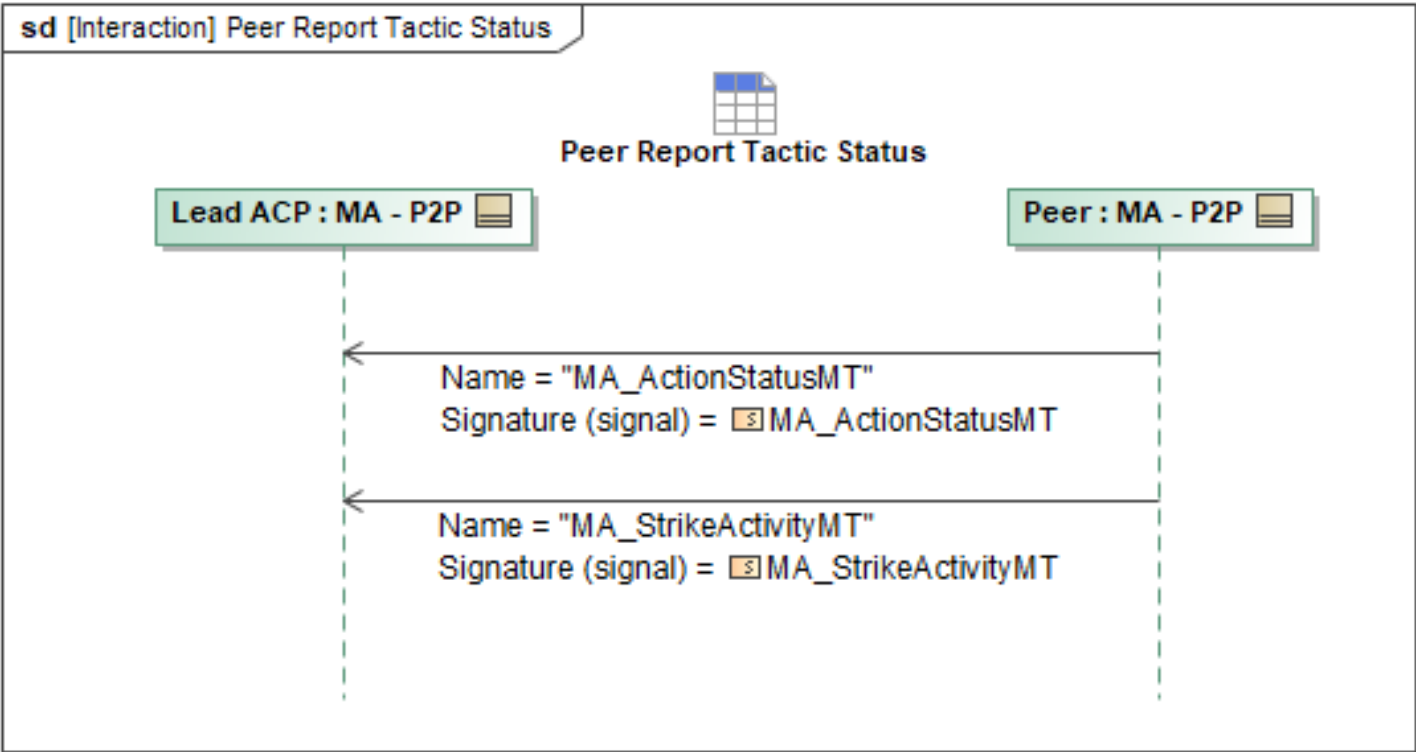


Figure 1-133: Peer Report Tactic Status

Table 1-132: Peer Report Tactic Status

| Message Name (Name:Type)                 | Required Fields (Name:Type)                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------|
| MA_ActionStatusMT: MA_ActionStatusMT     | SystemID: SystemID_Type<br>ActionID: ActionID_Type<br>ExecutingSystemOrPackage: MA_PackageSystemType |
| MA_StrikeActivityMT: MA_StrikeActivityMT | Activity: StrikeActivityType<br>CapabilityID: CapabilityID_Type<br>ActivityState: ActivityStateEnum  |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.6 Peer Send Tactic Parameters

The Peer Send Tactic Parameters sequence diagram specifies the interaction that enables a lead MA to provide tactic parameter updates to its followers within a package. The Lead MA will send a *MA\_ActionMT* message to all follower ACPs.

Specific tactic parameters are contained within the *MA\_ActionMT* message that enable sharing of data to inform specific tactics stored as an *Action*. For more information about these tactic parameters and what fields to utilize for detailed shot doctrine and intercept maneuver parameters, see the C2 behavior "Set Tactic Shot Doctrine and Intercept Maneuver Parameters".



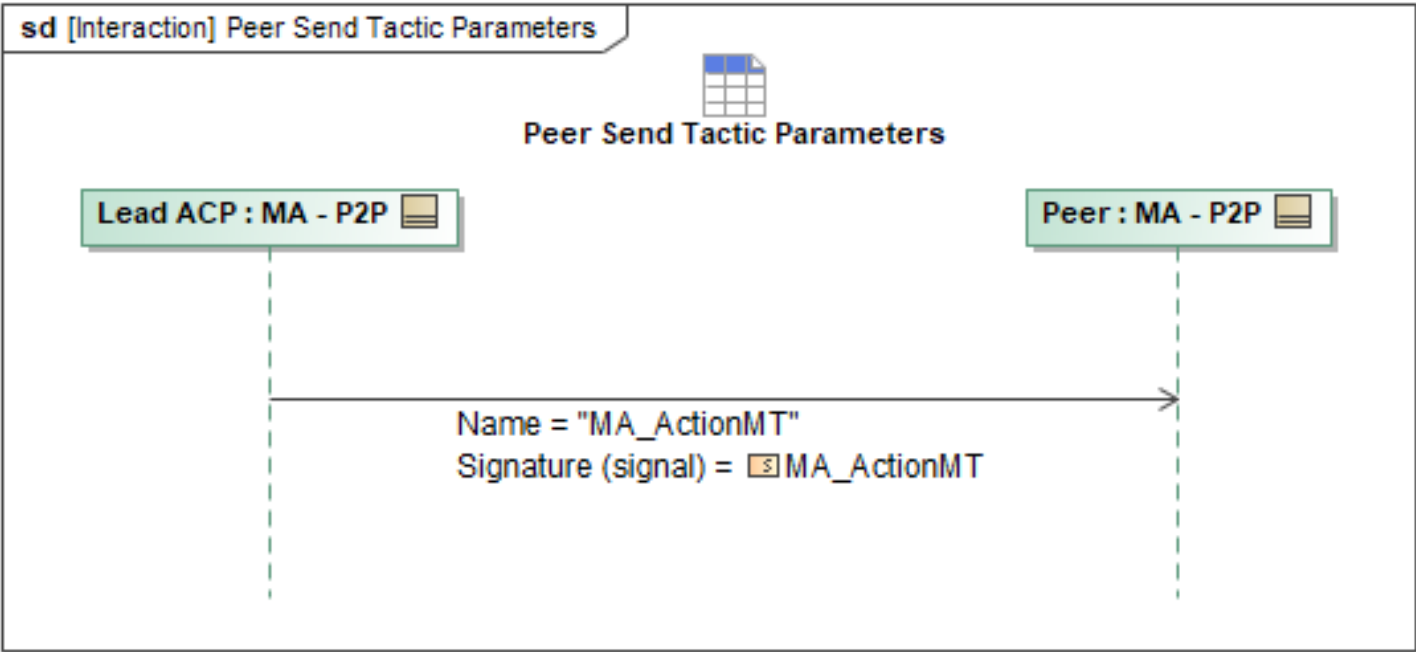


Figure 1-134: Peer Send Tactic Parameters

Table 1-133: Peer Send Tactic Parameters

| Message Name (Name:Type) | Required Fields (Name:Type) |
|--------------------------|-----------------------------|
| MA_ActionMT: MA_ActionMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.7 Peer Target Designation Request

The Peer Target Designation Request sequence diagram specifies the interaction that happens after a Follower MA identifies one or more targets to be Designated. The Target Designation Criteria or the Distributed Computation of Target Assignment sequence can be used to identify Targets for engagement. The MA would then send a *MA\_DesignationRequestMT* to the Package Lead. The Package Lead could then use the C2 Target Designation Request sequence to gain authorization from a C2 or respond directly if authorization has already been approved or is not needed. For an engagement, if an authorized C2 has already approved or commanded an engagement against that target to the Package Lead, the Lead may respond directly without another request. The *MA\_DesignationRequestMT* contains one or more *DesignationData* fields that contain the target designation information. The *DesignationID* field is a specific identifier for that specific target. The Target field can be set to many items, but for an engagement, the Target Designation Criteria is only applicable to *EntityMTs*. The *TypeOfDesignation* field would likely be set to *DESIGNATED\_TARGET* but could also be set to *DESIGNATED\_NO\_TARGET* if the service determines the target should no longer be a target. The *AppliesToID* field will specify the *SystemID* or *PackageID* which shall be assigned to the target. The use of *PackageID* allows a Lead ACP to delegate the target assignment, whereas *SystemID* allows for direct assignment to a specific ACP. If sending the message to C2, the package Lead would set the *AppliesToID* to the ID of the Package and will only send a single *MA\_DesignationRequestMT* to C2 when receiving the same Target Designation Request from

multiple package followers. The C2 Target Designation Request section describes the Package Lead and C2 messaging sequence. The optional *DesignationObjective* field allows for specification of a Requirement that the ACP should use against the target. For an engagement, these Requirement types would match what is in the *ApprovalPolicyMT* that is linked to the Target Authority. If C2 sends a *MA\_DesignationMT* with an *AppliesToID* set to the Package, the Package Lead can assign targeting as needed using the Peer Target Designation sequence. For an engagement, the Centralized Computation of Target Assignment sequence or the Distributed Computation of Target Assignment sequence can be used depending on the Package configured to assign the targets. The Package Lead will respond to the follower with a *DesignationRequestStatusMT* populating the *RequestID* field with the same ID as was in the request. As mentioned before for an engagement, the Distributed Computation of Target Assignment sequence could use *MA\_DesignationRequestMT(s)* messages to request the lead perform the final allocation of targets to followers, and the lead would then use the Peer Target Designation sequence. The *RequestProcessingState* field will be filled out as needed with the main options being *COMPLETED* (if C2 or Lead sent the *MA\_DesignationMT* messages) or *REJECTED* (if C2 or Lead did not send the *MA\_DesignationMT* messages). The *DesignationDataStatus* field will be populated with the relevant target data from the request message, including the associated *DesignationID* and the *RequestProcessingState* with similar options as stated above. There is an optional *RequestProcessingStateReason* field that can be filled out if desired.

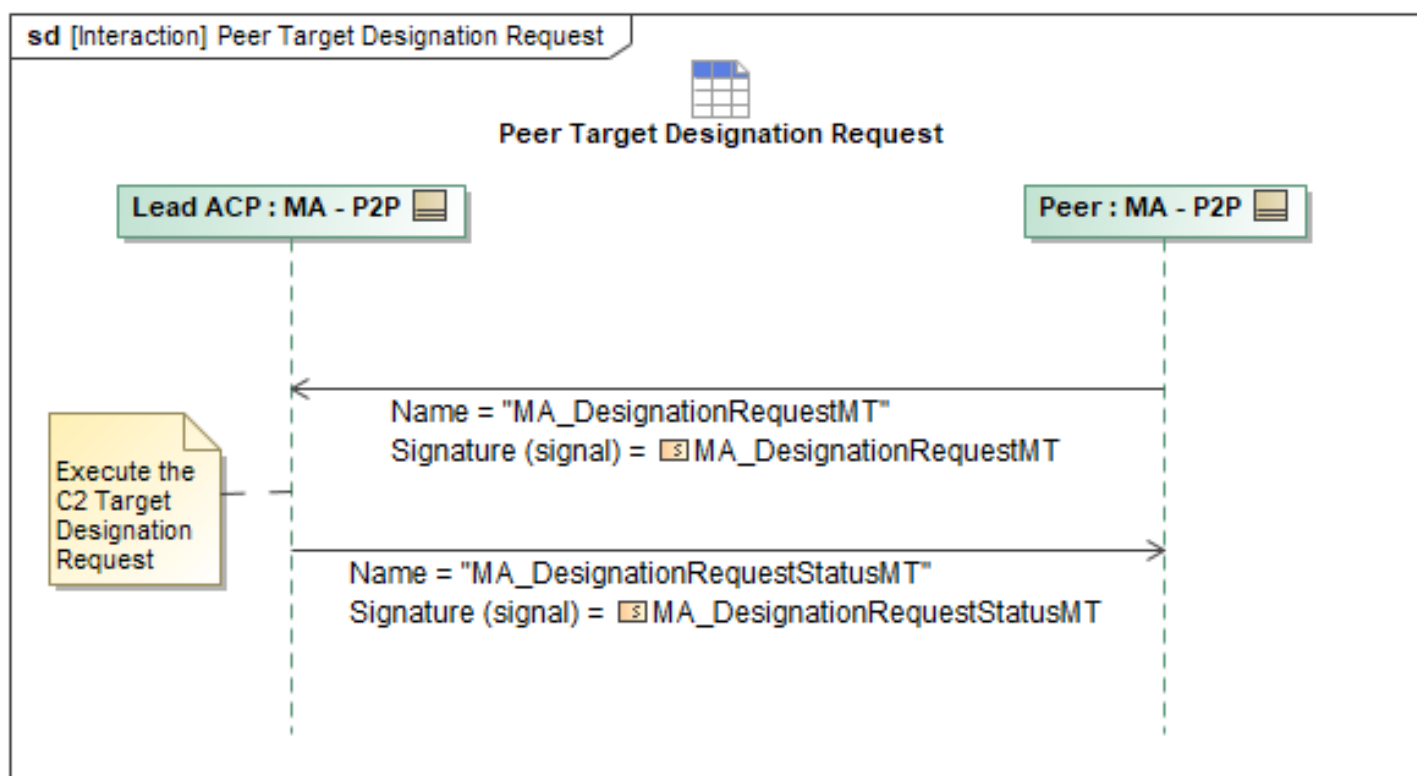


Figure 1-135: Peer Target Designation Request

Table 1-134: Peer Target Designation Request

| Message Name (Name:Type)                            | Required Fields (Name:Type) |
|-----------------------------------------------------|-----------------------------|
| MA_DesignationRequestMT:<br>MA_DesignationRequestMT | None                        |

| Message Name (Name:Type)                                        | Required Fields (Name:Type) |
|-----------------------------------------------------------------|-----------------------------|
| MA_DesignationRequestStatusMT:<br>MA_DesignationRequestStatusMT | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.15.8 Query Entity History

The following sequence is meant to standardize the interface between MA (Flight Lead) and Peer to query request for pulling entity track management historical or entity propagation data from the peer node.

The query for entity historical and propagation from MA to Peer node will be using the existing UCI message : *QueryDataRequest*.

- *ResultsInNative* extension needs *MessageType* & *ReceiverAddress* for querying *Entity* & *EntityPropagation* historical data from C2 to MA. This would not serve the purpose since it doesn't specify a *EntityId(s)* that need to be queried.
- *QueryMessage* extension needs *MessageType* & *QueryPET* (Polymorphic Extension Type) for querying *Entity* & *EntityPropagation* historical data. *QueryPET* a polymorphic extension type that is the base for *QueryType*. When creating a *QueryDataRequest* message, *QueryType* will replace the *QueryPET*.

To create a query, *QueryDataRequest* uses *QueryDataRequest.MessageData.QueryMessage*. An example *QueryDataRequest* (starting from the query element) shows querying for *EntityID* from the historical *Entity* messages stored by MA service (Entity Management Service). The query can be combined for querying both *Entity* & *EntityPropagation* messages (historical) in the *QueryType* for *EntityID*.

<!-- Example 1 --->

<QueryMessage>

<MessageType>ENTITY</MessageType>

<Query xsi:type="uci:QueryType">

<Equals>

<Step><Element><Name>MessageData</Name></Element></Step>

<Step><Element><Name>EntityID</Name></Element></Step>

<Value><UUID>xxxxxxx</UUID></Value>

</Equals>

</Query>

</QueryMessage>

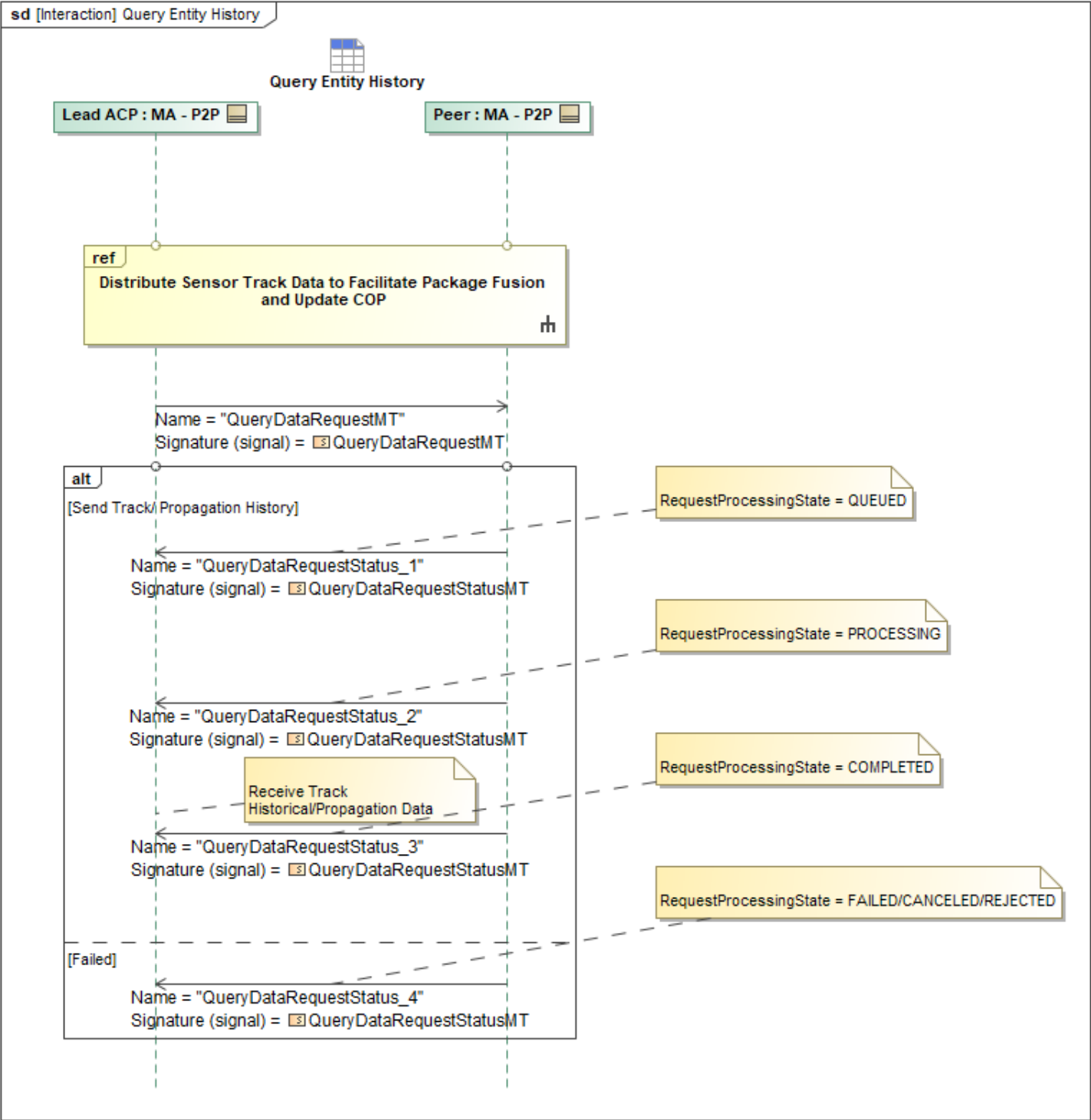


Figure 1-136: Query Entity History

Table 1-135: Query Entity History

| Message Name (Name:Type)               | Required Fields (Name:Type) |
|----------------------------------------|-----------------------------|
| QueryDataRequestMT: QueryDataRequestMT | None                        |

| Message Name (Name:Type)                              | Required Fields (Name:Type)                    |
|-------------------------------------------------------|------------------------------------------------|
| QueryDataRequestStatus_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatus_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatus_3:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |
| QueryDataRequestStatus_4:<br>QueryDataRequestStatusMT | None                                           |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.15.9 Query Entity Track History

The Track Management functionality for multi-ship/single-ship weapon employment is to maintain target tracks historical data along with entity predictions data communicated from Peer to MA when requested using a data query request message. MA queries the P2P nodes to get the historical and prediction track data. The track data may be used by other functions such as sending external track reports, stimulating alerts/notifications if rule sets are triggered, and updating the track display, history, and attribute data. MA will be getting Entity data from the COP, so the Track Management Data Fusion engine will take the COP data and its sensor data to build track history. Track propagation service takes the track state updates and predicts the extrapolated position (Kinematics) of an entity in the battle space and maintains these as a database within MA.

The query for entity and propagation data from MA to P2P will be using the existing UCI message : *QueryDataRequest*.

- *ResultsInNative* field needs *MessageType* & *ReceiverAddress* for querying *EntityPropagation* historical data from MA to P2P. This would not serve the purpose since it doesn't specify *EntityId(s)* that need to be queried.
- *QueryMessage* field needs *MessageType* & *Query* PET (Polymorphic Extension Type) for querying *Entity* and *EntityPropagation* historical data from MA to P2P.

To create a query, *QueryDataRequest* UCI message uses *QueryMessage* field. For example, *QueryDataRequestMT* (starting from the query element) shows querying for *entityID* from the historical *Entity* messages stored by MA's Entity Management Service. The query can be combined for querying both *Entity* & *EntityPropagation* messages (historical) in the *QueryType* for *entityid*.

<!-- Example 1 -->

<QueryMessage>

<MessageType>ENTITY</MessageType>

<Query xsi:type="uci:QueryType">

<Equals>

<Step><Element><Name>MessageData</Name></Element></Step>

<Step><Element><Name>EntityID</Name></Element></Step>

```
<Value><UUID> xxxxxxxxxx</UUID></Value>
</Equals>
</Query>
</QueryMessage>
```

MA receives *QueryDataRequestStatusMT\_1* response from the P2P node with intermediate response of *RequestProcessingState* field set to *QUEUED*, followed by *QueryDataRequestStatusMT\_2* response with *RequestProcessingState* set to *PROCESSING*. Upon successful query for the entity track management or propagation data, MA would receive *QueryDataRequestStatusMT\_3* message with the entity database along with the *RequestProcessingState* set to *COMPLETED*. A failure response can result in a *QueryDataRequestStatus\_4* with *RequestProcessingState* set to *FAILED* or *REJECTED* or *CANCELED* from P2P node to MA. For any non-compliance related issues, *QueryDataRequestStatusMT\_4* response is sent with *RequestProcessingStateReason* set to appropriate non-compliance code.

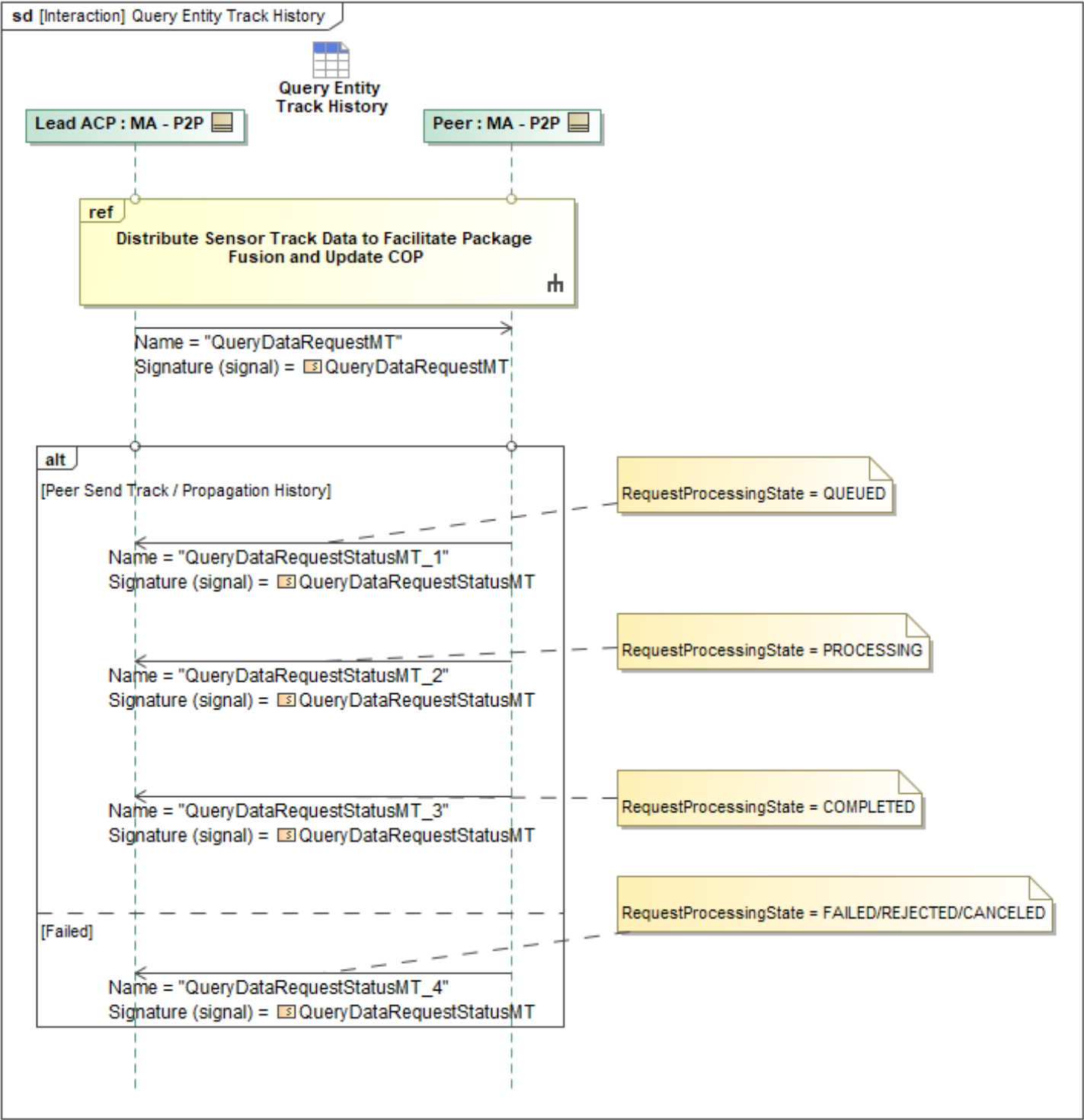


Figure 1-137: Query Entity Track History

Table 1-136: Query Entity Track History

| Message Name (Name:Type)               | Required Fields (Name:Type) |
|----------------------------------------|-----------------------------|
| QueryDataRequestMT: QueryDataRequestMT | Query: QueryMessageType     |

| Message Name (Name:Type)                                | Required Fields (Name:Type)                    |
|---------------------------------------------------------|------------------------------------------------|
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | None                                           |
| QueryDataRequestStatusMT_4:<br>QueryDataRequestStatusMT | RequestProcessingStateReason: CannotComplyType |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.15.10 Query Shot Queue Prioritization

Shot Queue Prioritization List is a list of mission targets ranked by assessing multiple factors to determine which target entities should be engaged first. The shot queue will utilize *MA\_PrioritizationListMT* with a *ListType* of *MISSION\_TARGET\_LIST* enumeration. The list requires subjects (*EntityID*) to be assigned a priority, where the lowest assigned value is the highest priority level. The message also includes an optional field, *ProbabilityOfKill*, that may be included with each assessed subject entity. P2P (Multi-Ship Lead) may query (*QueryDataRequest*) an ACP (Multi-Ship Peer) to request *MA\_PrioritizationListMT* in its native form. MA will respond with *QueryDataRequestStatusMT* to provide state changes (Queued "\_1", Processing "\_2", and Completed "\_3") in the processing of the request as depicted in the sequence diagram. MA will send *MA\_PrioritizationListMT* result in native form to the requesting *SystemID*.



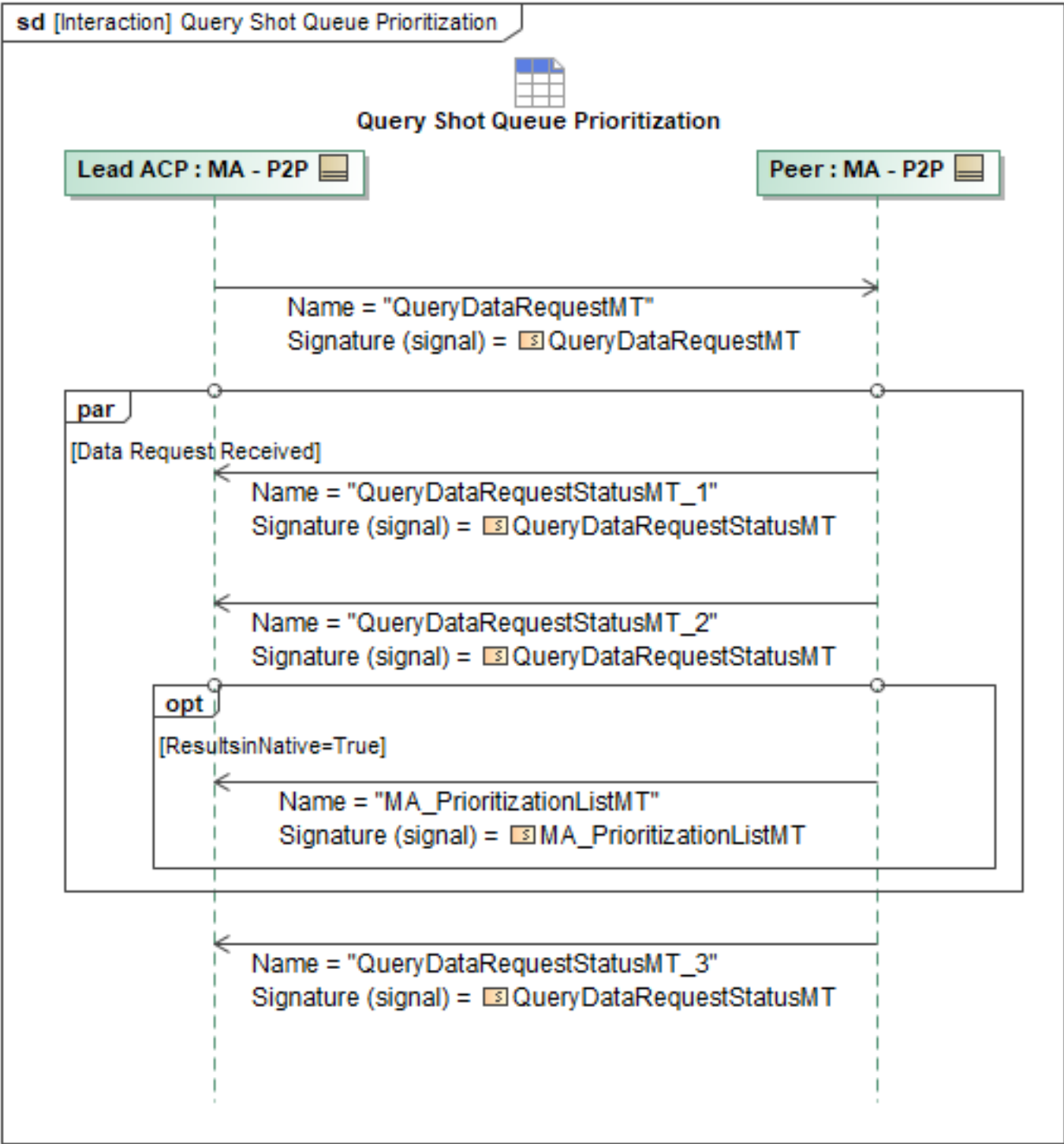


Figure 1-138: Query Shot Queue Prioritization

Table 1-137: Query Shot Queue Prioritization

| Message Name (Name:Type)                            | Required Fields (Name:Type)                          |
|-----------------------------------------------------|------------------------------------------------------|
| MA_PrioritizationListMT:<br>MA_PrioritizationListMT | MISSION_TARGET_LIST:<br>ObjectState: ObjectStateEnum |

| Message Name (Name:Type)                                | Required Fields (Name:Type)         |
|---------------------------------------------------------|-------------------------------------|
| QueryDataRequestMT: QueryDataRequestMT                  | ResultsInNative: QueryTopicPairType |
| QueryDataRequestStatusMT_1:<br>QueryDataRequestStatusMT | Result: QueryResultType             |
| QueryDataRequestStatusMT_2:<br>QueryDataRequestStatusMT | Result: QueryResultType             |
| QueryDataRequestStatusMT_3:<br>QueryDataRequestStatusMT | Result: QueryResultType             |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.2.15.11 Send Shot Queue Prioritization

Shot Queue Prioritization List is a list of mission targets ranked by assessing multiple factors to determine which target entities should be engaged first. The shot queue will utilize *MA\_PrioritizationListMT* with a *ListType* of *MISSION\_TARGET\_LIST* enumeration. The list requires subjects (*EntityID*) to be assigned a priority, where the lowest assigned value is the highest priority level. The message also includes an optional field, *ProbabilityOfKill*, that may be included with each assessed subject entity. MA (Multi-Ship Peer) may send an updated shot queue prioritization list to P2P (Multi-ship Lead) based on the active engagement criteria. Once *MA\_PrioritizationListMT* is received, P2P (Multi-ship Lead) will respond to the originating System with an *MA\_SystemNotificationMT*.

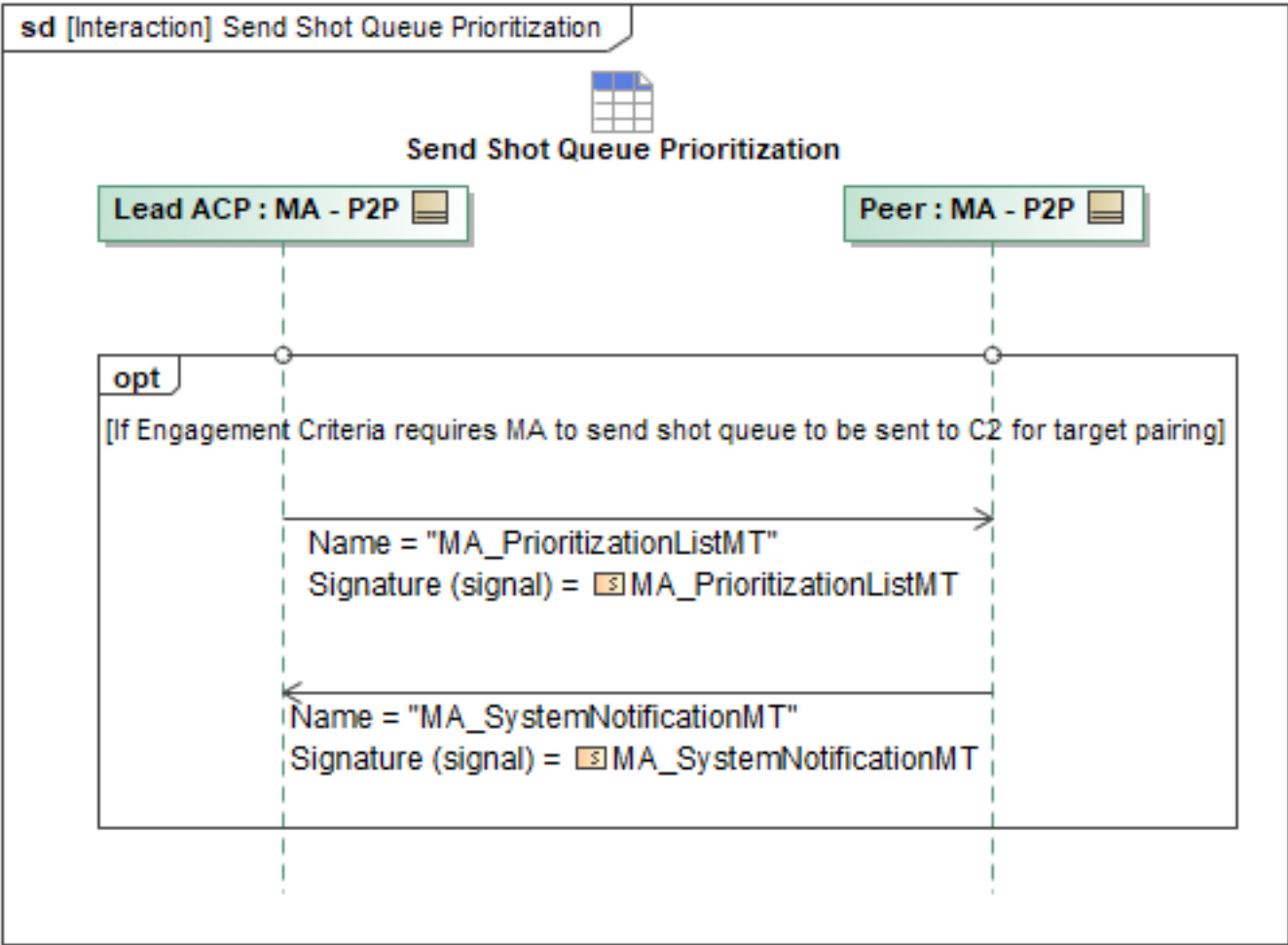


Figure 1-139: Send Shot Queue Prioritization

Table 1-138: Send Shot Queue Prioritization

| Message Name (Name:Type)                            | Required Fields (Name:Type)                          |
|-----------------------------------------------------|------------------------------------------------------|
| MA_PrioritizationListMT:<br>MA_PrioritizationListMT | MISSION_TARGET_LIST:<br>ObjectState: ObjectStateEnum |
| MA_SystemNotificationMT:<br>MA_SystemNotificationMT | None                                                 |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.12 Threat Prioritization

Mission Use Case: ALR

The Threat Prioritization List ranks entities by assessing the movements and behaviors to determine threat levels to a respective position, whether it is the ACP ownership or another platform. Threat Prioritization will utilize *MA\_PrioritizationListMT* with a ListType of THREAT\_TARGET\_PRIORITIZATION\_LIST enumeration. The list requires subjects (*EntityID*) to be assigned a priority, where the lowest assigned value is the highest threat level. Once an updated

*MA\_PrioritizationListMT* is received, P2P will respond to the originating System with an *MA\_SystemNotificationMT*.

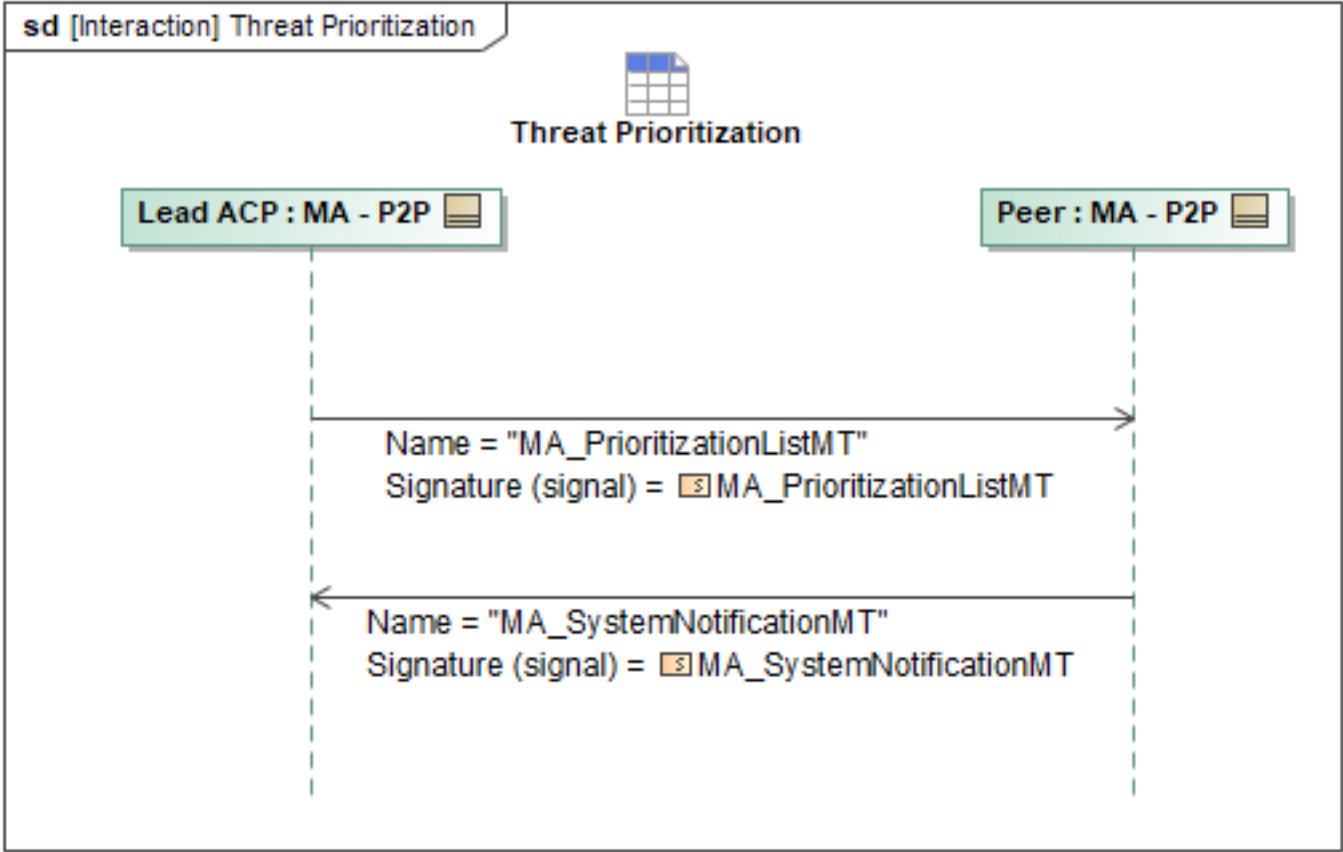


Figure 1-140: Threat Prioritization

Table 1-139: Threat Prioritization

| Message Name (Name:Type)                                          | Required Fields (Name:Type)                                                                                                                                         |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_PrioritizationListMT:</b><br><b>MA_PrioritizationListMT</b> | EntityID: EntityID_Type<br>ObjectState: ObjectStateEnum<br>THREAT_TARGET_PRIORITIZATION_LIST:<br>THREAT_TARGET_PRIORITIZATION_LIST:<br>ObjectState: ObjectStateEnum |
| <b>MA_SystemNotificationMT:</b><br><b>MA_SystemNotificationMT</b> | AssociatedMessage: AssociatedMessageType                                                                                                                            |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.13 Update Shot Queue Prioritization

Shot Queue Prioritization List is a list of mission targets ranked by assessing multiple factors to determine which target entities should be engaged first. The shot queue will utilize *MA\_PrioritizationListMT* with a *ListType* of MISSION\_TARGET\_LIST enumeration. The list requires

subjects (*EntityID*) to be assigned a priority, where the lowest assigned value is the highest priority level. The message also includes an optional field, *ProbabilityOfKill*, that may be included with each assessed subject entity. P2P (Multi-ship Lead) may send an updated shot queue prioritization list an ACP (Multi-Ship Peer). Once an updated *MA\_PrioritizationListMT* is received, MA will respond to the originating System with an *MA\_SystemNotificationMT*.

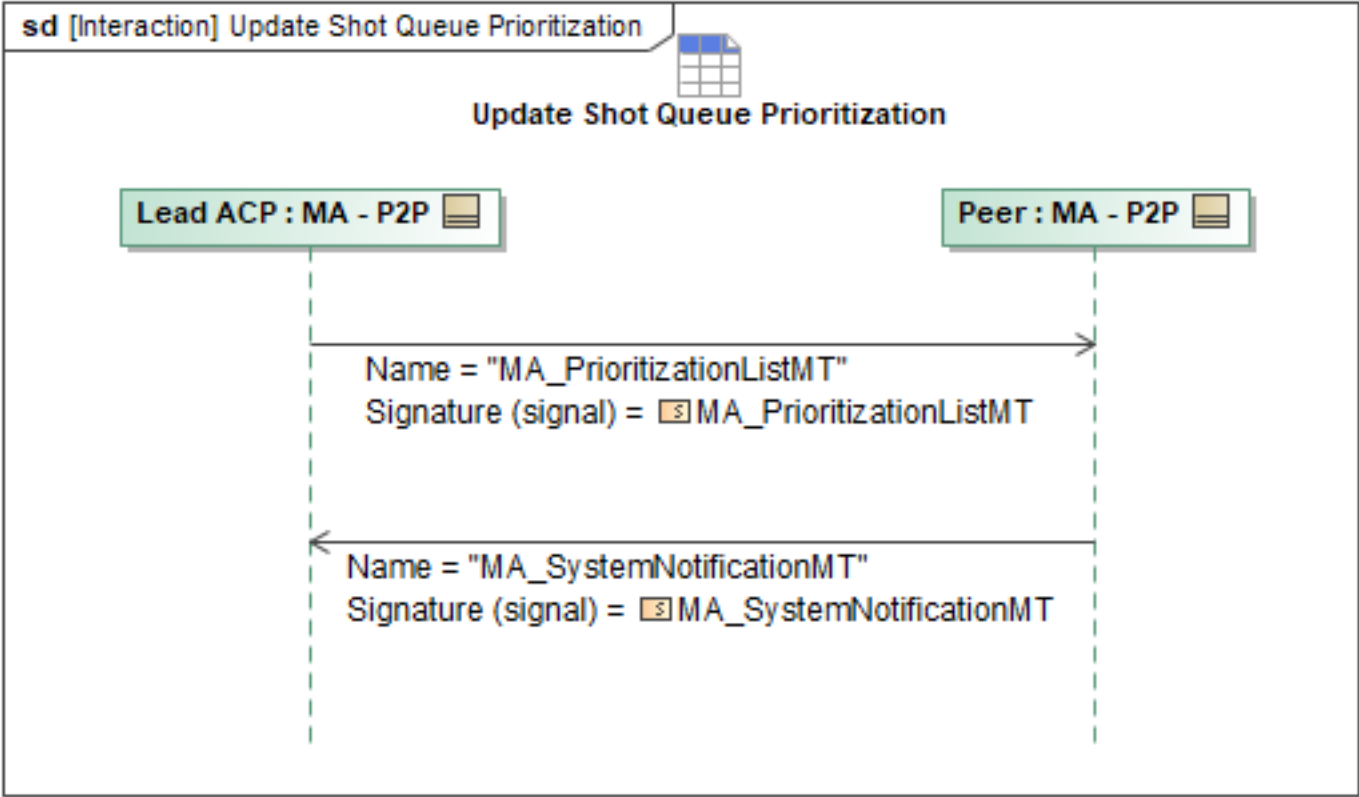


Figure 1-141: Update Shot Queue Prioritization

Table 1-140: Update Shot Queue Prioritization

| Message Name (Name:Type)                            | Required Fields (Name:Type)                          |
|-----------------------------------------------------|------------------------------------------------------|
| MA_PrioritizationListMT:<br>MA_PrioritizationListMT | MISSION_TARGET_LIST:<br>ObjectState: ObjectStateEnum |
| MA_SystemNotificationMT:<br>MA_SystemNotificationMT | AssociatedMessage: AssociatedMessageType             |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.14 WEZ Compound Convergence

WEZ Compound Convergence is a fusion of multiple WEZ from team members within a flight group. The fusion of WEZ enables P2P (Team Lead/QB) to generate a composite view of flight group’s weapon engagement capabilities to assess tasking for the best shot. Once ACPs are producing a WEZ in support of strike task, MA will send *MA\_WEZ\_MT* periodically to P2P (Team Lead /QB).

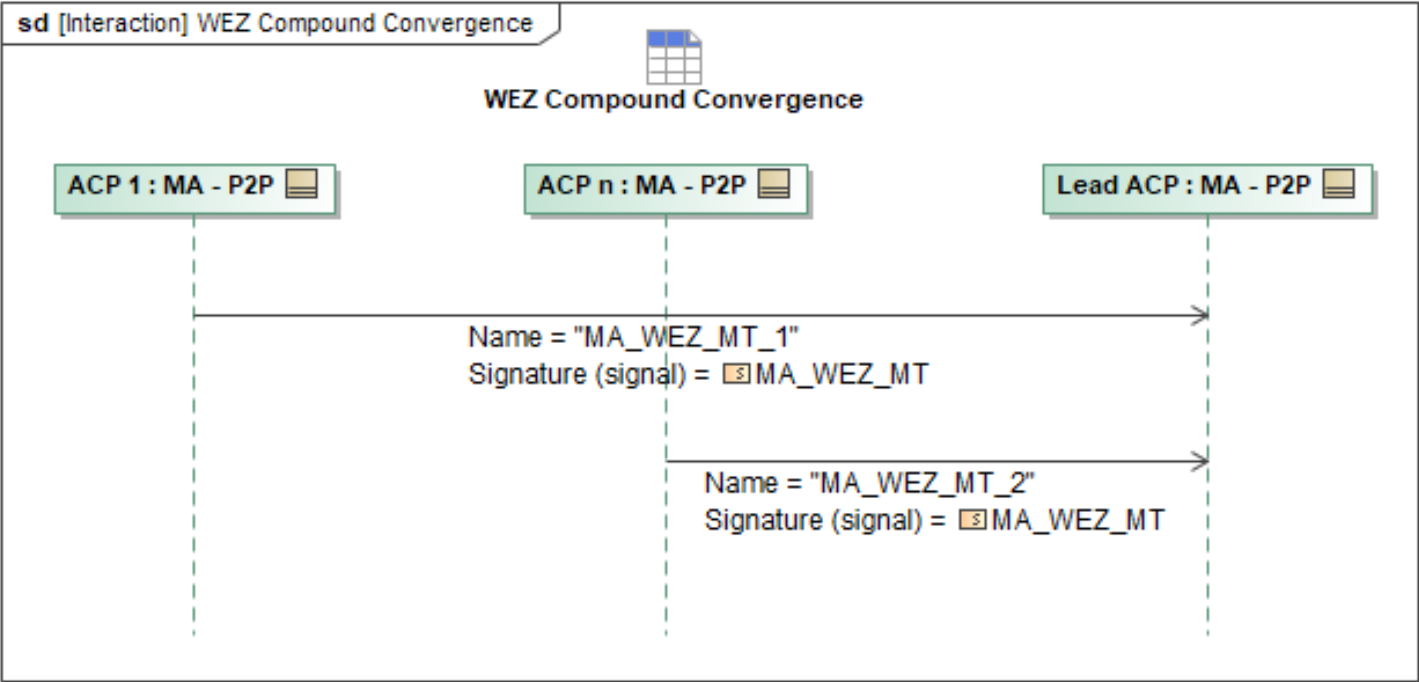


Figure 1-142: WEZ Compound Convergence

Table 1-141: WEZ Compound Convergence

| Message Name (Name:Type) | Required Fields (Name:Type) |
|--------------------------|-----------------------------|
| MA_WEZ_MT_1: MA_WEZ_MT   | None                        |
| MA_WEZ_MT_2: MA_WEZ_MT   | None                        |

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

### 1.3 Interface Compliance

The Interface Compliance table contains the L1 compliance requirements for the Peer interface.

Table 1-142: Peer Interface Compliance Requirements

| ID        | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MA-L1-003 | A Level 1 Peer interface shall support the Peer Interface Minimum Message Set defined in the Peer MMS table in the Peer feature profile (A-GRA location: 1 Problem Domain > 3 Exchange Items > Minimum Message Set (MMS)) for all messages in Core Mission Use Case sequences and all non-Core Mission Use Case sequences required for platform specific implementations.                                                  |
| MA-L1-004 | A Level 1 Peer interface shall implement all required messaging sequences in the Peer feature profile (A-GRA location: 1 Problem Domain > 2 White Box > 3 Functional Analysis > L1 Functional Analysis > L1 Interface Behaviors > Peer Interface Behavior), marked Core and all non-Core Mission Use Case sequences that apply to the platform specific implementation.                                                    |
| MA-L1-018 | Upon receipt of a Malformed Message (a message missing expected fields (i.e., required by the behavior but not the UCI schema) or with invalid context-dependent information within the scope of that individual message) from the P2P interface, MA shall respond to the error by notifying P2P of the malformed message and requesting the corrected message according to the Malformed Message Response (P2P) sequence. |
| MA-L1-024 | A Level 1 Peer interface shall follow state transitions for CommandProcessingState and RequestProcessingState in accordance with OAC-SPC-001 Rev C, Section 5.1.1.1 and 5.1.2.1. Representations of non-terminal statuses are optional and should align with the definitions and usage prescribed by the UCI Schema Style and Design Specification, ensuring consistent message state handling.                            |

### 1.3.1 P2P MMS

The Minimum Message Set referenced in the Interface Compliance Requirements is shown in this section. For an implementor to be compliant, all messages shown in the table shall be supported. The "Direction" column specifies the message direction relative to the P2P interface. The "MUC" column specifies the message's Mission Use Case. The message and its respective interaction(s) are only required for the particular Mission Use Case listed.

Mission Use Cases (MUCs) identify specific implementation needs for different ACP solutions. These include a Core set of behaviors that all A-GRA-compliant systems must meet, as well as select use cases such as Carrier-based or Electronic Support Measure-capable. MUCs are considered to be L0 as they describe the use of an ACP, not necessarily a single L1 system. For L1 interface compliance, all implementations must meet the interactions delegated to the Core MUC and all interactions delegated to other MUCs relevant to their implementation. Compliance to interactions not delegated to MUCs do not count towards A-GRA compliance.

Table 1-143: P2P MMS

| Message Type | Message Name                           | Interface | Direction | Applicable MUCs |
|--------------|----------------------------------------|-----------|-----------|-----------------|
| UCI 2.3      | <b>AccessAssessmentRequestMT</b>       | P2P       | inout     | Core            |
| UCI 2.3      | <b>AccessAssessmentRequestStatusMT</b> | P2P       | inout     | Core            |
| UCI 2.3      | <b>ActionCancelCommandMT</b>           | P2P       | inout     | Core            |
| UCI 2.3      | <b>ActionCancelCommandStatusMT</b>     | P2P       | inout     | Core            |

| Message Type | Message Name                        | Interface | Direction | Applicable MUCs |
|--------------|-------------------------------------|-----------|-----------|-----------------|
| UCI 2.3      | ActionCapabilityStatusMT            | P2P       | inout     | Core            |
| UCI 2.3      | ActionPlanValidationCommandMT       | P2P       | inout     | Core            |
| UCI 2.3      | ActionPlanValidationCommandStatusMT | P2P       | inout     | Core            |
| UCI 2.3      | ActivityPlanApprovalStatusMT        | P2P       | inout     | Core            |
| UCI 2.3      | AirfieldReportMT                    | P2P       | inout     | Core            |
| UCI 2.3      | AirSampleActivityMT                 | P2P       | inout     | Core            |
| UCI 2.3      | AirSampleCommandMT                  | P2P       | inout     | Core            |
| UCI 2.3      | AirSampleCommandStatusMT            | P2P       | inout     | Core            |
| UCI 2.3      | AirSampleSettingsCommandMT          | P2P       | inout     | Core            |
| UCI 2.3      | AirSampleSettingsCommandStatusMT    | P2P       | inout     | Core            |
| UCI 2.3      | AMTI_ActivityMT                     | P2P       | inout     | Core, AMTI      |
| UCI 2.3      | AMTI_CapabilityStatusMT             | P2P       | inout     | AMTI, Core      |
| UCI 2.3      | AMTI_CommandMT                      | P2P       | inout     | Core            |
| UCI 2.3      | AMTI_CommandStatusMT                | P2P       | inout     | Core            |
| UCI 2.3      | AMTI_SettingsCommandMT              | P2P       | inout     | Core            |
| UCI 2.3      | AMTI_SettingsCommandStatusMT        | P2P       | inout     | Core            |
| UCI 2.3      | AO_ActivityMT                       | P2P       | inout     | Core            |
| UCI 2.3      | AO_CapabilityMT                     | P2P       | inout     | Core            |
| UCI 2.3      | AO_CapabilityStatusMT               | P2P       | inout     | Core            |
| UCI 2.3      | AO_CommandMT                        | P2P       | inout     | Core            |
| UCI 2.3      | AO_CommandStatusMT                  | P2P       | inout     | Core            |
| UCI 2.3      | AO_SettingsCommandMT                | P2P       | inout     | Core            |
| UCI 2.3      | AO_SettingsCommandStatusMT          | P2P       | inout     | Core            |
| UCI 2.3      | CargoDeliveryCapabilityMT           | P2P       | inout     | Core            |
| UCI 2.3      | CargoDeliveryCapabilityStatusMT     | P2P       | inout     | Core            |
| UCI 2.3      | COMINT_ActivityMT                   | P2P       | inout     | Core            |
| UCI 2.3      | COMINT_CapabilityMT                 | P2P       | inout     | Core            |
| UCI 2.3      | COMINT_CapabilityStatusMT           | P2P       | inout     | Core            |
| UCI 2.3      | COMINT_CommandMT                    | P2P       | inout     | Core            |
| UCI 2.3      | COMINT_CommandStatusMT              | P2P       | inout     | Core            |
| UCI 2.3      | COMINT_SettingsCommandMT            | P2P       | inout     | Core            |
| UCI 2.3      | COMINT_SettingsCommandStatusMT      | P2P       | inout     | Core            |



| Message Type | Message Name                              | Interface | Direction | Applicable MUCs |
|--------------|-------------------------------------------|-----------|-----------|-----------------|
| UCI 2.3      | CommPointingCommandMT                     | P2P       | inout     | Core            |
| UCI 2.3      | CommPointingCommandStatusMT               | P2P       | inout     | Core            |
| UCI 2.3      | CommRelayActivityMT                       | P2P       | inout     | Core            |
| UCI 2.3      | CommRelayCommandMT                        | P2P       | inout     | Core            |
| UCI 2.3      | CommRelayCommandStatusMT                  | P2P       | inout     | Core            |
| UCI 2.3      | CommRelaySettingsCommandMT                | P2P       | inout     | Core            |
| UCI 2.3      | CommRelaySettingsCommandStatusMT          | P2P       | inout     | Core            |
| UCI 2.3      | CommTerminalCapabilityMT                  | P2P       | inout     | Core            |
| UCI 2.3      | CommTerminalCapabilityStatusMT            | P2P       | inout     | Core            |
| UCI 2.3      | CommTerminalCommandMT                     | P2P       | inout     | Core            |
| UCI 2.3      | CommTerminalCommandStatusMT               | P2P       | inout     | Core            |
| UCI 2.3      | CommTerminalPlanActivationCommandMT       | P2P       | inout     | Core            |
| UCI 2.3      | CommTerminalPlanActivationCommandStatusMT | P2P       | inout     | Core            |
| UCI 2.3      | CommTerminalSettingsCommandMT             | P2P       | inout     | Core            |
| UCI 2.3      | CommTerminalSettingsCommandStatusMT       | P2P       | inout     | Core            |
| UCI 2.3      | ComponentStatusMT                         | P2P       | inout     | Core            |
| UCI 2.3      | ControlStatusMT                           | P2P       | inout     | Core            |
| UCI 2.3      | DLZ_MT                                    | P2P       | inout     | Core            |
| UCI 2.3      | DMPI_CancelCommandMT                      | P2P       | inout     | Core            |
| UCI 2.3      | DMPI_CancelCommandStatusMT                | P2P       | inout     | Core            |
| UCI 2.3      | DoorCommandMT                             | P2P       | inout     | Core            |
| UCI 2.3      | DoorCommandStatusMT                       | P2P       | inout     | Core            |
| UCI 2.3      | EA_ActivityMT                             | P2P       | inout     | Core            |
| UCI 2.3      | EA_CapabilityMT                           | P2P       | inout     | Core            |
| UCI 2.3      | EA_CapabilityStatusMT                     | P2P       | inout     | Core            |
| UCI 2.3      | EA_CommandMT                              | P2P       | inout     | Core            |
| UCI 2.3      | EA_CommandStatusMT                        | P2P       | inout     | Core            |
| UCI 2.3      | EffectCapabilityStatusMT                  | P2P       | inout     | Core            |
| UCI 2.3      | EntityMT                                  | P2P       | inout     | Core            |
| UCI 2.3      | ESM_ActivityMT                            | P2P       | inout     | Core            |
| UCI 2.3      | ESM_CapabilityMT                          | P2P       | inout     | Core            |
| UCI 2.3      | ESM_CapabilityStatusMT                    | P2P       | inout     | Core            |

| Message Type           | Message Name                            | Interface | Direction | Applicable MUCs               |
|------------------------|-----------------------------------------|-----------|-----------|-------------------------------|
| UCI 2.3                | ESM_CommandMT                           | P2P       | inout     | Core                          |
| UCI 2.3                | ESM_CommandStatusMT                     | P2P       | inout     | Core                          |
| UCI 2.3                | ESM_SettingsCommandMT                   | P2P       | inout     | Core                          |
| UCI 2.3                | ESM_SettingsCommandStatusMT             | P2P       | inout     | Core                          |
| UCI 2.3                | GatewayCapabilityMT                     | P2P       | inout     | Additional Comms Capabilities |
| UCI 2.3                | GatewayCapabilityStatusMT               | P2P       | inout     | Additional Comms Capabilities |
| UCI 2.3                | GatewayCommandMT                        | P2P       | inout     | Core                          |
| UCI 2.3                | GatewayCommandStatusMT                  | P2P       | inout     | Core                          |
| UCI 2.3                | IFF_CapabilityMT                        | P2P       | inout     | Core                          |
| UCI 2.3                | IFF_CapabilityStatusMT                  | P2P       | inout     | Core                          |
| UCI 2.3                | IFF_CommandMT                           | P2P       | inout     | Core                          |
| UCI 2.3                | IFF_CommandStatusMT                     | P2P       | inout     | Core                          |
| UCI 2.3                | IFF_SettingsCommandMT                   | P2P       | inout     | Core                          |
| UCI 2.3                | IFF_SettingsCommandStatusMT             | P2P       | inout     | Core                          |
| A-GRA Custom Extension | MA_AcceptableLevelOfRiskCommandMT       | P2P       | inout     | ALR                           |
| A-GRA Custom Extension | MA_AcceptableLevelOfRiskCommandStatusMT | P2P       | inout     | ALR                           |
| A-GRA Custom Extension | MA_AcceptableLevelOfRiskStatusMT        | P2P       | inout     | ALR                           |
| A-GRA Custom Extension | MA_ActionCapabilityMT                   | P2P       | inout     | Core                          |
| A-GRA Custom Extension | MA_ActionCommandMT                      | P2P       | inout     | Core                          |
| A-GRA Custom Extension | MA_ActionCommandStatusMT                | P2P       | inout     | Core                          |
| A-GRA Custom Extension | MA_ActionMT                             | P2P       | inout     | Core                          |
| A-GRA Custom Extension | MA_ActionPlanMT                         | P2P       | inout     | Core                          |
| A-GRA Custom Extension | MA_ActionPlanValidationMT               | P2P       | inout     | Core                          |
| A-GRA Custom Extension | MA_ActionStatusMT                       | P2P       | inout     | Core                          |
| A-GRA Custom           | MA_ActivityPlanMT                       | P2P       | inout     | Core                          |

| Message Type           | Message Name                                       | Interface | Direction | Applicable MUCs |
|------------------------|----------------------------------------------------|-----------|-----------|-----------------|
| Extension              |                                                    |           |           |                 |
| A-GRA Custom Extension | <b>MA_AMTI_CapabilityMT</b>                        | P2P       | inout     | AMTI, Core      |
| A-GRA Custom Extension | <b>MA_ApprovalAuthorityMT</b>                      | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_ApprovalPolicyMT</b>                         | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_ApprovalRequestMT</b>                        | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_ApprovalRequestStatusMT</b>                  | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_AssessmentMT</b>                             | P2P       | inout     | ALR             |
| A-GRA Custom Extension | <b>MA_AssessmentRequestMT</b>                      | P2P       | inout     | ALR             |
| A-GRA Custom Extension | <b>MA_AssessmentRequestStatusMT</b>                | P2P       | inout     | ALR             |
| A-GRA Custom Extension | <b>MA_CAP_ExecutionStatusReportMT</b>              | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_CapabilitySetMT</b>                          | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_ConstraintPriorityListActivationDataMT</b>   | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_ConstraintPriorityListActivationStatusMT</b> | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_ConstraintPriorityListCommandMT</b>          | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_ConstraintPriorityListCommandStatusMT</b>    | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_ConstraintPriorityListMT</b>                 | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_COP_ConfigurationSettingsCommandMT</b>       | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_COP_ConfigurationSettingsCommandStatusMT</b> | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_COP_ConfigurationSettingsMT</b>              | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_COP_ConfigurationSettingsRequestMT</b>       | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_COP_ConfigurationSettingsRequestStatusMT</b> | P2P       | inout     | Core            |

| Message Type           | Message Name                                              | Interface | Direction | Applicable MUCs               |
|------------------------|-----------------------------------------------------------|-----------|-----------|-------------------------------|
| A-GRA Custom Extension | <b>MA_DataDisseminationPlanActivationComm andMT</b>       | P2P       | inout     | Additional Comms Capabilities |
| A-GRA Custom Extension | <b>MA_DataDisseminationPlanActivationComm andStatusMT</b> | P2P       | inout     | Additional Comms Capabilities |
| A-GRA Custom Extension | <b>MA_DataDisseminationPlanActivationStatus MT</b>        | P2P       | inout     | Additional Comms Capabilities |
| A-GRA Custom Extension | <b>MA_DataDisseminationPlanMT</b>                         | P2P       | inout     | Additional Comms Capabilities |
| A-GRA Custom Extension | <b>MA_DataDisseminationPlanOverrideRequest MT</b>         | P2P       | inout     | Additional Comms Capabilities |
| A-GRA Custom Extension | <b>MA_DataDisseminationPlanOverrideRequest StatusMT</b>   | P2P       | inout     | Additional Comms Capabilities |
| A-GRA Custom Extension | <b>MA_DeconflictionSolutionRequestMT</b>                  | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_DeconflictionSolutionRequestStatusMT</b>            | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_DesignationRequestMT</b>                            | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_DesignationRequestStatusMT</b>                      | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_EffectCapabilityMT</b>                              | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_EngagementCriteriaCommandMT</b>                     | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_EngagementCriteriaCommandStatusMT</b>               | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_EngagementCriteriaMT</b>                            | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_EngagementCriteriaStatusMT</b>                      | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_EscortActivityMT</b>                                | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_EscortCommandMT</b>                                 | P2P       | inout     | Core                          |
| A-GRA Custom Extension | <b>MA_EscortCommandStatusMT</b>                           | P2P       | inout     | Core                          |
| A-GRA Custom           | <b>MA_ExecutionPriorityListActivationDataMT</b>           | P2P       | inout     | Core                          |

| Message Type           | Message Name                                          | Interface | Direction | Applicable MUCs                                                                                  |
|------------------------|-------------------------------------------------------|-----------|-----------|--------------------------------------------------------------------------------------------------|
| Extension              |                                                       |           |           |                                                                                                  |
| A-GRA Custom Extension | <b>MA_ExecutionPriorityListActivationStatusMT</b>     | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_ExecutionPriorityListCommandMT</b>              | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_ExecutionPriorityListCommandStatusMT</b>        | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_ExecutionPriorityListMT</b>                     | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_FlightActivityMT</b>                            | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_FlightCapabilityMT</b>                          | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_FlightCapabilityStatusMT</b>                    | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_FlightCommandMT</b>                             | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_FlightCommandStatusMT</b>                       | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_GatewayActivityMT</b>                           | P2P       | inout     | Additional Comms Capabilities                                                                    |
| A-GRA Custom Extension | <b>MA_IgnoreConstraintCommandMT</b>                   | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_IgnoreConstraintCommandStatusMT</b>             | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_LeaderUpdateRequestMT</b>                       | P2P       | inout     | Core, Bully Leader Election, Max Consensus Leader Election, Static Fitness Score Leader Election |
| A-GRA Custom Extension | <b>MA_LeaderUpdateRequestStatusMT</b>                 | P2P       | inout     | Core, Max Consensus Leader Election, Bully Leader Election, Static Fitness Score Leader Election |
| A-GRA Custom Extension | <b>MA_MessageAcknowledgmentConfigurationMT</b>        | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_MessageAcknowledgmentConfigurationRequestMT</b> | P2P       | inout     | Core                                                                                             |

| Message Type           | Message Name                                                | Interface | Direction | Applicable MUCs |
|------------------------|-------------------------------------------------------------|-----------|-----------|-----------------|
| A-GRA Custom Extension | <b>MA_MessageAcknowledgmentConfigurationRequestStatusMT</b> | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_MissionPlanActivationCommandMT</b>                    | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_MissionPlanActivationCommandStatusMT</b>              | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_MissionPlanActivationStatusMT</b>                     | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_MissionPlanExecutionStatusMT</b>                      | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_MissionPlanMT</b>                                     | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_NavigationReportMT</b>                                | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpCommandMT</b>                                       | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpCommandStatusMT</b>                                 | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OperatorRoleMT</b>                                    | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpLineMT</b>                                          | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpPlanMT</b>                                          | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpPointMT</b>                                         | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpRoutingMT</b>                                       | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpStatusMT</b>                                        | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpVolumeMT</b>                                        | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_OpZoneMT</b>                                          | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_PackageManagementCommandMT</b>                        | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_PackageManagementCommandStatusMT</b>                  | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_PackagePartnerStatusReportMT</b>                      | P2P       | inout     | Core            |
| A-GRA Custom Extension | <b>MA_PlanModificationRequestMT</b>                         | P2P       | inout     | Core            |

| Message Type           | Message Name                                      | Interface | Direction | Applicable MUCs        |
|------------------------|---------------------------------------------------|-----------|-----------|------------------------|
| A-GRA Custom Extension | <b>MA_PlanModificationRequestStatusMT</b>         | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_PlanningFunctionSettingsCommandMT</b>       | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_PlanningFunctionSettingsCommandStatusMT</b> | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_PlatformOverrideStatusMT</b>                | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_PositionReportDetailedMT</b>                | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_PositionReportMT</b>                        | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_PrioritizationListMT</b>                    | P2P       | inout     | ALR, Core              |
| A-GRA Custom Extension | <b>MA_RaftAppendEntriesRequestMT</b>              | P2P       | inout     | Raft Leader Election   |
| A-GRA Custom Extension | <b>MA_RaftAppendEntriesRequestStatusMT</b>        | P2P       | inout     | Raft Leader Election   |
| A-GRA Custom Extension | <b>MA_RaftVoteRequestMT</b>                       | P2P       | inout     | Raft Leader Election   |
| A-GRA Custom Extension | <b>MA_RaftVoteRequestStatusMT</b>                 | P2P       | inout     | Raft Leader Election   |
| A-GRA Custom Extension | <b>MA_ResponseActivityMT</b>                      | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_ResponseCommandMT</b>                       | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_ResponseCommandStatusMT</b>                 | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_ResponseMT</b>                              | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_RouteActivityPlanMT</b>                     | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_RouteDeconflictionAlertMT</b>               | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_RoutePlanMT</b>                             | P2P       | inout     | Core                   |
| A-GRA Custom Extension | <b>MA_StrikeActivityMT</b>                        | P2P       | inout     | Core, Store Management |
| A-GRA Custom Extension | <b>MA_StrikeCapabilityMT</b>                      | P2P       | inout     | Store Management, Core |
| A-GRA Custom           | <b>MA_SystemManagementRequestMT</b>               | P2P       | inout     | Core                   |

| Message Type           | Message Name                              | Interface | Direction | Applicable MUCs                                                                                  |
|------------------------|-------------------------------------------|-----------|-----------|--------------------------------------------------------------------------------------------------|
| Extension              |                                           |           |           |                                                                                                  |
| A-GRA Custom Extension | <b>MA_SystemManagementRequestStatusMT</b> | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_SystemNotificationMT</b>            | P2P       | inout     | Core, ALR                                                                                        |
| A-GRA Custom Extension | <b>MA_TaskCommandMT</b>                   | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_TaskCommandStatusMT</b>             | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_TaskMT</b>                          | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_TaskPlanMT</b>                      | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_TaskStatusMT</b>                    | P2P       | inout     | Core                                                                                             |
| A-GRA Custom Extension | <b>MA_WEZ_MT</b>                          | P2P       | inout     | Core                                                                                             |
| UCI 2.3                | <b>MissionContingencyAlertMT</b>          | P2P       | inout     | Core                                                                                             |
| UCI 2.3                | <b>MissionPlanApprovalStatusMT</b>        | P2P       | inout     | Core                                                                                             |
| UCI 2.3                | <b>ObservationMeasurementReportMT</b>     | P2P       | inout     | Core                                                                                             |
| UCI 2.3                | <b>OpaqueCapabilityMT</b>                 | P2P       | inout     | Core                                                                                             |
| UCI 2.3                | <b>OpaqueCapabilityStatusMT</b>           | P2P       | inout     | Core                                                                                             |
| UCI 2.3                | <b>OperatorMT</b>                         | P2P       | inout     | Core                                                                                             |
| UCI 2.3                | <b>PackageMT</b>                          | P2P       | inout     | Bully Leader Election, Static Fitness Score Leader Election, Max Consensus Leader Election, Core |
| UCI 2.3                | <b>PackageStatusMT</b>                    | P2P       | inout     | Core, Bully Leader Election, Static Fitness Score Leader Election, Max Consensus Leader Election |
| UCI 2.3                | <b>PO_ActivityMT</b>                      | P2P       | inout     | Core, PO                                                                                         |
| UCI 2.3                | <b>PO_CapabilityMT</b>                    | P2P       | inout     | PO, Core                                                                                         |
| UCI 2.3                | <b>PO_CapabilityStatusMT</b>              | P2P       | inout     | PO, Core                                                                                         |
| UCI 2.3                | <b>PO_CommandMT</b>                       | P2P       | inout     | Core                                                                                             |



| Message Type | Message Name                          | Interface | Direction | Applicable MUCs |
|--------------|---------------------------------------|-----------|-----------|-----------------|
| UCI 2.3      | PO_CommandStatusMT                    | P2P       | inout     | Core            |
| UCI 2.3      | PO_SettingsCommandMT                  | P2P       | inout     | Core            |
| UCI 2.3      | PO_SettingsCommandStatusMT            | P2P       | inout     | Core            |
| UCI 2.3      | QueryDataRequestMT                    | P2P       | inout     | Core            |
| UCI 2.3      | QueryDataRequestStatusMT              | P2P       | inout     | Core            |
| UCI 2.3      | RadarAltimeterCapabilityMT            | P2P       | inout     | Core            |
| UCI 2.3      | RadarAltimeterCapabilityStatusMT      | P2P       | inout     | Core            |
| UCI 2.3      | RadarAltimeterCommandMT               | P2P       | inout     | Core            |
| UCI 2.3      | RadarAltimeterCommandStatusMT         | P2P       | inout     | Core            |
| UCI 2.3      | RadarAltimeterSettingsCommandMT       | P2P       | inout     | Core            |
| UCI 2.3      | RadarAltimeterSettingsCommandStatusMT | P2P       | inout     | Core            |
| UCI 2.3      | RefuelCapabilityMT                    | P2P       | inout     | Core            |
| UCI 2.3      | RefuelCapabilityStatusMT              | P2P       | inout     | Core            |
| UCI 2.3      | RefuelCommandMT                       | P2P       | inout     | Core            |
| UCI 2.3      | RefuelCommandStatusMT                 | P2P       | inout     | Core            |
| UCI 2.3      | RefuelSettingsCommandMT               | P2P       | inout     | Core            |
| UCI 2.3      | RefuelSettingsCommandStatusMT         | P2P       | inout     | Core            |
| UCI 2.3      | RelationshipDesignationMT             | P2P       | inout     | Core            |
| UCI 2.3      | ResponsePlanMT                        | P2P       | inout     | Core            |
| UCI 2.3      | RouteModificationRequestMT            | P2P       | inout     | Core            |
| UCI 2.3      | RouteModificationRequestStatusMT      | P2P       | inout     | Core            |
| UCI 2.3      | RoutePlanApprovalStatusMT             | P2P       | inout     | Core            |
| UCI 2.3      | RoutePlanExecutionStatusMT            | P2P       | inout     | Core            |
| UCI 2.3      | SAR_ActivityMT                        | P2P       | inout     | Core            |
| UCI 2.3      | SAR_CapabilityMT                      | P2P       | inout     | Core            |
| UCI 2.3      | SAR_CapabilityStatusMT                | P2P       | inout     | Core            |
| UCI 2.3      | SAR_CommandMT                         | P2P       | inout     | Core            |
| UCI 2.3      | SAR_CommandStatusMT                   | P2P       | inout     | Core            |
| UCI 2.3      | SAR_SettingsCommandMT                 | P2P       | inout     | Core            |
| UCI 2.3      | SAR_SettingsCommandStatusMT           | P2P       | inout     | Core            |
| UCI 2.3      | SMTI_ActivityMT                       | P2P       | inout     | Core            |
| UCI 2.3      | SMTI_CapabilityMT                     | P2P       | inout     | Core            |

| Message Type | Message Name                       | Interface | Direction | Applicable MUCs        |
|--------------|------------------------------------|-----------|-----------|------------------------|
| UCI 2.3      | SMTI_CapabilityStatusMT            | P2P       | inout     | Core                   |
| UCI 2.3      | SMTI_CommandMT                     | P2P       | inout     | Core                   |
| UCI 2.3      | SMTI_CommandStatusMT               | P2P       | inout     | Core                   |
| UCI 2.3      | SMTI_SettingsCommandMT             | P2P       | inout     | Core                   |
| UCI 2.3      | SMTI_SettingsCommandStatusMT       | P2P       | inout     | Core                   |
| UCI 2.3      | StoreManagementCommandMT           | P2P       | inout     | Core                   |
| UCI 2.3      | StoreManagementCommandStatusMT     | P2P       | inout     | Core                   |
| UCI 2.3      | StoreManagementStatusMT            | P2P       | inout     | Store Management       |
| UCI 2.3      | StrikeCapabilityStatusMT           | P2P       | inout     | Store Management, Core |
| UCI 2.3      | StrikeCommandMT                    | P2P       | inout     | Core                   |
| UCI 2.3      | StrikeCommandStatusMT              | P2P       | inout     | Core                   |
| UCI 2.3      | SubsystemStatusMT                  | P2P       | inout     | Core                   |
| UCI 2.3      | SurvivabilityRiskLevelMT           | P2P       | inout     | ALR                    |
| UCI 2.3      | SystemDeploymentCapabilityMT       | P2P       | inout     | Core                   |
| UCI 2.3      | SystemDeploymentCapabilityStatusMT | P2P       | inout     | Core                   |
| UCI 2.3      | SystemStatusMT                     | P2P       | inout     | Core                   |
| UCI 2.3      | TacticalOrderCapabilityMT          | P2P       | inout     | Core                   |
| UCI 2.3      | TacticalOrderCapabilityStatusMT    | P2P       | inout     | Core                   |
| UCI 2.3      | TaskCancelCommandMT                | P2P       | inout     | Core                   |
| UCI 2.3      | TaskCancelCommandStatusMT          | P2P       | inout     | Core                   |
| UCI 2.3      | TaskPlanApprovalStatusMT           | P2P       | inout     | Core                   |
| UCI 2.3      | TaskPlanExecutionStatusMT          | P2P       | inout     | Core                   |
| UCI 2.3      | TurretCommandMT                    | P2P       | inout     | Core                   |
| UCI 2.3      | TurretCommandStatusMT              | P2P       | inout     | Core                   |
| UCI 2.3      | WeatherMT                          | P2P       | inout     | Core                   |
| UCI 2.3      | WeatherRadarActivityMT             | P2P       | inout     | Core                   |
| UCI 2.3      | WeatherRadarCapabilityMT           | P2P       | inout     | Core                   |
| UCI 2.3      | WeatherRadarCapabilityStatusMT     | P2P       | inout     | Core                   |
| UCI 2.3      | WeatherRadarCommandMT              | P2P       | inout     | Core                   |
| UCI 2.3      | WeatherRadarCommandStatusMT        | P2P       | inout     | Core                   |
| UCI 2.3      | WeatherRadarSettingsCommandMT      | P2P       | inout     | Core                   |

| Message Type | Message Name                               | Interface | Direction | Applicable MUCs |
|--------------|--------------------------------------------|-----------|-----------|-----------------|
| UCI 2.3      | <b>WeatherRadarSettingsCommandStatusMT</b> | P2P       | inout     | Core            |

### 1.3.2 P2P UCI Extensions

The tables in the following sections provide the specifications for extended UCI 2.3 messages utilized by the A-GRA to communicate Autonomy L1 interface interactions. The tables include both extended messages and message types. Each table includes the message name, the functional description, and fields contained within. Top-level messages include images of the nested message structure. For a better view of the diagram and the message structure, it is strongly recommended to look at the schema in a XML viewer tool.

#### 1.3.2.1 MA\_WEZ\_MT UCI Extension

| MA_WEZ_MT          |                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | <p>This message represents a single Weapon Engagement Zone (WEZ). The WEZ represents a less dynamic capability than the DLZ for air to air engagements and is not reliant on target state data to be produced.</p> <p>See the XSD File for more info</p>                                                                                                                                                                   |
| <b>Ext. Fields</b> | <p>MessageData : MA_WEZ_MDT</p> <p>WEZ_ID : MA_WEZ_ID_Type</p> <p>WEZ_Data : MA_WEZ_DataType</p> <p>RangeTurnAndClose : MA_WeaponRangeType</p> <p>RangeMinimum : MA_WeaponRangeType</p> <p>RangeTurnAndRun : MA_WeaponRangeType</p> <p>RangeProbabilityOfIntercept : MA_WeaponRangeType</p> <p>RangeMaxGuidance : MA_WeaponRangeType</p> <p>RangeOptimal : MA_WeaponRangeType</p> <p>RangeMaxAero : MA_WeaponRangeType</p> |

**Table A-1-144: P2P UCI Extensions MA\_WEZ\_MT UCI Extension**

| MessageData       |                                                                                                       |             |
|-------------------|-------------------------------------------------------------------------------------------------------|-------------|
| Description       | This represents the characteristics that identify and describe a single Weapon Engagement Zone (WEZ). |             |
| Base Type         | MA_WEZ_MDT                                                                                            |             |
| Message Structure |                                                                                                       |             |
| Field Name        | Field Type                                                                                            | Field Usage |
| WEZ_ID            | MA_WEZ_ID_Type                                                                                        | Required    |
| Description       | This is an ID to uniquely identify the Weapon Engagement Zone (WEZ).                                  |             |
| WEZ_RequestID     | RequestID_Type                                                                                        | Required    |

|                          |                                                                            |          |
|--------------------------|----------------------------------------------------------------------------|----------|
| Description              | Indicates the unique ID of the request message.                            |          |
| <b>CapabilityID</b>      | CapabilityID_Type                                                          | Optional |
| Description              | Indicates the weapon Capability instance that this WEZ is associated with. |          |
| <b>GeneratedByWeapon</b> | boolean                                                                    | Optional |
| Description              | This element indicates if the WEZ was generated using payload algorithms.  |          |
| <b>WEZ_Data</b>          | MA_WEZ_DataType                                                            | Required |
| Description              | Indicates WEZ range calculations for air-to-air engagement assessments.    |          |

Table A-1-145: P2P UCI Extensions MA\_WEZ\_MT - MessageData UCI Extension Field

| WEZ_ID            |                                                                                                                                                                  |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | This is an ID to uniquely identify the Weapon Engagement Zone (WEZ).                                                                                             |             |
| Base Type         | MA_WEZ_ID_Type                                                                                                                                                   |             |
| Message Structure |                                                                                                                                                                  |             |
| Field Name        | Field Type                                                                                                                                                       | Field Usage |
| UUID              | UniversallyUniqueIdentifierType                                                                                                                                  | Required    |
| Description       | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122. |             |
| DescriptiveLabel  | VisibleString256Type                                                                                                                                             | Optional    |
| Description       | Human readable text label with content meaningful to human operators.                                                                                            |             |

Table A-1-146: P2P UCI Extensions MA\_WEZ\_MT - WEZ\_ID UCI Extension Field

| WEZ_Data          |                                                                                                                                                                                                                                                                                                                                                                       |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates WEZ range calculations for air-to-air engagement assessments.                                                                                                                                                                                                                                                                                               |             |
| Base Type         | MA_WEZ_DataType                                                                                                                                                                                                                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                       |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                            | Field Usage |
| RangeTurnAndClose | MA_WeaponRangeType                                                                                                                                                                                                                                                                                                                                                    | Repeated    |
| Description       | Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for:<br><br>1.Current steering<br><br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br><br>3.Target turn-and-close maneuver<br><br>RangeTurnAndClose is not applicable to ballistic trajectory targets. |             |
| RangeMinimum      | MA_WeaponRangeType                                                                                                                                                                                                                                                                                                                                                    | Repeated    |
| Description       | Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for:<br><br>1.Current steering                                                                                                                                                                                             |             |

|                                    |                                                                                                                                                                                                                                                                                                                                                   |                 |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
|                                    | 2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br>3.No target maneuvers<br><br>Note: ballistic trajectory velocity changes are not considered maneuvers.                                                                                                                                                                        |                 |
| <b>RangeTurnAndRun</b>             | <b>MA_WeaponRangeType</b>                                                                                                                                                                                                                                                                                                                         | <b>Repeated</b> |
| Description                        | Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for:<br>1.Current steering<br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br>3.Target turn-and-run maneuver<br><br>RangeTurnAndRun is not applicable to ballistic trajectory targets.   |                 |
| <b>RangeProbabilityOfIntercept</b> | <b>MA_WeaponRangeType</b>                                                                                                                                                                                                                                                                                                                         | <b>Repeated</b> |
| Description                        | Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for:<br>1.Current steering<br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br>3.No target maneuvers<br><br>Note: ballistic trajectory velocity changes are not considered maneuvers.     |                 |
| <b>RangeMaxGuidance</b>            | <b>MA_WeaponRangeType</b>                                                                                                                                                                                                                                                                                                                         | <b>Repeated</b> |
| Description                        | Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for:<br>1.Current steering<br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br>3.No target maneuvers,<br><br>Note: ballistic trajectory velocity changes are not considered maneuvers. |                 |
| <b>RangeOptimal</b>                | <b>MA_WeaponRangeType</b>                                                                                                                                                                                                                                                                                                                         | <b>Repeated</b> |
| Description                        | Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for:<br>1.Optimal steering<br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br>3.No target maneuvers,                                                                                     |                 |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                     |          |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                         | Note: ballistic trajectory velocity changes are not considered maneuvers.                                                                                                                                                                                                                                                                                                                                           |          |
| <b>RangeMaxAero</b>     | MA_WeaponRangeType                                                                                                                                                                                                                                                                                                                                                                                                  | Repeated |
| Description             | <p>Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for:</p> <ol style="list-style-type: none"> <li>1.Optimal steering</li> <li>2.Current shooter kinematic conditions (e.g., altitude/airspeed)</li> <li>3.No target maneuvers,</li> </ol> <p>Note: ballistic trajectory velocity changes are not considered maneuvers.</p> |          |
| <b>TimeToRelease</b>    | TimeType                                                                                                                                                                                                                                                                                                                                                                                                            | Required |
| Description             | The seconds time to activate for the proposed launch store type.                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>TimeOfActivation</b> | TimeType                                                                                                                                                                                                                                                                                                                                                                                                            | Optional |
| Description             | <p>The time of weapon seeker activation, assuming no platform guidance or weapon errors, for:</p> <ol style="list-style-type: none"> <li>1.Current steering</li> <li>2.Current shooter kinematic conditions (e.g., altitude/airspeed)</li> <li>3.No post-launch target maneuvers</li> </ol>                                                                                                                         |          |

Table A-1-147: P2P UCI Extensions MA\_WEZ\_MT - WEZ\_Data UCI Extension Field

| RangeTurnAndClose |                                                                                                                                                                                                                                                                                                                                                                           |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for:<br><br>1.Current steering<br><br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br><br>3.Target turn-and-close maneuver<br><br><br>RangeTurnAndClose is not applicable to ballistic trajectory targets. |             |
|                   | MA_WeaponRangeType                                                                                                                                                                                                                                                                                                                                                        |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                           |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                | Field Usage |
| Range             | DistanceType                                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description       | Indicates the slant range of the weapon.                                                                                                                                                                                                                                                                                                                                  |             |
| Azimuth           | AngleType                                                                                                                                                                                                                                                                                                                                                                 | Required    |
| Description       | Indicates the azimuth the slant range pertains to.                                                                                                                                                                                                                                                                                                                        |             |

Table A-1-148: P2P UCI Extensions MA\_WEZ\_MT - RangeTurnAndClose UCI Extension Field

| RangeMinimum      |                                                                                                                                                                                                                                                                                                                                                                     |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for:<br><br>1.Current steering<br><br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br><br>3.No target maneuvers<br><br><br>Note: ballistic trajectory velocity changes are not considered maneuvers. |             |
|                   | MA_WeaponRangeType                                                                                                                                                                                                                                                                                                                                                  |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                     |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                          | Field Usage |
| Range             | DistanceType                                                                                                                                                                                                                                                                                                                                                        | Required    |
| Description       | Indicates the slant range of the weapon.                                                                                                                                                                                                                                                                                                                            |             |
| Azimuth           | AngleType                                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description       | Indicates the azimuth the slant range pertains to.                                                                                                                                                                                                                                                                                                                  |             |

Table A-1-149: P2P UCI Extensions MA\_WEZ\_MT - RangeMinimum UCI Extension Field

| RangeTurnAndRun   |                                                                                                                                                                                                                                                                                                                                                                 |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for:<br><br>1.Current steering<br><br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br><br>3.Target turn-and-run maneuver<br><br><br>RangeTurnAndRun is not applicable to ballistic trajectory targets. |             |
|                   | MA_WeaponRangeType                                                                                                                                                                                                                                                                                                                                              |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                 |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                      | Field Usage |
| Range             | DistanceType                                                                                                                                                                                                                                                                                                                                                    | Required    |
| Description       | Indicates the slant range of the weapon.                                                                                                                                                                                                                                                                                                                        |             |
| Azimuth           | AngleType                                                                                                                                                                                                                                                                                                                                                       | Required    |
| Description       | Indicates the azimuth the slant range pertains to.                                                                                                                                                                                                                                                                                                              |             |

Table A-1-150: P2P UCI Extensions MA\_WEZ\_MT - RangeTurnAndRun UCI Extension Field

| RangeProbabilityOfIntercept |                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Description                 | Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: |
|                             | 1.Current steering                                                                                                                        |

|                   |                                                                                                                                                                            |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                   | 2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br>3.No target maneuvers<br><br>Note: ballistic trajectory velocity changes are not considered maneuvers. |             |
| Base Type         | MA_WeaponRangeType                                                                                                                                                         |             |
| Message Structure |                                                                                                                                                                            |             |
| Field Name        | Field Type                                                                                                                                                                 | Field Usage |
| Range             | DistanceType                                                                                                                                                               | Required    |
| Description       | Indicates the slant range of the weapon.                                                                                                                                   |             |
| Azimuth           | AngleType                                                                                                                                                                  | Required    |
| Description       | Indicates the azimuth the slant range pertains to.                                                                                                                         |             |

Table A-1-151: P2P UCI Extensions MA\_WEZ\_MT - RangeProbabilityOfIntercept UCI Extension Field

| RangeMaxGuidance  |                                                                                                                                                                                                                                                                                                                                                               |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for:<br><br>1.Current steering<br><br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br><br>3.No target maneuvers,<br><br>Note: ballistic trajectory velocity changes are not considered maneuvers. |             |
| Base Type         | MA_WeaponRangeType                                                                                                                                                                                                                                                                                                                                            |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                               |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                    | Field Usage |
| Range             | DistanceType                                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description       | Indicates the slant range of the weapon.                                                                                                                                                                                                                                                                                                                      |             |
| Azimuth           | AngleType                                                                                                                                                                                                                                                                                                                                                     | Required    |
| Description       | Indicates the azimuth the slant range pertains to.                                                                                                                                                                                                                                                                                                            |             |

Table A-1-152: P2P UCI Extensions MA\_WEZ\_MT - RangeMaxGuidance UCI Extension Field

|                     |                                                                                                                                                                                                                                                               |  |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>RangeOptimal</b> |                                                                                                                                                                                                                                                               |  |
| <b>Description</b>  | Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for:<br>1.Optimal steering<br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br>3.No target maneuvers, |  |



|                   |                                                                           |             |
|-------------------|---------------------------------------------------------------------------|-------------|
|                   | Note: ballistic trajectory velocity changes are not considered maneuvers. |             |
| Base Type         | MA_WeaponRangeType                                                        |             |
| Message Structure |                                                                           |             |
| Field Name        | Field Type                                                                | Field Usage |
| Range             | DistanceType                                                              | Required    |
| Description       | Indicates the slant range of the weapon.                                  |             |
| Azimuth           | AngleType                                                                 | Required    |
| Description       | Indicates the azimuth the slant range pertains to.                        |             |

Table A-1-153: P2P UCI Extensions MA\_WEZ\_MT - RangeOptimal UCI Extension Field

| RangeMaxAero      |                                                                                                                                                                                                                                                                                                                                                               |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for:<br><br>1.Optimal steering<br><br>2.Current shooter kinematic conditions (e.g., altitude/airspeed)<br><br>3.No target maneuvers,<br><br>Note: ballistic trajectory velocity changes are not considered maneuvers. |             |
|                   | MA_WeaponRangeType                                                                                                                                                                                                                                                                                                                                            |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                               |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                    | Field Usage |
| Range             | DistanceType                                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description       | Indicates the slant range of the weapon.                                                                                                                                                                                                                                                                                                                      |             |
| Azimuth           | AngleType                                                                                                                                                                                                                                                                                                                                                     | Required    |
| Description       | Indicates the azimuth the slant range pertains to.                                                                                                                                                                                                                                                                                                            |             |

Table A-1-154: P2P UCI Extensions MA\_WEZ\_MT - RangeMaxAero UCI Extension Field

### 1.3.2.2 MA\_COP\_ConfigurationSettingsRequestStatusMT UCI Extension

|                                                    |                                                                                                                                             |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_COP_ConfigurationSettingsRequestStatusMT</b> |                                                                                                                                             |
| <b>Description</b>                                 | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                                 | MessageData : MA_COP_ConfigurationSettingsRequestStatusMDT                                                                                  |

Table A-1-155: P2P UCI Extensions MA\_COP\_ConfigurationSettingsRequestStatusMT UCI Extension

|                    |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>MessageData</b> |                                                                                         |
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |

|                              |                                                                                                                                                                                                                                                                                             |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Base Type                    | MA_COP_ConfigurationSettingsRequestStatusMDT                                                                                                                                                                                                                                                |             |
| Message Structure            |                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                  | Field Usage |
| ConfigurationSettings        | MA_COP_ConfigurationSettingsMDT                                                                                                                                                                                                                                                             | Optional    |
| Description                  | The configuration settings                                                                                                                                                                                                                                                                  |             |
| RequestID                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |             |
| RequestProcessingState       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |             |
| RequestProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |

**Table A-1-156: P2P UCI Extensions MA\_COP\_ConfigurationSettingsRequestStatusMT - MessageData UCI Extension Field**

### 1.3.2.3 MA\_COP\_ConfigurationSettingsRequestMT UCI Extension

|                                              |                                                                                                                                             |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_COP_ConfigurationSettingsRequestMT</b> |                                                                                                                                             |
| <b>Description</b>                           | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                           | MessageData : MA_COP_ConfigurationSettingsRequestMDT                                                                                        |

**Table A-1-157: P2P UCI Extensions MA\_COP\_ConfigurationSettingsRequestMT UCI Extension**

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_COP_ConfigurationSettingsRequestMDT                                                  |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| ApplicableToIDs   | MA_PackageSystemType                                                                    | Repeated    |
| Description       | Indicates the ID of the Package or System that should respond to the request.           |             |
| RequestID         | RequestID_Type                                                                          | Required    |
| Description       | Indicates the unique ID of the request message.                                         |             |
| RequestState      | RequestStateEnum                                                                        | Required    |
| Description       | The state of the request message data. Indicates if a request message is a NEW          |             |

|  |                                                            |
|--|------------------------------------------------------------|
|  | request, an UPDATE to a request, or a CANCEL of a request. |
|--|------------------------------------------------------------|

**Table A-1-158: P2P UCI Extensions MA\_COP\_ConfigurationSettingsRequestMT - MessageData UCI Extension Field**

### 1.3.2.4 MA\_SystemNotificationMT UCI Extension

| MA_SystemNotificationMT |                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>      | MessageData : MA_SystemNotificationMDT<br>NotificationDetail : CannotComplyType<br>NotificationDetail : CannotComplyType      |

**Table A-1-159: P2P UCI Extensions MA\_SystemNotificationMT UCI Extension**

| MessageData        |                                                                                                                                                                                                                                         |             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                 |             |
| Base Type          | MA_SystemNotificationMDT                                                                                                                                                                                                                |             |
| Message Structure  |                                                                                                                                                                                                                                         |             |
| Field Name         | Field Type                                                                                                                                                                                                                              | Field Usage |
| SystemSubjectIDs   | MA_PackageSystemType                                                                                                                                                                                                                    | Required    |
| Description        | Indicates the subject of the Notification, the thing which the Notification is about.                                                                                                                                                   |             |
| SystemPerspective  | NotificationPerspectiveEnum                                                                                                                                                                                                             | Repeated    |
| Description        | Indicates a perspective (characteristic, behavior and/or change) of the Subject that is the basis of the Notification. List size for this element is based on "Select All That Apply" condition.                                        |             |
| AssociatedMessage  | AssociatedMessageType                                                                                                                                                                                                                   | Repeated    |
| Description        | Indicates a message that is associated with the Notification.                                                                                                                                                                           |             |
| NotificationDetail | CannotComplyType                                                                                                                                                                                                                        | Optional    |
| Description        | Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful |             |
| NotificationID     | NotificationID_Type                                                                                                                                                                                                                     | Required    |
| Description        | Indicates the unique ID of the Notification.                                                                                                                                                                                            |             |
| NotificationState  | NotificationStateEnum                                                                                                                                                                                                                   | Required    |
| Description        | Indicates the current state within an enumerated state machine that characterizes the life cycle of a notification.                                                                                                                     |             |
| Timestamp          | DateTimeType                                                                                                                                                                                                                            | Required    |
| Description        | The time at which the Notification was originally triggered (ex: event time).                                                                                                                                                           |             |
| Source             | NotificationSourceType                                                                                                                                                                                                                  | Required    |

|                              |                                                                                                                                                                                                                                                                                                                                                            |          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                  | Indicates the originating source of the Notification.                                                                                                                                                                                                                                                                                                      |          |
| <b>Severity</b>              | NotificationSeverityEnum                                                                                                                                                                                                                                                                                                                                   | Required |
| Description                  | Indicates the overall severity of the notification. Informational notifications are meant to provide informational context of events. Indicates the existence of a hazardous condition requiring immediate action to prevent loss of life, equipment damage, or abortion of the mission. Critical notifications indicate the need for immediate attention. |          |
| <b>AppliesTo</b>             | SubjectType                                                                                                                                                                                                                                                                                                                                                | Repeated |
| Description                  | Indicates an Asset that the Notification applies to and/or is affected by. Optional when the Notification characterizes the Subject without regard for Assets it might affect. Omission of this element doesn't necessarily mean the Notification isn't applicable to an Asset.                                                                            |          |
| <b>NotificationNarrative</b> | VisibleString1024Type                                                                                                                                                                                                                                                                                                                                      | Optional |
| Description                  | Human readable message for display to an operator.                                                                                                                                                                                                                                                                                                         |          |

Table A-1-160: P2P UCI Extensions MA\_SystemNotificationMT - MessageData UCI Extension Field

| NotificationDetail |                                                                                                                                                                                                                                                                               |             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful                                       |             |
| Base Type          | CannotComplyType                                                                                                                                                                                                                                                              |             |
| Message Structure  |                                                                                                                                                                                                                                                                               |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                    | Field Usage |
| Reason             | CannotComplyEnum                                                                                                                                                                                                                                                              | Required    |
| Description        | Indicates the reason why an action (Task, [Capability]Command, [SupportingCapability]Command, [Capability]Activity, etc.) was rejected, interrupted, unallocated, failed and/or generally can't be completed satisfactorily. See enumeration annotations for further details. |             |
| Description        | VisibleString1024Type                                                                                                                                                                                                                                                         | Optional    |
| Description        | Indicates a human-readable text description of the reason for the "can't comply".                                                                                                                                                                                             |             |
| AssociatedID       | ID_Type                                                                                                                                                                                                                                                                       | Optional    |
| Description        | Indicates an ID corresponding to another object that is associated with or the cause of the "can't comply".                                                                                                                                                                   |             |

Table A-1-161: P2P UCI Extensions MA\_SystemNotificationMT - NotificationDetail UCI Extension Field

| NotificationDetail |                                                                                                                                                                                                                                         |             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful |             |
| Base Type          | CannotComplyType                                                                                                                                                                                                                        |             |
| Message Structure  |                                                                                                                                                                                                                                         |             |
| Field Name         | Field Type                                                                                                                                                                                                                              | Field Usage |

| Reason       | CannotComplyEnum                                                                                                                                                                                                                                                              | Required |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description  | Indicates the reason why an action (Task, [Capability]Command, [SupportingCapability]Command, [Capability]Activity, etc.) was rejected, interrupted, unallocated, failed and/or generally can't be completed satisfactorily. See enumeration annotations for further details. |          |
| Description  | VisibleString1024Type                                                                                                                                                                                                                                                         | Optional |
| Description  | Indicates a human-readable text description of the reason for the "can't comply".                                                                                                                                                                                             |          |
| AssociatedID | ID_Type                                                                                                                                                                                                                                                                       | Optional |
| Description  | Indicates an ID corresponding to another object that is associated with or the cause of the "can't comply".                                                                                                                                                                   |          |

Table A-1-162: P2P UCI Extensions MA\_SystemNotificationMT - NotificationDetail UCI Extension Field

### 1.3.2.5 MA\_COP\_ConfigurationSettingsCommandMT UCI Extension

| MA_COP_ConfigurationSettingsCommandMT |                                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Description                           | Command message to adjust the configuration settings for the COP among ACPs.UCI_PRIMITIVE: Command-2.<br><br>See the XSD File for more info |
| Ext. Fields                           | MessageData : MA_COP_ConfigurationSettingsCommandMDT                                                                                        |

Table A-1-163: P2P UCI Extensions MA\_COP\_ConfigurationSettingsCommandMT UCI Extension

| MessageData       |                                                                                                                                 |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                         |             |
| Base Type         | MA_COP_ConfigurationSettingsCommandMDT                                                                                          |             |
| Message Structure |                                                                                                                                 |             |
| Field Name        | Field Type                                                                                                                      | Field Usage |
| ApplicableToIDs   | MA_PackageSystemType                                                                                                            | Required    |
| Description       | Indicates the ID of the Package or System being configured.                                                                     |             |
| COP_LeaderID      | SystemID_Type                                                                                                                   | Repeated    |
| Description       | Indicates a prioritized list of unique SystemIDs for leaders for the COP. If not specified, no change to leader requested       |             |
| RequestAuthority  | MA_COP_ConfigurationAuthorityEnum                                                                                               | Required    |
| Description       | Indicates the priority of the request being made. Generally C2 will set this to ORDER and the COP lead will set this to REQUEST |             |
| Configuration     | MA_COP_ConfigurationType                                                                                                        | Repeated    |
| Description       | A list of configuration items.                                                                                                  |             |
| CommandID         | CommandID_Type                                                                                                                  | Required    |
| Description       | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.      |             |
| CommandState      | CommandStateEnum                                                                                                                | Required    |

|             |                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command. |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Table A-1-164: P2P UCI Extensions MA\_COP\_ConfigurationSettingsCommandMT - MessageData UCI Extension Field**

### 1.3.2.6 MA\_EngagementCriteriaCommandStatusMT UCI Extension

| MA_EngagementCriteriaCommandStatusMT |                                                                                                                               |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description                          | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields                          | MessageData : MA_EngagementCriteriaCommandStatusMDT                                                                           |

**Table A-1-165: P2P UCI Extensions MA\_EngagementCriteriaCommandStatusMT UCI Extension**

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_EngagementCriteriaCommandStatusMDT                                                                                                                                                                                                                                                                                                                       |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| AppliesToIDs                 | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description                  | The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed.                                                                                                                                                                                                                |             |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

**Table A-1-166: P2P UCI Extensions MA\_EngagementCriteriaCommandStatusMT - MessageData UCI Extension Field**

### 1.3.2.7 MA\_EngagementCriteriaCommandMT UCI Extension

| MA_EngagementCriteriaCommandMT |                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------|
| Description                    | See annotations in child elements and messages/elements that use this type for details. |

|                    |                                               |
|--------------------|-----------------------------------------------|
|                    | See the XSD File for more info                |
| <b>Ext. Fields</b> | MessageData : MA_EngagementCriteriaCommandMDT |

Table A-1-167: P2P UCI Extensions MA\_EngagementCriteriaCommandMT UCI Extension

| MessageData          |                                                                                                                                                                                                                                                                                                                                           |             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                   |             |
| Base Type            | MA_EngagementCriteriaCommandMDT                                                                                                                                                                                                                                                                                                           |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                                                           |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                                                                | Field Usage |
| EngagementCriteriaID | MA_EngagementCriteriaID_Type                                                                                                                                                                                                                                                                                                              | Required    |
| Description          | Contains the ID of the Engagement Criteria message that should be active (for CommandState NEW or Update) or deactivated (for CommandState CANCEL).                                                                                                                                                                                       |             |
| ApplicableRegion     | MA_RegionChoiceType                                                                                                                                                                                                                                                                                                                       | Repeated    |
| Description          | The ID(s) of an OpZone/OpVolume or specified zone(s) that defines where this Engagement Criteria is applicable. If this field is not empty, an ACP and the Target must be within this region for the ACP to fire and the weapon to strike the intended target. The MA shall only start an engagement for targets inside of the Region(s). |             |
| TimeValid            | DateTimeRangeType                                                                                                                                                                                                                                                                                                                         | Optional    |
| Description          | The time window for which the Engagement Criteria is valid. If this field is left empty, the Engagement Criteria is valid until the Command is canceled or an updated.                                                                                                                                                                    |             |
| AppliesToIDs         | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                      | Repeated    |
| Description          | The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in this field as well.                                                                                                                                                                                   |             |
| CommandID            | CommandID_Type                                                                                                                                                                                                                                                                                                                            | Required    |
| Description          | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.                                                                                                                                                                                                                |             |
| CommandState         | CommandStateEnum                                                                                                                                                                                                                                                                                                                          | Required    |
| Description          | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.                                                                                                                                                                             |             |

Table A-1-168: P2P UCI Extensions MA\_EngagementCriteriaCommandMT - MessageData UCI Extension Field

### 1.3.2.8 MA\_COP\_ConfigurationSettingsCommandStatusMT UCI Extension

| MA_COP_ConfigurationSettingsCommandStatusMT |                                                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                          | Command status message in response to MA_COP_ConfigurationSettingsCommand.UCI_PRIMITIVE: Command-2.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                          | MessageData : MA_COP_ConfigurationSettingsCommandStatusMDT                                                                                |

Table A-1-169: P2P UCI Extensions MA\_COP\_ConfigurationSettingsCommandStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_COP_ConfigurationSettingsCommandStatusMDT                                                                                                                                                                                                                                                                                                                |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

**Table A-1-170: P2P UCI Extensions MA\_COP\_ConfigurationSettingsCommandStatusMT - MessageData UCI Extension Field**

### 1.3.2.9 MA\_DesignationRequestStatusMT UCI Extension

| MA_DesignationRequestStatusMT |                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------|
| Description                   | Reports the current status of a Designation.<br><br>See the XSD File for more info |
| Ext. Fields                   | MessageData : MA_DesignationRequestStatusMDT                                       |

**Table A-1-171: P2P UCI Extensions MA\_DesignationRequestStatusMT UCI Extension**

| MessageData           |                                                                                                       |             |
|-----------------------|-------------------------------------------------------------------------------------------------------|-------------|
| Description           | See the annotation in the associated message for an overall description of the message and this type. |             |
| Base Type             | MA_DesignationRequestStatusMDT                                                                        |             |
| Message Structure     |                                                                                                       |             |
| Field Name            | Field Type                                                                                            | Field Usage |
| DesignationDataStatus | DesignationDataStatusType                                                                             | Repeated    |
| Description           | Indicates the status for one or more designations.                                                    |             |
| RequestID             | RequestID_Type                                                                                        | Required    |
| Description           | Indicates the unique ID of the request message.                                                       |             |



|                     |                                                                                                                                           |          |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>RequestState</b> | RequestStateEnum                                                                                                                          | Required |
| Description         | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request. |          |

Table A-1-172: P2P UCI Extensions MA\_DesignationRequestStatusMT - MessageData UCI Extension Field

### 1.3.2.10 MA\_EngagementCriteriaStatusMT UCI Extension

| MA_EngagementCriteriaStatusMT |                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>            | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>            | MessageData : MA_EngagementCriteriaStatusMDT                                                                                  |

Table A-1-173: P2P UCI Extensions MA\_EngagementCriteriaStatusMT UCI Extension

| MessageData          |                                                                                                                                                                                                                                                                                                                                                         |             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                 |             |
| Base Type            | MA_EngagementCriteriaStatusMDT                                                                                                                                                                                                                                                                                                                          |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                                                                         |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                                                                              | Field Usage |
| EngagementCriteriaID | MA_EngagementCriteriaID_Type                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description          | Contains the ID of the Engagement Criteria message that is currently active. If this field is empty, that means there is no active Engagement Criteria and Engagement is not authorized.                                                                                                                                                                |             |
| ApplicableRegion     | MA_RegionChoiceType                                                                                                                                                                                                                                                                                                                                     | Repeated    |
| Description          | The ID(s) of an OpZone/OpVolume or specified zone(s) that defines where the Engagement Criteria is applicable. If this field is not empty, the ACP must be in this region when it fires, and the calculated weapon impact location of the Target must be within this region. The MA shall only start an engagement for targets inside of the Region(s). |             |
| TimeValid            | DateTimeRangeType                                                                                                                                                                                                                                                                                                                                       | Optional    |
| Description          | The time window for which the Engagement Criteria is valid. If this field is left empty, the Engagement Criteria is valid until the Command is canceled or an updated.                                                                                                                                                                                  |             |
| AppliesToIDs         | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                    | Repeated    |
| Description          | The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in the field as well.                                                                                                                                                                                                  |             |

Table A-1-174: P2P UCI Extensions MA\_EngagementCriteriaStatusMT - MessageData UCI Extension Field

### 1.3.2.11 MA\_AcceptableLevelOfRiskStatusMT UCI Extension

| MA_AcceptableLevelOfRiskStatusMT |                                                                                                                               |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>               | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |

|                    |                                                 |
|--------------------|-------------------------------------------------|
| <b>Ext. Fields</b> | MessageData : MA_AcceptableLevelOfRiskStatusMDT |
|--------------------|-------------------------------------------------|

Table A-1-175: P2P UCI Extensions MA\_AcceptableLevelOfRiskStatusMT UCI Extension

| MessageData              |                                                                                                                                                                     |             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | See annotations in child elements and messages/elements that use this type for details.                                                                             |             |
| Base Type                | MA_AcceptableLevelOfRiskStatusMDT                                                                                                                                   |             |
| Message Structure        |                                                                                                                                                                     |             |
| Field Name               | Field Type                                                                                                                                                          | Field Usage |
| AcceptableRiskSetting    | MA_AcceptableLevelOfRiskEnum                                                                                                                                        | Required    |
| Description              | This is the active acceptable level of risk permitted at the package or vehicle level, which applies to all tasks and actions for the vehicle/package while active. |             |
| AppliesToIDs             | MA_PackageSystemType                                                                                                                                                | Repeated    |
| Description              | The IDs, Package or ACP, that this message applies to. A lead is able to respond for the Package or individual ACPs as needed.                                      |             |
| SurvivabilityRiskLevelID | SurvivabilityRiskLevelID_Type                                                                                                                                       | Optional    |
| Description              | The unique identifier for the active Survivability Risk Level.                                                                                                      |             |

Table A-1-176: P2P UCI Extensions MA\_AcceptableLevelOfRiskStatusMT - MessageData UCI Extension Field

### 1.3.2.12 MA\_DesignationRequestMT UCI Extension

| MA_DesignationRequestMT |                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>      | MessageData : MA_DesignationRequestMDT                                                                                        |

Table A-1-177: P2P UCI Extensions MA\_DesignationRequestMT UCI Extension

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_DesignationRequestMDT                                                                |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| DesignationData   | MA_DesignationDataType                                                                  | Repeated    |
| Description       | Indicates the data for one or more designations.                                        |             |
| RequestID         | RequestID_Type                                                                          | Required    |
| Description       | Indicates the unique ID of the request message.                                         |             |
| RequestState      | RequestStateEnum                                                                        | Required    |

|             |                                                                                                                                           |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Description | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request. |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------|

Table A-1-178: P2P UCI Extensions MA\_DesignationRequestMT - MessageData UCI Extension Field

## 1.3.2.13 MA\_RaftVoteRequestMT UCI Extension

| MA_RaftVoteRequestMT |                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields          | MessageData : MA_RaftVoteRequestMDT                                                                                           |

Table A-1-179: P2P UCI Extensions MA\_RaftVoteRequestMT UCI Extension

| MessageData       |                                                                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                   |             |
| Base Type         | MA_RaftVoteRequestMDT                                                                                                                     |             |
| Message Structure |                                                                                                                                           |             |
| Field Name        | Field Type                                                                                                                                | Field Usage |
| PackageID         | PackageID_Type                                                                                                                            | Required    |
| Description       | The unique ID of the package that this vote request applies to.                                                                           |             |
| CandidateID       | SystemID_Type                                                                                                                             | Required    |
| Description       | The unique ID of the candidate requesting a vote.                                                                                         |             |
| Term              | unsignedInt                                                                                                                               | Required    |
| Description       | The candidate's term.                                                                                                                     |             |
| LastLogIndex      | unsignedInt                                                                                                                               | Optional    |
| Description       | The index of the candidate's last log entry.                                                                                              |             |
| LastLogTerm       | unsignedInt                                                                                                                               | Optional    |
| Description       | The term of the candidate's last log entry.                                                                                               |             |
| RequestID         | RequestID_Type                                                                                                                            | Required    |
| Description       | Indicates the unique ID of the request message.                                                                                           |             |
| RequestState      | RequestStateEnum                                                                                                                          | Required    |
| Description       | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request. |             |

Table A-1-180: P2P UCI Extensions MA\_RaftVoteRequestMT - MessageData UCI Extension Field

## 1.3.2.14 MA\_RaftVoteRequestStatusMT UCI Extension

| MA_RaftVoteRequestStatusMT |                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------|
| Description                | See annotations in child elements and messages/elements that use this type for details. |

|                    |                                           |
|--------------------|-------------------------------------------|
|                    | See the XSD File for more info            |
| <b>Ext. Fields</b> | MessageData : MA_RaftVoteRequestStatusMDT |

Table A-1-181: P2P UCI Extensions MA\_RaftVoteRequestStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                             |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                     |             |
| Base Type                    | MA_RaftVoteRequestStatusMDT                                                                                                                                                                                                                                                                 |             |
| Message Structure            |                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                  | Field Usage |
| Term                         | unsignedInt                                                                                                                                                                                                                                                                                 | Required    |
| Description                  | The candidate's current term.                                                                                                                                                                                                                                                               |             |
| VoteGranted                  | boolean                                                                                                                                                                                                                                                                                     | Required    |
| Description                  | Indicates whether the candidate received a vote from the follower.                                                                                                                                                                                                                          |             |
| RequestID                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |             |
| RequestProcessingState       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |             |
| RequestProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |

Table A-1-182: P2P UCI Extensions MA\_RaftVoteRequestStatusMT - MessageData UCI Extension Field

### 1.3.2.15 MA\_RaftAppendEntriesRequestMT UCI Extension

| MA_RaftAppendEntriesRequestMT |                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b>            | See annotations in child elements and messages/elements that use this type for details. |
|                               | See the XSD File for more info                                                          |
| <b>Ext. Fields</b>            | MessageData : MA_RaftAppendEntriesRequestMDT                                            |

Table A-1-183: P2P UCI Extensions MA\_RaftAppendEntriesRequestMT UCI Extension

| MessageData        |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |
| <b>Base Type</b>   | MA_RaftAppendEntriesRequestMDT                                                          |
| Message Structure  |                                                                                         |

| Field Name          | Field Type                                                                                                                                        | Field Usage |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>PackageID</b>    | PackageID_Type                                                                                                                                    | Required    |
| Description         | The unique ID of the package that this request applies to.                                                                                        |             |
| <b>LeaderID</b>     | SystemID_Type                                                                                                                                     | Required    |
| Description         | The unique ID of the current leader system.                                                                                                       |             |
| <b>Term</b>         | unsignedInt                                                                                                                                       | Required    |
| Description         | The leader's current term.                                                                                                                        |             |
| <b>PrevLogIndex</b> | unsignedInt                                                                                                                                       | Optional    |
| Description         | The index of the log entry immediately preceding new ones.                                                                                        |             |
| <b>PrevLogTerm</b>  | unsignedInt                                                                                                                                       | Optional    |
| Description         | The term of the PrevLogIndex entry.                                                                                                               |             |
| <b>LogEntry</b>     | MA_RaftLogEntryType                                                                                                                               | Repeated    |
| Description         | Log entries to store. For heartbeat, no log entry is included.                                                                                    |             |
| <b>LeaderCommit</b> | unsignedInt                                                                                                                                       | Optional    |
| Description         | The leader's commit index, which is the highest log entry index that the leader has committed and has been replicated on a majority of the nodes. |             |
| <b>RequestID</b>    | RequestID_Type                                                                                                                                    | Required    |
| Description         | Indicates the unique ID of the request message.                                                                                                   |             |
| <b>RequestState</b> | RequestStateEnum                                                                                                                                  | Required    |
| Description         | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.         |             |

Table A-1-184: P2P UCI Extensions MA\_RaftAppendEntriesRequestMT - MessageData UCI Extension Field

### 1.3.2.16 MA\_RaftAppendEntriesRequestStatusMT UCI Extension

| MA_RaftAppendEntriesRequestStatusMT |                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                  | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                  | MessageData : MA_RaftAppendEntriesRequestStatusMDT                                                                            |

Table A-1-185: P2P UCI Extensions MA\_RaftAppendEntriesRequestStatusMT UCI Extension

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_RaftAppendEntriesRequestStatusMDT                                                    |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| Term              | unsignedInt                                                                             | Required    |

|                                     |                                                                                                                                                                                                                                                                                             |          |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                         | The follower's current term, returned to the leader.                                                                                                                                                                                                                                        |          |
| <b>Success</b>                      | boolean                                                                                                                                                                                                                                                                                     | Optional |
| Description                         | Indicates whether the follower has a log entry matching PrevLogIndex and PrevLogTerm.                                                                                                                                                                                                       |          |
| <b>RequestID</b>                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required |
| Description                         | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |          |
| <b>RequestProcessingState</b>       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required |
| Description                         | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |          |
| <b>RequestProcessingStateReason</b> | CannotComplyType                                                                                                                                                                                                                                                                            | Optional |
| Description                         | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |          |

**Table A-1-186: P2P UCI Extensions MA\_RaftAppendEntriesRequestStatusMT - MessageData UCI Extension Field**

### 1.3.2.17 MA\_CAP\_ExecutionStatusReportMT UCI Extension

| MA_CAP_ExecutionStatusReportMT |                                                                                                                               |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>             | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>             | MessageData : MA_CAP_ExecutionStatusReportMDT                                                                                 |

**Table A-1-187: P2P UCI Extensions MA\_CAP\_ExecutionStatusReportMT UCI Extension**

| MessageData          |                                                                                                                                                                                                  |             |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                          |             |
| Base Type            | MA_CAP_ExecutionStatusReportMDT                                                                                                                                                                  |             |
| Message Structure    |                                                                                                                                                                                                  |             |
| Field Name           | Field Type                                                                                                                                                                                       | Field Usage |
| PackageID            | PackageID_Type                                                                                                                                                                                   | Optional    |
| Description          | The ID of the package executing the CAP.                                                                                                                                                         |             |
| PackageRequirementID | RequirementInstanceID_ChoiceType                                                                                                                                                                 | Repeated    |
| Description          | Contains the unique ID(s) of the package-level requirements associated with the ongoing CAP activity. This will typically be a package-level PROTECT_CAP Action and/or a package-level CAP Task. |             |
| AOI_Coverage         | PercentType                                                                                                                                                                                      | Optional    |
| Description          | Percent of the Area of Interest (AOI) currently covered by any vehicle's sensors.                                                                                                                |             |
| PlanMetrics          | MA_CAP_PlanMetrics_Type                                                                                                                                                                          | Optional    |

|                        |                                                                                               |          |
|------------------------|-----------------------------------------------------------------------------------------------|----------|
| Description            | Sensor coverage and time gap metrics of the planned CAP routes over the next racetrack cycle. |          |
| <b>ExecutionStatus</b> | MA_CAP_ExecutionStatus_Type                                                                   | Repeated |
| Description            | Contains a list of CAP statuses from vehicles executing the CAP.                              |          |

Table A-1-188: P2P UCI Extensions MA\_CAP\_ExecutionStatusReportMT - MessageData UCI Extension Field

## 1.3.2.18 MA\_MissionPlanMT UCI Extension

| MA_MissionPlanMT   |                                                                                         |  |
|--------------------|-----------------------------------------------------------------------------------------|--|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |  |
|                    | See the XSD File for more info                                                          |  |
| <b>Ext. Fields</b> | MessageData : MA_MissionPlanMDT                                                         |  |

Table A-1-189: P2P UCI Extensions MA\_MissionPlanMT UCI Extension

| MessageData          |                                                                                                                                                                                                                                                                                                                         |             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | Indicates the data being conveyed by the message.                                                                                                                                                                                                                                                                       |             |
| Base Type            | MA_MissionPlanMDT                                                                                                                                                                                                                                                                                                       |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                                         |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                                              | Field Usage |
| MissionPlanID        | MissionPlanID_Type                                                                                                                                                                                                                                                                                                      | Required    |
| Description          | Indicates a unique ID assigned to the MissionPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.                                                          |             |
| MissionPlanCommandID | MissionPlanCommandID_ChoiceType                                                                                                                                                                                                                                                                                         | Optional    |
| Description          | Indicates the unique ID of the *PlanCommand which this MissionPlan originated from.                                                                                                                                                                                                                                     |             |
| Applicability        | MA_PlanApplicabilityType                                                                                                                                                                                                                                                                                                | Required    |
| Description          | Indicates the System or Systems which have a direct relationship with the Mission Plan. See annotations for the underlying type and its child elements for details.                                                                                                                                                     |             |
| Window               | DateTimeRangeType                                                                                                                                                                                                                                                                                                       | Optional    |
| Description          | The time window for which the Plan is applicable.                                                                                                                                                                                                                                                                       |             |
| SubPlans             | MA_PlansReferenceBaseType                                                                                                                                                                                                                                                                                               | Optional    |
| Description          | Indicates the set of *Plans which are included as part of the MissionPlan. The included SubPlans share the same Applicability as the MissionPlan.                                                                                                                                                                       |             |
| ExecutionSequence    | ExecutionSequenceType                                                                                                                                                                                                                                                                                                   | Optional    |
| Description          | Indicates the MissionPlan's flow of execution when there are multiple kinematic and action plans. The kinematic and action plans are combined into sets and given unique IDs. An execution flow may be created by linking *ExecutionPlanSets of different types (e.g. OrbitExecutionPlanSet and RouteExecutionPlanSet). |             |
| ParentMissionPlanID  | MissionPlanID_Type                                                                                                                                                                                                                                                                                                      | Repeated    |
| Description          | Indicates a MissionPlan that is the parent of this MissionPlan.                                                                                                                                                                                                                                                         |             |

|                                 |                                                                                                                                                                                                                                                                                                                                                                   |          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>SubordinateMissionPlanID</b> | MissionPlanID_Type                                                                                                                                                                                                                                                                                                                                                | Repeated |
| Description                     | Indicates a MissionPlan that is subordinate to this MissionPlan.                                                                                                                                                                                                                                                                                                  |          |
| <b>ForPlanningUseOnly</b>       | boolean                                                                                                                                                                                                                                                                                                                                                           | Required |
| Description                     | When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc. |          |
| <b>PlanInputs</b>               | MA_MissionPlanInputsType                                                                                                                                                                                                                                                                                                                                          | Optional |
| Description                     | Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.                                                                                                                                                                                                                                                         |          |
| <b>Remarks</b>                  | RemarksType                                                                                                                                                                                                                                                                                                                                                       | Optional |
| Description                     | Indicates information about the mission plan from an operator intended for display. Both short and longer detailed information can be provided.                                                                                                                                                                                                                   |          |

Table A-1-190: P2P UCI Extensions MA\_MissionPlanMT - MessageData UCI Extension Field

## 1.3.2.19 MA\_ActionPlanMT UCI Extension

| MA_ActionPlanMT    |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_ActionPlanMDT                                                                                                |

Table A-1-191: P2P UCI Extensions MA\_ActionPlanMT UCI Extension

| MessageData        |                                                                                                                                                                                                                                                               |             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates the data being conveyed by the message.                                                                                                                                                                                                             |             |
| Base Type          | MA_ActionPlanMDT                                                                                                                                                                                                                                              |             |
| Message Structure  |                                                                                                                                                                                                                                                               |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                    | Field Usage |
| ActionPlanID       | ActionPlanID_Type                                                                                                                                                                                                                                             | Required    |
| Description        | Indicates a unique ID assigned to the ActionPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value. |             |
| PlanCommandID      | ActionPlanCommandID_ChoiceType                                                                                                                                                                                                                                | Optional    |
| Description        | Indicates the unique ID of the *PlanCommand which this ActionPlan originated from.                                                                                                                                                                            |             |
| Plan               | MA_ActionPlanType                                                                                                                                                                                                                                             | Required    |
| Description        | Indicates the details of the ActionPlan.                                                                                                                                                                                                                      |             |
| ForPlanningUseOnly | boolean                                                                                                                                                                                                                                                       | Required    |
| Description        | When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other     |             |



|                   |                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------|
|                   | messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.   |
| <b>PlanInputs</b> | ActionPlanInputsType Optional                                                                             |
| Description       | Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan. |

Table A-1-192: P2P UCI Extensions MA\_ActionPlanMT - MessageData UCI Extension Field

## 1.3.2.20 MA\_TaskPlanMT UCI Extension

| MA_TaskPlanMT      |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_TaskPlanMDT                                                                                                  |

Table A-1-193: P2P UCI Extensions MA\_TaskPlanMT UCI Extension

| MessageData        |                                                                                                                                                                                                                                                                                                                                                                   |             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates the data being conveyed by the message.                                                                                                                                                                                                                                                                                                                 |             |
| Base Type          | MA_TaskPlanMDT                                                                                                                                                                                                                                                                                                                                                    |             |
| Message Structure  |                                                                                                                                                                                                                                                                                                                                                                   |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                                                                                                        | Field Usage |
| TaskPlanID         | TaskPlanID_Type                                                                                                                                                                                                                                                                                                                                                   | Required    |
| Description        | Indicates a unique ID assigned to the TaskPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.                                                                                                       |             |
| PlanCommandID      | TaskPlanCommandID_ChoiceType                                                                                                                                                                                                                                                                                                                                      | Optional    |
| Description        | Indicates the unique ID of the *PlanCommand which the TaskPlan originated from.                                                                                                                                                                                                                                                                                   |             |
| Plan               | MA_TaskPlanType                                                                                                                                                                                                                                                                                                                                                   | Required    |
| Description        | Contains the details of the task allocation plan.                                                                                                                                                                                                                                                                                                                 |             |
| ForPlanningUseOnly | Boolean                                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description        | When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc. |             |
| PlanInputs         | MA_TaskPlanInputsType                                                                                                                                                                                                                                                                                                                                             | Optional    |
| Description        | Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.                                                                                                                                                                                                                                                         |             |

Table A-1-194: P2P UCI Extensions MA\_TaskPlanMT - MessageData UCI Extension Field

## 1.3.2.21 MA\_RoutePlanMT UCI Extension

| MA_RoutePlanMT |
|----------------|
|----------------|

|                    |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_RoutePlanMDT                                                                                                 |

Table A-1-195: P2P UCI Extensions MA\_RoutePlanMT UCI Extension

| MessageData        |                                                                                                                                                                                                                                                                                                                                                                   |             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates the data being conveyed by the message.                                                                                                                                                                                                                                                                                                                 |             |
| Base Type          | MA_RoutePlanMDT                                                                                                                                                                                                                                                                                                                                                   |             |
| Message Structure  |                                                                                                                                                                                                                                                                                                                                                                   |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                                                                                                        | Field Usage |
| RoutePlanID        | RoutePlanID_Type                                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description        | Indicates a unique ID assigned to the RoutePlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.                                                                                                      |             |
| PlanCommandID      | RoutePlanCommandID_ChoiceType                                                                                                                                                                                                                                                                                                                                     | Optional    |
| Description        | Indicates the unique ID of the *PlanCommand which the RoutePlan originated from.                                                                                                                                                                                                                                                                                  |             |
| Plan               | MA_RoutePlanType                                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description        | Contains the details of the route plan.                                                                                                                                                                                                                                                                                                                           |             |
| ForPlanningUseOnly | boolean                                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description        | When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc. |             |
| PlanInputs         | RoutePlanInputsType                                                                                                                                                                                                                                                                                                                                               | Optional    |
| Description        | Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.                                                                                                                                                                                                                                                         |             |

Table A-1-196: P2P UCI Extensions MA\_RoutePlanMT - MessageData UCI Extension Field

### 1.3.2.22 MA\_ActivityPlanMT UCI Extension

| MA_ActivityPlanMT  |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_ActivityPlanMDT                                                                                              |

Table A-1-197: P2P UCI Extensions MA\_ActivityPlanMT UCI Extension

| MessageData        |                                                                            |
|--------------------|----------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type |

|                    |                                                                                                                                                                                                                                                                                                                                                               |             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                    | for details.                                                                                                                                                                                                                                                                                                                                                  |             |
| Base Type          | MA_ActivityPlanMDT                                                                                                                                                                                                                                                                                                                                            |             |
| Message Structure  |                                                                                                                                                                                                                                                                                                                                                               |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                                                                                                    | Field Usage |
| ActivityPlanID     | ActivityPlanID_Type                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description        | Indicates a unique ID assigned to the ActivityPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.                                                                                               |             |
| PlanCommandID      | ActivityPlanCommandID_ChoiceType                                                                                                                                                                                                                                                                                                                              | Optional    |
| Description        | Indicates the unique ID of the *PlanCommand which this ActivityPlan originated from.                                                                                                                                                                                                                                                                          |             |
| Plan               | MA_ActivityPlanType                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description        | Indicates the details of the ActivityPlan.                                                                                                                                                                                                                                                                                                                    |             |
| ForPlanningUseOnly | boolean                                                                                                                                                                                                                                                                                                                                                       | Required    |
| Description        | When TRUE, indicates the *Plan is a used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc. |             |
| PlanInputs         | ActivityPlanInputsType                                                                                                                                                                                                                                                                                                                                        | Optional    |
| Description        | Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.                                                                                                                                                                                                                                                     |             |

Table A-1-198: P2P UCI Extensions MA\_ActivityPlanMT - MessageData UCI Extension Field

### 1.3.2.23 MA\_RouteActivityPlanMT UCI Extension

|                               |                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_RouteActivityPlanMT</b> |                                                                                                                               |
| <b>Description</b>            | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>            | MessageData : MA_RouteActivityPlanMDT                                                                                         |

Table A-1-199: P2P UCI Extensions MA\_RouteActivityPlanMT UCI Extension

| MessageData         |                                                                                                                                                                                                                                                          |             |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | Indicates the data being conveyed by the message.                                                                                                                                                                                                        |             |
| Base Type           | MA_RouteActivityPlanMDT                                                                                                                                                                                                                                  |             |
| Message Structure   |                                                                                                                                                                                                                                                          |             |
| Field Name          | Field Type                                                                                                                                                                                                                                               | Field Usage |
| RouteActivityPlanID | RouteActivityPlanID_Type                                                                                                                                                                                                                                 | Required    |
| Description         | Indicates a unique ID assigned to the RouteActivityPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child |             |

|                           |                                                                                                                                                                                                                                                                                                                                                                   |          |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                           | UUID value.                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>PlanCommandID</b>      | RouteActivityPlanCommandID_ChoiceType                                                                                                                                                                                                                                                                                                                             | Optional |
| Description               | Indicates the unique ID of the *PlanCommand which the RouteActivityPlan originated from.                                                                                                                                                                                                                                                                          |          |
| <b>Plan</b>               | MA_RouteActivityPlanType                                                                                                                                                                                                                                                                                                                                          | Required |
| Description               | Indicates the details of the RouteActivityPlan.                                                                                                                                                                                                                                                                                                                   |          |
| <b>ForPlanningUseOnly</b> | boolean                                                                                                                                                                                                                                                                                                                                                           | Required |
| Description               | When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc. |          |
| <b>PlanInputs</b>         | RouteActivityPlanInputsType                                                                                                                                                                                                                                                                                                                                       | Optional |
| Description               | Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.                                                                                                                                                                                                                                                         |          |

Table A-1-200: P2P UCI Extensions MA\_RouteActivityPlanMT - MessageData UCI Extension Field

### 1.3.2.24 MA\_ActionMT UCI Extension

| MA_ActionMT        |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_ActionMDT                                                                                                    |

Table A-1-201: P2P UCI Extensions MA\_ActionMT UCI Extension

| MessageData       |                                                                                                                                                                                                                                                                 |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                         |             |
| Base Type         | MA_ActionMDT                                                                                                                                                                                                                                                    |             |
| Message Structure |                                                                                                                                                                                                                                                                 |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                      | Field Usage |
| ActionID          | ActionID_Type                                                                                                                                                                                                                                                   | Required    |
| Description       | Indicates a unique ID assigned to the Action. Minor updates to the Action can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new instance with a different child UUID value. |             |
| ActionType        | MA_ActionTypeEnum                                                                                                                                                                                                                                               | Required    |
| Description       | Indicates the type of Action; the type of behavior to be performed when the Action is executed.<br><br>An Action is a representation of an operational planning sentence/statement whose most complex form is:                                                  |             |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                | <p>1. Do [ActionType] within [ActionConstraints] to [TargetObject] within [TargetObjectConstraint] to/on/against/toward [SecondaryObject]</p> <p>Many other forms are also possible, including:</p> <p>2. do [ActionType]</p> <p>3. do [ActionType] to [TargetObject]</p> <p>4. do [ActionType] to [TargetObject] doing [TargetObjectConstraint] to [SecondaryObject].</p>                                                                                                                                                                    |          |
| <b>ActionConstraints</b>       | MA_RequirementConstraintsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Optional |
| Description                    | Indicates constraints on the Action. The constraint type used here is common to all Requirement types: Effect, Action, Task, [Capability]Command, Response. Requirement constraints can be 1) given in a [Requirement] message, 2) overridden as a planning input in a [Requirement]PlanCommand message, 3) further refined/overridden during plan creation and reported in a [Requirement]Plan message to inform subsequent planning or execution. See sibling ActionType element annotation for an overall description of Action structure. |          |
| <b>ActionGuidance</b>          | RequirementGuidanceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Optional |
| Description                    | Indicates guidance on planning and C2 of the Action.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>TargetObject</b>            | IdentityKindInstanceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Repeated |
| Description                    | Indicates the target of the Action; the object the Action will be done to or for. If omitted, the Action places no explicit limitation on which TargetObject it can be done to. See sibling ActionType element annotation for an overall description of Action structure. Note that the term "target" is used in a general sense. For some Requirement types, a more suitable term might be "subject", "point", "location", etc. For example, the "target" of a COVER Action could be a blue/friendly System.                                 |          |
| <b>TargetObjectConstraints</b> | RequirementTargetConstraintsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Optional |
| Description                    | Indicates constraints "on" the TargetObject that must be satisfied before the Action can be done to or for it. See sibling ActionType element annotation for an overall description of Effect structure.                                                                                                                                                                                                                                                                                                                                      |          |
| <b>SecondaryObject</b>         | IdentityKindInstanceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Repeated |
| Description                    | Indicates a thing the sibling TargetObject must act on/against/toward to make the Action applicable. See sibling ActionType element annotation for an overall description of Action structure.                                                                                                                                                                                                                                                                                                                                                |          |
| <b>Metadata</b>                | RequirementMetadataType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Optional |
| Description                    | Indicates traceability and other metadata associated with the Action.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |

Table A-1-202: P2P UCI Extensions MA\_ActionMT - MessageData UCI Extension Field

### 1.3.2.25 MA\_ResponseMT UCI Extension

| MA_ResponseMT      |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |
|                    | See the XSD File for more info                                                          |

|                    |                              |
|--------------------|------------------------------|
| <b>Ext. Fields</b> | MessageData : MA_ResponseMDT |
|--------------------|------------------------------|

Table A-1-203: P2P UCI Extensions MA\_ResponseMT UCI Extension

| MessageData                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                   | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| Base Type                     | MA_ResponseMDT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
| Message Structure             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| Field Name                    | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Field Usage |
| ResponseID                    | ResponseID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required    |
| Description                   | Indicates the unique ID of the Response. Minor updates to the Response can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new instance with a different child UUID value.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| ResponseType                  | ResponseTypeEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Repeated    |
| Description                   | Indicates the types of Response options included in the message. List size for this element is based on "Select All That Apply" condition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| ResponseManagementConstraints | RequirementConstraintsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Optional    |
| Description                   | <p>Indicates monitoring constraints on the Response. The constraint type used here is common to all Requirement types: Effect, Action, Task, [Capability]Command, Response. Requirement constraints can be 1) given in a [Requirement] message, 2) overridden as a planning input in a [Requirement]PlanCommand message, 3) further refined/overridden during plan creation and reported in a [Requirement]Plan message to inform subsequent planning or execution. See sibling Option element annotation for an overall description of Response structure.</p> <p>Constraints given here aren't conveyed to instantiated Response options for a particular triggering event. The constraints given here apply to Response management ( planning, allocation and trigger monitoring prior to a triggering event).</p> <p>Constraints to apply to instantiated Response options, if any, are given elsewhere in the message.</p> |             |
| ResponseManagementGuidance    | RequirementGuidanceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Optional    |
| Description                   | Indicates guidance on planning and C2 of the Response.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |
| ResponsePlurality             | MA_TaskPluralityEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                   | Indicates whether the response applies a single entity or as a joint condition. A joint condition implies condiseration of the response trigger and activation of the response by all associated entities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| Option                        | MA_ResponseOptionDetailsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated    |
| Description                   | Indicates an Option of the Response. Responses are designed to be flexible. A Response is a set of prioritized Options to address an anticipated triggering event, events or event sequence in the battlespace. Responses are allocated                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                             | to Systems via a ResponsePlan. When a ResponsePlan is activated, monitoring for the triggering events in the Options begins. When a triggering event happens, the applicable Option(s) are evaluated.                                                                                                                                                                                                                                                                                       |          |
| <b>RequirementsTemplate</b> | MA_RequirementsTemplateType                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Repeated |
| Description                 | The sibling Option element associates triggering events with a desired Response one of which is to use this element, a Requirements template, to generate or change one or more Requirements messages/objects (Effect, Action, Task, [Capability]Command). An instance of this element is a RequirementsTemplate; it indicates the types of Requirements to generate or update and their details. Each RequirementsTemplate has a unique ID and therefore can be used for multiple Options. |          |
| <b>Metadata</b>             | RequirementMetadataType                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Optional |
| Description                 | Indicates traceability and other metadata associated with the Response.                                                                                                                                                                                                                                                                                                                                                                                                                     |          |

Table A-1-204: P2P UCI Extensions MA\_ResponseMT - MessageData UCI Extension Field

## 1.3.2.26 MA\_OpVolumeMT UCI Extension

| MA_OpVolumeMT      |                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_OpVolumeMDT                                                                                                                |

Table A-1-205: P2P UCI Extensions MA\_OpVolumeMT UCI Extension

| MessageData          |                                                                                             |             |
|----------------------|---------------------------------------------------------------------------------------------|-------------|
| Description          | This element represents the details regarding the operational volume.                       |             |
| Base Type            | MA_OpVolumeMDT                                                                              |             |
| Message Structure    |                                                                                             |             |
| Field Name           | Field Type                                                                                  | Field Usage |
| OpVolumeID           | OpVolumeID_Type                                                                             | Required    |
| Description          | This element represents the unique identifier of the OpVolume associated with this message. |             |
| Category             | MA_OpZoneCategoryEnum                                                                       | Required    |
| Description          | This element indicates the type of volume.                                                  |             |
| CategoryUniqueData   | MA_OpZoneCategoryType                                                                       | Optional    |
| Description          | Contains parameters unique to certain OpZoneCategoryEnum values.                            |             |
| Volume               | MA_OpVolumeChoiceType                                                                       | Required    |
| Description          | This element represents the characteristics of the operational volume.                      |             |
| SituationalAwareness | boolean                                                                                     | Required    |
| Description          | Volume used only for situational awareness. Not used for planning purposes.                 |             |
| OpDescription        | OpDescriptionType                                                                           | Optional    |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                   | A human-readable descriptive name with free text remarks.                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>MissionTraceability</b>    | MissionTraceabilityType                                                                                                                                                                                                                                                                                                                                                                                          | Optional |
| Description                   | Allows the creator of this operational item to specify what the source or motivation for it originated from. This could be ATO, ACO, etc.                                                                                                                                                                                                                                                                        |          |
| <b>Source</b>                 | CreationSourceEnum                                                                                                                                                                                                                                                                                                                                                                                               | Optional |
| Description                   | Indicates the source of the information.                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Schedule</b>               | ScheduleType                                                                                                                                                                                                                                                                                                                                                                                                     | Optional |
| Description                   | This element represents a date/time interval associated with the operational item during which the item is in effect.                                                                                                                                                                                                                                                                                            |          |
| <b>AssociatedTime</b>         | TimeFunctionType                                                                                                                                                                                                                                                                                                                                                                                                 | Repeated |
| Description                   | Indicates time related events that are associated to the Op data type such as when activated or when operational. The multiplicity is limited to N-1 options where NONE is an event that has no value, thus removed from list of available options.                                                                                                                                                              |          |
| <b>DataLinkIdentifier</b>     | DataLinkIdentifierPET                                                                                                                                                                                                                                                                                                                                                                                            | Repeated |
| Description                   | List of data link ID. Multiple data link IDs can be reported for the same network type.                                                                                                                                                                                                                                                                                                                          |          |
| <b>Priority</b>               | unsignedInt                                                                                                                                                                                                                                                                                                                                                                                                      | Optional |
| Description                   | This field represents the priority to be considered when choosing between multiple available points.                                                                                                                                                                                                                                                                                                             |          |
| <b>QualifyingTags</b>         | QualifyingTagsType                                                                                                                                                                                                                                                                                                                                                                                               | Optional |
| Description                   | Provides identity qualifications to augment identification and associated processing of the OpPoint.                                                                                                                                                                                                                                                                                                             |          |
| <b>AppliesToIDs</b>           | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                                                                             | Repeated |
| Description                   | One or more systems to which this item applies. If omitted, item applies to all systems.                                                                                                                                                                                                                                                                                                                         |          |
| <b>SystemScheduleOverride</b> | SystemScheduleStateType                                                                                                                                                                                                                                                                                                                                                                                          | Repeated |
| Description                   | System-specific schedule that will override any part of the main schedule. This allows for system-specific schedule different than main schedule. When adding a SystemScheduleOverride, the child SystemID also must be added to the sibling SystemID list if the sibling list is not empty. The SystemScheduleOverride is independent of time planning. For the system, this override can be done at execution. |          |

Table A-1-206: P2P UCI Extensions MA\_OpVolumeMT - MessageData UCI Extension Field

### 1.3.2.27 MA\_ApprovalAuthorityMT UCI Extension

| MA_ApprovalAuthorityMT |                                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>     | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>     | MessageData : MA_ApprovalAuthorityMDT<br>ApplicableIDs : MA_PackageSystemType                                                 |

Table A-1-207: P2P UCI Extensions MA\_ApprovalAuthorityMT UCI Extension

| MessageData |
|-------------|
|-------------|



|                      |                                                                                                                                                                                                                                                                                                          |             |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                  |             |
| Base Type            | MA_ApprovalAuthorityMDT                                                                                                                                                                                                                                                                                  |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                          |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                               | Field Usage |
| ApprovalAuthorityID  | ApprovalAuthorityID_Type                                                                                                                                                                                                                                                                                 | Required    |
| Description          | Indicates associated Approval Authority identifier for applicable system and approval policy.                                                                                                                                                                                                            |             |
| ApplicableIDs        | MA_PackageSystemType                                                                                                                                                                                                                                                                                     | Repeated    |
| Description          | Indicates to which system this Approval Authority applies.                                                                                                                                                                                                                                               |             |
| SystemApprovalPolicy | SystemApprovalPolicyType                                                                                                                                                                                                                                                                                 | Repeated    |
| Description          | Indicates an ApprovalPolicy and one or more approval authorities ("approvers").                                                                                                                                                                                                                          |             |
| DataRecordInstanceID | DataRecordInstanceID_Type                                                                                                                                                                                                                                                                                | Optional    |
| Description          | Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee. |             |

Table A-1-208: P2P UCI Extensions MA\_ApprovalAuthorityMT - MessageData UCI Extension Field

| ApplicableIDs     |                                                            |             |
|-------------------|------------------------------------------------------------|-------------|
| Description       | Indicates to which system this Approval Authority applies. |             |
| Base Type         | MA_PackageSystemType                                       |             |
| Message Structure |                                                            |             |
| Field Name        | Field Type                                                 | Field Usage |
| ALL               | EmptyType                                                  | Required    |
| Description       | Indicates applies to ALL entities.                         |             |
| NONE              | EmptyType                                                  | Required    |
| Description       | Indicates applies to NO entities.                          |             |
| PackageID         | PackageID_Type                                             | Required    |
| Description       | The ID of a Package.                                       |             |
| SystemID          | SystemID_Type                                              | Required    |
| Description       | The ID of an ACP or C2 Operator.                           |             |

Table A-1-209: P2P UCI Extensions MA\_ApprovalAuthorityMT - ApplicableIDs UCI Extension Field

### 1.3.2.28 MA\_CapabilitySetMT UCI Extension

|                           |                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------|
| <b>MA_CapabilitySetMT</b> |                                                                                         |
| <b>Description</b>        | See annotations in child elements and messages/elements that use this type for details. |

|                    |                                   |
|--------------------|-----------------------------------|
|                    | See the XSD File for more info    |
| <b>Ext. Fields</b> | MessageData : MA_CapabilitySetMDT |

Table A-1-210: P2P UCI Extensions MA\_CapabilitySetMT UCI Extension

| MessageData          |                                                                                                                                                                                                                                                                                                          |             |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                  |             |
| Base Type            | MA_CapabilitySetMDT                                                                                                                                                                                                                                                                                      |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                          |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                               | Field Usage |
| CapabilitySetID      | MA_CapabilitySetID_Type                                                                                                                                                                                                                                                                                  | Required    |
| Description          | Indicates the unique ID of the CapabilitySet.                                                                                                                                                                                                                                                            |             |
| CapabilityID         | CapabilityID_Type                                                                                                                                                                                                                                                                                        | Repeated    |
| Description          | Indicates a Capability that exists in the set.                                                                                                                                                                                                                                                           |             |
| ApplicableSystemID   | SystemID_Type                                                                                                                                                                                                                                                                                            | Optional    |
| Description          | Indicates the System that the CapabilitySet is relevant to.                                                                                                                                                                                                                                              |             |
| MissionPlanID        | MissionPlanID_Type                                                                                                                                                                                                                                                                                       | Optional    |
| Description          | Indicates the MissionPlan that the CapabilitySet is relevant to.                                                                                                                                                                                                                                         |             |
| MissionID            | MissionID_Type                                                                                                                                                                                                                                                                                           | Optional    |
| Description          | Indicates the Mission that the CapabilitySet is relevant to.                                                                                                                                                                                                                                             |             |
| DataRecordInstanceID | DataRecordInstanceID_Type                                                                                                                                                                                                                                                                                | Optional    |
| Description          | Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee. |             |

Table A-1-211: P2P UCI Extensions MA\_CapabilitySetMT - MessageData UCI Extension Field

### 1.3.2.29 MA\_MissionPlanActivationCommandStatusMT UCI Extension

| MA_MissionPlanActivationCommandStatusMT |                                                                                                                                                   |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                      | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info       |
| <b>Ext. Fields</b>                      | Plans : MA_PlansReferenceType<br>MessageData : MA_MissionPlanActivationCommandStatusMDT<br>ActivationCommandByState : MA_PlanActivationStatusType |

Table A-1-212: P2P UCI Extensions MA\_MissionPlanActivationCommandStatusMT UCI Extension

| Plans |
|-------|
|-------|

|                     |                                                          |             |
|---------------------|----------------------------------------------------------|-------------|
| Description         | Indicates the Plan or Plans in the sibling CommandState. |             |
| Base Type           | MA_PlansReferenceType                                    |             |
| Message Structure   |                                                          |             |
| Field Name          | Field Type                                               | Field Usage |
| MissionPlanID       | MissionPlanID_Type                                       | Repeated    |
| Description         | Indicates a reference to a MissionPlan.                  |             |
| TaskPlanID          | TaskPlanID_Type                                          | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                     |             |
| OrbitPlanID         | OrbitPlanID_Type                                         | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.                   |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                                 | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.           |             |
| RoutePlanID         | RoutePlanID_Type                                         | Repeated    |
| Description         | Indicates a reference to a RoutePlan.                    |             |
| RouteActivityPlanID | RouteActivityPlanID_Type                                 | Repeated    |
| Description         | Indicates a reference to a RouteActivityPlan.            |             |
| ActivityPlanID      | ActivityPlanID_Type                                      | Repeated    |
| Description         | Indicates a reference to an ActivityPlan.                |             |
| EffectPlanID        | EffectPlanID_Type                                        | Repeated    |
| Description         | Indicates a reference to an EffectPlan.                  |             |
| ActionPlanID        | ActionPlanID_Type                                        | Repeated    |
| Description         | Indicates a reference to an ActionPlan.                  |             |
| ResponsePlanID      | ResponsePlanID_Type                                      | Repeated    |
| Description         | Indicates a reference to an ResponsePlan.                |             |
| OpPlanID            | MA_OpPlanID_Type                                         | Repeated    |
| Description         | Indicates a reference to an OpPlan.                      |             |

**Table A-1-213: P2P UCI Extensions MA\_MissionPlanActivationCommandStatusMT - Plans UCI Extension Field**

| MessageData       |                                                                     |             |
|-------------------|---------------------------------------------------------------------|-------------|
| Description       | Indicates the data being conveyed by the message.                   |             |
| Base Type         | MA_MissionPlanActivationCommandStatusMDT                            |             |
| Message Structure |                                                                     |             |
| Field Name        | Field Type                                                          | Field Usage |
| CommandID         | MissionPlanActivationCommandID_Type                                 | Required    |
| Description       | The ID of the MissionPlanActivationCommand this status pertains to. |             |
| ActivationStatus  | PlanCommandStatusType                                               | Required    |

|                                 |                                                                                                                                          |          |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                     | Indicates the status of the MissionPlanActivationCommand.                                                                                |          |
| <b>ActivationCommandByState</b> | MA_PlanActivationStatusType                                                                                                              | Repeated |
| Description                     | Indicates plans in a given commanded activation state. List size allows for each available PlanActivationCommandState to be represented. |          |

**Table A-1-214: P2P UCI Extensions MA\_MissionPlanActivationCommandStatusMT - MessageData UCI Extension Field**

| ActivationCommandByState   |                                                                                                                                          |             |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                | Indicates plans in a given commanded activation state. List size allows for each available PlanActivationCommandState to be represented. |             |
| Base Type                  | MA_PlanActivationStatusType                                                                                                              |             |
| Message Structure          |                                                                                                                                          |             |
| Field Name                 | Field Type                                                                                                                               | Field Usage |
| PlanActivationCommandState | PlanActivationStateEnum                                                                                                                  | Required    |
| Description                | Indicates an activation command state that one or more *Plans is in.                                                                     |             |
| Plans                      | MA_PlansReferenceType                                                                                                                    | Required    |
| Description                | Indicates the Plan or Plans in the sibling CommandState.                                                                                 |             |

**Table A-1-215: P2P UCI Extensions MA\_MissionPlanActivationCommandStatusMT - ActivationCommandByState UCI Extension Field**

### 1.3.2.30 MA\_MissionPlanExecutionStatusMT UCI Extension

| MA_MissionPlanExecutionStatusMT |                                                                                                                                                                            |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>              | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info                                              |
| <b>Ext. Fields</b>              | ExecutionPlanSetStatus : MA_ExecutionPlanSetExecutionStateType<br>PlanExecutionStatus : MA_MissionPlanExecutionStateType<br>MessageData : MA_MissionPlanExecutionStatusMDT |

**Table A-1-216: P2P UCI Extensions MA\_MissionPlanExecutionStatusMT UCI Extension**

| ExecutionPlanSetStatus         |                                                                                                                           |             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                    | Indicates the execution state of an execution plan set of the MissionPlan indicated by the sibling MissionPlanID element. |             |
| Base Type                      | MA_ExecutionPlanSetExecutionStateType                                                                                     |             |
| Message Structure              |                                                                                                                           |             |
| Field Name                     | Field Type                                                                                                                | Field Usage |
| ExecutionPlanSetID             | ExecutionPlanSetID_Type                                                                                                   | Required    |
| Description                    | Indicates the unique ID of the execution plan set.                                                                        |             |
| ExecutionPlanSetExecutionState | PlanExecutionStateEnum                                                                                                    | Required    |
| Description                    | The current state of the execution plan set.                                                                              |             |

**Table A-1-217: P2P UCI Extensions MA\_MissionPlanExecutionStatusMT - ExecutionPlanSetStatus UCI Extension Field**

| PlanExecutionStatus    |                                                                                                                           |             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | Indicates the execution state of a MissionPlan of the System indicated by the sibling SystemID element.                   |             |
| Base Type              | MA_MissionPlanExecutionStateType                                                                                          |             |
| Message Structure      |                                                                                                                           |             |
| Field Name             | Field Type                                                                                                                | Field Usage |
| MissionPlanID          | MissionPlanID_Type                                                                                                        | Required    |
| Description            | The ID of the MissionPlan this status pertains to.                                                                        |             |
| PlanExecutionState     | PlanExecutionStateEnum                                                                                                    | Required    |
| Description            | The current state of the plan.                                                                                            |             |
| ExecutionPlanSetStatus | MA_ExecutionPlanSetExecutionStateType                                                                                     | Repeated    |
| Description            | Indicates the execution state of an execution plan set of the MissionPlan indicated by the sibling MissionPlanID element. |             |

**Table A-1-218: P2P UCI Extensions MA\_MissionPlanExecutionStatusMT - PlanExecutionStatus UCI Extension Field**

| MessageData         |                                                                                                         |             |
|---------------------|---------------------------------------------------------------------------------------------------------|-------------|
| Description         | Indicates the data being conveyed by the message.                                                       |             |
| Base Type           | MA_MissionPlanExecutionStatusMDT                                                                        |             |
| Message Structure   |                                                                                                         |             |
| Field Name          | Field Type                                                                                              | Field Usage |
| SystemID            | SystemID_Type                                                                                           | Required    |
| Description         | The unique ID of the System whose Plan is being statused.                                               |             |
| Source              | SystemSourceEnum                                                                                        | Required    |
| Description         | Indicates the source of the data in sibling elements. See enumeration annotations for further details.  |             |
| PlanExecutionStatus | MA_MissionPlanExecutionStateType                                                                        | Repeated    |
| Description         | Indicates the execution state of a MissionPlan of the System indicated by the sibling SystemID element. |             |

**Table A-1-219: P2P UCI Extensions MA\_MissionPlanExecutionStatusMT - MessageData UCI Extension Field****1.3.2.31 MA\_ConstraintPriorityListActivationStatusMT UCI Extension**

| MA_ConstraintPriorityListActivationStatusMT |                                                                                                                               |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                          | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                          | MessageData : MA_ConstraintPriorityListActivationStatusMDT                                                                    |

Table A-1-220: P2P UCI Extensions MA\_ConstraintPriorityListActivationStatusMT UCI Extension

| MessageData                            |                                                                                                                                |             |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                            | See annotations in child elements and messages/elements that use this type for details.                                        |             |
| Base Type                              | MA_ConstraintPriorityListActivationStatusMDT                                                                                   |             |
| Message Structure                      |                                                                                                                                |             |
| Field Name                             | Field Type                                                                                                                     | Field Usage |
| ConstraintPriorityListActivationDataID | MA_ConstraintPriorityListActivationDataID_Type                                                                                 | Required    |
| Description                            | The unique identifier for the CPL Activation Data message.                                                                     |             |
| ActiveConstraintPriorityListID         | MA_ConstraintPriorityListID_Type                                                                                               | Required    |
| Description                            | The unique identifier for the active Constraint Priority List.                                                                 |             |
| AppliesToIDs                           | MA_PackageSystemType                                                                                                           | Repeated    |
| Description                            | The IDs, Package or ACP, that this message applies to. A lead is able to respond for the Package or individual ACPs as needed. |             |

Table A-1-221: P2P UCI Extensions MA\_ConstraintPriorityListActivationStatusMT - MessageData UCI Extension Field

### 1.3.2.32 MA\_ConstraintPriorityListCommandMT UCI Extension

| MA_ConstraintPriorityListCommandMT |                                                                                                                               |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description                        | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields                        | MessageData : MA_ConstraintPriorityListCommandMDT                                                                             |

Table A-1-222: P2P UCI Extensions MA\_ConstraintPriorityListCommandMT UCI Extension

| MessageData                            |                                                                                                                                                         |             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                            | See annotations in child elements and messages/elements that use this type for details.                                                                 |             |
| Base Type                              | MA_ConstraintPriorityListCommandMDT                                                                                                                     |             |
| Message Structure                      |                                                                                                                                                         |             |
| Field Name                             | Field Type                                                                                                                                              | Field Usage |
| ConstraintPriorityListActivationDataID | MA_ConstraintPriorityListActivationDataID_Type                                                                                                          | Required    |
| Description                            | This ID of the Activated message that should be active.                                                                                                 |             |
| AppliesToIDs                           | MA_PackageSystemType                                                                                                                                    | Repeated    |
| Description                            | The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in this field as well. |             |
| CommandID                              | CommandID_Type                                                                                                                                          | Required    |
| Description                            | Indicates the unique ID of the Command corresponding to this message.                                                                                   |             |

|                     |                                                                                                                                                               |          |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                     | Used to reference the command when providing status.                                                                                                          |          |
| <b>CommandState</b> | CommandStateEnum                                                                                                                                              | Required |
| Description         | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command. |          |

**Table A-1-223: P2P UCI Extensions MA\_ConstraintPriorityListCommandMT - MessageData UCI Extension Field**

### 1.3.2.33 MA\_ConstraintPriorityListCommandStatusMT UCI Extension

| MA_ConstraintPriorityListCommandStatusMT |                                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                       | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                       | MessageData : MA_ConstraintPriorityListCommandStatusMDT                                                                       |

**Table A-1-224: P2P UCI Extensions MA\_ConstraintPriorityListCommandStatusMT UCI Extension**

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_ConstraintPriorityListCommandStatusMDT                                                                                                                                                                                                                                                                                                                   |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| AppliesToIDs                 | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description                  | The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed.                                                                                                                                                                                                                |             |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

**Table A-1-225: P2P UCI Extensions MA\_ConstraintPriorityListCommandStatusMT - MessageData UCI Extension Field**

### 1.3.2.34 MA\_EscortCommandMT UCI Extension

| MA_EscortCommandMT |
|--------------------|
|--------------------|

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | <p>This message is the Escort form of the general UCI Capability Command. It is used to manually invoke a Escort Capability to perform a new Escort Activity or change an existing interactive Escort Activity. It differs from a Escort Task because it is intended for direct interaction with the host of the Capability, not a planning Service, and therefore is not expected to impact the route of the host System or otherwise interrupt the planned MissionPlan. Additionally, Capability Commands are intended to allow multiple clients to use a single Capability as a shared resource; see the Control* series of messages. Capability Commands can be used in parallel with Tasks, but are generally deferential to them. However, exceptions are allowed and even expected; the relationship between Tasks and Capability Commands is not rigidly dictated by the UCI schema. Several mechanisms are provided in other messages to tailor the relationship and relative importance of Tasks and [Capability]Commands including Capability precedence, Task rank, [Capability]Command rank, [Capability]Activity rank and control Authorization.</p> <p>See the XSD File for more info</p> |
| <b>Ext. Fields</b> | MessageData : MA_EscortCommandMDT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Table A-1-226: P2P UCI Extensions MA\_EscortCommandMT UCI Extension

| MessageData       |                                                                                                                                                                                                                                                                                                  |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                          |             |
| Base Type         | MA_EscortCommandMDT                                                                                                                                                                                                                                                                              |             |
| Message Structure |                                                                                                                                                                                                                                                                                                  |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                       | Field Usage |
| Command           | MA_EscortCommandType                                                                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates an instance of an Escort Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message. |             |

Table A-1-227: P2P UCI Extensions MA\_EscortCommandMT - MessageData UCI Extension Field

### 1.3.2.35 MA\_EscortCommandStatusMT UCI Extension

| MA_EscortCommandStatusMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>       | <p>This message indicates the result of an EscortCommand. This message will reflect whether the command was received, accepted and/or rejected. This message will only be used to status the command until the recipient determines whether or not it can be started or scheduled. The details of the specific activity or activities that result from the command will report status via the EscortActivity message.</p> <p>See the XSD File for more info</p> |
| <b>Ext. Fields</b>       | MessageData : MA_EscortCommandStatusMDT                                                                                                                                                                                                                                                                                                                                                                                                                         |

Table A-1-228: P2P UCI Extensions MA\_EscortCommandStatusMT UCI Extension

| MessageData        |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |



|                              |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Base Type                    | MA_EscortCommandStatusMDT                                                                                                                                                                                                                                                                                                                                   |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| Activity                     | ResultingActivityType                                                                                                                                                                                                                                                                                                                                       | Repeated    |
| Description                  | Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.                                                                                                                                                                                                                    |             |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

Table A-1-229: P2P UCI Extensions MA\_EscortCommandStatusMT - MessageData UCI Extension Field

### 1.3.2.36 MA\_ExecutionPriorityListActivationStatusMT UCI Extension

|                                                   |                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_ExecutionPriorityListActivationStatusMT</b> |                                                                                                                               |
| <b>Description</b>                                | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                                | MessageData : MA_ExecutionPriorityListActivationStatusMDT                                                                     |

Table A-1-230: P2P UCI Extensions MA\_ExecutionPriorityListActivationStatusMT UCI Extension

| MessageData                           |                                                                                         |             |
|---------------------------------------|-----------------------------------------------------------------------------------------|-------------|
| Description                           | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type                             | MA_ExecutionPriorityListActivationStatusMDT                                             |             |
| Message Structure                     |                                                                                         |             |
| Field Name                            | Field Type                                                                              | Field Usage |
| ExecutionPriorityListActivationDataID | MA_ExecutionPriorityListActivationDataID_Type                                           | Required    |
| Description                           | The unique identifier for the EPL Activation Data message.                              |             |
| ActiveExecutionPriorityListID         | MA_ExecutionPriorityListID_Type                                                         | Required    |
| Description                           | The unique identifier for the active EPL message.                                       |             |
| AppliesToIDs                          | MA_PackageSystemType                                                                    | Repeated    |

|             |                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| Description | The IDs, Package or ACP, that this message applies to. A lead is able to respond for the Package or individual ACPs as needed. |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|

**Table A-1-231: P2P UCI Extensions MA\_ExecutionPriorityListActivationStatusMT - MessageData UCI Extension Field**

### 1.3.2.37 MA\_ExecutionPriorityListCommandMT UCI Extension

| MA_ExecutionPriorityListCommandMT |                                                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description                       | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields                       | MessageData : MA_ExecutionPriorityListCommandMDT                                                                              |

**Table A-1-232: P2P UCI Extensions MA\_ExecutionPriorityListCommandMT UCI Extension**

| MessageData                           |                                                                                                                                                               |             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                           | See annotations in child elements and messages/elements that use this type for details.                                                                       |             |
| Base Type                             | MA_ExecutionPriorityListCommandMDT                                                                                                                            |             |
| Message Structure                     |                                                                                                                                                               |             |
| Field Name                            | Field Type                                                                                                                                                    | Field Usage |
| ExecutionPriorityListActivationDataID | MA_ExecutionPriorityListActivationDataID_Type                                                                                                                 | Required    |
| Description                           | The ID of the Activation Data message that should be used.                                                                                                    |             |
| AppliesToIDs                          | MA_PackageSystemType                                                                                                                                          | Repeated    |
| Description                           | The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in this field as well.       |             |
| CommandID                             | CommandID_Type                                                                                                                                                | Required    |
| Description                           | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.                                    |             |
| CommandState                          | CommandStateEnum                                                                                                                                              | Required    |
| Description                           | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command. |             |

**Table A-1-233: P2P UCI Extensions MA\_ExecutionPriorityListCommandMT - MessageData UCI Extension Field**

### 1.3.2.38 MA\_ExecutionPriorityListCommandStatusMT UCI Extension

| MA_ExecutionPriorityListCommandStatusMT |                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description                             | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields                             | MessageData : MA_ExecutionPriorityListCommandStatusMDT                                                                        |

Table A-1-234: P2P UCI Extensions MA\_ExecutionPriorityListCommandStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_ExecutionPriorityListCommandStatusMDT                                                                                                                                                                                                                                                                                                                    |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| AppliesToIDs                 | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description                  | The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed.                                                                                                                                                                                                                |             |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

Table A-1-235: P2P UCI Extensions MA\_ExecutionPriorityListCommandStatusMT - MessageData UCI Extension Field

### 1.3.2.39 MA\_FlightActivityMT UCI Extension

| MA_FlightActivityMT |                                                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Description         | See annotations in child elements and messages/elements that use this type for details.                                               |
|                     | See the XSD File for more info                                                                                                        |
| Ext. Fields         | VehicleCommandState :<br>Activity : MA_FlightActivityType<br>Speed : PathSegmentSpeedValueType<br>Heading : MA_DirectionReferenceType |

Table A-1-236: P2P UCI Extensions MA\_FlightActivityMT UCI Extension

| VehicleCommandState |  |
|---------------------|--|
| Description         |  |
| Base Type           |  |
| Message Structure   |  |

| Field Name | Field Type | Field Usage |
|------------|------------|-------------|
|------------|------------|-------------|

Table A-1-237: P2P UCI Extensions MA\_FlightActivityMT - VehicleCommandState UCI Extension Field

| Activity                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | Indicates the dynamic operational characteristics of an Activity of a System's Flight Capability.                                                                                                                                                                                                                                                                                                                                                                                      |             |
| Base Type                | MA_FlightActivityType                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
| Message Structure        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |
| Field Name               | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Field Usage |
| ActualEndPoint           | MA_EndPointType                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description              | Indicates end points for the Flight activity. The current end point (the end point the System is currently flying to) is listed first followed by subsequent end points if known. If the Flight Activity was planned and the actual Activity is following the plan, the RouteActivityPlan* messages provide details relative to the plan. For planned Flight Activities against moving targets the actual, dynamically calculated kinematic details can be provided with this element. |             |
| ActualCompletionTime     | DateTimeType                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Optional    |
| Description              | This field specifies the actual time when the Flight Activity completed.                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| VehicleCommandState      | MA_VehicleCommandStateType                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Required    |
| Description              | Indicates the currently commanded parameters per the Flight Autonomy's execution of the request FlightCommand.                                                                                                                                                                                                                                                                                                                                                                         |             |
| CurveFollowingControl    | MA_CurveFollowingControlType                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Optional    |
| Description              | Indicates the curve following parameters, validity and status.                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| RoutePlanInterceptStatus | MA_RoutePlanInterceptStatusType                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Optional    |
| Description              | Status of the executing RoutePlan Intercept.                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
| ActivityID               | ActivityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required    |
| Description              | Indicates the unique ID of the Activity.                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| CapabilityID             | CapabilityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Repeated    |
| Description              | Indicates a Capability which the Activity is an instance of. Generally an Activity is an instance of a single Capability. An Activity can be associated with multiple Capabilities when, for example, a Subsystem has distinct Capabilities for automated and manual. In this case, if a manual command is issued while an automated Activity that achieves the desired result already exists, a single Activity represents both Capabilities.                                         |             |
| Interactive              | boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Required    |
| Description              | Indicates whether or not the Activity can be edited/modified via a [Capability]Command...Activity message.                                                                                                                                                                                                                                                                                                                                                                             |             |
| ActivityState            | ActivityStateEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Required    |
| Description              | Indicates the current state of the Activity its state machine. See enumeration annotations for further details.                                                                                                                                                                                                                                                                                                                                                                        |             |
| Basis                    | ActivityBasisEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Optional    |
| Description              | Indicates the basis of the Activity. See enumerated type annotations for further details. If omitted, a basis of ACTUAL should be assumed.                                                                                                                                                                                                                                                                                                                                             |             |

| ActivityRank             | ComparableRankingType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Optional |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description              | Indicates the rank of the Activity amongst all other Activities of the Capability and/or Subsystem. This isn't necessarily the same as the Rank associated with the sibling ActivitySource. This element is meant to convey the Subsystem determined relative ranking of all Activities of a Capability category (ESM, EA, etc.). For Subsystems with multiple Capability categories this element is meant to convey relative ranking across all Capability categories. In addition to the rank of the ActivitySource, an Activity's rank can be influenced by Capability priority (see CapabilitySettings), RF_ControlCommands and other factors that influence the Subsystem's internal scheduling and resource allocation. |          |
| ActivityReason           | CannotComplyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Optional |
| Description              | Indicates details regarding the reason for and/or a contributing factor to the sibling ActivityState element. This element is only expected when the Activity is in a potentially undesirable state such as constrained, failed or completed with less than optimum results.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| Source                   | ActivitySourceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Repeated |
| Description              | Indicates an originating "source" of the Activity; the command or other item which invoked this specific Activity of a Capability. A single Activity can fully or partially satisfy Activity sources.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| EstimatedStartTime       | DateTimeType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description              | Indicates the estimated time when the Activity started or is expected to start.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| EstimatedCompletionTime  | DateTimeType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description              | Indicates the estimated time when the Activity will complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| EstimatedPercentComplete | PercentType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Optional |
| Description              | Indicates the current estimated percent complete of the Activity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |

Table A-1-238: P2P UCI Extensions MA\_FlightActivityMT - Activity UCI Extension Field

| Speed             |                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------|-------------|
| Description       | Defines speed type (reference frame) for all speed related fields in this message.        |             |
| Base Type         | PathSegmentSpeedValueType                                                                 |             |
| Message Structure |                                                                                           |             |
| Field Name        | Field Type                                                                                | Field Usage |
| Value             | SpeedType                                                                                 | Required    |
| Description       | Indicates the speed along the PathSegment at the EndPoint.                                |             |
| Reference         | SpeedReferenceEnum                                                                        | Required    |
| Description       | Indicates whether the given vehicle speed is reference to ground speed or true air speed. |             |

Table A-1-239: P2P UCI Extensions MA\_FlightActivityMT - Speed UCI Extension Field

| Heading     |                                   |
|-------------|-----------------------------------|
| Description | Intended heading of the aircraft. |
| Base Type   | MA_DirectionReferenceType         |

| Message Structure |                                                |             |
|-------------------|------------------------------------------------|-------------|
| Field Name        | Field Type                                     | Field Usage |
| Value             | AngleType                                      | Required    |
| Description       | Indicates the heading or course value to hold. |             |
| Reference         | MA_HeadingReferenceEnum                        | Required    |
| Description       | Reference to type of heading or course given.  |             |

Table A-1-240: P2P UCI Extensions MA\_FlightActivityMT - Heading UCI Extension Field

### 1.3.2.40 MA\_FlightCommandMT UCI Extension

| MA_FlightCommandMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description        | <p>Message extension included to invoke a flight capability to perform a new flight activity or change and existing flight activity.</p> <p>See the XSD File for more info</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Ext. Fields        | <p>Command : MA_FlightCommandType</p> <p>Capability : MA_FlightCapabilityCommandType</p> <p>HSA_CSA : MA_HSA_CSA_Type</p> <p>WaypointFollowing : MA_WaypointFollowingType</p> <p>CurveFollowing : MA_CurveControlType</p> <p>Speed : PathSegmentSpeedType</p> <p>Heading : MA_DirectionReferenceType</p> <p>SpeedChoice : MA_HSA_SpeedChoiceType</p> <p>SpeedValue : PathSegmentSpeedValueType</p> <p>CurveFitMethod : MA_CurveFitMethodEnum</p> <p>Reference : MA_HeadingReferenceEnum</p> <p>ControlPoint : MA_CurveFollowingControlPointType</p> <p>RelativeAcceleration : MA_AccelerationChoiceType</p> <p>RelativeVelocity : MA_VelocityChoiceType</p> <p>OrientationRate : MA_OrientationRateChoiceType</p> <p>Direction : MA_DirectionChoiceType</p> <p>Course : MA_DirectionReferenceType</p> |

Table A-1-241: P2P UCI Extensions MA\_FlightCommandMT UCI Extension

| Command     |                                                                                                                                                                                                                                                                                                |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | Indicates an instance of a Flight Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate independent Capability CommandStatus message. |
| Base Type   | MA_FlightCommandType                                                                                                                                                                                                                                                                           |

| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Field Usage |
| <b>Capability</b> | MA_FlightCapabilityCommandType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Required    |
| Description       | Indicates a new invocation of a Flight Capability. Generally, if accepted, the command will result in one or more new Flight Activities being created and reported via the Flight Activity message. The request/response interaction terminates as soon as the command is accepted or rejected; this element is not used to interact with an "active" command. Updates and/or additional interaction with the resulting Activity (after the command is accepted) is accomplished via the sibling Activity element. |             |
| <b>Activity</b>   | MA_FlightActivityCommandType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Required    |
| Description       | Indicates a command to modify an existing Flight Activity (which was previously reported via the Flight Activity message and was marked as "interactive"). The request/response interaction terminates as soon as the modification is accepted or rejected. The modifications are reflected in subsequent Flight Activity messages.                                                                                                                                                                                |             |

Table A-1-242: P2P UCI Extensions MA\_FlightCommandMT - Command UCI Extension Field

| Capability              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description             | Indicates a new invocation of a Flight Capability. Generally, if accepted, the command will result in one or more new Flight Activities being created and reported via the Flight Activity message. The request/response interaction terminates as soon as the command is accepted or rejected; this element is not used to interact with an "active" command. Updates and/or additional interaction with the resulting Activity (after the command is accepted) is accomplished via the sibling Activity element. |             |
| Base Type               | MA_FlightCapabilityCommandType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| Message Structure       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| Field Name              | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Field Usage |
| Loiter                  | MA_LoiterType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Optional    |
| Description             | Indicates the parameters for a loiter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
| MustFly                 | MustFlyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Optional    |
| Description             | Indicates the parameters for a must fly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
| Formation               | MA_FormationType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Repeated    |
| Description             | Indicates the relative positions for the entities in the formation.                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
| FlightControlMode       | MA_FlightControlModesChoiceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Optional    |
| Description             | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
| AltitudeStackedMarshall | MA_AltitudeStackedMarshallType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Optional    |
| Description             | Indicates kinematics for an altitude stacked marshal.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |
| Departure               | MA_DepartureType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Optional    |
| Description             | This element identifies the parameters for a departure/take-off task.                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |
| Recovery                | MA_RecoveryType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Optional    |
| Description             | This element identifies the parameters for a recovery/land task.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |

|                            |                                                                                                                                                                                                                                                                                                                                                                          |          |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>RoutePlanIntercept</b>  | MA_RoutePlanInterceptType                                                                                                                                                                                                                                                                                                                                                | Optional |
| Description                | This element defines the parameters for a Route intercept                                                                                                                                                                                                                                                                                                                |          |
| <b>CapabilityID</b>        | CapabilityID_Type                                                                                                                                                                                                                                                                                                                                                        | Required |
| Description                | Indicates the Capability instance being commanded.                                                                                                                                                                                                                                                                                                                       |          |
| <b>Ranking</b>             | CapabilityCommandRankingType                                                                                                                                                                                                                                                                                                                                             | Required |
| Description                | Indicates the command's rank relative to other commands and rank-related overrides and tailoring.                                                                                                                                                                                                                                                                        |          |
| <b>TemporalConstraints</b> | CapabilityCommandTemporalConstraintsType                                                                                                                                                                                                                                                                                                                                 | Optional |
| Description                | Indicates temporal constraints for the command. If omitted, the command should be scheduled immediately while satisfying other applicable constraints.                                                                                                                                                                                                                   |          |
| <b>OverrideRejection</b>   | boolean                                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                | Indicates that the Command shall be performed even though it was previously rejected.                                                                                                                                                                                                                                                                                    |          |
| <b>Traceability</b>        | TraceabilityType                                                                                                                                                                                                                                                                                                                                                         | Optional |
| Description                | C2 sources work at different levels of detail and abstraction, progressing from what to do to how specifically to do it. Inputs from abstract tasking sources and Tasks can be decomposed into discrete, actionable Tasks and/or Capability commands. This element provides traceability to the external tasking or Task from which this Capability command was derived. |          |
| <b>Classification</b>      | SecurityInformationType                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                | This element provides the classification label that the Subsystem should place on messages and other data resulting from the Command.                                                                                                                                                                                                                                    |          |
| <b>CommandID</b>           | CommandID_Type                                                                                                                                                                                                                                                                                                                                                           | Required |
| Description                | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.                                                                                                                                                                                                                                               |          |
| <b>CommandState</b>        | CommandStateEnum                                                                                                                                                                                                                                                                                                                                                         | Required |
| Description                | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.                                                                                                                                                                                                            |          |

Table A-1-243: P2P UCI Extensions MA\_FlightCommandMT - Capability UCI Extension Field

| HSA_CSA           |                                                                                                   |             |
|-------------------|---------------------------------------------------------------------------------------------------|-------------|
| Description       | This message shall be used to provide the ability to command a new flight vector to the Platform. |             |
| Base Type         | MA_HSA_CSA_Type                                                                                   |             |
| Message Structure |                                                                                                   |             |
| Field Name        | Field Type                                                                                        | Field Usage |
| Altitude          | AltitudeReferenceType                                                                             | Optional    |
| Description       | Altitude hold value to be achieved and reference to altitude measurement source.                  |             |
| Speed             | PathSegmentSpeedType                                                                              | Optional    |
| Description       | Defines speed type (reference frame) for all speed related fields in this message.                |             |
| Direction         | MA_DirectionChoiceType                                                                            | Optional    |



|             |                                                      |
|-------------|------------------------------------------------------|
| Description | Defines heading or course type and reference to hold |
|-------------|------------------------------------------------------|

**Table A-1-244: P2P UCI Extensions MA\_FlightCommandMT - HSA\_CSA UCI Extension Field**

| WaypointFollowing |                                                                                                                      |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | This message shall be used to provide the ability to send a new waypoint path or segment for the Platform to follow. |             |
| Base Type         | MA_WaypointFollowingType                                                                                             |             |
| Message Structure |                                                                                                                      |             |
| Field Name        | Field Type                                                                                                           | Field Usage |
| Route             | MA_RouteType                                                                                                         | Required    |
| Description       | Indicates the Paths and Segments that make up the Route to be flown.                                                 |             |

**Table A-1-245: P2P UCI Extensions MA\_FlightCommandMT - WaypointFollowing UCI Extension Field**

| CurveFollowing     |                                                                                                                                      |             |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates the parameters for curve following control.                                                                                |             |
| Base Type          | MA_CurveControlType                                                                                                                  |             |
| Message Structure  |                                                                                                                                      |             |
| Field Name         | Field Type                                                                                                                           | Field Usage |
| CurveSegments      | MA_NURBS_PointType                                                                                                                   | Repeated    |
| Description        | Specifies the curve segments as a set of NURBS points.                                                                               |             |
| AppendCurve        | EmptyType                                                                                                                            | Optional    |
| Description        | Specifies this curve will be appended to a previous curve command. This is only valid if the vehicle is currently executing a curve. |             |
| EndOfCurveBehavior | MA_EndOfCurveBehaviorEnum                                                                                                            | Optional    |
| Description        | Specifies the vehicle behavior at the completion of the curve.                                                                       |             |

**Table A-1-246: P2P UCI Extensions MA\_FlightCommandMT - CurveFollowing UCI Extension Field**

| Speed             |                                                                                                                                                                              |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Defines speed type (reference frame) for all speed related fields in this message.                                                                                           |             |
| Base Type         | PathSegmentSpeedType                                                                                                                                                         |             |
| Message Structure |                                                                                                                                                                              |             |
| Field Name        | Field Type                                                                                                                                                                   | Field Usage |
| SpeedChoice       | PathSegmentSpeedChoiceType                                                                                                                                                   | Optional    |
| Description       | Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number. |             |
| SpeedOptimization | SpeedOptimizationEnum                                                                                                                                                        | Optional    |
| Description       | Indicates that the System will vary its speed autonomously according to the specified optimization algorithm.                                                                |             |

Table A-1-247: P2P UCI Extensions MA\_FlightCommandMT - Speed UCI Extension Field

| Heading           |                                                |             |
|-------------------|------------------------------------------------|-------------|
| Description       | Defines heading type and reference to hold.    |             |
| Base Type         | MA_DirectionReferenceType                      |             |
| Message Structure |                                                |             |
| Field Name        | Field Type                                     | Field Usage |
| Value             | AngleType                                      | Required    |
| Description       | Indicates the heading or course value to hold. |             |
| Reference         | MA_HeadingReferenceEnum                        | Required    |
| Description       | Reference to type of heading or course given.  |             |

Table A-1-248: P2P UCI Extensions MA\_FlightCommandMT - Heading UCI Extension Field

| SpeedChoice       |                                                                                                                                                                              |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number. |             |
| Base Type         | MA_HSA_SpeedChoiceType                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                              |             |
| Field Name        | Field Type                                                                                                                                                                   | Field Usage |
| SpeedValue        | PathSegmentSpeedValueType                                                                                                                                                    | Required    |
| Description       | Indicates speed in meters per second (m/s).                                                                                                                                  |             |
| MachValue         | MachType                                                                                                                                                                     | Required    |
| Description       | Indicates the speed reference for the associated speed value.                                                                                                                |             |

Table A-1-249: P2P UCI Extensions MA\_FlightCommandMT - SpeedChoice UCI Extension Field

| SpeedValue        |                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates speed in meters per second (m/s).                                               |             |
| Base Type         | PathSegmentSpeedValueType                                                                 |             |
| Message Structure |                                                                                           |             |
| Field Name        | Field Type                                                                                | Field Usage |
| Value             | SpeedType                                                                                 | Required    |
| Description       | Indicates the speed along the PathSegment at the EndPoint.                                |             |
| Reference         | SpeedReferenceEnum                                                                        | Required    |
| Description       | Indicates whether the given vehicle speed is reference to ground speed or true air speed. |             |

Table A-1-250: P2P UCI Extensions MA\_FlightCommandMT - SpeedValue UCI Extension Field

| CurveFitMethod     |                                                                            |
|--------------------|----------------------------------------------------------------------------|
| <b>Description</b> | Method through which a curve should be fitted to the discretized points in |

|                            |                              |
|----------------------------|------------------------------|
|                            | ControlPoint.                |
| <b>Base Type</b>           | MA_CurveFitMethodEnum        |
| <b>Message Structure</b>   |                              |
| <b>Enumeration Literal</b> | <b>Description</b>           |
| B_SPLINE                   | B-Spline (basis spline) fit. |
| BEZIER                     | Bezier curve fit.            |
| CATMULL_ROM                | Catmull-Rom spline fit.      |
| NATURAL_CUBIC              | Natural Cubic curve fit.     |

Table A-1-251: P2P UCI Extensions MA\_FlightCommandMT - CurveFitMethod UCI Extension Field

|                            |                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------|
| <b>Reference</b>           |                                                                                                   |
| <b>Description</b>         | Reference to type of heading or course given.                                                     |
| <b>Base Type</b>           | MA_HeadingReferenceEnum                                                                           |
| <b>Message Structure</b>   |                                                                                                   |
| <b>Enumeration Literal</b> | <b>Description</b>                                                                                |
| TRUE_NORTH                 | Heading value according to Geographic North.                                                      |
| MAGNETIC_NORTH             | Heading value according to location where the planet's magnetic field points vertically downward. |

Table A-1-252: P2P UCI Extensions MA\_FlightCommandMT - Reference UCI Extension Field

| ControlPoint        |                                                                                                                                                                                                                                                                                          |             |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | Control Points for the desired discretized curve or spline path.                                                                                                                                                                                                                         |             |
| Base Type           | MA_CurveFollowingControlPointType                                                                                                                                                                                                                                                        |             |
| Message Structure   |                                                                                                                                                                                                                                                                                          |             |
| Field Name          | Field Type                                                                                                                                                                                                                                                                               | Field Usage |
| Position            | PointChoice4D_Type                                                                                                                                                                                                                                                                       | Required    |
| Description         | Physical location of the referenced item in geospatial coordinates. The use of the timestamp in this context represents the time the positional data was last measured. Position data may be specified as either a geospatial point or relative to a separately defined reference frame. |             |
| PositionUncertainty | UncertaintyType                                                                                                                                                                                                                                                                          | Optional    |
| Description         | Uncertainty of the physical location of the referenced item.                                                                                                                                                                                                                             |             |
| DomainVelocity      | Velocity3D_Type                                                                                                                                                                                                                                                                          | Optional    |
| Description         | Vehicle's velocity in the current operating domain of the system. Air vehicle components are in True Air Speed.                                                                                                                                                                          |             |
| GroundVelocity      | Velocity2D_Type                                                                                                                                                                                                                                                                          | Optional    |
| Description         | Representation of the vehicle's ground velocity as directional speed components.                                                                                                                                                                                                         |             |
| RelativeVelocity    | MA_VelocityChoiceType                                                                                                                                                                                                                                                                    | Optional    |
| Description         | Represents the vehicle's velocity, in units of m/s, in the separately defined reference frame referenced in the sibling Position element. This element should only be used                                                                                                               |             |

|                             |                                                                                                                                                                                                                  |          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                             | when the RelativePoint choice is selected and the separately defined reference frame is itself moving.                                                                                                           |          |
| <b>DomainAcceleration</b>   | Acceleration3D_Type                                                                                                                                                                                              | Optional |
| Description                 | The vehicle's acceleration in the current operating domain of the system.                                                                                                                                        |          |
| <b>Orientation</b>          | OrientationType                                                                                                                                                                                                  | Optional |
| Description                 | Euler Angle Sequence describing the orientation of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle.                |          |
| <b>OrientationRate</b>      | MA_OrientationRateChoiceType                                                                                                                                                                                     | Optional |
| Description                 | Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle. |          |
| <b>RelativeAcceleration</b> | MA_AccelerationChoiceType                                                                                                                                                                                        | Optional |
| Description                 |                                                                                                                                                                                                                  |          |

Table A-1-253: P2P UCI Extensions MA\_FlightCommandMT - ControlPoint UCI Extension Field

| RelativeAcceleration      |                                               |             |
|---------------------------|-----------------------------------------------|-------------|
| Description               |                                               |             |
| Base Type                 | MA_AccelerationChoiceType                     |             |
| Message Structure         |                                               |             |
| Field Name                | Field Type                                    | Field Usage |
| BodyReferenceAcceleration | MA_BodyReferenceAccelerationType              | Required    |
| Description               | Body reference accelerations.                 |             |
| NED_Acceleration          | Acceleration3D_Type                           | Required    |
| Description               | Inertial, NED coordinate frame accelerations. |             |

Table A-1-254: P2P UCI Extensions MA\_FlightCommandMT - RelativeAcceleration UCI Extension Field

| RelativeVelocity      |                                                                                                                                                                                                                                                                                   |             |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description           | Represents the vehicle's velocity, in units of m/s, in the separately defined reference frame referenced in the sibling Position element. This element should only be used when the RelativePoint choice is selected and the seperately defined reference frame is itself moving. |             |
| Base Type             | MA_VelocityChoiceType                                                                                                                                                                                                                                                             |             |
| Message Structure     |                                                                                                                                                                                                                                                                                   |             |
| Field Name            | Field Type                                                                                                                                                                                                                                                                        | Field Usage |
| BodyReferenceVelocity | MA_BodyReferenceVelocityType                                                                                                                                                                                                                                                      | Required    |
| Description           | Velocity in body reference frame.                                                                                                                                                                                                                                                 |             |
| NED_Velocity          | Velocity3D_Type                                                                                                                                                                                                                                                                   | Required    |
| Description           | Velocity in North East Down reference frame.                                                                                                                                                                                                                                      |             |

Table A-1-255: P2P UCI Extensions MA\_FlightCommandMT - RelativeVelocity UCI Extension Field

| OrientationRate              |                                                                                                                                                                                                                  |             |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle. |             |
| Base Type                    | MA_OrientationRateChoiceType                                                                                                                                                                                     |             |
| Message Structure            |                                                                                                                                                                                                                  |             |
| Field Name                   | Field Type                                                                                                                                                                                                       | Field Usage |
| BodyReferenceOrientationRate | MA_BodyReferenceOrientationRateType                                                                                                                                                                              | Required    |
| Description                  | Orientation rate in body reference.                                                                                                                                                                              |             |
| NED_OrientationRate          | OrientationRateType                                                                                                                                                                                              | Required    |
| Description                  | Orientation rate in inertial reference.                                                                                                                                                                          |             |

Table A-1-256: P2P UCI Extensions MA\_FlightCommandMT - OrientationRate UCI Extension Field

| Direction         |                                                      |             |
|-------------------|------------------------------------------------------|-------------|
| Description       | Defines heading or course type and reference to hold |             |
| Base Type         | MA_DirectionChoiceType                               |             |
| Message Structure |                                                      |             |
| Field Name        | Field Type                                           | Field Usage |
| Heading           | MA_DirectionReferenceType                            | Optional    |
| Description       | Defines heading type and reference to hold.          |             |
| Course            | MA_DirectionReferenceType                            | Optional    |
| Description       | Defines course type and reference to hold.           |             |

Table A-1-257: P2P UCI Extensions MA\_FlightCommandMT - Direction UCI Extension Field

| Course            |                                                |             |
|-------------------|------------------------------------------------|-------------|
| Description       | Defines course type and reference to hold.     |             |
| Base Type         | MA_DirectionReferenceType                      |             |
| Message Structure |                                                |             |
| Field Name        | Field Type                                     | Field Usage |
| Value             | AngleType                                      | Required    |
| Description       | Indicates the heading or course value to hold. |             |
| Reference         | MA_HeadingReferenceEnum                        | Required    |
| Description       | Reference to type of heading or course given.  |             |

Table A-1-258: P2P UCI Extensions MA\_FlightCommandMT - Course UCI Extension Field

### 1.3.2.41 MA\_FlightCommandStatusMT UCI Extension

| MA_FlightCommandStatusMT |                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------|
| Description              | See annotations in child elements and messages/elements that use this type for details. |

|                    |                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
|                    | See the XSD File for more info                                                                                                 |
| <b>Ext. Fields</b> | MessageData : MA_FlightCommandStatusMDT<br>CannotComplyDetails : MA_CannotComplyDetailsType<br>Severity : MA_FaultSeverityEnum |

Table A-1-259: P2P UCI Extensions MA\_FlightCommandStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_FlightCommandStatusMDT                                                                                                                                                                                                                                                                                                                                   |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| CannotComplyDetails          | MA_CannotComplyDetailsType                                                                                                                                                                                                                                                                                                                                  | Optional    |
| Description                  | Provides detail for why FA cannot comply with the flight command.                                                                                                                                                                                                                                                                                           |             |
| Activity                     | ResultingActivityType                                                                                                                                                                                                                                                                                                                                       | Repeated    |
| Description                  | Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.                                                                                                                                                                                                                    |             |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

Table A-1-260: P2P UCI Extensions MA\_FlightCommandStatusMT - MessageData UCI Extension Field

| CannotComplyDetails |                                                                                                                                                                                                                                               |             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | Provides detail for why FA cannot comply with the flight command.                                                                                                                                                                             |             |
| Base Type           | MA_CannotComplyDetailsType                                                                                                                                                                                                                    |             |
| Message Structure   |                                                                                                                                                                                                                                               |             |
| Field Name          | Field Type                                                                                                                                                                                                                                    | Field Usage |
| ValidationResult    | MA_ValidationResultEnum                                                                                                                                                                                                                       | Repeated    |
| Description         | In case of flight command rejection, provides the reason for Flight Autonomy rejection. Further details can be provided by Flight Autonomy through the ValidationResultReason. List size for this element is based on "Select All That Apply" |             |

|                               |                                                                                                 |          |
|-------------------------------|-------------------------------------------------------------------------------------------------|----------|
|                               | condition.                                                                                      |          |
| <b>ValidationResultReason</b> | MA_ValidationResultReasonType                                                                   | Repeated |
| Description                   | In case of flight command rejection, provides additional context for Flight Autonomy rejection. |          |

**Table A-1-261: P2P UCI Extensions MA\_FlightCommandStatusMT - CannotComplyDetails UCI Extension Field**

| Severity            |                                                                                                                                                                                                                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>  | The severity of the fault. This enumeration is modified for Mission Autonomy adaptation.                                                                                                                                                                                                                                                     |
| <b>Base Type</b>    | MA_FaultSeverityEnum                                                                                                                                                                                                                                                                                                                         |
| Message Structure   |                                                                                                                                                                                                                                                                                                                                              |
| Enumeration Literal | Description                                                                                                                                                                                                                                                                                                                                  |
| <b>NOMINAL</b>      | A fault that carries the nominal level indicates that fault does not impact the current mission but may indicate something that should be checked in maintenance. Faults should be tagged as nominal if they are targeting maintenance. An example might be air filter hasn't been changed in 5000 hours or queue length reached 60% of max. |
| <b>ADVISORY</b>     | A fault that is important but is not a Caution or Warning; this can be a configuration change, a condition that is contributing to a degraded aircraft or subsystem operation, or an anomaly that is being automatically controlled or corrected by a higher level system.                                                                   |
| <b>CAUTION</b>      | A fault that is indicative of a condition that, without a timely and correct response, could result in equipment damage or failure of mission.                                                                                                                                                                                               |
| <b>WARNING</b>      | A fault that is indicative of a condition requiring immediate response or attention to prevent loss of aircraft, or failure of mission.                                                                                                                                                                                                      |
| <b>FAILED</b>       | A fault that carries a failed level indicates the failure of a component which may preclude the use of a capability. Faults should be tagged with failed if the component state will be marked faulted. Failed components may or may not prevent the completion of the mission.                                                              |

**Table A-1-262: P2P UCI Extensions MA\_FlightCommandStatusMT - Severity UCI Extension Field**

### 1.3.2.42 MA\_IgnoreConstraintCommandMT UCI Extension

| MA_IgnoreConstraintCommandMT |                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>           | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>           | MessageData : MA_IgnoreConstraintCommandMDT                                                                                   |

**Table A-1-263: P2P UCI Extensions MA\_IgnoreConstraintCommandMT UCI Extension**

| MessageData        |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |

|                           |                                                                                                                                                               |             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Base Type                 | MA_IgnoreConstraintCommandMDT                                                                                                                                 |             |
| Message Structure         |                                                                                                                                                               |             |
| Field Name                | Field Type                                                                                                                                                    | Field Usage |
| ReferenceType             | RequirementsReferenceType                                                                                                                                     | Required    |
| Description               | ID of the task(s) that the command applies to.                                                                                                                |             |
| ListOfConstraintsToIgnore | MA_RegionChoiceType                                                                                                                                           | Repeated    |
| Description               | Indicates the constraint(s) being overridden.                                                                                                                 |             |
| IgnoreAllConstraints      | boolean                                                                                                                                                       | Optional    |
| Description               | If True, ignores all constraints.                                                                                                                             |             |
| AppliesToIDs              | MA_PackageSystemType                                                                                                                                          | Repeated    |
| Description               | The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in this field as well.       |             |
| CommandID                 | CommandID_Type                                                                                                                                                | Required    |
| Description               | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.                                    |             |
| CommandState              | CommandStateEnum                                                                                                                                              | Required    |
| Description               | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command. |             |

Table A-1-264: P2P UCI Extensions MA\_IgnoreConstraintCommandMT - MessageData UCI Extension Field

### 1.3.2.43 MA\_IgnoreConstraintCommandStatusMT UCI Extension

|                                           |                                                                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_IgnoreConstraintCommandStatusMT</b> |                                                                                                                               |
| <b>Description</b>                        | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                        | MessageData : MA_IgnoreConstraintCommandStatusMDT                                                                             |

Table A-1-265: P2P UCI Extensions MA\_IgnoreConstraintCommandStatusMT UCI Extension

| MessageData       |                                                                                                                                              |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                      |             |
| Base Type         | MA_IgnoreConstraintCommandStatusMDT                                                                                                          |             |
| Message Structure |                                                                                                                                              |             |
| Field Name        | Field Type                                                                                                                                   | Field Usage |
| AppliesToIDs      | MA_PackageSystemType                                                                                                                         | Repeated    |
| Description       | The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed. |             |
| CommandID         | CommandID_Type                                                                                                                               | Required    |



|                                     |                                                                                                                                                                                                                                                                                                                                                             |          |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                         | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |          |
| <b>CommandProcessingState</b>       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required |
| Description                         | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |          |
| <b>CommandProcessingStateReason</b> | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional |
| Description                         | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |          |

**Table A-1-266: P2P UCI Extensions MA\_IgnoreConstraintCommandStatusMT - MessageData UCI Extension Field**

### 1.3.2.44 MA\_LeaderUpdateRequestMT UCI Extension

| MA_LeaderUpdateRequestMT |                                                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>       | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>       | MessageData : MA_LeaderUpdateRequestMDT                                                                                                     |

**Table A-1-267: P2P UCI Extensions MA\_LeaderUpdateRequestMT UCI Extension**

| MessageData                   |                                                                                                                                                       |             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                   | The message data relating to LeaderUpdateRequest message specifically.                                                                                |             |
| Base Type                     | MA_LeaderUpdateRequestMDT                                                                                                                             |             |
| Message Structure             |                                                                                                                                                       |             |
| Field Name                    | Field Type                                                                                                                                            | Field Usage |
| Status                        | MA_NominationEnum                                                                                                                                     | Required    |
| Description                   | The related status of this message, whether it is a nomination or an assignment.                                                                      |             |
| Request                       | MA_LeaderUpdateRequestSetType                                                                                                                         | Required    |
| Description                   | Indicates the details of the request.                                                                                                                 |             |
| StaticLeadershipFitnessScore  | MA_PartnerLeadershipFitnessScoreType                                                                                                                  | Repeated    |
| Description                   | Indicates the static leadership fitness score for the partners in the package, which can be shared to trigger a change in leader from another System. |             |
| DynamicLeadershipFitnessScore | MA_PartnerLeadershipFitnessScoreType                                                                                                                  | Optional    |
| Description                   | Indicates the dynamic leadership fitness score and determination type for a partner in the package.                                                   |             |
| RequestID                     | RequestID_Type                                                                                                                                        | Required    |
| Description                   | Indicates the unique ID of the request message.                                                                                                       |             |

| RequestState | RequestStateEnum                                                                                                                          | Required |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description  | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request. |          |

Table A-1-268: P2P UCI Extensions MA\_LeaderUpdateRequestMT - MessageData UCI Extension Field

### 1.3.2.45 MA\_LeaderUpdateRequestStatusMT UCI Extension

| MA_LeaderUpdateRequestStatusMT |                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Description                    | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| Ext. Fields                    | MessageData : MA_LeaderUpdateRequestStatusMDT                                                                                               |

Table A-1-269: P2P UCI Extensions MA\_LeaderUpdateRequestStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                             |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | The message data relating to LeaderUpdateRequestStatus specifically.                                                                                                                                                                                                                        |             |
| Base Type                    | MA_LeaderUpdateRequestStatusMDT                                                                                                                                                                                                                                                             |             |
| Message Structure            |                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                  | Field Usage |
| ApplicableIDs                | MA_LeaderUpdateRequestSetType                                                                                                                                                                                                                                                               | Required    |
| Description                  | The unique IDs of the systems, packages or tasks applicable to the original request message. The status of the message can be found at the RequestProcessingState enum.                                                                                                                     |             |
| RequestID                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |             |
| RequestProcessingState       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |             |
| RequestProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |

Table A-1-270: P2P UCI Extensions MA\_LeaderUpdateRequestStatusMT - MessageData UCI Extension Field

### 1.3.2.46 MA\_AcceptableLevelOfRiskCommandMT UCI Extension

| MA_AcceptableLevelOfRiskCommandMT |                                                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description                       | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |

|                    |                                                  |
|--------------------|--------------------------------------------------|
| <b>Ext. Fields</b> | MessageData : MA_AcceptableLevelOfRiskCommandMDT |
|--------------------|--------------------------------------------------|

Table A-1-271: P2P UCI Extensions MA\_AcceptableLevelOfRiskCommandMT UCI Extension

| MessageData              |                                                                                                                                                                                                                                                                                                                           |             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                   |             |
| Base Type                | MA_AcceptableLevelOfRiskCommandMDT                                                                                                                                                                                                                                                                                        |             |
| Message Structure        |                                                                                                                                                                                                                                                                                                                           |             |
| Field Name               | Field Type                                                                                                                                                                                                                                                                                                                | Field Usage |
| AcceptableRiskSetting    | MA_AcceptableLevelOfRiskEnum                                                                                                                                                                                                                                                                                              | Required    |
| Description              | This is the acceptable level of risk permitted at the package or vehicle level, which applies to all tasks and actions for the vehicle/package while active.                                                                                                                                                              |             |
| AppliesToIDs             | MA_PackageSystemType                                                                                                                                                                                                                                                                                                      | Repeated    |
| Description              | Identifies the Package ID or ACP System ID for which the acceptable risk parameter applies. If Package ID is used, this includes all package member System IDs, so a member ACP's System ID does not need to be separately included. This value overrides the AppliesToSystemID in the SurvivabilityRiskLevelMDT message. |             |
| SurvivabilityRiskLevelID | SurvivabilityRiskLevelID_Type                                                                                                                                                                                                                                                                                             | Required    |
| Description              | The unique identifier for the Survivability Risk Level that should be active as the default setting.                                                                                                                                                                                                                      |             |
| CommandID                | CommandID_Type                                                                                                                                                                                                                                                                                                            | Required    |
| Description              | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.                                                                                                                                                                                                |             |
| CommandState             | CommandStateEnum                                                                                                                                                                                                                                                                                                          | Required    |
| Description              | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.                                                                                                                                                             |             |

Table A-1-272: P2P UCI Extensions MA\_AcceptableLevelOfRiskCommandMT - MessageData UCI Extension Field

### 1.3.2.47 MA\_AcceptableLevelOfRiskCommandStatusMT UCI Extension

| MA_AcceptableLevelOfRiskCommandStatusMT |                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                      | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                      | MessageData : MA_AcceptableLevelOfRiskCommandStatusMDT                                                                        |

Table A-1-273: P2P UCI Extensions MA\_AcceptableLevelOfRiskCommandStatusMT UCI Extension

| MessageData        |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |
| <b>Base Type</b>   | MA_AcceptableLevelOfRiskCommandStatusMDT                                                |

| Message Structure                   |                                                                                                                                                                                                                                                                                                                                                             |             |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Field Name                          | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| <b>AppliesToIDs</b>                 | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description                         | The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed.                                                                                                                                                                                                                |             |
| <b>CommandID</b>                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                         | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| <b>CommandProcessingState</b>       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                         | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| <b>CommandProcessingStateReason</b> | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                         | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

**Table A-1-274: P2P UCI Extensions MA\_AcceptableLevelOfRiskCommandStatusMT - MessageData UCI Extension Field**

### 1.3.2.48 MA\_MissionPlanActivationCommandMT UCI Extension

| MA_MissionPlanActivationCommandMT |                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info                                                                                                                                                                                                                                     |
| <b>Ext. Fields</b>                | OpPlanID : MA_OpPlanID_Type<br>ApplicableToIDs : MA_PackageSystemType<br>ByExecutionPlanSet : MA_ExecutionPlanSetActivationType<br>MessageData : MA_MissionPlanActivationCommandMDT<br>ActivationDetails : MA_MissionPlanActivationDetailsType<br>Applicability : MA_PlanApplicabilityType<br>OpPlan : MA_OpPlanActivationType<br>Command : MA_MissionPlanActivationCommandType |

**Table A-1-275: P2P UCI Extensions MA\_MissionPlanActivationCommandMT UCI Extension**

| OpPlanID           |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| <b>Description</b> | Indicates the unique ID of the TaskPlan to perform the sibling activation command on. |
| <b>Base Type</b>   | MA_OpPlanID_Type                                                                      |

| Message Structure       |                                                                                                                                                                                                                                                                |             |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Field Name              | Field Type                                                                                                                                                                                                                                                     | Field Usage |
| <b>Version</b>          | unsignedInt                                                                                                                                                                                                                                                    | Optional    |
| Description             | Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval. |             |
| <b>UUID</b>             | UniversallyUniqueIdentifierType                                                                                                                                                                                                                                | Required    |
| Description             | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.                                                                                               |             |
| <b>DescriptiveLabel</b> | VisibleString256Type                                                                                                                                                                                                                                           | Optional    |
| Description             | Human readable text label with content meaningful to human operators.                                                                                                                                                                                          |             |

Table A-1-276: P2P UCI Extensions MA\_MissionPlanActivationCommandMT - OpPlanID UCI Extension Field

| ApplicableToIDs   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the unique ID of the System (e.g. aircraft, ground-based sensor, ship, satellite, site, etc.) or Package whose Capabilities, performance, position, etc. the plan is based on. If the plan is a fully detailed one that can be activated and executed, this element indicates the System intended/responsible for executing the plan and producing the intended effects. See annotations in sibling element PlannedForID which provide additional context. |             |
| Base Type         | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Field Usage |
| ALL               | EmptyType                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Required    |
| Description       | Indicates applies to ALL entities.                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| NONE              | EmptyType                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Required    |
| Description       | Indicates applies to NO entities.                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| PackageID         | PackageID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Required    |
| Description       | The ID of a Package.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| SystemID          | SystemID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required    |
| Description       | The ID of an ACP or C2 Operator.                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |

Table A-1-277: P2P UCI Extensions MA\_MissionPlanActivationCommandMT - ApplicableToIDs UCI Extension Field

| ByExecutionPlanSet |                                                                                                                                                   |             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates simultaneous activation of all sub-*Plans (ActivityPlan, RoutePlan for example) of an *ExecutionPlanSet into the same activation state. |             |
| Base Type          | MA_ExecutionPlanSetActivationType                                                                                                                 |             |
| Message Structure  |                                                                                                                                                   |             |
| Field Name         | Field Type                                                                                                                                        | Field Usage |

|                           |                                                                           |          |
|---------------------------|---------------------------------------------------------------------------|----------|
| <b>ExecutionPlanSetID</b> | ExecutionPlanSetID_Type                                                   | Required |
| Description               | Indicates the unique ID of the execution plan set.                        |          |
| <b>CommandType</b>        | PlanActivationCommandEnum                                                 | Required |
| Description               | Indicates the activation to perform on the associated execution plan set. |          |

**Table A-1-278: P2P UCI Extensions MA\_MissionPlanActivationCommandMT - ByExecutionPlanSet UCI Extension Field**

| MessageData                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                 | Indicates the data being conveyed by the message.                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| Base Type                   | MA_MissionPlanActivationCommandMDT                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Message Structure           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| Field Name                  | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Field Usage |
| CommandID                   | MissionPlanActivationCommandID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                 | The ID of the Mission Plan Activation Command. Used to reference the command when providing status.                                                                                                                                                                                                                                                                                                                                                                  |             |
| ApprovalManagementCommandID | CommandID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Repeated    |
| Description                 | Indicates the command(s) that initiated the Activation Approvals for the *Plans identified in the sibling Command element. The approval status can be found by locating the corresponding MissionPlanActivationApprovalStatus with the ID identified here.<br><br>The ActivationCommand identified in the sibling Command\ActivationDetails should be verified against the ActivationType that is identified in the associated MissionPlanActivationApprovalCommand. |             |
| CommandState                | CommandStateEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Required    |
| Description                 | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.                                                                                                                                                                                                                                                                                                        |             |
| Command                     | MA_MissionPlanActivationCommandType                                                                                                                                                                                                                                                                                                                                                                                                                                  | Repeated    |
| Description                 | Indicates command or commands to change the activation state of a MissionPlan and/or it subplans or subordinate plans. If more than one instance of this element is given, each should correspond to a different MissionPlanID. For example, if the intent is to transition from one MissionPlan to another, the new MissionPlan can be activated in one instance and the old MissionPlan can be deactivated in another instance.                                    |             |
| Applicability               | MA_PlanApplicabilityType                                                                                                                                                                                                                                                                                                                                                                                                                                             | Optional    |
| Description                 | Indicates the System or Systems who are being commanded to activate a specified plan.                                                                                                                                                                                                                                                                                                                                                                                |             |

**Table A-1-279: P2P UCI Extensions MA\_MissionPlanActivationCommandMT - MessageData UCI Extension Field**

| ActivationDetails  |                                             |
|--------------------|---------------------------------------------|
| <b>Description</b> | Indicates the commanded activation details. |

| Base Type          | MA_MissionPlanActivationDetailsType                                                                                                               |             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Message Structure  |                                                                                                                                                   |             |
| Field Name         | Field Type                                                                                                                                        | Field Usage |
| ByMissionPlan      | MissionPlanActivationType                                                                                                                         | Required    |
| Description        | Indicates simultaneous activation of all sub-*Plans (RoutePlan, RouteActivityPlan for example) of a MissionPlan into the same activation state.   |             |
| BySubPlan          | MA_MissionPlanSubplanActivationType                                                                                                               | Required    |
| Description        | Indicates activation by sub-*Plan (RoutePlan or OrbitPlan for example) of the MissionPlan, with potentially different states for each.            |             |
| ByExecutionPlanSet | MA_ExecutionPlanSetActivationType                                                                                                                 | Required    |
| Description        | Indicates simultaneous activation of all sub-*Plans (ActivityPlan, RoutePlan for example) of an *ExecutionPlanSet into the same activation state. |             |

**Table A-1-280: P2P UCI Extensions MA\_MissionPlanActivationCommandMT - ActivationDetails UCI Extension Field**

| Applicability     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the System or Systems who are being commanded to activate a specified plan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |
| Base Type         | MA_PlanApplicabilityType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Field Usage |
| PlannedForID      | SystemID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Optional    |
| Description       | <p>Indicates the System which is the intended user of the plan. This type can have multiple meanings, but primarily enables multiple systems, possibly at different levels of warfare or C2 hierarchy levels, to maintain independent plans for the same System identified in the sibling ApplicableToID element.</p> <p>For instance, during planning at the operational level of warfare, a non-tactical planning service may create and simulate execution of plans internally. Such plans would identify the operational level System as the PlannedFor System; they would never be activated or executed by a real world tactical System.</p> <p>Alternatively, the operational level System could have a tactical planning service that is functionally capable and responsible for creating fully detailed plans that can be activated and executed by real world tactical Systems. Such a tactical planning service could be used even if the operational level System does not have control authority over the tactical System at the time the plan is created. Such plans would identify the tactical level System as the PlannedFor System.</p> <p>As a final example, a plan which is intended to be used by the same System which produces its intended effects would have a common SystemID between the sibling PlannedForID and ApplicableToID elements.</p> <p>This element does not replace existing approval policies, authorization or control status</p> |             |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                        | and therefore does not automatically imply that the PlannedFor System has any kind of authority over the receiving system.                                                                                                                                                                                                                                                                                                                                           |          |
| <b>ApplicableToIDs</b> | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required |
| Description            | Indicates the unique ID of the System (e.g. aircraft, ground-based sensor, ship, satellite, site, etc.) or Package whose Capabilities, performance, position, etc. the plan is based on. If the plan is a fully detailed one that can be activated and executed, this element indicates the System intended/responsible for executing the plan and producing the intended effects. See annotations in sibling element PlannedForID which provide additional context. |          |

**Table A-1-281: P2P UCI Extensions MA\_MissionPlanActivationCommandMT - Applicability UCI Extension Field**

| OpPlan            |                                                                                       |             |
|-------------------|---------------------------------------------------------------------------------------|-------------|
| Description       | Indicates commanded details for activation of an OpPlan.                              |             |
| Base Type         | MA_OpPlanActivationType                                                               |             |
| Message Structure |                                                                                       |             |
| Field Name        | Field Type                                                                            | Field Usage |
| OpPlanID          | MA_OpPlanID_Type                                                                      | Required    |
| Description       | Indicates the unique ID of the TaskPlan to perform the sibling activation command on. |             |
| CommandType       | PlanActivationCommandEnum                                                             | Required    |
| Description       | Indicates the activation to perform on the associated TaskPlan.                       |             |

**Table A-1-282: P2P UCI Extensions MA\_MissionPlanActivationCommandMT - OpPlan UCI Extension Field**

| Command           |                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates command or commands to change the activation state of a MissionPlan and/or it subplans or subordinate plans. If more than one instance of this element is given, each should correspond to a different MissionPlanID. For example, if the intent is to transition from one MissionPlan to another, the new MissionPlan can be activated in one instance and the old MissionPlan can be deactivated in another instance. |             |
| Base Type         | MA_MissionPlanActivationCommandType                                                                                                                                                                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                        | Field Usage |
| MissionPlanID     | MissionPlanID_Type                                                                                                                                                                                                                                                                                                                                                                                                                | Required    |
| Description       | Indicates the unique ID of the MissionPlan whose commanded activation state or states are summarized in the sibling element.                                                                                                                                                                                                                                                                                                      |             |
| ActivationDetails | MA_MissionPlanActivationDetailsType                                                                                                                                                                                                                                                                                                                                                                                               | Required    |
| Description       | Indicates the commanded activation details.                                                                                                                                                                                                                                                                                                                                                                                       |             |

**Table A-1-283: P2P UCI Extensions MA\_MissionPlanActivationCommandMT - Command UCI Extension Field**



### 1.3.2.49 MA\_OpZoneMT UCI Extension

| MA_OpZoneMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | See the annotation in the associated message for an overall description of the message and this type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|             | See the XSD File for more info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Ext. Fields | <p>           MessageData : MA_OpZoneMDT<br/>           Category : MA_OpZoneCategoryEnum<br/>           CategoryUniqueData : MA_OpZoneCategoryType<br/>           SpeedLimits : MA_OpZoneSpeedLimitsType<br/>           VehicleConfiguration : MA_VehicleCommandDataType<br/>           Zone : MA_OpZoneChoiceType<br/>           GeometryReference : MA_ReferentialOpZoneType<br/>           ReferenceObjectsFilter : MA_GeoLocatedObjectFilterType<br/>           OpPointFilter : MA_OpPointFilterType<br/>           OpLineFilter : MA_OpLineFilterType<br/>           OpLineCategory : MA_OpLineCategoryEnum<br/>           OpZoneFilter : MA_OpZoneFilterType<br/>           OpZoneCategory : MA_OpZoneCategoryEnum<br/>           OpVolumeFilter : MA_OpVolumeFilterType<br/>           OpVolumeCategory : MA_OpZoneCategoryEnum<br/>           DMPI_Filter : MA_DMPI_FilterType<br/>           SignalReportFilter : MA_SignalReportFilterType<br/>           OpVolumeFilter : MA_OpVolumeFilterType<br/>           OpLineFilter : MA_OpLineFilterType<br/>           OpLineCategory : MA_OpLineCategoryEnum<br/>           SpeedLimits : MA_OpZoneSpeedLimitsType<br/>           SignalReportFilter : MA_SignalReportFilterType<br/>           OpPointFilter : MA_OpPointFilterType<br/>           VehicleConfiguration : MA_VehicleCommandDataType<br/>           DMPI_Filter : MA_DMPI_FilterType<br/>           Zone : MA_OpZoneChoiceType<br/>           Category : MA_OpZoneCategoryEnum<br/>           CategoryUniqueData : MA_OpZoneCategoryType<br/>           ReferenceObjectsFilter : MA_GeoLocatedObjectFilterType<br/>           OpVolumeCategory : MA_OpZoneCategoryEnum<br/>           GeometryReference : MA_ReferentialOpZoneType         </p> |

OpZoneCategory : MA\_OpZoneCategoryEnum  
OpZoneFilter : MA\_OpZoneFilterType

Table A-1-284: P2P UCI Extensions MA\_OpZoneMT UCI Extension

| MessageData          |                                                                                                                                                                                                                                                     |             |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                             |             |
| Base Type            | MA_OpZoneMDT                                                                                                                                                                                                                                        |             |
| Message Structure    |                                                                                                                                                                                                                                                     |             |
| Field Name           | Field Type                                                                                                                                                                                                                                          | Field Usage |
| OpZoneID             | OpZoneID_Type                                                                                                                                                                                                                                       | Required    |
| Description          | This element represents the unique identifier of the zone associated with this message.                                                                                                                                                             |             |
| Category             | MA_OpZoneCategoryEnum                                                                                                                                                                                                                               | Required    |
| Description          | This element indicates the type of zone.                                                                                                                                                                                                            |             |
| CategoryUniqueData   | MA_OpZoneCategoryType                                                                                                                                                                                                                               | Optional    |
| Description          | Contains parameters unique to certain OpZoneCategoryEnum values.                                                                                                                                                                                    |             |
| Zone                 | MA_OpZoneChoiceType                                                                                                                                                                                                                                 | Required    |
| Description          | This element represents the geospatial and temporal characteristics of the OpZone.                                                                                                                                                                  |             |
| SituationalAwareness | boolean                                                                                                                                                                                                                                             | Required    |
| Description          | Zone used only for situational awareness. Not used for planning purposes.                                                                                                                                                                           |             |
| OpDescription        | OpDescriptionType                                                                                                                                                                                                                                   | Optional    |
| Description          | A human-readable descriptive name with free text remarks.                                                                                                                                                                                           |             |
| MissionTraceability  | MissionTraceabilityType                                                                                                                                                                                                                             | Optional    |
| Description          | Allows the creator of this operational item to specify what the source or motivation for it originated from. This could be ATO, ACO, etc.                                                                                                           |             |
| Source               | CreationSourceEnum                                                                                                                                                                                                                                  | Optional    |
| Description          | Indicates the source of the information.                                                                                                                                                                                                            |             |
| Schedule             | ScheduleType                                                                                                                                                                                                                                        | Optional    |
| Description          | This element represents a date/time interval associated with the operational item during which the item is in effect.                                                                                                                               |             |
| AssociatedTime       | TimeFunctionType                                                                                                                                                                                                                                    | Repeated    |
| Description          | Indicates time related events that are associated to the Op data type such as when activated or when operational. The multiplicity is limited to N-1 options where NONE is an event that has no value, thus removed from list of available options. |             |
| DataLinkIdIdentifier | DataLinkIdIdentifierPET                                                                                                                                                                                                                             | Repeated    |
| Description          | List of data link ID. Multiple data link IDs can be reported for the same network type.                                                                                                                                                             |             |
| Priority             | unsignedInt                                                                                                                                                                                                                                         | Optional    |
| Description          | This field represents the priority to be considered when choosing between multiple                                                                                                                                                                  |             |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                               | available points.                                                                                                                                                                                                                                                                                                                                                                                                |          |
| <b>QualifyingTags</b>         | QualifyingTagsType                                                                                                                                                                                                                                                                                                                                                                                               | Optional |
| Description                   | Provides identity qualifications to augment identification and associated processing of the OpPoint.                                                                                                                                                                                                                                                                                                             |          |
| <b>AppliesToIDs</b>           | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                                                                             | Repeated |
| Description                   | One or more systems to which this item applies. If omitted, item applies to all systems.                                                                                                                                                                                                                                                                                                                         |          |
| <b>SystemScheduleOverride</b> | SystemScheduleStateType                                                                                                                                                                                                                                                                                                                                                                                          | Repeated |
| Description                   | System-specific schedule that will override any part of the main schedule. This allows for system-specific schedule different than main schedule. When adding a SystemScheduleOverride, the child SystemID also must be added to the sibling SystemID list if the sibling list is not empty. The SystemScheduleOverride is independent of time planning. For the system, this override can be done at execution. |          |

Table A-1-285: P2P UCI Extensions MA\_OpZoneMT - MessageData UCI Extension Field

| Category                      |                                                                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>            | This element indicates the type of zone.                                                                                                                                                              |
| <b>Base Type</b>              | MA_OpZoneCategoryEnum                                                                                                                                                                                 |
| Message Structure             |                                                                                                                                                                                                       |
| Enumeration Literal           | Description                                                                                                                                                                                           |
| <b>AIRBORNE_EARLY_WARNING</b> | A point at which AEW aircraft are or will be orbiting or maneuvering.                                                                                                                                 |
| <b>COMBAT_AIR_PATROL</b>      | A point at which CAP aircraft are or will be orbiting or maneuvering.                                                                                                                                 |
| <b>CONSTRAINED_ENTRY_EXIT</b> | Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.                |
| <b>CONTAMINATED</b>           | Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.  |
| <b>DANGER</b>                 | A specified area above, below, or within which there may be a potential danger.                                                                                                                       |
| <b>DEFENDED_AREA</b>          | An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).                        |
| <b>DIRECTED_SEARCH_AREA</b>   | Indicates the zone is a directed search area.                                                                                                                                                         |
| <b>EOB</b>                    | Indicates the zone is the boundary for an electronic order of battle request.                                                                                                                         |
| <b>EXERCISE</b>               | An area for military maneuver or simulated wartime operation involving planning, preparation and execution.                                                                                           |
| <b>FIGHTER_ENGAGE_AOR</b>     | Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.                  |
| <b>FILTER_AREA</b>            | Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType. |
| <b>GENERAL_HAZARD</b>         | Indicates the zone is a hazard area that doesn't have a specified operational purpose.                                                                                                                |

|                               |                                                                                                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GENERAL_STATION</b>        | A general zone describing an air station that must be amplified by voice or predetermined procedures.                                                                                                                                       |
| <b>GROUND_AOR</b>             | Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.                                                   |
| <b>HOSTILE_TACTICAL_AREA</b>  | Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.                                                                         |
| <b>HOSTILE_WEAPON</b>         | Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.                                                                                                                        |
| <b>IMPACT_POINT</b>           | The real or expected point that a device has or will penetrate or contact a surface.                                                                                                                                                        |
| <b>JAMMING</b>                | Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.                                                                     |
| <b>KEEP_IN</b>                | Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.                                       |
| <b>KILL</b>                   | Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.                                                                                                |
| <b>MISSILE_ENGAGEMENT</b>     | Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.                                                                                                                 |
| <b>MISSILE_LAUNCH_POINT</b>   | A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.                      |
| <b>MISSION_AREA</b>           | Indicates the zone is an area in which a mission will take place.                                                                                                                                                                           |
| <b>NAMED_AREA_OF_INTEREST</b> | An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.                                                                                                               |
| <b>NO_COMM</b>                | Indicates the zone is an area where communications use is restricted.                                                                                                                                                                       |
| <b>NO_FIRE</b>                | Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType. |
| <b>NO_FLY</b>                 | Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.                                                           |
| <b>NO_IMAGE</b>               | Indicates the zone is an area where image collection is restricted.                                                                                                                                                                         |
| <b>ORBIT_FIGURE_EIGHT</b>     | A figure eight pattern within a rectangle that the aircraft is or will be tracking.                                                                                                                                                         |
| <b>ORBIT_RACE_TRACK</b>       | An oval within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                        |
| <b>ORBIT_RANDOM_CLOSED</b>    | A rectangle in which the aircraft will track randomly.                                                                                                                                                                                      |
| <b>REFERENCE</b>              | Indicates the zone is an area that doesn't have a specified operational purpose.                                                                                                                                                            |
| <b>SEARCH</b>                 | An area where a search is being or will be conducted. normally associated with a SAR mission.                                                                                                                                               |
| <b>SEPARATION</b>             | Indicates an update in reservation volume for an individual vehicle.                                                                                                                                                                        |
| <b>SHORAD</b>                 | Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.                                                                                                               |

|                                |                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SPEED_LIMITS</b>            | An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.                                                                                                    |
| <b>SUBMARINE_PATROL_AREA</b>   | A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.                                                   |
| <b>TARGET_AREA_OF_INTEREST</b> | An engagement area usually along an enemy mobility corridor.                                                                                                                                                                                                               |
| <b>VEHICLE_CONFIGURATION</b>   | Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.                                           |
| <b>WEAPON_RESTRICTION</b>      | Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType. |
| <b>WEATHER_CONDITIONS</b>      | Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.                                               |

Table A-1-286: P2P UCI Extensions MA\_OpZoneMT - Category UCI Extension Field

| CategoryUniqueData   |                                                                                            |             |
|----------------------|--------------------------------------------------------------------------------------------|-------------|
| Description          | Contains parameters unique to certain OpZoneCategoryEnum values.                           |             |
| Base Type            | MA_OpZoneCategoryType                                                                      |             |
| Message Structure    |                                                                                            |             |
| Field Name           | Field Type                                                                                 | Field Usage |
| ConstrainedEntryExit | ConstrainedEntryExitType                                                                   | Required    |
| Description          | Defines boundaries which applicable Systems can only enter and exit through defined edges. |             |
| FilterArea           | OpZoneFilterAreaPET                                                                        | Repeated    |
| Description          | Indicates that the OpZone can be a zone filter type.                                       |             |
| Jamming              | OpZoneJammingType                                                                          | Required    |
| Description          | Indicates that the OpZone is a jamming control zone.                                       |             |
| KeepIn               | IngressEgressType                                                                          | Required    |
| Description          | Defines boundaries to which applicable Systems must stay inside.                           |             |
| MissileLaunchPoint   | OpZoneMissileDataType                                                                      | Required    |
| Description          | Data defining a missile type, related track, and source of launch position.                |             |
| NoFire               | OpZoneNoFireType                                                                           | Required    |
| Description          | Defines areas where strike impact is restricted. Does not restrict the launch of weapons.  |             |
| NoFly                | OpZoneNoFlyType                                                                            | Required    |
| Description          | Defines area where flight is restricted. Equivalent to MIL-STD-6016 restricted zone.       |             |
| SpeedLimits          | MA_OpZoneSpeedLimitsType                                                                   | Required    |

|                             |                                                                                                         |          |
|-----------------------------|---------------------------------------------------------------------------------------------------------|----------|
| Description                 | Defines area where speeds are restricted.                                                               |          |
| <b>VehicleConfiguration</b> | MA_VehicleCommandDataType                                                                               | Required |
| Description                 | Defines vehicle configuration parameters that should change based on the planned location of a vehicle. |          |
| <b>WeaponRestriction</b>    | OpZoneWeaponRestrictionType                                                                             | Required |
| Description                 | Set of restricted weapons that cannot be used against a target type and or in a zone.                   |          |
| <b>WeatherConditions</b>    | OpZoneWeatherType                                                                                       | Required |
| Description                 | Defines area of weather conditions with potential of mission impact.                                    |          |

Table A-1-287: P2P UCI Extensions MA\_OpZoneMT - CategoryUniqueData UCI Extension Field

| SpeedLimits       |                                                                              |             |
|-------------------|------------------------------------------------------------------------------|-------------|
| Description       | Defines area where speeds are restricted.                                    |             |
| Base Type         | MA_OpZoneSpeedLimitsType                                                     |             |
| Message Structure |                                                                              |             |
| Field Name        | Field Type                                                                   | Field Usage |
| MinSpeedLimit     | PathSegmentSpeedValueType                                                    | Optional    |
| Description       | Indicates the minimum speed limit in meters per second within a zone/region. |             |
| MaxSpeedLimit     | PathSegmentSpeedValueType                                                    | Optional    |
| Description       | Indicates the maximum speed limit in meters per second within a zone/region. |             |

Table A-1-288: P2P UCI Extensions MA\_OpZoneMT - SpeedLimits UCI Extension Field

| VehicleConfiguration |                                                                                                                                                                                                                                                                                                                               |             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | Defines vehicle configuration parameters that should change based on the planned location of a vehicle.                                                                                                                                                                                                                       |             |
| Base Type            | MA_VehicleCommandDataType                                                                                                                                                                                                                                                                                                     |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                                               |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                                                    | Field Usage |
| VehicleAction        | VehicleActionEnum                                                                                                                                                                                                                                                                                                             | Optional    |
| Description          | Defines vehicle actions that can be done along a path segment of a route.                                                                                                                                                                                                                                                     |             |
| IFF_Settings         | IFF_Type                                                                                                                                                                                                                                                                                                                      | Optional    |
| Description          | If present the IFF settings for this vehicle will be changed along this path segment.                                                                                                                                                                                                                                         |             |
| SurvivabilityMode    | VehicleSurvivabilityModeEnum                                                                                                                                                                                                                                                                                                  | Optional    |
| Description          | Specifies that the vehicle should put itself into a more survivable mode. Different vehicles will have specific behaviors that reduce the signature of the vehicle (doors closed, minimal use of flaps, etc.) If this element is not present then it is assumed that the aircraft being configured has no survivability mode. |             |
| LostCommTimeout      | DurationType                                                                                                                                                                                                                                                                                                                  | Optional    |
| Description          | Defines the lost communication contingency timeout. When this timeout value is exceeded, the AV considers that lost communication has occurred and begins to perform contingency measures.                                                                                                                                    |             |

|                             |                                                                                                                                                                                           |          |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>LOS</b>                  | boolean                                                                                                                                                                                   | Optional |
| Description                 | Use Line Of Sight (LOS) communication as backup contingency.                                                                                                                              |          |
| <b>LossOfLinkProcessing</b> | VehicleLossOfLinkProcessingEnum                                                                                                                                                           | Optional |
| Description                 | Specifies whether the vehicle should invoke special Loss of Link processing. This could be turned off if the vehicle is going to be in a known and accepted lost communication situation. |          |
| <b>CommAllocationAction</b> | CommAllocationActionType                                                                                                                                                                  | Optional |
| Description                 | Indicates a change of communication allocation.                                                                                                                                           |          |
| <b>QNH_Setting</b>          | PressureType                                                                                                                                                                              | Optional |
| Description                 | Represents the barometric altimeter setting for correct MSL reading, commonly known as QNH, requested. Expressed in kilopascals (kPa).                                                    |          |

Table A-1-289: P2P UCI Extensions MA\_OpZoneMT - VehicleConfiguration UCI Extension Field

| Zone                  |                                                                                    |             |
|-----------------------|------------------------------------------------------------------------------------|-------------|
| Description           | This element represents the geospatial and temporal characteristics of the OpZone. |             |
| Base Type             | MA_OpZoneChoiceType                                                                |             |
| Message Structure     |                                                                                    |             |
| Field Name            | Field Type                                                                         | Field Usage |
| GeometrySpecification | OpZoneType                                                                         | Required    |
| Description           | Defines an OpZone geometry                                                         |             |
| GeometryReference     | MA_ReferentialOpZoneType                                                           | Required    |
| Description           | Defines an OpZone geometry via reference to other filtered objects                 |             |

Table A-1-290: P2P UCI Extensions MA\_OpZoneMT - Zone UCI Extension Field

| GeometryReference      |                                                                                                                                                                        |             |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | Defines an OpZone geometry via reference to other filtered objects                                                                                                     |             |
| Base Type              | MA_ReferentialOpZoneType                                                                                                                                               |             |
| Message Structure      |                                                                                                                                                                        |             |
| Field Name             | Field Type                                                                                                                                                             | Field Usage |
| ReferenceObjectsFilter | MA_GeoLocatedObjectFilterType                                                                                                                                          | Repeated    |
| Description            | The geometry of all objects returned by this filter will be used for the OpZone being defined.                                                                         |             |
| HorizontalBuffer       | DistanceType                                                                                                                                                           | Required    |
| Description            | The area of the resulting OpZone is the area of the referenced object, expanded by this horizontal buffer along the plane orthogonal to the NED frame's down direction |             |
| MinAltitudeBuffer      | DistanceType                                                                                                                                                           | Required    |
| Description            | The min altitude of the resulting OpZone is the min altitude of the referenced object, decreased by this min altitude buffer                                           |             |
| MaxAltitudeBuffer      | DistanceType                                                                                                                                                           | Required    |

|             |                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------|
| Description | The max altitude of the resulting OpZone is the max altitude of the referenced object, increased by this max altitude buffer |
|-------------|------------------------------------------------------------------------------------------------------------------------------|

Table A-1-291: P2P UCI Extensions MA\_OpZoneMT - GeometryReference UCI Extension Field

| ReferenceObjectsFilter |                                                                                                                                     |             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | The geometry of all objects returned by this filter will be used for the OpZone being defined.                                      |             |
| Base Type              | MA_GeoLocatedObjectFilterType                                                                                                       |             |
| Message Structure      |                                                                                                                                     |             |
| Field Name             | Field Type                                                                                                                          | Field Usage |
| EntityFilter           | EntityFilterType                                                                                                                    | Optional    |
| Description            | Indicates a name/type of Entity required to match the filter. If omitted, any name/type of Entity satisfies the filter.             |             |
| SystemFilter           | SystemFilterType                                                                                                                    | Optional    |
| Description            | Indicates a name/type of System required to match the filter. If omitted, any name/type of System satisfies the filter.             |             |
| OpPointFilter          | MA_OpPointFilterType                                                                                                                | Optional    |
| Description            | Indicates a name/type of OpPoint required to match the filter. If omitted, any name/type of OpPoint satisfies the filter.           |             |
| OpLineFilter           | MA_OpLineFilterType                                                                                                                 | Optional    |
| Description            | Indicates a name/type of OpLine required to match the filter. If omitted, any name/type of OpLine satisfies the filter.             |             |
| OpZoneFilter           | MA_OpZoneFilterType                                                                                                                 | Optional    |
| Description            | Indicates a name/type of OpZone required to match the filter. If omitted, any name/type of OpZone satisfies the filter.             |             |
| OpVolumeFilter         | MA_OpVolumeFilterType                                                                                                               | Optional    |
| Description            | Indicates a name/type of OpVolume required to match the filter. If omitted, any name/type of OpVolume satisfies the filter.         |             |
| DMPI_Filter            | MA_DMPI_FilterType                                                                                                                  | Optional    |
| Description            | Indicates a name/type of DMPI required to match the filter. If omitted, any name/type of DMPI satisfies the filter.                 |             |
| SignalReportFilter     | MA_SignalReportFilterType                                                                                                           | Optional    |
| Description            | Indicates a name/type of SignalReport required to match the filter. If omitted, any name/type of SignalReport satisfies the filter. |             |

Table A-1-292: P2P UCI Extensions MA\_OpZoneMT - ReferenceObjectsFilter UCI Extension Field

| OpPointFilter     |                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| Description       | Indicates a name/type of OpPoint required to match the filter. If omitted, any name/type of OpPoint satisfies the filter. |
| Base Type         | MA_OpPointFilterType                                                                                                      |
| Message Structure |                                                                                                                           |



| Field Name             | Field Type                                                                                                                                                                                                                                             | Field Usage |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>OpPointID</b>       | OpPointID_Type                                                                                                                                                                                                                                         | Repeated    |
| Description            | Indicates a specific OpPointID that is required to match the filter. If omitted, any OpPointID satisfies the filter. Multiple instances are logically ORed.                                                                                            |             |
| <b>OpPointCategory</b> | OpPointCategoriesType                                                                                                                                                                                                                                  | Repeated    |
| Description            | Indicates an OpPoint category that is required to match the filter. If omitted, any OpPoint category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.            |             |
| <b>GeoFilter</b>       | GeoFiltersQueryType                                                                                                                                                                                                                                    | Repeated    |
| Description            | Indicates a zone the OpPoint must fall within to satisfy the filter. Multiple instances are logically ORed. Indicates that the OpRouting type is applicable. List size for this element is based on there being no category type for OpRouting objects |             |

Table A-1-293: P2P UCI Extensions MA\_OpZoneMT - OpPointFilter UCI Extension Field

| OpLineFilter      |                                                                                                                                                                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a name/type of OpLine required to match the filter. If omitted, any name/type of OpLine satisfies the filter.                                                                                                                   |             |
| Base Type         | MA_OpLineFilterType                                                                                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                                                                                           |             |
| Field Name        | Field Type                                                                                                                                                                                                                                | Field Usage |
| OpLineID          | OpLineID_Type                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates a specific OpLineID that is required to match the filter. If omitted, any OpLineID satisfies the filter. Multiple instances are logically ORed.                                                                                 |             |
| OpLineCategory    | MA_OpLineCategoryEnum                                                                                                                                                                                                                     | Repeated    |
| Description       | Indicates an OpLine category that is required to match the filter. If omitted, any OpLine category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |             |
| GeoFilter         | GeoFiltersQueryType                                                                                                                                                                                                                       | Repeated    |
| Description       | Indicates a zone the OpLine must fall within to satisfy the filter. Multiple instances are logically ORed.                                                                                                                                |             |

Table A-1-294: P2P UCI Extensions MA\_OpZoneMT - OpLineFilter UCI Extension Field

| OpLineCategory                     |                                                                                                                                                                                                                                           |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                 | Indicates an OpLine category that is required to match the filter. If omitted, any OpLine category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |
| <b>Base Type</b>                   | MA_OpLineCategoryEnum                                                                                                                                                                                                                     |
| Message Structure                  |                                                                                                                                                                                                                                           |
| Enumeration Literal                | Description                                                                                                                                                                                                                               |
| <b>GENERAL_LINE</b>                | A general line where the function must be described by other procedures.                                                                                                                                                                  |
| <b>FORWARD_EDGE_OF_BATTLE_AREA</b> | The foremost limits of a series of areas in which ground combat units are                                                                                                                                                                 |

|                                       |                                                                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | deployed, excluding the areas in which the covering or screening forces are operating.                                                                                                                                                                                 |
| <b>GUN_TARGET_LINE</b>                | Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.                                                                                                                                                                |
| <b>CORRIDOR</b>                       | Line with width describing the passageway designated for travel by friendly aircraft.                                                                                                                                                                                  |
| <b>HOSTILE_BOUNDARY</b>               | Line that indicates a limit beyond which exists a hostile environment.                                                                                                                                                                                                 |
| <b>BUFFER_ZONE_BOUNDARY</b>           | Line beyond which friendly aircraft usually do not fly in peacetime.                                                                                                                                                                                                   |
| <b>LOW_LEVEL_TRANSIT_ROUTE</b>        | Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.                                                                           |
| <b>TACTICAL_ACTION_LINE</b>           | Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.                                                                                                                                                    |
| <b>FIRE_SUPPORT_COORDINATION_LINE</b> | Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander. |
| <b>FORWARD_LINE_OF_OWN_TROOPS</b>     | Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.                                                                                                          |
| <b>LINE_OF_COMMUNICATIONS</b>         | Line the route that connects an operating military unit with its supply base.                                                                                                                                                                                          |
| <b>JETWAY</b>                         | Line with width that defines areas where commercial aircraft are flying through.                                                                                                                                                                                       |
| <b>DEPARTURE_RADIAL</b>               | Line the defines the departure radial when performing Carrier admin departures.                                                                                                                                                                                        |
| <b>RECOVERY_RADIAL</b>                | Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.                                                                                                                                |
| <b>FINAL_BEARING</b>                  | Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.                                                                                                                                    |
| <b>BRC</b>                            | Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.                                                                                                             |

Table A-1-295: P2P UCI Extensions MA\_OpZoneMT - OpLineCategory UCI Extension Field

| OpZoneFilter      |                                                                                                                                                           |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a name/type of OpZone required to match the filter. If omitted, any name/type of OpZone satisfies the filter.                                   |             |
| Base Type         | MA_OpZoneFilterType                                                                                                                                       |             |
| Message Structure |                                                                                                                                                           |             |
| Field Name        | Field Type                                                                                                                                                | Field Usage |
| OpZoneID          | OpZoneID_Type                                                                                                                                             | Repeated    |
| Description       | Indicates a specific OpZoneID that is required to match the filter. If omitted, any OpZoneID satisfies the filter. Multiple instances are logically ORed. |             |

| OpZoneCategory | MA_OpZoneCategoryEnum                                                                                                                                                                                                                     | Repeated |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description    | Indicates an OpZone category that is required to match the filter. If omitted, any OpZone category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |          |
| GeoFilter      | GeoFiltersQueryType                                                                                                                                                                                                                       | Repeated |
| Description    | Indicates a zone the OpZone must fall within to satisfy the filter. Multiple instances are logically ORed.                                                                                                                                |          |

Table A-1-296: P2P UCI Extensions MA\_OpZoneMT - OpZoneFilter UCI Extension Field

| OpZoneCategory         |                                                                                                                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description            | Indicates an OpZone category that is required to match the filter. If omitted, any OpZone category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |
| Base Type              | MA_OpZoneCategoryEnum                                                                                                                                                                                                                     |
| Message Structure      |                                                                                                                                                                                                                                           |
| Enumeration Literal    | Description                                                                                                                                                                                                                               |
| AIRBORNE_EARLY_WARNING | A point at which AEW aircraft are or will be orbiting or maneuvering.                                                                                                                                                                     |
| COMBAT_AIR_PATROL      | A point at which CAP aircraft are or will be orbiting or maneuvering.                                                                                                                                                                     |
| CONSTRAINED_ENTRY_EXIT | Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.                                                    |
| CONTAMINATED           | Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.                                      |
| DANGER                 | A specified area above, below, or within which there may be a potential danger.                                                                                                                                                           |
| DEFENDED_AREA          | An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).                                                            |
| DIRECTED_SEARCH_AREA   | Indicates the zone is a directed search area.                                                                                                                                                                                             |
| EOB                    | Indicates the zone is the boundary for an electronic order of battle request.                                                                                                                                                             |
| EXERCISE               | An area for military maneuver or simulated wartime operation involving planning, preparation and execution.                                                                                                                               |
| FIGHTER_ENGAGE_AOR     | Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.                                                      |
| FILTER_AREA            | Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.                                     |
| GENERAL_HAZARD         | Indicates the zone is a hazard area that doesn't have a specified operational purpose.                                                                                                                                                    |
| GENERAL_STATION        | A general zone describing an air station that must be amplified by voice or predetermined procedures.                                                                                                                                     |

|                               |                                                                                                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GROUND_AOR</b>             | Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.                                                   |
| <b>HOSTILE_TACTICAL_AREA</b>  | Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.                                                                         |
| <b>HOSTILE_WEAPON</b>         | Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.                                                                                                                        |
| <b>IMPACT_POINT</b>           | The real or expected point that a device has or will penetrate or contact a surface.                                                                                                                                                        |
| <b>JAMMING</b>                | Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.                                                                     |
| <b>KEEP_IN</b>                | Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.                                       |
| <b>KILL</b>                   | Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.                                                                                                |
| <b>MISSILE_ENGAGEMENT</b>     | Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.                                                                                                                 |
| <b>MISSILE_LAUNCH_POINT</b>   | A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.                      |
| <b>MISSION_AREA</b>           | Indicates the zone is an area in which a mission will take place.                                                                                                                                                                           |
| <b>NAMED_AREA_OF_INTEREST</b> | An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.                                                                                                               |
| <b>NO_COMM</b>                | Indicates the zone is an area where communications use is restricted.                                                                                                                                                                       |
| <b>NO_FIRE</b>                | Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType. |
| <b>NO_FLY</b>                 | Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.                                                           |
| <b>NO_IMAGE</b>               | Indicates the zone is an area where image collection is restricted.                                                                                                                                                                         |
| <b>ORBIT_FIGURE_EIGHT</b>     | A figure eight pattern within a rectangle that the aircraft is or will be tracking.                                                                                                                                                         |
| <b>ORBIT_RACE_TRACK</b>       | An oval within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                        |
| <b>ORBIT_RANDOM_CLOSED</b>    | A rectangle in which the aircraft will track randomly.                                                                                                                                                                                      |
| <b>REFERENCE</b>              | Indicates the zone is an area that doesn't have a specified operational purpose.                                                                                                                                                            |
| <b>SEARCH</b>                 | An area where a search is being or will be conducted. normally associated with a SAR mission.                                                                                                                                               |
| <b>SEPARATION</b>             | Indicates an update in reservation volume for an individual vehicle.                                                                                                                                                                        |
| <b>SHORAD</b>                 | Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.                                                                                                               |
| <b>SPEED_LIMITS</b>           | An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.                                                                     |

|                                |                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SUBMARINE_PATROL_AREA</b>   | A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.                                                   |
| <b>TARGET_AREA_OF_INTEREST</b> | An engagement area usually along an enemy mobility corridor.                                                                                                                                                                                                               |
| <b>VEHICLE_CONFIGURATION</b>   | Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.                                           |
| <b>WEAPON_RESTRICTION</b>      | Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType. |
| <b>WEATHER_CONDITIONS</b>      | Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.                                               |

Table A-1-297: P2P UCI Extensions MA\_OpZoneMT - OpZoneCategory UCI Extension Field

| OpVolumeFilter    |                                                                                                                                                                                                                                               |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a name/type of OpVolume required to match the filter. If omitted, any name/type of OpVolume satisfies the filter.                                                                                                                   |             |
| Base Type         | MA_OpVolumeFilterType                                                                                                                                                                                                                         |             |
| Message Structure |                                                                                                                                                                                                                                               |             |
| Field Name        | Field Type                                                                                                                                                                                                                                    | Field Usage |
| OpVolumeID        | OpVolumeID_Type                                                                                                                                                                                                                               | Repeated    |
| Description       | Indicates a specific OpVolumeID that is required to match the filter. If omitted, any OpVolumeID satisfies the filter. Multiple instances are logically OREd.                                                                                 |             |
| OpVolumeCategory  | MA_OpZoneCategoryEnum                                                                                                                                                                                                                         | Repeated    |
| Description       | Indicates an OpVolume category that is required to match the filter. If omitted, any OpVolume category satisfies the filter. Multiple instances are logically OREd. List size for this element is based on "Select All That Apply" condition. |             |
| GeoFilter         | GeoFiltersQueryType                                                                                                                                                                                                                           | Repeated    |
| Description       | Indicates a zone the OpVolume must fall within to satisfy the filter. Multiple instances are logically OREd.                                                                                                                                  |             |

Table A-1-298: P2P UCI Extensions MA\_OpZoneMT - OpVolumeFilter UCI Extension Field

| OpVolumeCategory              |                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>            | Indicates an OpVolume category that is required to match the filter. If omitted, any OpVolume category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |
| <b>Base Type</b>              | MA_OpZoneCategoryEnum                                                                                                                                                                                                                         |
| Message Structure             |                                                                                                                                                                                                                                               |
| Enumeration Literal           | Description                                                                                                                                                                                                                                   |
| <b>AIRBORNE_EARLY_WARNING</b> | A point at which AEW aircraft are or will be orbiting or maneuvering.                                                                                                                                                                         |

|                               |                                                                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>COMBAT_AIR_PATROL</b>      | A point at which CAP aircraft are or will be orbiting or maneuvering.                                                                                                                                 |
| <b>CONSTRAINED_ENTRY_EXIT</b> | Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.                |
| <b>CONTAMINATED</b>           | Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.  |
| <b>DANGER</b>                 | A specified area above, below, or within which there may be a potential danger.                                                                                                                       |
| <b>DEFENDED_AREA</b>          | An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).                        |
| <b>DIRECTED_SEARCH_AREA</b>   | Indicates the zone is a directed search area.                                                                                                                                                         |
| <b>EOB</b>                    | Indicates the zone is the boundary for an electronic order of battle request.                                                                                                                         |
| <b>EXERCISE</b>               | An area for military maneuver or simulated wartime operation involving planning, preparation and execution.                                                                                           |
| <b>FIGHTER_ENGAGE_AOR</b>     | Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.                  |
| <b>FILTER_AREA</b>            | Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType. |
| <b>GENERAL_HAZARD</b>         | Indicates the zone is a hazard area that doesn't have a specified operational purpose.                                                                                                                |
| <b>GENERAL_STATION</b>        | A general zone describing an air station that must be amplified by voice or predetermined procedures.                                                                                                 |
| <b>GROUND_AOR</b>             | Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.             |
| <b>HOSTILE_TACTICAL_AREA</b>  | Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.                                   |
| <b>HOSTILE_WEAPON</b>         | Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.                                                                                  |
| <b>IMPACT_POINT</b>           | The real or expected point that a device has or will penetrate or contact a surface.                                                                                                                  |
| <b>JAMMING</b>                | Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.                               |
| <b>KEEP_IN</b>                | Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType. |
| <b>KILL</b>                   | Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.                                                          |
| <b>MISSILE_ENGAGEMENT</b>     | Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.                                                                           |
| <b>MISSILE_LAUNCH_POINT</b>   | A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the                                          |



|                                |                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | OpZoneMissileLaunchDataType choice in OpZoneCategoryType.                                                                                                                                                                                                                  |
| <b>MISSION_AREA</b>            | Indicates the zone is an area in which a mission will take place.                                                                                                                                                                                                          |
| <b>NAMED_AREA_OF_INTEREST</b>  | An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.                                                                                                                                              |
| <b>NO_COMM</b>                 | Indicates the zone is an area where communications use is restricted.                                                                                                                                                                                                      |
| <b>NO_FIRE</b>                 | Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.                                |
| <b>NO_FLY</b>                  | Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.                                                                                          |
| <b>NO_IMAGE</b>                | Indicates the zone is an area where image collection is restricted.                                                                                                                                                                                                        |
| <b>ORBIT_FIGURE_EIGHT</b>      | A figure eight pattern within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                                        |
| <b>ORBIT_RACE_TRACK</b>        | An oval within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                                                       |
| <b>ORBIT_RANDOM_CLOSED</b>     | A rectangle in which the aircraft will track randomly.                                                                                                                                                                                                                     |
| <b>REFERENCE</b>               | Indicates the zone is an area that doesn't have a specified operational purpose.                                                                                                                                                                                           |
| <b>SEARCH</b>                  | An area where a search is being or will be conducted. normally associated with a SAR mission.                                                                                                                                                                              |
| <b>SEPARATION</b>              | Indicates an update in reservation volume for an individual vehicle.                                                                                                                                                                                                       |
| <b>SHORAD</b>                  | Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.                                                                                                                                              |
| <b>SPEED_LIMITS</b>            | An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.                                                                                                    |
| <b>SUBMARINE_PATROL_AREA</b>   | A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.                                                   |
| <b>TARGET_AREA_OF_INTEREST</b> | An engagement area usually along an enemy mobility corridor.                                                                                                                                                                                                               |
| <b>VEHICLE_CONFIGURATION</b>   | Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.                                           |
| <b>WEAPON_RESTRICTION</b>      | Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType. |
| <b>WEATHER_CONDITIONS</b>      | Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.                                               |

Table A-1-299: P2P UCI Extensions MA\_OpZoneMT - OpVolumeCategory UCI Extension Field

| DMPI_Filter        |                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Indicates a name/type of DMPI required to match the filter. If omitted, any name/type of DMPI satisfies the filter. |
| <b>Base Type</b>   | MA_DMPI_FilterType                                                                                                  |

| Message Structure |                                                                                                                                                         |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Field Name        | Field Type                                                                                                                                              | Field Usage |
| <b>DMPI_ID</b>    | DMPI_ID_Type                                                                                                                                            | Repeated    |
| Description       | Indicates a specific DMPI_ID that is required to match the filter. If omitted, any DMPI_ID satisfies the filter. Multiple instances are logically ORed. |             |
| <b>GeoFilter</b>  | GeoFiltersQueryType                                                                                                                                     | Repeated    |
| Description       | Indicates a zone the DMPI must fall within to satisfy the filter. Multiple instances are logically ORed.                                                |             |

Table A-1-300: P2P UCI Extensions MA\_OpZoneMT - DMPI\_Filter UCI Extension Field

| SignalReportFilter |                                                                                                                                                                               |             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates a name/type of SignalReport required to match the filter. If omitted, any name/type of SignalReport satisfies the filter.                                           |             |
| Base Type          | MA_SignalReportFilterType                                                                                                                                                     |             |
| Message Structure  |                                                                                                                                                                               |             |
| Field Name         | Field Type                                                                                                                                                                    | Field Usage |
| SignalReportID     | SignalReportID_Type                                                                                                                                                           | Repeated    |
| Description        | Indicates a specific SignalReportID that is required to match the filter. If omitted, any SignalReportID satisfies the filter. Multiple instances are logically ORed.         |             |
| SystemID           | SystemID_Type                                                                                                                                                                 | Repeated    |
| Description        | Indicates a specific source SystemID that is required to match the filter. If omitted, any source SystemID satisfies the filter. Multiple instances are logically ORed.       |             |
| SubsystemID        | SubsystemID_Type                                                                                                                                                              | Repeated    |
| Description        | Indicates a specific source SubsystemID that is required to match the filter. If omitted, any source SubsystemID satisfies the filter. Multiple instances are logically ORed. |             |
| GeoFilter          | GeoFiltersQueryType                                                                                                                                                           | Repeated    |
| Description        | Indicates a zone the SignalReport must fall within to satisfy the filter. Multiple instances are logically ORed.                                                              |             |

Table A-1-301: P2P UCI Extensions MA\_OpZoneMT - SignalReportFilter UCI Extension Field

| OpVolumeFilter    |                                                                                                                                                               |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a name/type of OpVolume required to match the filter. If omitted, any name/type of OpVolume satisfies the filter.                                   |             |
| Base Type         | MA_OpVolumeFilterType                                                                                                                                         |             |
| Message Structure |                                                                                                                                                               |             |
| Field Name        | Field Type                                                                                                                                                    | Field Usage |
| OpVolumeID        | OpVolumeID_Type                                                                                                                                               | Repeated    |
| Description       | Indicates a specific OpVolumeID that is required to match the filter. If omitted, any OpVolumeID satisfies the filter. Multiple instances are logically ORed. |             |
| OpVolumeCategory  | MA_OpZoneCategoryEnum                                                                                                                                         | Repeated    |



|                  |                                                                                                                                                                                                                                               |          |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description      | Indicates an OpVolume category that is required to match the filter. If omitted, any OpVolume category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |          |
| <b>GeoFilter</b> | GeoFiltersQueryType                                                                                                                                                                                                                           | Repeated |
| Description      | Indicates a zone the OpVolume must fall within to satisfy the filter. Multiple instances are logically ORed.                                                                                                                                  |          |

Table A-1-302: P2P UCI Extensions MA\_OpZoneMT - OpVolumeFilter UCI Extension Field

| OpLineFilter      |                                                                                                                                                                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a name/type of OpLine required to match the filter. If omitted, any name/type of OpLine satisfies the filter.                                                                                                                   |             |
| Base Type         | MA_OpLineFilterType                                                                                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                                                                                           |             |
| Field Name        | Field Type                                                                                                                                                                                                                                | Field Usage |
| OpLineID          | OpLineID_Type                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates a specific OpLineID that is required to match the filter. If omitted, any OpLineID satisfies the filter. Multiple instances are logically ORed.                                                                                 |             |
| OpLineCategory    | MA_OpLineCategoryEnum                                                                                                                                                                                                                     | Repeated    |
| Description       | Indicates an OpLine category that is required to match the filter. If omitted, any OpLine category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |             |
| GeoFilter         | GeoFiltersQueryType                                                                                                                                                                                                                       | Repeated    |
| Description       | Indicates a zone the OpLine must fall within to satisfy the filter. Multiple instances are logically ORed.                                                                                                                                |             |

Table A-1-303: P2P UCI Extensions MA\_OpZoneMT - OpLineFilter UCI Extension Field

| OpLineCategory              |                                                                                                                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                 | Indicates an OpLine category that is required to match the filter. If omitted, any OpLine category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |
| Base Type                   | MA_OpLineCategoryEnum                                                                                                                                                                                                                     |
| Message Structure           |                                                                                                                                                                                                                                           |
| Enumeration Literal         | Description                                                                                                                                                                                                                               |
| GENERAL_LINE                | A general line where the function must be described by other procedures.                                                                                                                                                                  |
| FORWARD_EDGE_OF_BATTLE_AREA | The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.                                                                          |
| GUN_TARGET_LINE             | Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.                                                                                                                                   |
| CORRIDOR                    | Line with width describing the passageway designated for travel by friendly aircraft.                                                                                                                                                     |

|                                       |                                                                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HOSTILE_BOUNDARY</b>               | Line that indicates a limit beyond which exists a hostile environment.                                                                                                                                                                                                 |
| <b>BUFFER_ZONE_BOUNDARY</b>           | Line beyond which friendly aircraft usually do not fly in peacetime.                                                                                                                                                                                                   |
| <b>LOW_LEVEL_TRANSIT_ROUTE</b>        | Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.                                                                           |
| <b>TACTICAL_ACTION_LINE</b>           | Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.                                                                                                                                                    |
| <b>FIRE_SUPPORT_COORDINATION_LINE</b> | Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander. |
| <b>FORWARD_LINE_OF_OWN_TROOPS</b>     | Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.                                                                                                          |
| <b>LINE_OF_COMMUNICATIONS</b>         | Line the route that connects an operating military unit with its supply base.                                                                                                                                                                                          |
| <b>JETWAY</b>                         | Line with width that defines areas where commercial aircraft are flying through.                                                                                                                                                                                       |
| <b>DEPARTURE_RADIAL</b>               | Line the defines the departure radial when performing Carrier admin departures.                                                                                                                                                                                        |
| <b>RECOVERY_RADIAL</b>                | Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.                                                                                                                                |
| <b>FINAL_BEARING</b>                  | Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.                                                                                                                                    |
| <b>BRC</b>                            | Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.                                                                                                             |

Table A-1-304: P2P UCI Extensions MA\_OpZoneMT - OpLineCategory UCI Extension Field

| SpeedLimits       |                                                                              |             |
|-------------------|------------------------------------------------------------------------------|-------------|
| Description       | Defines area where speeds are restricted.                                    |             |
| Base Type         | MA_OpZoneSpeedLimitsType                                                     |             |
| Message Structure |                                                                              |             |
| Field Name        | Field Type                                                                   | Field Usage |
| MinSpeedLimit     | PathSegmentSpeedValueType                                                    | Optional    |
| Description       | Indicates the minimum speed limit in meters per second within a zone/region. |             |
| MaxSpeedLimit     | PathSegmentSpeedValueType                                                    | Optional    |
| Description       | Indicates the maximum speed limit in meters per second within a zone/region. |             |

Table A-1-305: P2P UCI Extensions MA\_OpZoneMT - SpeedLimits UCI Extension Field

| SignalReportFilter |                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Indicates a name/type of SignalReport required to match the filter. If omitted, any name/type of SignalReport satisfies the filter. |
| <b>Base Type</b>   | MA_SignalReportFilterType                                                                                                           |

| Message Structure     |                                                                                                                                                                               |             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Field Name            | Field Type                                                                                                                                                                    | Field Usage |
| <b>SignalReportID</b> | SignalReportID_Type                                                                                                                                                           | Repeated    |
| Description           | Indicates a specific SignalReportID that is required to match the filter. If omitted, any SignalReportID satisfies the filter. Multiple instances are logically ORed.         |             |
| <b>SystemID</b>       | SystemID_Type                                                                                                                                                                 | Repeated    |
| Description           | Indicates a specific source SystemID that is required to match the filter. If omitted, any source SystemID satisfies the filter. Multiple instances are logically ORed.       |             |
| <b>SubsystemID</b>    | SubsystemID_Type                                                                                                                                                              | Repeated    |
| Description           | Indicates a specific source SubsystemID that is required to match the filter. If omitted, any source SubsystemID satisfies the filter. Multiple instances are logically ORed. |             |
| <b>GeoFilter</b>      | GeoFiltersQueryType                                                                                                                                                           | Repeated    |
| Description           | Indicates a zone the SignalReport must fall within to satisfy the filter. Multiple instances are logically ORed.                                                              |             |

Table A-1-306: P2P UCI Extensions MA\_OpZoneMT - SignalReportFilter UCI Extension Field

| OpPointFilter     |                                                                                                                                                                                                                                                       |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a name/type of OpPoint required to match the filter. If omitted, any name/type of OpPoint satisfies the filter.                                                                                                                             |             |
| Base Type         | MA_OpPointFilterType                                                                                                                                                                                                                                  |             |
| Message Structure |                                                                                                                                                                                                                                                       |             |
| Field Name        | Field Type                                                                                                                                                                                                                                            | Field Usage |
| OpPointID         | OpPointID_Type                                                                                                                                                                                                                                        | Repeated    |
| Description       | Indicates a specific OpPointID that is required to match the filter. If omitted, any OpPointID satisfies the filter. Multiple instances are logically OREd.                                                                                           |             |
| OpPointCategory   | OpPointCategoriesType                                                                                                                                                                                                                                 | Repeated    |
| Description       | Indicates an OpPoint category that is required to match the filter. If omitted, any OpPoint category satisfies the filter. Multiple instances are logically OREd. List size for this element is based on "Select All That Apply" condition.           |             |
| GeoFilter         | GeoFiltersQueryType                                                                                                                                                                                                                                   | Repeated    |
| Description       | Indicates a zone the OpPoint must fall within to satisfy the filter. Multiple instances are logically OREd.Indicates that the OpRouting type is applicable. List size for this element is based on there being no category type for OpRouting objects |             |

Table A-1-307: P2P UCI Extensions MA\_OpZoneMT - OpPointFilter UCI Extension Field

| VehicleConfiguration |                                                                                                         |             |
|----------------------|---------------------------------------------------------------------------------------------------------|-------------|
| Description          | Defines vehicle configuration parameters that should change based on the planned location of a vehicle. |             |
| Base Type            | MA_VehicleCommandDataType                                                                               |             |
| Message Structure    |                                                                                                         |             |
| Field Name           | Field Type                                                                                              | Field Usage |

|                             |                                                                                                                                                                                                                                                                                                                               |          |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>VehicleAction</b>        | VehicleActionEnum                                                                                                                                                                                                                                                                                                             | Optional |
| Description                 | Defines vehicle actions that can be done along a path segment of a route.                                                                                                                                                                                                                                                     |          |
| <b>IFF_Settings</b>         | IFF_Type                                                                                                                                                                                                                                                                                                                      | Optional |
| Description                 | If present the IFF settings for this vehicle will be changed along this path segment.                                                                                                                                                                                                                                         |          |
| <b>SurvivabilityMode</b>    | VehicleSurvivabilityModeEnum                                                                                                                                                                                                                                                                                                  | Optional |
| Description                 | Specifies that the vehicle should put itself into a more survivable mode. Different vehicles will have specific behaviors that reduce the signature of the vehicle (doors closed, minimal use of flaps, etc.) If this element is not present then it is assumed that the aircraft being configured has no survivability mode. |          |
| <b>LostCommTimeout</b>      | DurationType                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                 | Defines the lost communication contingency timeout. When this timeout value is exceeded, the AV considers that lost communication has occurred and begins to perform contingency measures.                                                                                                                                    |          |
| <b>LOS</b>                  | boolean                                                                                                                                                                                                                                                                                                                       | Optional |
| Description                 | Use Line Of Sight (LOS) communication as backup contingency.                                                                                                                                                                                                                                                                  |          |
| <b>LossOfLinkProcessing</b> | VehicleLossOfLinkProcessingEnum                                                                                                                                                                                                                                                                                               | Optional |
| Description                 | Specifies whether the vehicle should invoke special Loss of Link processing. This could be turned off if the vehicle is going to be in a known and accepted lost communication situation.                                                                                                                                     |          |
| <b>CommAllocationAction</b> | CommAllocationActionType                                                                                                                                                                                                                                                                                                      | Optional |
| Description                 | Indicates a change of communication allocation.                                                                                                                                                                                                                                                                               |          |
| <b>QNH_Setting</b>          | PressureType                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                 | Represents the barometric altimeter setting for correct MSL reading, commonly known as QNH, requested. Expressed in kilopascals (kPa).                                                                                                                                                                                        |          |

Table A-1-308: P2P UCI Extensions MA\_OpZoneMT - VehicleConfiguration UCI Extension Field

| DMPI_Filter       |                                                                                                                                                         |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a name/type of DMPI required to match the filter. If omitted, any name/type of DMPI satisfies the filter.                                     |             |
| Base Type         | MA_DMPI_FilterType                                                                                                                                      |             |
| Message Structure |                                                                                                                                                         |             |
| Field Name        | Field Type                                                                                                                                              | Field Usage |
| DMPI_ID           | DMPI_ID_Type                                                                                                                                            | Repeated    |
| Description       | Indicates a specific DMPI_ID that is required to match the filter. If omitted, any DMPI_ID satisfies the filter. Multiple instances are logically ORed. |             |
| GeoFilter         | GeoFiltersQueryType                                                                                                                                     | Repeated    |
| Description       | Indicates a zone the DMPI must fall within to satisfy the filter. Multiple instances are logically ORed.                                                |             |

Table A-1-309: P2P UCI Extensions MA\_OpZoneMT - DMPI\_Filter UCI Extension Field

| Zone |
|------|
|------|

|                       |                                                                                    |             |
|-----------------------|------------------------------------------------------------------------------------|-------------|
| Description           | This element represents the geospatial and temporal characteristics of the OpZone. |             |
| Base Type             | MA_OpZoneChoiceType                                                                |             |
| Message Structure     |                                                                                    |             |
| Field Name            | Field Type                                                                         | Field Usage |
| GeometrySpecification | OpZoneType                                                                         | Required    |
| Description           | Defines an OpZone geometry                                                         |             |
| GeometryReference     | MA_ReferentialOpZoneType                                                           | Required    |
| Description           | Defines an OpZone geometry via reference to other filtered objects                 |             |

Table A-1-310: P2P UCI Extensions MA\_OpZoneMT - Zone UCI Extension Field

|                               |                                                                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Category</b>               |                                                                                                                                                                                                       |
| <b>Description</b>            | This element indicates the type of zone.                                                                                                                                                              |
| <b>Base Type</b>              | MA_OpZoneCategoryEnum                                                                                                                                                                                 |
| <b>Message Structure</b>      |                                                                                                                                                                                                       |
| <b>Enumeration Literal</b>    | <b>Description</b>                                                                                                                                                                                    |
| <b>AIRBORNE_EARLY_WARNING</b> | A point at which AEW aircraft are or will be orbiting or maneuvering.                                                                                                                                 |
| <b>COMBAT_AIR_PATROL</b>      | A point at which CAP aircraft are or will be orbiting or maneuvering.                                                                                                                                 |
| <b>CONSTRAINED_ENTRY_EXIT</b> | Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.                |
| <b>CONTAMINATED</b>           | Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.  |
| <b>DANGER</b>                 | A specified area above, below, or within which there may be a potential danger.                                                                                                                       |
| <b>DEFENDED_AREA</b>          | An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).                        |
| <b>DIRECTED_SEARCH_AREA</b>   | Indicates the zone is a directed search area.                                                                                                                                                         |
| <b>EOB</b>                    | Indicates the zone is the boundary for an electronic order of battle request.                                                                                                                         |
| <b>EXERCISE</b>               | An area for military maneuver or simulated wartime operation involving planning, preparation and execution.                                                                                           |
| <b>FIGHTER_ENGAGE_AOR</b>     | Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.                  |
| <b>FILTER_AREA</b>            | Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType. |
| <b>GENERAL_HAZARD</b>         | Indicates the zone is a hazard area that doesn't have a specified operational purpose.                                                                                                                |
| <b>GENERAL_STATION</b>        | A general zone describing an air station that must be amplified by voice or                                                                                                                           |

|                               |                                                                                                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | predetermined procedures.                                                                                                                                                                                                                   |
| <b>GROUND_AOR</b>             | Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.                                                   |
| <b>HOSTILE_TACTICAL_AREA</b>  | Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.                                                                         |
| <b>HOSTILE_WEAPON</b>         | Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.                                                                                                                        |
| <b>IMPACT_POINT</b>           | The real or expected point that a device has or will penetrate or contact a surface.                                                                                                                                                        |
| <b>JAMMING</b>                | Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.                                                                     |
| <b>KEEP_IN</b>                | Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.                                       |
| <b>KILL</b>                   | Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.                                                                                                |
| <b>MISSILE_ENGAGEMENT</b>     | Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.                                                                                                                 |
| <b>MISSILE_LAUNCH_POINT</b>   | A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.                      |
| <b>MISSION_AREA</b>           | Indicates the zone is an area in which a mission will take place.                                                                                                                                                                           |
| <b>NAMED_AREA_OF_INTEREST</b> | An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.                                                                                                               |
| <b>NO_COMM</b>                | Indicates the zone is an area where communications use is restricted.                                                                                                                                                                       |
| <b>NO_FIRE</b>                | Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType. |
| <b>NO_FLY</b>                 | Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.                                                           |
| <b>NO_IMAGE</b>               | Indicates the zone is an area where image collection is restricted.                                                                                                                                                                         |
| <b>ORBIT_FIGURE_EIGHT</b>     | A figure eight pattern within a rectangle that the aircraft is or will be tracking.                                                                                                                                                         |
| <b>ORBIT_RACE_TRACK</b>       | An oval within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                        |
| <b>ORBIT_RANDOM_CLOSED</b>    | A rectangle in which the aircraft will track randomly.                                                                                                                                                                                      |
| <b>REFERENCE</b>              | Indicates the zone is an area that doesn't have a specified operational purpose.                                                                                                                                                            |
| <b>SEARCH</b>                 | An area where a search is being or will be conducted. normally associated with a SAR mission.                                                                                                                                               |
| <b>SEPARATION</b>             | Indicates an update in reservation volume for an individual vehicle.                                                                                                                                                                        |
| <b>SHORAD</b>                 | Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.                                                                                                               |
| <b>SPEED_LIMITS</b>           | An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in                                                                                         |

|                                |                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | OpZoneCategoryType.                                                                                                                                                                                                                                                        |
| <b>SUBMARINE_PATROL_AREA</b>   | A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.                                                   |
| <b>TARGET_AREA_OF_INTEREST</b> | An engagement area usually along an enemy mobility corridor.                                                                                                                                                                                                               |
| <b>VEHICLE_CONFIGURATION</b>   | Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.                                           |
| <b>WEAPON_RESTRICTION</b>      | Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType. |
| <b>WEATHER_CONDITIONS</b>      | Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.                                               |

Table A-1-311: P2P UCI Extensions MA\_OpZoneMT - Category UCI Extension Field

| CategoryUniqueData   |                                                                                            |             |
|----------------------|--------------------------------------------------------------------------------------------|-------------|
| Description          | Contains parameters unique to certain OpZoneCategoryEnum values.                           |             |
| Base Type            | MA_OpZoneCategoryType                                                                      |             |
| Message Structure    |                                                                                            |             |
| Field Name           | Field Type                                                                                 | Field Usage |
| ConstrainedEntryExit | ConstrainedEntryExitType                                                                   | Required    |
| Description          | Defines boundaries which applicable Systems can only enter and exit through defined edges. |             |
| FilterArea           | OpZoneFilterAreaPET                                                                        | Repeated    |
| Description          | Indicates that the OpZone can be a zone filter type.                                       |             |
| Jamming              | OpZoneJammingType                                                                          | Required    |
| Description          | Indicates that the OpZone is a jamming control zone.                                       |             |
| KeepIn               | IngressEgressType                                                                          | Required    |
| Description          | Defines boundaries to which applicable Systems must stay inside.                           |             |
| MissileLaunchPoint   | OpZoneMissileDataType                                                                      | Required    |
| Description          | Data defining a missile type, related track, and source of launch position.                |             |
| NoFire               | OpZoneNoFireType                                                                           | Required    |
| Description          | Defines areas where strike impact is restricted. Does not restrict the launch of weapons.  |             |
| NoFly                | OpZoneNoFlyType                                                                            | Required    |
| Description          | Defines area where flight is restricted. Equivalent to MIL-STD-6016 restricted zone.       |             |
| SpeedLimits          | MA_OpZoneSpeedLimitsType                                                                   | Required    |
| Description          | Defines area where speeds are restricted.                                                  |             |
| VehicleConfiguration | MA_VehicleCommandDataType                                                                  | Required    |



|                          |                                                                                                         |          |
|--------------------------|---------------------------------------------------------------------------------------------------------|----------|
| Description              | Defines vehicle configuration parameters that should change based on the planned location of a vehicle. |          |
| <b>WeaponRestriction</b> | OpZoneWeaponRestrictionType                                                                             | Required |
| Description              | Set of restricted weapons that cannot be used against a target type and or in a zone.                   |          |
| <b>WeatherConditions</b> | OpZoneWeatherType                                                                                       | Required |
| Description              | Defines area of weather conditions with potential of mission impact.                                    |          |

Table A-1-312: P2P UCI Extensions MA\_OpZoneMT - CategoryUniqueData UCI Extension Field

| ReferenceObjectsFilter |                                                                                                                                     |             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | The geometry of all objects returned by this filter will be used for the OpZone being defined.                                      |             |
| Base Type              | MA_GeoLocatedObjectFilterType                                                                                                       |             |
| Message Structure      |                                                                                                                                     |             |
| Field Name             | Field Type                                                                                                                          | Field Usage |
| EntityFilter           | EntityFilterType                                                                                                                    | Optional    |
| Description            | Indicates a name/type of Entity required to match the filter. If omitted, any name/type of Entity satisfies the filter.             |             |
| SystemFilter           | SystemFilterType                                                                                                                    | Optional    |
| Description            | Indicates a name/type of System required to match the filter. If omitted, any name/type of System satisfies the filter.             |             |
| OpPointFilter          | MA_OpPointFilterType                                                                                                                | Optional    |
| Description            | Indicates a name/type of OpPoint required to match the filter. If omitted, any name/type of OpPoint satisfies the filter.           |             |
| OpLineFilter           | MA_OpLineFilterType                                                                                                                 | Optional    |
| Description            | Indicates a name/type of OpLine required to match the filter. If omitted, any name/type of OpLine satisfies the filter.             |             |
| OpZoneFilter           | MA_OpZoneFilterType                                                                                                                 | Optional    |
| Description            | Indicates a name/type of OpZone required to match the filter. If omitted, any name/type of OpZone satisfies the filter.             |             |
| OpVolumeFilter         | MA_OpVolumeFilterType                                                                                                               | Optional    |
| Description            | Indicates a name/type of OpVolume required to match the filter. If omitted, any name/type of OpVolume satisfies the filter.         |             |
| DMPI_Filter            | MA_DMPI_FilterType                                                                                                                  | Optional    |
| Description            | Indicates a name/type of DMPI required to match the filter. If omitted, any name/type of DMPI satisfies the filter.                 |             |
| SignalReportFilter     | MA_SignalReportFilterType                                                                                                           | Optional    |
| Description            | Indicates a name/type of SignalReport required to match the filter. If omitted, any name/type of SignalReport satisfies the filter. |             |

Table A-1-313: P2P UCI Extensions MA\_OpZoneMT - ReferenceObjectsFilter UCI Extension Field

| OpVolumeCategory |
|------------------|
|------------------|



|                               |                                                                                                                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>            | Indicates an OpVolume category that is required to match the filter. If omitted, any OpVolume category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |
| <b>Base Type</b>              | MA_OpZoneCategoryEnum                                                                                                                                                                                                                         |
| <b>Message Structure</b>      |                                                                                                                                                                                                                                               |
| <b>Enumeration Literal</b>    | <b>Description</b>                                                                                                                                                                                                                            |
| <b>AIRBORNE_EARLY_WARNING</b> | A point at which AEW aircraft are or will be orbiting or maneuvering.                                                                                                                                                                         |
| <b>COMBAT_AIR_PATROL</b>      | A point at which CAP aircraft are or will be orbiting or maneuvering.                                                                                                                                                                         |
| <b>CONSTRAINED_ENTRY_EXIT</b> | Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.                                                        |
| <b>CONTAMINATED</b>           | Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.                                          |
| <b>DANGER</b>                 | A specified area above, below, or within which there may be a potential danger.                                                                                                                                                               |
| <b>DEFENDED_AREA</b>          | An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).                                                                |
| <b>DIRECTED_SEARCH_AREA</b>   | Indicates the zone is a directed search area.                                                                                                                                                                                                 |
| <b>EOB</b>                    | Indicates the zone is the boundary for an electronic order of battle request.                                                                                                                                                                 |
| <b>EXERCISE</b>               | An area for military maneuver or simulated wartime operation involving planning, preparation and execution.                                                                                                                                   |
| <b>FIGHTER_ENGAGE_AOR</b>     | Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.                                                          |
| <b>FILTER_AREA</b>            | Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.                                         |
| <b>GENERAL_HAZARD</b>         | Indicates the zone is a hazard area that doesn't have a specified operational purpose.                                                                                                                                                        |
| <b>GENERAL_STATION</b>        | A general zone describing an air station that must be amplified by voice or predetermined procedures.                                                                                                                                         |
| <b>GROUND_AOR</b>             | Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.                                                     |
| <b>HOSTILE_TACTICAL_AREA</b>  | Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.                                                                           |
| <b>HOSTILE_WEAPON</b>         | Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.                                                                                                                          |
| <b>IMPACT_POINT</b>           | The real or expected point that a device has or will penetrate or contact a surface.                                                                                                                                                          |
| <b>JAMMING</b>                | Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.                                                                       |

|                                |                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>KEEP_IN</b>                 | Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.                                                                      |
| <b>KILL</b>                    | Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.                                                                                                                               |
| <b>MISSILE_ENGAGEMENT</b>      | Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.                                                                                                                                                |
| <b>MISSILE_LAUNCH_POINT</b>    | A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.                                                     |
| <b>MISSION_AREA</b>            | Indicates the zone is an area in which a mission will take place.                                                                                                                                                                                                          |
| <b>NAMED_AREA_OF_INTEREST</b>  | An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.                                                                                                                                              |
| <b>NO_COMM</b>                 | Indicates the zone is an area where communications use is restricted.                                                                                                                                                                                                      |
| <b>NO_FIRE</b>                 | Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.                                |
| <b>NO_FLY</b>                  | Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.                                                                                          |
| <b>NO_IMAGE</b>                | Indicates the zone is an area where image collection is restricted.                                                                                                                                                                                                        |
| <b>ORBIT_FIGURE_EIGHT</b>      | A figure eight pattern within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                                        |
| <b>ORBIT_RACE_TRACK</b>        | An oval within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                                                       |
| <b>ORBIT_RANDOM_CLOSED</b>     | A rectangle in which the aircraft will track randomly.                                                                                                                                                                                                                     |
| <b>REFERENCE</b>               | Indicates the zone is an area that doesn't have a specified operational purpose.                                                                                                                                                                                           |
| <b>SEARCH</b>                  | An area where a search is being or will be conducted. normally associated with a SAR mission.                                                                                                                                                                              |
| <b>SEPARATION</b>              | Indicates an update in reservation volume for an individual vehicle.                                                                                                                                                                                                       |
| <b>SHORAD</b>                  | Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.                                                                                                                                              |
| <b>SPEED_LIMITS</b>            | An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.                                                                                                    |
| <b>SUBMARINE_PATROL_AREA</b>   | A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.                                                   |
| <b>TARGET_AREA_OF_INTEREST</b> | An engagement area usually along an enemy mobility corridor.                                                                                                                                                                                                               |
| <b>VEHICLE_CONFIGURATION</b>   | Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.                                           |
| <b>WEAPON_RESTRICTION</b>      | Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType. |

|                           |                                                                                                                                                                                                                              |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEATHER_CONDITIONS</b> | Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType. |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table A-1-314: P2P UCI Extensions MA\_OpZoneMT - OpVolumeCategory UCI Extension Field

| GeometryReference      |                                                                                                                                                                        |             |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | Defines an OpZone geometry via reference to other filtered objects                                                                                                     |             |
| Base Type              | MA_ReferentialOpZoneType                                                                                                                                               |             |
| Message Structure      |                                                                                                                                                                        |             |
| Field Name             | Field Type                                                                                                                                                             | Field Usage |
| ReferenceObjectsFilter | MA_GeoLocatedObjectFilterType                                                                                                                                          | Repeated    |
| Description            | The geometry of all objects returned by this filter will be used for the OpZone being defined.                                                                         |             |
| HorizontalBuffer       | DistanceType                                                                                                                                                           | Required    |
| Description            | The area of the resulting OpZone is the area of the referenced object, expanded by this horizontal buffer along the plane orthogonal to the NED frame's down direction |             |
| MinAltitudeBuffer      | DistanceType                                                                                                                                                           | Required    |
| Description            | The min altitude of the resulting OpZone is the min altitude of the referenced object, decreased by this min altitude buffer                                           |             |
| MaxAltitudeBuffer      | DistanceType                                                                                                                                                           | Required    |
| Description            | The max altitude of the resulting OpZone is the max altitude of the referenced object, increased by this max altitude buffer                                           |             |

Table A-1-315: P2P UCI Extensions MA\_OpZoneMT - GeometryReference UCI Extension Field

| OpZoneCategory                |                                                                                                                                                                                                                                           |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>            | Indicates an OpZone category that is required to match the filter. If omitted, any OpZone category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |
| <b>Base Type</b>              | MA_OpZoneCategoryEnum                                                                                                                                                                                                                     |
| Message Structure             |                                                                                                                                                                                                                                           |
| Enumeration Literal           | Description                                                                                                                                                                                                                               |
| <b>AIRBORNE_EARLY_WARNING</b> | A point at which AEW aircraft are or will be orbiting or maneuvering.                                                                                                                                                                     |
| <b>COMBAT_AIR_PATROL</b>      | A point at which CAP aircraft are or will be orbiting or maneuvering.                                                                                                                                                                     |
| <b>CONSTRAINED_ENTRY_EXIT</b> | Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.                                                    |
| <b>CONTAMINATED</b>           | Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.                                      |
| <b>DANGER</b>                 | A specified area above, below, or within which there may be a potential danger.                                                                                                                                                           |
| <b>DEFENDED_AREA</b>          | An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for                                                                      |

|                               |                                                                                                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | defense).                                                                                                                                                                                                                                   |
| <b>DIRECTED_SEARCH_AREA</b>   | Indicates the zone is a directed search area.                                                                                                                                                                                               |
| <b>EOB</b>                    | Indicates the zone is the boundary for an electronic order of battle request.                                                                                                                                                               |
| <b>EXERCISE</b>               | An area for military maneuver or simulated wartime operation involving planning, preparation and execution.                                                                                                                                 |
| <b>FIGHTER_ENGAGE_AOR</b>     | Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.                                                        |
| <b>FILTER_AREA</b>            | Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.                                       |
| <b>GENERAL_HAZARD</b>         | Indicates the zone is a hazard area that doesn't have a specified operational purpose.                                                                                                                                                      |
| <b>GENERAL_STATION</b>        | A general zone describing an air station that must be amplified by voice or predetermined procedures.                                                                                                                                       |
| <b>GROUND_AOR</b>             | Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.                                                   |
| <b>HOSTILE_TACTICAL_AREA</b>  | Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.                                                                         |
| <b>HOSTILE_WEAPON</b>         | Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.                                                                                                                        |
| <b>IMPACT_POINT</b>           | The real or expected point that a device has or will penetrate or contact a surface.                                                                                                                                                        |
| <b>JAMMING</b>                | Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.                                                                     |
| <b>KEEP_IN</b>                | Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.                                       |
| <b>KILL</b>                   | Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.                                                                                                |
| <b>MISSILE_ENGAGEMENT</b>     | Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.                                                                                                                 |
| <b>MISSILE_LAUNCH_POINT</b>   | A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.                      |
| <b>MISSION_AREA</b>           | Indicates the zone is an area in which a mission will take place.                                                                                                                                                                           |
| <b>NAMED_AREA_OF_INTEREST</b> | An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.                                                                                                               |
| <b>NO_COMM</b>                | Indicates the zone is an area where communications use is restricted.                                                                                                                                                                       |
| <b>NO_FIRE</b>                | Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType. |
| <b>NO_FLY</b>                 | Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in                                                                               |

|                                |                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | OpZoneCategoryType.                                                                                                                                                                                                                                                        |
| <b>NO_IMAGE</b>                | Indicates the zone is an area where image collection is restricted.                                                                                                                                                                                                        |
| <b>ORBIT_FIGURE_EIGHT</b>      | A figure eight pattern within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                                        |
| <b>ORBIT_RACE_TRACK</b>        | An oval within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                                                       |
| <b>ORBIT_RANDOM_CLOSED</b>     | A rectangle in which the aircraft will track randomly.                                                                                                                                                                                                                     |
| <b>REFERENCE</b>               | Indicates the zone is an area that doesn't have a specified operational purpose.                                                                                                                                                                                           |
| <b>SEARCH</b>                  | An area where a search is being or will be conducted. normally associated with a SAR mission.                                                                                                                                                                              |
| <b>SEPARATION</b>              | Indicates an update in reservation volume for an individual vehicle.                                                                                                                                                                                                       |
| <b>SHORAD</b>                  | Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.                                                                                                                                              |
| <b>SPEED_LIMITS</b>            | An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.                                                                                                    |
| <b>SUBMARINE_PATROL_AREA</b>   | A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.                                                   |
| <b>TARGET_AREA_OF_INTEREST</b> | An engagement area usually along an enemy mobility corridor.                                                                                                                                                                                                               |
| <b>VEHICLE_CONFIGURATION</b>   | Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.                                           |
| <b>WEAPON_RESTRICTION</b>      | Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType. |
| <b>WEATHER_CONDITIONS</b>      | Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.                                               |

Table A-1-316: P2P UCI Extensions MA\_OpZoneMT - OpZoneCategory UCI Extension Field

| OpZoneFilter      |                                                                                                                                                                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a name/type of OpZone required to match the filter. If omitted, any name/type of OpZone satisfies the filter.                                                                                                                   |             |
| Base Type         | MA_OpZoneFilterType                                                                                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                                                                                           |             |
| Field Name        | Field Type                                                                                                                                                                                                                                | Field Usage |
| OpZoneID          | OpZoneID_Type                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates a specific OpZoneID that is required to match the filter. If omitted, any OpZoneID satisfies the filter. Multiple instances are logically ORed.                                                                                 |             |
| OpZoneCategory    | MA_OpZoneCategoryEnum                                                                                                                                                                                                                     | Repeated    |
| Description       | Indicates an OpZone category that is required to match the filter. If omitted, any OpZone category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |             |

|                  |                                                                                                            |          |
|------------------|------------------------------------------------------------------------------------------------------------|----------|
| <b>GeoFilter</b> | GeoFiltersQueryType                                                                                        | Repeated |
| Description      | Indicates a zone the OpZone must fall within to satisfy the filter. Multiple instances are logically ORed. |          |

Table A-1-317: P2P UCI Extensions MA\_OpZoneMT - OpZoneFilter UCI Extension Field

### 1.3.2.50 MA\_PositionReportMT UCI Extension

| MA_PositionReportMT |                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>  | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>  | MessageData : MA_PositionReportMDT                                                                                                          |

Table A-1-318: P2P UCI Extensions MA\_PositionReportMT UCI Extension

| MessageData            |                                                                                                                                                                                                                                                                                                                                         |             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                 |             |
| Base Type              | MA_PositionReportMDT                                                                                                                                                                                                                                                                                                                    |             |
| Message Structure      |                                                                                                                                                                                                                                                                                                                                         |             |
| Field Name             | Field Type                                                                                                                                                                                                                                                                                                                              | Field Usage |
| SystemID               | SystemID_Type                                                                                                                                                                                                                                                                                                                           | Required    |
| Description            | The unique ID of the System corresponding to this PositionReport.                                                                                                                                                                                                                                                                       |             |
| DisplayName            | VisibleString256Type                                                                                                                                                                                                                                                                                                                    | Optional    |
| Description            | This is the free-text, generally human-readable, name of the System or Resource corresponding to this PositionReport. This name isn't necessarily unique.                                                                                                                                                                               |             |
| Source                 | SystemSourceEnum                                                                                                                                                                                                                                                                                                                        | Required    |
| Description            | Indicates the source of the data in sibling elements. See enumeration annotations for further details.                                                                                                                                                                                                                                  |             |
| CurrentOperatingDomain | EnvironmentEnum                                                                                                                                                                                                                                                                                                                         | Required    |
| Description            | This enumerated element represents the domain/environment in which the system is currently operating.                                                                                                                                                                                                                                   |             |
| InertialState          | MA_InertialStateType                                                                                                                                                                                                                                                                                                                    | Required    |
| Description            | This element represents the inertial state of the system or resource.                                                                                                                                                                                                                                                                   |             |
| WanderAngle            | AngleType                                                                                                                                                                                                                                                                                                                               | Optional    |
| Description            | The angle by which the Inertial Navigation System (INS) rotates the geodetic frame about the vertical axis of the North-East-Down (NED) coordinate frame to avoid the introduction of a singularity at the North pole when it is calculating vehicle location. A rotation from true North towards East is reported as a positive angle. |             |
| MagneticHeading        | AngleType                                                                                                                                                                                                                                                                                                                               | Optional    |
| Description            | Indicates the heading in relationship to magnetic North.                                                                                                                                                                                                                                                                                |             |

Table A-1-319: P2P UCI Extensions MA\_PositionReportMT - MessageData UCI Extension Field

### 1.3.2.51 MA\_PackageManagementCommandMT UCI Extension

| MA_PackageManagementCommandMT |                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>            | This message is used to command one or more entities to form, join or leave a Package.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>            | MessageData : MA_PackageManagementCommandMDT                                                                                 |

**Table A-1-320: P2P UCI Extensions MA\_PackageManagementCommandMT UCI Extension**

| MessageData           |                                                                                                                                                                |             |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description           | See annotations in child elements and messages/elements that use this type for details.                                                                        |             |
| Base Type             | MA_PackageManagementCommandMDT                                                                                                                                 |             |
| Message Structure     |                                                                                                                                                                |             |
| Field Name            | Field Type                                                                                                                                                     | Field Usage |
| PackageID             | PackageID_Type                                                                                                                                                 | Required    |
| Description           | Indicates the unique identifier of the Package to form, join or leave.                                                                                         |             |
| ParentPackageID       | PackageID_Type                                                                                                                                                 | Optional    |
| Description           | If the sibling PackageID element is a sub-package, indicates the unique identifier of the parent package.                                                      |             |
| ChangeType            | MA_PackageChangeEnum                                                                                                                                           | Required    |
| Description           | Indicates the type of package change to perform.                                                                                                               |             |
| SystemID              | SystemID_Type                                                                                                                                                  | Repeated    |
| Description           | Indicates the System(s) commanded to perform a package change.                                                                                                 |             |
| ConstraintID          | MA_ConstraintID_Type                                                                                                                                           | Repeated    |
| Description           | Indicates any Constraints that may apply to this package (e.g FormationConstraint).                                                                            |             |
| PackageLeaderID       | SystemID_Type                                                                                                                                                  | Optional    |
| Description           | Indicates the System commanded to assume package leader role. Allows for the explicit specification of a leader without needing MA to conduct leader election. |             |
| PackagePartnerMetrics | MA_PartnerMetricsType                                                                                                                                          | Repeated    |
| Description           | Indicates the leadership fitness metrics for the partners in the package.                                                                                      |             |



| PackageLeaderElectionMethod                       |  | Integer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Required |
|---------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                                       |  | <p>Specifies which election method/algorithm the package will utilize for electing a Package Leader. Implementors are expected to set a default election method. If blank, the default election method will be used.</p> <p>Integer values and corresponding election methods are as follows:</p> <p>0 - Bully Election Method. Supports Dynamic Fitness Scores and cases where Leadership Fitness Scores need to be shared as part of an election process.</p> <p>1- Static Fitness Score Election Method. Supports nomination of a leader assuming leadership fitness scores remain static once they have been defined by MP/C2.</p> <p>2 - Maximum Consensus Election Method. Supports utilizing consensus algorithms to reach team consensus on the leader while preserving communication bandwidth.</p> <p>3 - Raft Leader Election Method. Supports leader election using the Raft consensus algorithm, which ensures a distributed and fault-tolerant approach to electing a leader.</p> |          |
| PackageLeadershipFitnessScoreDetermination Method |  | MA_LeadershipFitnessScoreDeterminationMethodChoiceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Optional |
| Description                                       |  | This field specifies the chosen fitness score determination method to be used for fitness scoring for the package.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| CommandID                                         |  | CommandID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Required |
| Description                                       |  | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| CommandState                                      |  | CommandStateEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Required |
| Description                                       |  | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |

**Table A-1-321: P2P UCI Extensions MA\_PackageManagementCommandMT - MessageData UCI Extension Field**

### 1.3.2.52 MA\_PackageManagementCommandStatusMT UCI Extension

| MA_PackageManagementCommandStatusMT |                                                                                                                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                         | This message is sent in response to a PackageManagementCommand and indicates if the command was accepted and if not why it was rejected. The resulting status of the Package Change is reported in PackagePartnerStatusReport for each system in the original command. |



|                    |                                                    |
|--------------------|----------------------------------------------------|
|                    | See the XSD File for more info                     |
| <b>Ext. Fields</b> | MessageData : MA_PackageManagementCommandStatusMDT |

Table A-1-322: P2P UCI Extensions MA\_PackageManagementCommandStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_PackageManagementCommandStatusMDT                                                                                                                                                                                                                                                                                                                        |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

Table A-1-323: P2P UCI Extensions MA\_PackageManagementCommandStatusMT - MessageData UCI Extension Field

### 1.3.2.53 MA\_PackagePartnerStatusReportMT UCI Extension

| MA_PackagePartnerStatusReportMT |                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>              | This message reports the resulting status of a package change for a PackageManagementCommand for individual entities in a package.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>              | MessageData : MA_PackagePartnerStatusReportMDT                                                                                                                           |

Table A-1-324: P2P UCI Extensions MA\_PackagePartnerStatusReportMT UCI Extension

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_PackagePartnerStatusReportMDT                                                        |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| CommandID         | CommandID_Type                                                                          | Required    |

|                                      |             |                                                                                                                                   |          |
|--------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------|----------|
|                                      | Description | Indicates the unique ID of the command that formed the Package specified in the sibling PackageID element.                        |          |
| <b>PackageID</b>                     |             | PackageID_Type                                                                                                                    | Required |
|                                      | Description | Indicates the unique ID of the package that this message is reporting a change status for.                                        |          |
| <b>PackageLeaderID</b>               |             | SystemID_Type                                                                                                                     | Optional |
|                                      | Description | Indicates the unique ID of the system that represents the Package lead.                                                           |          |
| <b>PackageStatus</b>                 |             | RequestProcessingStateEnum                                                                                                        | Required |
|                                      | Description | The status of the Package as a whole.                                                                                             |          |
| <b>ConstraintID</b>                  |             | MA_ConstraintID_Type                                                                                                              | Repeated |
|                                      | Description | Indicates any constraints that were applied to the original PackageManagementCommand, indicated in the sibling CommandID element. |          |
| <b>SystemStatus</b>                  |             | MA_PackagePartnerStatusType                                                                                                       | Repeated |
|                                      | Description | The status for an individual system within the package.                                                                           |          |
| <b>StaticLeadershipFitnessScore</b>  |             | FloatMinMaxType                                                                                                                   | Optional |
|                                      | Description | Indicates the static leadership fitness score for the partners in the package.                                                    |          |
| <b>DynamicLeadershipFitnessScore</b> |             | MA_PartnerLeadershipFitnessScoreType                                                                                              | Optional |
|                                      | Description | Indicates the dynamic leadership fitness score and determination type for a partner in the package.                               |          |

Table A-1-325: P2P UCI Extensions MA\_PackagePartnerStatusReportMT - MessageData UCI Extension Field

### 1.3.2.54 MA\_SystemManagementRequestMT UCI Extension

| MA_SystemManagementRequestMT |                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>           | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>           | MessageData : MA_SystemManagementRequestMDT<br>RequestType : MA_SystemManagementRequestType<br>VehicleSettings : MA_VehicleCommandDataType  |

Table A-1-326: P2P UCI Extensions MA\_SystemManagementRequestMT UCI Extension

| MessageData       |                                                                                                                                                  |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | This contains the message data.                                                                                                                  |             |
| Base Type         | MA_SystemManagementRequestMDT                                                                                                                    |             |
| Message Structure |                                                                                                                                                  |             |
| Field Name        | Field Type                                                                                                                                       | Field Usage |
| AuthorityType     | EntityManagementRequestAuthorityEnum                                                                                                             | Optional    |
| Description       | Indicates whether this SystemManagementRequest is a request to change data or an order that must be followed. Defaults to REQUEST if not filled. |             |

|                                  |                                                                                                                                                       |          |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>RequestSource</b>             | RequestSourceEnum                                                                                                                                     | Optional |
| Description                      | This element indicates the type of service requesting the change. Defaults to AUTOMATIC_INTERNAL if not filled.                                       |          |
| <b>ReferenceSystemID</b>         | SystemID_Type                                                                                                                                         | Required |
| Description                      | Indicates the system to be acted upon by this message.                                                                                                |          |
| <b>RequestType</b>               | MA_SystemManagementRequestType                                                                                                                        | Repeated |
| Description                      | Indicates the type of request to perform on a system.                                                                                                 |          |
| <b>UpdateFromControllingUnit</b> | boolean                                                                                                                                               | Optional |
| Description                      | Indicates if this SystemManagementRequest came from the system controlling the referenced system. This indicates a higher confidence in the new data. |          |
| <b>Source</b>                    | SystemMessageIdentifierType                                                                                                                           | Optional |
| Description                      | Indicates the system initiating this request and allows the requesting Service, Activity, or Network to be specified.                                 |          |
| <b>RequestID</b>                 | RequestID_Type                                                                                                                                        | Required |
| Description                      | Indicates the unique ID of the request message.                                                                                                       |          |
| <b>RequestState</b>              | RequestStateEnum                                                                                                                                      | Required |
| Description                      | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.             |          |

Table A-1-327: P2P UCI Extensions MA\_SystemManagementRequestMT - MessageData UCI Extension Field

| RequestType              |                                                                                                                                                     |             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | Indicates the type of request to perform on a system.                                                                                               |             |
| Base Type                | MA_SystemManagementRequestType                                                                                                                      |             |
| Message Structure        |                                                                                                                                                     |             |
| Field Name               | Field Type                                                                                                                                          | Field Usage |
| SetMode                  | MessageModeEnum                                                                                                                                     | Required    |
| Description              | Indicates a request to modify the mode of a system.                                                                                                 |             |
| SetIdentity              | SystemIdentityType                                                                                                                                  | Required    |
| Description              | Indicates a request to set the identity of a system.                                                                                                |             |
| SetLink16Metadata        | Link16MetadataType                                                                                                                                  | Required    |
| Description              | Indicates a request to modify the Link16Metadata of a system.                                                                                       |             |
| SetVoiceControl          | VoiceControlType                                                                                                                                    | Required    |
| Description              | Set the voice control frequency of a system.                                                                                                        |             |
| SetSensorEntityReporting | boolean                                                                                                                                             | Required    |
| Description              | When TRUE, indicates an order to report all locally derived sensor, signal, track or Entity data. When FALSE, indicates an order to stop reporting. |             |
| VehicleSettings          | MA_VehicleCommandDataType                                                                                                                           | Required    |
| Description              | Indicates a request to modify vehicle settings.                                                                                                     |             |

Table A-1-328: P2P UCI Extensions MA\_SystemManagementRequestMT - RequestType UCI Extension Field

| VehicleSettings      |                                                                                                                                                                                                                                                                                                                               |             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | Indicates a request to modify vehicle settings.                                                                                                                                                                                                                                                                               |             |
| Base Type            | MA_VehicleCommandDataType                                                                                                                                                                                                                                                                                                     |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                                               |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                                                    | Field Usage |
| VehicleAction        | VehicleActionEnum                                                                                                                                                                                                                                                                                                             | Optional    |
| Description          | Defines vehicle actions that can be done along a path segment of a route.                                                                                                                                                                                                                                                     |             |
| IFF_Settings         | IFF_Type                                                                                                                                                                                                                                                                                                                      | Optional    |
| Description          | If present the IFF settings for this vehicle will be changed along this path segment.                                                                                                                                                                                                                                         |             |
| SurvivabilityMode    | VehicleSurvivabilityModeEnum                                                                                                                                                                                                                                                                                                  | Optional    |
| Description          | Specifies that the vehicle should put itself into a more survivable mode. Different vehicles will have specific behaviors that reduce the signature of the vehicle (doors closed, minimal use of flaps, etc.) If this element is not present then it is assumed that the aircraft being configured has no survivability mode. |             |
| LostCommTimeout      | DurationType                                                                                                                                                                                                                                                                                                                  | Optional    |
| Description          | Defines the lost communication contingency timeout. When this timeout value is exceeded, the AV considers that lost communication has occurred and begins to perform contingency measures.                                                                                                                                    |             |
| LOS                  | boolean                                                                                                                                                                                                                                                                                                                       | Optional    |
| Description          | Use Line Of Sight (LOS) communication as backup contingency.                                                                                                                                                                                                                                                                  |             |
| LossOfLinkProcessing | VehicleLossOfLinkProcessingEnum                                                                                                                                                                                                                                                                                               | Optional    |
| Description          | Specifies whether the vehicle should invoke special Loss of Link processing. This could be turned off if the vehicle is going to be in a known and accepted lost communication situation.                                                                                                                                     |             |
| CommAllocationAction | CommAllocationActionType                                                                                                                                                                                                                                                                                                      | Optional    |
| Description          | Indicates a change of communication allocation.                                                                                                                                                                                                                                                                               |             |
| QNH_Setting          | PressureType                                                                                                                                                                                                                                                                                                                  | Optional    |
| Description          | Represents the barometric altimeter setting for correct MSL reading, commonly known as QNH, requested. Expressed in kilopascals (kPa).                                                                                                                                                                                        |             |

**Table A-1-329: P2P UCI Extensions MA\_SystemManagementRequestMT - VehicleSettings UCI Extension Field**

### 1.3.2.55 MA\_SystemManagementRequestStatusMT UCI Extension

| MA_SystemManagementRequestStatusMT |                                                                                                                                             |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                 | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                 | MessageData : MA_SystemManagementRequestStatusMDT                                                                                           |

**Table A-1-330: P2P UCI Extensions MA\_SystemManagementRequestStatusMT UCI Extension**

| MessageData                  |                                                                                                                                                                                                                                                                                             |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                     |             |
| Base Type                    | MA_SystemManagementRequestStatusMDT                                                                                                                                                                                                                                                         |             |
| Message Structure            |                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                  | Field Usage |
| SystemID                     | SystemID_Type                                                                                                                                                                                                                                                                               | Required    |
| Description                  | Indicates the ID of the system responding to the SystemManagementRequest.                                                                                                                                                                                                                   |             |
| ServiceID                    | ServiceID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the ID of the service responding to the SystemManagementRequest.                                                                                                                                                                                                                  |             |
| RequestID                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |             |
| RequestProcessingState       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |             |
| RequestProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |

**Table A-1-331: P2P UCI Extensions MA\_SystemManagementRequestStatusMT - MessageData UCI Extension Field**

### 1.3.2.56 MA\_AMTI\_CapabilityMT UCI Extension

| MA_AMTI_CapabilityMT |                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields          | MessageData : MA_AMTI_CapabilityMDT<br>Capability : MA_AMTI_CapabilityType                                                    |

**Table A-1-332: P2P UCI Extensions MA\_AMTI\_CapabilityMT UCI Extension**

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_AMTI_CapabilityMDT                                                                   |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| Capability        | MA_AMTI_CapabilityType                                                                  | Repeated    |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description         | Indicates an installed AMTI Capability, its configurable characteristics and its controllability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>AvailableMDF</b> | MDF_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated |
| Description         | Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability. |          |

Table A-1-333: P2P UCI Extensions MA\_AMTI\_CapabilityMT - MessageData UCI Extension Field

| Capability             |                                                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | Indicates an installed AMTI Capability, its configurable characteristics and its controllability.                                                                                                                                                                                                                                                                                           |             |
| Base Type              | MA_AMTI_CapabilityType                                                                                                                                                                                                                                                                                                                                                                      |             |
| Message Structure      |                                                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name             | Field Type                                                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| MT_IIRS_Level          | NIIRS_Type                                                                                                                                                                                                                                                                                                                                                                                  | Repeated    |
| Description            | Indicates a Moving Target Indication Interpretability Rating Scale level supported by the Capability. See annotations of the underlying type for further details.                                                                                                                                                                                                                           |             |
| ComponentFieldOfRegard | ComponentFieldOfRegardType                                                                                                                                                                                                                                                                                                                                                                  | Repeated    |
| Description            | Fields of Regard for each component in the capability.                                                                                                                                                                                                                                                                                                                                      |             |
| CapabilityType         | AMTI_CapabilityEnum                                                                                                                                                                                                                                                                                                                                                                         | Required    |
| Description            | Indicates the Capability's first tier type in the AMTI taxonomy. For AMTI, the first tier is an indication of the physical area covered. See enumerated type annotations for further details.                                                                                                                                                                                               |             |
| SubCapabilityType      | AMTI_SubCapabilityEnum                                                                                                                                                                                                                                                                                                                                                                      | Repeated    |
| Description            | Indicates a SubCapability of the AMTI Capability; the second tier in the taxonomy of AMTI. For AMTI, the second tier is waveform category. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition. The selection of AUTO_SUBCAPABILITY_SELECT still requires the evaluation of all sibling attributes by the Capability. |             |
| CapabilityOptions      | AMTI_CapabilityOptionsType                                                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description            | Indicates the Capability's support for select optional elements in its accepted control interface messages. Support for these elements is expected to vary across Capability implementations but must be known in order to adapt and configure upstream Systems and Services which use the Capability.                                                                                      |             |
| SupportedFrequencies   | FrequencyRangeType                                                                                                                                                                                                                                                                                                                                                                          | Repeated    |
| Description            | Indicates a frequency range used by the Capability.                                                                                                                                                                                                                                                                                                                                         |             |
| MessageOutput          | AMTI_MessageOutputsEnum                                                                                                                                                                                                                                                                                                                                                                     | Repeated    |
| Description            | Indicates a message that is an output of the Capability. See enumerated type annotations for details. List size for this element is based on "Select All That Apply"                                                                                                                                                                                                                        |             |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                          | condition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>AcceptedInterface</b> | CapabilityControlInterfacesEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Repeated |
| Description              | Indicates an accepted control interface for the Capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>CommsRequired</b>     | boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Optional |
| Description              | Indicates whether (TRUE) or not (FALSE) communications are required to use the Capability. Omitted is equivalent to FALSE. For example, an unmanned system with no onboard storage would require communications to disseminate the outputs of the Capability.                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>CapabilityID</b>      | CapabilityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Required |
| Description              | Indicates the unique ID of this Capability. This ID is used to invoke the Capability via Command messages, report its activity via Activity messages, etc.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>DomainPairing</b>     | EnvironmentPairingEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Repeated |
| Description              | Indicates the domain/environment which the Capability is employed from and the "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>ProductOutput</b>     | ProductOutputType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Repeated |
| Description              | Indicates a Product that can be an output of the Capability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>PackageOperation</b>  | PackageOperationEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Optional |
| Description              | Indicates if and how the operation of the Capability can be coordinated with a partner Capability of another System in a multi-System Package. Omitting this element is equivalent to the enumerate NO_PACKAGE_OPERATION. The actual Package configuration including per-Capability partnering between Systems is given in the PackageStatus message. For example, a System that can perform cooperative geolocation could advertise with this element in the ESM_Capability message. A System that can navigate in a swarm with other Systems could advertise with this element in the FlightCapability message. |          |

Table A-1-334: P2P UCI Extensions MA\_AMTI\_CapabilityMT - Capability UCI Extension Field

### 1.3.2.57 MA\_ActionCapabilityMT UCI Extension

| MA_ActionCapabilityMT |                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>    | This message represents the Action category of UCI Capability.<br><br>See the XSD File for more info                 |
| <b>Ext. Fields</b>    | AssociatedTaskType : MA_TaskTypeEnum<br>Capability : MA_ActionCapabilityType<br>MessageData : MA_ActionCapabilityMDT |

Table A-1-335: P2P UCI Extensions MA\_ActionCapabilityMT UCI Extension

| AssociatedTaskType |                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Indicates a Task type associated with the Action. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition. |



|                             |                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------|
| <b>Base Type</b>            | MA_TaskTypeEnum                                                                         |
| <b>Message Structure</b>    |                                                                                         |
| <b>Enumeration Literal</b>  | <b>Description</b>                                                                      |
| AIR_SAMPLE                  | Indicates the task is to sample and analyze the air around the vehicle.                 |
| AMTI                        | Indicates the task is to track air targets using an Airborne Moving Target Indicator.   |
| AO                          | Indicates the task is to employ an Active Optical capability.                           |
| CARGO_DELIVERY              | Indicates the task is to deliver cargo.                                                 |
| CAP                         | Indicates the task is to execute a Combat Air Patrol.                                   |
| COMINT                      | Indicates the task is to employ a communications intelligence capability.               |
| COMM_RELAY                  | Indicates the task is to relay communications from other battlespace participants.      |
| COUNTER_SPACE               | Indicates the task is to employ a Counter Space Capability.                             |
| EA                          | Indicates the task is to employ an electronic attack capability.                        |
| ESCORT                      | Indicates the task is to escort another entity.                                         |
| ESM                         | Indicates the task is to employ an Electronic Support Measure.                          |
| FLIGHT                      | Indicates the task is to fly.                                                           |
| ORBIT_CHANGE                | Indicates a task to perform an orbit change via a spacecraft maneuver.                  |
| ORBITAL_SURVEILLANCE        | Indicates a task to perform orbital surveillance.                                       |
| ORBITAL_SURVEILLANCE_SENSOR | Indicates a task to perform orbital surveillance via sensor collection capabilities.    |
| PO                          | Indicates the task is to employ a Passive Optical capability.                           |
| REFUEL                      | Indicates the task is to refuel another System or Entity.                               |
| SAR                         | Indicates the task is to employ a Synthetic Aperture Radar capability.                  |
| SMTI                        | Indicates the task is to track surface targets using a Surface Moving Target Indicator. |
| STRIKE                      | Indicates the task is to strike a target.                                               |
| SYSTEM_DEPLOYMENT           | Indicates the task is to deploy a system.                                               |
| TACTICAL_ORDER              | Indicates the task is to perform a tactical order.                                      |
| WEATHER_RADAR               | Indicates the task is to collect weather data.                                          |

Table A-1-336: P2P UCI Extensions MA\_ActionCapabilityMT - AssociatedTaskType UCI Extension Field

| Capability        |                                                                                                     |             |
|-------------------|-----------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates an installed Action Capability, its configurable characteristics and its controllability. |             |
| Base Type         | MA_ActionCapabilityType                                                                             |             |
| Message Structure |                                                                                                     |             |
| Field Name        | Field Type                                                                                          | Field Usage |



|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>CapabilityType</b>           | MA_ActionTypeEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required |
| Description                     | Indicates the type of Action. See enumerated type annotations for further details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |
| <b>AssociatedTaskType</b>       | MA_TaskTypeEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Repeated |
| Description                     | Indicates a Task type associated with the Action. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>AssociatedCapabilityType</b> | CapabilityTypeEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Repeated |
| Description                     | Indicates a Capability type associated with the Action. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>CapabilityOptions</b>        | ActionCapabilityOptionsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Required |
| Description                     | Indicates the Capability's support for select optional elements in its accepted control interface messages. Support for these elements is expected to vary across Capability implementations but must be known in order to adapt and configure upstream Systems and Services which use the Capability.                                                                                                                                                                                                                                                                   |          |
| <b>MessageOutput</b>            | ActionMessageOutputsEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Repeated |
| Description                     | Indicates a message that is an output of the Capability. See enumerated type annotations for details. List size for this element is based on "Select All That Apply" condition.                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>AcceptedInterface</b>        | CapabilityControlInterfacesEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Repeated |
| Description                     | Indicates an accepted control interface for the Capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>CommsRequired</b>            | boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                     | Indicates whether (TRUE) or not (FALSE) communications are required to use the Capability. Omitted is equivalent to FALSE. For example, an unmanned system with no onboard storage would require communications to disseminate the outputs of the Capability.                                                                                                                                                                                                                                                                                                            |          |
| <b>CapabilityID</b>             | CapabilityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required |
| Description                     | Indicates the unique ID of this Capability. This ID is used to invoke the Capability via Command messages, report its activity via Activity messages, etc.                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>DomainPairing</b>            | EnvironmentPairingEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Repeated |
| Description                     | Indicates the domain/environment which the Capability is employed from and the "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.                                                                                                                                                                                                                                                                                                                               |          |
| <b>ProductOutput</b>            | ProductOutputType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated |
| Description                     | Indicates a Product that can be an output of the Capability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>PackageOperation</b>         | PackageOperationEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Optional |
| Description                     | Indicates if and how the operation of the Capability can be coordinated with a partner Capability of another System in a multi-System Package. Omitting this element is equivalent to the enumerate NO_PACKAGE_OPERATION. The actual Package configuration including per-Capability partnering between Systems is given in the PackageStatus message. For example, a System that can perform cooperative geolocation could advertise with this element in the ESM_Capability message. A System that can navigate in a swarm with other Systems could advertise with this |          |

|  |                                          |
|--|------------------------------------------|
|  | element in the FlightCapability message. |
|--|------------------------------------------|

**Table A-1-337: P2P UCI Extensions MA\_ActionCapabilityMT - Capability UCI Extension Field**

| MessageData       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| Base Type         | MA_ActionCapabilityMDT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Field Usage |
| Capability        | MA_ActionCapabilityType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Repeated    |
| Description       | Indicates an installed Action Capability, its configurable characteristics and its controllability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
| AvailableMDF      | MDF_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description       | Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability. |             |

**Table A-1-338: P2P UCI Extensions MA\_ActionCapabilityMT - MessageData UCI Extension Field**

### 1.3.2.58 MA\_ApprovalRequestMT UCI Extension

| MA_ApprovalRequestMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>   | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info                                                                                                                                                                                                                                                                                                                      |
| <b>Ext. Fields</b>   | ActivationDetails : MA_MissionPlanActivationDetailsType<br>OpPlanID : MA_OpPlanID_Type<br>MissionPlanActivationApproval : MA_MissionPlanActivationCommandType<br>ApprovalReferences : ApprovalRequestPolicyReferenceType<br>ApprovalItem : ApprovalRequestItemReferenceType<br>ByExecutionPlanSet : MA_ExecutionPlanSetActivationType<br>MessageData : ApprovalRequestMDT<br>BySubPlan : MA_MissionPlanSubplanActivationType<br>OpPlan : MA_OpPlanActivationType |

**Table A-1-339: P2P UCI Extensions MA\_ApprovalRequestMT UCI Extension**

| ActivationDetails  |                                             |
|--------------------|---------------------------------------------|
| <b>Description</b> | Indicates the commanded activation details. |

| Base Type          | MA_MissionPlanActivationDetailsType                                                                                                               |             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Message Structure  |                                                                                                                                                   |             |
| Field Name         | Field Type                                                                                                                                        | Field Usage |
| ByMissionPlan      | MissionPlanActivationType                                                                                                                         | Required    |
| Description        | Indicates simultaneous activation of all sub-*Plans (RoutePlan, RouteActivityPlan for example) of a MissionPlan into the same activation state.   |             |
| BySubPlan          | MA_MissionPlanSubplanActivationType                                                                                                               | Required    |
| Description        | Indicates activation by sub-*Plan (RoutePlan or OrbitPlan for example) of the MissionPlan, with potentially different states for each.            |             |
| ByExecutionPlanSet | MA_ExecutionPlanSetActivationType                                                                                                                 | Required    |
| Description        | Indicates simultaneous activation of all sub-*Plans (ActivityPlan, RoutePlan for example) of an *ExecutionPlanSet into the same activation state. |             |

Table A-1-340: P2P UCI Extensions MA\_ApprovalRequestMT - ActivationDetails UCI Extension Field

| OpPlanID          |                                                                                                                                                                                                                                                                |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the unique ID of the TaskPlan to perform the sibling activation command on.                                                                                                                                                                          |             |
| Base Type         | MA_OpPlanID_Type                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                     | Field Usage |
| Version           | unsignedInt                                                                                                                                                                                                                                                    | Optional    |
| Description       | Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval. |             |
| UUID              | UniversallyUniqueIdentifierType                                                                                                                                                                                                                                | Required    |
| Description       | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.                                                                                               |             |
| DescriptiveLabel  | VisibleString256Type                                                                                                                                                                                                                                           | Optional    |
| Description       | Human readable text label with content meaningful to human operators.                                                                                                                                                                                          |             |

Table A-1-341: P2P UCI Extensions MA\_ApprovalRequestMT - OpPlanID UCI Extension Field

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MissionPlanActivationApproval</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Description</b>                   | Indicates the MissionPlanActivationCommand details that are under review for approval. If more than one instance of this element is given, each should correspond to a different MissionPlanID. For example, if the intent is to transition from one MissionPlan to another, the new MissionPlan can be activated in one instance and the old MissionPlan can be deactivated in another instance. This transition would be subject to approval. |
| <b>Base Type</b>                     | MA_MissionPlanActivationCommandType                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Message Structure</b>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Field Name               | Field Type                                                                                                                   | Field Usage |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>MissionPlanID</b>     | MissionPlanID_Type                                                                                                           | Required    |
| Description              | Indicates the unique ID of the MissionPlan whose commanded activation state or states are summarized in the sibling element. |             |
| <b>ActivationDetails</b> | MA_MissionPlanActivationDetailsType                                                                                          | Required    |
| Description              | Indicates the commanded activation details.                                                                                  |             |

**Table A-1-342: P2P UCI Extensions MA\_ApprovalRequestMT - MissionPlanActivationApproval UCI Extension Field**

| ApprovalReferences |                                                                                                           |             |
|--------------------|-----------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates the policy that resulted in the ApprovalRequest along with references to the data under review. |             |
| Base Type          | ApprovalRequestPolicyReferenceType                                                                        |             |
| Message Structure  |                                                                                                           |             |
| Field Name         | Field Type                                                                                                | Field Usage |
| ApprovalPolicyID   | ApprovalPolicyID_Type                                                                                     | Required    |
| Description        | Indicates the unique ID of the ApprovalPolicy that resulted in this ApprovalRequest.                      |             |
| ApprovalItem       | ApprovalRequestItemReferenceType                                                                          | Required    |
| Description        | Indicates the ID of the specific item that is being requested for approval.                               |             |

**Table A-1-343: P2P UCI Extensions MA\_ApprovalRequestMT - ApprovalReferences UCI Extension Field**

| ApprovalItem                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                   | Indicates the ID of the specific item that is being requested for approval.                                                                                                                                                                                                                                                                                                                                                                     |             |
| Base Type                     | ApprovalRequestItemReferenceType                                                                                                                                                                                                                                                                                                                                                                                                                |             |
| Message Structure             |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| Field Name                    | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                      | Field Usage |
| PlanApproval                  | PlanReferenceID_ChoiceType                                                                                                                                                                                                                                                                                                                                                                                                                      | Required    |
| Description                   | Indicates the unique ID of the *Plan that is under review for approval.                                                                                                                                                                                                                                                                                                                                                                         |             |
| RequirementExecutionApproval  | ApprovalRequestItemType                                                                                                                                                                                                                                                                                                                                                                                                                         | Required    |
| Description                   | Indicates the Requirement, including any associated DMPIs, that is under review for approval to execute.                                                                                                                                                                                                                                                                                                                                        |             |
| MissionPlanActivationApproval | MissionPlanActivationCommandType                                                                                                                                                                                                                                                                                                                                                                                                                | Repeated    |
| Description                   | Indicates the MissionPlanActivationCommand details that are under review for approval. If more than one instance of this element is given, each should correspond to a different MissionPlanID. For example, if the intent is to transition from one MissionPlan to another, the new MissionPlan can be activated in one instance and the old MissionPlan can be deactivated in another instance. This transition would be subject to approval. |             |

**Table A-1-344: P2P UCI Extensions MA\_ApprovalRequestMT - ApprovalItem UCI Extension Field**

| ByExecutionPlanSet |
|--------------------|
|--------------------|

|                    |                                                                                                                                                   |             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates simultaneous activation of all sub-*Plans (ActivityPlan, RoutePlan for example) of an *ExecutionPlanSet into the same activation state. |             |
| Base Type          | MA_ExecutionPlanSetActivationType                                                                                                                 |             |
| Message Structure  |                                                                                                                                                   |             |
| Field Name         | Field Type                                                                                                                                        | Field Usage |
| ExecutionPlanSetID | ExecutionPlanSetID_Type                                                                                                                           | Required    |
| Description        | Indicates the unique ID of the execution plan set.                                                                                                |             |
| CommandType        | PlanActivationCommandEnum                                                                                                                         | Required    |
| Description        | Indicates the activation to perform on the associated execution plan set.                                                                         |             |

Table A-1-345: P2P UCI Extensions MA\_ApprovalRequestMT - ByExecutionPlanSet UCI Extension Field

| MessageData        |                                                                                                                                                                                                                                                      |             |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                              |             |
| Base Type          | ApprovalRequestMDT                                                                                                                                                                                                                                   |             |
| Message Structure  |                                                                                                                                                                                                                                                      |             |
| Field Name         | Field Type                                                                                                                                                                                                                                           | Field Usage |
| Approver           | OperatorRoleType                                                                                                                                                                                                                                     | Required    |
| Description        | An Operator Role or non-Operator identifier that is allowed to approve the included approval policy and approval item.                                                                                                                               |             |
| ApprovalReferences | ApprovalRequestPolicyReferenceType                                                                                                                                                                                                                   | Required    |
| Description        | Indicates the policy that resulted in the ApprovalRequest along with references to the data under review.                                                                                                                                            |             |
| RespondBy          | DateTimeType                                                                                                                                                                                                                                         | Required    |
| Description        | If a response is not received from this approval authority by the given time, the approval policy's default response will be used. This field should be driven based on the timeout specified within the approval policy for approvals of this type. |             |
| RequestID          | RequestID_Type                                                                                                                                                                                                                                       | Required    |
| Description        | Indicates the unique ID of the request message.                                                                                                                                                                                                      |             |
| RequestState       | RequestStateEnum                                                                                                                                                                                                                                     | Required    |
| Description        | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.                                                                                                            |             |

Table A-1-346: P2P UCI Extensions MA\_ApprovalRequestMT - MessageData UCI Extension Field

| BySubPlan         |                                                                                                                                        |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates activation by sub-*Plan (RoutePlan or OrbitPlan for example) of the MissionPlan, with potentially different states for each. |             |
| Base Type         | MA_MissionPlanSubplanActivationType                                                                                                    |             |
| Message Structure |                                                                                                                                        |             |
| Field Name        | Field Type                                                                                                                             | Field Usage |

|                          |                                                                     |          |
|--------------------------|---------------------------------------------------------------------|----------|
| <b>TaskPlan</b>          | TaskPlanActivationType                                              | Repeated |
| Description              | Indicates commanded details for activation of a TaskPlan.           |          |
| <b>RoutePlan</b>         | MA_RoutePlanActivationType                                          | Repeated |
| Description              | Indicates commanded details for activation of a RoutePlan.          |          |
| <b>RouteActivityPlan</b> | RouteActivityPlanActivationType                                     | Repeated |
| Description              | Indicates commanded details for activation of a RouteActivityPlan.  |          |
| <b>OrbitPlan</b>         | OrbitPlanActivationType                                             | Repeated |
| Description              | Indicates commanded details for activation of an OrbitPlan.         |          |
| <b>OrbitActivityPlan</b> | OrbitActivityPlanActivationType                                     | Repeated |
| Description              | Indicates commanded details for activation of an OrbitActivityPlan. |          |
| <b>ActivityPlan</b>      | ActivityPlanActivationType                                          | Repeated |
| Description              | Indicates commanded details for activation of an ActivityPlan.      |          |
| <b>EffectPlan</b>        | EffectPlanActivationType                                            | Repeated |
| Description              | Indicates commanded details for activation of an EffectPlan.        |          |
| <b>ActionPlan</b>        | ActionPlanActivationType                                            | Repeated |
| Description              | Indicates commanded details for activation of an ActionPlan.        |          |
| <b>ResponsePlan</b>      | ResponsePlanActivationType                                          | Repeated |
| Description              | Indicates commanded details for activation of a ResponsePlan.       |          |
| <b>OpPlan</b>            | MA_OpPlanActivationType                                             | Repeated |
| Description              | Indicates commanded details for activation of an OpPlan.            |          |

Table A-1-347: P2P UCI Extensions MA\_ApprovalRequestMT - BySubPlan UCI Extension Field

| OpPlan            |                                                                                       |             |
|-------------------|---------------------------------------------------------------------------------------|-------------|
| Description       | Indicates commanded details for activation of an OpPlan.                              |             |
| Base Type         | MA_OpPlanActivationType                                                               |             |
| Message Structure |                                                                                       |             |
| Field Name        | Field Type                                                                            | Field Usage |
| OpPlanID          | MA_OpPlanID_Type                                                                      | Required    |
| Description       | Indicates the unique ID of the TaskPlan to perform the sibling activation command on. |             |
| CommandType       | PlanActivationCommandEnum                                                             | Required    |
| Description       | Indicates the activation to perform on the associated TaskPlan.                       |             |

Table A-1-348: P2P UCI Extensions MA\_ApprovalRequestMT - OpPlan UCI Extension Field

### 1.3.2.59 MA\_ApprovalRequestStatusMT UCI Extension

|                                   |                                                                                                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>MA_ApprovalRequestStatusMT</b> |                                                                                                       |
| <b>Description</b>                | See the annotation in the associated message for an overall description of the message and this type. |

|                    |                                           |
|--------------------|-------------------------------------------|
|                    | See the XSD File for more info            |
| <b>Ext. Fields</b> | MessageData : MA_ApprovalRequestStatusMDT |

Table A-1-349: P2P UCI Extensions MA\_ApprovalRequestStatusMT UCI Extension

| MessageData                          |                                                                                                                                                                                                                                                                                                     |             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                             |             |
| Base Type                            | MA_ApprovalRequestStatusMDT                                                                                                                                                                                                                                                                         |             |
| Message Structure                    |                                                                                                                                                                                                                                                                                                     |             |
| Field Name                           | Field Type                                                                                                                                                                                                                                                                                          | Field Usage |
| ApprovalRequestProcessingState       | ApprovalStatusEnum                                                                                                                                                                                                                                                                                  | Required    |
| Description                          | The ApprovalState indicates the current state of the approval for an approval request.                                                                                                                                                                                                              |             |
| ApprovalRequestProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                    | Optional    |
| Description                          | Indicates details regarding the reason for and/or a contributing factor to the sibling ApprovalRequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |
| OperatorID                           | OperatorID_Type                                                                                                                                                                                                                                                                                     | Optional    |
| Description                          | Indicates the Operator that provided the approval/rejection, if available. This element can be beneficial when systems require two-person integrity.                                                                                                                                                |             |
| ApprovedDMPI_ID                      | DMPI_ID_Type                                                                                                                                                                                                                                                                                        | Repeated    |
| Description                          | If the ApprovalRequest was for DMPI approval then this element contains the list of approved DMPIs. If any DMPIs are approved, the sibling ApprovalState will indicate that the request is approved.                                                                                                |             |
| RequestID                            | RequestID_Type                                                                                                                                                                                                                                                                                      | Required    |
| Description                          | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                                |             |
| RequestProcessingState               | RequestProcessingStateEnum                                                                                                                                                                                                                                                                          | Required    |
| Description                          | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                        |             |
| RequestProcessingStateReason         | CannotComplyType                                                                                                                                                                                                                                                                                    | Optional    |
| Description                          | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.         |             |

Table A-1-350: P2P UCI Extensions MA\_ApprovalRequestStatusMT - MessageData UCI Extension Field

### 1.3.2.60 MA\_EffectCapabilityMT UCI Extension

| MA_EffectCapabilityMT |                                                                |
|-----------------------|----------------------------------------------------------------|
| <b>Description</b>    | This message represents the Effect category of UCI Capability. |



|                    |                                   |
|--------------------|-----------------------------------|
|                    | See the XSD File for more info    |
| <b>Ext. Fields</b> | AssociatedTaskType : TaskTypeEnum |

Table A-1-351: P2P UCI Extensions MA\_EffectCapabilityMT UCI Extension

| AssociatedTaskType          |                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>          | Indicates a Task type associated with the Effect. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition. |
| <b>Base Type</b>            | TaskTypeEnum                                                                                                                                                                     |
| Message Structure           |                                                                                                                                                                                  |
| Enumeration Literal         | Description                                                                                                                                                                      |
| AIR_SAMPLE                  | Indicates the task is to sample and analyze the air around the vehicle.                                                                                                          |
| AMTI                        | Indicates the task is to track air targets using an Airborne Moving Target Indicator.                                                                                            |
| AO                          | Indicates the task is to employ an Active Optical capability.                                                                                                                    |
| CARGO_DELIVERY              | Indicates the task is to deliver cargo.                                                                                                                                          |
| COMINT                      | Indicates the task is to employ a communications intelligence capability.                                                                                                        |
| COMM_RELAY                  | Indicates the task is to relay communications from other battlespace participants.                                                                                               |
| COUNTER_SPACE               | Indicates the task is to employ a Counter Space Capability.                                                                                                                      |
| EA                          | Indicates the task is to employ an electronic attack capability.                                                                                                                 |
| ESM                         | Indicates the task is to employ an Electronic Support Measure.                                                                                                                   |
| FLIGHT                      | Indicates the task is to fly.                                                                                                                                                    |
| ORBIT_CHANGE                | Indicates a task to perform an orbit change via a spacecraft maneuver.                                                                                                           |
| ORBITAL_SURVEILLANCE        | Indicates a task to perform orbital surveillance.                                                                                                                                |
| ORBITAL_SURVEILLANCE_SENSOR | Indicates a task to perform orbital surveillance via sensor collection capabilities.                                                                                             |
| PO                          | Indicates the task is to employ a Passive Optical capability.                                                                                                                    |
| REFUEL                      | Indicates the task is to refuel another System or Entity.                                                                                                                        |
| SAR                         | Indicates the task is to employ a Synthetic Aperture Radar capability.                                                                                                           |
| SMTI                        | Indicates the task is to track surface targets using a Surface Moving Target Indicator.                                                                                          |
| STRIKE                      | Indicates the task is to strike a target.                                                                                                                                        |
| SYSTEM_DEPLOYMENT           | Indicates the task is to deploy a system.                                                                                                                                        |
| TACTICAL_ORDER              | Indicates the task is to perform a tactical order.                                                                                                                               |
| WEATHER_RADAR               | Indicates the task is to collect weather data.                                                                                                                                   |

Table A-1-352: P2P UCI Extensions MA\_EffectCapabilityMT - AssociatedTaskType UCI Extension Field



### 1.3.2.61 MA\_FlightCapabilityMT UCI Extension

| MA_FlightCapabilityMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>    | Extended to indicate the type of flight capability via the FlightCapabilityEnum and provide per-control mode performance parameters.<br><br>See the XSD File for more info                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Ext. Fields</b>    | HSA_CSA_PerformanceProfile : MA_FlightControlModesPerformanceProfileType<br>WaypointFollowingPerformanceProfile : MA_FlightControlModesPerformanceProfileType<br>CurveFollowingPerformanceProfile : MA_FlightControlModesPerformanceProfileType<br>AccelerationLimit : MA_BodyReferenceAccelerationType<br>AccelerationLimitValue : MA_AccelerationLimitPairType<br>BodyReferenceOrientationRate : MA_BodyReferenceOrientationRateType<br>OrientationRateLimits : MA_BodyReferenceOrientationRateType<br>AirspeedPair : PathSegmentSpeedValueType |

**Table A-1-353: P2P UCI Extensions MA\_FlightCapabilityMT UCI Extension**

| HSA_CSA_PerformanceProfile |                                                                                                                                                                                                                                                                                                                       |             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                | Indicates performance profile for HSA_CSA flight control mode.                                                                                                                                                                                                                                                        |             |
| Base Type                  | MA_FlightControlModesPerformanceProfileType                                                                                                                                                                                                                                                                           |             |
| Message Structure          |                                                                                                                                                                                                                                                                                                                       |             |
| Field Name                 | Field Type                                                                                                                                                                                                                                                                                                            | Field Usage |
| MinAirspeed                | MA_AirspeedLimitType                                                                                                                                                                                                                                                                                                  | Repeated    |
| Description                | Minimum allowable airspeed given as a function of altitude.                                                                                                                                                                                                                                                           |             |
| MaxAirspeed                | MA_AirspeedLimitType                                                                                                                                                                                                                                                                                                  | Repeated    |
| Description                | Maximum allowable airspeed given as a function of altitude.                                                                                                                                                                                                                                                           |             |
| BestEnduranceAirspeed      | MA_AirspeedLimitType                                                                                                                                                                                                                                                                                                  | Repeated    |
| Description                | Airspeed of maximum endurance given as a function of altitude.                                                                                                                                                                                                                                                        |             |
| BestRangeAirspeed          | MA_AirspeedLimitType                                                                                                                                                                                                                                                                                                  | Repeated    |
| Description                | Airspeed of maximum range given as a function of altitude.                                                                                                                                                                                                                                                            |             |
| MinAltitude                | AltitudeReferenceType                                                                                                                                                                                                                                                                                                 | Optional    |
| Description                | Minimum allowable altitude.                                                                                                                                                                                                                                                                                           |             |
| MaxAltitude                | AltitudeReferenceType                                                                                                                                                                                                                                                                                                 | Optional    |
| Description                | Maximum allowable altitude.                                                                                                                                                                                                                                                                                           |             |
| MinAccelerationLimits      | MA_AccelerationLimitsType                                                                                                                                                                                                                                                                                             | Repeated    |
| Description                | The vehicle's acceleration limits in the current operating domain of the system as a function of time. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories. |             |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>ExcessPowerOrAcceleration</b> | MA_ExcessPowerOrAccelerationType                                                                                                                                                                                                                                                                                                                                                                                                | Optional |
| Description                      | Provides either specific excess power or paired acceleration and climb rate data for the platform.                                                                                                                                                                                                                                                                                                                              |          |
| <b>MaxOrientationLimits</b>      | MA_OrientationLimitType                                                                                                                                                                                                                                                                                                                                                                                                         | Repeated |
| Description                      | Euler Angle Sequence describing the maximum orientation of the vehicle in the order yaw, pitch, roll as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                           |          |
| <b>MaxOrientationRateLimits</b>  | MA_OrientationRateLimitsType                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated |
| Description                      | Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories. |          |
| <b>MaxTurnRate</b>               | AngleRateType                                                                                                                                                                                                                                                                                                                                                                                                                   | Optional |
| Description                      | Maximum allowable turn rate in radians/sec.                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>MaxDescentRate</b>            | MA_SpeedType                                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated |
| Description                      | Maximum allowable descent rate as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>MaxDeceleration</b>           | MA_AccelerationLimitsType                                                                                                                                                                                                                                                                                                                                                                                                       | Repeated |
| Description                      | Maximum allowable deceleration limits as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                          |          |
| <b>FuelBurnRate</b>              | MA_FuelBurnRateType                                                                                                                                                                                                                                                                                                                                                                                                             | Repeated |
| Description                      | Fuel burn rate as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                                                 |          |

**Table A-1-354: P2P UCI Extensions MA\_FlightCapabilityMT - HSA\_CSA\_PerformanceProfile UCI Extension Field**

| WaypointFollowingPerformanceProfile |                                                                           |             |
|-------------------------------------|---------------------------------------------------------------------------|-------------|
| Description                         | Indicates performance profile for Waypoint Following flight control mode. |             |
| Base Type                           | MA_FlightControlModesPerformanceProfileType                               |             |
| Message Structure                   |                                                                           |             |
| Field Name                          | Field Type                                                                | Field Usage |
| MinAirspeed                         | MA_AirspeedLimitType                                                      | Repeated    |
| Description                         | Minimum allowable airspeed given as a function of altitude.               |             |
| MaxAirspeed                         | MA_AirspeedLimitType                                                      | Repeated    |
| Description                         | Maximum allowable airspeed given as a function of altitude.               |             |
| BestEnduranceAirspeed               | MA_AirspeedLimitType                                                      | Repeated    |
| Description                         | Airspeed of maximum endurance given as a function of altitude.            |             |
| BestRangeAirspeed                   | MA_AirspeedLimitType                                                      | Repeated    |
| Description                         | Airspeed of maximum range given as a function of altitude.                |             |
| MinAltitude                         | AltitudeReferenceType                                                     | Optional    |
| Description                         | Minimum allowable altitude.                                               |             |
| MaxAltitude                         | AltitudeReferenceType                                                     | Optional    |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                      | Maximum allowable altitude.                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>MinAccelerationLimits</b>     | MA_AccelerationLimitsType                                                                                                                                                                                                                                                                                                                                                                                                       | Repeated |
| Description                      | The vehicle's acceleration limits in the current operating domain of the system as a function of time. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories.                                                                                                           |          |
| <b>ExcessPowerOrAcceleration</b> | MA_ExcessPowerOrAccelerationType                                                                                                                                                                                                                                                                                                                                                                                                | Optional |
| Description                      | Provides either specific excess power or paired acceleration and climb rate data for the platform.                                                                                                                                                                                                                                                                                                                              |          |
| <b>MaxOrientationLimits</b>      | MA_OrientationLimitType                                                                                                                                                                                                                                                                                                                                                                                                         | Repeated |
| Description                      | Euler Angle Sequence describing the maximum orientation of the vehicle in the order yaw, pitch, roll as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                           |          |
| <b>MaxOrientationRateLimits</b>  | MA_OrientationRateLimitsType                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated |
| Description                      | Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories. |          |
| <b>MaxTurnRate</b>               | AngleRateType                                                                                                                                                                                                                                                                                                                                                                                                                   | Optional |
| Description                      | Maximum allowable turn rate in radians/sec.                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>MaxDescentRate</b>            | MA_SpeedType                                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated |
| Description                      | Maximum allowable descent rate as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>MaxDeceleration</b>           | MA_AccelerationLimitsType                                                                                                                                                                                                                                                                                                                                                                                                       | Repeated |
| Description                      | Maximum allowable deceleration limits as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                          |          |
| <b>FuelBurnRate</b>              | MA_FuelBurnRateType                                                                                                                                                                                                                                                                                                                                                                                                             | Repeated |
| Description                      | Fuel burn rate as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                                                 |          |

**Table A-1-355: P2P UCI Extensions MA\_FlightCapabilityMT - WaypointFollowingPerformanceProfile UCI Extension Field**

| CurveFollowingPerformanceProfile |                                                                        |             |
|----------------------------------|------------------------------------------------------------------------|-------------|
| Description                      | Indicates performance profile for Curve Following flight control mode. |             |
| Base Type                        | MA_FlightControlModesPerformanceProfileType                            |             |
| Message Structure                |                                                                        |             |
| Field Name                       | Field Type                                                             | Field Usage |
| MinAirspeed                      | MA_AirspeedLimitType                                                   | Repeated    |
| Description                      | Minimum allowable airspeed given as a function of altitude.            |             |
| MaxAirspeed                      | MA_AirspeedLimitType                                                   | Repeated    |
| Description                      | Maximum allowable airspeed given as a function of altitude.            |             |
| BestEnduranceAirspeed            | MA_AirspeedLimitType                                                   | Repeated    |
| Description                      | Airspeed of maximum endurance given as a function of altitude.         |             |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>BestRangeAirspeed</b>         | MA_AirspeedLimitType                                                                                                                                                                                                                                                                                                                                                                                                            | Repeated |
| Description                      | Airspeed of maximum range given as a function of altitude.                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>MinAltitude</b>               | AltitudeReferenceType                                                                                                                                                                                                                                                                                                                                                                                                           | Optional |
| Description                      | Minimum allowable altitude.                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>MaxAltitude</b>               | AltitudeReferenceType                                                                                                                                                                                                                                                                                                                                                                                                           | Optional |
| Description                      | Maximum allowable altitude.                                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>MinAccelerationLimits</b>     | MA_AccelerationLimitsType                                                                                                                                                                                                                                                                                                                                                                                                       | Repeated |
| Description                      | The vehicle's acceleration limits in the current operating domain of the system as a function of time. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories.                                                                                                           |          |
| <b>ExcessPowerOrAcceleration</b> | MA_ExcessPowerOrAccelerationType                                                                                                                                                                                                                                                                                                                                                                                                | Optional |
| Description                      | Provides either specific excess power or paired acceleration and climb rate data for the platform.                                                                                                                                                                                                                                                                                                                              |          |
| <b>MaxOrientationLimits</b>      | MA_OrientationLimitType                                                                                                                                                                                                                                                                                                                                                                                                         | Repeated |
| Description                      | Euler Angle Sequence describing the maximum orientation of the vehicle in the order yaw, pitch, roll as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                           |          |
| <b>MaxOrientationRateLimits</b>  | MA_OrientationRateLimitsType                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated |
| Description                      | Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories. |          |
| <b>MaxTurnRate</b>               | AngleRateType                                                                                                                                                                                                                                                                                                                                                                                                                   | Optional |
| Description                      | Maximum allowable turn rate in radians/sec.                                                                                                                                                                                                                                                                                                                                                                                     |          |
| <b>MaxDescentRate</b>            | MA_SpeedType                                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated |
| Description                      | Maximum allowable descent rate as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>MaxDeceleration</b>           | MA_AccelerationLimitsType                                                                                                                                                                                                                                                                                                                                                                                                       | Repeated |
| Description                      | Maximum allowable deceleration limits as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                          |          |
| <b>FuelBurnRate</b>              | MA_FuelBurnRateType                                                                                                                                                                                                                                                                                                                                                                                                             | Repeated |
| Description                      | Fuel burn rate as a function of airspeed, altitude, and weight.                                                                                                                                                                                                                                                                                                                                                                 |          |

**Table A-1-356: P2P UCI Extensions MA\_FlightCapabilityMT - CurveFollowingPerformanceProfile UCI Extension Field**

| AccelerationLimit |                                                                          |             |
|-------------------|--------------------------------------------------------------------------|-------------|
| Description       | Specified acceleration limits in body reference or NED coordinate frame. |             |
| Base Type         | MA_BodyReferenceAccelerationType                                         |             |
| Message Structure |                                                                          |             |
| Field Name        | Field Type                                                               | Field Usage |
| X_Accel           | AccelerationType                                                         | Required    |

|                  |                                                                                                                                                                                                       |          |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description      | Body-referenced acceleration component along the longitudinal (x or roll) axis, measured in meters/sec^2                                                                                              |          |
| <b>Y_Accel</b>   | AccelerationType                                                                                                                                                                                      | Required |
| Description      | Body-referenced acceleration component along the lateral (y or pitch) axis, measured in meters/sec^2                                                                                                  |          |
| <b>Z_Accel</b>   | AccelerationType                                                                                                                                                                                      | Required |
| Description      | Body-referenced acceleration component along the vertical (z or yaw) axis, measured in meters/sec^2.                                                                                                  |          |
| <b>Timestamp</b> | DateTimeType                                                                                                                                                                                          | Optional |
| Description      | Indicates the time when the acceleration data was measured or last known to be correct. The element can also indicate the time at which a future velocity of a System, Entity, etc. will be achieved. |          |

Table A-1-357: P2P UCI Extensions MA\_FlightCapabilityMT - AccelerationLimit UCI Extension Field

| AccelerationLimitValue       |                                                                                                                                      |             |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | Value for the specified acceleration limit, provided either as a set of body referenced orientation rates, or a unitless Mach value. |             |
| Base Type                    | MA_AccelerationLimitPairType                                                                                                         |             |
| Message Structure            |                                                                                                                                      |             |
| Field Name                   | Field Type                                                                                                                           | Field Usage |
| BodyReferenceOrientationRate | MA_BodyReferenceOrientationRateType                                                                                                  | Required    |
| Description                  | Body reference orientation rates to be paired with the specified acceleration limit.                                                 |             |
| MachValue                    | MachType                                                                                                                             | Required    |
| Description                  | Unitless Mach value to be paired with the specified acceleration limit.                                                              |             |

Table A-1-358: P2P UCI Extensions MA\_FlightCapabilityMT - AccelerationLimitValue UCI Extension Field

| BodyReferenceOrientationRate |                                                                                                           |             |
|------------------------------|-----------------------------------------------------------------------------------------------------------|-------------|
| Description                  | Body reference orientation rates to be paired with the specified acceleration limit.                      |             |
| Base Type                    | MA_BodyReferenceOrientationRateType                                                                       |             |
| Message Structure            |                                                                                                           |             |
| Field Name                   | Field Type                                                                                                | Field Usage |
| RollRate                     | AngleRateType                                                                                             | Required    |
| Description                  | Body-referenced attitude rate component along the longitudinal (x or roll) axis, measured in radians/sec. |             |
| PitchRate                    | AngleRateType                                                                                             | Required    |
| Description                  | Body-referenced attitude rate component along the lateral (y or pitch) axis, measured in radians/sec.     |             |
| YawRate                      | AngleRateType                                                                                             | Required    |
| Description                  | Body-referenced attitude rate component along the vertical (z or yaw) axis, measured in radians/sec.      |             |

| Timestamp   | DateTimeType                                                                                                                                                                                      | Optional |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description | Indicates the time when the velocity data was measured or last known to be correct. The element can also indicate the time at which a future velocity of a System, Entity, etc. will be achieved. |          |

**Table A-1-359: P2P UCI Extensions MA\_FlightCapabilityMT - BodyReferenceOrientationRate UCI Extension Field**

| OrientationRateLimits |                                                                                                                                                                                                   |             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description           | Specified orientation rate limits in body reference or NED coordinate frame.                                                                                                                      |             |
| Base Type             | MA_BodyReferenceOrientationRateType                                                                                                                                                               |             |
| Message Structure     |                                                                                                                                                                                                   |             |
| Field Name            | Field Type                                                                                                                                                                                        | Field Usage |
| RollRate              | AngleRateType                                                                                                                                                                                     | Required    |
| Description           | Body-referenced attitude rate component along the longitudinal (x or roll) axis, measured in radians/sec.                                                                                         |             |
| PitchRate             | AngleRateType                                                                                                                                                                                     | Required    |
| Description           | Body-referenced attitude rate component along the lateral (y or pitch) axis, measured in radians/sec.                                                                                             |             |
| YawRate               | AngleRateType                                                                                                                                                                                     | Required    |
| Description           | Body-referenced attitude rate component along the vertical (z or yaw) axis, measured in radians/sec.                                                                                              |             |
| Timestamp             | DateTimeType                                                                                                                                                                                      | Optional    |
| Description           | Indicates the time when the velocity data was measured or last known to be correct. The element can also indicate the time at which a future velocity of a System, Entity, etc. will be achieved. |             |

**Table A-1-360: P2P UCI Extensions MA\_FlightCapabilityMT - OrientationRateLimits UCI Extension Field**

| AirspeedPair      |                                                                                                                 |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Airspeed pair value for associated orientation rate limit, specified in body reference or NED coordinate frame. |             |
| Base Type         | PathSegmentSpeedValueType                                                                                       |             |
| Message Structure |                                                                                                                 |             |
| Field Name        | Field Type                                                                                                      | Field Usage |
| Value             | SpeedType                                                                                                       | Required    |
| Description       | Indicates the speed along the PathSegment at the EndPoint.                                                      |             |
| Reference         | SpeedReferenceEnum                                                                                              | Required    |
| Description       | Indicates whether the given vehicle speed is reference to ground speed or true air speed.                       |             |

**Table A-1-361: P2P UCI Extensions MA\_FlightCapabilityMT - AirspeedPair UCI Extension Field**

## 1.3.2.62 MA\_FlightCapabilityStatusMT UCI Extension

| MA_FlightCapabilityStatusMT |                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------|
| Description                 | See annotations in child elements and messages/elements that use this type for details. |
|                             | See the XSD File for more info                                                          |
| Ext. Fields                 | Capability : MA_FlightCapabilityType<br>MessageData : MA_FlightCapabilityMDT            |

Table A-1-362: P2P UCI Extensions MA\_FlightCapabilityStatusMT UCI Extension

| Capability                         |                                                                                                                                                                                                                                                               |             |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                        | Provides a list of the specific Flight Capabilities as defined by the FlightCapabilityEnum.                                                                                                                                                                   |             |
| Base Type                          | MA_FlightCapabilityType                                                                                                                                                                                                                                       |             |
| Message Structure                  |                                                                                                                                                                                                                                                               |             |
| Field Name                         | Field Type                                                                                                                                                                                                                                                    | Field Usage |
| CapabilityType                     | MA_FlightCapabilityEnum                                                                                                                                                                                                                                       | Repeated    |
| Description                        | Indicates the type of Flight capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.                                                                                           |             |
| FlightCapabilityPerformanceProfile | MA_FlightCapabilityPerformanceProfileType                                                                                                                                                                                                                     | Optional    |
| Description                        | Provides per-control mode performance parameters. Enables autonomy commands to properly conform to envelope limits or limit summary values before sending a FlightCommand to the vehicle.                                                                     |             |
| AcceptedInterface                  | CapabilityControlInterfacesEnum                                                                                                                                                                                                                               | Repeated    |
| Description                        | Indicates an accepted control interface for the Capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.                                                                        |             |
| CommsRequired                      | boolean                                                                                                                                                                                                                                                       | Optional    |
| Description                        | Indicates whether (TRUE) or not (FALSE) communications are required to use the Capability. Omitted is equivalent to FALSE. For example, an unmanned system with no onboard storage would require communications to disseminate the outputs of the Capability. |             |
| CapabilityID                       | CapabilityID_Type                                                                                                                                                                                                                                             | Required    |
| Description                        | Indicates the unique ID of this Capability. This ID is used to invoke the Capability via Command messages, report its activity via Activity messages, etc.                                                                                                    |             |
| DomainPairing                      | EnvironmentPairingEnum                                                                                                                                                                                                                                        | Repeated    |
| Description                        | Indicates the domain/environment which the Capability is employed from and the "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.                    |             |
| ProductOutput                      | ProductOutputType                                                                                                                                                                                                                                             | Repeated    |
| Description                        | Indicates a Product that can be an output of the Capability.                                                                                                                                                                                                  |             |

| PackageOperation | PackageOperationEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Optional |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description      | Indicates if and how the operation of the Capability can be coordinated with a partner Capability of another System in a multi-System Package. Omitting this element is equivalent to the enumerate NO_PACKAGE_OPERATION. The actual Package configuration including per-Capability partnering between Systems is given in the PackageStatus message. For example, a System that can perform cooperative geolocation could advertise with this element in the ESM_Capability message. A System that can navigate in a swarm with other Systems could advertise with this element in the FlightCapability message. |          |

Table A-1-363: P2P UCI Extensions MA\_FlightCapabilityStatusMT - Capability UCI Extension Field

| MessageData       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| Base Type         | MA_FlightCapabilityMDT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Field Usage |
| Capability        | MA_FlightCapabilityType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Repeated    |
| Description       | Provides a list of the specific Flight Capabilities as defined by the FlightCapabilityEnum.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| AvailableMDF      | MDF_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description       | Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability. |             |

Table A-1-364: P2P UCI Extensions MA\_FlightCapabilityStatusMT - MessageData UCI Extension Field

### 1.3.2.63 MA\_TaskMT UCI Extension

| MA_TaskMT   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | <p>This message specifies a Task that a System will perform against a specific entity or geographical location. Tasks include flight, collection, strike, electronic attack, directed energy, laser designation, cargo delivery and communication relay. They are abstracted to a level of C2 and autonomy derived from UCI operational vignettes. Tasks enable services, operators and other actors to do automated, dynamic, machine-to-machine allocation of Tasks to Systems and subsequent mission planning. If necessary, authorization of Users to Task Systems can be controlled using the Authorization message.</p> <p>See the XSD File for more info</p> |
| Ext. Fields | <p>TaskType : MA_TaskType</p> <p>TaskPlurality : MA_TaskPluralityEnum</p> <p>ConstraintID : MA_ConstraintID_Type</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |



|  |                                              |
|--|----------------------------------------------|
|  | HSA_CSA : MA_HSA_CSA_Type                    |
|  | WaypointFollowing : MA_WaypointFollowingType |
|  | CurveFollowing : MA_CurveControlType         |
|  | Speed : PathSegmentSpeedType                 |
|  | Heading : MA_DirectionReferenceType          |
|  | SpeedChoice : MA_HSA_SpeedChoiceType         |
|  | SpeedValue : PathSegmentSpeedValueType       |
|  | CurveFitMethod : MA_CurveFitMethodEnum       |

Table A-1-365: P2P UCI Extensions MA\_TaskMT UCI Extension

| TaskType          |                                                                                                                                                                                               |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Identifies the type of this Task instance.                                                                                                                                                    |             |
| Base Type         | MA_TaskType                                                                                                                                                                                   |             |
| Message Structure |                                                                                                                                                                                               |             |
| Field Name        | Field Type                                                                                                                                                                                    | Field Usage |
| AirSample         | AirSampleTaskType                                                                                                                                                                             | Required    |
| Description       | Air sample includes direct sampling of the air (SAMPLE) and remote sensing with spectral analysis (SPECTROMETER) with the intent of detecting NBC events.                                     |             |
| AMTI              | AMTI_TaskType                                                                                                                                                                                 | Required    |
| Description       | Indicates a Task to collect Air Moving Target Indicator (AMTI) data.                                                                                                                          |             |
| AO                | AO_TaskType                                                                                                                                                                                   | Required    |
| Description       | Indicates a Task to perform an optical emission such as laser designation.                                                                                                                    |             |
| CAP               | MA_CAP_TaskType                                                                                                                                                                               | Required    |
| Description       | Indicates a Task to perform Combat Air Patrol.                                                                                                                                                |             |
| CargoDelivery     | CargoDeliveryTaskType                                                                                                                                                                         | Required    |
| Description       | Indicates a Task to transfer cargo between locations.                                                                                                                                         |             |
| COMINT            | COMINT_TaskType                                                                                                                                                                               | Required    |
| Description       | Indicates a Task to provide COMINT.                                                                                                                                                           |             |
| CommRelay         | CommRelayTaskType                                                                                                                                                                             | Required    |
| Description       | Indicates a Task to provide communications relay support.                                                                                                                                     |             |
| CounterSpace      | CounterSpaceTaskType                                                                                                                                                                          | Required    |
| Description       | Indicates a Task to employ a CounterSpace capability.                                                                                                                                         |             |
| Escort            | MA_EscortTaskType                                                                                                                                                                             | Required    |
| Description       | Indicates a Task to escort another Entity or System.                                                                                                                                          |             |
| EA                | EA_TaskType                                                                                                                                                                                   | Required    |
| Description       | Indicates a Task to provide electronic attack support to another System. It guides/constrains the EA System by specifying where it should fly, what it should protect and what the threat is. |             |

|                                  |                                                                                                                          |          |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|----------|
| <b>ESM</b>                       | ESM_TaskType                                                                                                             | Required |
| Description                      | Indicates a Task to collect ESM data.                                                                                    |          |
| <b>Flight</b>                    | MA_FlightTaskType                                                                                                        | Required |
| Description                      | Indicates a Task to effect the flight path/plan of the System.                                                           |          |
| <b>Jettison</b>                  | MA_JettisonTaskType                                                                                                      | Required |
| Description                      | Indicates a Task to a jettisonable weapon from the system.                                                               |          |
| <b>OrbitChange</b>               | OrbitChangeTaskType                                                                                                      | Required |
| Description                      | Indicates a task to perform an orbit change via a spacecraft maneuver.                                                   |          |
| <b>OrbitalSurveillance</b>       | OrbitalSurveillanceTaskType                                                                                              | Required |
| Description                      | Indicates an Orbital Surveillance Task.                                                                                  |          |
| <b>OrbitalSurveillanceSensor</b> | OrbitalSurveillanceSensorTaskType                                                                                        | Required |
| Description                      | Indicates a task to perform orbital surveillance sensor tasking.                                                         |          |
| <b>PO</b>                        | PO_TaskType                                                                                                              | Required |
| Description                      | Indicates a Task to collect Passive Optical data, imagery and video as well as perform PO search and track capabilities. |          |
| <b>Refuel</b>                    | RefuelTaskType                                                                                                           | Required |
| Description                      | Indicates a Task for one System to refuel another.                                                                       |          |
| <b>SAR</b>                       | SAR_TaskType                                                                                                             | Required |
| Description                      | Indicates a Task to collect a Synthetic Aperture Radar (SAR) image.                                                      |          |
| <b>SMTI</b>                      | SMTI_TaskType                                                                                                            | Required |
| Description                      | Indicates a Task to collect Moving Target Indicator (MTI) data.                                                          |          |
| <b>Strike</b>                    | StrikeTaskType                                                                                                           | Required |
| Description                      | Indicates a Task to kinetically attack/strike, with a weapon that can be released from the System.                       |          |
| <b>SystemDeployment</b>          | SystemDeploymentTaskType                                                                                                 | Required |
| Description                      | Indicates a task to perform a deployment or release of a system at a specified location.                                 |          |
| <b>TacticalOrder</b>             | TacticalOrderTaskType                                                                                                    | Required |
| Description                      | Indicates a task to perform a tactical order.                                                                            |          |
| <b>WeatherRadar</b>              | WeatherRadarTaskType                                                                                                     | Required |
| Description                      | Indicates a task to collect weather radar data.                                                                          |          |

Table A-1-366: P2P UCI Extensions MA\_TaskMT - TaskType UCI Extension Field

| TaskPlurality      |                                                                                                                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | Indicates whether the task is meant to be executed by a single entity or as a joint operation. A join operation implies forming a group prior to execution. A single entity could be a single system or a single pre-formed group. |
| <b>Base Type</b>   | MA_TaskPluralityEnum                                                                                                                                                                                                               |
| Message Structure  |                                                                                                                                                                                                                                    |

| Enumeration Literal  | Description                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>UNSPECIFIED</b>   | Indicates the task plurality is unspecified, and should be defaulted or determined by the receiving system.                       |
| <b>JOINT</b>         | Indicates the task is to be executed as a joint task. A group will be formed when this task is assigned.                          |
| <b>SINGLE_ENTITY</b> | Indicates the task is to be executed by a single entity. A single entity could be a single system or a single pre-existing group. |

Table A-1-367: P2P UCI Extensions MA\_TaskMT - TaskPlurality UCI Extension Field

| ConstraintID            |                                                                                                                                                                  |                                                                                                                                                                                                                         |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      |                                                                                                                                                                  | Indicates any Constraints that apply to the Task (e.g FormationConstraint). A FormationConstraint implies that if a group is formed to execute this task, the referenced Constraint(s) should be applied to that group. |
| <b>Base Type</b>        |                                                                                                                                                                  | MA_ConstraintID_Type                                                                                                                                                                                                    |
| Message Structure       |                                                                                                                                                                  |                                                                                                                                                                                                                         |
| Field Name              | Field Type                                                                                                                                                       | Field Usage                                                                                                                                                                                                             |
| <b>UUID</b>             | UniversallyUniqueIdentifierType                                                                                                                                  | Required                                                                                                                                                                                                                |
| Description             | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122. |                                                                                                                                                                                                                         |
| <b>DescriptiveLabel</b> | VisibleString256Type                                                                                                                                             | Optional                                                                                                                                                                                                                |
| Description             | Human readable text label with content meaningful to human operators.                                                                                            |                                                                                                                                                                                                                         |

Table A-1-368: P2P UCI Extensions MA\_TaskMT - ConstraintID UCI Extension Field

| HSA_CSA            |                                                                                    |                                                                                                   |
|--------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Description</b> |                                                                                    | This message shall be used to provide the ability to command a new flight vector to the Platform. |
| <b>Base Type</b>   |                                                                                    | MA_HSA_CSA_Type                                                                                   |
| Message Structure  |                                                                                    |                                                                                                   |
| Field Name         | Field Type                                                                         | Field Usage                                                                                       |
| <b>Altitude</b>    | AltitudeReferenceType                                                              | Optional                                                                                          |
| Description        | Altitude hold value to be achieved and reference to altitude measurement source.   |                                                                                                   |
| <b>Speed</b>       | PathSegmentSpeedType                                                               | Optional                                                                                          |
| Description        | Defines speed type (reference frame) for all speed related fields in this message. |                                                                                                   |
| <b>Direction</b>   | MA_DirectionChoiceType                                                             | Optional                                                                                          |
| Description        | Defines heading or course type and reference to hold                               |                                                                                                   |

Table A-1-369: P2P UCI Extensions MA\_TaskMT - HSA\_CSA UCI Extension Field

| WaypointFollowing                                                                                                    |  |
|----------------------------------------------------------------------------------------------------------------------|--|
| <b>Description</b>                                                                                                   |  |
| This message shall be used to provide the ability to send a new waypoint path or segment for the Platform to follow. |  |
| <b>Base Type</b>                                                                                                     |  |
| MA_WaypointFollowingType                                                                                             |  |

| Message Structure |                                                                      |             |
|-------------------|----------------------------------------------------------------------|-------------|
| Field Name        | Field Type                                                           | Field Usage |
| <b>Route</b>      | MA_RouteType                                                         | Required    |
| Description       | Indicates the Paths and Segments that make up the Route to be flown. |             |

Table A-1-370: P2P UCI Extensions MA\_TaskMT - WaypointFollowing UCI Extension Field

| CurveFollowing     |                                                                                                                                      |             |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates the parameters for curve following control.                                                                                |             |
| Base Type          | MA_CurveControlType                                                                                                                  |             |
| Message Structure  |                                                                                                                                      |             |
| Field Name         | Field Type                                                                                                                           | Field Usage |
| CurveSegments      | MA_NURBS_PointType                                                                                                                   | Repeated    |
| Description        | Specifies the curve segments as a set of NURBS points.                                                                               |             |
| AppendCurve        | EmptyType                                                                                                                            | Optional    |
| Description        | Specifies this curve will be appended to a previous curve command. This is only valid if the vehicle is currently executing a curve. |             |
| EndOfCurveBehavior | MA_EndOfCurveBehaviorEnum                                                                                                            | Optional    |
| Description        | Specifies the vehicle behavior at the completion of the curve.                                                                       |             |

Table A-1-371: P2P UCI Extensions MA\_TaskMT - CurveFollowing UCI Extension Field

| Speed             |                                                                                                                                                                              |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Defines speed type (reference frame) for all speed related fields in this message.                                                                                           |             |
| Base Type         | PathSegmentSpeedType                                                                                                                                                         |             |
| Message Structure |                                                                                                                                                                              |             |
| Field Name        | Field Type                                                                                                                                                                   | Field Usage |
| SpeedChoice       | PathSegmentSpeedChoiceType                                                                                                                                                   | Optional    |
| Description       | Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number. |             |
| SpeedOptimization | SpeedOptimizationEnum                                                                                                                                                        | Optional    |
| Description       | Indicates that the System will vary its speed autonomously according to the specified optimization algorithm.                                                                |             |

Table A-1-372: P2P UCI Extensions MA\_TaskMT - Speed UCI Extension Field

| Heading           |                                             |
|-------------------|---------------------------------------------|
| Description       | Defines heading type and reference to hold. |
| Base Type         | MA_DirectionReferenceType                   |
| Message Structure |                                             |

| Field Name       | Field Type                                     | Field Usage |
|------------------|------------------------------------------------|-------------|
| <b>Value</b>     | AngleType                                      | Required    |
| Description      | Indicates the heading or course value to hold. |             |
| <b>Reference</b> | MA_HeadingReferenceEnum                        | Required    |
| Description      | Reference to type of heading or course given.  |             |

Table A-1-373: P2P UCI Extensions MA\_TaskMT - Heading UCI Extension Field

| SpeedChoice       |                                                                                                                                                                              |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number. |             |
| Base Type         | MA_HSA_SpeedChoiceType                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                              |             |
| Field Name        | Field Type                                                                                                                                                                   | Field Usage |
| SpeedValue        | PathSegmentSpeedValueType                                                                                                                                                    | Required    |
| Description       | Indicates speed in meters per second (m/s).                                                                                                                                  |             |
| MachValue         | MachType                                                                                                                                                                     | Required    |
| Description       | Indicates the speed reference for the associated speed value.                                                                                                                |             |

Table A-1-374: P2P UCI Extensions MA\_TaskMT - SpeedChoice UCI Extension Field

| SpeedValue        |                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates speed in meters per second (m/s).                                               |             |
| Base Type         | PathSegmentSpeedValueType                                                                 |             |
| Message Structure |                                                                                           |             |
| Field Name        | Field Type                                                                                | Field Usage |
| Value             | SpeedType                                                                                 | Required    |
| Description       | Indicates the speed along the PathSegment at the EndPoint.                                |             |
| Reference         | SpeedReferenceEnum                                                                        | Required    |
| Description       | Indicates whether the given vehicle speed is reference to ground speed or true air speed. |             |

Table A-1-375: P2P UCI Extensions MA\_TaskMT - SpeedValue UCI Extension Field

| CurveFitMethod      |                                                                                          |
|---------------------|------------------------------------------------------------------------------------------|
| <b>Description</b>  | Method through which a curve should be fitted to the discretized points in ControlPoint. |
| <b>Base Type</b>    | MA_CurveFitMethodEnum                                                                    |
| Message Structure   |                                                                                          |
| Enumeration Literal | Description                                                                              |
| <b>B_SPLINE</b>     | B-Spline (basis spline) fit.                                                             |

|                      |                          |
|----------------------|--------------------------|
| <b>BEZIER</b>        | Bezier curve fit.        |
| <b>CATMULL_ROM</b>   | Catmull-Rom spline fit.  |
| <b>NATURAL_CUBIC</b> | Natural Cubic curve fit. |

Table A-1-376: P2P UCI Extensions MA\_TaskMT - CurveFitMethod UCI Extension Field

## 1.3.2.64 MA\_NavigationReportMT UCI Extension

| MA_NavigationReportMT |                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>    | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>    | MessageData : MA_NavigationReportMDT                                                                                                        |

Table A-1-377: P2P UCI Extensions MA\_NavigationReportMT UCI Extension

| MessageData       |                                                                                                                                                                    |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                            |             |
| Base Type         | MA_NavigationReportMDT                                                                                                                                             |             |
| Message Structure |                                                                                                                                                                    |             |
| Field Name        | Field Type                                                                                                                                                         | Field Usage |
| SystemID          | SystemID_Type                                                                                                                                                      | Required    |
| Description       | Indicates the unique ID of the System corresponding to this NavigationReport.                                                                                      |             |
| Source            | SystemSourceEnum                                                                                                                                                   | Required    |
| Description       | Indicates the source of the data in sibling elements. See enumeration annotations for further details.                                                             |             |
| ContingencyLevel  | SystemContingencyLevelEnum                                                                                                                                         | Required    |
| Description       | Designates the current contingency state of the System. Contingencies can be an indication that the System will/has autonomously modified its means of navigation. |             |
| Endurance         | EnduranceType                                                                                                                                                      | Required    |
| Description       | Indicates the System's remaining endurance. It is expected that at least one of the optional children are populated in this context.                               |             |
| Playtime          | DurationType                                                                                                                                                       | Optional    |
| Description       | Estimated remaining playtime of operation/mission when taking into account RTB routing, fuel overhead, current fuel usage, current tasking, etc.                   |             |
| Navigation        | NavigationType                                                                                                                                                     | Optional    |
| Description       | Vehicle navigation method currently employed.                                                                                                                      |             |

Table A-1-378: P2P UCI Extensions MA\_NavigationReportMT - MessageData UCI Extension Field

## 1.3.2.65 MA\_StrikeActivityMT UCI Extension

| MA_StrikeActivityMT |
|---------------------|
|---------------------|

|                    |                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                             |
|                    | See the XSD File for more info                                                                                                                                                                                                                                                                                                                                      |
| <b>Ext. Fields</b> | ActiveWeaponHSA_CSA : MA_HSA_CSA_Type<br>Direction : MA_DirectionChoiceType<br>MunitionActive : MA_MunitionActiveType<br>MessageData : MA_StrikeActivityMDT<br>ActiveWeaponState : MA_ActiveWeaponStateEnum<br>Activity : MA_StrikeActivityType<br>Course : MA_DirectionReferenceType<br>Heading : MA_DirectionReferenceType<br>Reference : MA_HeadingReferenceEnum |

Table A-1-379: P2P UCI Extensions MA\_StrikeActivityMT UCI Extension

| ActiveWeaponHSA_CSA |                                                                                                  |             |
|---------------------|--------------------------------------------------------------------------------------------------|-------------|
| Description         | Provides the heading/course, speed, and altitude of the munition while the Activity is ACTIVE_*. |             |
| Base Type           | MA_HSA_CSA_Type                                                                                  |             |
| Message Structure   |                                                                                                  |             |
| Field Name          | Field Type                                                                                       | Field Usage |
| Altitude            | AltitudeReferenceType                                                                            | Optional    |
| Description         | Altitude hold value to be achieved and reference to altitude measurement source.                 |             |
| Speed               | PathSegmentSpeedType                                                                             | Optional    |
| Description         | Defines speed type (reference frame) for all speed related fields in this message.               |             |
| Direction           | MA_DirectionChoiceType                                                                           | Optional    |
| Description         | Defines heading or course type and reference to hold                                             |             |

Table A-1-380: P2P UCI Extensions MA\_StrikeActivityMT - ActiveWeaponHSA\_CSA UCI Extension Field

| Direction         |                                                      |             |
|-------------------|------------------------------------------------------|-------------|
| Description       | Defines heading or course type and reference to hold |             |
| Base Type         | MA_DirectionChoiceType                               |             |
| Message Structure |                                                      |             |
| Field Name        | Field Type                                           | Field Usage |
| Heading           | MA_DirectionReferenceType                            | Optional    |
| Description       | Defines heading type and reference to hold.          |             |
| Course            | MA_DirectionReferenceType                            | Optional    |
| Description       | Defines course type and reference to hold.           |             |

Table A-1-381: P2P UCI Extensions MA\_StrikeActivityMT - Direction UCI Extension Field

| MunitionActive      |                                                                                                                                                                                                                                                                                        |             |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | Data regarding the munition while it is active or in flight.                                                                                                                                                                                                                           |             |
| Base Type           | MA_MunitionActiveType                                                                                                                                                                                                                                                                  |             |
| Message Structure   |                                                                                                                                                                                                                                                                                        |             |
| Field Name          | Field Type                                                                                                                                                                                                                                                                             | Field Usage |
| ActiveWeaponState   | MA_ActiveWeaponStateEnum                                                                                                                                                                                                                                                               | Repeated    |
| Description         | Provides further munition state data detail if the Activity is ENABLED, ACTIVE_* or COMPLETED as reported by the ActivityState. This message can be omitted if ActivityState is DISABLED, FAILED or DELETED. List size for this element is based on "Select All That Apply" condition. |             |
| ActiveWeaponHSA_CSA | MA_HSA_CSA_Type                                                                                                                                                                                                                                                                        | Optional    |
| Description         | Provides the heading/course, speed, and altitude of the munition while the Activity is ACTIVE_*.                                                                                                                                                                                       |             |

Table A-1-382: P2P UCI Extensions MA\_StrikeActivityMT - MunitionActive UCI Extension Field

| MessageData       |                                                                                                         |             |
|-------------------|---------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                 |             |
| Base Type         | MA_StrikeActivityMDT                                                                                    |             |
| Message Structure |                                                                                                         |             |
| Field Name        | Field Type                                                                                              | Field Usage |
| SubsystemID       | SubsystemID_Type                                                                                        | Required    |
| Description       | Indicates the Subsystem which hosts the Strike Capabilities whose Activities are given in this message. |             |
| Activity          | MA_StrikeActivityType                                                                                   | Repeated    |
| Description       | Indicates the dynamic operational characteristics of an Activity of a Strike Capability.                |             |

Table A-1-383: P2P UCI Extensions MA\_StrikeActivityMT - MessageData UCI Extension Field

| ActiveWeaponState      |                                                                                                                                                                                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>     | Provides further munition state data detail if the Activity is ENABLED, ACTIVE_* or COMPLETED as reported by the ActivityState. This message can be omitted if ActivityState is DISABLED, FAILED or DELETED. List size for this element is based on "Select All That Apply" condition. |
| <b>Base Type</b>       | MA_ActiveWeaponStateEnum                                                                                                                                                                                                                                                               |
| Message Structure      |                                                                                                                                                                                                                                                                                        |
| Enumeration Literal    | Description                                                                                                                                                                                                                                                                            |
| <b>TARGETED</b>        | Indicates that a target has been selected and the munition has not been released yet. Report with the ENABLED state generated by ActivityStateEnum.                                                                                                                                    |
| <b>REQUESTED</b>       | Indicates the munition has been requested for release. Report with the ENABLED state generated by ActivityStateEnum.                                                                                                                                                                   |
| <b>SEEKER_INACTIVE</b> | Indicates the munition has been released but its local seeker has not been activated.                                                                                                                                                                                                  |



|                             |                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | Report with the ACTIVE_* state generated by ActivityStateEnum.                                                                                                                                  |
| <b>SEEKER_ACTIVE</b>        | Indicates the munition has been released and its local seeker has been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.                                                |
| <b>FUZE_INACTIVE</b>        | Indicates the munition has been released but its fuze has not been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.                                                    |
| <b>FUZE_ACTIVE</b>          | Indicates the munition has been released and its fuze has been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.                                                        |
| <b>NOT_FULLY_AUTONOMOUS</b> | Indicates the munition has been released not fully autonomous and thus not able to guide without datalink or target destination. Report with the ACTIVE_* state generated by ActivityStateEnum. |
| <b>FULLY_AUTONOMOUS</b>     | Indicates the munition has been released and is fully autonomous and able guide without datalink or target destination. Report with the ACTIVE_* state generated by ActivityStateEnum.          |
| <b>DEFEATED</b>             | Indicates the munition has been released and but it has been out maneuvered by the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.                             |
| <b>TIMED_OUT</b>            | Indicates the munition flight has ended, and it did not intercept the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.                                          |
| <b>SUCCESS</b>              | Indicates the munition flight has ended, and it intercepted the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.                                                |

**Table A-1-384: P2P UCI Extensions MA\_StrikeActivityMT - ActiveWeaponState UCI Extension Field**

| Activity            |                                                                                                                                                                                                                                                                                                |             |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | Indicates the dynamic operational characteristics of an Activity of a Strike Capability.                                                                                                                                                                                                       |             |
| Base Type           | MA_StrikeActivityType                                                                                                                                                                                                                                                                          |             |
| Message Structure   |                                                                                                                                                                                                                                                                                                |             |
| Field Name          | Field Type                                                                                                                                                                                                                                                                                     | Field Usage |
| SelectedForKeyLoad  | boolean                                                                                                                                                                                                                                                                                        | Required    |
| Description         | Key Load indicates the store has been put into a state where it is ready to receive cryptographic keys.                                                                                                                                                                                        |             |
| WeaponArmed         | boolean                                                                                                                                                                                                                                                                                        | Required    |
| Description         | When true, the weapon is in the armed state.                                                                                                                                                                                                                                                   |             |
| ReadyForRelease     | boolean                                                                                                                                                                                                                                                                                        | Required    |
| Description         | The store has been prepared for launch such that when released the store will function properly.                                                                                                                                                                                               |             |
| SelectedForRelease  | boolean                                                                                                                                                                                                                                                                                        | Required    |
| Description         | When Master Arm, Release Consent, and any LAR or vehicle Attitude parameters enforced by the carriage are all met, this store will be released.                                                                                                                                                |             |
| SelectedForJettison | boolean                                                                                                                                                                                                                                                                                        | Required    |
| Description         | Jettison indicates an ejection from the host system without powering or preparing the delivery vehicle. The store will be in an inert state. When Master Arm, Release Consent, and any LAR or vehicle attitude parameters enforced by the carriage are all met, this store will be jettisoned. |             |

|                           |                                                                                                                                                                                                                                       |          |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>AssignedTarget</b>     | GeoLocatedObjectType                                                                                                                                                                                                                  | Optional |
| Description               | Indicates the Entity target assigned to the StrikeCapability.                                                                                                                                                                         |          |
| <b>AO_Code</b>            | AO_CodeType                                                                                                                                                                                                                           | Optional |
| Description               | Indicates the current/set AO/laser PRF and/or PIM code.                                                                                                                                                                               |          |
| <b>LAR_ID</b>             | LAR_ID_Type                                                                                                                                                                                                                           | Optional |
| Description               | Indicates the ID of the associated LAR and/or LAR_Report associated with the Strike capability.                                                                                                                                       |          |
| <b>ConsentRequired</b>    | boolean                                                                                                                                                                                                                               | Optional |
| Description               | Indicates whether (TRUE) or not (FALSE) the Activity requires consent to transition from the ENABLED state to an ACTIVE state. This element is required and shall be TRUE for Activities associated with "mother may I" Capabilities. |          |
| <b>ConsentState</b>       | ConsentEnum                                                                                                                                                                                                                           | Optional |
| Description               | Indicates the state of consent for the Activity. This element is required for Activities which require Consent.                                                                                                                       |          |
| <b>KineticFootprint</b>   | EnduranceFootprintType                                                                                                                                                                                                                | Optional |
| Description               | The estimated footprint of where the store may fall if released from the current position.                                                                                                                                            |          |
| <b>TargetLockAcquired</b> | boolean                                                                                                                                                                                                                               | Optional |
| Description               | Indicated whether (TRUE) or not (FALSE) a lock on target has been acquired.                                                                                                                                                           |          |
| <b>MunitionActive</b>     | MA_MunitionActiveType                                                                                                                                                                                                                 | Optional |
| Description               | Data regarding the munition while it is active or in flight.                                                                                                                                                                          |          |
| <b>TimeToRelease</b>      | DurationType                                                                                                                                                                                                                          | Optional |
| Description               | Time interval before strike capability is released.                                                                                                                                                                                   |          |

Table A-1-385: P2P UCI Extensions MA\_StrikeActivityMT - Activity UCI Extension Field

| Course            |                                                |             |
|-------------------|------------------------------------------------|-------------|
| Description       | Defines course type and reference to hold.     |             |
| Base Type         | MA_DirectionReferenceType                      |             |
| Message Structure |                                                |             |
| Field Name        | Field Type                                     | Field Usage |
| Value             | AngleType                                      | Required    |
| Description       | Indicates the heading or course value to hold. |             |
| Reference         | MA_HeadingReferenceEnum                        | Required    |
| Description       | Reference to type of heading or course given.  |             |

Table A-1-386: P2P UCI Extensions MA\_StrikeActivityMT - Course UCI Extension Field

| Heading           |                                             |
|-------------------|---------------------------------------------|
| Description       | Defines heading type and reference to hold. |
| Base Type         | MA_DirectionReferenceType                   |
| Message Structure |                                             |

| Field Name       | Field Type                                     | Field Usage |
|------------------|------------------------------------------------|-------------|
| <b>Value</b>     | AngleType                                      | Required    |
| Description      | Indicates the heading or course value to hold. |             |
| <b>Reference</b> | MA_HeadingReferenceEnum                        | Required    |
| Description      | Reference to type of heading or course given.  |             |

Table A-1-387: P2P UCI Extensions MA\_StrikeActivityMT - Heading UCI Extension Field

| Reference             |                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------|
| <b>Description</b>    | Reference to type of heading or course given.                                                     |
| <b>Base Type</b>      | MA_HeadingReferenceEnum                                                                           |
| Message Structure     |                                                                                                   |
| Enumeration Literal   | Description                                                                                       |
| <b>TRUE_NORTH</b>     | Heading value according to Geographic North.                                                      |
| <b>MAGNETIC_NORTH</b> | Heading value according to location where the planet's magnetic field points vertically downward. |

Table A-1-388: P2P UCI Extensions MA\_StrikeActivityMT - Reference UCI Extension Field

### 1.3.2.66 MA\_StrikeCapabilityMT UCI Extension

| MA_StrikeCapabilityMT |                                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>    | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>    | MessageData : MA_StrikeCapabilityMDT                                                                                          |

Table A-1-389: P2P UCI Extensions MA\_StrikeCapabilityMT UCI Extension

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_StrikeCapabilityMDT                                                                  |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| Capability        | MA_StrikeCapabilityType                                                                 | Repeated    |
| Description       | Indicates a Strike Capability provided by the Subsystem.                                |             |
| Carriage          | StoreCarriageCapabilityType                                                             | Repeated    |
| Description       | Optional list of carriage related to the StrikeCapability.                              |             |
| Verification      | StoreVerificationStatusType                                                             | Optional    |
| Description       | This is a rollup of all verification of each capability in this message.                |             |
| LoadoutMnemonic   | VisibleString256Type                                                                    | Optional    |
| Description       | Mnemonic representation of the store loadout that the reported strike capabilities and  |             |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                     | carriage support capabilities are derived from.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>AvailableMDF</b> | MDF_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated |
| Description         | Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability. |          |

Table A-1-390: P2P UCI Extensions MA\_StrikeCapabilityMT - MessageData UCI Extension Field

## 1.3.2.67 MA\_TaskCommandMT UCI Extension

| MA_TaskCommandMT   |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_TaskCommandMDT                                                                                               |

Table A-1-391: P2P UCI Extensions MA\_TaskCommandMT UCI Extension

| MessageData       |                                                                                                                                                                                                                                                                                               |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                       |             |
| Base Type         | MA_TaskCommandMDT                                                                                                                                                                                                                                                                             |             |
| Message Structure |                                                                                                                                                                                                                                                                                               |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                    | Field Usage |
| Command           | MA_TaskCommandType                                                                                                                                                                                                                                                                            | Repeated    |
| Description       | Indicates an instance of a Task Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message. |             |

Table A-1-392: P2P UCI Extensions MA\_TaskCommandMT - MessageData UCI Extension Field

## 1.3.2.68 MA\_TaskCommandStatusMT UCI Extension

| MA_TaskCommandStatusMT |                                                                                                                               |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>     | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>     | MessageData : MA_TaskCommandStatusMDT                                                                                         |

Table A-1-393: P2P UCI Extensions MA\_TaskCommandStatusMT UCI Extension

| MessageData        |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this |

|                              |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                              | type for details.                                                                                                                                                                                                                                                                                                                                           |             |
| Base Type                    | MA_TaskCommandStatusMDT                                                                                                                                                                                                                                                                                                                                     |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| Activity                     | ResultingActivityType                                                                                                                                                                                                                                                                                                                                       | Repeated    |
| Description                  | Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.                                                                                                                                                                                                                    |             |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

Table A-1-394: P2P UCI Extensions MA\_TaskCommandStatusMT - MessageData UCI Extension Field

### 1.3.2.69 MA\_COP\_ConfigurationSettingsMT UCI Extension

| MA_COP_ConfigurationSettingsMT |                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>             | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>             | MessageData : MA_COP_ConfigurationSettingsMDT                                                                                               |

Table A-1-395: P2P UCI Extensions MA\_COP\_ConfigurationSettingsMT UCI Extension

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_COP_ConfigurationSettingsMDT                                                         |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| ApplicableToIDs   | MA_PackageSystemType                                                                    | Required    |
| Description       | Indicates the ID of the Package or System with the COP settings.                        |             |
| COP_LeaderID      | SystemID_Type                                                                           | Optional    |
| Description       | Indicates the unique SystemID of the current leader of the COP.                         |             |

|                           |                                                                  |          |
|---------------------------|------------------------------------------------------------------|----------|
| <b>COP_LeaderBackupID</b> | SystemID_Type                                                    | Repeated |
| Description               | Indicates the unique SystemIDs of the backup leaders of the COP. |          |
| <b>COP_FusionCapable</b>  | boolean                                                          | Required |
| Description               | Indicates if the COP is capable of fusing tracks.                |          |
| <b>Configuration</b>      | MA_COP_ConfigurationType                                         | Repeated |
| Description               | A list of the configuration items.                               |          |

Table A-1-396: P2P UCI Extensions MA\_COP\_ConfigurationSettingsMT - MessageData UCI Extension Field

### 1.3.2.70 MA\_OpCommandMT UCI Extension

| MA_OpCommandMT     |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_OpCommandMDT                                                                                                 |

Table A-1-397: P2P UCI Extensions MA\_OpCommandMT UCI Extension

| MessageData       |                                                                                                                                                                                                                                                                       |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                               |             |
| Base Type         | MA_OpCommandMDT                                                                                                                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                                                                                                                       |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                            | Field Usage |
| Command           | MA_OpCommandType                                                                                                                                                                                                                                                      | Repeated    |
| Description       | Indicates an instance of an Op* command. An Op* command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Op* CommandStatus message. |             |

Table A-1-398: P2P UCI Extensions MA\_OpCommandMT - MessageData UCI Extension Field

### 1.3.2.71 MA\_OpCommandStatusMT UCI Extension

| MA_OpCommandStatusMT |                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>   | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>   | MessageData : MA_OpCommandStatusMDT                                                                                           |

Table A-1-399: P2P UCI Extensions MA\_OpCommandStatusMT UCI Extension

| MessageData        |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this |

|                              |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                              | type for details.                                                                                                                                                                                                                                                                                                                                           |             |
| Base Type                    | MA_OpCommandStatusMDT                                                                                                                                                                                                                                                                                                                                       |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

Table A-1-400: P2P UCI Extensions MA\_OpCommandStatusMT - MessageData UCI Extension Field

### 1.3.2.72 MA\_DeconflictionSolutionRequestStatusMT UCI Extension

|                                                |                                                                                                                               |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_DeconflictionSolutionRequestStatusMT</b> |                                                                                                                               |
| <b>Description</b>                             | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                             | MessageData : MA_DeconflictionSolutionRequestStatusMDT                                                                        |

Table A-1-401: P2P UCI Extensions MA\_DeconflictionSolutionRequestStatusMT UCI Extension

| MessageData                 |                                                                             |             |
|-----------------------------|-----------------------------------------------------------------------------|-------------|
| Description                 | Indicates the data being conveyed by the message.                           |             |
| Base Type                   | MA_DeconflictionSolutionRequestStatusMDT                                    |             |
| Message Structure           |                                                                             |             |
| Field Name                  | Field Type                                                                  | Field Usage |
| ConflictID                  | MA_ConflictID_Type                                                          | Repeated    |
| Description                 | List of conflict identifiers this message applies to                        |             |
| RoutePlanID                 | RoutePlanID_Type                                                            | Optional    |
| Description                 | A reference to the specific route plan being responded to                   |             |
| DeconflictionProposalStatus | MA_DeconflictionProposalStatusEnum                                          | Optional    |
| Description                 | Indicates the status of response to the referenced proposed route plan      |             |
| RequestID                   | RequestID_Type                                                              | Required    |
| Description                 | Indicates the unique ID of the request message corresponding to this status |             |

|                                     |                                                                                                                                                                                                                                                                                             |          |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                     | message.                                                                                                                                                                                                                                                                                    |          |
| <b>RequestProcessingState</b>       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required |
| Description                         | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |          |
| <b>RequestProcessingStateReason</b> | CannotComplyType                                                                                                                                                                                                                                                                            | Optional |
| Description                         | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |          |

**Table A-1-402: P2P UCI Extensions MA\_DeconflictionSolutionRequestStatusMT - MessageData UCI Extension Field**

### 1.3.2.73 MA\_OpPlanMT UCI Extension

| MA_OpPlanMT        |                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_OpPlanMDT                                                                                                    |

**Table A-1-403: P2P UCI Extensions MA\_OpPlanMT UCI Extension**

| MessageData        |                                                                                                                                                                                                                                                                                                                                                                   |             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates the data being conveyed by the message.                                                                                                                                                                                                                                                                                                                 |             |
| Base Type          | MA_OpPlanMDT                                                                                                                                                                                                                                                                                                                                                      |             |
| Message Structure  |                                                                                                                                                                                                                                                                                                                                                                   |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                                                                                                        | Field Usage |
| OpPlanID           | MA_OpPlanID_Type                                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description        | Indicates a unique ID assigned to the OpPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.                                                                                                         |             |
| PlanCommandID      | MA_OpPlanCommandID_ChoiceType                                                                                                                                                                                                                                                                                                                                     | Optional    |
| Description        | Indicates the unique ID of the *PlanCommand which the OpPlan originated from.                                                                                                                                                                                                                                                                                     |             |
| Plan               | MA_OpPlanType                                                                                                                                                                                                                                                                                                                                                     | Required    |
| Description        | Contains the details of the Op* allocation plan.                                                                                                                                                                                                                                                                                                                  |             |
| ForPlanningUseOnly | boolean                                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description        | When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc. |             |

**Table A-1-404: P2P UCI Extensions MA\_OpPlanMT - MessageData UCI Extension Field**



### 1.3.2.74 MA\_RouteDeconflictionAlertMT UCI Extension

| MA_RouteDeconflictionAlertMT |                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>           | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>           | MessageData : MA_RouteDeconflictionAlertMDT                                                                                   |

**Table A-1-405: P2P UCI Extensions MA\_RouteDeconflictionAlertMT UCI Extension**

| MessageData       |                                                                                    |             |
|-------------------|------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the data being conveyed by the message.                                  |             |
| Base Type         | MA_RouteDeconflictionAlertMDT                                                      |             |
| Message Structure |                                                                                    |             |
| Field Name        | Field Type                                                                         | Field Usage |
| ConflictID        | MA_ConflictID_Type                                                                 | Required    |
| Description       | Indicates a unique identifier ID related to the path conflict for future reference |             |
| ConflictData      | MA_RouteDeconflictionAlertDataType                                                 | Repeated    |
| Description       | Conflict meta data providing relevant route IDs and location(s) of the conflict.   |             |
| Resolved          | boolean                                                                            | Required    |
| Description       | This element specifies whether this route conflict was resolved or not.            |             |
| ConflictPoint     | Point2D_Type                                                                       | Repeated    |
| Description       | This element identifies the location where a route conflict occurs.                |             |

**Table A-1-406: P2P UCI Extensions MA\_RouteDeconflictionAlertMT - MessageData UCI Extension Field**

### 1.3.2.75 MA\_OpStatusMT UCI Extension

| MA_OpStatusMT      |                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_OpStatusMDT                                                                                                                |

**Table A-1-407: P2P UCI Extensions MA\_OpStatusMT UCI Extension**

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_OpStatusMDT                                                                          |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |

|                                  |                                                                                                                                                                                                                                                          |          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>ExecutingPackageSystemIDs</b> | MA_PackageSystemType                                                                                                                                                                                                                                     | Required |
| Description                      | Identifies the Package ID or ACP System ID whose interpretation of the Op* status is reflected in this message. If Package ID is used, this includes all package member System IDs, so a member ACP's System ID does not need to be separately included. |          |
| <b>CurrentState</b>              | OpStateEnum                                                                                                                                                                                                                                              | Required |
| Description                      | The state of the schedule (active or inactive) for which the defined schedule timeline applies.                                                                                                                                                          |          |
| <b>OpID</b>                      | OpID_ChoiceType                                                                                                                                                                                                                                          | Required |
| Description                      | Indicates the Op* being commanded. Leaving this blank makes the command apply to all Op* IDs                                                                                                                                                             |          |
| <b>CommandID</b>                 | CommandID_Type                                                                                                                                                                                                                                           | Repeated |
| Description                      | Indicates the unique ID of the Op* command resulting in this status.                                                                                                                                                                                     |          |

Table A-1-408: P2P UCI Extensions MA\_OpStatusMT - MessageData UCI Extension Field

### 1.3.2.76 MA\_DeconflictionSolutionRequestMT UCI Extension

| MA_DeconflictionSolutionRequestMT |                                                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                | MessageData : MA_DeconflictionSolutionRequestMDT                                                                              |

Table A-1-409: P2P UCI Extensions MA\_DeconflictionSolutionRequestMT UCI Extension

| MessageData       |                                                                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the data being conveyed by the message.                                                                                         |             |
| Base Type         | MA_DeconflictionSolutionRequestMDT                                                                                                        |             |
| Message Structure |                                                                                                                                           |             |
| Field Name        | Field Type                                                                                                                                | Field Usage |
| ConflictID        | MA_ConflictID_Type                                                                                                                        | Repeated    |
| Description       | List of conflict identifiers this message applies to                                                                                      |             |
| ConflictStatus    | MA_ConflictStatusEnum                                                                                                                     | Required    |
| Description       | Indicates the status of the current conflict                                                                                              |             |
| Plan              | MA_RoutePlanMDT                                                                                                                           | Required    |
| Description       | Contains the details of the proposed route plan.                                                                                          |             |
| RequestID         | RequestID_Type                                                                                                                            | Required    |
| Description       | Indicates the unique ID of the request message.                                                                                           |             |
| RequestState      | RequestStateEnum                                                                                                                          | Required    |
| Description       | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request. |             |

Table A-1-410: P2P UCI Extensions MA\_DeconflictionSolutionRequestMT - MessageData UCI Extension Field

### 1.3.2.77 MA\_PrioritizationListMT UCI Extension

| MA_PrioritizationListMT |                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>      | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>      | ListType : MA_PrioritizationListEnum<br>ListItem : MA_PrioritizationListItemType<br>MessageData : MA_PrioritizationListMDT                  |

**Table A-1-411: P2P UCI Extensions MA\_PrioritizationListMT UCI Extension**

| ListType                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                       | Indicates the type of prioritization list. For example, battlespace objects can be prioritized differently for targeting vs. protection as indicated by enumerates given in this element.                                                                                                                                                                                                                                                                                            |
| <b>Base Type</b>                         | MA_PrioritizationListEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Message Structure                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Enumeration Literal                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>THREAT_TARGET_PRIORITIZATION_LIST</b> | A prioritized list of battlespace objects or areas that are prioritized by an operational or tactical planner. For planning purposes, the ListItem associated with this enumeration could include, for example, an entity, a signal, a point or region.                                                                                                                                                                                                                              |
| <b>CRITICAL_ASSET_LIST</b>               | A prioritized list of battlespace objects or areas that are prioritized by the joint force commander to be defended with the resources available. Critical assets are generally specific assets of such extraordinary importance that their loss or degradation would have a significant and debilitating effect on operations or the mission. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system or a point. |
| <b>JIPTL</b>                             | The Joint Integrated Prioritization Target List (JPITL) is a prioritized list of targets and associated data approved by the Joint Forces Commander or designated representative. An approved JIPTL is the central product of the target development cycle. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity or a system.                                                                                             |
| <b>JIPTL_NOMINATION</b>                  | A product of the target development phase is the JPITL nomination list. Targeters determine this list during the objectives, effects, and guidance phase and analyze which targets should be struck (or otherwise affected) to accomplish them. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity or a system.                                                                                                         |
| <b>JOINT_TARGET_LIST</b>                 | A consolidated list of selected targets considered to have military significance in the combatant commander's area of responsibility. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system, a point, or area.                                                                                                                                                                                                   |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MISSION_TARGET_LIST</b>    | The list of battlespace targets that are considered to have military significance and are required for the successful completion of the mission. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system, a point, or area.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>NO_STRIKE_LIST</b>         | A list of battlespace objects characterized as protected from the effects of military operations under international law or Engagement Criteria. Attacking these may violate Law of Armed Conflict (LOAC) or Engagement Criteria or interfere with friendly relations with indigenous personnel or governments. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system, a point, or area.                                                                                                                                                                                                                                                                                                        |
| <b>RESTRICTED_TARGET_LIST</b> | A list of targets that have specific restrictions imposed upon them. Actions on restricted targets are prohibited until coordinated and approved by the establishing authority. Targets are restricted because certain types of actions against them may have negative political, cultural, or propaganda implications, or may interfere with projected friendly operations. The Restricted Target List (RTL) is nominated by elements of the joint force and approved by the JFC. Targets on this list may only be struck with JFC or higher approval. Actions taken by an opponent may remove a target from the RTL. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system, a point, or area. |
| <b>DEFENDED_ASSET_LIST</b>    | The Defended Asset List (DAL) is a listing of those assets from the critical asset list prioritized by the joint force commander to be defended with the resources available. Critical assets that are reinforced with additional protection capabilities or capabilities from other combat power elements become part of the DAL. It represents what can be protected, by priority. The DAL allows commanders to apply finite protection capabilities to the most valuable assets.                                                                                                                                                                                                                                                                                 |
| <b>PLATFORM_TARGET_LIST</b>   | A list of active entity platform targets that meet custody requirements. This list type is used in support of presenting a shot queue prioritization list with List Item of EntityID in accordance with the mission common operating picture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Table A-1-412: P2P UCI Extensions MA\_PrioritizationListMT - ListType UCI Extension Field

| ListItem          |                                                                                                                                                                                                                                                                                                                                |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates an item in a prioritization list of the sibling ListType.                                                                                                                                                                                                                                                            |             |
| Base Type         | MA_PrioritizationListItemType                                                                                                                                                                                                                                                                                                  |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                     | Field Usage |
| Subject           | IdentityKindInstanceType                                                                                                                                                                                                                                                                                                       | Required    |
| Description       | Indicates the subject of the prioritization list item.                                                                                                                                                                                                                                                                         |             |
| Priority          | PrioritizationType                                                                                                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates the priority of the item. If priority for any child PriorityType is given, then omitted PriorityTypes for the sibling Subject have no/zero priority. No duplicates allowed; at most one instance of each of the child PriorityTypes is allowed. If the child PriorityType is not used, only one instance is allowed. |             |

|                          |                                                                                                                                                                                                                                       |          |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>ProbabilityOfKill</b> | PercentType                                                                                                                                                                                                                           | Optional |
| Description              | Indicates the probability of kill, in a percentage where a value of 100.0 = 100%, of a selected target with an available capability.                                                                                                  |          |
| <b>ObjectCorrelation</b> | ObjectCorrelationType                                                                                                                                                                                                                 | Optional |
| Description              | Object Correlation contains a description of the correlation of this object with another object as well as whether the correlating system has C2 responsibilities or not, to indicate a rough level of confidence in the information. |          |

Table A-1-413: P2P UCI Extensions MA\_PrioritizationListMT - ListItem UCI Extension Field

| MessageData          |                                                                                                                                                                                                                                                                               |             |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                       |             |
| Base Type            | MA_PrioritizationListMDT                                                                                                                                                                                                                                                      |             |
| Message Structure    |                                                                                                                                                                                                                                                                               |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                    | Field Usage |
| PrioritizationListID | PrioritizationListID_Type                                                                                                                                                                                                                                                     | Required    |
| Description          | Indicates the unique ID of the prioritization list.                                                                                                                                                                                                                           |             |
| ListType             | MA_PrioritizationListEnum                                                                                                                                                                                                                                                     | Required    |
| Description          | Indicates the type of prioritization list. For example, battlespace objects can be prioritized differently for targeting vs. protection as indicated by enumerates given in this element.                                                                                     |             |
| ListItem             | MA_PrioritizationListItemType                                                                                                                                                                                                                                                 | Repeated    |
| Description          | Indicates an item in a prioritization list of the sibling ListType.                                                                                                                                                                                                           |             |
| OpOrderTraceability  | OrderTraceabilityType                                                                                                                                                                                                                                                         | Repeated    |
| Description          | Indicates an operational order which this PrioritizationList is traceable to. This does not mean ranking. This element refers to a Planning Order, Commander's Intent, or other Order that came from a higher echelon. The higher echelon is identified by the OrderSourceID. |             |
| OpDescription        | OpDescriptionType                                                                                                                                                                                                                                                             | Optional    |
| Description          | Indicates a human readable description of the purpose of the prioritization list.                                                                                                                                                                                             |             |
| ApplicableSystemID   | SystemID_Type                                                                                                                                                                                                                                                                 | Repeated    |
| Description          | One or more Systems to which the prioritization applies. If omitted, the prioritization applies to all Systems.                                                                                                                                                               |             |
| ApplicableServiceID  | ServiceID_Type                                                                                                                                                                                                                                                                | Repeated    |
| Description          | One or more Services to which the mission response list applies. If omitted, the settings apply to all Services of the Systems given in the sibling ApplicableSystemID.                                                                                                       |             |
| ApplicableZone       | ZoneType                                                                                                                                                                                                                                                                      | Optional    |
| Description          | Zones to which the prioritization applies.                                                                                                                                                                                                                                    |             |
| Schedule             | ScheduleType                                                                                                                                                                                                                                                                  | Optional    |
| Description          | Scheduled times to which the prioritization applies.                                                                                                                                                                                                                          |             |
| DataRecordInstanceID | DataRecordInstanceID_Type                                                                                                                                                                                                                                                     | Optional    |

|             |                                                                                                                                                                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee. |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table A-1-414: P2P UCI Extensions MA\_PrioritizationListMT - MessageData UCI Extension Field

## 1.3.2.78 MA\_MissionPlanActivationStatusMT UCI Extension

| MA_MissionPlanActivationStatusMT |                                                                                                                                                                            |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                      | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info                                              |
| Ext. Fields                      | OpPlanID : MA_OpPlanID_Type<br>MessageData : MA_MissionPlanActivationStatusMDT<br>Plans : MA_PlansReferenceBaseType<br>SubPlanActivationState : MA_PlanActivationStateType |

Table A-1-415: P2P UCI Extensions MA\_MissionPlanActivationStatusMT UCI Extension

| OpPlanID          |                                                                                                                                                                                                                                                                |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a reference to an OpPlan.                                                                                                                                                                                                                            |             |
| Base Type         | MA_OpPlanID_Type                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                     | Field Usage |
| Version           | unsignedInt                                                                                                                                                                                                                                                    | Optional    |
| Description       | Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval. |             |
| UUID              | UniversallyUniqueIdentifierType                                                                                                                                                                                                                                | Required    |
| Description       | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.                                                                                               |             |
| DescriptiveLabel  | VisibleString256Type                                                                                                                                                                                                                                           | Optional    |
| Description       | Human readable text label with content meaningful to human operators.                                                                                                                                                                                          |             |

Table A-1-416: P2P UCI Extensions MA\_MissionPlanActivationStatusMT - OpPlanID UCI Extension Field

| MessageData       |                                                   |             |
|-------------------|---------------------------------------------------|-------------|
| Description       | Indicates the data being conveyed by the message. |             |
| Base Type         | MA_MissionPlanActivationStatusMDT                 |             |
| Message Structure |                                                   |             |
| Field Name        | Field Type                                        | Field Usage |
| MissionPlanID     | MissionPlanID_Type                                | Required    |

|                               |                                                                                                                                                                                                                                                                              |          |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                   | The ID of the MissionPlan this status pertains to.                                                                                                                                                                                                                           |          |
| <b>PlanActivationState</b>    | PlanActivationStateEnum                                                                                                                                                                                                                                                      | Required |
| Description                   | Indicates the activation state of the MissionPlan.                                                                                                                                                                                                                           |          |
| <b>SubPlanActivationState</b> | MA_PlanActivationStateType                                                                                                                                                                                                                                                   | Repeated |
| Description                   | Indicates the activation state of a SubPlan or SubPlans of the associated MissionPlan. An instance of this element should include all of the SubPlans in a given state; the cardinality is based in the number of states in the associated activation state enumerated type. |          |

Table A-1-417: P2P UCI Extensions MA\_MissionPlanActivationStatusMT - MessageData UCI Extension Field

| Plans               |                                                             |             |
|---------------------|-------------------------------------------------------------|-------------|
| Description         | Indicates the Plan or Plans in the sibling ActivationState. |             |
| Base Type           | MA_PlansReferenceBaseType                                   |             |
| Message Structure   |                                                             |             |
| Field Name          | Field Type                                                  | Field Usage |
| TaskPlanID          | TaskPlanID_Type                                             | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                        |             |
| OrbitPlanID         | OrbitPlanID_Type                                            | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.                      |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                                    | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.              |             |
| RoutePlanID         | RoutePlanID_Type                                            | Repeated    |
| Description         | Indicates a reference to a RoutePlan.                       |             |
| RouteActivityPlanID | RouteActivityPlanID_Type                                    | Repeated    |
| Description         | Indicates a reference to a RouteActivityPlan.               |             |
| ActivityPlanID      | ActivityPlanID_Type                                         | Repeated    |
| Description         | Indicates a reference to an ActivityPlan.                   |             |
| EffectPlanID        | EffectPlanID_Type                                           | Repeated    |
| Description         | Indicates a reference to an EffectPlan.                     |             |
| ActionPlanID        | ActionPlanID_Type                                           | Repeated    |
| Description         | Indicates a reference to an ActionPlan.                     |             |
| ResponsePlanID      | ResponsePlanID_Type                                         | Repeated    |
| Description         | Indicates a reference to an ResponsePlan.                   |             |
| OpPlanID            | MA_OpPlanID_Type                                            | Repeated    |
| Description         | Indicates a reference to an OpPlan.                         |             |

Table A-1-418: P2P UCI Extensions MA\_MissionPlanActivationStatusMT - Plans UCI Extension Field

| SubPlanActivationState |                                                                           |
|------------------------|---------------------------------------------------------------------------|
| <b>Description</b>     | Indicates the activation state of a SubPlan or SubPlans of the associated |

|                   |                                                                                                                                                                                                    |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                   | MissionPlan. An instance of this element should include all of the SubPlans in a given state; the cardinality is based in the number of states in the associated activation state enumerated type. |             |
| Base Type         | MA_PlanActivationStateType                                                                                                                                                                         |             |
| Message Structure |                                                                                                                                                                                                    |             |
| Field Name        | Field Type                                                                                                                                                                                         | Field Usage |
| ActivationState   | PlanActivationStateEnum                                                                                                                                                                            | Required    |
| Description       | Indicates an activation state that one or more *Plans is in.                                                                                                                                       |             |
| Plans             | MA_PlansReferenceBaseType                                                                                                                                                                          | Required    |
| Description       | Indicates the Plan or Plans in the sibling ActivationState.                                                                                                                                        |             |

**Table A-1-419: P2P UCI Extensions MA\_MissionPlanActivationStatusMT - SubPlanActivationState UCI Extension Field**

### 1.3.2.79 MA\_ActionCommandMT UCI Extension

| MA_ActionCommandMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Ext. Fields</b> | ActionConstraintOverride : MA_ActionConstraintType<br>RequiredAllocation : MA_RequirementAllocationConstraintType<br>System : MA_SystemFilterType<br>PreferredAllocation : MA_RequirementAllocationConstraintType<br>Analytic : MA_AnalyticConstraintsType<br>Designation : MA_DesignationFilterType<br>AssessedObject : MA_AssetFilterType<br>Characteristics : MA_EntityComparativeType<br>Allocation : MA_AllocationChoiceType<br>ConstraintOverride : MA_RequirementConstraintsType<br>Allocation : MA_RequirementAllocationParametersType<br>AccessAssessmentThreshold : MA_AccessAssessmentFilterType<br>Command : MA_ActionCommandType<br>ExcludedRequirementTypes : MA_RequirementTaxonomyType<br>Kinematic : MA_RequirementKinematicConstraintsType<br>EngagementParameters : MA_EngagementParametersType<br>Subject : MA_AssetFilterType<br>MessageData : MA_ActionCommandMDT<br>AllowedRequirementTypes : MA_RequirementTaxonomyType<br>Entity : MA_EntityFilterType |



Table A-1-420: P2P UCI Extensions MA\_ActionCommandMT UCI Extension

| ActionConstraintOverride      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                   | Indicates refined/overridden values of constraints for the Action. Added to allow similar override options in direct commanding as is available in planning.                                                                                                                                                                                                                                                                                                                                                  |             |
| Base Type                     | MA_ActionConstraintType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
| Message Structure             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| Field Name                    | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Field Usage |
| TargetObject                  | IdentityKindInstanceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Repeated    |
| Description                   | Indicates the target of the Action; the object the Action will be done to or for. If omitted, the Action places no explicit limitation on which TargetObject it can be done to. See sibling ActionType element annotation for an overall description of Action structure. Note that the term "target" is used in a general sense. For some Requirement types, a more suitable term might be "subject", "point", "location", etc. For example, the "target" of a COVER Action could be a blue/friendly System. |             |
| TargetObjectConstraints       | RequirementTargetConstraintsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Optional    |
| Description                   | Indicates constraints "on" the TargetObject that must be satisfied before the Action can be done to or for it. See sibling ActionType element annotation for an overall description of Effect structure.                                                                                                                                                                                                                                                                                                      |             |
| SecondaryObject               | IdentityKindInstanceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Repeated    |
| Description                   | Indicates a thing the sibling TargetObject must act on/against/toward to make the Action applicable. See sibling ActionType element annotation for an overall description of Action structure.                                                                                                                                                                                                                                                                                                                |             |
| PerTargetEngagementParameters | MA_EngagementParametersOrDefaultChoiceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated    |
| Description                   | Indicates a command to change the engagement parameters corresponding to an ongoing tactic. This field is meant to be used on a per target basis. Setting this field overrides the engagement parameters sent in MA_Action, and Activity modification by ChangeEngagement, and any previous audibles that modified MA_EngagementParameters for the sibling TargetObject.                                                                                                                                      |             |

Table A-1-421: P2P UCI Extensions MA\_ActionCommandMT - ActionConstraintOverride UCI Extension Field

| RequiredAllocation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description        | Indicates the system and optionally Capability that the Requirement (Effect, Action, Task, [Capability]Command, Response) MUST be allocated to during the allocation process. The system can be an explicit UCI System or implied UCI System (by Identity, hosted Capability, kinematics, etc.) in which case the associated UCI System must be determined as part of the allocation process. If multiple child elements are given: 1) allocation to any one of the associated Systems is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the associated Systems is acceptable, 3) there is no expectation that each associated System must perform the Requirement. This element allows the Requirement creator to force an allocation to a specific system at the risk of 1) a less than optimal overall solution or 2) failure of the Requirement being allocated despite availability of another suitable system. |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Base Type         | MA_RequirementAllocationConstraintType                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                | Field Usage |
| Allocation        | MA_AllocationChoiceType                                                                                                                                                                                                                                                                                                                                                                                                                   | Required    |
| Description       | Indicates the type of Allocation, either by direct assignment of SystemIDs or by PackageID definition for delegation.                                                                                                                                                                                                                                                                                                                     |             |
| Identity          | IdentityType                                                                                                                                                                                                                                                                                                                                                                                                                              | Repeated    |
| Description       | Indicates a kind/type or instance of platform/system that is the target of the allocation constraint. For example, this element could be used to allocate a Requirement to any fighter aircraft platform, or a specific kind of fighter aircraft, or a specific tail number of aircraft, or an aircraft with an aircrew member with a certain voice call sign.                                                                            |             |
| CapabilityID      | CapabilityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                         | Repeated    |
| Description       | Indicates the CapabilityID of the Capability that is the target of the allocation constraint. If multiple required Capabilities are given: 1) allocation to any one of the Capabilities is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the multiple required Capabilities is acceptable, 3) there is no expectation that each Capability must perform the Requirement. |             |

Table A-1-422: P2P UCI Extensions MA\_ActionCommandMT - RequiredAllocation UCI Extension Field

| System            |                                                                                                                                                                                                      |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a filter on characteristics of a System asset. The asset filter is satisfied if a match is found in this element OR the sibling Entity element.                                            |             |
| Base Type         | MA_SystemFilterType                                                                                                                                                                                  |             |
| Message Structure |                                                                                                                                                                                                      |             |
| Field Name        | Field Type                                                                                                                                                                                           | Field Usage |
| Characteristics   | MA_SystemComparativeType                                                                                                                                                                             | Repeated    |
| Description       | This element indicates a System filter requiring a variable logical comparator test between a value in a System-associated message and the value given here. Multiple instances are logically ANDed. |             |
| SystemID          | SystemID_Type                                                                                                                                                                                        | Repeated    |
| Description       | Indicates a specific SystemID that is required to match the filter. If omitted, any SystemID satisfies the filter. Multiple instances are logically ORed.                                            |             |
| GeoFilter         | GeoFiltersQueryType                                                                                                                                                                                  | Repeated    |
| Description       | Indicates a zone the System must fall within to satisfy the clause. Multiple instances are logically ORed.                                                                                           |             |
| OrbitalFilter     | OrbitalFiltersQueryType                                                                                                                                                                              | Repeated    |
| Description       | Indicates a zone the System must fall within to satisfy the clause. Multiple instances are logically ORed.                                                                                           |             |

Table A-1-423: P2P UCI Extensions MA\_ActionCommandMT - System UCI Extension Field

|                            |                                                                                                                                                         |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PreferredAllocation</b> |                                                                                                                                                         |
| <b>Description</b>         | Indicates the system and optionally Capability that the Requirement (Effect, Action, Task, [Capability]Command, Response) should be allocated to during |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                   | the allocation process. The system can be an explicit UCI System or implied UCI System (by Identity, hosted Capability, kinematics, etc.) in which case the associated UCI System must be determined as part of the allocation process. If multiple child elements are given: 1) allocation to any one of the associated Systems is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the associated Systems is acceptable, 3) there is no expectation that each associated System must perform the Requirement. This element allows the Requirement creator to suggest an allocation to a specific system at the risk of 1) a less than optimal overall solution or 2) failure of the Requirement being allocated despite availability of another suitable system. |             |
| Base Type         | MA_RequirementAllocationConstraintType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Field Usage |
| Allocation        | MA_AllocationChoiceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Required    |
| Description       | Indicates the type of Allocation, either by direct assignment of SystemIDs or by PackageID definition for delegation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
| Identity          | IdentityType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Repeated    |
| Description       | Indicates a kind/type or instance of platform/system that is the target of the allocation constraint. For example, this element could be used to allocate a Requirement to any fighter aircraft platform, or a specific kind of fighter aircraft, or a specific tail number of aircraft, or an aircraft with an aircrew member with a certain voice call sign.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| CapabilityID      | CapabilityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Repeated    |
| Description       | Indicates the CapabilityID of the Capability that is the target of the allocation constraint. If multiple required Capabilities are given: 1) allocation to any one of the Capabilities is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the multiple required Capabilities is acceptable, 3) there is no expectation that each Capability must perform the Requirement.                                                                                                                                                                                                                                                                                                                                                                                        |             |

Table A-1-424: P2P UCI Extensions MA\_ActionCommandMT - PreferredAllocation UCI Extension Field

| Analytic                  |                                                                                                                       |             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------|
| Description               | Indicates a constraint in the form of an analytic measure that the Requirement must satisfy when planned/executed.    |             |
| Base Type                 | MA_AnalyticConstraintsType                                                                                            |             |
| Message Structure         |                                                                                                                       |             |
| Field Name                | Field Type                                                                                                            | Field Usage |
| OpConstraint              | OpConstraintWeightingType                                                                                             | Repeated    |
| Description               | Indicates an operational constraint to achieve. The multiplicity is based on the list of available OpConstraintEnums. |             |
| RiskSetting               | SurvivabilityRiskSettingType                                                                                          | Repeated    |
| Description               | Identifies the Risk Level constraint for each provided RiskPrioritization (max of 4).                                 |             |
| AccessAssessmentThreshold | MA_AccessAssessmentFilterType                                                                                         | Optional    |
| Description               | Indicates access assessment analysis event thresholds.                                                                |             |

|                           |                                            |          |
|---------------------------|--------------------------------------------|----------|
| <b>RF_TaskPerformance</b> | RF_TaskPerformanceType                     | Optional |
| Description               | Mission specific RF performance objective. |          |

Table A-1-425: P2P UCI Extensions MA\_ActionCommandMT - Analytic UCI Extension Field

| Designation          |                                                                                                                                                                |             |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | Indicates a type of Entity designation, established by the Designation message, required to satisfy the filter.                                                |             |
| Base Type            | MA_DesignationFilterType                                                                                                                                       |             |
| Message Structure    |                                                                                                                                                                |             |
| Field Name           | Field Type                                                                                                                                                     | Field Usage |
| DesignationType      | DesignationEnum                                                                                                                                                | Repeated    |
| Description          | Indicates a type of designation required to match the filter. List size for this element is based on "Select All That Apply" condition.                        |             |
| DesignationObjective | MA_RequirementTaxonomyType                                                                                                                                     | Optional    |
| Description          | Indicates objective or objectives that the designation must include to match the filter. When omitted, the designation objective is irrelevant for the filter. |             |

Table A-1-426: P2P UCI Extensions MA\_ActionCommandMT - Designation UCI Extension Field

| AssessedObject    |                                                                                                                                                            |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates Assessed Objects of the AccessAssessment that are required to match the filter. If omitted, any Assessed Object satisfies the filter.            |             |
| Base Type         | MA_AssetFilterType                                                                                                                                         |             |
| Message Structure |                                                                                                                                                            |             |
| Field Name        | Field Type                                                                                                                                                 | Field Usage |
| System            | MA_SystemFilterType                                                                                                                                        | Optional    |
| Description       | Indicates a filter on characteristics of a System asset. The asset filter is satisfied if a match is found in this element OR the sibling Entity element.  |             |
| Entity            | MA_EntityFilterType                                                                                                                                        | Optional    |
| Description       | Indicates a filter on characteristics of an Entity asset. The asset filter is satisfied if a match is found in this element OR the sibling System element. |             |

Table A-1-427: P2P UCI Extensions MA\_ActionCommandMT - AssessedObject UCI Extension Field

| Characteristics   |                                                                                                                                                                                                        |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | This element indicates an Entity filter requiring a variable logical comparator test between a value in an Entity-associated message and the value given here. Multiple instances are logically ANDed. |             |
| Base Type         | MA_EntityComparativeType                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                        |             |
| Field Name        | Field Type                                                                                                                                                                                             | Field Usage |
| Characteristic    | MA_EntityCharacteristicType                                                                                                                                                                            | Required    |

|                   |                                                                                                                                                                                                                                                                                                                            |          |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description       | Indicates a characteristic/element value to be used to compare to the corresponding value of the Entity or related message.                                                                                                                                                                                                |          |
| <b>Comparator</b> | EqualityExpressionEnum                                                                                                                                                                                                                                                                                                     | Required |
| Description       | Indicates the logical comparator to use when comparing the target (Entity or related message) Characteristic to the Response Characteristic. The logical expression is: (target Characteristic) (Comparator) (Response Characteristic). When the logical expression is TRUE then the parent TargetClause instance is TRUE. |          |

Table A-1-428: P2P UCI Extensions MA\_ActionCommandMT - Characteristics UCI Extension Field

| Allocation        |                                                                                                                       |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the type of Allocation, either by direct assignment of SystemIDs or by PackageID definition for delegation. |             |
| Base Type         | MA_AllocationChoiceType                                                                                               |             |
| Message Structure |                                                                                                                       |             |
| Field Name        | Field Type                                                                                                            | Field Usage |
| SystemID          | SystemID_Type                                                                                                         | Repeated    |
| Description       | Indicates the SystemID(s) of the System that is the target of the allocation constraint.                              |             |
| PackageID         | PackageID_Type                                                                                                        | Required    |
| Description       | Indicates the unique ID of the package for delegation of Systems.                                                     |             |

Table A-1-429: P2P UCI Extensions MA\_ActionCommandMT - Allocation UCI Extension Field

| ConstraintOverride |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates refined/overridden values of constraints for the Action. Added to allow similar override options in direct commanding as is available in planning.                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |
| Base Type          | MA_RequirementConstraintsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| Message Structure  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Field Usage |
| Rank               | ComparableRankingType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Optional    |
| Description        | Indicates a constraint in the form of the Rank of the associated Requirement (Effect, Action, Task, [Capability]Command or Response).                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| InterruptLowerRank | boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Optional    |
| Description        | Indicates whether a) an in-work/planned item or items of a lower rank should be interrupted/replanned in order to address this item (TRUE) or b) this item should be simply queued/planned according to rank without interrupting in-work/planned items (FALSE). For example, a Task being added to an existing TaskPlan (TaskPlanCommand constrained by an existing TaskPlan) would potentially cause unlocked Tasks of lower rank in the existing TaskPlan to be unallocated if 1) this element indicated TRUE and 2) the lower ranked Tasks prevented the higher ranked Task from being allocated. |             |
| Allocation         | MA_RequirementAllocationParametersType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Optional    |

|                                                   |             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|---------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                                   | Description | Indicates allocation related constraints which can vary by Requirement at the discretion of the creator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>Timing</b>                                     |             | RequirementTimingType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Repeated |
|                                                   | Description | Indicates a temporal constraint on the Requirement. Allows for defining for each timing type (RequirementTimingEnum).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |
| <b>Kinematic</b>                                  |             | MA_RequirementKinematicConstraintsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Repeated |
|                                                   | Description | Indicates a constraint in the form of kinematic position of Requirement actor or actors. When more than one is given they are logically ANDed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>Dependency</b>                                 |             | RequirementDependencyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated |
|                                                   | Description | <p>Indicates a constraint in the form of a temporal dependency between Requirements; between Effect, Action, Task, [Capability]Command or Commands or Response. If more than one is given, they are logically ANDed.</p> <p>For example, a pre-strike image collection (PO Task) might be required several minutes prior to a strike (Strike Task) to determine combat ID and target coordinates. A post-strike image collection (PO Task) might be required several minutes after the strike to inform battle damage assessment. The dependency between the three Tasks can be established with this element. Assuming the Strike Task is the most difficult to plan/coordinate, the other two Tasks would likely each have dependencies on the Strike Task in order to orchestrate the desired "look-shoot-look" sequence.</p> |          |
| <b>MaxProductDisseminationClassificationLevel</b> |             | DataProductClassificationLevelType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Optional |
|                                                   | Description | <p>Indicates the maximum allowed classification level of the networks that the allocated system can disseminate products to for this task. The allocated system must have a dissemination route for data at this classification or lower.</p> <p>For example, when trying to collect data to hand-off to an unclassified system or network, the allocated system has to be able to generate products at no higher than the specified level and have the authority to disseminate on a network at no higher than the specified classification level.</p>                                                                                                                                                                                                                                                                          |          |
| <b>Analytic</b>                                   |             | MA_AnalyticConstraintsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Optional |
|                                                   | Description | Indicates a constraint in the form of an analytic measure that the Requirement must satisfy when planned/executed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>AcceptableClassificationLevel</b>              |             | SecurityInformationType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Optional |
|                                                   | Description | Indicates a constraint on the highest acceptable classification level at which the Response can be planned and executed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>CommsRequired</b>                              |             | boolean                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Optional |
|                                                   | Description | Indicates whether or not communications are required when executing the Requirement. In this context, "required" means the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |

|                                 |                                                                                                                                                                                                                                                                                                   |          |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                 | creator or a Requirement considers communications as necessary for C2 and/or Product dissemination related to this Requirement, independent of the type of vehicle that it is allocated to. This element may result in consideration of communications allocations during Requirement allocation. |          |
| <b>AllowedRequirementTypes</b>  | MA_RequirementTaxonomyType                                                                                                                                                                                                                                                                        | Optional |
| Description                     | Indicates other types of Requirements that can be decomposed from, used to plan, used to implement, inherited from traced parents or otherwise associated with the subject Requirement.                                                                                                           |          |
| <b>ExcludedRequirementTypes</b> | MA_RequirementTaxonomyType                                                                                                                                                                                                                                                                        | Optional |
| Description                     | Indicates other types of Requirements that cannot be decomposed from, used to plan, used to implement, inherited from traced parents or otherwise be associated with the subject Requirement.                                                                                                     |          |
| <b>ConstraintsNarrative</b>     | VisibleString1024Type                                                                                                                                                                                                                                                                             | Optional |
| Description                     | Free-text explanation of sibling constraints, additional extraordinary constraints, etc.                                                                                                                                                                                                          |          |
| <b>AllowedDomainPairing</b>     | EnvironmentPairingEnum                                                                                                                                                                                                                                                                            | Repeated |
| Description                     | Indicates an allowed domain/environment which the Requirement can be employed from and allowed "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.                                        |          |
| <b>ExcludedDomainPairing</b>    | EnvironmentPairingEnum                                                                                                                                                                                                                                                                            | Repeated |
| Description                     | Indicates an excluded domain/environment which the Requirement can't be employed from an excluded "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.                                     |          |
| <b>EngagementParameters</b>     | MA_EngagementParametersType                                                                                                                                                                                                                                                                       | Optional |
| Description                     | Provides engagement parameters that this requirement should adhere to.                                                                                                                                                                                                                            |          |

Table A-1-430: P2P UCI Extensions MA\_ActionCommandMT - ConstraintOverride UCI Extension Field

| Allocation         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates allocation related constraints which can vary by Requirement at the discretion of the creator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
| Base Type          | MA_RequirementAllocationParametersType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| Message Structure  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Field Usage |
| RequiredAllocation | MA_RequirementAllocationConstraintType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Optional    |
| Description        | Indicates the system and optionally Capability that the Requirement (Effect, Action, Task, [Capability]Command, Response) MUST be allocated to during the allocation process. The system can be an explicit UCI System or implied UCI System (by Identity, hosted Capability, kinematics, etc.) in which case the associated UCI System must be determined as part of the allocation process. If multiple child elements are given: 1) allocation to any one of the associated Systems is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the associated Systems is acceptable, 3) there is no expectation that each associated |             |



|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                            | System must perform the Requirement. This element allows the Requirement creator to force an allocation to a specific system at the risk of 1) a less than optimal overall solution or 2) failure of the Requirement being allocated despite availability of another suitable system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>PreferredAllocation</b> | MA_RequirementAllocationConstraintType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Optional |
| Description                | Indicates the system and optionally Capability that the Requirement (Effect, Action, Task, [Capability]Command, Response) should be allocated to during the allocation process. The system can be an explicit UCI System or implied UCI System (by Identity, hosted Capability, kinematics, etc.) in which case the associated UCI System must be determined as part of the allocation process. If multiple child elements are given: 1) allocation to any one of the associated Systems is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the associated Systems is acceptable, 3) there is no expectation that each associated System must perform the Requirement. This element allows the Requirement creator to suggest an allocation to a specific system at the risk of 1) a less than optimal overall solution or 2) failure of the Requirement being allocated despite availability of another suitable system. |          |

Table A-1-431: P2P UCI Extensions MA\_ActionCommandMT - Allocation UCI Extension Field

| AccessAssessmentThreshold |                                                                                                                                                                 |             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description               | Indicates access assessment analysis event thresholds.                                                                                                          |             |
| Base Type                 | MA_AccessAssessmentFilterType                                                                                                                                   |             |
| Message Structure         |                                                                                                                                                                 |             |
| Field Name                | Field Type                                                                                                                                                      | Field Usage |
| AssessmentName            | VisibleString32Type                                                                                                                                             | Repeated    |
| Description               | Indicates a name/type of AccessAssessment algorithm required to match the filter. If omitted, any name/type of AccessAssessment algorithm satisfies the filter. |             |
| AssessedCapabilities      | CapabilityTaxonomyUniversalType                                                                                                                                 | Optional    |
| Description               | Indicates assessed Capabilities of the AccessAssessment that are required to match the filter. If omitted, any assessed Capability satisfies the filter.        |             |
| Subject                   | MA_AssetFilterType                                                                                                                                              | Optional    |
| Description               | Indicates Subjects of the AccessAssessment that are required to match the filter. If omitted, any Subject satisfies the filter.                                 |             |
| AssessedObject            | MA_AssetFilterType                                                                                                                                              | Optional    |
| Description               | Indicates Assessed Objects of the AccessAssessment that are required to match the filter. If omitted, any Assessed Object satisfies the filter.                 |             |
| AccessEvent               | AccessEventFilterType                                                                                                                                           | Repeated    |
| Description               | Indicates Access Event details necessary to match the filter. If omitted, any Access Event satisfies the filter. Multiple instances are logically ORed.         |             |

Table A-1-432: P2P UCI Extensions MA\_ActionCommandMT - AccessAssessmentThreshold UCI Extension Field

| Command     |                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description | Indicates an instance of an Action Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each |



|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                   | command instance reports status via a separate, independent Capability CommandStatus message.                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| Base Type         | MA_ActionCommandType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Field Usage |
| Capability        | MA_ActionCapabilityCommandType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Required    |
| Description       | Indicates a new invocation of an Action Capability. Generally, if accepted, the command will result in one or more new Action Activities being created and reported via the ActionActivity message. The request/response interaction terminates as soon as the command is accepted or rejected; this element is not used to interact with an "active" command. Updates and/or additional interaction with the resulting Activity (after the command is accepted) is accomplished via the sibling Activity element. |             |
| Activity          | MA_ActivityCommandBaseType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Required    |
| Description       | Indicates a command to modify an existing Action Activity (which was previously reported via the ActionActivity message and was marked as "interactive"). The request/response interaction terminates as soon as the modification is accepted or rejected. The modifications are reflected in subsequent ActionActivity messages.                                                                                                                                                                                  |             |

Table A-1-433: P2P UCI Extensions MA\_ActionCommandMT - Command UCI Extension Field

| ExcludedRequirementTypes |                                                                                                                                                                                               |             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | Indicates other types of Requirements that cannot be decomposed from, used to plan, used to implement, inherited from traced parents or otherwise be associated with the subject Requirement. |             |
| Base Type                | MA_RequirementTaxonomyType                                                                                                                                                                    |             |
| Message Structure        |                                                                                                                                                                                               |             |
| Field Name               | Field Type                                                                                                                                                                                    | Field Usage |
| Effect                   | EffectTypeEnum                                                                                                                                                                                | Repeated    |
| Description              | Indicates Effects applicable via the Effect* message set. List size for this element is based on "Select All That Apply" condition.                                                           |             |
| Action                   | MA_ActionTypeEnum                                                                                                                                                                             | Repeated    |
| Description              | Indicates Actions applicable via the Action* message set. List size for this element is based on "Select All That Apply" condition.                                                           |             |
| Task                     | MA_TaskTypeEnum                                                                                                                                                                               | Repeated    |
| Description              | Indicates Tasks applicable via the Task* message set. List size for this element is based on "Select All That Apply" condition.                                                               |             |
| CapabilityCommand        | MA_CapabilityTypeEnum                                                                                                                                                                         | Repeated    |
| Description              | Indicates CapabilityCommands applicable via a [Capability]Command message set. List size for this element is based on "Select All That Apply" condition.                                      |             |
| Response                 | ResponseTypeEnum                                                                                                                                                                              | Repeated    |
| Description              | Indicates Responses applicable via the Response* message set. List size for this element is based on "Select All That Apply" condition.                                                       |             |

Table A-1-434: P2P UCI Extensions MA\_ActionCommandMT - ExcludedRequirementTypes UCI Extension Field

| Kinematic            |                                                                                                                                                |             |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | Indicates a constraint in the form of kinematic position of Requirement actor or actors. When more than one is given they are logically ANDed. |             |
| Base Type            | MA_RequirementKinematicConstraintsType                                                                                                         |             |
| Message Structure    |                                                                                                                                                |             |
| Field Name           | Field Type                                                                                                                                     | Field Usage |
| ApplicableObject     | RequirementObjectEnum                                                                                                                          | Required    |
| Description          | Indicates which Requirement actor the sibling kinematic constraints are applicable to.                                                         |             |
| Geo                  | GeoFiltersQueryType                                                                                                                            | Repeated    |
| Description          | This element gives a geographical filter to the query.                                                                                         |             |
| Orbital              | OrbitalFiltersQueryType                                                                                                                        | Repeated    |
| Description          | This element gives an orbital filter to the query.                                                                                             |             |
| LookAngleConstraints | AnglePairType                                                                                                                                  | Optional    |
| Description          | The angle from true north to the line-of-sight vector from the platform to the target tangent plane.                                           |             |

Table A-1-435: P2P UCI Extensions MA\_ActionCommandMT - Kinematic UCI Extension Field

| EngagementParameters      |                                                                                                                               |             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description               | Provides engagement parameters that this requirement should adhere to.                                                        |             |
| Base Type                 | MA_EngagementParametersType                                                                                                   |             |
| Message Structure         |                                                                                                                               |             |
| Field Name                | Field Type                                                                                                                    | Field Usage |
| InterceptTactic           | MA_InterceptTacticChoiceType                                                                                                  | Optional    |
| Description               | The type of intercept tactic to be employed. Used in combination with shot parameters to define tactics like Skate or Banzai. |             |
| ShotParameters            | MA_ShotParametersType                                                                                                         | Optional    |
| Description               | Provides shot parameters that this requirement should adhere to.                                                              |             |
| EnableTargetRecalculation | boolean                                                                                                                       | Optional    |
| Description               | This field specifies that MA should recalculate optimal targets during the engagement.                                        |             |

Table A-1-436: P2P UCI Extensions MA\_ActionCommandMT - EngagementParameters UCI Extension Field

| Subject           |                                                                                                                                 |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates Subjects of the AccessAssessment that are required to match the filter. If omitted, any Subject satisfies the filter. |             |
| Base Type         | MA_AssetFilterType                                                                                                              |             |
| Message Structure |                                                                                                                                 |             |
| Field Name        | Field Type                                                                                                                      | Field Usage |
| System            | MA_SystemFilterType                                                                                                             | Optional    |

|               |                                                                                                                                                            |          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description   | Indicates a filter on characteristics of a System asset. The asset filter is satisfied if a match is found in this element OR the sibling Entity element.  |          |
| <b>Entity</b> | MA_EntityFilterType                                                                                                                                        | Optional |
| Description   | Indicates a filter on characteristics of an Entity asset. The asset filter is satisfied if a match is found in this element OR the sibling System element. |          |

Table A-1-437: P2P UCI Extensions MA\_ActionCommandMT - Subject UCI Extension Field

| MessageData       |                                                                                                                                                                                                                                                                                                  |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                          |             |
| Base Type         | MA_ActionCommandMDT                                                                                                                                                                                                                                                                              |             |
| Message Structure |                                                                                                                                                                                                                                                                                                  |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                       | Field Usage |
| Command           | MA_ActionCommandType                                                                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates an instance of an Action Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message. |             |

Table A-1-438: P2P UCI Extensions MA\_ActionCommandMT - MessageData UCI Extension Field

| AllowedRequirementTypes |                                                                                                                                                                                         |             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description             | Indicates other types of Requirements that can be decomposed from, used to plan, used to implement, inherited from traced parents or otherwise associated with the subject Requirement. |             |
| Base Type               | MA_RequirementTaxonomyType                                                                                                                                                              |             |
| Message Structure       |                                                                                                                                                                                         |             |
| Field Name              | Field Type                                                                                                                                                                              | Field Usage |
| Effect                  | EffectTypeEnum                                                                                                                                                                          | Repeated    |
| Description             | Indicates Effects applicable via the Effect* message set. List size for this element is based on "Select All That Apply" condition.                                                     |             |
| Action                  | MA_ActionTypeEnum                                                                                                                                                                       | Repeated    |
| Description             | Indicates Actions applicable via the Action* message set. List size for this element is based on "Select All That Apply" condition.                                                     |             |
| Task                    | MA_TaskTypeEnum                                                                                                                                                                         | Repeated    |
| Description             | Indicates Tasks applicable via the Task* message set. List size for this element is based on "Select All That Apply" condition.                                                         |             |
| CapabilityCommand       | MA_CapabilityTypeEnum                                                                                                                                                                   | Repeated    |
| Description             | Indicates CapabilityCommands applicable via a [Capability]Command message set. List size for this element is based on "Select All That Apply" condition.                                |             |
| Response                | ResponseTypeEnum                                                                                                                                                                        | Repeated    |
| Description             | Indicates Responses applicable via the Response* message set. List size for this element is based on "Select All That Apply" condition.                                                 |             |

**Table A-1-439: P2P UCI Extensions MA\_ActionCommandMT - AllowedRequirementTypes UCI Extension Field**

| Entity            |                                                                                                                                                                                                                                       |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a filter on characteristics of an Entity asset. The asset filter is satisfied if a match is found in this element OR the sibling System element.                                                                            |             |
| Base Type         | MA_EntityFilterType                                                                                                                                                                                                                   |             |
| Message Structure |                                                                                                                                                                                                                                       |             |
| Field Name        | Field Type                                                                                                                                                                                                                            | Field Usage |
| Designation       | MA_DesignationFilterType                                                                                                                                                                                                              | Optional    |
| Description       | Indicates a type of Entity designation, established by the Designation message, required to satisfy the filter.                                                                                                                       |             |
| Characteristics   | MA_EntityComparativeType                                                                                                                                                                                                              | Repeated    |
| Description       | This element indicates an Entity filter requiring a variable logical comparator test between a value in an Entity-associated message and the value given here. Multiple instances are logically ANDed.                                |             |
| EntityID          | EntityID_Type                                                                                                                                                                                                                         | Repeated    |
| Description       | Indicates a specific EntityID that is required to match the filter. If omitted, any EntityID satisfies the filter. Multiple instances are logically ORed.                                                                             |             |
| Source            | EntitySourceEnum                                                                                                                                                                                                                      | Repeated    |
| Description       | Indicates an Entity source that is required to match the filter. If omitted, any Entity source satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition. |             |
| GeoFilter         | GeoFiltersQueryType                                                                                                                                                                                                                   | Repeated    |
| Description       | Indicates a zone the Entity must fall within to satisfy the filter. Multiple instances are logically ORed.                                                                                                                            |             |
| OrbitalFilter     | OrbitalFiltersQueryType                                                                                                                                                                                                               | Repeated    |
| Description       | Indicates a zone the Entity must fall within to satisfy the filter. Multiple instances are logically ORed.                                                                                                                            |             |

**Table A-1-440: P2P UCI Extensions MA\_ActionCommandMT - Entity UCI Extension Field****1.3.2.80 MA\_ActionCommandStatusMT UCI Extension**

| MA_ActionCommandStatusMT |                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>       | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>       | MessageData : MA_ActionCommandStatusMDT                                                                                       |

**Table A-1-441: P2P UCI Extensions MA\_ActionCommandStatusMT UCI Extension**

| MessageData        |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |

|                                     |                                                                                                                                                                                                                                                                                                                                                             |                           |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Base Type</b>                    |                                                                                                                                                                                                                                                                                                                                                             | MA_ActionCommandStatusMDT |
| <b>Message Structure</b>            |                                                                                                                                                                                                                                                                                                                                                             |                           |
| <b>Field Name</b>                   | <b>Field Type</b>                                                                                                                                                                                                                                                                                                                                           | <b>Field Usage</b>        |
| <b>Activity</b>                     | ResultingActivityType                                                                                                                                                                                                                                                                                                                                       | Repeated                  |
| Description                         | Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.                                                                                                                                                                                                                    |                           |
| <b>CommandID</b>                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required                  |
| Description                         | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |                           |
| <b>CommandProcessingState</b>       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required                  |
| Description                         | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |                           |
| <b>CommandProcessingStateReason</b> | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional                  |
| Description                         | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |                           |

Table A-1-442: P2P UCI Extensions MA\_ActionCommandStatusMT - MessageData UCI Extension Field

### 1.3.2.81 MA\_ActionStatusMT UCI Extension

|                          |                                                                                                                                                                                                            |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_ActionStatusMT</b> |                                                                                                                                                                                                            |
| <b>Description</b>       | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info                                                                              |
| <b>Ext. Fields</b>       | Traceability : MA_RequirementStatusTraceabilityType<br>Plans : MA_PlansReferenceType<br>OpPlanID : MA_OpPlanID_Type<br>MessageData : MA_ActionStatusMDT<br>ExecutingSystemOrPackage : MA_PackageSystemType |

Table A-1-443: P2P UCI Extensions MA\_ActionStatusMT UCI Extension

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Traceability</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Description</b>  | Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture.<br><br>For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for |

|                   |                                                                                                                                                                                                                                                                                                                                    |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                   | architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands. |             |
| Base Type         | MA_RequirementStatusTraceabilityType                                                                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                    |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                         | Field Usage |
| Plans             | MA_PlansReferenceType                                                                                                                                                                                                                                                                                                              | Optional    |
| Description       | Indicates *Plans the Requirement is associated with.                                                                                                                                                                                                                                                                               |             |
| PlannedActivityID | PlannedActivityID_Type                                                                                                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates a specific Planned Activity the Requirement is associated with.                                                                                                                                                                                                                                                          |             |
| CapabilityID      | CapabilityID_Type                                                                                                                                                                                                                                                                                                                  | Repeated    |
| Description       | Identifies a specific Capability the Requirement is associated with.                                                                                                                                                                                                                                                               |             |
| CommandID         | CommandID_Type                                                                                                                                                                                                                                                                                                                     | Repeated    |
| Description       | Indicates the ID of an associated [Capability]Command.                                                                                                                                                                                                                                                                             |             |
| ActivityID        | ActivityID_Type                                                                                                                                                                                                                                                                                                                    | Repeated    |
| Description       | Indicates the ID of an associated actual/executing/executed Activity.                                                                                                                                                                                                                                                              |             |

Table A-1-444: P2P UCI Extensions MA\_ActionStatusMT - Traceability UCI Extension Field

| Plans               |                                                      |             |
|---------------------|------------------------------------------------------|-------------|
| Description         | Indicates *Plans the Requirement is associated with. |             |
| Base Type           | MA_PlansReferenceType                                |             |
| Message Structure   |                                                      |             |
| Field Name          | Field Type                                           | Field Usage |
| MissionPlanID       | MissionPlanID_Type                                   | Repeated    |
| Description         | Indicates a reference to a MissionPlan.              |             |
| TaskPlanID          | TaskPlanID_Type                                      | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                 |             |
| OrbitPlanID         | OrbitPlanID_Type                                     | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.               |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                             | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.       |             |
| RoutePlanID         | RoutePlanID_Type                                     | Repeated    |
| Description         | Indicates a reference to a RoutePlan.                |             |
| RouteActivityPlanID | RouteActivityPlanID_Type                             | Repeated    |
| Description         | Indicates a reference to a RouteActivityPlan.        |             |
| ActivityPlanID      | ActivityPlanID_Type                                  | Repeated    |

|                       |                                           |          |
|-----------------------|-------------------------------------------|----------|
| Description           | Indicates a reference to an ActivityPlan. |          |
| <b>EffectPlanID</b>   | EffectPlanID_Type                         | Repeated |
| Description           | Indicates a reference to an EffectPlan.   |          |
| <b>ActionPlanID</b>   | ActionPlanID_Type                         | Repeated |
| Description           | Indicates a reference to an ActionPlan.   |          |
| <b>ResponsePlanID</b> | ResponsePlanID_Type                       | Repeated |
| Description           | Indicates a reference to an ResponsePlan. |          |
| <b>OpPlanID</b>       | MA_OpPlanID_Type                          | Repeated |
| Description           | Indicates a reference to an OpPlan.       |          |

Table A-1-445: P2P UCI Extensions MA\_ActionStatusMT - Plans UCI Extension Field

| OpPlanID          |                                                                                                                                                                                                                                                                |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a reference to an OpPlan.                                                                                                                                                                                                                            |             |
| Base Type         | MA_OpPlanID_Type                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                     | Field Usage |
| Version           | unsignedInt                                                                                                                                                                                                                                                    | Optional    |
| Description       | Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval. |             |
| UUID              | UniversallyUniqueIdentifierType                                                                                                                                                                                                                                | Required    |
| Description       | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.                                                                                               |             |
| DescriptiveLabel  | VisibleString256Type                                                                                                                                                                                                                                           | Optional    |
| Description       | Human readable text label with content meaningful to human operators.                                                                                                                                                                                          |             |

Table A-1-446: P2P UCI Extensions MA\_ActionStatusMT - OpPlanID UCI Extension Field

| MessageData              |                                                                                                                                                                |             |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | See annotations in child elements and messages/elements that use this type for details.                                                                        |             |
| Base Type                | MA_ActionStatusMDT                                                                                                                                             |             |
| Message Structure        |                                                                                                                                                                |             |
| Field Name               | Field Type                                                                                                                                                     | Field Usage |
| ActionID                 | ActionID_Type                                                                                                                                                  | Required    |
| Description              | Indicates the unique ID of the Action whose execution status is given by sibling elements.                                                                     |             |
| ExecutingSystemOrPackage | MA_PackageSystemType                                                                                                                                           | Required    |
| Description              | Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling |             |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                             | elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>ExecutionState</b>       | RequirementExecutionStateEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Required |
| Description                 | Identifies the state of the Requirement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
| <b>ExecutionStateReason</b> | CannotComplyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                 | Indicates details regarding the reason for and/or a contributing factor to the sibling State element. This element is only expected when the State is in a potentially undesirable state such as failed or completed with less than optimum results. It is strongly encouraged for transitions to failed or dropped.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>ActualTiming</b>         | RequirementTimingType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Repeated |
| Description                 | Indicates actual timing status for the Requirement. Allows for defining for each timing type (RequirementTimingEnum).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
| <b>PercentCompleted</b>     | PercentType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Optional |
| Description                 | Percentage of assigned Requirement completion. The percentage increments until the Requirement has COMPLETED or FAILED. A percent complete of 0 indicates the Requirement has not yet begun execution, and a failed Requirement may have any percentage between 1 - 99.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>Traceability</b>         | MA_RequirementStatusTraceabilityType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Optional |
| Description                 | <p>Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture.</p> <p>For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands.</p> |          |
| <b>Metrics</b>              | MetricsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Optional |
| Description                 | Specifies the actual metrics associated with the Requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |

Table A-1-447: P2P UCI Extensions MA\_ActionStatusMT - MessageData UCI Extension Field

| ExecutingSystemOrPackage |                                                                                                                                                                                                                                                                                          |             |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement. |             |
| Base Type                | MA_PackageSystemType                                                                                                                                                                                                                                                                     |             |
| Message Structure        |                                                                                                                                                                                                                                                                                          |             |
| Field Name               | Field Type                                                                                                                                                                                                                                                                               | Field Usage |
| ALL                      | EmptyType                                                                                                                                                                                                                                                                                | Required    |
| Description              | Indicates applies to ALL entities.                                                                                                                                                                                                                                                       |             |



|                  |                                   |          |
|------------------|-----------------------------------|----------|
| <b>NONE</b>      | EmptyType                         | Required |
| Description      | Indicates applies to NO entities. |          |
| <b>PackageID</b> | PackageID_Type                    | Required |
| Description      | The ID of a Package.              |          |
| <b>SystemID</b>  | SystemID_Type                     | Required |
| Description      | The ID of an ACP or C2 Operator.  |          |

Table A-1-448: P2P UCI Extensions MA\_ActionStatusMT - ExecutingSystemOrPackage UCI Extension Field

## 1.3.2.82 MA\_AssessmentMT UCI Extension

| MA_AssessmentMT    |                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_AssessmentMDT<br>Assessment : MA_AssessmentType                                                                            |

Table A-1-449: P2P UCI Extensions MA\_AssessmentMT UCI Extension

| MessageData         |                                                                                                                                                      |             |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | See annotations in child elements and messages/elements that use this type for details.                                                              |             |
| Base Type           | MA_AssessmentMDT                                                                                                                                     |             |
| Message Structure   |                                                                                                                                                      |             |
| Field Name          | Field Type                                                                                                                                           | Field Usage |
| AssessmentID        | AssessmentID_Type                                                                                                                                    | Required    |
| Description         | This element references the assessment ID.                                                                                                           |             |
| AssessmentRequestID | RequestID_Type                                                                                                                                       | Optional    |
| Description         | This element is required when the message is generated in response to an AssessmentRequest. It is omitted when the message is generated unsolicited. |             |
| AnalysisRouteID     | AnalysisRouteID_Type                                                                                                                                 | Optional    |
| Description         | This element indicates the AnalysisRoute that was used to perform the assessment.                                                                    |             |
| Assessment          | MA_AssessmentType                                                                                                                                    | Required    |
| Description         | This element indicates the results of the assessment.                                                                                                |             |

Table A-1-450: P2P UCI Extensions MA\_AssessmentMT - MessageData UCI Extension Field

| Assessment         |                                                       |
|--------------------|-------------------------------------------------------|
| <b>Description</b> | This element indicates the results of the assessment. |
| <b>Base Type</b>   | MA_AssessmentType                                     |
| Message Structure  |                                                       |

| Field Name                        | Field Type                                                                                                 | Field Usage |
|-----------------------------------|------------------------------------------------------------------------------------------------------------|-------------|
| <b>CommPointingPlan</b>           | CommPointingPlanAssessmentType                                                                             | Required    |
| Description                       | This element defines the response to a communications pointing plan assessment request.                    |             |
| <b>CapabilityUtilization</b>      | CapabilityUtilizationAssessmentType                                                                        | Required    |
| Description                       | This assessment type is utilized to assess predicted capability utilization along a mission planned route. |             |
| <b>RouteDeconfliction</b>         | RouteDeconflictionAssessmentType                                                                           | Required    |
| Description                       | This element defines the response to a route deconfliction assessment request.                             |             |
| <b>RouteVulnerabilityMetrics</b>  | RouteVulnerabilityMetricsAssessmentType                                                                    | Required    |
| Description                       | Indicates the results of assessment of vulnerability along a route.                                        |             |
| <b>RouteThreatAssessment</b>      | RouteThreatAssessmentType                                                                                  | Required    |
| Description                       | Indicates the results of assessment of threats along a route.                                              |             |
| <b>TargetMobility</b>             | TargetMobilityAssessmentType                                                                               | Required    |
| Description                       | This element defines the response to a target mobility assessment request.                                 |             |
| <b>VehicleThreatAssessment</b>    | VehicleThreatAssessmentType                                                                                | Required    |
| Description                       | This element defines the response to a vehicle threat assessment request.                                  |             |
| <b>ThreatNominationAssessment</b> | ThreatNominationAssessmentType                                                                             | Required    |
| Description                       | This element defines the response to a threat nomination assessment.                                       |             |
| <b>AchievabilityAssessment</b>    | AchievabilityAssessmentPET                                                                                 | Required    |
| Description                       | This element defines the response to an achievability assessment.                                          |             |
| <b>CoverageAssessment</b>         | MA_CoverageAssessmentType                                                                                  | Required    |
| Description                       | This element defines the response to a coverage assessment.                                                |             |

Table A-1-451: P2P UCI Extensions MA\_AssessmentMT - Assessment UCI Extension Field

## 1.3.2.83 MA\_AssessmentRequestMT UCI Extension

| MA_AssessmentRequestMT |                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>     | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>     | MessageData : MA_AssessmentRequestMDT<br>Category : MA_AssessmentRequestType                                                                |

Table A-1-452: P2P UCI Extensions MA\_AssessmentRequestMT UCI Extension

| MessageData        |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |
| <b>Base Type</b>   | MA_AssessmentRequestMDT                                                                 |

| Message Structure                 |                                                                                                                                                                                                                                                                                                                               |             |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Field Name                        | Field Type                                                                                                                                                                                                                                                                                                                    | Field Usage |
| <b>AnalysisRouteID</b>            | AnalysisRouteID_Type                                                                                                                                                                                                                                                                                                          | Optional    |
| Description                       | This element indicates the AnalysisRoute to use when performing the Assessment specified in sibling Category.                                                                                                                                                                                                                 |             |
| <b>RequestFrequencyType</b>       | RepeatEnum                                                                                                                                                                                                                                                                                                                    | Required    |
| Description                       | Indicates whether this request for assessment is a request for a single reply, or a request for periodic assessments with a corresponding request to cease periodic assessments.                                                                                                                                              |             |
| <b>RequestFrequencyPeriod</b>     | DurationType                                                                                                                                                                                                                                                                                                                  | Optional    |
| Description                       | If requesting periodic assessments, this field specifies the amount of time that should elapse between the periodic assessments, i.e. the repetition time interval of the assessment.                                                                                                                                         |             |
| <b>ResultsInAssessmentMessage</b> | boolean                                                                                                                                                                                                                                                                                                                       | Required    |
| Description                       | If TRUE this element specifies that the response to this message should be an AssessmentRequest and an Assessment message. If FALSE the assessment data that is generated should be bundled inside the response. In some cases there isn't a native message and the results can only be reported via AssessmentRequestStatus. |             |
| <b>Category</b>                   | MA_AssessmentRequestType                                                                                                                                                                                                                                                                                                      | Required    |
| Description                       | Indicates the assessment category for requested assessment.                                                                                                                                                                                                                                                                   |             |
| <b>RequestID</b>                  | RequestID_Type                                                                                                                                                                                                                                                                                                                | Required    |
| Description                       | Indicates the unique ID of the request message.                                                                                                                                                                                                                                                                               |             |
| <b>RequestState</b>               | RequestStateEnum                                                                                                                                                                                                                                                                                                              | Required    |
| Description                       | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.                                                                                                                                                                                     |             |

Table A-1-453: P2P UCI Extensions MA\_AssessmentRequestMT - MessageData UCI Extension Field

| Category              |                                                                                                                                                                                        |             |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description           | Indicates the assessment category for requested assessment.                                                                                                                            |             |
| Base Type             | MA_AssessmentRequestType                                                                                                                                                               |             |
| Message Structure     |                                                                                                                                                                                        |             |
| Field Name            | Field Type                                                                                                                                                                             | Field Usage |
| CommPointingPlan      | CommPointingPlanRequestType                                                                                                                                                            | Required    |
| Description           | This element is used to specify that communication requests should be assessed, and a Pointing Plan should be generated based on the requests. The native message is CommPointingPlan. |             |
| CapabilityUtilization | CapabilityUtilizationRequestType                                                                                                                                                       | Required    |
| Description           | This assessment type is utilized to assess predicted Capability utilization along a mission planned route.                                                                             |             |
| RouteDeconfliction    | RouteDeconflictionRequestType                                                                                                                                                          | Required    |

|                                   |                                                                                                                                                                                                                                                                                                                                          |          |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                       | This element is used to specify that an assessment of the conflicts along a route is requested.                                                                                                                                                                                                                                          |          |
| <b>RouteVulnerabilityMetrics</b>  | RouteVulnerabilityMetricsRequestType                                                                                                                                                                                                                                                                                                     | Required |
| Description                       | This element is used to specify that detection metrics need to be recomputed along a route due to threats. The data generated for detection metrics are different from threat assessments because detection metrics break down the exposure numbers to a much greater detail in order to generate a route than a threat assessment will. |          |
| <b>RouteThreatAssessment</b>      | ThreatAssessmentRequestType                                                                                                                                                                                                                                                                                                              | Required |
| Description                       | This element is used to specify that an assessment of the threat exposure along a route is requested.                                                                                                                                                                                                                                    |          |
| <b>TargetMobility</b>             | TargetMobilityRequestType                                                                                                                                                                                                                                                                                                                | Required |
| Description                       | This element is used to specify that an assessment of the possible zone an Entity could have moved within some given period of time.                                                                                                                                                                                                     |          |
| <b>VehicleThreatAssessment</b>    | VehicleThreatAssessmentRequestType                                                                                                                                                                                                                                                                                                       | Required |
| Description                       | This element is used to specify that a threat assessment is being requested for the current position of a vehicle.                                                                                                                                                                                                                       |          |
| <b>ThreatNominationAssessment</b> | ThreatNominationAssessmentRequestType                                                                                                                                                                                                                                                                                                    | Required |
| Description                       | This element is used to specify that a threat nomination assessment is being requested for the given mission plans and entities.                                                                                                                                                                                                         |          |
| <b>AchievabilityAssessment</b>    | AchievabilityAssessmentRequestPET                                                                                                                                                                                                                                                                                                        | Required |
| Description                       | This element is used to specify that achievability assessment is being requested for the given type of AchievabilityRequest.                                                                                                                                                                                                             |          |
| <b>CoverageAssessment</b>         | MA_CoverageAssessmentRequestType                                                                                                                                                                                                                                                                                                         | Required |
| Description                       | This element is used to specify that assessment is being requested for sensor coverage.                                                                                                                                                                                                                                                  |          |

Table A-1-454: P2P UCI Extensions MA\_AssessmentRequestMT - Category UCI Extension Field

### 1.3.2.84 MA\_AssessmentRequestStatusMT UCI Extension

| MA_AssessmentRequestStatusMT |                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>           | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>           | Assessment : MA_AssessmentType<br>MessageData : MA_AssessmentRequestStatusMDT                                                               |

Table A-1-455: P2P UCI Extensions MA\_AssessmentRequestStatusMT UCI Extension

| Assessment         |                                                                                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | This element indicates the results of the assessment. It is optional based on whether the assessment request specified that the response should result in native message. |
| <b>Base Type</b>   | MA_AssessmentType                                                                                                                                                         |

| Message Structure                 |                                                                                                            |             |
|-----------------------------------|------------------------------------------------------------------------------------------------------------|-------------|
| Field Name                        | Field Type                                                                                                 | Field Usage |
| <b>CommPointingPlan</b>           | CommPointingPlanAssessmentType                                                                             | Required    |
| Description                       | This element defines the response to a communications pointing plan assessment request.                    |             |
| <b>CapabilityUtilization</b>      | CapabilityUtilizationAssessmentType                                                                        | Required    |
| Description                       | This assessment type is utilized to assess predicted capability utilization along a mission planned route. |             |
| <b>RouteDeconfliction</b>         | RouteDeconflictionAssessmentType                                                                           | Required    |
| Description                       | This element defines the response to a route deconfliction assessment request.                             |             |
| <b>RouteVulnerabilityMetrics</b>  | RouteVulnerabilityMetricsAssessmentType                                                                    | Required    |
| Description                       | Indicates the results of assessment of vulnerability along a route.                                        |             |
| <b>RouteThreatAssessment</b>      | RouteThreatAssessmentType                                                                                  | Required    |
| Description                       | Indicates the results of assessment of threats along a route.                                              |             |
| <b>TargetMobility</b>             | TargetMobilityAssessmentType                                                                               | Required    |
| Description                       | This element defines the response to a target mobility assessment request.                                 |             |
| <b>VehicleThreatAssessment</b>    | VehicleThreatAssessmentType                                                                                | Required    |
| Description                       | This element defines the response to a vehicle threat assessment request.                                  |             |
| <b>ThreatNominationAssessment</b> | ThreatNominationAssessmentType                                                                             | Required    |
| Description                       | This element defines the response to a threat nomination assessment.                                       |             |
| <b>AchievabilityAssessment</b>    | AchievabilityAssessmentPET                                                                                 | Required    |
| Description                       | This element defines the response to an achievability assessment.                                          |             |
| <b>CoverageAssessment</b>         | MA_CoverageAssessmentType                                                                                  | Required    |
| Description                       | This element defines the response to a coverage assessment.                                                |             |

Table A-1-456: P2P UCI Extensions MA\_AssessmentRequestStatusMT - Assessment UCI Extension Field

| MessageData       |                                                                                                                                                                                     |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                                                                                             |             |
| Base Type         | MA_AssessmentRequestStatusMDT                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                                     |             |
| Field Name        | Field Type                                                                                                                                                                          | Field Usage |
| CompletionStatus  | CompletionStatusType                                                                                                                                                                | Optional    |
| Description       | Intermediate status information for an assessment. Typically populated when sending periodic statuses of PROCESSING, but can also store start and finish times for this assessment. |             |
| Assessment        | MA_AssessmentType                                                                                                                                                                   | Optional    |
| Description       | This element indicates the results of the assessment. It is optional based on whether the assessment request specified that the response should result in                           |             |

|                                     |                                                                                                                                                                                                                                                                                             |          |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                     | native message.                                                                                                                                                                                                                                                                             |          |
| <b>IncompleteAssessment</b>         | IncompleteProcessingType                                                                                                                                                                                                                                                                    | Repeated |
| Description                         | This element indicates any assessments that failed to complete. It is used to indicate details about why a specific assessment target failed to process.                                                                                                                                    |          |
| <b>RequestID</b>                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required |
| Description                         | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |          |
| <b>RequestProcessingState</b>       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required |
| Description                         | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |          |
| <b>RequestProcessingStateReason</b> | CannotComplyType                                                                                                                                                                                                                                                                            | Optional |
| Description                         | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |          |

Table A-1-457: P2P UCI Extensions MA\_AssessmentRequestStatusMT - MessageData UCI Extension Field

### 1.3.2.85 MA\_OperatorRoleMT UCI Extension

| MA_OperatorRoleMT  |                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | SystemID : SystemID_Type                                                                                                                    |

Table A-1-458: P2P UCI Extensions MA\_OperatorRoleMT UCI Extension

| SystemID          |                                                                                                                                                                  |             |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the SystemID that the OperatorRole belongs to. If not populated, it is assumed to be the SystemID specified in the MessageHeader.                      |             |
| Base Type         | SystemID_Type                                                                                                                                                    |             |
| Message Structure |                                                                                                                                                                  |             |
| Field Name        | Field Type                                                                                                                                                       | Field Usage |
| UUID              | UniversallyUniqueIdentifierType                                                                                                                                  | Required    |
| Description       | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122. |             |
| DescriptiveLabel  | VisibleString256Type                                                                                                                                             | Optional    |
| Description       | Human readable text label with content meaningful to human operators.                                                                                            |             |

Table A-1-459: P2P UCI Extensions MA\_OperatorRoleMT - SystemID UCI Extension Field

### 1.3.2.86 MA\_ActionPlanValidationMT UCI Extension

| MA_ActionPlanValidationMT |
|---------------------------|
|---------------------------|

|                    |                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info                                       |
| <b>Ext. Fields</b> | AllocationResult : MA_ActionPlanningResultType<br>OpPlanID : MA_OpPlanID_Type<br>MessageData : MA_ActionPlanValidationMDT<br>AssociatedPlans : MA_PlanReferenceType |

Table A-1-460: P2P UCI Extensions MA\_ActionPlanValidationMT UCI Extension

| AllocationResult   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | Indicates allocation results per Action. This element is only expected when the ActionPlan was found to be invalid and the associated ActionPlanValidationCommand indicated "modify to validate" was allowed.                                                                                                                                                                                                                                                                                                                                           |             |
| Base Type          | MA_ActionPlanningResultType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
| Message Structure  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Field Usage |
| ActionID           | ActionID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description        | Unique object identifier for the Action whose planning results are given by sibling elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
| PlanningResult     | PlanningResultEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Required    |
| Description        | Indicates the outcome of attempting to allocate or plan activities for the Requirement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| SystemID           | SystemID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description        | Indicates the SystemID of the System whose *Plan are associated with the sibling PlanningResult element.                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
| AssociatedPlans    | MA_PlanReferenceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Optional    |
| Description        | Indicates *Plan associated with the sibling PlanningResult element. If the sibling PlanningResult indicates the Requirement was/remained allocated then this element indicates the specific *Plan it is/remains allocated in. If the PlanningResult indicates the Requirement was DEALLOCATED then this element indicates the constraining *Plan it was DEALLOCATED from. This element isn't expected when the PlanningResult indicates UNALLOCATED.                                                                                                    |             |
| NotAllocatedReason | UnallocatedReasonType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Repeated    |
| Description        | Indicates a reason why the Requirement wasn't allocated or reallocated. If the sibling PlanningResult element indicates the Requirement was left not allocated then this element indicates why. Multiple instances of this element can provide the reason for each System that was attempted and even many reasons for each.<br><br>If the PlanningResult indicates the Requirement was allocated then this element can still be used to indicate reasons for failed attempts to allocate to other Systems or plan, prior to the successful allocation. |             |

Table A-1-461: P2P UCI Extensions MA\_ActionPlanValidationMT - AllocationResult UCI Extension Field

| OpPlanID          |                                                                                                                                                                                                                                                                |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a reference to an OpPlan.                                                                                                                                                                                                                            |             |
| Base Type         | MA_OpPlanID_Type                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                     | Field Usage |
| Version           | unsignedInt                                                                                                                                                                                                                                                    | Optional    |
| Description       | Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval. |             |
| UUID              | UniversallyUniqueIdentifierType                                                                                                                                                                                                                                | Required    |
| Description       | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.                                                                                               |             |
| DescriptiveLabel  | VisibleString256Type                                                                                                                                                                                                                                           | Optional    |
| Description       | Human readable text label with content meaningful to human operators.                                                                                                                                                                                          |             |

Table A-1-462: P2P UCI Extensions MA\_ActionPlanValidationMT - OpPlanID UCI Extension Field

| MessageData            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |
| Base Type              | MA_ActionPlanValidationMDT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
| Message Structure      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| Field Name             | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Field Usage |
| ActionPlanValidationID | ActionPlanValidationID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Required    |
| Description            | The ID of this ActionPlanValidation message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| PlanID                 | ActionPlanID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Required    |
| Description            | The ID of the Plan this status pertains to.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |
| Validator              | ValidatorType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required    |
| Description            | The ID of the Validator responsible for this Status message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| ValidationState        | PlanValidationStateEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description            | Indicates the determined validation state of the plan. The variables a given planning service, operator or system considers when validating a plan varies. Different variables might be considered for a single given System depending on the level of detail of the plan. The scope and meaning of "valid" comes from context of the services, operator and systems creating the plan. The value in this element can change during validation processing. The final validity determination is given when the sibling ValidationStatus element reaches an end state. |             |
| CompletionStatus       | CompletionStatusType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Optional    |
| Description            | Indicates actual and predicted metrics for completion of the validation process.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| IncompleteItem         | IncompleteProcessingType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Repeated    |



|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description              | Indicates items associated with command processing that failed to complete. For example if a particular required Capability wasn't available, it could be referenced here. This element/type is highly flexible. It can reference almost any message/concept in UCI as a planning issue. The wide range of messages/concepts it can reference makes it 1) difficult to use as a basis for software/services to resolve the issue but 2) an excellent general method for processing algorithms to convey issues to operators. This element shouldn't be used to report processing issues/results that are more directly indicated by other processing status elements. For example, for planning processing, other elements focus specifically on per-Requirement allocation results therefore this element shouldn't be used for per-Requirement allocation results. |          |
| <b>InvalidAllocation</b> | ActionAllocationInvalidType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Repeated |
| Description              | Indicates an Action whose allocation was found to be invalid. If the associated ActionPlanValidationCommand indicated "modify to validate", results of attempts to modify the allocation can be found in sibling elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>AllocationResult</b>  | MA_ActionPlanningResultType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Repeated |
| Description              | Indicates allocation results per Action. This element is only expected when the ActionPlan was found to be invalid and the associated ActionPlanValidationCommand indicated "modify to validate" was allowed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
| <b>ModifiedPlanID</b>    | ActionPlanID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated |
| Description              | Indicates a modified, valid version of the evaluated ActionPlan. This element should only be given when the evaluated ActionPlan was found to be invalid, the ActionPlanValidationCommand indicated "modify to validate" and the modifications resulted in a valid ActionPlan.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |

Table A-1-463: P2P UCI Extensions MA\_ActionPlanValidationMT - MessageData UCI Extension Field

| AssociatedPlans     |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | Indicates *Plan associated with the sibling PlanningResult element. If the sibling PlanningResult indicates the Requirement was/remained allocated then this element indicates the specific *Plan it is/remains allocated in. If the PlanningResult indicates the Requirement was DEALLOCATED then this element indicates the constraining *Plan it was DEALLOCATED from. This element isn't expected when the PlanningResult indicates UNALLOCATED. |             |
| Base Type           | MA_PlanReferenceType                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| Message Structure   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| Field Name          | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                           | Field Usage |
| MissionPlanID       | MissionPlanID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                   | Optional    |
| Description         | Indicates a reference to a MissionPlan.                                                                                                                                                                                                                                                                                                                                                                                                              |             |
| TaskPlanID          | TaskPlanID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                      | Optional    |
| Description         | Indicates a reference to a TaskPlan.                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| OrbitPlanID         | OrbitPlanID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                     | Optional    |
| Description         | Indicates a reference to an OrbitPlan.                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                                                                                                                                                                                                                                                                                                                                                                                                                             | Optional    |
| Description         | Indicates a reference to an OrbitActivityPlan.                                                                                                                                                                                                                                                                                                                                                                                                       |             |
| RoutePlanID         | RoutePlanID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                     | Optional    |

|                                 |                                               |          |
|---------------------------------|-----------------------------------------------|----------|
| Description                     | Indicates a reference to a RoutePlan.         |          |
| <b>RouteActivityPlanID</b>      | RouteActivityPlanID_Type                      | Optional |
| Description                     | Indicates a reference to a RouteActivityPlan. |          |
| <b>CommScheduleAllocationID</b> | CommScheduleAllocationID_Type                 | Optional |
| Description                     | Indicates a reference to a Comms allocation.  |          |
| <b>ActivityPlanID</b>           | ActivityPlanID_Type                           | Optional |
| Description                     | Indicates a reference to an ActivityPlan.     |          |
| <b>EffectPlanID</b>             | EffectPlanID_Type                             | Optional |
| Description                     | Indicates a reference to an EffectPlan.       |          |
| <b>ActionPlanID</b>             | ActionPlanID_Type                             | Optional |
| Description                     | Indicates a reference to an ActionPlan.       |          |
| <b>ResponsePlanID</b>           | ResponsePlanID_Type                           | Optional |
| Description                     | Indicates a reference to a ResponsePlan.      |          |
| <b>OpPlanID</b>                 | MA_OpPlanID_Type                              | Optional |
| Description                     | Indicates a reference to an OpPlan.           |          |

Table A-1-464: P2P UCI Extensions MA\_ActionPlanValidationMT - AssociatedPlans UCI Extension Field

### 1.3.2.87 MA\_ApprovalPolicyMT UCI Extension

| MA_ApprovalPolicyMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>  | <p>See the annotation in the associated message for an overall description of the message and this type.</p> <p>See the XSD File for more info</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Ext. Fields</b>  | <p>TaskPlan : MA_TaskPlanPartsType</p> <p>OpLineCategory : MA_OpLineCategoryEnum</p> <p>ActivityPlan : MA_ActivityPlanPartsType</p> <p>ApplicablePlanningActivity : MA_PlanPolicyApplicablePlanType</p> <p>OpVolumeCategory : MA_OpZoneCategoryEnum</p> <p>TaskType : MA_TaskTypeEnum</p> <p>OrbitActivityPlan : MA_ActivityPlanPartsType</p> <p>MessageData : MA_ApprovalPolicyMDT</p> <p>OpZoneCategory : MA_OpZoneCategoryEnum</p> <p>ByPlanParts : MA_PlanPartsType</p> <p>CapabilityCommand : MA_CapabilityTypeEnum</p> <p>PlanParts : MA_PlanPartsType</p> <p>Plans : MA_PlanPolicyType</p> <p>RouteActivityPlan : MA_ActivityPlanPartsType</p> <p>OpPlan : MA_OpPlanPartsType</p> |

PlanActivation : MA\_PlanActivationPolicyType

Table A-1-465: P2P UCI Extensions MA\_ApprovalPolicyMT UCI Extension

| TaskPlan          |                                                                                                                                     |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates applicable TaskPlan parts.                                                                                                |             |
| Base Type         | MA_TaskPlanPartsType                                                                                                                |             |
| Message Structure |                                                                                                                                     |             |
| Field Name        | Field Type                                                                                                                          | Field Usage |
| TaskType          | MA_TaskTypeEnum                                                                                                                     | Repeated    |
| Description       | Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition. |             |

Table A-1-466: P2P UCI Extensions MA\_ApprovalPolicyMT - TaskPlan UCI Extension Field

| OpLineCategory                 |                                                                                                                                                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                    | Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                                                                                                                                 |
| Base Type                      | MA_OpLineCategoryEnum                                                                                                                                                                                                                                                  |
| Message Structure              |                                                                                                                                                                                                                                                                        |
| Enumeration Literal            | Description                                                                                                                                                                                                                                                            |
| GENERAL_LINE                   | A general line where the function must be described by other procedures.                                                                                                                                                                                               |
| FORWARD_EDGE_OF_BATTLE_AREA    | The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.                                                                                                       |
| GUN_TARGET_LINE                | Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.                                                                                                                                                                |
| CORRIDOR                       | Line with width describing the passageway designated for travel by friendly aircraft.                                                                                                                                                                                  |
| HOSTILE_BOUNDARY               | Line that indicates a limit beyond which exists a hostile environment.                                                                                                                                                                                                 |
| BUFFER_ZONE_BOUNDARY           | Line beyond which friendly aircraft usually do not fly in peacetime.                                                                                                                                                                                                   |
| LOW_LEVEL_TRANSIT_ROUTE        | Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.                                                                           |
| TACTICAL_ACTION_LINE           | Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.                                                                                                                                                    |
| FIRE_SUPPORT_COORDINATION_LINE | Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander. |
| FORWARD_LINE_OF_OWN_TROOPS     | Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.                                                                                                          |
| LINE_OF_COMMUNICATIONS         | Line the route that connects an operating military unit with its supply base.                                                                                                                                                                                          |

|                         |                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>JETWAY</b>           | Line with width that defines areas where commercial aircraft are flying through.                                                                           |
| <b>DEPARTURE_RADIAL</b> | Line the defines the departure radial when performing Carrier admin departures.                                                                            |
| <b>RECOVERY_RADIAL</b>  | Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.                    |
| <b>FINAL_BEARING</b>    | Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.                        |
| <b>BRC</b>              | Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries. |

Table A-1-467: P2P UCI Extensions MA\_ApprovalPolicyMT - OpLineCategory UCI Extension Field

| ActivityPlan                |                                                                                                                                                                                                                                     |             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                 | Indicates applicable ActivityPlan parts.                                                                                                                                                                                            |             |
| Base Type                   | MA_ActivityPlanPartsType                                                                                                                                                                                                            |             |
| Message Structure           |                                                                                                                                                                                                                                     |             |
| Field Name                  | Field Type                                                                                                                                                                                                                          | Field Usage |
| CapabilityCommand           | MA_CapabilityTypeEnum                                                                                                                                                                                                               | Repeated    |
| Description                 | Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.                       |             |
| SupportingCapabilityCommand | SupportCapabilityTypeEnum                                                                                                                                                                                                           | Repeated    |
| Description                 | Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition. |             |
| VehicleSettings             | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates VehicleSettings are applicable to the *ActivityPlan.                                                                                                                                                                      |             |
| CommsUsage                  | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates Comms usage is applicable to the *ActivityPlan.                                                                                                                                                                           |             |
| ProductTasks                | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.                                                                                     |             |
| EffectPlan                  | EffectPlanPartsType                                                                                                                                                                                                                 | Optional    |
| Description                 | Indicates applicable EffectPlan parts.                                                                                                                                                                                              |             |
| ActionPlan                  | ActionPlanPartsType                                                                                                                                                                                                                 | Optional    |
| Description                 | Indicates applicable ActionPlan parts.                                                                                                                                                                                              |             |
| TaskPlan                    | MA_TaskPlanPartsType                                                                                                                                                                                                                | Optional    |
| Description                 | Indicates applicable TaskPlan parts.                                                                                                                                                                                                |             |
| ResponsePlan                | ResponsePlanPartsType                                                                                                                                                                                                               | Optional    |
| Description                 | Indicates applicable ResponsePlan parts.                                                                                                                                                                                            |             |

|               |                                    |          |
|---------------|------------------------------------|----------|
| <b>OpPlan</b> | MA_OpPlanPartsType                 | Optional |
| Description   | Indicates applicable OpPlan parts. |          |

Table A-1-468: P2P UCI Extensions MA\_ApprovalPolicyMT - ActivityPlan UCI Extension Field

| ApplicablePlanningActivity   |                                                                                                                                                                                                         |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | Indicates the set of planning activities whose generated plans are subject to the sibling *Policy elements.                                                                                             |             |
| Base Type                    | MA_PlanPolicyApplicablePlanType                                                                                                                                                                         |             |
| Message Structure            |                                                                                                                                                                                                         |             |
| Field Name                   | Field Type                                                                                                                                                                                              | Field Usage |
| PlanType                     | PlanTypeEnum                                                                                                                                                                                            | Required    |
| Description                  | Indicates the PlanType that is generated by the planning activity, which is applicable to the associated *Policy elements.                                                                              |             |
| PlanParts                    | MA_PlanPartsType                                                                                                                                                                                        | Optional    |
| Description                  | Indicates the parts of the sibling PlanType that are generated by the planning activity, which is applicable to the associated *Policy elements. If omitted, it indicates all PlanParts are applicable. |             |
| ConstrainingPlans            | AutonomousPlanningConstrainingPlansType                                                                                                                                                                 | Optional    |
| Description                  | Indicates a set of constraining plans (applicable to same system than the one planned for) that are used in the planning activity, which is applicable to the associated *Policy elements.              |             |
| OtherSystemConstrainingPlans | AutonomousPlanningOtherSystemConstrainingPlansType                                                                                                                                                      | Repeated    |
| Description                  | Indicates a set of constraining plans (applicable to different system than the one planned for) that are used in the planning activity, which is applicable to the associated *Policy elements.         |             |

Table A-1-469: P2P UCI Extensions MA\_ApprovalPolicyMT - ApplicablePlanningActivity UCI Extension Field

| OpVolumeCategory              |                                                                                                                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>            | Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                                                             |
| <b>Base Type</b>              | MA_OpZoneCategoryEnum                                                                                                                                                                                |
| Message Structure             |                                                                                                                                                                                                      |
| Enumeration Literal           | Description                                                                                                                                                                                          |
| <b>AIRBORNE_EARLY_WARNING</b> | A point at which AEW aircraft are or will be orbiting or maneuvering.                                                                                                                                |
| <b>COMBAT_AIR_PATROL</b>      | A point at which CAP aircraft are or will be orbiting or maneuvering.                                                                                                                                |
| <b>CONSTRAINED_ENTRY_EXIT</b> | Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.               |
| <b>CONTAMINATED</b>           | Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects. |
| <b>DANGER</b>                 | A specified area above, below, or within which there may be a potential danger.                                                                                                                      |

|                               |                                                                                                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DEFENDED_AREA</b>          | An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).                                                              |
| <b>DIRECTED_SEARCH_AREA</b>   | Indicates the zone is a directed search area.                                                                                                                                                                                               |
| <b>EOB</b>                    | Indicates the zone is the boundary for an electronic order of battle request.                                                                                                                                                               |
| <b>EXERCISE</b>               | An area for military maneuver or simulated wartime operation involving planning, preparation and execution.                                                                                                                                 |
| <b>FIGHTER_ENGAGE_AOR</b>     | Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.                                                        |
| <b>FILTER_AREA</b>            | Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.                                       |
| <b>GENERAL_HAZARD</b>         | Indicates the zone is a hazard area that doesn't have a specified operational purpose.                                                                                                                                                      |
| <b>GENERAL_STATION</b>        | A general zone describing an air station that must be amplified by voice or predetermined procedures.                                                                                                                                       |
| <b>GROUND_AOR</b>             | Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.                                                   |
| <b>HOSTILE_TACTICAL_AREA</b>  | Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.                                                                         |
| <b>HOSTILE_WEAPON</b>         | Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.                                                                                                                        |
| <b>IMPACT_POINT</b>           | The real or expected point that a device has or will penetrate or contact a surface.                                                                                                                                                        |
| <b>JAMMING</b>                | Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.                                                                     |
| <b>KEEP_IN</b>                | Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.                                       |
| <b>KILL</b>                   | Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.                                                                                                |
| <b>MISSILE_ENGAGEMENT</b>     | Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.                                                                                                                 |
| <b>MISSILE_LAUNCH_POINT</b>   | A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.                      |
| <b>MISSION_AREA</b>           | Indicates the zone is an area in which a mission will take place.                                                                                                                                                                           |
| <b>NAMED_AREA_OF_INTEREST</b> | An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.                                                                                                               |
| <b>NO_COMM</b>                | Indicates the zone is an area where communications use is restricted.                                                                                                                                                                       |
| <b>NO_FIRE</b>                | Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType. |

|                                |                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NO_FLY</b>                  | Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.                                                                                          |
| <b>NO_IMAGE</b>                | Indicates the zone is an area where image collection is restricted.                                                                                                                                                                                                        |
| <b>ORBIT_FIGURE_EIGHT</b>      | A figure eight pattern within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                                        |
| <b>ORBIT_RACE_TRACK</b>        | An oval within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                                                       |
| <b>ORBIT_RANDOM_CLOSED</b>     | A rectangle in which the aircraft will track randomly.                                                                                                                                                                                                                     |
| <b>REFERENCE</b>               | Indicates the zone is an area that doesn't have a specified operational purpose.                                                                                                                                                                                           |
| <b>SEARCH</b>                  | An area where a search is being or will be conducted. normally associated with a SAR mission.                                                                                                                                                                              |
| <b>SEPARATION</b>              | Indicates an update in reservation volume for an individual vehicle.                                                                                                                                                                                                       |
| <b>SHORAD</b>                  | Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.                                                                                                                                              |
| <b>SPEED_LIMITS</b>            | An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.                                                                                                    |
| <b>SUBMARINE_PATROL_AREA</b>   | A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.                                                   |
| <b>TARGET_AREA_OF_INTEREST</b> | An engagement area usually along an enemy mobility corridor.                                                                                                                                                                                                               |
| <b>VEHICLE_CONFIGURATION</b>   | Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.                                           |
| <b>WEAPON_RESTRICTION</b>      | Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType. |
| <b>WEATHER_CONDITIONS</b>      | Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.                                               |

Table A-1-470: P2P UCI Extensions MA\_ApprovalPolicyMT - OpVolumeCategory UCI Extension Field

| TaskType              |                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>    | Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition. |
| <b>Base Type</b>      | MA_TaskTypeEnum                                                                                                                     |
| Message Structure     |                                                                                                                                     |
| Enumeration Literal   | Description                                                                                                                         |
| <b>AIR_SAMPLE</b>     | Indicates the task is to sample and analyze the air around the vehicle.                                                             |
| <b>AMTI</b>           | Indicates the task is to track air targets using an Airborne Moving Target Indicator.                                               |
| <b>AO</b>             | Indicates the task is to employ an Active Optical capability.                                                                       |
| <b>CARGO_DELIVERY</b> | Indicates the task is to deliver cargo.                                                                                             |



|                                    |                                                                                         |
|------------------------------------|-----------------------------------------------------------------------------------------|
| <b>CAP</b>                         | Indicates the task is to execute a Combat Air Patrol.                                   |
| <b>COMINT</b>                      | Indicates the task is to employ a communications intelligence capability.               |
| <b>COMM_RELAY</b>                  | Indicates the task is to relay communications from other battlespace participants.      |
| <b>COUNTER_SPACE</b>               | Indicates the task is to employ a Counter Space Capability.                             |
| <b>EA</b>                          | Indicates the task is to employ an electronic attack capability.                        |
| <b>ESCORT</b>                      | Indicates the task is to escort another entity.                                         |
| <b>ESM</b>                         | Indicates the task is to employ an Electronic Support Measure.                          |
| <b>FLIGHT</b>                      | Indicates the task is to fly.                                                           |
| <b>ORBIT_CHANGE</b>                | Indicates a task to perform an orbit change via a spacecraft maneuver.                  |
| <b>ORBITAL_SURVEILLANCE</b>        | Indicates a task to perform orbital surveillance.                                       |
| <b>ORBITAL_SURVEILLANCE_SENSOR</b> | Indicates a task to perform orbital surveillance via sensor collection capabilities.    |
| <b>PO</b>                          | Indicates the task is to employ a Passive Optical capability.                           |
| <b>REFUEL</b>                      | Indicates the task is to refuel another System or Entity.                               |
| <b>SAR</b>                         | Indicates the task is to employ a Synthetic Aperture Radar capability.                  |
| <b>SMTI</b>                        | Indicates the task is to track surface targets using a Surface Moving Target Indicator. |
| <b>STRIKE</b>                      | Indicates the task is to strike a target.                                               |
| <b>SYSTEM_DEPLOYMENT</b>           | Indicates the task is to deploy a system.                                               |
| <b>TACTICAL_ORDER</b>              | Indicates the task is to perform a tactical order.                                      |
| <b>WEATHER_RADAR</b>               | Indicates the task is to collect weather data.                                          |

Table A-1-471: P2P UCI Extensions MA\_ApprovalPolicyMT - TaskType UCI Extension Field

| OrbitActivityPlan           |                                                                                                                                                                                                                                     |             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                 | Indicates applicable OrbitActivityPlan parts.                                                                                                                                                                                       |             |
| Base Type                   | MA_ActivityPlanPartsType                                                                                                                                                                                                            |             |
| Message Structure           |                                                                                                                                                                                                                                     |             |
| Field Name                  | Field Type                                                                                                                                                                                                                          | Field Usage |
| CapabilityCommand           | MA_CapabilityTypeEnum                                                                                                                                                                                                               | Repeated    |
| Description                 | Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.                       |             |
| SupportingCapabilityCommand | SupportCapabilityTypeEnum                                                                                                                                                                                                           | Repeated    |
| Description                 | Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition. |             |
| VehicleSettings             | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates VehicleSettings are applicable to the *ActivityPlan.                                                                                                                                                                      |             |



|                     |                                                                                                                                                 |          |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>CommsUsage</b>   | EmptyType                                                                                                                                       | Optional |
| Description         | Indicates Comms usage is applicable to the *ActivityPlan.                                                                                       |          |
| <b>ProductTasks</b> | EmptyType                                                                                                                                       | Optional |
| Description         | Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan. |          |
| <b>EffectPlan</b>   | EffectPlanPartsType                                                                                                                             | Optional |
| Description         | Indicates applicable EffectPlan parts.                                                                                                          |          |
| <b>ActionPlan</b>   | ActionPlanPartsType                                                                                                                             | Optional |
| Description         | Indicates applicable ActionPlan parts.                                                                                                          |          |
| <b>TaskPlan</b>     | MA_TaskPlanPartsType                                                                                                                            | Optional |
| Description         | Indicates applicable TaskPlan parts.                                                                                                            |          |
| <b>ResponsePlan</b> | ResponsePlanPartsType                                                                                                                           | Optional |
| Description         | Indicates applicable ResponsePlan parts.                                                                                                        |          |
| <b>OpPlan</b>       | MA_OpPlanPartsType                                                                                                                              | Optional |
| Description         | Indicates applicable OpPlan parts.                                                                                                              |          |

Table A-1-472: P2P UCI Extensions MA\_ApprovalPolicyMT - OrbitActivityPlan UCI Extension Field

| MessageData          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| Base Type            | MA_ApprovalPolicyMDT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Field Usage |
| ApprovalPolicyID     | ApprovalPolicyID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Required    |
| Description          | Indicates the unique ID of the approval policy.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |
| Plans                | MA_PlanPolicyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Repeated    |
| Description          | Indicates a set of approval policies for the results of planning activities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
| PlanActivation       | MA_PlanActivationPolicyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Repeated    |
| Description          | Indicates approval policies which govern Plan Activation commands.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| RequirementExecution | RequirementExecutionPolicyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Optional    |
| Description          | Indicates which Requirement (Effect, Action, Task, [Capability]Command, Response) execution/performance related activities require approval.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
| TimedZone            | TimedZoneType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Optional    |
| Description          | Indicates the geospatial zone and optionally time/date span or spans when the policy is active. For task allocation, the policy is applicable to allocations that 1) complete within the time/date span and 2) include Task or Tasks whose target falls within the zone, regardless of when the Task will be executed. For route planning, the policy is applicable to route plans with segments that fall within the zone during the time/date span, regardless of when the route plan was completed or the target of Tasks associated with the segments. For task execution, the policy is applicable to Tasks that 1) are scheduled to complete within the time/date span and 2) have a target which |             |

|                             |                                                                                                                                                                                                                                                                                                                                                     |          |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                             | falls within the zone. This element supersedes the sibling Expires element.                                                                                                                                                                                                                                                                         |          |
| <b>Expires</b>              | DateTimeType                                                                                                                                                                                                                                                                                                                                        | Optional |
| Description                 | Indicates the time and date when the policy expires. The policy is applicable to task allocation and route planning activities completed prior to this time. The policy is applicable to task execution activities initiated prior to this time. When the sibling TimedZone element is used, the time spans specified there supersede this element. |          |
| <b>MissionTraceability</b>  | MissionTraceabilityType                                                                                                                                                                                                                                                                                                                             | Optional |
| Description                 | Allows the creator of the approval policy to specify the source or motivation for this ApprovalPolicy such as ATO, ACO, etc.                                                                                                                                                                                                                        |          |
| <b>DataRecordInstanceID</b> | DataRecordInstanceID_Type                                                                                                                                                                                                                                                                                                                           | Optional |
| Description                 | Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee.                                            |          |

**Table A-1-473: P2P UCI Extensions MA\_ApprovalPolicyMT - MessageData UCI Extension Field**

| OpZoneCategory                |                                                                                                                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>            | Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                                                               |
| <b>Base Type</b>              | MA_OpZoneCategoryEnum                                                                                                                                                                                |
| Message Structure             |                                                                                                                                                                                                      |
| Enumeration Literal           | Description                                                                                                                                                                                          |
| <b>AIRBORNE_EARLY_WARNING</b> | A point at which AEW aircraft are or will be orbiting or maneuvering.                                                                                                                                |
| <b>COMBAT_AIR_PATROL</b>      | A point at which CAP aircraft are or will be orbiting or maneuvering.                                                                                                                                |
| <b>CONSTRAINED_ENTRY_EXIT</b> | Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.               |
| <b>CONTAMINATED</b>           | Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects. |
| <b>DANGER</b>                 | A specified area above, below, or within which there may be a potential danger.                                                                                                                      |
| <b>DEFENDED_AREA</b>          | An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).                       |
| <b>DIRECTED_SEARCH_AREA</b>   | Indicates the zone is a directed search area.                                                                                                                                                        |
| <b>EOB</b>                    | Indicates the zone is the boundary for an electronic order of battle request.                                                                                                                        |
| <b>EXERCISE</b>               | An area for military maneuver or simulated wartime operation involving planning, preparation and execution.                                                                                          |
| <b>FIGHTER_ENGAGE_AOR</b>     | Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.                 |
| <b>FILTER_AREA</b>            | Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the                                                  |

|                               |                                                                                                                                                                                                                                             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                               | OpZoneFilterAreaPET choice in OpZoneCategoryType.                                                                                                                                                                                           |
| <b>GENERAL_HAZARD</b>         | Indicates the zone is a hazard area that doesn't have a specified operational purpose.                                                                                                                                                      |
| <b>GENERAL_STATION</b>        | A general zone describing an air station that must be amplified by voice or predetermined procedures.                                                                                                                                       |
| <b>GROUND_AOR</b>             | Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.                                                   |
| <b>HOSTILE_TACTICAL_AREA</b>  | Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.                                                                         |
| <b>HOSTILE_WEAPON</b>         | Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.                                                                                                                        |
| <b>IMPACT_POINT</b>           | The real or expected point that a device has or will penetrate or contact a surface.                                                                                                                                                        |
| <b>JAMMING</b>                | Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.                                                                     |
| <b>KEEP_IN</b>                | Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.                                       |
| <b>KILL</b>                   | Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.                                                                                                |
| <b>MISSILE_ENGAGEMENT</b>     | Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.                                                                                                                 |
| <b>MISSILE_LAUNCH_POINT</b>   | A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.                      |
| <b>MISSION_AREA</b>           | Indicates the zone is an area in which a mission will take place.                                                                                                                                                                           |
| <b>NAMED_AREA_OF_INTEREST</b> | An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.                                                                                                               |
| <b>NO_COMM</b>                | Indicates the zone is an area where communications use is restricted.                                                                                                                                                                       |
| <b>NO_FIRE</b>                | Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType. |
| <b>NO_FLY</b>                 | Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.                                                           |
| <b>NO_IMAGE</b>               | Indicates the zone is an area where image collection is restricted.                                                                                                                                                                         |
| <b>ORBIT_FIGURE_EIGHT</b>     | A figure eight pattern within a rectangle that the aircraft is or will be tracking.                                                                                                                                                         |
| <b>ORBIT_RACE_TRACK</b>       | An oval within a rectangle that the aircraft is or will be tracking.                                                                                                                                                                        |
| <b>ORBIT_RANDOM_CLOSED</b>    | A rectangle in which the aircraft will track randomly.                                                                                                                                                                                      |
| <b>REFERENCE</b>              | Indicates the zone is an area that doesn't have a specified operational purpose.                                                                                                                                                            |
| <b>SEARCH</b>                 | An area where a search is being or will be conducted. normally associated with a SAR mission.                                                                                                                                               |

|                                |                                                                                                                                                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SEPARATION</b>              | Indicates an update in reservation volume for an individual vehicle.                                                                                                                                                                                                       |
| <b>SHORAD</b>                  | Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.                                                                                                                                              |
| <b>SPEED_LIMITS</b>            | An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.                                                                                                    |
| <b>SUBMARINE_PATROL_AREA</b>   | A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.                                                   |
| <b>TARGET_AREA_OF_INTEREST</b> | An engagement area usually along an enemy mobility corridor.                                                                                                                                                                                                               |
| <b>VEHICLE_CONFIGURATION</b>   | Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.                                           |
| <b>WEAPON_RESTRICTION</b>      | Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType. |
| <b>WEATHER_CONDITIONS</b>      | Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.                                               |

Table A-1-474: P2P UCI Extensions MA\_ApprovalPolicyMT - OpZoneCategory UCI Extension Field

| ByPlanParts       |                                                                                                                                                                             |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates *Plan parts which the Plan to be activated must contain in order for the Policy to be applicable. If omitted, the Policy is applicable regardless of *Plan parts. |             |
| Base Type         | MA_PlanPartsType                                                                                                                                                            |             |
| Message Structure |                                                                                                                                                                             |             |
| Field Name        | Field Type                                                                                                                                                                  | Field Usage |
| ActivityPlan      | MA_ActivityPlanPartsType                                                                                                                                                    | Optional    |
| Description       | Indicates applicable ActivityPlan parts.                                                                                                                                    |             |
| RoutePlan         | RoutePlanPartsType                                                                                                                                                          | Optional    |
| Description       | Indicates applicable RoutePlan parts.                                                                                                                                       |             |
| RouteActivityPlan | MA_ActivityPlanPartsType                                                                                                                                                    | Optional    |
| Description       | Indicates applicable RouteActivityPlan parts.                                                                                                                               |             |
| OrbitPlan         | OrbitPlanPartsType                                                                                                                                                          | Optional    |
| Description       | Indicates applicable OrbitPlan parts.                                                                                                                                       |             |
| OrbitActivityPlan | MA_ActivityPlanPartsType                                                                                                                                                    | Optional    |
| Description       | Indicates applicable OrbitActivityPlan parts.                                                                                                                               |             |
| EffectPlan        | EffectPlanPartsType                                                                                                                                                         | Optional    |
| Description       | Indicates applicable EffectPlan parts.                                                                                                                                      |             |
| ActionPlan        | ActionPlanPartsType                                                                                                                                                         | Optional    |

|                     |                                          |          |
|---------------------|------------------------------------------|----------|
| Description         | Indicates applicable ActionPlan parts.   |          |
| <b>TaskPlan</b>     | MA_TaskPlanPartsType                     | Optional |
| Description         | Indicates applicable TaskPlan parts.     |          |
| <b>ResponsePlan</b> | ResponsePlanPartsType                    | Optional |
| Description         | Indicates applicable ResponsePlan parts. |          |
| <b>OpPlan</b>       | MA_OpPlanPartsType                       | Optional |
| Description         | Indicates applicable OpPlan parts.       |          |

Table A-1-475: P2P UCI Extensions MA\_ApprovalPolicyMT - ByPlanParts UCI Extension Field

| CapabilityCommand                  |                                                                                                                                                                                                               |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                 | Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition. |
| <b>Base Type</b>                   | MA_CapabilityTypeEnum                                                                                                                                                                                         |
| Message Structure                  |                                                                                                                                                                                                               |
| Enumeration Literal                | Description                                                                                                                                                                                                   |
| <b>ACTION</b>                      | Indicates an action capability.                                                                                                                                                                               |
| <b>AIR_SAMPLE</b>                  | Indicates an air sampling capability.                                                                                                                                                                         |
| <b>AMTI</b>                        | Indicates an airborne moving target indication (AMTI) capability.                                                                                                                                             |
| <b>AO</b>                          | Indicates an active optical capability.                                                                                                                                                                       |
| <b>CAP</b>                         | Indicates a cargo delivery capability.                                                                                                                                                                        |
| <b>CARGO_DELIVERY</b>              | Indicates a cargo delivery capability.                                                                                                                                                                        |
| <b>COMINT</b>                      | Indicates a Communications Intelligence (COMINT) capability.                                                                                                                                                  |
| <b>COMM_RELAY</b>                  | Indicates a communications relay capability.                                                                                                                                                                  |
| <b>COMM_TERMINAL</b>               | Indicates a communications terminal capability                                                                                                                                                                |
| <b>COUNTER_SPACE</b>               | Indicates a Counter Space Capability.                                                                                                                                                                         |
| <b>EA</b>                          | Indicates an electronic attack (EA) capability.                                                                                                                                                               |
| <b>EFFECT</b>                      | Indicates an effect capability.                                                                                                                                                                               |
| <b>ESM</b>                         | Indicates an electronic support measures (ESM) capability.                                                                                                                                                    |
| <b>FLIGHT</b>                      | Indicates a capability for waypoint-based flight tasks/commands.                                                                                                                                              |
| <b>IFF</b>                         | Indicates an Identification Friend or Foe (IFF) capability.                                                                                                                                                   |
| <b>OPAQUE</b>                      | Indicates a Opaque Capability                                                                                                                                                                                 |
| <b>ORBIT_CHANGE</b>                | Indicates this space system has the ability to change orbits.                                                                                                                                                 |
| <b>ORBITAL_SURVEILLANCE</b>        | Indicates a task to perform Orbital Surveillance.                                                                                                                                                             |
| <b>ORBITAL_SURVEILLANCE_SENSOR</b> | Indicates an Orbital Surveillance Sensor Capability.                                                                                                                                                          |
| <b>PO</b>                          | Indicates a passive optical capability.                                                                                                                                                                       |
| <b>RADAR_ALTIMETER</b>             | Indicates a Radar Altimeter capability                                                                                                                                                                        |

|                          |                                                                 |
|--------------------------|-----------------------------------------------------------------|
| <b>REFUEL</b>            | Indicates a capability to refuel another asset.                 |
| <b>RESPONSE</b>          | Indicates an response capability.                               |
| <b>SAR</b>               | Indicates a synthetic aperture radar (SAR) capability.          |
| <b>SMTI</b>              | Indicates a surface moving target indication (SMTI) capability. |
| <b>STRIKE</b>            | Indicates a strike capability.                                  |
| <b>SYSTEM_DEPLOYMENT</b> | Indicates a deployment capability.                              |
| <b>TACTICAL_ORDER</b>    | Indicates a tactical order capability.                          |
| <b>TASK</b>              | Indicates a task capability.                                    |
| <b>WEATHER_RADAR</b>     | Indicates a weather radar capability.                           |

Table A-1-476: P2P UCI Extensions MA\_ApprovalPolicyMT - CapabilityCommand UCI Extension Field

| PlanParts         |                                                                                                                                                                                                         |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the parts of the sibling PlanType that are generated by the planning activity, which is applicable to the associated *Policy elements. If omitted, it indicates all PlanParts are applicable. |             |
| Base Type         | MA_PlanPartsType                                                                                                                                                                                        |             |
| Message Structure |                                                                                                                                                                                                         |             |
| Field Name        | Field Type                                                                                                                                                                                              | Field Usage |
| ActivityPlan      | MA_ActivityPlanPartsType                                                                                                                                                                                | Optional    |
| Description       | Indicates applicable ActivityPlan parts.                                                                                                                                                                |             |
| RoutePlan         | RoutePlanPartsType                                                                                                                                                                                      | Optional    |
| Description       | Indicates applicable RoutePlan parts.                                                                                                                                                                   |             |
| RouteActivityPlan | MA_ActivityPlanPartsType                                                                                                                                                                                | Optional    |
| Description       | Indicates applicable RouteActivityPlan parts.                                                                                                                                                           |             |
| OrbitPlan         | OrbitPlanPartsType                                                                                                                                                                                      | Optional    |
| Description       | Indicates applicable OrbitPlan parts.                                                                                                                                                                   |             |
| OrbitActivityPlan | MA_ActivityPlanPartsType                                                                                                                                                                                | Optional    |
| Description       | Indicates applicable OrbitActivityPlan parts.                                                                                                                                                           |             |
| EffectPlan        | EffectPlanPartsType                                                                                                                                                                                     | Optional    |
| Description       | Indicates applicable EffectPlan parts.                                                                                                                                                                  |             |
| ActionPlan        | ActionPlanPartsType                                                                                                                                                                                     | Optional    |
| Description       | Indicates applicable ActionPlan parts.                                                                                                                                                                  |             |
| TaskPlan          | MA_TaskPlanPartsType                                                                                                                                                                                    | Optional    |
| Description       | Indicates applicable TaskPlan parts.                                                                                                                                                                    |             |
| ResponsePlan      | ResponsePlanPartsType                                                                                                                                                                                   | Optional    |
| Description       | Indicates applicable ResponsePlan parts.                                                                                                                                                                |             |
| OpPlan            | MA_OpPlanPartsType                                                                                                                                                                                      | Optional    |

|             |                                    |
|-------------|------------------------------------|
| Description | Indicates applicable OpPlan parts. |
|-------------|------------------------------------|

**Table A-1-477: P2P UCI Extensions MA\_ApprovalPolicyMT - PlanParts UCI Extension Field**

| Plans                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                | Indicates a set of approval policies for the results of planning activities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| Base Type                  | MA_PlanPolicyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| Message Structure          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
| Field Name                 | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Field Usage |
| ApplicablePlanningActivity | MA_PlanPolicyApplicablePlanType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Repeated    |
| Description                | Indicates the set of planning activities whose generated plans are subject to the sibling *Policy elements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| DefaultPolicy              | ApprovalPolicyBaseType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Optional    |
| Description                | Indicates the default approval policy for the sibling ApplicablePlanningActivity. The default policy can be overridden in specific conditions as indicated by the ByResultPolicy or ByTriggerPolicy siblings of this element. For example, this element could indicate that approval is required by default for all TaskPlans and then use ByResultPolicy or ByTriggerPolicy siblings to specify conditions when task allocation approval is not required. Alternately, this element could indicate that approval is not required by default and then use ByResultPolicy or ByTriggerPolicy siblings to specify conditions when task allocation approval is required. Creators of this message are expected to carefully avoid creating logically inconsistent ByResultPolicy and ByTriggerPolicy instances and therefore no guidance is given on deconflicting illogical instances. For example, an ApprovalPolicy that indicates approval is not required by default with conflicting ByResultPolicy or ByTriggerPolicy overrides of 1) AUTO_REJECT for DROPPED_TASKS and 2) AUTO_APPROVE for operator triggered DROPPED_TASKS would be considered illogical. |             |
| ByResultPolicy             | ByResultPolicyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Repeated    |
| Description                | Indicates approval policies when planning activity results match certain conditions. Conditions specified here are considered to be overrides to the default planning activity approval policy given in the sibling Default element. For example, if the default policy indicates task allocation approval is never required, this element could override the default and require approval whenever a task allocation results in previously allocated Task or Tasks being dropped. See the annotations in the sibling DefaultPolicy element for an overview of how the siblings relate to each other. If multiple instances are given, each should be a different Result instance as indicated by the child Result element.                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| ByTriggerPolicy            | ByTriggerPolicyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Repeated    |
| Description                | Indicates approval policies for re-plan activities initiated by triggered events, overriding the default approval policy for the planning activity given in the sibling DefaultPolicy element. Additionally, this element allows trigger-specific policies for a specific planning activities. For example, if the default policy indicates route planning approval is never required, this element could override, requiring approval for route plans affecting all Systems triggered by new threats if the route re-plan results in dropped tasks. If multiple instances are given, each should be of a different trigger type as indicated by the child EventTrigger element.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |

**Table A-1-478: P2P UCI Extensions MA\_ApprovalPolicyMT - Plans UCI Extension Field**

| RouteActivityPlan |                                               |
|-------------------|-----------------------------------------------|
| Description       | Indicates applicable RouteActivityPlan parts. |



|                             |                                                                                                                                                                                                                                     |             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Base Type                   | MA_ActivityPlanPartsType                                                                                                                                                                                                            |             |
| Message Structure           |                                                                                                                                                                                                                                     |             |
| Field Name                  | Field Type                                                                                                                                                                                                                          | Field Usage |
| CapabilityCommand           | MA_CapabilityTypeEnum                                                                                                                                                                                                               | Repeated    |
| Description                 | Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.                       |             |
| SupportingCapabilityCommand | SupportCapabilityTypeEnum                                                                                                                                                                                                           | Repeated    |
| Description                 | Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition. |             |
| VehicleSettings             | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates VehicleSettings are applicable to the *ActivityPlan.                                                                                                                                                                      |             |
| CommsUsage                  | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates Comms usage is applicable to the *ActivityPlan.                                                                                                                                                                           |             |
| ProductTasks                | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.                                                                                     |             |
| EffectPlan                  | EffectPlanPartsType                                                                                                                                                                                                                 | Optional    |
| Description                 | Indicates applicable EffectPlan parts.                                                                                                                                                                                              |             |
| ActionPlan                  | ActionPlanPartsType                                                                                                                                                                                                                 | Optional    |
| Description                 | Indicates applicable ActionPlan parts.                                                                                                                                                                                              |             |
| TaskPlan                    | MA_TaskPlanPartsType                                                                                                                                                                                                                | Optional    |
| Description                 | Indicates applicable TaskPlan parts.                                                                                                                                                                                                |             |
| ResponsePlan                | ResponsePlanPartsType                                                                                                                                                                                                               | Optional    |
| Description                 | Indicates applicable ResponsePlan parts.                                                                                                                                                                                            |             |
| OpPlan                      | MA_OpPlanPartsType                                                                                                                                                                                                                  | Optional    |
| Description                 | Indicates applicable OpPlan parts.                                                                                                                                                                                                  |             |

Table A-1-479: P2P UCI Extensions MA\_ApprovalPolicyMT - RouteActivityPlan UCI Extension Field

| OpPlan            |                                                                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates applicable OpPlan parts.                                                                                                      |             |
| Base Type         | MA_OpPlanPartsType                                                                                                                      |             |
| Message Structure |                                                                                                                                         |             |
| Field Name        | Field Type                                                                                                                              | Field Usage |
| OpPointCategory   | OpPointCategoriesType                                                                                                                   | Repeated    |
| Description       | Indicates an OpPoint type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition. |             |
| OpLineCategory    | MA_OpLineCategoryEnum                                                                                                                   | Repeated    |



|                         |                                                                                                                                                         |          |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description             | Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                  |          |
| <b>OpZoneCategory</b>   | MA_OpZoneCategoryEnum                                                                                                                                   | Repeated |
| Description             | Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                  |          |
| <b>OpVolumeCategory</b> | MA_OpZoneCategoryEnum                                                                                                                                   | Repeated |
| Description             | Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                |          |
| <b>OpRouting</b>        | EmptyType                                                                                                                                               | Optional |
| Description             | Indicates an OpRouting is applicable to the associated *Plan. List size for this element is based on there being no category type for OpRouting objects |          |

Table A-1-480: P2P UCI Extensions MA\_ApprovalPolicyMT - OpPlan UCI Extension Field

| PlanActivation    |                                                                                                                                                                                                                                                                                                   |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates approval policies which govern Plan Activation commands.                                                                                                                                                                                                                                |             |
| Base Type         | MA_PlanActivationPolicyType                                                                                                                                                                                                                                                                       |             |
| Message Structure |                                                                                                                                                                                                                                                                                                   |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                        | Field Usage |
| ByMissionPlan     | MissionPlanActivationSettingType                                                                                                                                                                                                                                                                  | Repeated    |
| Description       | Indicates the set of ByMissionPlan activation commands that are applicable to the sibling Policy element. List size allows for identifying those MissionPlans whose subordinate MissionPlans are either included in the activation (TRUE) or not (FALSE) via the CommandSubordinatePlans setting. |             |
| BySubPlan         | SubPlanActivationSettingType                                                                                                                                                                                                                                                                      | Repeated    |
| Description       | Indicates the set of BySubPlan activation commands that are applicable to the sibling Policy element.                                                                                                                                                                                             |             |
| ByPlanParts       | MA_PlanPartsType                                                                                                                                                                                                                                                                                  | Optional    |
| Description       | Indicates *Plan parts which the Plan to be activated must contain in order for the Policy to be applicable. If omitted, the Policy is applicable regardless of *Plan parts.                                                                                                                       |             |
| Policy            | ApprovalPolicyBaseType                                                                                                                                                                                                                                                                            | Optional    |
| Description       | Indicates the approval policy for the sibling applicable activation activities.                                                                                                                                                                                                                   |             |

Table A-1-481: P2P UCI Extensions MA\_ApprovalPolicyMT - PlanActivation UCI Extension Field

### 1.3.2.88 MA\_OpLineMT UCI Extension

| MA_OpLineMT        |                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | Category : MA_OpLineCategoryEnum<br>AppliesToIDs : MA_PackageSystemType                                                                     |

|                            |
|----------------------------|
| MessageData : MA_OpLineMDT |
|----------------------------|

Table A-1-482: P2P UCI Extensions MA\_OpLineMT UCI Extension

| Category                       |                                                                                                                                                                                                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                    | This element indicates the type of operational line.                                                                                                                                                                                                                   |
| Base Type                      | MA_OpLineCategoryEnum                                                                                                                                                                                                                                                  |
| Message Structure              |                                                                                                                                                                                                                                                                        |
| Enumeration Literal            | Description                                                                                                                                                                                                                                                            |
| GENERAL_LINE                   | A general line where the function must be described by other procedures.                                                                                                                                                                                               |
| FORWARD_EDGE_OF_BATTLE_AREA    | The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.                                                                                                       |
| GUN_TARGET_LINE                | Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.                                                                                                                                                                |
| CORRIDOR                       | Line with width describing the passageway designated for travel by friendly aircraft.                                                                                                                                                                                  |
| HOSTILE_BOUNDARY               | Line that indicates a limit beyond which exists a hostile environment.                                                                                                                                                                                                 |
| BUFFER_ZONE_BOUNDARY           | Line beyond which friendly aircraft usually do not fly in peacetime.                                                                                                                                                                                                   |
| LOW_LEVEL_TRANSIT_ROUTE        | Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.                                                                           |
| TACTICAL_ACTION_LINE           | Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.                                                                                                                                                    |
| FIRE_SUPPORT_COORDINATION_LINE | Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander. |
| FORWARD_LINE_OF_OWN_TROOPS     | Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.                                                                                                          |
| LINE_OF_COMMUNICATIONS         | Line the route that connects an operating military unit with its supply base.                                                                                                                                                                                          |
| JETWAY                         | Line with width that defines areas where commercial aircraft are flying through.                                                                                                                                                                                       |
| DEPARTURE_RADIAL               | Line the defines the departure radial when performing Carrier admin departures.                                                                                                                                                                                        |
| RECOVERY_RADIAL                | Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.                                                                                                                                |
| FINAL_BEARING                  | Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.                                                                                                                                    |
| BRC                            | Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.                                                                                                             |

Table A-1-483: P2P UCI Extensions MA\_OpLineMT - Category UCI Extension Field

| AppliesToIDs      |                                                                                          |             |
|-------------------|------------------------------------------------------------------------------------------|-------------|
| Description       | One or more systems to which this item applies. If omitted, item applies to all systems. |             |
| Base Type         | MA_PackageSystemType                                                                     |             |
| Message Structure |                                                                                          |             |
| Field Name        | Field Type                                                                               | Field Usage |
| ALL               | EmptyType                                                                                | Required    |
| Description       | Indicates applies to ALL entities.                                                       |             |
| NONE              | EmptyType                                                                                | Required    |
| Description       | Indicates applies to NO entities.                                                        |             |
| PackageID         | PackageID_Type                                                                           | Required    |
| Description       | The ID of a Package.                                                                     |             |
| SystemID          | SystemID_Type                                                                            | Required    |
| Description       | The ID of an ACP or C2 Operator.                                                         |             |

**Table A-1-484: P2P UCI Extensions MA\_OpLineMT - AppliesToIDs UCI Extension Field**

| MessageData         |                                                                                                                                           |             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | See annotations in child elements and messages/elements that use this type for details.                                                   |             |
| Base Type           | MA_OpLineMDT                                                                                                                              |             |
| Message Structure   |                                                                                                                                           |             |
| Field Name          | Field Type                                                                                                                                | Field Usage |
| OpLineID            | OpLineID_Type                                                                                                                             | Required    |
| Description         | Indicates the unique ID of the OpLine.                                                                                                    |             |
| Category            | MA_OpLineCategoryEnum                                                                                                                     | Required    |
| Description         | This element indicates the type of operational line.                                                                                      |             |
| Line                | OpLineType                                                                                                                                | Required    |
| Description         | Indicates the geometric description of the OpLine. In practice, operational lines can be geometric lines, areas or in some cases volumes. |             |
| OpDescription       | OpDescriptionType                                                                                                                         | Optional    |
| Description         | A human-readable descriptive name with free text remarks.                                                                                 |             |
| MissionTraceability | MissionTraceabilityType                                                                                                                   | Optional    |
| Description         | Allows the creator of this operational item to specify what the source or motivation for it originated from. This could be ATO, ACO, etc. |             |
| Source              | CreationSourceEnum                                                                                                                        | Optional    |
| Description         | Indicates the source of the information.                                                                                                  |             |
| Schedule            | ScheduleType                                                                                                                              | Optional    |
| Description         | This element represents a date/time interval associated with the operational item                                                         |             |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                               | during which the item is in effect.                                                                                                                                                                                                                                                                                                                                                                              |          |
| <b>AssociatedTime</b>         | TimeFunctionType                                                                                                                                                                                                                                                                                                                                                                                                 | Repeated |
| Description                   | Indicates time related events that are associated to the Op data type such as when activated or when operational. The multiplicity is limited to N-1 options where NONE is an event that has no value, thus removed from list of available options.                                                                                                                                                              |          |
| <b>DataLinkIdentifier</b>     | DataLinkIdentifierPET                                                                                                                                                                                                                                                                                                                                                                                            | Repeated |
| Description                   | List of data link ID. Multiple data link IDs can be reported for the same network type.                                                                                                                                                                                                                                                                                                                          |          |
| <b>Priority</b>               | unsignedInt                                                                                                                                                                                                                                                                                                                                                                                                      | Optional |
| Description                   | This field represents the priority to be considered when choosing between multiple available points.                                                                                                                                                                                                                                                                                                             |          |
| <b>QualifyingTags</b>         | QualifyingTagsType                                                                                                                                                                                                                                                                                                                                                                                               | Optional |
| Description                   | Provides identity qualifications to augment identification and associated processing of the OpPoint.                                                                                                                                                                                                                                                                                                             |          |
| <b>AppliesToIDs</b>           | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                                                                             | Repeated |
| Description                   | One or more systems to which this item applies. If omitted, item applies to all systems.                                                                                                                                                                                                                                                                                                                         |          |
| <b>SystemScheduleOverride</b> | SystemScheduleStateType                                                                                                                                                                                                                                                                                                                                                                                          | Repeated |
| Description                   | System-specific schedule that will override any part of the main schedule. This allows for system-specific schedule different than main schedule. When adding a SystemScheduleOverride, the child SystemID also must be added to the sibling SystemID list if the sibling list is not empty. The SystemScheduleOverride is independent of time planning. For the system, this override can be done at execution. |          |

**Table A-1-485: P2P UCI Extensions MA\_OpLineMT - MessageData UCI Extension Field**

### 1.3.2.89 MA\_OpPointMT UCI Extension

| MA_OpPointMT       |                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b> | MessageData : MA_OpPointMDT<br>AppliesToIDs : MA_PackageSystemType                                                                          |

**Table A-1-486: P2P UCI Extensions MA\_OpPointMT UCI Extension**

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_OpPointMDT                                                                           |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| Category          | OpPointCategoriesType                                                                   | Required    |
| Description       | This element indicates the type of operational point.                                   |             |

|                            |                                                                                                                                                                                                                                                                                                                                                                                             |          |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>CategoryUniqueData</b>  | OpPointCategoriesUniqueDataType                                                                                                                                                                                                                                                                                                                                                             | Optional |
| Description                | This element indicates the unique data for the type of operational point.                                                                                                                                                                                                                                                                                                                   |          |
| <b>OpPointID</b>           | OpPointID_Type                                                                                                                                                                                                                                                                                                                                                                              | Required |
| Description                | This element represents the unique identifier of the point associated with this message.                                                                                                                                                                                                                                                                                                    |          |
| <b>PointChoice</b>         | OpPointChoiceType                                                                                                                                                                                                                                                                                                                                                                           | Required |
| Description                | Choice of either geospatial or relative position of the OpPoint.                                                                                                                                                                                                                                                                                                                            |          |
| <b>SystemConfiguration</b> | SystemConfigurationType                                                                                                                                                                                                                                                                                                                                                                     | Repeated |
| Description                | Indicates a System which the OpPoint is applicable for and its planned configuration/state upon arriving at the OpPoint. If omitted, the OpPoint applies to all Systems.                                                                                                                                                                                                                    |          |
| <b>TurnDirection</b>       | RelativeDirectionEnum                                                                                                                                                                                                                                                                                                                                                                       | Optional |
| Description                | This field provides the turn direction for orbit.                                                                                                                                                                                                                                                                                                                                           |          |
| <b>IngressConstraint</b>   | AnglePairType                                                                                                                                                                                                                                                                                                                                                                               | Optional |
| Description                | When present this field constrains the vehicle ingress path toward the OpPoint. The allowable ingress range is defined clockwise from Min to Max.                                                                                                                                                                                                                                           |          |
| <b>SafeAltitude</b>        | SafeAltitudeType                                                                                                                                                                                                                                                                                                                                                                            | Optional |
| Description                | This field specifies a set of safe minimum approach altitudes and an emergency safe altitude. This field also specifies the cylindrical area, defined by a center point and radius, where the altitudes apply. This field defines minimum approach altitude and emergency safe altitude consistent with standard FAA instrument approach chart definitions (using UCI standard data types). |          |
| <b>OpDescription</b>       | OpDescriptionType                                                                                                                                                                                                                                                                                                                                                                           | Optional |
| Description                | A human-readable descriptive name with free text remarks.                                                                                                                                                                                                                                                                                                                                   |          |
| <b>MissionTraceability</b> | MissionTraceabilityType                                                                                                                                                                                                                                                                                                                                                                     | Optional |
| Description                | Allows the creator of this operational item to specify what the source or motivation for it originated from. This could be ATO, ACO, etc.                                                                                                                                                                                                                                                   |          |
| <b>Source</b>              | CreationSourceEnum                                                                                                                                                                                                                                                                                                                                                                          | Optional |
| Description                | Indicates the source of the information.                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>Schedule</b>            | ScheduleType                                                                                                                                                                                                                                                                                                                                                                                | Optional |
| Description                | This element represents a date/time interval associated with the operational item during which the item is in effect.                                                                                                                                                                                                                                                                       |          |
| <b>AssociatedTime</b>      | TimeFunctionType                                                                                                                                                                                                                                                                                                                                                                            | Repeated |
| Description                | Indicates time related events that are associated to the Op data type such as when activated or when operational. The multiplicity is limited to N-1 options where NONE is an event that has no value, thus removed from list of available options.                                                                                                                                         |          |
| <b>DataLinkIdentifier</b>  | DataLinkIdentifierPET                                                                                                                                                                                                                                                                                                                                                                       | Repeated |
| Description                | List of data link ID. Multiple data link IDs can be reported for the same network type.                                                                                                                                                                                                                                                                                                     |          |
| <b>Priority</b>            | unsignedInt                                                                                                                                                                                                                                                                                                                                                                                 | Optional |
| Description                | This field represents the priority to be considered when choosing between multiple available points.                                                                                                                                                                                                                                                                                        |          |

| QualifyingTags         | QualifyingTagsType                                                                                                                                                                                                                                                                                                                                                                                               | Optional |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description            | Provides identity qualifications to augment identification and associated processing of the OpPoint.                                                                                                                                                                                                                                                                                                             |          |
| AppliesToIDs           | MA_PackageSystemType                                                                                                                                                                                                                                                                                                                                                                                             | Repeated |
| Description            | One or more systems to which this item applies. If omitted, item applies to all systems.                                                                                                                                                                                                                                                                                                                         |          |
| SystemScheduleOverride | SystemScheduleStateType                                                                                                                                                                                                                                                                                                                                                                                          | Repeated |
| Description            | System-specific schedule that will override any part of the main schedule. This allows for system-specific schedule different than main schedule. When adding a SystemScheduleOverride, the child SystemID also must be added to the sibling SystemID list if the sibling list is not empty. The SystemScheduleOverride is independent of time planning. For the system, this override can be done at execution. |          |

Table A-1-487: P2P UCI Extensions MA\_OpPointMT - MessageData UCI Extension Field

| AppliesToIDs      |                                                                                          |             |
|-------------------|------------------------------------------------------------------------------------------|-------------|
| Description       | One or more systems to which this item applies. If omitted, item applies to all systems. |             |
| Base Type         | MA_PackageSystemType                                                                     |             |
| Message Structure |                                                                                          |             |
| Field Name        | Field Type                                                                               | Field Usage |
| ALL               | EmptyType                                                                                | Required    |
| Description       | Indicates applies to ALL entities.                                                       |             |
| NONE              | EmptyType                                                                                | Required    |
| Description       | Indicates applies to NO entities.                                                        |             |
| PackageID         | PackageID_Type                                                                           | Required    |
| Description       | The ID of a Package.                                                                     |             |
| SystemID          | SystemID_Type                                                                            | Required    |
| Description       | The ID of an ACP or C2 Operator.                                                         |             |

Table A-1-488: P2P UCI Extensions MA\_OpPointMT - AppliesToIDs UCI Extension Field

### 1.3.2.90 MA\_OpRoutingMT UCI Extension

| MA_OpRoutingMT |                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Description    | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| Ext. Fields    | ApplicableToIDs : MA_PackageSystemType<br>MessageData : MA_OpRoutingMDT                                                                     |

Table A-1-489: P2P UCI Extensions MA\_OpRoutingMT UCI Extension

| ApplicableToIDs |                                                                             |
|-----------------|-----------------------------------------------------------------------------|
| Description     | One or more Systems to which the routing constraints apply. If omitted, the |

|                   |                                    |             |
|-------------------|------------------------------------|-------------|
|                   | constraints apply to all Systems.  |             |
| Base Type         | MA_PackageSystemType               |             |
| Message Structure |                                    |             |
| Field Name        | Field Type                         | Field Usage |
| ALL               | EmptyType                          | Required    |
| Description       | Indicates applies to ALL entities. |             |
| NONE              | EmptyType                          | Required    |
| Description       | Indicates applies to NO entities.  |             |
| PackageID         | PackageID_Type                     | Required    |
| Description       | The ID of a Package.               |             |
| SystemID          | SystemID_Type                      | Required    |
| Description       | The ID of an ACP or C2 Operator.   |             |

Table A-1-490: P2P UCI Extensions MA\_OpRoutingMT - ApplicableToIDs UCI Extension Field

| MessageData            |                                                                                                                                                                                                                                                                                                                        |             |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description            | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                |             |
| Base Type              | MA_OpRoutingMDT                                                                                                                                                                                                                                                                                                        |             |
| Message Structure      |                                                                                                                                                                                                                                                                                                                        |             |
| Field Name             | Field Type                                                                                                                                                                                                                                                                                                             | Field Usage |
| OpRoutingID            | OpRoutingID_Type                                                                                                                                                                                                                                                                                                       | Required    |
| Description            | The unique identifier of the set of routing constraints during operation associated with this message.                                                                                                                                                                                                                 |             |
| DefaultBlueSeparation  | SeparationParametersType                                                                                                                                                                                                                                                                                               | Optional    |
| Description            | Indicates default minimum geospatial separation to be maintained when planning or analyzing routes or manually navigating. The separation is between the planned/navigated route and other routes or conflicting items. This is the default separation. It can be overridden in specific cases by the sibling element. |             |
| DefaultRedSeparation   | SeparationParametersType                                                                                                                                                                                                                                                                                               | Optional    |
| Description            | Indicates default minimum geospatial separation to be maintained when planning or analyzing routes or manually navigating. The separation is between the planned/navigated route and other routes or conflicting items. This is the default separation. It can be overridden in specific cases by the sibling element. |             |
| SpecificBlueSeparation | SpecificBlueSeparationType                                                                                                                                                                                                                                                                                             | Repeated    |
| Description            | Defines parameters to use when deconflicting blue player routes or generating routes to ensure proper separation from other blue players. These parameters apply at the vehicle type level.                                                                                                                            |             |
| SpecificRedSeparation  | SpecificRedSeparationType                                                                                                                                                                                                                                                                                              | Repeated    |
| Description            | Defines parameters to use when deconflicting blue player routes or generating routes to ensure proper separation from other red players. These parameters apply at the vehicle type level.                                                                                                                             |             |

| OpDescription       | OpDescriptionType                                                                                                                    | Optional |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description         | Represents the name and description of the set of associated routing constraints.                                                    |          |
| ApplicableToIDs     | MA_PackageSystemType                                                                                                                 | Repeated |
| Description         | One or more Systems to which the routing constraints apply. If omitted, the constraints apply to all Systems.                        |          |
| ApplicableZone      | ZoneType                                                                                                                             | Optional |
| Description         | Indicates the geospatial zone in which the routing constraints apply.                                                                |          |
| Schedule            | ScheduleType                                                                                                                         | Optional |
| Description         | Indicates the time schedule during which the routing constraints apply.                                                              |          |
| MissionTraceability | MissionTraceabilityType                                                                                                              | Optional |
| Description         | Allows the creator of this zone to specify what the source or motivation for this zone originated from. This could be ATO, ACO, etc. |          |

Table A-1-491: P2P UCI Extensions MA\_OpRoutingMT - MessageData UCI Extension Field

## 1.3.2.91 MA\_PlanningFunctionSettingsCommandMT UCI Extension

| MA_PlanningFunctionSettingsCommandMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description                          | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Ext. Fields                          | AutonomousPlanningAction : MA_PlanningAllowedEscalationType<br>OpPlan : MA_OpPlanPartsType<br>AutonomousPlanningResponseChoice : MA_MissionPlanningAutonomyResponseChoiceType<br>MissionPlanningAutonomy : MA_MissionPlanningAutonomySettingType<br>OrbitActivityPlan : MA_ActivityPlanPartsType<br>RouteActivityPlan : MA_ActivityPlanPartsType<br>ByResultSetting : MA_MissionPlanningAutonomySettingByResultType<br>PlanningAllowed : MA_PlanningAllowedType<br>OutputPlanParts : MA_PlanPartsType<br>AllowedType : MA_PlanningAllowedType<br>AutonomousAction : MA_MissionPlanningByResultAutonomousActionType<br>MessageData : MA_PlanningFunctionSettingsCommandMDT<br>ActivityPlan : MA_ActivityPlanPartsType<br>TaskPlan : MA_TaskPlanPartsType |

Table A-1-492: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT UCI Extension

| AutonomousPlanningAction |                                                                                                                                                                                                                           |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Description              | This element indicates the allowed autonomous mission planning for the ancestor Trigger. It also indicates the expected MissionContingencyAlert for the ancestor Trigger; whenever autonomous mission planning is allowed |



|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                            | and triggered, a MissionContingencyAlert is expected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
| Base Type                  | MA_PlanningAllowedEscalationType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |
| Message Structure          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Field Name                 | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Field Usage |
| AutonomousPlanningActionID | AutonomousPlanningActionID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Required    |
| Description                | Assigns a unique ID to an allowed planning action. This ID can be reference in *Plans, alerts, etc. that result when the planning action is triggered.                                                                                                                                                                                                                                                                                                                                                                                            |             |
| ByResultSetting            | MA_MissionPlanningAutonomySettingByResultType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Repeated    |
| Description                | This element allows control of the escalation of the autonomous replanning. If the autonomous function indicated by the sibling AllowedType leads to a "by result" trigger described here, then additional autonomous planning is allowed as described here. For example, if Tasks are dropped as a result of initial autonomous replanning then this element can be used to enable/disable a secondary "escalated" planning iteration that includes global Task allocation. If omitted, the sibling AllowedType is allowed regardless of result. |             |
| AllowedType                | MA_PlanningAllowedType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Required    |
| Description                | This element defines the mission planning type that is allowed to be autonomously triggered.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |

**Table A-1-493: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - AutonomousPlanningAction UCI Extension Field**

| OpPlan            |                                                                                                                                                         |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates applicable OpPlan parts.                                                                                                                      |             |
| Base Type         | MA_OpPlanPartsType                                                                                                                                      |             |
| Message Structure |                                                                                                                                                         |             |
| Field Name        | Field Type                                                                                                                                              | Field Usage |
| OpPointCategory   | OpPointCategoriesType                                                                                                                                   | Repeated    |
| Description       | Indicates an OpPoint type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                 |             |
| OpLineCategory    | MA_OpLineCategoryEnum                                                                                                                                   | Repeated    |
| Description       | Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                  |             |
| OpZoneCategory    | MA_OpZoneCategoryEnum                                                                                                                                   | Repeated    |
| Description       | Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                  |             |
| OpVolumeCategory  | MA_OpZoneCategoryEnum                                                                                                                                   | Repeated    |
| Description       | Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.                |             |
| OpRouting         | EmptyType                                                                                                                                               | Optional    |
| Description       | Indicates an OpRouting is applicable to the associated *Plan. List size for this element is based on there being no category type for OpRouting objects |             |

**Table A-1-494: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - OpPlan UCI Extension**

## Field

| AutonomousPlanningResponseChoice |                                                                                                                                                                                                                                                                                 |             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                      | Indicates the autonomous mission planning type or types allowed for the sibling Trigger.                                                                                                                                                                                        |             |
| Base Type                        | MA_MissionPlanningAutonomyResponseChoiceType                                                                                                                                                                                                                                    |             |
| Message Structure                |                                                                                                                                                                                                                                                                                 |             |
| Field Name                       | Field Type                                                                                                                                                                                                                                                                      | Field Usage |
| AutonomousPlanningAction         | MA_PlanningAllowedEscalationType                                                                                                                                                                                                                                                | Repeated    |
| Description                      | This element indicates the allowed autonomous mission planning for the ancestor Trigger. It also indicates the expected MissionContingencyAlert for the ancestor Trigger; whenever autonomous mission planning is allowed and triggered, a MissionContingencyAlert is expected. |             |
| AlertOnly                        | EmptyType                                                                                                                                                                                                                                                                       | Required    |
| Description                      | This element indicates that autonomous mission planning isn't allowed for the ancestor Trigger but a MissionContingencyAlert is expected.                                                                                                                                       |             |

**Table A-1-495: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - AutonomousPlanningResponseChoice UCI Extension Field**

| MissionPlanningAutonomy          |                                                                                                                                                                                                                                                                                         |             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                      | Indicates commanded settings for autonomous mission planning capabilities of the System.                                                                                                                                                                                                |             |
| Base Type                        | MA_MissionPlanningAutonomySettingType                                                                                                                                                                                                                                                   |             |
| Message Structure                |                                                                                                                                                                                                                                                                                         |             |
| Field Name                       | Field Type                                                                                                                                                                                                                                                                              | Field Usage |
| Trigger                          | PlanningByCaseTriggerType                                                                                                                                                                                                                                                               | Required    |
| Description                      | Indicates an allowed trigger for autonomous planning.                                                                                                                                                                                                                                   |             |
| AutonomousPlanningResponseChoice | MA_MissionPlanningAutonomyResponseChoiceType                                                                                                                                                                                                                                            | Required    |
| Description                      | Indicates the autonomous mission planning type or types allowed for the sibling Trigger.                                                                                                                                                                                                |             |
| PlanningProcessID                | PlanningProcessID_Type                                                                                                                                                                                                                                                                  | Optional    |
| Description                      | Indicates the unique ID of a planning process which the settings are applicable to. If omitted, the setting is applicable to all processes of the corresponding System.                                                                                                                 |             |
| MinimumResponseInterval          | DurationType                                                                                                                                                                                                                                                                            | Optional    |
| Description                      | Indicates the minimum time interval between autonomous responses for the sibling Trigger. In other words, regardless of when and how frequently the Trigger occurs, the sibling Response shouldn't be initiated until the minimum time duration given here after the previous Response. |             |

**Table A-1-496: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - MissionPlanningAutonomy UCI Extension Field**

| OrbitActivityPlan |
|-------------------|
|-------------------|

| Description                 | Indicates applicable OrbitActivityPlan parts.                                                                                                                                                                                       |             |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Base Type                   | MA_ActivityPlanPartsType                                                                                                                                                                                                            |             |
| Message Structure           |                                                                                                                                                                                                                                     |             |
| Field Name                  | Field Type                                                                                                                                                                                                                          | Field Usage |
| CapabilityCommand           | MA_CapabilityTypeEnum                                                                                                                                                                                                               | Repeated    |
| Description                 | Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.                       |             |
| SupportingCapabilityCommand | SupportCapabilityTypeEnum                                                                                                                                                                                                           | Repeated    |
| Description                 | Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition. |             |
| VehicleSettings             | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates VehicleSettings are applicable to the *ActivityPlan.                                                                                                                                                                      |             |
| CommsUsage                  | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates Comms usage is applicable to the *ActivityPlan.                                                                                                                                                                           |             |
| ProductTasks                | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                 | Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.                                                                                     |             |
| EffectPlan                  | EffectPlanPartsType                                                                                                                                                                                                                 | Optional    |
| Description                 | Indicates applicable EffectPlan parts.                                                                                                                                                                                              |             |
| ActionPlan                  | ActionPlanPartsType                                                                                                                                                                                                                 | Optional    |
| Description                 | Indicates applicable ActionPlan parts.                                                                                                                                                                                              |             |
| TaskPlan                    | MA_TaskPlanPartsType                                                                                                                                                                                                                | Optional    |
| Description                 | Indicates applicable TaskPlan parts.                                                                                                                                                                                                |             |
| ResponsePlan                | ResponsePlanPartsType                                                                                                                                                                                                               | Optional    |
| Description                 | Indicates applicable ResponsePlan parts.                                                                                                                                                                                            |             |
| OpPlan                      | MA_OpPlanPartsType                                                                                                                                                                                                                  | Optional    |
| Description                 | Indicates applicable OpPlan parts.                                                                                                                                                                                                  |             |

**Table A-1-497: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - OrbitActivityPlan UCI Extension Field**

| RouteActivityPlan |                                               |             |
|-------------------|-----------------------------------------------|-------------|
| Description       | Indicates applicable RouteActivityPlan parts. |             |
| Base Type         | MA_ActivityPlanPartsType                      |             |
| Message Structure |                                               |             |
| Field Name        | Field Type                                    | Field Usage |
| CapabilityCommand | MA_CapabilityTypeEnum                         | Repeated    |

|                                    |                                                                                                                                                                                                                                     |          |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                        | Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.                       |          |
| <b>SupportingCapabilityCommand</b> | SupportCapabilityTypeEnum                                                                                                                                                                                                           | Repeated |
| Description                        | Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition. |          |
| <b>VehicleSettings</b>             | EmptyType                                                                                                                                                                                                                           | Optional |
| Description                        | Indicates VehicleSettings are applicable to the *ActivityPlan.                                                                                                                                                                      |          |
| <b>CommsUsage</b>                  | EmptyType                                                                                                                                                                                                                           | Optional |
| Description                        | Indicates Comms usage is applicable to the *ActivityPlan.                                                                                                                                                                           |          |
| <b>ProductTasks</b>                | EmptyType                                                                                                                                                                                                                           | Optional |
| Description                        | Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.                                                                                     |          |
| <b>EffectPlan</b>                  | EffectPlanPartsType                                                                                                                                                                                                                 | Optional |
| Description                        | Indicates applicable EffectPlan parts.                                                                                                                                                                                              |          |
| <b>ActionPlan</b>                  | ActionPlanPartsType                                                                                                                                                                                                                 | Optional |
| Description                        | Indicates applicable ActionPlan parts.                                                                                                                                                                                              |          |
| <b>TaskPlan</b>                    | MA_TaskPlanPartsType                                                                                                                                                                                                                | Optional |
| Description                        | Indicates applicable TaskPlan parts.                                                                                                                                                                                                |          |
| <b>ResponsePlan</b>                | ResponsePlanPartsType                                                                                                                                                                                                               | Optional |
| Description                        | Indicates applicable ResponsePlan parts.                                                                                                                                                                                            |          |
| <b>OpPlan</b>                      | MA_OpPlanPartsType                                                                                                                                                                                                                  | Optional |
| Description                        | Indicates applicable OpPlan parts.                                                                                                                                                                                                  |          |

**Table A-1-498: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - RouteActivityPlan UCI Extension Field**

| ByResultSetting   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | This element allows control of the escalation of the autonomous replanning. If the autonomous function indicated by the sibling AllowedType leads to a "by result" trigger described here, then additional autonomous planning is allowed as described here. For example, if Tasks are dropped as a result of initial autonomous replanning then this element can be used to enable/disable a secondary "escalated" planning iteration that includes global Task allocation. If omitted, the sibling AllowedType is allowed regardless of result. |             |
| Base Type         | MA_MissionPlanningAutonomySettingByResultType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Field Usage |
| Trigger           | PlanningByResultTriggerType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Required    |
| Description       | Indicates a trigger based on the results of a previously initiated autonomous mission planning function.                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |

|                         |                                                                  |          |
|-------------------------|------------------------------------------------------------------|----------|
| <b>AutonomousAction</b> | MA_MissionPlanningByResultAutonomousActionType                   | Required |
| Description             | Indicates the action or actions allowed for the sibling trigger. |          |

**Table A-1-499: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - ByResultSetting UCI Extension Field**

| PlanningAllowed              |                                                                                                                                                                                                                                                                                                                                                                   |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | This element defines the mission planning type to be triggered. If multiple instances are given, each should be of a different planning type as indicated by the child element.                                                                                                                                                                                   |             |
| Base Type                    | MA_PlanningAllowedType                                                                                                                                                                                                                                                                                                                                            |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                                   |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                        | Field Usage |
| PlanAheadDuration            | DurationType                                                                                                                                                                                                                                                                                                                                                      | Required    |
| Description                  | Required duration in the future when the Plan should start. Existing *Plan will continue during this time. This duration should account for everything that must happen from replan initiation to new plan activation.                                                                                                                                            |             |
| PlanningProcessID            | PlanningProcessID_Type                                                                                                                                                                                                                                                                                                                                            | Required    |
| Description                  | Indicates the unique ID of the planning process whose autonomous invocation is described the sibling elements. Planning process are declared in the ControlStatus message.                                                                                                                                                                                        |             |
| ConstrainingPlans            | AutonomousPlanningConstrainingPlansType                                                                                                                                                                                                                                                                                                                           | Optional    |
| Description                  | Indicates existing types of *Plans that can be used as constraints when invoking the planning process indicated by the sibling PlanningProcessID. The types of *Plans indicated here should be consistent with those supported by the planning process as indicated in the ControlStatus message.                                                                 |             |
| OtherSystemConstrainingPlans | AutonomousPlanningOtherSystemConstrainingPlansType                                                                                                                                                                                                                                                                                                                | Repeated    |
| Description                  | Indicates existing plans of another System that constrain planning. One for each descendant ConstraintUsage.                                                                                                                                                                                                                                                      |             |
| OutputPlanParts              | MA_PlanPartsType                                                                                                                                                                                                                                                                                                                                                  | Optional    |
| Description                  | Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the Plan parts for the sibling PlanningType.                                                                        |             |
| ResultsInMissionPlan         | boolean                                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description                  | In some cases, the commanding service or HMI may not want to receive the planning process summarized in a MissionPlan message. If TRUE, the results will be published in individual *Plan messages and summarized in a published MissionPlan message. If FALSE, the results will be published in individual *Plan messages without a summary MissionPlan message. |             |
| ForPlanningUseOnly           | boolean                                                                                                                                                                                                                                                                                                                                                           | Required    |
| Description                  | Indicates the state to use for the corresponding element in the *PlanCommand when invoking the planning process.                                                                                                                                                                                                                                                  |             |

**Table A-1-500: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - PlanningAllowed UCI Extension Field**

| OutputPlanParts   |                                                                                                                                                                                                                                                                                            |             |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the Plan parts for the sibling PlanningType. |             |
| Base Type         | MA_PlanPartsType                                                                                                                                                                                                                                                                           |             |
| Message Structure |                                                                                                                                                                                                                                                                                            |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                 | Field Usage |
| ActivityPlan      | MA_ActivityPlanPartsType                                                                                                                                                                                                                                                                   | Optional    |
| Description       | Indicates applicable ActivityPlan parts.                                                                                                                                                                                                                                                   |             |
| RoutePlan         | RoutePlanPartsType                                                                                                                                                                                                                                                                         | Optional    |
| Description       | Indicates applicable RoutePlan parts.                                                                                                                                                                                                                                                      |             |
| RouteActivityPlan | MA_ActivityPlanPartsType                                                                                                                                                                                                                                                                   | Optional    |
| Description       | Indicates applicable RouteActivityPlan parts.                                                                                                                                                                                                                                              |             |
| OrbitPlan         | OrbitPlanPartsType                                                                                                                                                                                                                                                                         | Optional    |
| Description       | Indicates applicable OrbitPlan parts.                                                                                                                                                                                                                                                      |             |
| OrbitActivityPlan | MA_ActivityPlanPartsType                                                                                                                                                                                                                                                                   | Optional    |
| Description       | Indicates applicable OrbitActivityPlan parts.                                                                                                                                                                                                                                              |             |
| EffectPlan        | EffectPlanPartsType                                                                                                                                                                                                                                                                        | Optional    |
| Description       | Indicates applicable EffectPlan parts.                                                                                                                                                                                                                                                     |             |
| ActionPlan        | ActionPlanPartsType                                                                                                                                                                                                                                                                        | Optional    |
| Description       | Indicates applicable ActionPlan parts.                                                                                                                                                                                                                                                     |             |
| TaskPlan          | MA_TaskPlanPartsType                                                                                                                                                                                                                                                                       | Optional    |
| Description       | Indicates applicable TaskPlan parts.                                                                                                                                                                                                                                                       |             |
| ResponsePlan      | ResponsePlanPartsType                                                                                                                                                                                                                                                                      | Optional    |
| Description       | Indicates applicable ResponsePlan parts.                                                                                                                                                                                                                                                   |             |
| OpPlan            | MA_OpPlanPartsType                                                                                                                                                                                                                                                                         | Optional    |
| Description       | Indicates applicable OpPlan parts.                                                                                                                                                                                                                                                         |             |

**Table A-1-501: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - OutputPlanParts UCI Extension Field**

| AllowedType       |                                                                                              |             |
|-------------------|----------------------------------------------------------------------------------------------|-------------|
| Description       | This element defines the mission planning type that is allowed to be autonomously triggered. |             |
| Base Type         | MA_PlanningAllowedType                                                                       |             |
| Message Structure |                                                                                              |             |
| Field Name        | Field Type                                                                                   | Field Usage |
| PlanAheadDuration | DurationType                                                                                 | Required    |

|                                     |                                                                                                                                                                                                                                                                                                                                                                   |          |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description                         | Required duration in the future when the Plan should start. Existing *Plan will continue during this time. This duration should account for everything that must happen from replan initiation to new plan activation.                                                                                                                                            |          |
| <b>PlanningProcessID</b>            | PlanningProcessID_Type                                                                                                                                                                                                                                                                                                                                            | Required |
| Description                         | Indicates the unique ID of the planning process whose autonomous invocation is described the sibling elements. Planning process are declared in the ControlStatus message.                                                                                                                                                                                        |          |
| <b>ConstrainingPlans</b>            | AutonomousPlanningConstrainingPlansType                                                                                                                                                                                                                                                                                                                           | Optional |
| Description                         | Indicates existing types of *Plans that can be used as constraints when invoking the planning process indicated by the sibling PlanningProcessID. The types of *Plans indicated here should be consistent with those supported by the planning process as indicated in the ControlStatus message.                                                                 |          |
| <b>OtherSystemConstrainingPlans</b> | AutonomousPlanningOtherSystemConstrainingPlansType                                                                                                                                                                                                                                                                                                                | Repeated |
| Description                         | Indicates existing plans of another System that constrain planning. One for each descendant ConstraintUsage.                                                                                                                                                                                                                                                      |          |
| <b>OutputPlanParts</b>              | MA_PlanPartsType                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                         | Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the Plan parts for the sibling PlanningType.                                                                        |          |
| <b>ResultsInMissionPlan</b>         | boolean                                                                                                                                                                                                                                                                                                                                                           | Required |
| Description                         | In some cases, the commanding service or HMI may not want to receive the planning process summarized in a MissionPlan message. If TRUE, the results will be published in individual *Plan messages and summarized in a published MissionPlan message. If FALSE, the results will be published in individual *Plan messages without a summary MissionPlan message. |          |
| <b>ForPlanningUseOnly</b>           | boolean                                                                                                                                                                                                                                                                                                                                                           | Required |
| Description                         | Indicates the state to use for the corresponding element in the *PlanCommand when invoking the planning process.                                                                                                                                                                                                                                                  |          |

**Table A-1-502: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - AllowedType UCI Extension Field**

| AutonomousAction  |                                                                                                                                                                                 |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates the action or actions allowed for the sibling trigger.                                                                                                                |             |
| Base Type         | MA_MissionPlanningByResultAutonomousActionType                                                                                                                                  |             |
| Message Structure |                                                                                                                                                                                 |             |
| Field Name        | Field Type                                                                                                                                                                      | Field Usage |
| PlanningAllowed   | MA_PlanningAllowedType                                                                                                                                                          | Repeated    |
| Description       | This element defines the mission planning type to be triggered. If multiple instances are given, each should be of a different planning type as indicated by the child element. |             |
| AlertOnly         | EmptyType                                                                                                                                                                       | Required    |
| Description       | This element indicates that autonomous mission planning isn't allowed for the ancestor Trigger but a MissionContingencyAlert is expected.                                       |             |



**Table A-1-503: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - AutonomousAction UCI Extension Field**

| MessageData                  |                                                                                                                                                                                                                                  |             |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | Indicates the data being conveyed by the PlanningFunctionSettingsCommand message.                                                                                                                                                |             |
| Base Type                    | MA_PlanningFunctionSettingsCommandMDT                                                                                                                                                                                            |             |
| Message Structure            |                                                                                                                                                                                                                                  |             |
| Field Name                   | Field Type                                                                                                                                                                                                                       | Field Usage |
| SystemID                     | SystemID_Type                                                                                                                                                                                                                    | Required    |
| Description                  | Indicates the unique ID of the System whose planning functions settings are being commanded/changed.                                                                                                                             |             |
| TransitionDroppedToFailed    | boolean                                                                                                                                                                                                                          | Required    |
| Description                  | This element specifies whether or not dropped Tasks that cannot be reallocated within the System can be automatically transitioned to the FAILED state for higher level interaction.                                             |             |
| PlanningInterface            | PlanInterfaceCommandType                                                                                                                                                                                                         | Repeated    |
| Description                  | Indicates commanded settings for planning interfaces. If multiple instances of this element are given, each should be of a different type of *Plan as enumerated by the child PlanType element.                                  |             |
| MissionPlanningAutonomy      | MA_MissionPlanningAutonomySettingType                                                                                                                                                                                            | Repeated    |
| Description                  | Indicates commanded settings for autonomous mission planning capabilities of the System.                                                                                                                                         |             |
| PlanActivationAutonomy       | PlanActivationAutonomyType                                                                                                                                                                                                       | Optional    |
| Description                  | Indicates commanded settings for autonomous plan activation capabilities of the System.                                                                                                                                          |             |
| RoutePlanningContingencyPath | ContingencyPathAutonomyType                                                                                                                                                                                                      | Repeated    |
| Description                  | Indicates commanded route planning settings for the autonomous generation of contingency paths along the primary path. If multiple instances are given, each must be of a different path type as indicated by the child element. |             |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                   | Required    |
| Description                  | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.                                                                                                       |             |
| CommandState                 | CommandStateEnum                                                                                                                                                                                                                 | Required    |
| Description                  | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.                                                                    |             |

**Table A-1-504: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - MessageData UCI Extension Field**

| ActivityPlan       |                                          |
|--------------------|------------------------------------------|
| <b>Description</b> | Indicates applicable ActivityPlan parts. |
| <b>Base Type</b>   | MA_ActivityPlanPartsType                 |



| Message Structure                  |                                                                                                                                                                                                                                     |             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Field Name                         | Field Type                                                                                                                                                                                                                          | Field Usage |
| <b>CapabilityCommand</b>           | MA_CapabilityTypeEnum                                                                                                                                                                                                               | Repeated    |
| Description                        | Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.                       |             |
| <b>SupportingCapabilityCommand</b> | SupportCapabilityTypeEnum                                                                                                                                                                                                           | Repeated    |
| Description                        | Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition. |             |
| <b>VehicleSettings</b>             | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                        | Indicates VehicleSettings are applicable to the *ActivityPlan.                                                                                                                                                                      |             |
| <b>CommsUsage</b>                  | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                        | Indicates Comms usage is applicable to the *ActivityPlan.                                                                                                                                                                           |             |
| <b>ProductTasks</b>                | EmptyType                                                                                                                                                                                                                           | Optional    |
| Description                        | Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.                                                                                     |             |
| <b>EffectPlan</b>                  | EffectPlanPartsType                                                                                                                                                                                                                 | Optional    |
| Description                        | Indicates applicable EffectPlan parts.                                                                                                                                                                                              |             |
| <b>ActionPlan</b>                  | ActionPlanPartsType                                                                                                                                                                                                                 | Optional    |
| Description                        | Indicates applicable ActionPlan parts.                                                                                                                                                                                              |             |
| <b>TaskPlan</b>                    | MA_TaskPlanPartsType                                                                                                                                                                                                                | Optional    |
| Description                        | Indicates applicable TaskPlan parts.                                                                                                                                                                                                |             |
| <b>ResponsePlan</b>                | ResponsePlanPartsType                                                                                                                                                                                                               | Optional    |
| Description                        | Indicates applicable ResponsePlan parts.                                                                                                                                                                                            |             |
| <b>OpPlan</b>                      | MA_OpPlanPartsType                                                                                                                                                                                                                  | Optional    |
| Description                        | Indicates applicable OpPlan parts.                                                                                                                                                                                                  |             |

**Table A-1-505: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - ActivityPlan UCI Extension Field**

| TaskPlan          |                                                                                                                                     |             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates applicable TaskPlan parts.                                                                                                |             |
| Base Type         | MA_TaskPlanPartsType                                                                                                                |             |
| Message Structure |                                                                                                                                     |             |
| Field Name        | Field Type                                                                                                                          | Field Usage |
| TaskType          | MA_TaskTypeEnum                                                                                                                     | Repeated    |
| Description       | Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition. |             |

**Table A-1-506: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandMT - TaskPlan UCI Extension Field**

### 1.3.2.92 MA\_PlanningFunctionSettingsCommandStatusMT UCI Extension

| MA_PlanningFunctionSettingsCommandStatusMT |                                                                                                                                             |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                         | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                         | MessageData : MA_PlanningFunctionSettingsCommandStatusMDT                                                                                   |

Table A-1-507: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | Indicates the data being conveyed by the PlanningFunctionSettingsCommandStatus message.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_PlanningFunctionSettingsCommandStatusMDT                                                                                                                                                                                                                                                                                                                 |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

Table A-1-508: P2P UCI Extensions MA\_PlanningFunctionSettingsCommandStatusMT - MessageData UCI Extension Field

### 1.3.2.93 MA\_ResponseActivityMT UCI Extension

| MA_ResponseActivityMT |                                                                                                                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>    | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info                                                                                                        |
| <b>Ext. Fields</b>    | RequirementInstantiation : MA_RequirementTemplateInstantiationType<br>ActivatedPlans : MA_PlansReferenceType<br>Activity : MA_ResponseActivityType<br>MessageData : MA_ResponseActivityMDT<br>ResultingPlans : MA_PlansReferenceType |

OpPlanID : MA\_OpPlanID\_Type

Table A-1-509: P2P UCI Extensions MA\_ResponseActivityMT UCI Extension

| RequirementInstantiation |                                                                                                                                                                                                                                                   |             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | Indicates an independent set of Requirement or Requirements and *Plans that were instantiated for the Response. Subsequent progress in planning and/or execution is given in [Requirement]Plan, [Requirement]Status, MissionPlan*, etc. messages. |             |
| Base Type                | MA_RequirementTemplateInstantiationType                                                                                                                                                                                                           |             |
| Message Structure        |                                                                                                                                                                                                                                                   |             |
| Field Name               | Field Type                                                                                                                                                                                                                                        | Field Usage |
| RequirementID            | RequirementInstanceID_ChoiceType                                                                                                                                                                                                                  | Repeated    |
| Description              | Indicates the unique ID of a Requirement that was instantiated as a result of the Response. Subsequent progress in planning and/or execution of the Requirement should be given in [Requirement]Plan, [Requirement]Status, etc. messages.         |             |
| ResultingPlans           | MA_PlansReferenceType                                                                                                                                                                                                                             | Optional    |
| Description              | Indicates *Plans that were updated or created to include the sibling Requirements.                                                                                                                                                                |             |

Table A-1-510: P2P UCI Extensions MA\_ResponseActivityMT - RequirementInstantiation UCI Extension Field

| ActivatedPlans      |                                                        |             |
|---------------------|--------------------------------------------------------|-------------|
| Description         | Indicates *Plans that were activated for the Response. |             |
| Base Type           | MA_PlansReferenceType                                  |             |
| Message Structure   |                                                        |             |
| Field Name          | Field Type                                             | Field Usage |
| MissionPlanID       | MissionPlanID_Type                                     | Repeated    |
| Description         | Indicates a reference to a MissionPlan.                |             |
| TaskPlanID          | TaskPlanID_Type                                        | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                   |             |
| OrbitPlanID         | OrbitPlanID_Type                                       | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.                 |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                               | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.         |             |
| RoutePlanID         | RoutePlanID_Type                                       | Repeated    |
| Description         | Indicates a reference to a RoutePlan.                  |             |
| RouteActivityPlanID | RouteActivityPlanID_Type                               | Repeated    |
| Description         | Indicates a reference to a RouteActivityPlan.          |             |
| ActivityPlanID      | ActivityPlanID_Type                                    | Repeated    |
| Description         | Indicates a reference to an ActivityPlan.              |             |

|                       |                                           |          |
|-----------------------|-------------------------------------------|----------|
| <b>EffectPlanID</b>   | EffectPlanID_Type                         | Repeated |
| Description           | Indicates a reference to an EffectPlan.   |          |
| <b>ActionPlanID</b>   | ActionPlanID_Type                         | Repeated |
| Description           | Indicates a reference to an ActionPlan.   |          |
| <b>ResponsePlanID</b> | ResponsePlanID_Type                       | Repeated |
| Description           | Indicates a reference to an ResponsePlan. |          |
| <b>OpPlanID</b>       | MA_OpPlanID_Type                          | Repeated |
| Description           | Indicates a reference to an OpPlan.       |          |

Table A-1-511: P2P UCI Extensions MA\_ResponseActivityMT - ActivatedPlans UCI Extension Field

| Activity                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | Indicates the dynamic operational characteristics of an Activity of an Response Capability.                                                                                                                                                                                                                                                                                                                                                    |             |
| Base Type                | MA_ResponseActivityType                                                                                                                                                                                                                                                                                                                                                                                                                        |             |
| Message Structure        |                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |
| Field Name               | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                     | Field Usage |
| InstantiationTime        | DateTimeType                                                                                                                                                                                                                                                                                                                                                                                                                                   | Required    |
| Description              | Indicates the time at which the Response was instantiated.                                                                                                                                                                                                                                                                                                                                                                                     |             |
| OptionIndex              | unsignedInt                                                                                                                                                                                                                                                                                                                                                                                                                                    | Optional    |
| Description              | Indicates the specific Option within the Response that was instantiated.                                                                                                                                                                                                                                                                                                                                                                       |             |
| RequirementInstantiation | MA_RequirementTemplateInstantiationType                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description              | Indicates an independent set of Requirement or Requirements and *Plans that were instantiated for the Response. Subsequent progress in planning and/or execution is given in [Requirement]Plan, [Requirement]Status, MissionPlan*, etc. messages.                                                                                                                                                                                              |             |
| ActivatedPlans           | MA_PlansReferenceType                                                                                                                                                                                                                                                                                                                                                                                                                          | Optional    |
| Description              | Indicates *Plans that were activated for the Response.                                                                                                                                                                                                                                                                                                                                                                                         |             |
| AlertID                  | MissionContingencyAlertID_Type                                                                                                                                                                                                                                                                                                                                                                                                                 | Optional    |
| Description              | Indicates a MissionContingencyAlert that was generated for the Response.                                                                                                                                                                                                                                                                                                                                                                       |             |
| ActivityID               | ActivityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                | Required    |
| Description              | Indicates the unique ID of the Activity.                                                                                                                                                                                                                                                                                                                                                                                                       |             |
| CapabilityID             | CapabilityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                              | Repeated    |
| Description              | Indicates a Capability which the Activity is an instance of. Generally an Activity is an instance of a single Capability. An Activity can be associated with multiple Capabilities when, for example, a Subsystem has distinct Capabilities for automated and manual. In this case, if a manual command is issued while an automated Activity that achieves the desired result already exists, a single Activity represents both Capabilities. |             |
| Interactive              | boolean                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required    |
| Description              | Indicates whether or not the Activity can be edited/modified via a [Capability]Command...Activity message.                                                                                                                                                                                                                                                                                                                                     |             |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>ActivityState</b>            | ActivityStateEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Required |
| Description                     | Indicates the current state of the Activity its state machine. See enumeration annotations for further details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>Basis</b>                    | ActivityBasisEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Optional |
| Description                     | Indicates the basis of the Activity. See enumerated type annotations for further details. If omitted, a basis of ACTUAL should be assumed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>ActivityRank</b>             | ComparableRankingType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Optional |
| Description                     | Indicates the rank of the Activity amongst all other Activities of the Capability and/or Subsystem. This isn't necessarily the same as the Rank associated with the sibling ActivitySource. This element is meant to convey the Subsystem determined relative ranking of all Activities of a Capability category (ESM, EA, etc.). For Subsystems with multiple Capability categories this element is meant to convey relative ranking across all Capability categories. In addition to the rank of the ActivitySource, an Activity's rank can be influenced by Capability priority (see CapabilitySettings), RF_ControlCommands and other factors that influence the Subsystem's internal scheduling and resource allocation. |          |
| <b>ActivityReason</b>           | CannotComplyType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Optional |
| Description                     | Indicates details regarding the reason for and/or a contributing factor to the sibling ActivityState element. This element is only expected when the Activity is in a potentially undesirable state such as constrained, failed or completed with less than optimum results.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |
| <b>Source</b>                   | ActivitySourceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Repeated |
| Description                     | Indicates an originating "source" of the Activity; the command or other item which invoked this specific Activity of a Capability. A single Activity can fully or partially satisfy Activity sources.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>EstimatedStartTime</b>       | DateTimeType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                     | Indicates the estimated time when the Activity started or is expected to start.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
| <b>EstimatedCompletionTime</b>  | DateTimeType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Optional |
| Description                     | Indicates the estimated time when the Activity will complete.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |
| <b>EstimatedPercentComplete</b> | PercentType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Optional |
| Description                     | Indicates the current estimated percent complete of the Activity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |

Table A-1-512: P2P UCI Extensions MA\_ResponseActivityMT - Activity UCI Extension Field

| MessageData       |                                                                                                        |             |
|-------------------|--------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                |             |
| Base Type         | MA_ResponseActivityMDT                                                                                 |             |
| Message Structure |                                                                                                        |             |
| Field Name        | Field Type                                                                                             | Field Usage |
| SystemID          | SystemID_Type                                                                                          | Required    |
| Description       | Indicates the System which hosts the Response Capabilities whose Activities are given in this message. |             |
| Activity          | MA_ResponseActivityType                                                                                | Repeated    |

|             |                                                                                             |
|-------------|---------------------------------------------------------------------------------------------|
| Description | Indicates the dynamic operational characteristics of an Activity of an Response Capability. |
|-------------|---------------------------------------------------------------------------------------------|

Table A-1-513: P2P UCI Extensions MA\_ResponseActivityMT - MessageData UCI Extension Field

| ResultingPlans      |                                                                                    |             |
|---------------------|------------------------------------------------------------------------------------|-------------|
| Description         | Indicates *Plans that were updated or created to include the sibling Requirements. |             |
| Base Type           | MA_PlansReferenceType                                                              |             |
| Message Structure   |                                                                                    |             |
| Field Name          | Field Type                                                                         | Field Usage |
| MissionPlanID       | MissionPlanID_Type                                                                 | Repeated    |
| Description         | Indicates a reference to a MissionPlan.                                            |             |
| TaskPlanID          | TaskPlanID_Type                                                                    | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                                               |             |
| OrbitPlanID         | OrbitPlanID_Type                                                                   | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.                                             |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                                                           | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.                                     |             |
| RoutePlanID         | RoutePlanID_Type                                                                   | Repeated    |
| Description         | Indicates a reference to a RoutePlan.                                              |             |
| RouteActivityPlanID | RouteActivityPlanID_Type                                                           | Repeated    |
| Description         | Indicates a reference to a RouteActivityPlan.                                      |             |
| ActivityPlanID      | ActivityPlanID_Type                                                                | Repeated    |
| Description         | Indicates a reference to an ActivityPlan.                                          |             |
| EffectPlanID        | EffectPlanID_Type                                                                  | Repeated    |
| Description         | Indicates a reference to an EffectPlan.                                            |             |
| ActionPlanID        | ActionPlanID_Type                                                                  | Repeated    |
| Description         | Indicates a reference to an ActionPlan.                                            |             |
| ResponsePlanID      | ResponsePlanID_Type                                                                | Repeated    |
| Description         | Indicates a reference to an ResponsePlan.                                          |             |
| OpPlanID            | MA_OpPlanID_Type                                                                   | Repeated    |
| Description         | Indicates a reference to an OpPlan.                                                |             |

Table A-1-514: P2P UCI Extensions MA\_ResponseActivityMT - ResultingPlans UCI Extension Field

| OpPlanID          |                                     |
|-------------------|-------------------------------------|
| Description       | Indicates a reference to an OpPlan. |
| Base Type         | MA_OpPlanID_Type                    |
| Message Structure |                                     |

| Field Name              | Field Type                                                                                                                                                                                                                                                     | Field Usage |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Version</b>          | unsignedInt                                                                                                                                                                                                                                                    | Optional    |
| Description             | Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval. |             |
| <b>UUID</b>             | UniversallyUniqueIdentifierType                                                                                                                                                                                                                                | Required    |
| Description             | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.                                                                                               |             |
| <b>DescriptiveLabel</b> | VisibleString256Type                                                                                                                                                                                                                                           | Optional    |
| Description             | Human readable text label with content meaningful to human operators.                                                                                                                                                                                          |             |

Table A-1-515: P2P UCI Extensions MA\_ResponseActivityMT - OpPlanID UCI Extension Field

### 1.3.2.94 MA\_TaskStatusMT UCI Extension

| MA_TaskStatusMT    |                                                                                                                                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b> | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info                                                              |
| <b>Ext. Fields</b> | Plans : MA_PlansReferenceType<br>OpPlanID : MA_OpPlanID_Type<br>ExecutingSystemOrPackage : MA_PackageSystemType<br>MessageData : MA_TaskStatusMDT<br>Traceability : MA_RequirementStatusTraceabilityType |

Table A-1-516: P2P UCI Extensions MA\_TaskStatusMT UCI Extension

| Plans               |                                                      |             |
|---------------------|------------------------------------------------------|-------------|
| Description         | Indicates *Plans the Requirement is associated with. |             |
| Base Type           | MA_PlansReferenceType                                |             |
| Message Structure   |                                                      |             |
| Field Name          | Field Type                                           | Field Usage |
| MissionPlanID       | MissionPlanID_Type                                   | Repeated    |
| Description         | Indicates a reference to a MissionPlan.              |             |
| TaskPlanID          | TaskPlanID_Type                                      | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                 |             |
| OrbitPlanID         | OrbitPlanID_Type                                     | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.               |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                             | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.       |             |

|                            |                                               |          |
|----------------------------|-----------------------------------------------|----------|
| <b>RoutePlanID</b>         | RoutePlanID_Type                              | Repeated |
| Description                | Indicates a reference to a RoutePlan.         |          |
| <b>RouteActivityPlanID</b> | RouteActivityPlanID_Type                      | Repeated |
| Description                | Indicates a reference to a RouteActivityPlan. |          |
| <b>ActivityPlanID</b>      | ActivityPlanID_Type                           | Repeated |
| Description                | Indicates a reference to an ActivityPlan.     |          |
| <b>EffectPlanID</b>        | EffectPlanID_Type                             | Repeated |
| Description                | Indicates a reference to an EffectPlan.       |          |
| <b>ActionPlanID</b>        | ActionPlanID_Type                             | Repeated |
| Description                | Indicates a reference to an ActionPlan.       |          |
| <b>ResponsePlanID</b>      | ResponsePlanID_Type                           | Repeated |
| Description                | Indicates a reference to an ResponsePlan.     |          |
| <b>OpPlanID</b>            | MA_OpPlanID_Type                              | Repeated |
| Description                | Indicates a reference to an OpPlan.           |          |

Table A-1-517: P2P UCI Extensions MA\_TaskStatusMT - Plans UCI Extension Field

| OpPlanID          |                                                                                                                                                                                                                                                                |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a reference to an OpPlan.                                                                                                                                                                                                                            |             |
| Base Type         | MA_OpPlanID_Type                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                     | Field Usage |
| Version           | unsignedInt                                                                                                                                                                                                                                                    | Optional    |
| Description       | Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval. |             |
| UUID              | UniversallyUniqueIdentifierType                                                                                                                                                                                                                                | Required    |
| Description       | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.                                                                                               |             |
| DescriptiveLabel  | VisibleString256Type                                                                                                                                                                                                                                           | Optional    |
| Description       | Human readable text label with content meaningful to human operators.                                                                                                                                                                                          |             |

Table A-1-518: P2P UCI Extensions MA\_TaskStatusMT - OpPlanID UCI Extension Field

|                                 |                                                                                                                                                                                                                                                                                          |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ExecutingSystemOrPackage</b> |                                                                                                                                                                                                                                                                                          |
| <b>Description</b>              | Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement. |
| <b>Base Type</b>                | MA_PackageSystemType                                                                                                                                                                                                                                                                     |
| <b>Message Structure</b>        |                                                                                                                                                                                                                                                                                          |



| Field Name       | Field Type                         | Field Usage |
|------------------|------------------------------------|-------------|
| <b>ALL</b>       | EmptyType                          | Required    |
| Description      | Indicates applies to ALL entities. |             |
| <b>NONE</b>      | EmptyType                          | Required    |
| Description      | Indicates applies to NO entities.  |             |
| <b>PackageID</b> | PackageID_Type                     | Required    |
| Description      | The ID of a Package.               |             |
| <b>SystemID</b>  | SystemID_Type                      | Required    |
| Description      | The ID of an ACP or C2 Operator.   |             |

Table A-1-519: P2P UCI Extensions MA\_TaskStatusMT - ExecutingSystemOrPackage UCI Extension Field

| MessageData              |                                                                                                                                                                                                                                                                                                                      |             |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                              |             |
| Base Type                | MA_TaskStatusMDT                                                                                                                                                                                                                                                                                                     |             |
| Message Structure        |                                                                                                                                                                                                                                                                                                                      |             |
| Field Name               | Field Type                                                                                                                                                                                                                                                                                                           | Field Usage |
| TaskID                   | TaskID_Type                                                                                                                                                                                                                                                                                                          | Required    |
| Description              | Indicates the unique ID of the Task whose execution status is being reported.                                                                                                                                                                                                                                        |             |
| ExecutingSystemOrPackage | MA_PackageSystemType                                                                                                                                                                                                                                                                                                 | Required    |
| Description              | Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.                             |             |
| ExecutionState           | RequirementExecutionStateEnum                                                                                                                                                                                                                                                                                        | Required    |
| Description              | Identifies the state of the Requirement.                                                                                                                                                                                                                                                                             |             |
| ExecutionStateReason     | CannotComplyType                                                                                                                                                                                                                                                                                                     | Optional    |
| Description              | Indicates details regarding the reason for and/or a contributing factor to the sibling State element. This element is only expected when the State is in a potentially undesirable state such as failed or completed with less than optimum results. It is strongly encouraged for transitions to failed or dropped. |             |
| ActualTiming             | RequirementTimingType                                                                                                                                                                                                                                                                                                | Repeated    |
| Description              | Indicates actual timing status for the Requirement. Allows for defining for each timing type (RequirementTimingEnum).                                                                                                                                                                                                |             |
| PercentCompleted         | PercentType                                                                                                                                                                                                                                                                                                          | Optional    |
| Description              | Percentage of assigned Requirement completion. The percentage increments until the Requirement has COMPLETED or FAILED. A percent complete of 0 indicates the Requirement has not yet begun execution, and a failed Requirement may have any percentage between 1 - 99.                                              |             |
| Traceability             | MA_RequirementStatusTraceabilityType                                                                                                                                                                                                                                                                                 | Optional    |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Description | <p>Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture.</p> <p>For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands.</p> |          |
| Metrics     | MetricsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Optional |
| Description | Specifies the actual metrics associated with the Requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |

Table A-1-520: P2P UCI Extensions MA\_TaskStatusMT - MessageData UCI Extension Field

| Traceability      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture.                                                                                                                                                                                                                                                                                                     |             |
|                   | For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands. |             |
| Base Type         | MA_RequirementStatusTraceabilityType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |
| Message Structure |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| Plans             | MA_PlansReferenceType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Optional    |
| Description       | Indicates *Plans the Requirement is associated with.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |
| PlannedActivityID | PlannedActivityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Repeated    |
| Description       | Indicates a specific Planned Activity the Requirement is associated with.                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| CapabilityID      | CapabilityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Repeated    |
| Description       | Identifies a specific Capability the Requirement is associated with.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |             |
| CommandID         | CommandID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Repeated    |
| Description       | Indicates the ID of an associated [Capability]Command.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
| ActivityID        | ActivityID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates the ID of an associated actual/executing/executed Activity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |

Table A-1-521: P2P UCI Extensions MA\_TaskStatusMT - Traceability UCI Extension Field

## 1.3.2.95 MA\_PlanModificationRequestMT UCI Extension

| MA_PlanModificationRequestMT |                                                                                                                                                                                                                                                                                                                                   |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>           | Indicates the contents of the PlanModificationRequest message details, and also includes the message header.<br><br>See the XSD File for more info                                                                                                                                                                                |
| <b>Ext. Fields</b>           | SubPlansAdditions : MA_PlansReferenceBaseType<br>SubPlansRetractions : MA_PlansReferenceBaseType<br>SubPlansModification : MA_SubPlansModificationType<br>PlanModificationDetails : MA_PlanModificationDetailsType<br>MessageData : MA_PlanModificationRequestMDT<br>Plans : MA_PlansReferenceType<br>OpPlanID : MA_OpPlanID_Type |

Table A-1-522: P2P UCI Extensions MA\_PlanModificationRequestMT UCI Extension

| SubPlansAdditions   |                                                                                                                                                             |             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | Indicates an additional set of *Plans which are included as part of the MissionPlan. The included SubPlans share the same Applicability as the MissionPlan. |             |
| Base Type           | MA_PlansReferenceBaseType                                                                                                                                   |             |
| Message Structure   |                                                                                                                                                             |             |
| Field Name          | Field Type                                                                                                                                                  | Field Usage |
| TaskPlanID          | TaskPlanID_Type                                                                                                                                             | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                                                                                                                        |             |
| OrbitPlanID         | OrbitPlanID_Type                                                                                                                                            | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.                                                                                                                      |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                                                                                                                                    | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.                                                                                                              |             |
| RoutePlanID         | RoutePlanID_Type                                                                                                                                            | Repeated    |
| Description         | Indicates a reference to a RoutePlan.                                                                                                                       |             |
| RouteActivityPlanID | RouteActivityPlanID_Type                                                                                                                                    | Repeated    |
| Description         | Indicates a reference to a RouteActivityPlan.                                                                                                               |             |
| ActivityPlanID      | ActivityPlanID_Type                                                                                                                                         | Repeated    |
| Description         | Indicates a reference to an ActivityPlan.                                                                                                                   |             |
| EffectPlanID        | EffectPlanID_Type                                                                                                                                           | Repeated    |
| Description         | Indicates a reference to an EffectPlan.                                                                                                                     |             |
| ActionPlanID        | ActionPlanID_Type                                                                                                                                           | Repeated    |

|                       |                                           |          |
|-----------------------|-------------------------------------------|----------|
| Description           | Indicates a reference to an ActionPlan.   |          |
| <b>ResponsePlanID</b> | ResponsePlanID_Type                       | Repeated |
| Description           | Indicates a reference to an ResponsePlan. |          |
| <b>OpPlanID</b>       | MA_OpPlanID_Type                          | Repeated |
| Description           | Indicates a reference to an OpPlan.       |          |

**Table A-1-523: P2P UCI Extensions MA\_PlanModificationRequestMT - SubPlansAdditions UCI Extension Field**

| SubPlansRetractions |                                                                                       |             |
|---------------------|---------------------------------------------------------------------------------------|-------------|
| Description         | Indicates the removal of set of *Plans which are included as part of the MissionPlan. |             |
| Base Type           | MA_PlansReferenceBaseType                                                             |             |
| Message Structure   |                                                                                       |             |
| Field Name          | Field Type                                                                            | Field Usage |
| TaskPlanID          | TaskPlanID_Type                                                                       | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                                                  |             |
| OrbitPlanID         | OrbitPlanID_Type                                                                      | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.                                                |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                                                              | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.                                        |             |
| RoutePlanID         | RoutePlanID_Type                                                                      | Repeated    |
| Description         | Indicates a reference to a RoutePlan.                                                 |             |
| RouteActivityPlanID | RouteActivityPlanID_Type                                                              | Repeated    |
| Description         | Indicates a reference to a RouteActivityPlan.                                         |             |
| ActivityPlanID      | ActivityPlanID_Type                                                                   | Repeated    |
| Description         | Indicates a reference to an ActivityPlan.                                             |             |
| EffectPlanID        | EffectPlanID_Type                                                                     | Repeated    |
| Description         | Indicates a reference to an EffectPlan.                                               |             |
| ActionPlanID        | ActionPlanID_Type                                                                     | Repeated    |
| Description         | Indicates a reference to an ActionPlan.                                               |             |
| ResponsePlanID      | ResponsePlanID_Type                                                                   | Repeated    |
| Description         | Indicates a reference to an ResponsePlan.                                             |             |
| OpPlanID            | MA_OpPlanID_Type                                                                      | Repeated    |
| Description         | Indicates a reference to an OpPlan.                                                   |             |

**Table A-1-524: P2P UCI Extensions MA\_PlanModificationRequestMT - SubPlansRetractions UCI Extension Field**

### SubPlansModification

|                     |                                                                                                                                                                    |             |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description         | Indicates a change in the set of *Plans which are included as part of the MissionPlan. See annotations for the underlying type and its child elements for details. |             |
| Base Type           | MA_SubPlansModificationType                                                                                                                                        |             |
| Message Structure   |                                                                                                                                                                    |             |
| Field Name          | Field Type                                                                                                                                                         | Field Usage |
| SubPlansAdditions   | MA_PlansReferenceBaseType                                                                                                                                          | Optional    |
| Description         | Indicates an additional set of *Plans which are included as part of the MissionPlan. The included SubPlans share the same Applicability as the MissionPlan.        |             |
| SubPlansRetractions | MA_PlansReferenceBaseType                                                                                                                                          | Optional    |
| Description         | Indicates the removal of set of *Plans which are included as part of the MissionPlan.                                                                              |             |

**Table A-1-525: P2P UCI Extensions MA\_PlanModificationRequestMT - SubPlansModification UCI Extension Field**

| PlanModificationDetails       |                                                                                                                                                                                                                                                                                                                                                               |             |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                   | Specifies details of the plan modification.                                                                                                                                                                                                                                                                                                                   |             |
| Base Type                     | MA_PlanModificationDetailsType                                                                                                                                                                                                                                                                                                                                |             |
| Message Structure             |                                                                                                                                                                                                                                                                                                                                                               |             |
| Field Name                    | Field Type                                                                                                                                                                                                                                                                                                                                                    | Field Usage |
| ForPlanningUseOnly            | boolean                                                                                                                                                                                                                                                                                                                                                       | Optional    |
| Description                   | When TRUE, indicates the *Plan is a used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc. |             |
| WindowModification            | PlanWindowModificationTypeChoiceType                                                                                                                                                                                                                                                                                                                          | Optional    |
| Description                   | Specifies an absolute time Window or alternatively a TimeOffset by which the Plan's, or set of Plans', Windows should be shifted.                                                                                                                                                                                                                             |             |
| ApplicabilityModification     | PlanApplicabilityModificationType                                                                                                                                                                                                                                                                                                                             | Optional    |
| Description                   | Indicates a change in the System or Systems which have a direct relationship with the Mission Plan. See annotations for the underlying type and its child elements for details.                                                                                                                                                                               |             |
| SubPlansModification          | MA_SubPlansModificationType                                                                                                                                                                                                                                                                                                                                   | Optional    |
| Description                   | Indicates a change in the set of *Plans which are included as part of the MissionPlan. See annotations for the underlying type and its child elements for details.                                                                                                                                                                                            |             |
| HierarchyModification         | HierarchyModificationType                                                                                                                                                                                                                                                                                                                                     | Optional    |
| Description                   | Indicates a change in the hierarchy of the set of MissionPlans.                                                                                                                                                                                                                                                                                               |             |
| ExecutionSequenceModification | ExecutionSequenceReplaceOrModifyChoiceType                                                                                                                                                                                                                                                                                                                    | Optional    |
| Description                   | Indicates a change in the execution sequence of the set of *Plans.                                                                                                                                                                                                                                                                                            |             |
| RequirementAllocationLock     | RequirementAllocationLockDetailsType                                                                                                                                                                                                                                                                                                                          | Optional    |

|             |                                                                           |
|-------------|---------------------------------------------------------------------------|
| Description | Indicates a requested change in Allocated Requirements to Lock or Unlock. |
|-------------|---------------------------------------------------------------------------|

**Table A-1-526: P2P UCI Extensions MA\_PlanModificationRequestMT - PlanModificationDetails UCI Extension Field**

| MessageData             |                                                                                                                                           |             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description             | See annotations in child elements and messages/elements that use this type for details.                                                   |             |
| Base Type               | MA_PlanModificationRequestMDT                                                                                                             |             |
| Message Structure       |                                                                                                                                           |             |
| Field Name              | Field Type                                                                                                                                | Field Usage |
| Source                  | PlanInitiationSourceEnum                                                                                                                  | Required    |
| Description             | Source of the plan modification request.                                                                                                  |             |
| Plans                   | MA_PlansReferenceType                                                                                                                     | Required    |
| Description             | Indicates the Plan, or set of Plans, for which this Plan Modification Request is applicable.                                              |             |
| PlanModificationDetails | MA_PlanModificationDetailsType                                                                                                            | Required    |
| Description             | Specifies details of the plan modification.                                                                                               |             |
| RequestID               | RequestID_Type                                                                                                                            | Required    |
| Description             | Indicates the unique ID of the request message.                                                                                           |             |
| RequestState            | RequestStateEnum                                                                                                                          | Required    |
| Description             | The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request. |             |

**Table A-1-527: P2P UCI Extensions MA\_PlanModificationRequestMT - MessageData UCI Extension Field**

| Plans               |                                                                                              |             |
|---------------------|----------------------------------------------------------------------------------------------|-------------|
| Description         | Indicates the Plan, or set of Plans, for which this Plan Modification Request is applicable. |             |
| Base Type           | MA_PlansReferenceType                                                                        |             |
| Message Structure   |                                                                                              |             |
| Field Name          | Field Type                                                                                   | Field Usage |
| MissionPlanID       | MissionPlanID_Type                                                                           | Repeated    |
| Description         | Indicates a reference to a MissionPlan.                                                      |             |
| TaskPlanID          | TaskPlanID_Type                                                                              | Repeated    |
| Description         | Indicates a reference to a TaskPlan.                                                         |             |
| OrbitPlanID         | OrbitPlanID_Type                                                                             | Repeated    |
| Description         | Indicates a reference to an OrbitPlan.                                                       |             |
| OrbitActivityPlanID | OrbitActivityPlanID_Type                                                                     | Repeated    |
| Description         | Indicates a reference to an OrbitActivityPlan.                                               |             |
| RoutePlanID         | RoutePlanID_Type                                                                             | Repeated    |

|                            |                                               |          |
|----------------------------|-----------------------------------------------|----------|
| Description                | Indicates a reference to a RoutePlan.         |          |
| <b>RouteActivityPlanID</b> | RouteActivityPlanID_Type                      | Repeated |
| Description                | Indicates a reference to a RouteActivityPlan. |          |
| <b>ActivityPlanID</b>      | ActivityPlanID_Type                           | Repeated |
| Description                | Indicates a reference to an ActivityPlan.     |          |
| <b>EffectPlanID</b>        | EffectPlanID_Type                             | Repeated |
| Description                | Indicates a reference to an EffectPlan.       |          |
| <b>ActionPlanID</b>        | ActionPlanID_Type                             | Repeated |
| Description                | Indicates a reference to an ActionPlan.       |          |
| <b>ResponsePlanID</b>      | ResponsePlanID_Type                           | Repeated |
| Description                | Indicates a reference to an ResponsePlan.     |          |
| <b>OpPlanID</b>            | MA_OpPlanID_Type                              | Repeated |
| Description                | Indicates a reference to an OpPlan.           |          |

Table A-1-528: P2P UCI Extensions MA\_PlanModificationRequestMT - Plans UCI Extension Field

| OpPlanID          |                                                                                                                                                                                                                                                                |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description       | Indicates a reference to an OpPlan.                                                                                                                                                                                                                            |             |
| Base Type         | MA_OpPlanID_Type                                                                                                                                                                                                                                               |             |
| Message Structure |                                                                                                                                                                                                                                                                |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                     | Field Usage |
| Version           | unsignedInt                                                                                                                                                                                                                                                    | Optional    |
| Description       | Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval. |             |
| UUID              | UniversallyUniqueIdentifierType                                                                                                                                                                                                                                | Required    |
| Description       | Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.                                                                                               |             |
| DescriptiveLabel  | VisibleString256Type                                                                                                                                                                                                                                           | Optional    |
| Description       | Human readable text label with content meaningful to human operators.                                                                                                                                                                                          |             |

Table A-1-529: P2P UCI Extensions MA\_PlanModificationRequestMT - OpPlanID UCI Extension Field

### 1.3.2.96 MA\_PlanModificationRequestStatusMT UCI Extension

|                                           |                                                                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_PlanModificationRequestStatusMT</b> |                                                                                                                               |
| <b>Description</b>                        | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                        | MessageData : MA_PlanModificationRequestStatusMDT                                                                             |

Table A-1-530: P2P UCI Extensions MA\_PlanModificationRequestStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                             |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                     |             |
| Base Type                    | MA_PlanModificationRequestStatusMDT                                                                                                                                                                                                                                                         |             |
| Message Structure            |                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                  | Field Usage |
| RequestID                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |             |
| RequestProcessingState       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |             |
| RequestProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |

**Table A-1-531: P2P UCI Extensions MA\_PlanModificationRequestStatusMT - MessageData UCI Extension Field**

### 1.3.2.97 MA\_ConstraintPriorityListActivationDataMT UCI Extension

| MA_ConstraintPriorityListActivationDataMT |                                                                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                        | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                        | MessageData : MA_ConstraintPriorityListActivationDataMDT                                                                      |

**Table A-1-532: P2P UCI Extensions MA\_ConstraintPriorityListActivationDataMT UCI Extension**

| MessageData                            |                                                                                         |             |
|----------------------------------------|-----------------------------------------------------------------------------------------|-------------|
| Description                            | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type                              | MA_ConstraintPriorityListActivationDataMDT                                              |             |
| Message Structure                      |                                                                                         |             |
| Field Name                             | Field Type                                                                              | Field Usage |
| ConstraintPriorityListActivationDataID | MA_ConstraintPriorityListActivationDataID_Type                                          | Required    |
| Description                            | The unique identifier for the message.                                                  |             |
| DefaultConstraintPriorityListID        | MA_ConstraintPriorityListID_Type                                                        | Required    |
| Description                            | Contains the ID of the Constraint Priority List that should be set                      |             |



|                                                               |                                                                                                                                                                                 |          |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                                               | as the default.                                                                                                                                                                 |          |
| <b>ListOfApplicableOpConstrainedConstraintPriorityListIDs</b> | MA_ListOfApplicableOpConstrainedConstraintPriorityListIDType                                                                                                                    | Repeated |
| Description                                                   | When an Package/ACP is within this geozone, the linked Constraint Priority List will be used instead of the default. OpZone and OpVolume IDs can only appear once in this list. |          |

**Table A-1-533: P2P UCI Extensions MA\_ConstraintPriorityListActivationDataMT - MessageData UCI Extension Field**

### 1.3.2.98 MA\_ConstraintPriorityListMT UCI Extension

| MA_ConstraintPriorityListMT |                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>          | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>          | MessageData : MA_ConstraintPriorityListMDT                                                                                    |

**Table A-1-534: P2P UCI Extensions MA\_ConstraintPriorityListMT UCI Extension**

| MessageData              |                                                                                                                                                                                                                                                                          |             |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                  |             |
| Base Type                | MA_ConstraintPriorityListMDT                                                                                                                                                                                                                                             |             |
| Message Structure        |                                                                                                                                                                                                                                                                          |             |
| Field Name               | Field Type                                                                                                                                                                                                                                                               | Field Usage |
| ConstraintPriorityListID | MA_ConstraintPriorityListID_Type                                                                                                                                                                                                                                         | Required    |
| Description              | The unique identifier for the message.                                                                                                                                                                                                                                   |             |
| DefaultPriority          | unsignedShort                                                                                                                                                                                                                                                            | Optional    |
| Description              | This field, if set, is the value that would be assigned to any ConstraintType that is not listed in the ListItem field. If this field is not set, the default value of ConstraintTypes not listed the ListItem field would be blank or the highest priority ("above" 0). |             |
| ListItem                 | MA_ConstraintPriorityListItemType                                                                                                                                                                                                                                        | Repeated    |
| Description              | Identifies a constraint type and it's priority in the Constraint Priority List.                                                                                                                                                                                          |             |

**Table A-1-535: P2P UCI Extensions MA\_ConstraintPriorityListMT - MessageData UCI Extension Field**

### 1.3.2.99 MA\_EscortActivityMT UCI Extension

| MA_EscortActivityMT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>  | This message indicates the combined in-work and to-do list of the Escort Capability of a Subsystem of a System. In general, a [Capability]Activity message describes what invoked Capabilities are doing. It is generally a complete, coherent snapshot of all Activities of a Subsystem for an entire Capability category (such as ESM, AirSample, EA, etc.). In other words, an Activity is the dynamic, per-use/per-invocation status of a Capability. An Activity is an abstraction that includes 1) the autonomous (MDF-driven or MissionPlan-driven) activity of a Capability, 2) the activity resulting from manual commands (EscortCommand) from an Operator or Capability management Service, 3) currently in-work Escort, 4) Escort scheduled for the future and 5) Escort that was scheduled and/or attempted but couldn't be completed |

|                    |                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | due to resource limitations or other reasons. A single EscortCommand or other triggering event can result in multiple activities within the EscortActivity message. |
|                    | See the XSD File for more info                                                                                                                                      |
| <b>Ext. Fields</b> | MessageData : MA_EscortActivityMDT                                                                                                                                  |

Table A-1-536: P2P UCI Extensions MA\_EscortActivityMT UCI Extension

| MessageData       |                                                                                                         |             |
|-------------------|---------------------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.                 |             |
| Base Type         | MA_EscortActivityMDT                                                                                    |             |
| Message Structure |                                                                                                         |             |
| Field Name        | Field Type                                                                                              | Field Usage |
| SubsystemID       | SubsystemID_Type                                                                                        | Required    |
| Description       | Indicates the Subsystem which hosts the Escort Capabilities whose Activities are given in this message. |             |
| Activity          | MA_EscortActivityType                                                                                   | Repeated    |
| Description       | Indicates the dynamic operational characteristics of an Activity of an Escort Capability.               |             |

Table A-1-537: P2P UCI Extensions MA\_EscortActivityMT - MessageData UCI Extension Field

### 1.3.2.100 MA\_PositionReportDetailedMT UCI Extension

| MA_PositionReportDetailedMT |                                                                                              |
|-----------------------------|----------------------------------------------------------------------------------------------|
| <b>Description</b>          | See annotations in child elements and messages/elements that use this type for details.      |
|                             | See the XSD File for more info                                                               |
| <b>Ext. Fields</b>          | PositionReportData : MA_PositionReportDataType<br>MessageData : MA_PositionReportDetailedMDT |

Table A-1-538: P2P UCI Extensions MA\_PositionReportDetailedMT UCI Extension

| PositionReportData |                                                                                                                                                                         |             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | If multiple instances are given, each should be of a different navigation solution type as indicated by the child element.                                              |             |
| Base Type          | MA_PositionReportDataType                                                                                                                                               |             |
| Message Structure  |                                                                                                                                                                         |             |
| Field Name         | Field Type                                                                                                                                                              | Field Usage |
| PositionSource     | PositionSourceID_ChoiceType                                                                                                                                             | Required    |
| Description        | The source of the positional data.                                                                                                                                      |             |
| ComponentID        | ComponentID_Type                                                                                                                                                        | Optional    |
| Description        | The ID of the specific component within the subsystem that is the source of this data. This can be used if multiple redundant INS/GPS units are represented as a single |             |

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                                | subsystem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>NavigationSolutionState</b> | NavigationSolutionStateEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Required |
| Description                    | The navigation solution state indicates what sensor inputs the navigation unit is using to generate its kinematic solution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |
| <b>FigureOfMerit</b>           | BytePositiveType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Optional |
| Description                    | <p>A FOM is typically calculated as the root-sum-square of the estimated errors contributing to the solution accuracy. Example criteria include:</p> <ul style="list-style-type: none"> <li>a. GPS receiver state (e.g., carrier tracking, code tracking, acquisition)</li> <li>b. Carrier to noise ratio</li> <li>c. Satellite geometry (DOP value)</li> <li>d. Satellite range accuracy (URA value)</li> <li>e. Ionospheric measurement or modeling error</li> <li>f. Receiver aiding used</li> <li>g. Kalman filter error estimates.</li> </ul> <p>The resultant FOM is presented as a numerical value from 1 to 9, where 1 indicates the best navigation performance.</p> |          |
| <b>Kinematics</b>              | MA_DetailedKinematicsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Required |
| Description                    | The kinematic state of the vehicle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
| <b>KinematicsError</b>         | DetailedKinematicsErrorType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Required |
| Description                    | Indicates the covariance matrix of the vehicle, containing detailed kinematic errors of the position, velocity, and orientation. See the DetailedKinematicsErrorType element annotations for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |
| <b>SolutionCorrections</b>     | NavigationSolutionCorrectionsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Optional |
| Description                    | This element contains the corrections applied by the measurement unit to the reported values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |

Table A-1-539: P2P UCI Extensions MA\_PositionReportDetailedMT - PositionReportData UCI Extension Field

| MessageData        |                                                                                         |             |
|--------------------|-----------------------------------------------------------------------------------------|-------------|
| Description        | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type          | MA_PositionReportDetailedMDT                                                            |             |
| Message Structure  |                                                                                         |             |
| Field Name         | Field Type                                                                              | Field Usage |
| PositionReportData | MA_PositionReportDataType                                                               | Repeated    |

|             |                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------|
| Description | If multiple instances are given, each should be of a different navigation solution type as indicated by the child element. |
|-------------|----------------------------------------------------------------------------------------------------------------------------|

Table A-1-540: P2P UCI Extensions MA\_PositionReportDetailedMT - MessageData UCI Extension Field

## 1.3.2.101 MA\_EngagementCriteriaMT UCI Extension

| MA_EngagementCriteriaMT |                                                                                                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description             | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields             | MessageData : MA_EngagementCriteriaMDT                                                                                        |

Table A-1-541: P2P UCI Extensions MA\_EngagementCriteriaMT UCI Extension

| MessageData               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description               | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |
| Base Type                 | MA_EngagementCriteriaMDT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
| Message Structure         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| Field Name                | Field Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Field Usage |
| EngagementCriteriaID      | MA_EngagementCriteriaID_Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Required    |
| Description               | The unique identifier for the message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |
| IdMatrixCriteria          | MA_IdMatrixCriteriaType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Optional    |
| Description               | These markers are used to identify an Entity as a Friend, Neutral, or Enemy and which enables or confirms the Entity's Standard Identity type to be set to FRIEND, NEUTRAL, SUSPECT or HOSTILE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |
| TargetDesignationCriteria | MA_TargetDesignationCriteriaType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Repeated    |
| Description               | This provides a list of criteria that when met can trigger a MA_DesignationRequestMT to be sent to a Target Authority who can use the info in this message to send a MA_DesignationMT that will designate an Entity as a Target.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |             |
| TargetAuthorityCriteria   | MA_AuthorityCriteriaType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description               | This provides a list of Authorized Operators that are allowed to set an Entity as a Target and to respond to a StrikeConsentRequestMT. An Entity becomes a Target when a Designation message declares that entity to be a Target. This can also be done by an Action or Task specifying an Entity as a Target (TargetType). The associated Approval Policy will specify which Actions or Tasks make an Entity a Target and who is authorized to command them or approve a response that activates them. MA can send a MA_DesignationRequestMT to Operators identified in the Approval Authority. An Authorized operator can also directly send a MA_DesignationMT. An MA can be an authorized approver if specified, but an MA cannot send a MA_DesignationMT for itself. A Lead can send a MA_DesignationMT to a follower if a C2 has already approved the Target. |             |
| TargetCustodyCriteria     | MA_TargetCustodyCriteriaType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Repeated    |
| Description               | This describes how the Custody of a Target is confirmed and must be held through a Weapons Engagement with that Target.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |             |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                  |          |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>StrikeConsentRequired</b> | boolean                                                                                                                                                                                                                                                                                                                                                                                          | Required |
| Description                  | Indicates whether (TRUE) or not (FALSE) the Strike Activities requires consent to transition from the ENABLED state to an ACTIVE state. If TRUE, each Strike Activity would be required to send a StrikeConsentRequestMT to an Authorized Operator (defined in the TargetAuthorityCriteria field) and receive a StrikeConsentRequestStatusMT before each Strike Activity can execute the Strike. |          |

Table A-1-542: P2P UCI Extensions MA\_EngagementCriteriaMT - MessageData UCI Extension Field

## 1.3.2.102 MA\_ExecutionPriorityListMT UCI Extension

| MA_ExecutionPriorityListMT |                                                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description                | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields                | MessageData : MA_ExecutionPriorityListMDT                                                                                     |

Table A-1-543: P2P UCI Extensions MA\_ExecutionPriorityListMT UCI Extension

| MessageData             |                                                                                         |             |
|-------------------------|-----------------------------------------------------------------------------------------|-------------|
| Description             | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type               | MA_ExecutionPriorityListMDT                                                             |             |
| Message Structure       |                                                                                         |             |
| Field Name              | Field Type                                                                              | Field Usage |
| ExecutionPriorityListID | MA_ExecutionPriorityListID_Type                                                         | Required    |
| Description             | The unique identifier for the message.                                                  |             |
| ListItem                | MA_ExecutionPriorityListItemType                                                        | Repeated    |
| Description             | Identifies an item (task or action) and its rank in the Execution Priority List.        |             |

Table A-1-544: P2P UCI Extensions MA\_ExecutionPriorityListMT - MessageData UCI Extension Field

## 1.3.2.103 MA\_ExecutionPriorityListActivationDataMT UCI Extension

| MA_ExecutionPriorityListActivationDataMT |                                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Description                              | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| Ext. Fields                              | MessageData : MA_ExecutionPriorityListActivationDataMDT                                                                       |

Table A-1-545: P2P UCI Extensions MA\_ExecutionPriorityListActivationDataMT UCI Extension

| MessageData       |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |
| Base Type         | MA_ExecutionPriorityListActivationDataMDT                                               |
| Message Structure |                                                                                         |

| Field Name                                                   | Field Type                                                                                                                                             | Field Usage |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>ExecutionPriorityListActivationDataID</b>                 | MA_ExecutionPriorityListActivationDataID_Type                                                                                                          | Required    |
| Description                                                  | The unique identifier for the message.                                                                                                                 |             |
| <b>DefaultExecutionPriorityListID</b>                        | MA_ExecutionPriorityListID_Type                                                                                                                        | Required    |
| Description                                                  | Contains the ID of the message that should be set as the default.                                                                                      |             |
| <b>ListOfApplicableOpConstrainedExecutionPriorityListIDs</b> | MA_ListOfApplicableOpConstrainedExecutionPriorityListIDType                                                                                            | Repeated    |
| Description                                                  | When an ACP is within this geozone, the linked message will be used instead of the default. OpZone and OpVolume IDs can only appear once in this list. |             |

**Table A-1-546: P2P UCI Extensions MA\_ExecutionPriorityListActivationDataMT - MessageData UCI Extension Field**

### 1.3.2.104 MA\_MessageAcknowledgmentConfigurationMT UCI Extension

| MA_MessageAcknowledgmentConfigurationMT |                                                                                                                               |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                      | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                      | MessageData : MA_MessageAcknowledgmentConfigurationMDT                                                                        |

**Table A-1-547: P2P UCI Extensions MA\_MessageAcknowledgmentConfigurationMT UCI Extension**

| MessageData              |                                                                                                                                                                           |             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | See annotations in child elements and messages/elements that use this type for details.                                                                                   |             |
| Base Type                | MA_MessageAcknowledgmentConfigurationMDT                                                                                                                                  |             |
| Message Structure        |                                                                                                                                                                           |             |
| Field Name               | Field Type                                                                                                                                                                | Field Usage |
| DefaultGroupMessageFreq  | DurationType                                                                                                                                                              | Optional    |
| Description              | If set, this provides the frequency at which the Grouped Acknowledgment message is set at. If not set there is no grouping.                                               |             |
| DefaultMessagesToAck     | MA_AckMessageEnum                                                                                                                                                         | Repeated    |
| Description              | If set, the list of messages that an Acknowledgment message will be sent for when received. If not set all messages of the type will be responded to.                     |             |
| ReceiverSpecificSettings | MA_AckReceiverSpecificConfigType                                                                                                                                          | Repeated    |
| Description              | This is the list of individual configuration settings that a remote system is configured. Values in this will override the Default settings.                              |             |
| ConfigurationSupported   | boolean                                                                                                                                                                   | Required    |
| Description              | If FALSE, the publisher does not support grouping, they expect acknowledgments for messages covered by this configuration, and will reject all configuration requests. If |             |

|  |                                                                                                                       |
|--|-----------------------------------------------------------------------------------------------------------------------|
|  | TRUE, the publisher supports individual configurations, grouping, and the other items in this collection of messages. |
|--|-----------------------------------------------------------------------------------------------------------------------|

**Table A-1-548: P2P UCI Extensions MA\_MessageAcknowledgmentConfigurationMT - MessageData UCI Extension Field**

### 1.3.2.105 MA\_MessageAcknowledgmentConfigurationRequestMT UCI Extension

| MA_MessageAcknowledgmentConfigurationRequestMT |                                                                                                                               |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                             | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                             | MessageData : MA_MessageAcknowledgmentConfigurationRequestMDT                                                                 |

**Table A-1-549: P2P UCI Extensions MA\_MessageAcknowledgmentConfigurationRequestMT UCI Extension**

| MessageData                      |                                                                                                                                                                                                                                                                                             |             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                      | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                     |             |
| Base Type                        | MA_MessageAcknowledgmentConfigurationRequestMDT                                                                                                                                                                                                                                             |             |
| Message Structure                |                                                                                                                                                                                                                                                                                             |             |
| Field Name                       | Field Type                                                                                                                                                                                                                                                                                  | Field Usage |
| DefaultGroupMessageFreq          | DurationType                                                                                                                                                                                                                                                                                | Optional    |
| Description                      | This will override the current DefaultGroupMessageFreq setting.                                                                                                                                                                                                                             |             |
| DefaultMessagesToAck             | MA_AckMessageEnum                                                                                                                                                                                                                                                                           | Repeated    |
| Description                      | This will override the current DefaultMessageToAck setting.                                                                                                                                                                                                                                 |             |
| RequestedReceiverSpecificSetting | MA_AckReceiverSpecificConfigType                                                                                                                                                                                                                                                            | Optional    |
| Description                      | This is an individual configuration settings that is being requested to be configured.                                                                                                                                                                                                      |             |
| RequestID                        | RequestID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                      | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |             |
| RequestProcessingState           | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required    |
| Description                      | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |             |
| RequestProcessingStateReason     | CannotComplyType                                                                                                                                                                                                                                                                            | Optional    |
| Description                      | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |

**Table A-1-550: P2P UCI Extensions MA\_MessageAcknowledgmentConfigurationRequestMT - MessageData UCI Extension Field**

### 1.3.2.106 MA\_MessageAcknowledgmentConfigurationRequestStatusMT UCI Extension

| MA_MessageAcknowledgmentConfigurationRequestStatusMT |                                                                                                                               |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                                   | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                                   | MessageData : MA_MessageAcknowledgmentConfigurationRequestStatusMDT                                                           |

**Table A-1-551: P2P UCI Extensions MA\_MessageAcknowledgmentConfigurationRequestStatusMT UCI Extension**

| MessageData                  |                                                                                                                                                                                                                                                                                             |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                     |             |
| Base Type                    | MA_MessageAcknowledgmentConfigurationRequestStatusMDT                                                                                                                                                                                                                                       |             |
| Message Structure            |                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                  | Field Usage |
| RequestID                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |             |
| RequestProcessingState       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |             |
| RequestProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |

**Table A-1-552: P2P UCI Extensions MA\_MessageAcknowledgmentConfigurationRequestStatusMT - MessageData UCI Extension Field**

### 1.3.2.107 MA\_ResponseCommandMT UCI Extension

| MA_ResponseCommandMT |                                                                                                                               |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>   | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>   | MessageData : MA_ResponseCommandMDT                                                                                           |

**Table A-1-553: P2P UCI Extensions MA\_ResponseCommandMT UCI Extension**

| MessageData        |                                                                                         |
|--------------------|-----------------------------------------------------------------------------------------|
| <b>Description</b> | See annotations in child elements and messages/elements that use this type for details. |



|                   |                                                                                                                                                                                                                                                                                                    |             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Base Type         | MA_ResponseCommandMDT                                                                                                                                                                                                                                                                              |             |
| Message Structure |                                                                                                                                                                                                                                                                                                    |             |
| Field Name        | Field Type                                                                                                                                                                                                                                                                                         | Field Usage |
| Command           | MA_ResponseCommandType                                                                                                                                                                                                                                                                             | Repeated    |
| Description       | Indicates an instance of an Response Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message. |             |

Table A-1-554: P2P UCI Extensions MA\_ResponseCommandMT - MessageData UCI Extension Field

### 1.3.2.108 MA\_ResponseCommandStatusMT UCI Extension

|                                   |                                                                                                                               |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>MA_ResponseCommandStatusMT</b> |                                                                                                                               |
| <b>Description</b>                | See annotations in child elements and messages/elements that use this type for details.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                | MessageData : MA_ResponseCommandStatusMDT                                                                                     |

Table A-1-555: P2P UCI Extensions MA\_ResponseCommandStatusMT UCI Extension

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_ResponseCommandStatusMDT                                                                                                                                                                                                                                                                                                                                 |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| Activity                     | ResultingActivityType                                                                                                                                                                                                                                                                                                                                       | Repeated    |
| Description                  | Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.                                                                                                                                                                                                                    |             |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

Table A-1-556: P2P UCI Extensions MA\_ResponseCommandStatusMT - MessageData UCI Extension Field

## 1.3.2.109 MA\_PlatformOverrideStatusMT UCI Extension

| MA_PlatformOverrideStatusMT |                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------|
| Description                 | See annotations in child elements and messages/elements that use this type for details. |
|                             | See the XSD File for more info                                                          |
| Ext. Fields                 | MessageData : MA_PlatformOverrideStatusMDT                                              |

Table A-1-557: P2P UCI Extensions MA\_PlatformOverrideStatusMT UCI Extension

| MessageData              |                                                                                                                                                                                                                                                                                                                      |             |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description              | Indicates the data being conveyed by the message.                                                                                                                                                                                                                                                                    |             |
| Base Type                | MA_PlatformOverrideStatusMDT                                                                                                                                                                                                                                                                                         |             |
| Message Structure        |                                                                                                                                                                                                                                                                                                                      |             |
| Field Name               | Field Type                                                                                                                                                                                                                                                                                                           | Field Usage |
| PlatformOverrideStatus   | MA_PlatformDefaultAndOverrideDataType                                                                                                                                                                                                                                                                                | Repeated    |
| Description              | Indicates the current platform override setting(s) applied.                                                                                                                                                                                                                                                          |             |
| ActiveOverrides          | boolean                                                                                                                                                                                                                                                                                                              | Required    |
| Description              | When TRUE, indicates one or more overrides are active. When FALSE, indicates no overrides are active.                                                                                                                                                                                                                |             |
| ExecutingSystemOrPackage | MA_PackageSystemType                                                                                                                                                                                                                                                                                                 | Required    |
| Description              | Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.                             |             |
| ExecutionState           | RequirementExecutionStateEnum                                                                                                                                                                                                                                                                                        | Required    |
| Description              | Identifies the state of the Requirement.                                                                                                                                                                                                                                                                             |             |
| ExecutionStateReason     | CannotComplyType                                                                                                                                                                                                                                                                                                     | Optional    |
| Description              | Indicates details regarding the reason for and/or a contributing factor to the sibling State element. This element is only expected when the State is in a potentially undesirable state such as failed or completed with less than optimum results. It is strongly encouraged for transitions to failed or dropped. |             |
| ActualTiming             | RequirementTimingType                                                                                                                                                                                                                                                                                                | Repeated    |
| Description              | Indicates actual timing status for the Requirement. Allows for defining for each timing type (RequirementTimingEnum).                                                                                                                                                                                                |             |
| PercentCompleted         | PercentType                                                                                                                                                                                                                                                                                                          | Optional    |
| Description              | Percentage of assigned Requirement completion. The percentage increments until the Requirement has COMPLETED or FAILED. A percent complete of 0 indicates the Requirement has not yet begun execution, and a failed Requirement may have any percentage between 1 - 99.                                              |             |
| Traceability             | MA_RequirementStatusTraceabilityType                                                                                                                                                                                                                                                                                 | Optional    |
| Description              | Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update.                                                                                                                                                |             |

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|                | Traceability can vary depending on a given System's architecture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|                | For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands. |          |
| <b>Metrics</b> | MetricsType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Optional |
| Description    | Specifies the actual metrics associated with the Requirements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |

Table A-1-558: P2P UCI Extensions MA\_PlatformOverrideStatusMT - MessageData UCI Extension Field

## 1.3.2.110 MA\_GatewayActivityMT UCI Extension

| MA_GatewayActivityMT |                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>   | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>   | ObjectState : ObjectStateEnum<br>MessageData : MA_GatewayActivityMDT                                                                        |

Table A-1-559: P2P UCI Extensions MA\_GatewayActivityMT UCI Extension

| ObjectState         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>  | Indicates the state of the internal software object that the data in the message is based upon. The state of the software object is determined by the publisher. The publisher could be the original/master publisher or an intermediary re-publisher such as a persistence/data manager. This element is not a commanded CRUD operation. It is not an indication of the state of an instance of a message. It is not a mechanism for partial, incremental, or fragmentary publication of a message. It is a signal from a publisher to subscribers. For REMOVED state, in particular, some subscribers will mirror the publisher and treat the message as if the object no longer exists, but others, such as mission recorders, might not (in order to support retaining historical records of activity, for example). |
| <b>Base Type</b>    | ObjectStateEnum                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Message Structure   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Enumeration Literal | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>NEW</b>          | Indicates the internal software object corresponding to this message currently exists and this message is the first publication corresponding to the object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>UPDATED</b>      | Indicates the internal software object corresponding to this message currently exists and has been updated since the last publication of the message; this publication of the message may contain updates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                |                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>REMOVED</b> | Indicates the internal software object corresponding to this message no longer exists; there should be no further publications of this message corresponding to the object. |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table A-1-560: P2P UCI Extensions MA\_GatewayActivityMT - ObjectState UCI Extension Field

| MessageData       |                                                                                          |             |
|-------------------|------------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details.  |             |
| Base Type         | MA_GatewayActivityMDT                                                                    |             |
| Message Structure |                                                                                          |             |
| Field Name        | Field Type                                                                               | Field Usage |
| GatewayServiceID  | ServiceID_Type                                                                           | Required    |
| Description       | This element represents the ServiceID of the Gateway Service that sent this message.     |             |
| Activity          | MA_GatewayActivityType                                                                   | Repeated    |
| Description       | Indicates the dynamic operational characteristics of an Activity of a GatewayCapability. |             |

Table A-1-561: P2P UCI Extensions MA\_GatewayActivityMT - MessageData UCI Extension Field

### 1.3.2.111 MA\_DataDisseminationPlanOverrideRequestMT UCI Extension

| MA_DataDisseminationPlanOverrideRequestMT |                                                                                                                                             |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                        | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                        | MessageData : MA_DataDisseminationPlanOverrideRequestMDT                                                                                    |

Table A-1-562: P2P UCI Extensions MA\_DataDisseminationPlanOverrideRequestMT UCI Extension

| MessageData          |                                                                                                                                                                                                                                                                                                               |             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description          | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                       |             |
| Base Type            | MA_DataDisseminationPlanOverrideRequestMDT                                                                                                                                                                                                                                                                    |             |
| Message Structure    |                                                                                                                                                                                                                                                                                                               |             |
| Field Name           | Field Type                                                                                                                                                                                                                                                                                                    | Field Usage |
| AllocatedToSystemIDs | MA_PackageSystemType                                                                                                                                                                                                                                                                                          | Repeated    |
| Description          | Indicates the unique ID of the system(s) or package that is the recipient of this command.                                                                                                                                                                                                                    |             |
| AllocatedToServiceID | ServiceID_Type                                                                                                                                                                                                                                                                                                | Optional    |
| Description          | Indicates the unique ID of the service that is the recipient of this command.                                                                                                                                                                                                                                 |             |
| Request              | MA_DataDisseminationPlanOverrideRequestType                                                                                                                                                                                                                                                                   | Repeated    |
| Description          | Indicates an instance of a data plan override request. A data plan override request message can have multiple request instances to allow a sequence of related requests in a single coherent message. Each request instance reports status via a separate, independent DataPlanOverrideRequestStatus message. |             |

**Table A-1-563: P2P UCI Extensions MA\_DataDisseminationPlanOverrideRequestMT - MessageData UCI Extension Field**

### 1.3.2.112 MA\_DataDisseminationPlanOverrideRequestStatusMT UCI Extension

| MA_DataDisseminationPlanOverrideRequestStatusMT |                                                                                                                                             |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                              | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                              | MessageData : MA_DataDisseminationPlanOverrideRequestStatusMDT                                                                              |

**Table A-1-564: P2P UCI Extensions MA\_DataDisseminationPlanOverrideRequestStatusMT UCI Extension**

| MessageData                  |                                                                                                                                                                                                                                                                                             |             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                     |             |
| Base Type                    | MA_DataDisseminationPlanOverrideRequestStatusMDT                                                                                                                                                                                                                                            |             |
| Message Structure            |                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                  | Field Usage |
| PlanID                       | MA_DataDisseminationPlanID_Type                                                                                                                                                                                                                                                             | Required    |
| Description                  | Indicates the unique ID of the DataDisseminationPlan that this override modification status applies to.                                                                                                                                                                                     |             |
| EffectivityActivationStatus  | DataPlanEffectivityActivationStatusType                                                                                                                                                                                                                                                     | Repeated    |
| Description                  | This element indicates the status of pre-mission planned Data-Link gateway settings. It is used to indicate details about specific planned effectivity activations.                                                                                                                         |             |
| OperatorModificationStatus   | OperatorGatewayDesignationStatusType                                                                                                                                                                                                                                                        | Repeated    |
| Description                  | This element indicates the status of operator designations. It is used to indicate details about specific designation configurations.                                                                                                                                                       |             |
| RequestID                    | RequestID_Type                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the request message corresponding to this status message.                                                                                                                                                                                                        |             |
| RequestProcessingState       | RequestProcessingStateEnum                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.                                                                                                                                                                |             |
| RequestProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected. |             |

**Table A-1-565: P2P UCI Extensions MA\_DataDisseminationPlanOverrideRequestStatusMT - MessageData UCI Extension Field**

### 1.3.2.113 MA\_DataDisseminationPlanActivationCommandStatusMT UCI Extension

| MA_DataDisseminationPlanActivationCommandStatusMT |                                                                                                                                             |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                                | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                                | MessageData : MA_DataDisseminationPlanActivationCommandStatusMDT                                                                            |

**Table A-1-566: P2P UCI Extensions MA\_DataDisseminationPlanActivationCommandStatusMT UCI Extension**

| MessageData                  |                                                                                                                                                                                                                                                                                                                                                             |             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description                  | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                                                                                     |             |
| Base Type                    | MA_DataDisseminationPlanActivationCommandStatusMDT                                                                                                                                                                                                                                                                                                          |             |
| Message Structure            |                                                                                                                                                                                                                                                                                                                                                             |             |
| Field Name                   | Field Type                                                                                                                                                                                                                                                                                                                                                  | Field Usage |
| CommandID                    | CommandID_Type                                                                                                                                                                                                                                                                                                                                              | Required    |
| Description                  | Indicates the unique ID of the command corresponding to this status message.                                                                                                                                                                                                                                                                                |             |
| CommandProcessingState       | CommandProcessingStateEnum                                                                                                                                                                                                                                                                                                                                  | Required    |
| Description                  | Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details. |             |
| CommandProcessingStateReason | CannotComplyType                                                                                                                                                                                                                                                                                                                                            | Optional    |
| Description                  | Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.                                                                 |             |

**Table A-1-567: P2P UCI Extensions MA\_DataDisseminationPlanActivationCommandStatusMT - MessageData UCI Extension Field**

### 1.3.2.114 MA\_DataDisseminationPlanActivationCommandMT UCI Extension

| MA_DataDisseminationPlanActivationCommandMT |                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                          | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                          | MessageData : MA_DataDisseminationPlanActivationCommandMDT                                                                                  |

**Table A-1-568: P2P UCI Extensions MA\_DataDisseminationPlanActivationCommandMT UCI Extension**

| MessageData |
|-------------|
|-------------|

|                       |                                                                                                                                                               |             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description           | See annotations in child elements and messages/elements that use this type for details.                                                                       |             |
| Base Type             | MA_DataDisseminationPlanActivationCommandMDT                                                                                                                  |             |
| Message Structure     |                                                                                                                                                               |             |
| Field Name            | Field Type                                                                                                                                                    | Field Usage |
| PlanID                | MA_DataDisseminationPlanID_Type                                                                                                                               | Required    |
| Description           | The DataDisseminationPlan that this command pertains to.                                                                                                      |             |
| CommandedState        | PlanActivationCommandEnum                                                                                                                                     | Required    |
| Description           | Indicates the activation to perform on the associated DataDisseminationPlan. See enumerate annotations for further details.                                   |             |
| Applicability         | MA_PackageSystemType                                                                                                                                          | Repeated    |
| Description           | The System or Package IDs, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual Systems as needed.             |             |
| EffectivityActivation | MA_DataDisseminationPlanEffectivityActivationType                                                                                                             | Repeated    |
| Description           | Indicates an Effectivity of the DataDisseminationPlan to activate.                                                                                            |             |
| CommandID             | CommandID_Type                                                                                                                                                | Required    |
| Description           | Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.                                    |             |
| CommandState          | CommandStateEnum                                                                                                                                              | Required    |
| Description           | The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command. |             |

**Table A-1-569: P2P UCI Extensions MA\_DataDisseminationPlanActivationCommandMT - MessageData UCI Extension Field**

### 1.3.2.115 MA\_DataDisseminationPlanMT UCI Extension

|                                   |                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------|
| <b>MA_DataDisseminationPlanMT</b> |                                                                                         |
| <b>Description</b>                | See annotations in child elements and messages/elements that use this type for details. |
|                                   | See the XSD File for more info                                                          |
| <b>Ext. Fields</b>                | ObjectState : ObjectStateEnum<br>MessageData : MA_DataDisseminationPlanMDT              |

**Table A-1-570: P2P UCI Extensions MA\_DataDisseminationPlanMT UCI Extension**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ObjectState</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Description</b> | Indicates the state of the internal software object that the data in the message is based upon. The state of the software object is determined by the publisher. The publisher could be the original/master publisher or an intermediary re-publisher such as a persistence/data manager. This element is not a commanded CRUD operation. It is not an indication of the state of an instance of a message. It is not a mechanism for partial, incremental, or fragmentary publication of a message. It is a signal from a publisher to subscribers. For REMOVED state, in particular, some subscribers will mirror the publisher and |



|                            |                                                                                                                                                                                                            |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                            | treat the message as if the object no longer exists, but others, such as mission recorders, might not (in order to support retaining historical records of activity, for example).                         |
| <b>Base Type</b>           | ObjectStateEnum                                                                                                                                                                                            |
| <b>Message Structure</b>   |                                                                                                                                                                                                            |
| <b>Enumeration Literal</b> | <b>Description</b>                                                                                                                                                                                         |
| <b>NEW</b>                 | Indicates the internal software object corresponding to this message currently exists and this message is the first publication corresponding to the object.                                               |
| <b>UPDATED</b>             | Indicates the internal software object corresponding to this message currently exists and has been updated since the last publication of the message; this publication of the message may contain updates. |
| <b>REMOVED</b>             | Indicates the internal software object corresponding to this message no longer exists; there should be no further publications of this message corresponding to the object.                                |

Table A-1-571: P2P UCI Extensions **MA\_DataDisseminationPlanMT - ObjectState UCI Extension Field**

| MessageData        |                                                                                                                                                                                                                                                                                                  |             |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description        | See annotations in child elements and messages/elements that use this type for details.                                                                                                                                                                                                          |             |
| Base Type          | MA_DataDisseminationPlanMDT                                                                                                                                                                                                                                                                      |             |
| Message Structure  |                                                                                                                                                                                                                                                                                                  |             |
| Field Name         | Field Type                                                                                                                                                                                                                                                                                       | Field Usage |
| PlanID             | MA_DataDisseminationPlanID_Type                                                                                                                                                                                                                                                                  | Required    |
| Description        | Indicates the unique ID of the DataDisseminationPlan.                                                                                                                                                                                                                                            |             |
| AssociatedPlanID   | MA_DataDisseminationPlanID_Type                                                                                                                                                                                                                                                                  | Repeated    |
| Description        | Indicates the unique ID of a DataDisseminationPlan that is a parent to this one.                                                                                                                                                                                                                 |             |
| Applicability      | MA_PlanApplicabilityType                                                                                                                                                                                                                                                                         | Repeated    |
| Description        | Indicates the System or Systems which have a direct relationship with the DataDisseminationPlan. See annotations for the underlying type and its child elements for details. If Applicability is not defined then it indicates this DataDisseminationPlan applies to ALL systems.                |             |
| IER                | MA_CommIER_Planned_Type                                                                                                                                                                                                                                                                          | Repeated    |
| Description        | Indicates a planned Information Exchange Requirement (IER). An IER can reflect the information needs of local or remote Systems or third party Systems coordinating on behalf of other Systems. The planned assignment of an IER to comms resources is given in the sibling Effectivity element. |             |
| IER_DesiredService | MA_CommIER_DesiredServicesType                                                                                                                                                                                                                                                                   | Repeated    |
| Description        | Defines the desired service constraints for an IER (Information Exchange Requirement), allowing multi-constrained solutions. All child elements are optional; when omitted software should apply no explicit bound for that metric.                                                              |             |
| RateGroup          | MA_CommRateGroupType                                                                                                                                                                                                                                                                             | Repeated    |
| Description        | Indicates a RateGroup that can be associated with an IER to configure its transmit rate.                                                                                                                                                                                                         |             |



|                                    |                                                                                                                                                                                                                                      |          |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Effectivity</b>                 | MA_CommDataDisseminationEffectivityType                                                                                                                                                                                              | Repeated |
| Description                        | Indicates a DataDisseminationPlan "effectivity set" of items that are used together.                                                                                                                                                 |          |
| <b>Vector</b>                      | MA_CommVectorPlannedType                                                                                                                                                                                                             | Repeated |
| Description                        | Indicates a planned Vector. A vector describes a logical connection between nodes initiating on a specific NetworkLink (comms resource).                                                                                             |          |
| <b>DistributionGroup</b>           | MA_DistributionGroupType                                                                                                                                                                                                             | Repeated |
| Description                        | Indicates a grouping of tiles or node groups and associated ip address.                                                                                                                                                              |          |
| <b>DistributionGroupMembership</b> | MA_DistributionGroupMembershipType                                                                                                                                                                                                   | Repeated |
| Description                        | Indicates a system's membership to a multicast group.                                                                                                                                                                                |          |
| <b>Gateway</b>                     | MA_GatewayReferenceType                                                                                                                                                                                                              | Repeated |
| Description                        | Indicates a planned Gateway local to the system for which this DataDisseminationPlan is applicable to, with a summary of its Capability to be used for dissemination of IERs in the DataDisseminationPlan.                           |          |
| <b>Trigger</b>                     | MA_CommTriggerType                                                                                                                                                                                                                   | Repeated |
| Description                        | Signifies available triggers which would automatically invoke this DataDisseminationPlan or parts within this DataDisseminationPlan. These Triggers that can be associated with activation of an Effectivity by using the TriggerID. |          |

Table A-1-572: P2P UCI Extensions MA\_DataDisseminationPlanMT - MessageData UCI Extension Field

## 1.3.2.116 MA\_DataDisseminationPlanActivationStatusMT UCI Extension

| MA_DataDisseminationPlanActivationStatusMT |                                                                                                                                             |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>                         | See the annotation in the associated message for an overall description of the message and this type.<br><br>See the XSD File for more info |
| <b>Ext. Fields</b>                         | MessageData : MA_DataDisseminationPlanActivationStatusMDT                                                                                   |

Table A-1-573: P2P UCI Extensions MA\_DataDisseminationPlanActivationStatusMT UCI Extension

| MessageData       |                                                                                         |             |
|-------------------|-----------------------------------------------------------------------------------------|-------------|
| Description       | See annotations in child elements and messages/elements that use this type for details. |             |
| Base Type         | MA_DataDisseminationPlanActivationStatusMDT                                             |             |
| Message Structure |                                                                                         |             |
| Field Name        | Field Type                                                                              | Field Usage |
| PlanID            | MA_DataDisseminationPlanID_Type                                                         | Required    |
| Description       | The ID of the DataDisseminationPlan this status pertains to.                            |             |
| ActivationState   | PlanActivationStateEnum                                                                 | Required    |
| Description       | Indicates the activation state of the associated DataDisseminationPlan.                 |             |
| StateReason       | MA_CannotComplyType                                                                     | Optional    |
| Description       | Indicates details regarding the reason for and/or a contributing factor to the sibling  |             |

|                            |                                                                                                                                       |          |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------|
|                            | ActivationState element.                                                                                                              |          |
| EffectivityActivationState | MA_DataDisseminationPlanEffectivityStateType                                                                                          | Repeated |
| Description                | Indicates the activation state of an Effectivity of the associated DataDisseminationPlan.                                             |          |
| OverrideModificationState  | MA_OvrerriderModificationStatusType                                                                                                   | Repeated |
| Description                | This element indicates the status of operator designations. It is used to indicate details about specific designation configurations. |          |

Table A-1-574: P2P UCI Extensions MA\_DataDisseminationPlanActivationStatusMT - MessageData UCI Extension Field